GreenHub integrated model report

Non-PEF BioSTEAM screening assessment

Complete scientific interpretation and source-linked figure library

Prepared 2 November 2026

All 266 figure PNGs are preserved unchanged. A full-page figure atlas follows this report in the PDF.

Evidence status: screening process model; partial TEA; electricity-only attributional climate burden. No validated joint optimum is claimed.

1. Executive summary

This report asks which GreenHub valorization route has the strongest joint technical, economic and environmental performance. It integrates the current 266-figure non-PEF library, the saved BioSTEAM decision batch, exact selected reruns, and the repository documentation. The complete figure atlas follows the main report in the PDF. Each source PNG is copied without graphical changes.

The F1 screening economic endpoint (full-front rank 0) is F1 PR3 AD1 BG0 NR3 A at 12,173 t/year wet grass. Its candidate-adjusted profit objective is EUR 75,511/year and its electricity-only attributional burden is 23.29 t CO2-eq/year (1.91 kg/t). The lowest-burden endpoint (rank 10) bypasses protein, digestion, gas upgrading and targeted nutrients. It has EUR -9,048/year screening profit and 6.41 t CO2-eq/year. Lower annual burden also reflects lower throughput.

These endpoints are screening candidates, not validated plant optima. The selected reports fail the technical ranking gate because of unresolved storage evidence; installed CAPEX, complete OPEX, full LCA and product quality are incomplete. The most economic selected route exports raw gas under a product ID that can receive a biomethane benchmark price. That quality-price mismatch is material to its apparent advantage. The model therefore cannot yet defend a single technical-economic-environmental winner.

1. Executive summary

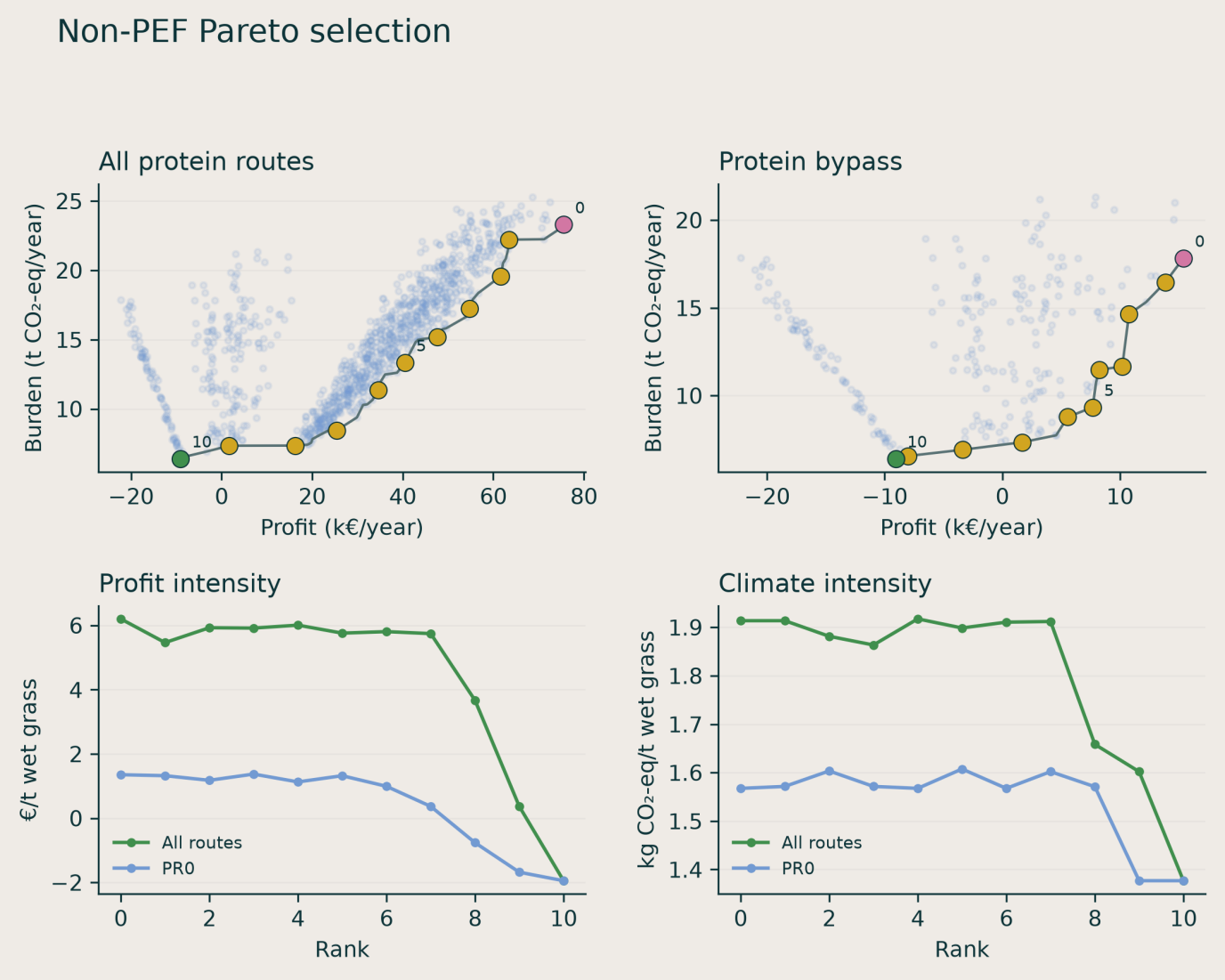


Figure: 07 optimization/non pef pareto dashboard.png

2. Scope and model architecture

The physical system begins with wet roadside grass, then preparation and mechanical separation into liquid, fibre cake and rejects. The liquid pathway can recover protein, pass through anaerobic digestion, condition or upgrade gas, and recover nitrogen, phosphorus and other nutrient-rich solids from digestate. Press cake leaves this modeled liquid-valorization boundary; downstream panel production is outside the process. The application forms one connected BioSTEAM flowsheet and applies TEA and LCA services after simulation [D1, D7-D11].

This edition fixes feed preparation at F1, excluding PEF before Pareto selection. The reference route is F1 PR3 AD1 BG2 NR4 A at 1,000 kg wet grass/hour and 8,760 hours/year. All 33 selected candidates use F1. The package also contains the earlier F2 manuscript and its data as historical documentation [H1]; it must not be used as the current baseline.

The current model is structural. It conserves material and estimates utilities and some equipment sizes, but selected yield, kinetic, transfer, product-quality and cost assumptions lack project validation. A numerical balance pass is not proof of industrial performance [D2, D3].

2. Scope and model architecture

Figure: 01 baseline/combined/01 grass to products.png

3. Sources, functional units and calculations

Every figure is indexed in portable_figure_index.csv with its scenario, units, source and interpretation notes. The process functional unit for the flow maps is 1 kg wet grass at the plant inlet. First-stage grass = liquid fraction + fibre cake + preparation rejects closes exactly. Later columns show selected intermediate and product stream masses and cannot be summed into a closed Sankey without omitted water, reagents, evaporation and side streams [D6].

Steady rates are annualized using the explicit 8,760-hour schedule. Dry gas in the dense figures is converted from saved component masses to Nm3 at 273.15 K and 101,325 Pa. The older group comparison uses a TEA gas sales quantity at 293.15 K, so those m3 figures require temperature conversion before comparing with Nm3. Raw BG0 export and upgraded gas are distinct streams. Solid nutrient-product amounts are stream masses, not pure elemental N, P or K [D6].

The 20-year financial projection repeats the saved nominal annual operating account and subtracts subsidy-adjusted partial direct capital at year zero. NPV discounts that same cash account at the report's 8% rate. It omits financing, replacement, working capital, inflation, terminal value and uncaptured cost lines. Electricity-only climate burden equals modeled purchased kWh multiplied by the screening factor 0.20 kg CO2-eq/kWh. Heat and other environmental inventory remain separate and uncharacterized [D4, D6].

3. Sources, functional units and calculations

Figure: 01 baseline/combined/02 material and products.png

4. F1 baseline performance

The F1 reference processes 8,760 t/year at 1,000 kg/hour. The saved simulation produces 40.39 t/year dry protein stream, 73.99 t/year ammonium stream, and 0.829 t/year phosphorus product stream. Electricity is 85.43 MWh/year and positive heating duty is 335.01 MWh/year.

For an explicit benchmark-price screen with no candidate savings or subsidy, the baseline records EUR 78,156/year revenue, EUR 17,009/year included OPEX and EUR 50,000 partial net capital. Its electricity-only climate screen is 17.09 t CO2-eq/year. These accounts do not complete the model's full TEA or LCA.

The baseline process route contains protein recovery, mesophilic digestion, water scrubbing and combined P/N recovery. Several recovered streams are small relative to wet grass because water remains in the inlet and because some material exits as fibre cake or residual liquid. Stream recovery should be interpreted against measured product composition and quality before sale or land application [D2, D3].

Indicator	F1 baseline	Unit
Wet grass processed	8,760	t/year
Protein stream	40.39	t/year
Ammonium product stream	73.99	t/year
Phosphorus product stream	0.829	t/year
Electricity	85.43	MWh/year
Heating	335.01	MWh/year
Screening revenue	78.16	EUR thousand/year
Included OPEX	17.01	EUR thousand/year
Electricity burden	17.09	t CO2-eq/year

4. F1 baseline performance

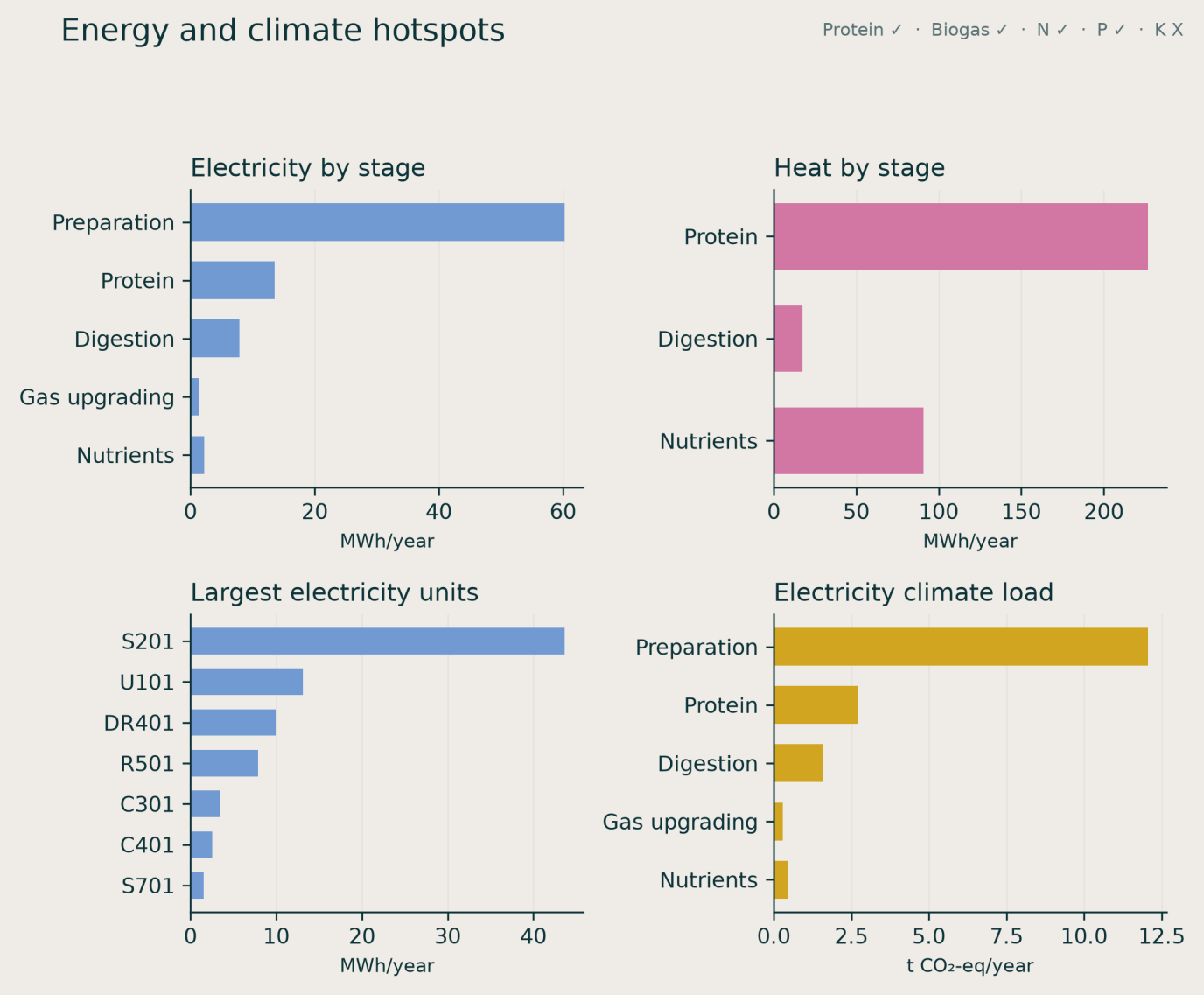


Figure: 01 baseline/combined/03 energy and climate.png

4. F1 baseline performance

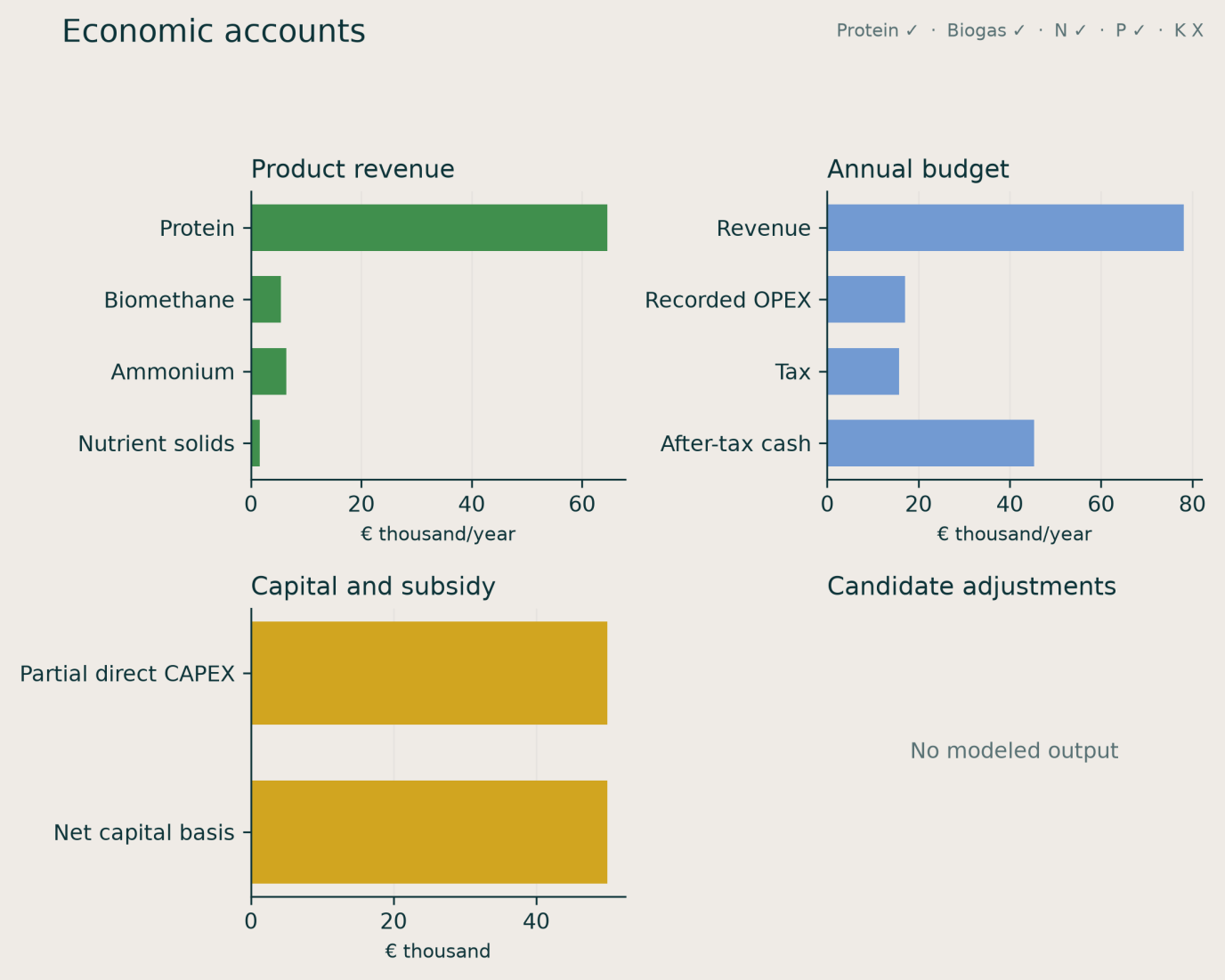


Figure: 01 baseline/combined/04 economic accounts.png

5. Technical pathways and hotspots

The six-page set for every candidate pairs feed split, product amounts, utility consumption, partial financial accounts, cash projection and model evidence. Cross-route technical panels normalize solid and gas production, electricity and heat to wet grass throughput. This avoids attributing differences caused solely by sampled plant size to technology selection.

The modeled preparation stage is a prominent electricity hotspot in the selected F1 cases. Protein and nutrient operations can add substantial thermal duty. Individual equipment power is shown in the candidate energy page. Reagent mass is intentionally absent from the hotspot plots because a reliable cross-route cost and environmental characterization has not been assembled [D3, D6].

No selected candidate contains a calibrated time trajectory. The digestion documentation explicitly uses steady-state fate fractions where kinetic and time-course evidence is missing. Consequently, this report does not infer VFA or microbial growth curves [D9].

5. Technical pathways and hotspots

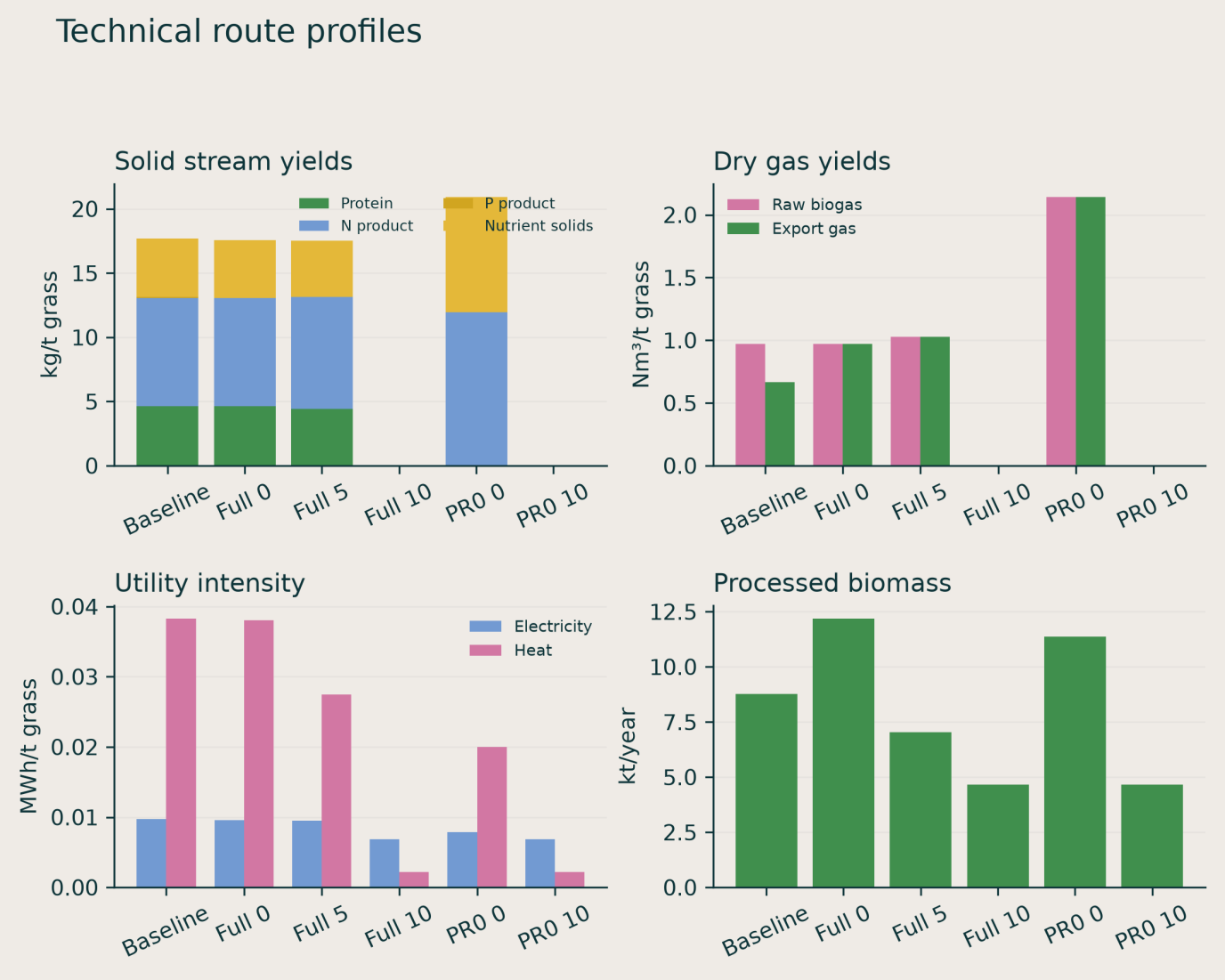


Figure: 02 technical/non pef route profiles.png

5. Technical pathways and hotspots

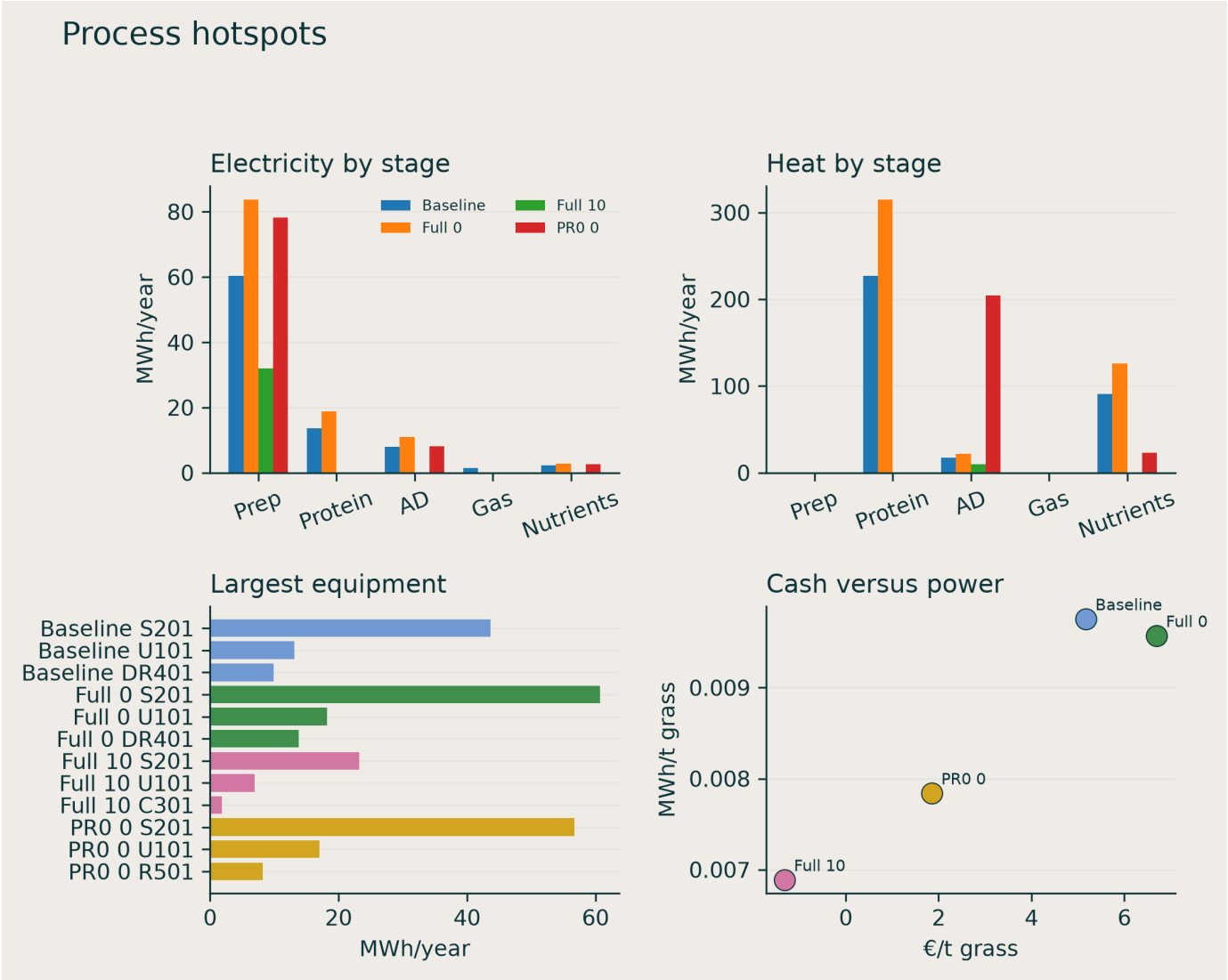


Figure: 05 hotspots/non pef utility cost hotspots.png

6. Techno-economic screening

Across the full F1 frontier, rank 0 has EUR 129,335/year modeled benchmark revenue, EUR 19,399/year recorded OPEX, EUR 59,514 subsidy-adjusted partial capital and a repeated-account NPV of EUR 741,382. Rank 10 has no priced recovered product and a negative screening profit. These numbers are incomplete accounts, not bankable returns.

Product-level revenue is shown where a saved price line exists. Shared equipment cost is not allocated artificially to individual products. Paired protein-bypass simulations estimate incremental pathway effects with throughput and sampled economics held fixed. The economic sensitivity chart varies price, included OPEX and capital by +/-10%, and discount from 6% to 10%; these are hypothetical arithmetic changes, not new process simulations [D6].

A specific price-to-quality problem affects BG0: its raw gas export is stored under the product ID biomethane and can receive the biomethane benchmark price in the saved TEA. Figures now label the stream Raw gas export but retain recorded revenue. No sale, injection qualification or raw-gas market price is demonstrated. Economic rankings involving BG0 must be revisited after the product and pricing boundary is corrected.

6. Techno-economic screening

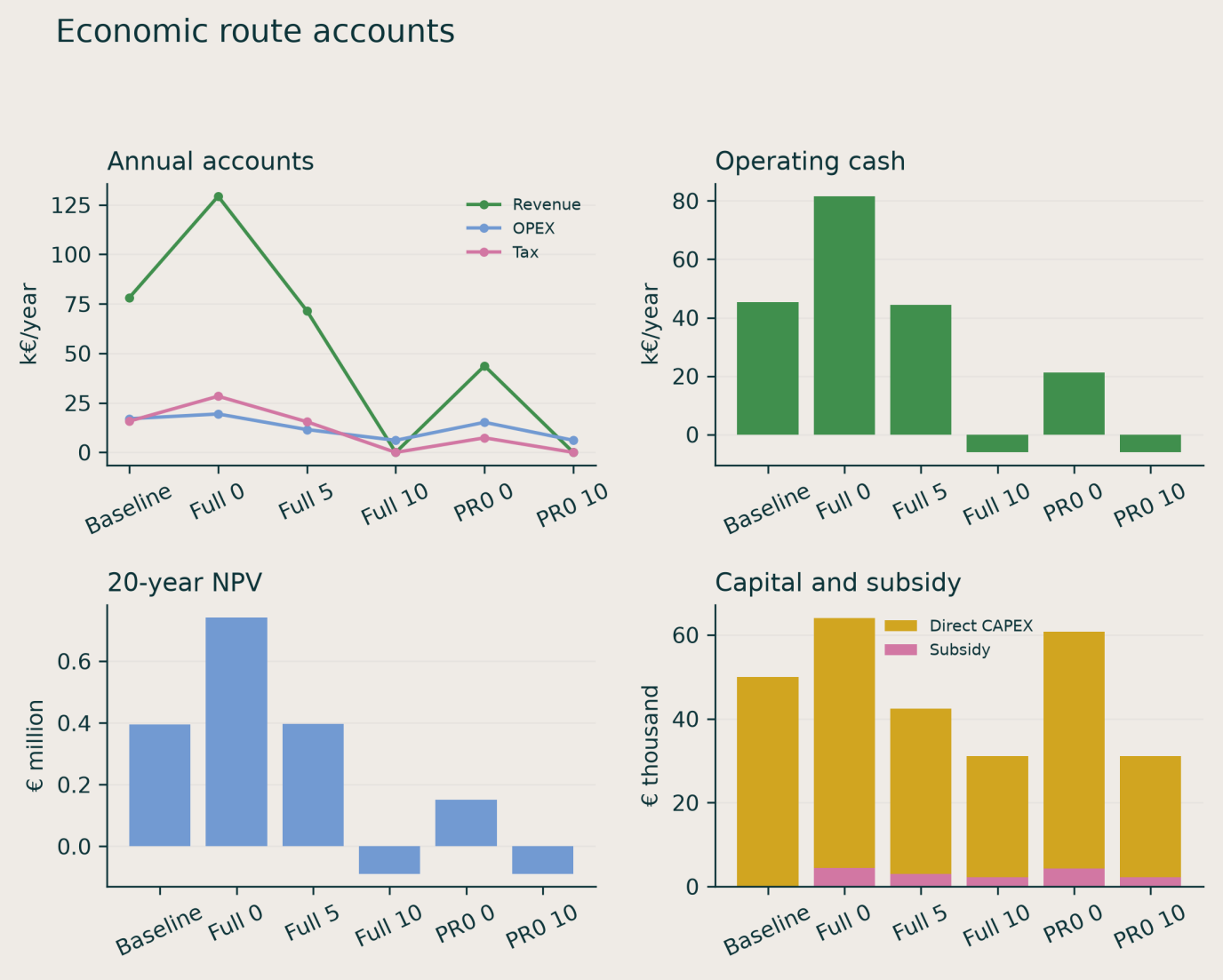


Figure: 03 tea/non pef economic accounts.png

6. Techno-economic screening

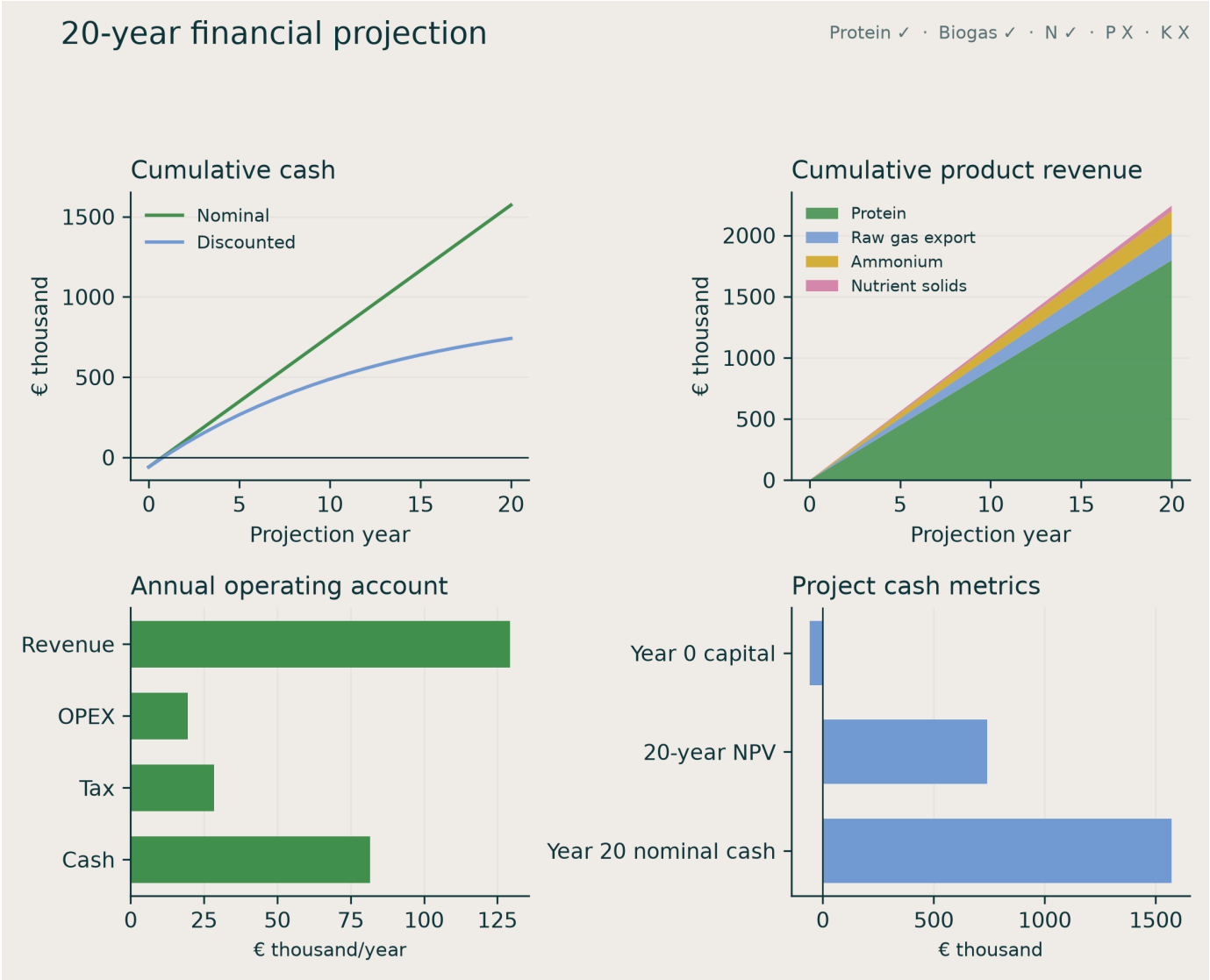


Figure: 08 candidate comparison/biosteam/all/rank 00/combined/05 cashflow and product revenue.png

6. Techno-economic screening

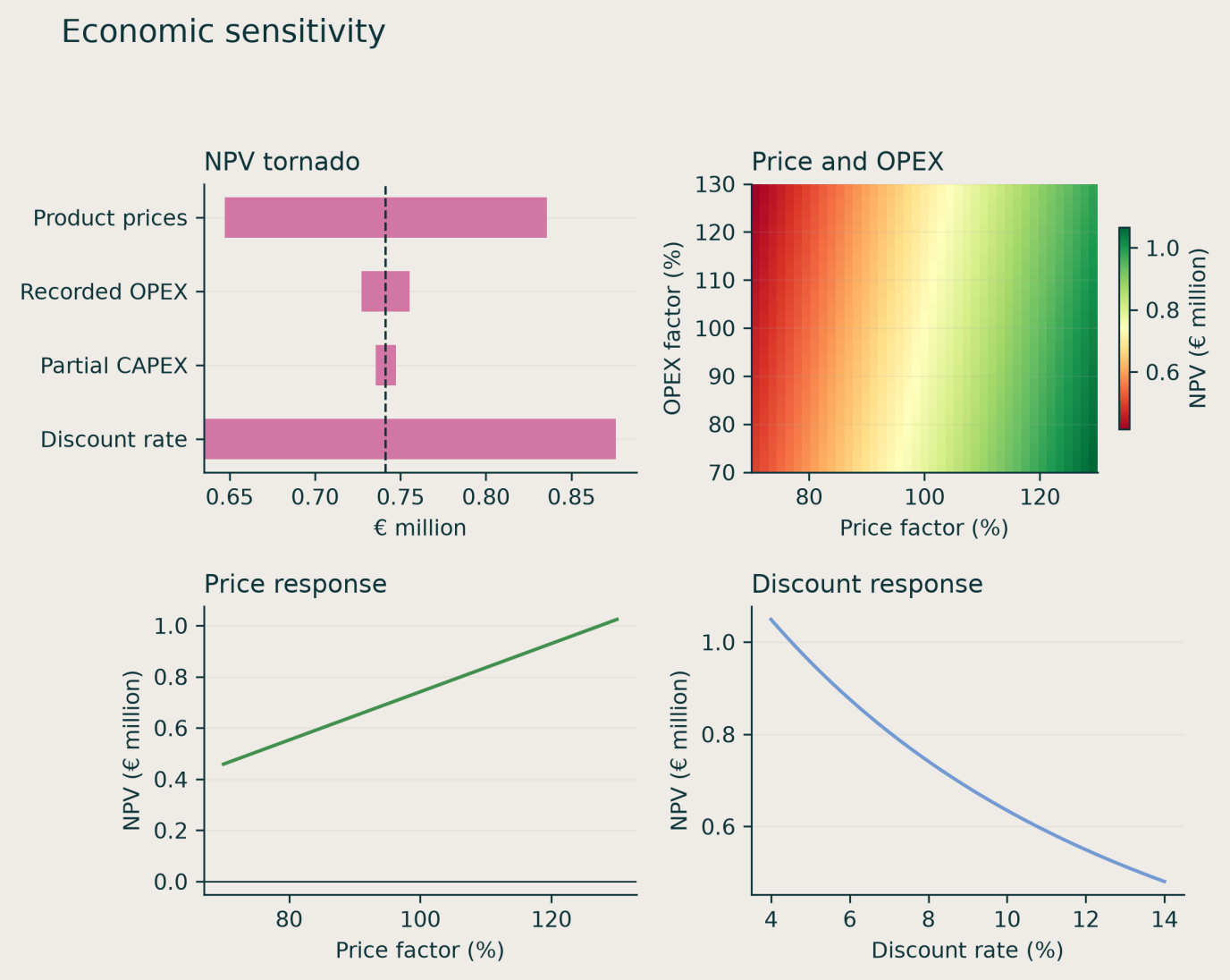


Figure: 09 uncertainty/non pef economic tornado.png

7. Attributional climate and LCA boundary

The displayed climate numbers characterize modeled purchased electricity only at 0.20 kg CO2-eq/kWh. This is an attributional burden, never a product displacement credit or climate offset. The saved LCA inventory also records heat and other flows, but factors, fate and boundary evidence remain incomplete. The report therefore keeps heat in MWh and counts missing factor records instead of inventing CO2-equivalent totals [D4, D6].

Full-front rank 0 has 23.29 t CO2-eq/year annual electricity burden at 12,173 t/year feed. Rank 10 has 6.41 t CO2-eq/year at 4,652 t/year. Their feed-normalized burdens are 1.91 and 1.38 kg/t, respectively. The environmental endpoint is a low-activity bypass configuration, not evidence that abandoning product recovery has lower whole-life-cycle impact.

Factor-sensitivity panels vary the electricity factor explicitly. These sweeps show how strongly the partial burden depends on electricity sourcing, but do not fill the missing heat, materials, transport, storage, direct-emission, product-use or substitution inventories. No comparative full-LCA claim is made.

7. Attributional climate and LCA boundary

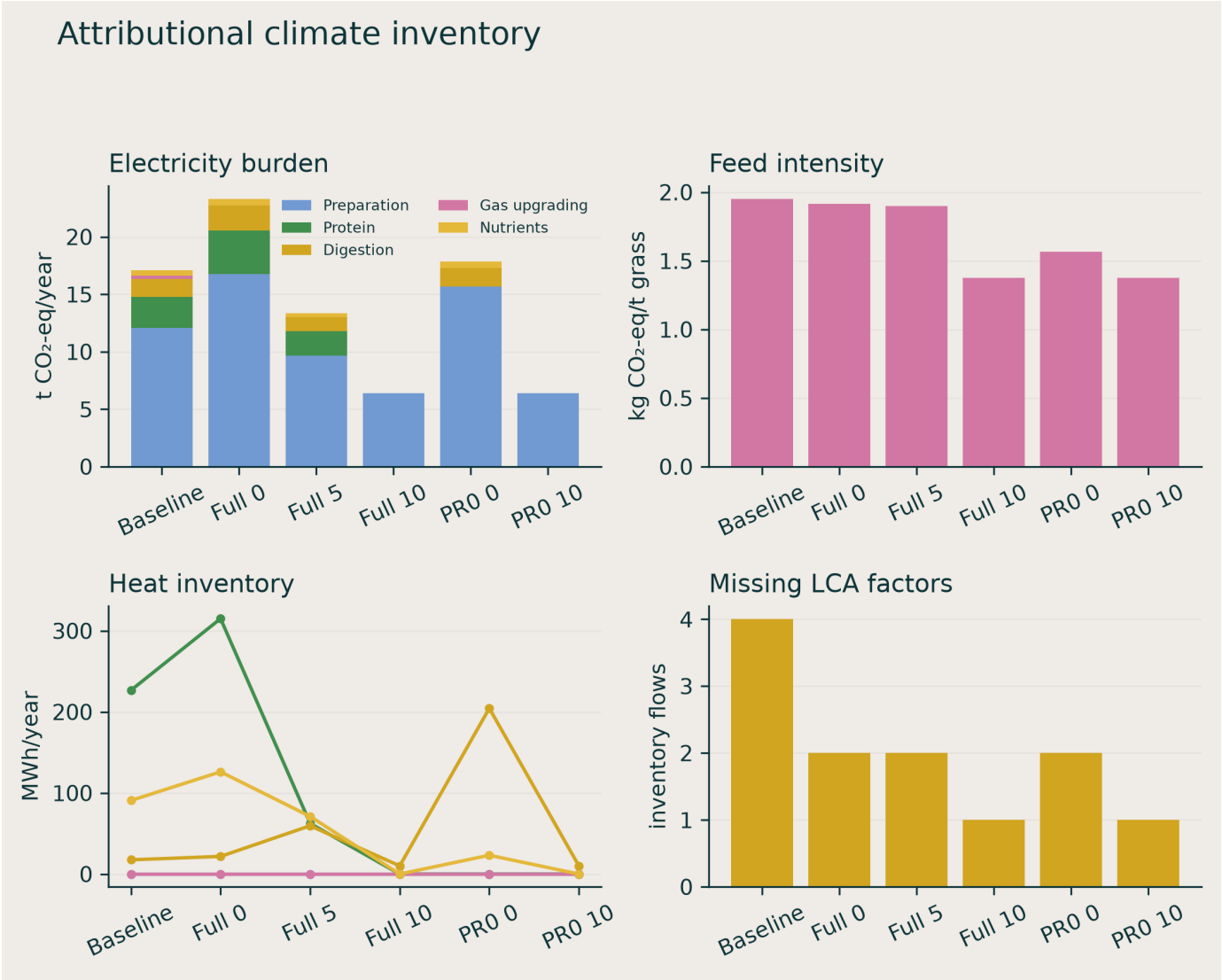


Figure: 04 lca/non pef attributional burden.png

7. Attributional climate and LCA boundary

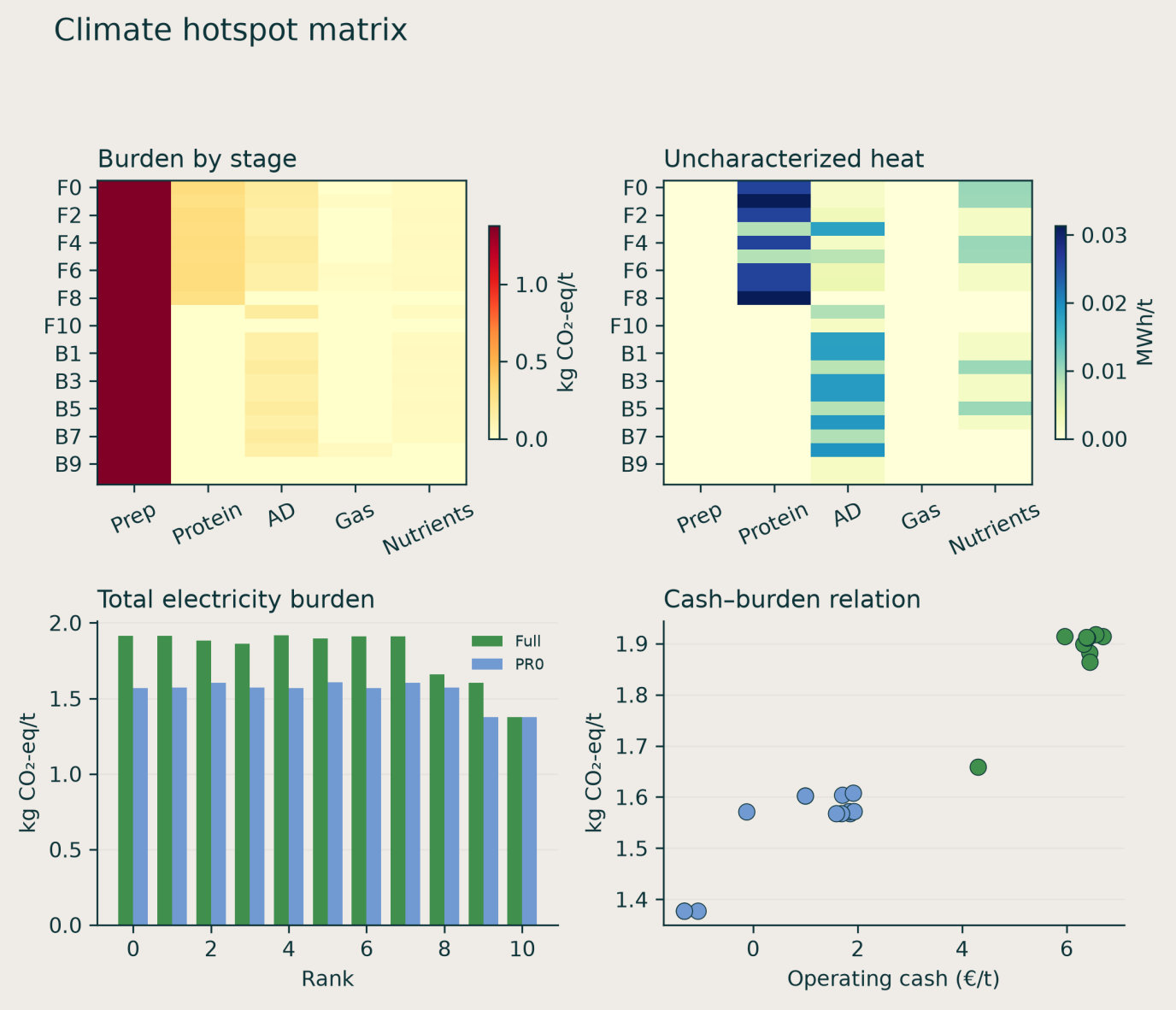


Figure: 04 lca/non pef climate hotspot matrix.png

7. Attributional climate and LCA boundary

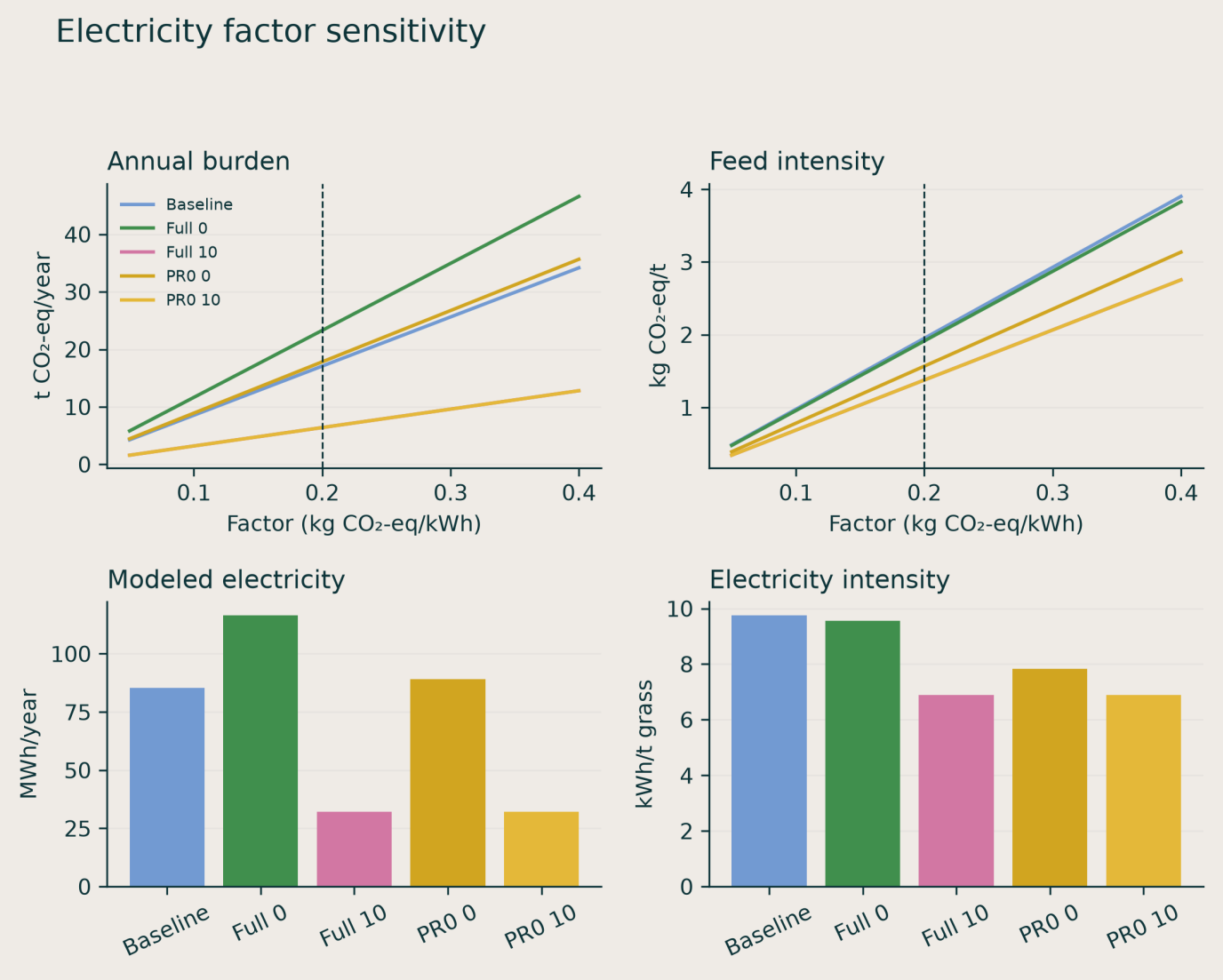


Figure: 04 lca/non pef electricity factor sensitivity.png

8. Pareto selection and candidate ranking

The completed saved batch contains 2,000 sampled configurations. Filtering to F1 before objective eligibility leaves 1049 eligible full-route cases and 247 eligible PR0 cases. Their two-objective screening fronts contain 42 and 15 points. Eleven points are selected on each front by normalized arc length. Rank 0 is the most economic selected point (rose); rank 10 has the lowest electricity-only burden (green) [D5, D6].

The full-front rank 5 is F1 PR1 AD1 BG0 NR3 A at 7,026 t/year with EUR 40,468/year screening profit and 13.34 t CO2-eq/year electricity burden. It is a useful intermediate trade-off example, not a formally validated MCDA optimum. The independent PR0 front has different throughput and route choices, so matching rank numbers across fronts does not isolate protein effects.

Selected candidate reports carry eligible_for_ranking=false because of unresolved storage blockers even though their screening objectives were finite in the original batch. This distinction is retained. Pareto status expresses non-domination within a partial model and sampled design space; it does not certify engineering feasibility or a complete economic-environmental optimum.

Set	Eligible F1 designs	Front points	Selected ranks
Full routes	1049	42	0-10
Protein bypass	247	15	0-10

8. Pareto selection and candidate ranking

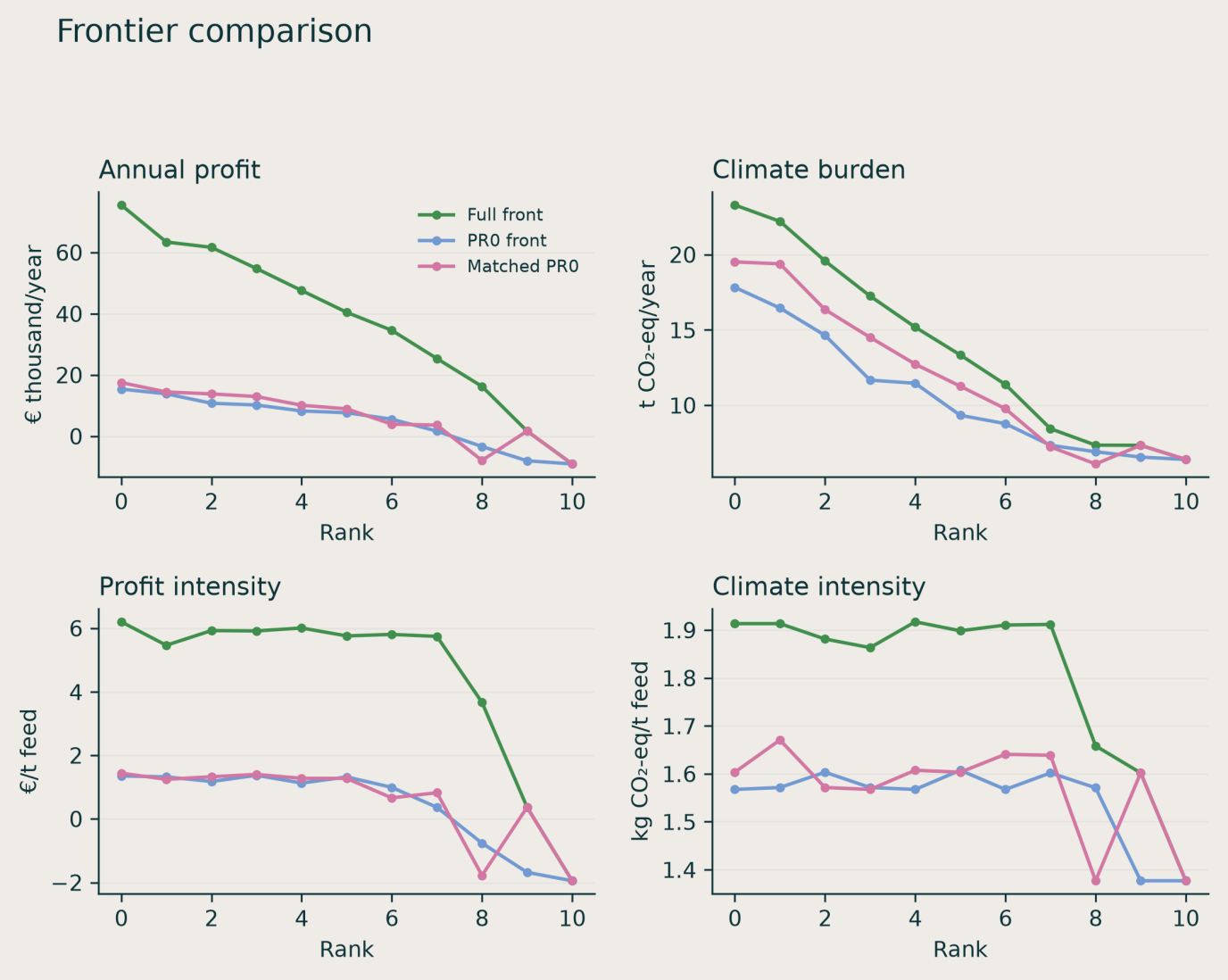


Figure: 06 scenario comparison/biosteam dense/three series profit climate.png

9. Protein bypass intervention

At the full-front economic endpoint, changing only PR3 to PR0 while retaining throughput and sampled economic assumptions reduces the saved profit objective by EUR 58 thousand/year and lowers electricity-only burden by 3.78 t CO2-eq/year. The matched route is F1 PR0 AD1 BG0 NR3 A. This is an incremental simulation comparison, not a shared-equipment allocation.

The paired differences vary across ranks because the original protein technologies, downstream routes and feed rates differ. Some high ranks already use PR0, producing zero intervention difference. The volcano plot has no statistical significance axis; it maps paired changes in screening profit and burden [D6].

9. Protein bypass intervention

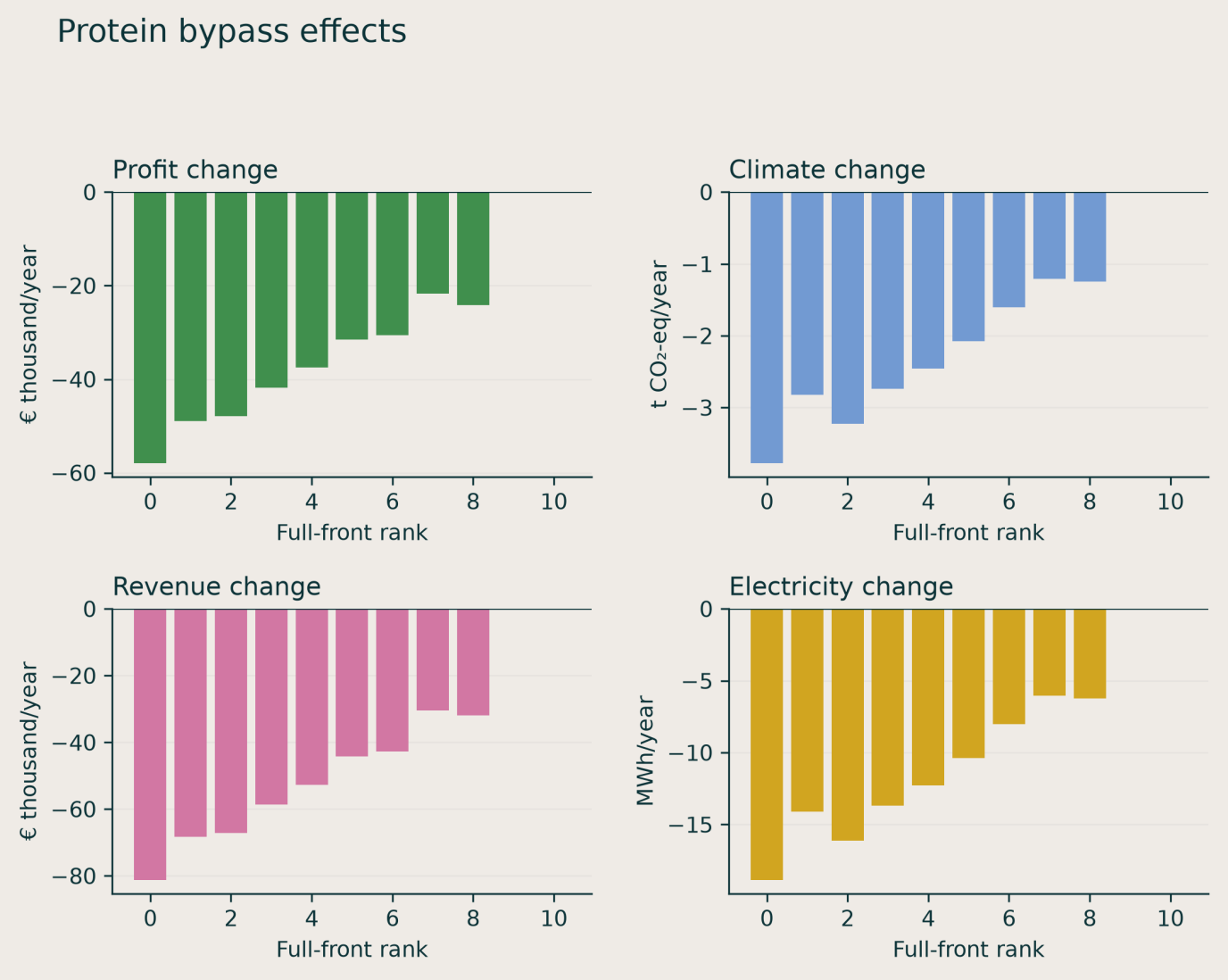


Figure: 06 scenario comparison/biosteam dense/matched protein bypass effects.png

9. Protein bypass intervention

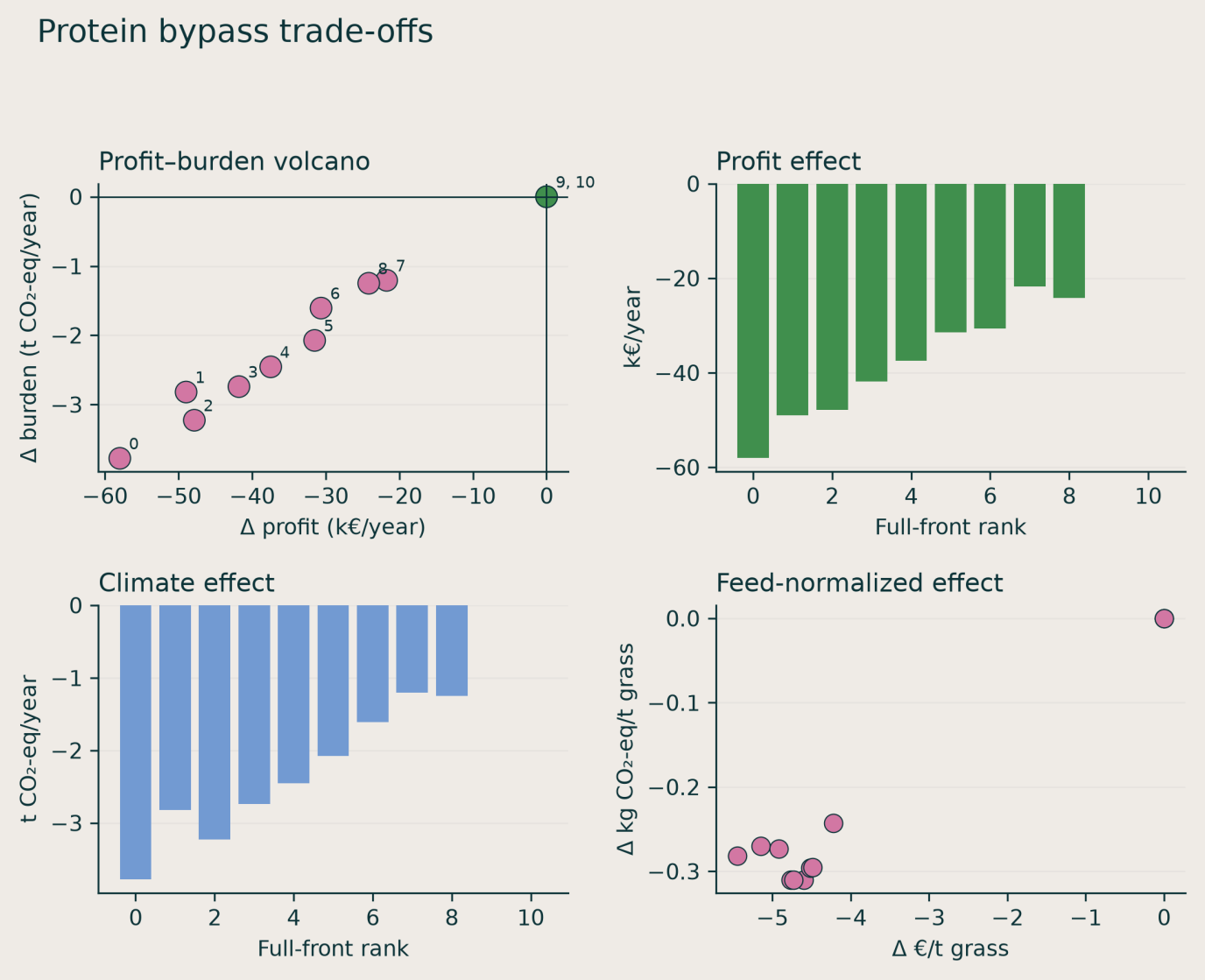


Figure: 09 uncertainty/non pef protein bypass volcano.png

10. Design sweep, sensitivity and evidence quality

The Monte Carlo panels describe the saved random design sweep: technology options, throughput and independent sampled economic assumptions vary simultaneously. Histogram frequencies are frequencies of sampled designs, not probabilities of future plant outcomes. Parameter scatter plots are descriptive associations and do not isolate causal response. The route violins stratify this design space by protein route, without claiming experimental confidence intervals [D6].

The economic and environmental tornado charts are one-factor arithmetic sweeps around saved outputs. The latter uses a stated electricity factor range, +/-10% purchased power and +/-5% operating hours. No synthetic process-parameter uncertainty, confidence band, candidate kinetic curve or full-LCA distribution is generated.

The model audit records no unit at E5 validated-and-costed maturity. Custom equipment costs, kinetic and separation calibration, product specifications, end-use eligibility, storage fate, full delivered-energy factors and complete inventory remain priority gaps [D2-D4]. Each candidate evidence panel shows balance residuals, evidence records, calculated account lines and warnings. Numerical closure establishes bookkeeping consistency only.

10. Design sweep, sensitivity and evidence quality

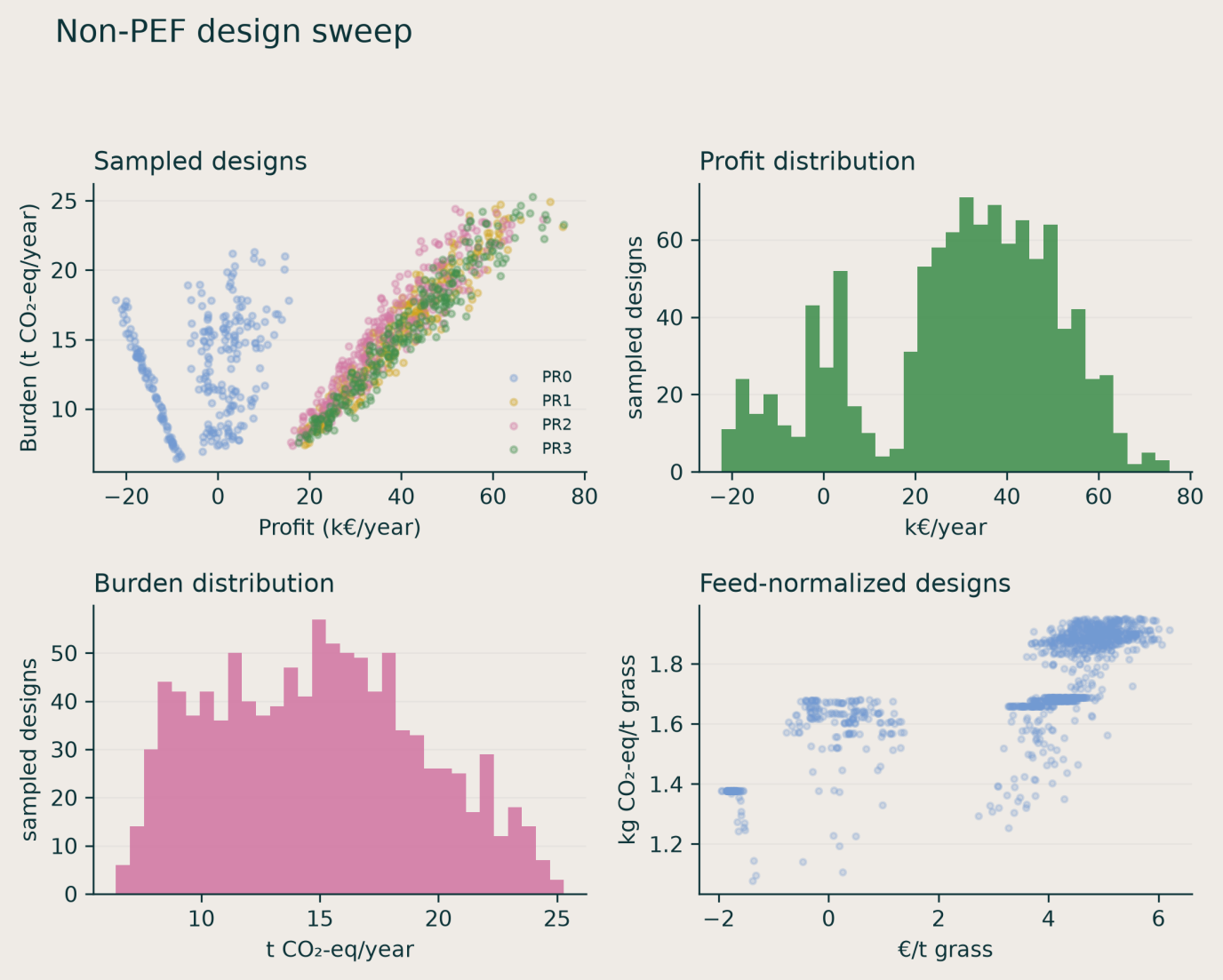


Figure: 09 uncertainty/non pef monte carlo design sweep.png

10. Design sweep, sensitivity and evidence quality

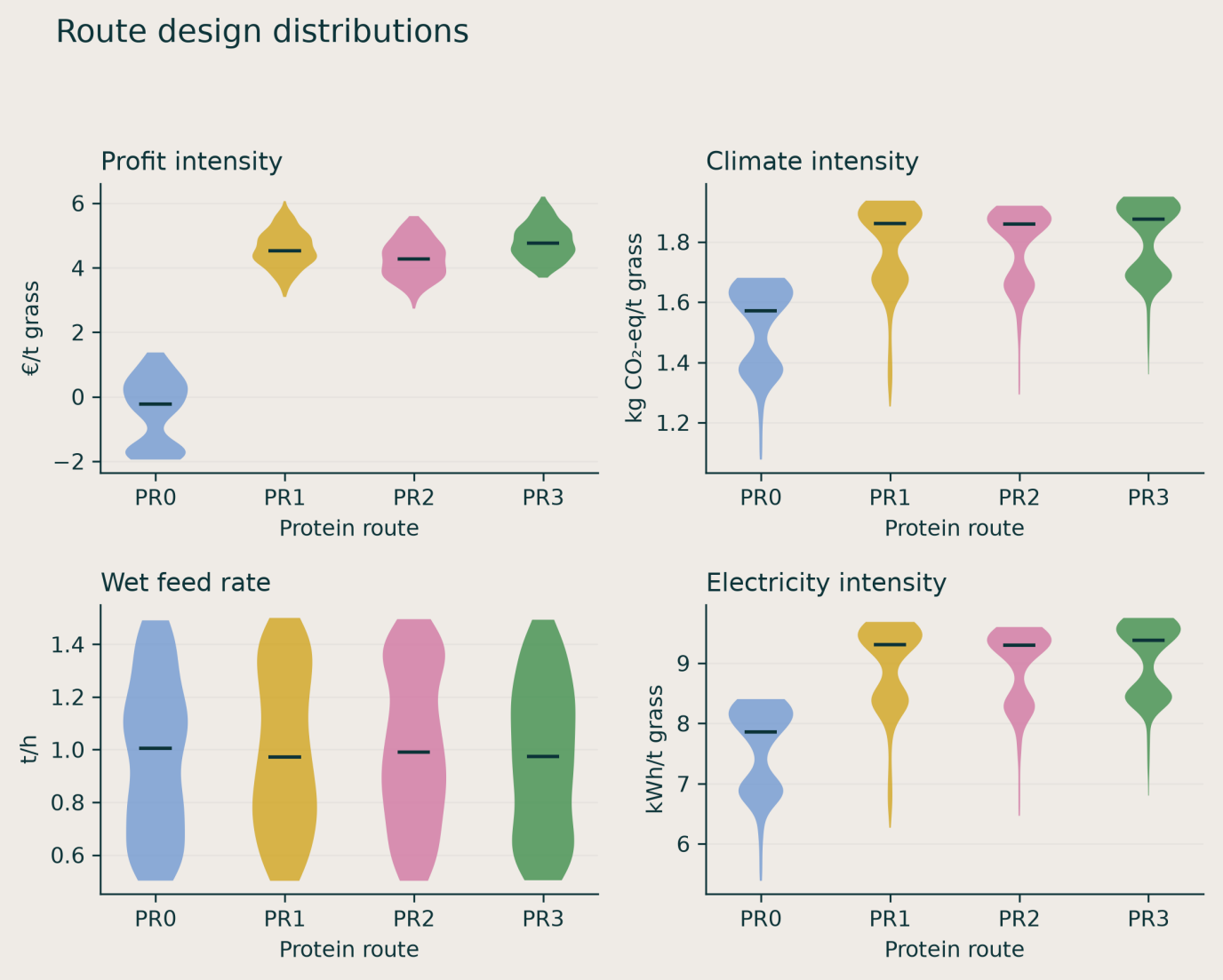


Figure: 09 uncertainty/non pef route design distributions.png

10. Design sweep, sensitivity and evidence quality

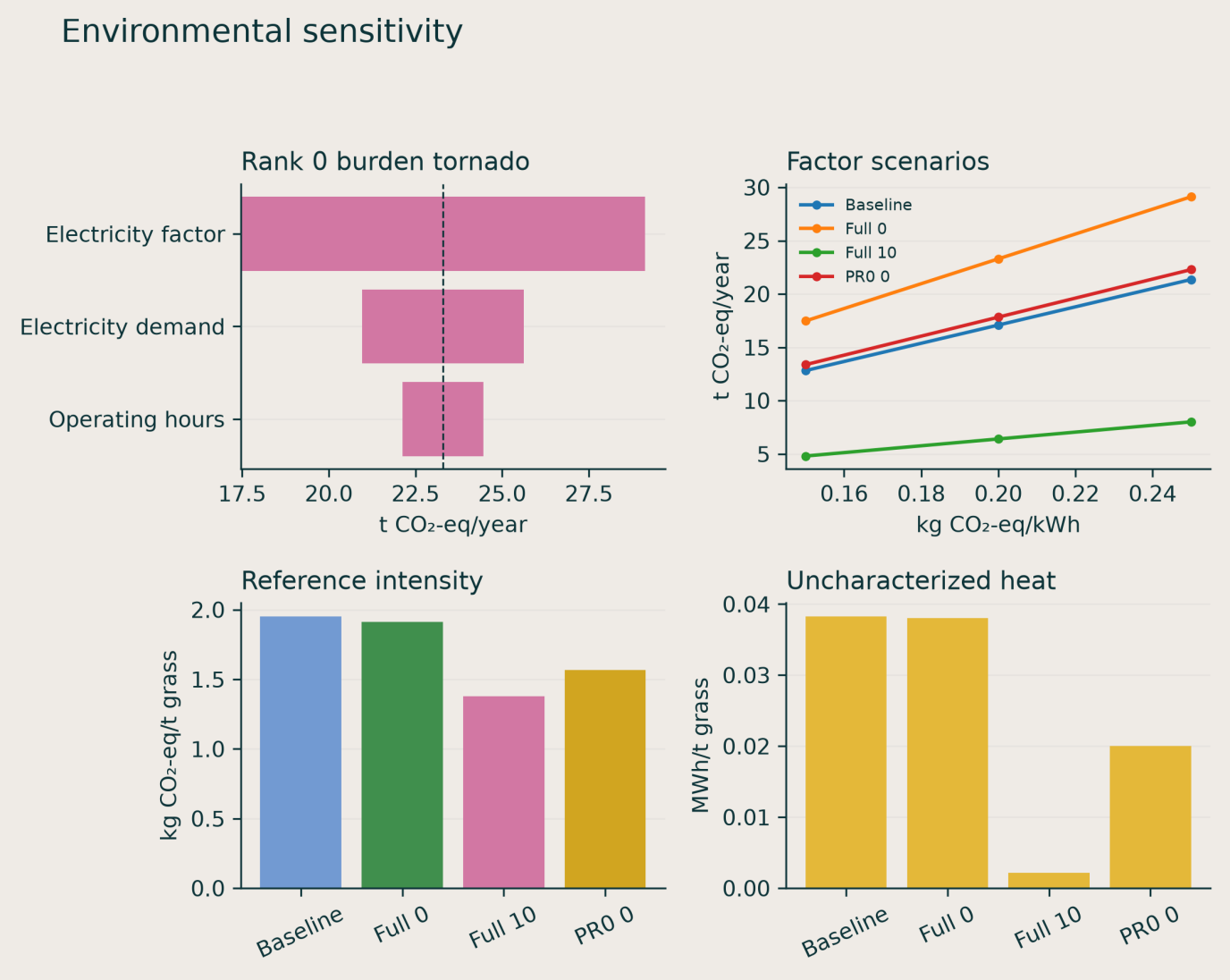


Figure: 09 uncertainty/non pef environmental tornado.png

11. Decision interpretation and next evidence

For a screening question focused on recorded financial return, full-front rank 0 is the current economic endpoint. For the narrow electricity-only annual burden, rank 10 is the minimum but recovers no protein, biogas or targeted nutrients. A configuration with meaningful recovery and intermediate objectives, such as full-front rank 5, is a process-study candidate. None should be declared the best GreenHub investment or full environmental option from the present evidence.

The next decision gate should reconcile BG0 raw gas identity and sale price; measure feed, protein and digestion yields over seasons; validate storage reactions and emissions; obtain vendor equipment quotations and labor/maintenance estimates; establish product quality and end-use routes; add location/year-matched electricity and heat factors plus transport, materials and direct emissions; then rerun matched-throughput TEA/LCA and multi-objective screening. These needs follow the model documentation and the selected reports [D2-D4].

Historical F2 results remain available in the documentation archive for provenance, but are not merged into the non-PEF candidate ranking. This prevents unlike preparation routes from being presented as one baseline.

11. Decision interpretation and next evidence

Figure: 08 candidate comparison/biosteam/all/rank 00/combined/06 balance and evidence.png

12. Complete candidate ledger

The table below lists every selected F1 full-front and protein-bypass point. Profit and NPV use partial, assumption-dependent financial accounts. Climate is electricity-only attributional burden. The independent fronts have different throughput; matched intervention values are supplied in evidence/biosteam_frontiers/matched_protein_bypass_manifest.json and the corresponding candidate pages.

Set	Rank	Route	Feed t/y	Profit kEUR/y	Burden t/y	kg/t	NPV kEUR
Full	0	F1 PR3 AD1 BG0 NR3 A	12,173	75.5	23.3	1.91	741
Full	1	F1 PR2 AD1 BG3 NR3 A	11,606	63.4	22.2	1.91	623
Full	2	F1 PR3 AD3 BG0 NR4 A	10,406	61.7	19.6	1.88	606
Full	3	F1 PR1 AD3 BG0 NR3 A	9,257	54.8	17.2	1.86	538
Full	4	F1 PR3 AD1 BG0 NR4 A	7,921	47.6	15.2	1.92	467
Full	5	F1 PR1 AD1 BG0 NR3 A	7,026	40.5	13.3	1.90	397
Full	6	F1 PR3 AD3 BG2 NR3 A	5,956	34.6	11.4	1.91	340
Full	7	F1 PR3 AD3 BG3 NR4 A	4,418	25.4	8.4	1.91	249
Full	8	F1 PR2 AD0 BG0 NR0 A	4,436	16.3	7.4	1.66	160
Full	9	F1 PR0 AD1 BG0 NR1 A	4,585	1.7	7.3	1.60	17
Full	10	F1 PR0 AD0 BG0 NR0 A	4,652	-9.0	6.4	1.38	-89
PR0	0	F1 PR0 AD3 BG0 NR3 A	11,374	15.4	17.8	1.57	151
PR0	1	F1 PR0 AD3 BG0 NR4 A	10,472	13.9	16.5	1.57	136
PR0	2	F1 PR0 AD1 BG0 NR3 A	9,132	10.8	14.6	1.60	106
PR0	3	F1 PR0 AD3 BG0 NR4 A	7,427	10.2	11.7	1.57	100
PR0	4	F1 PR0 AD3 BG0 NR3 A	7,313	8.3	11.5	1.57	81
PR0	5	F1 PR0 AD1 BG0 NR4 A	5,808	7.7	9.3	1.61	75
PR0	6	F1 PR0 AD3 BG0 NR3 A	5,603	5.6	8.8	1.57	55
PR0	7	F1 PR0 AD1 BG0 NR1 A	4,585	1.7	7.3	1.60	17
PR0	8	F1 PR0 AD3 BG4 NR0 A	4,405	-3.4	6.9	1.57	-33
PR0	9	F1 PR0 AD0 BG0 NR0 A	4,767	-8.0	6.6	1.38	-79
PR0	10	F1 PR0 AD0 BG0 NR0 A	4,652	-9.0	6.4	1.38	-89

13. Documentation and reproducibility

The complete PDF appends one full-size page for each of the 266 unchanged PNGs. An offline figure_atlas.html provides category and text filtering, while figure_atlas.md and portable_figure_index.csv record every figure with source, scenario, units and notes. The evidence folder contains the 33 exact candidate reports, the F1 baseline screening report, selected run streams, generator scripts and a compressed copy of the complete 2,000-case batch.

The documentation folder preserves the model docs and audits, phase methodologies, frontend docs, data registers and the earlier scientific report. A SHA-256 manifest records every copied source path, size and hash. The historical F2 manuscript is retained as source history, not as the baseline or a current conclusion. The package README states the reproduction sequence and evidence hierarchy.

The model and figure data are a research screening resource. Full installed costs, project validation, market acceptance, environmental characterization and probabilistic parameter evidence remain unresolved. The archive enables review of each plotted number without silently promoting those gaps into certainty.

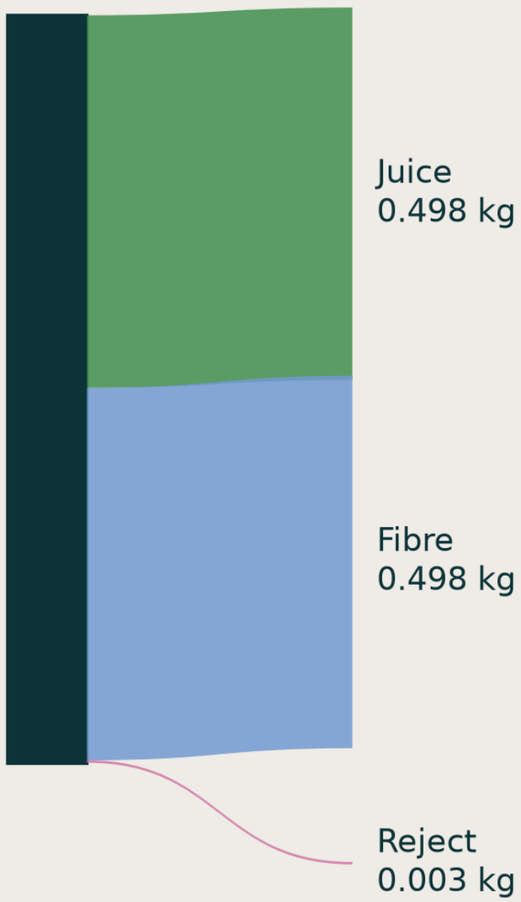
Reference	Document	Packaged path
D1	Architecture	documentation/model_docs/architecture.md
D2	Scientific model audit	documentation/model_docs/audit/scientific_model_audit.md
D3	Model limitations	documentation/model_docs/model_limitations.md
D4	LCA inventory method	documentation/model_docs/lca_inventory.md
D5	Backend Pareto jobs	documentation/model_docs/backend_pareto_jobs.md
D6	Figure methodology	documentation/figure_methodology.md
D7	Phase 3 methodology	documentation/model_docs/phase3_methodology.md
D8	Phase 4 methodology	documentation/model_docs/phase4_methodology.md
D9	Phase 5 methodology	documentation/model_docs/phase5_methodology.md
D10	Phase 6 methodology	documentation/model_docs/phase6_methodology.md
D11	Phase 7 methodology	documentation/model_docs/phase7_methodology.md
H1	Earlier F2 scientific report (historical)	documentation/historical_scientific_report/GreenHub_scientific_report.md

Grass to products

Protein ✓ · Biogas ✓ · N ✓ · P ✓ · K X

Grass split

Grass 1.000 kg



Juice pathway

Juice 0.498 kg

Protein depleted 0.358 kg

Digester feed 0.360 kg

Digestate 0.358 kg

Selected intermediate streams

Recovered outputs

Protein 0.0046 kg

N product 0.0084 kg

P product 0.0001 kg

Nutrient solids 0.0045 kg

Raw gas 0.0010 Nm³

Upgraded 0.0007 Nm³

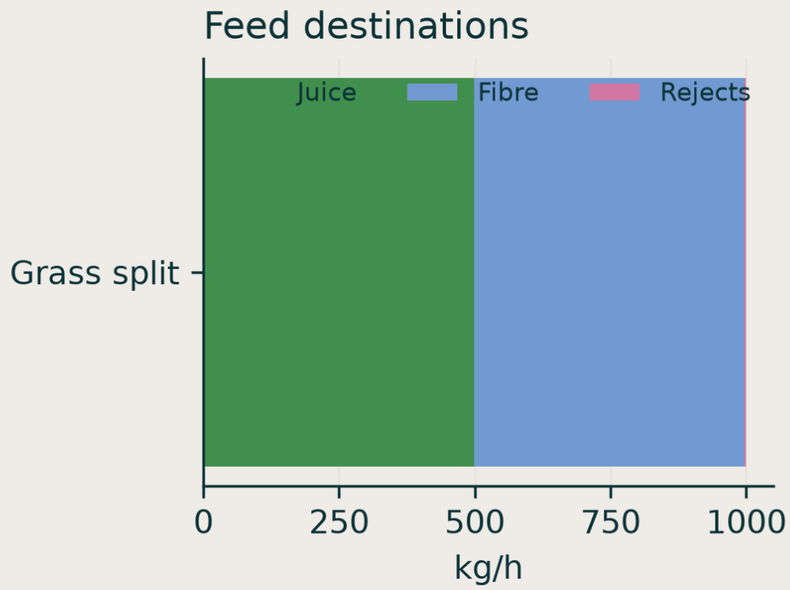
per kg wet grass

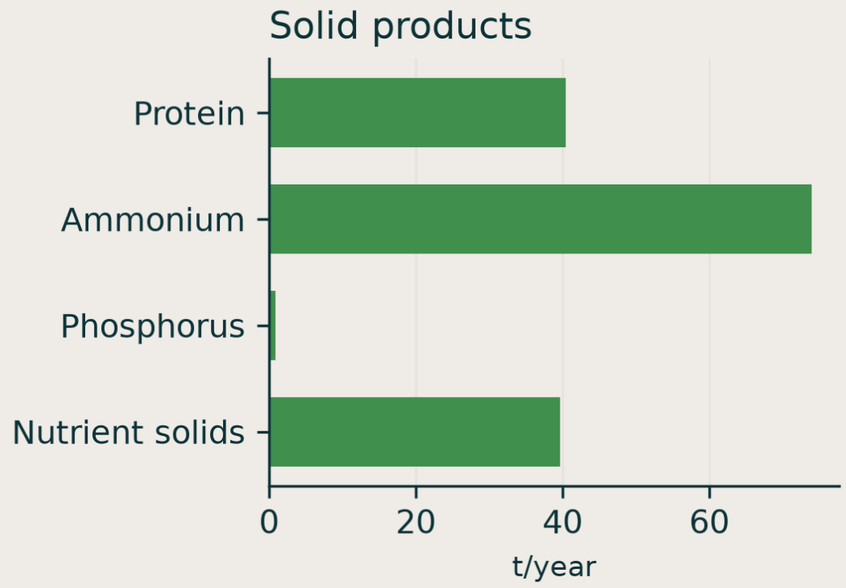
Mass-proportional grass split plus modeled downstream stream/product map

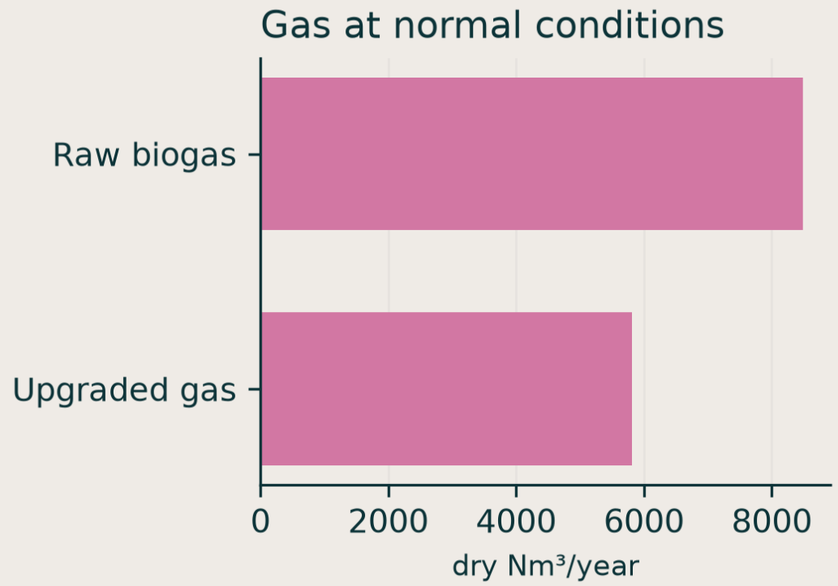
Scenario: F1 PR3 AD1 BG2 NR4 A reference; 0483640f1c194132b22e95fba0eaea1e; 1000 kg/h; 8760 h/year; zero savings/subsidy, benchmark product prices
Units: kg/kg wet grass; dry Nm³/kg wet grass
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

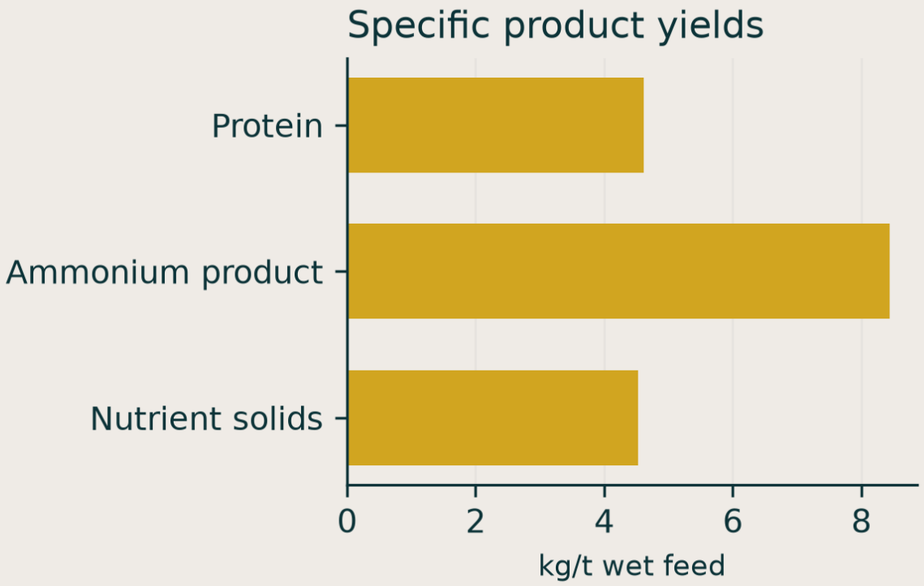
Material and products

Protein ✓ · Biogas ✓ · N ✓ · P ✓ · K X







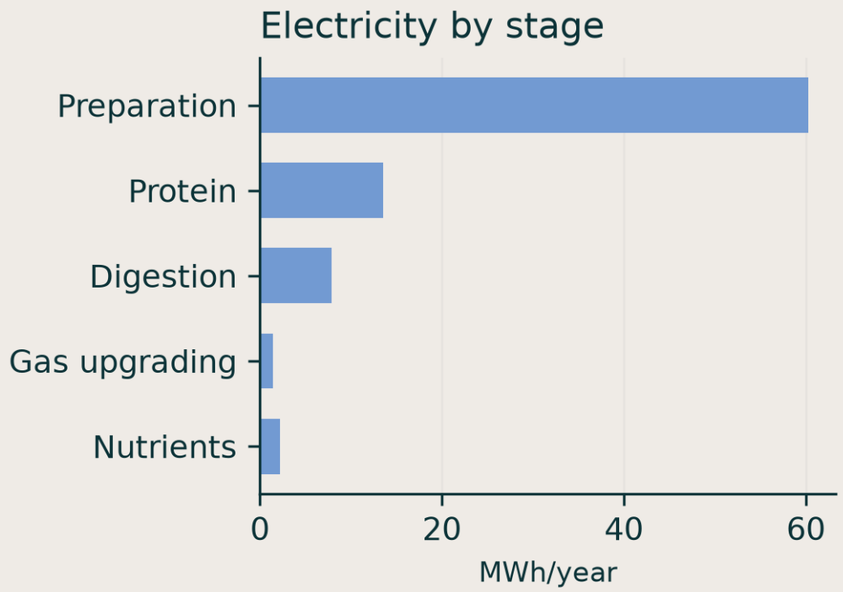


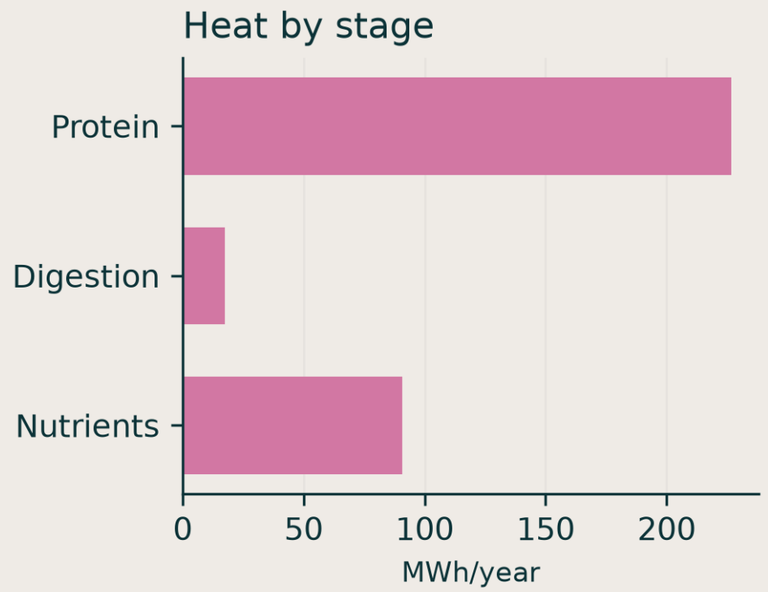
Front-end split, solid products, dry gas volumes and specific yields

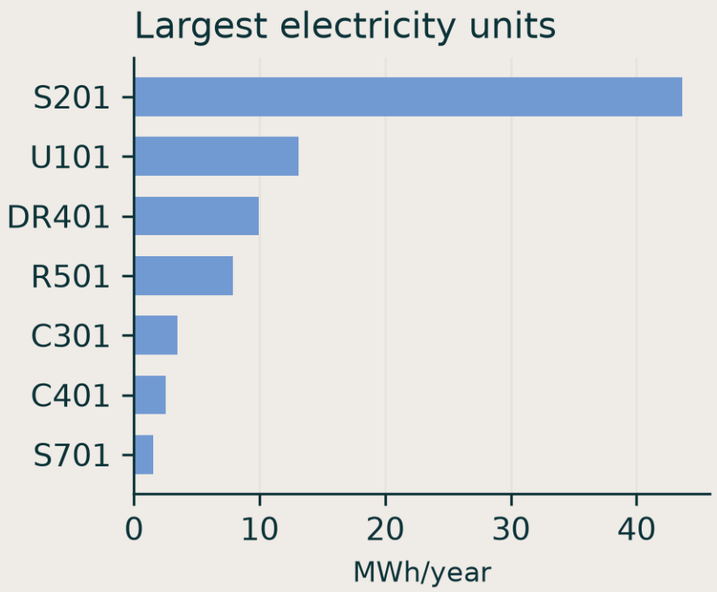
Scenario: F1 PR3 AD1 BG2 NR4 A reference; 0483640f1c194132b22e95fba0eaea1e; 1000 kg/h; 8760 h/year; zero savings/subsidy, benchmark product prices
Units: kg/h; t/year; dry Nm³/year; kg/t feed
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

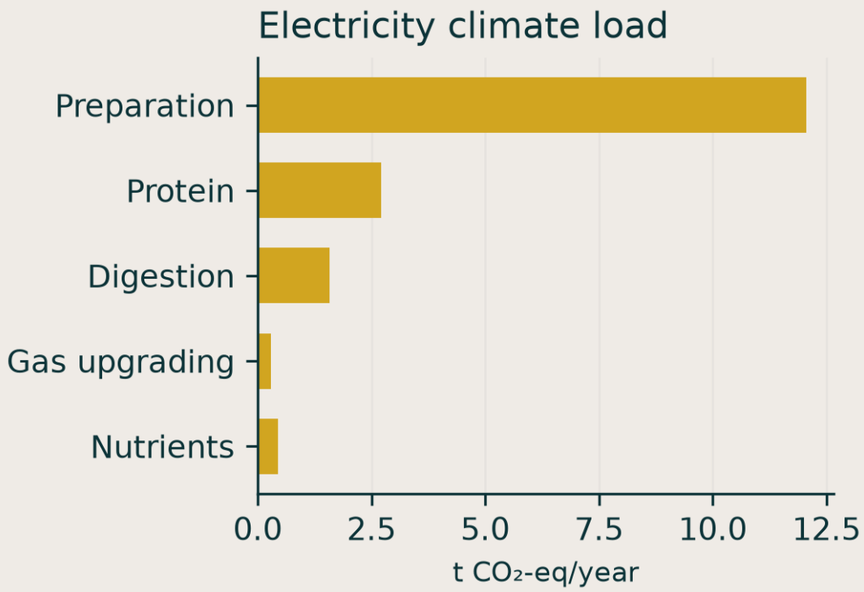
Energy and climate hotspots

Protein ✓ · Biogas ✓ · N ✓ · P ✓ · K X







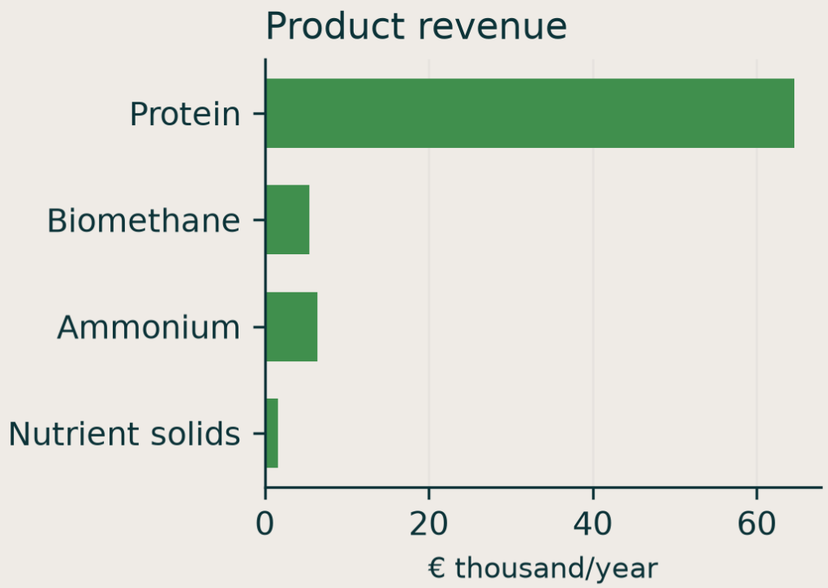


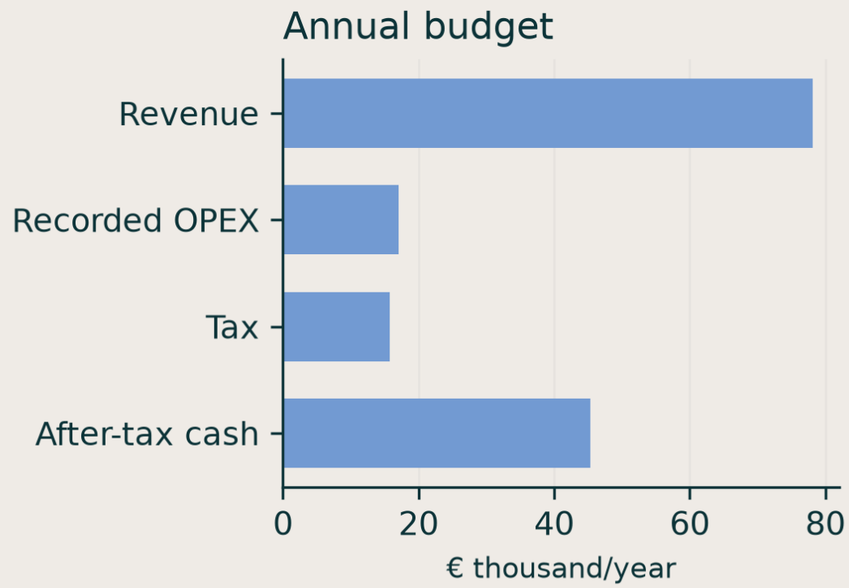
Stage and unit utility hotspots and included climate load

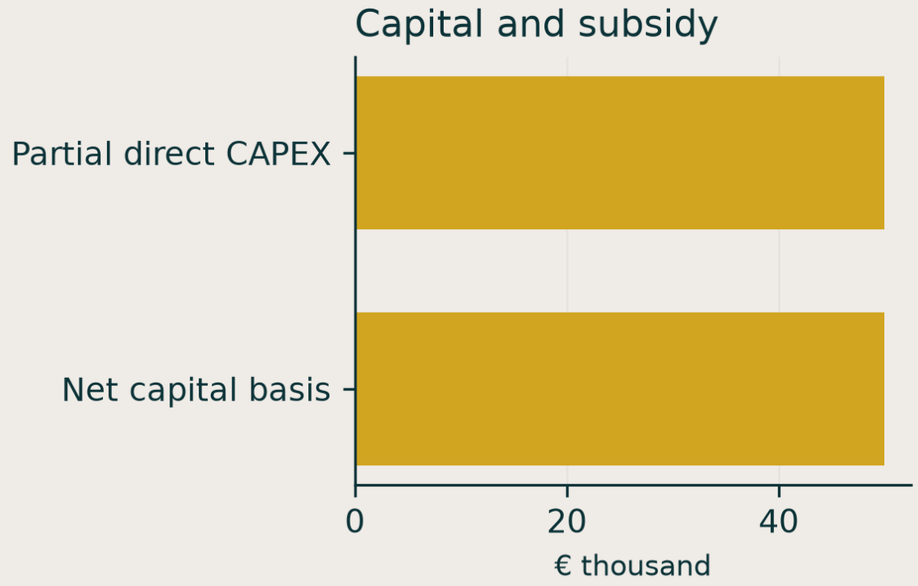
Scenario: F1 PR3 AD1 BG2 NR4 A reference; 0483640f1c194132b22e95fba0eaea1e; 1000 kg/h; 8760 h/year; zero savings/subsidy, benchmark product prices
Units: MWh/year; t CO_2-eq/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

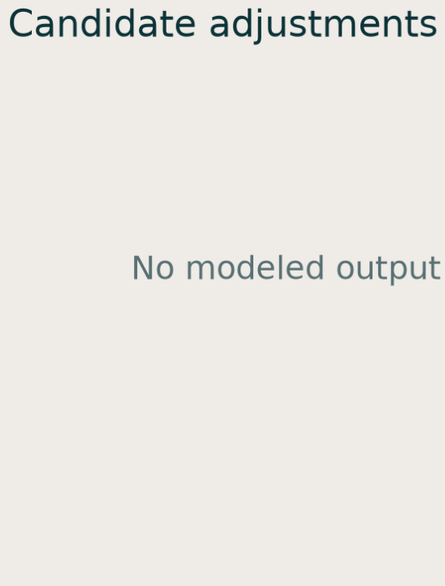
Economic accounts

Protein ✓ · Biogas ✓ · N ✓ · P ✓ · K X







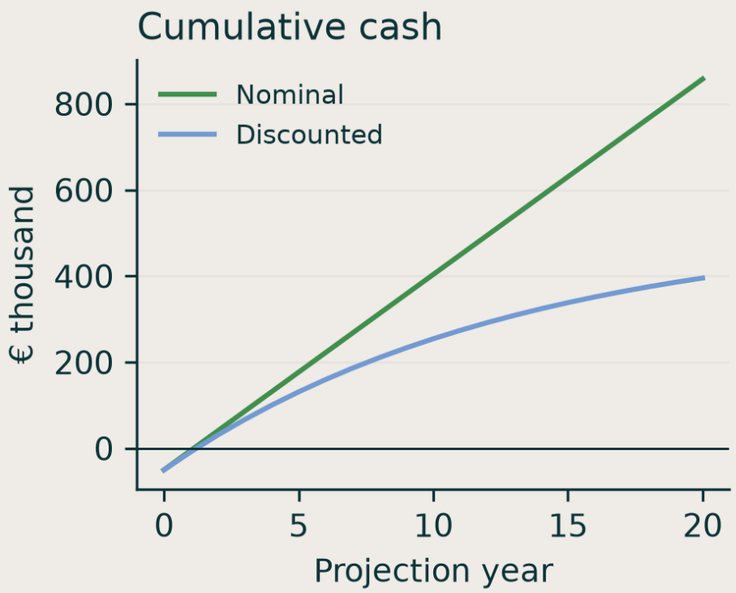


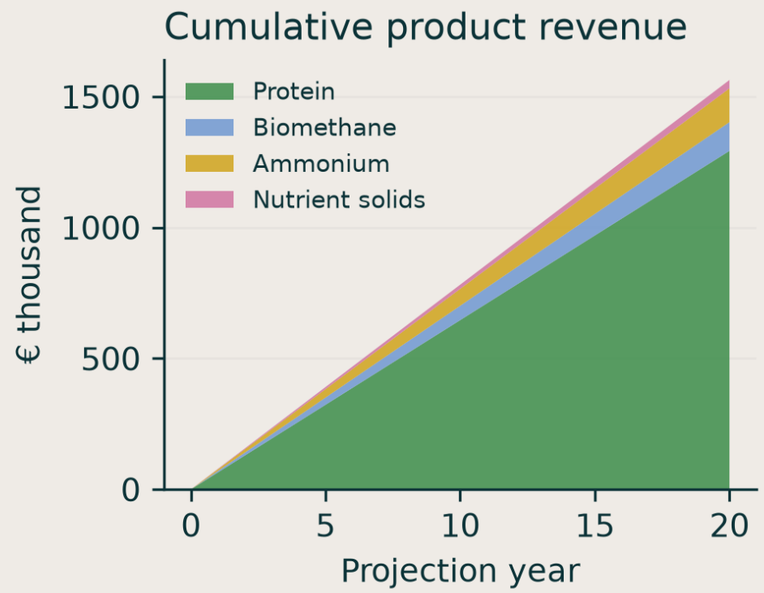
Product revenue, partial annual budget, capital subsidy and candidate overlays

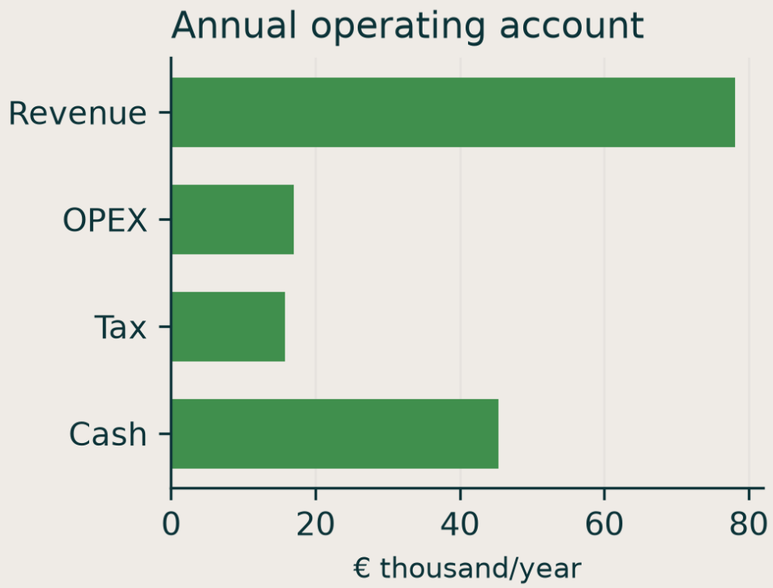
Scenario: F1 PR3 AD1 BG2 NR4 A reference; 0483640f1c194132b22e95fba0eaea1e; 1000 kg/h; 8760 h/year; zero savings/subsidy, benchmark product prices
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

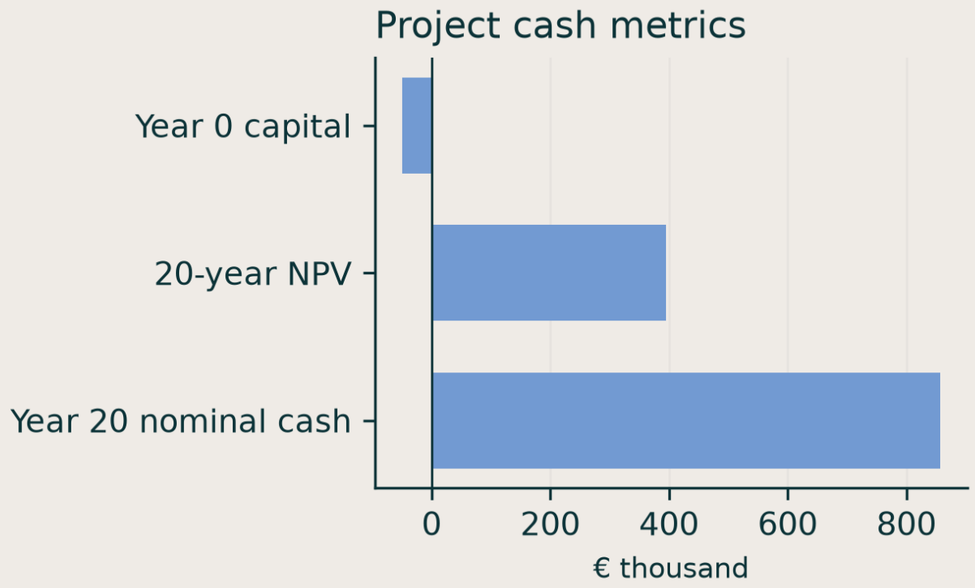
20-year financial projection

Protein ✓ · Biogas ✓ · N ✓ · P ✓ · K X









Constant-account cash flow and cumulative product revenue over 20 years

Scenario: F1 PR3 AD1 BG2 NR4 A reference; 0483640f1c194132b22e95fba0eaea1e; 1000 kg/h; 8760 h/year; zero savings/subsidy, benchmark product prices
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Balance and evidence

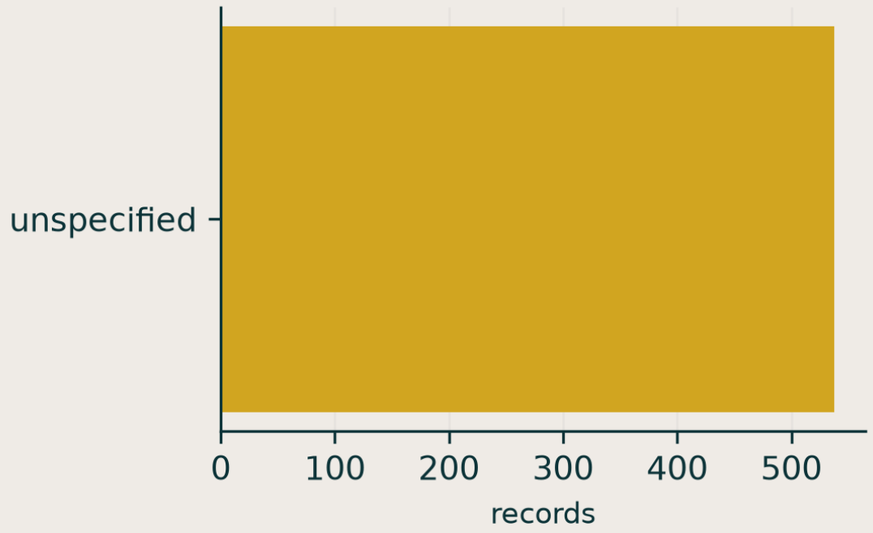
Protein ✓ · Biogas ✓ · N ✓ · P ✓ · K X

Element closure

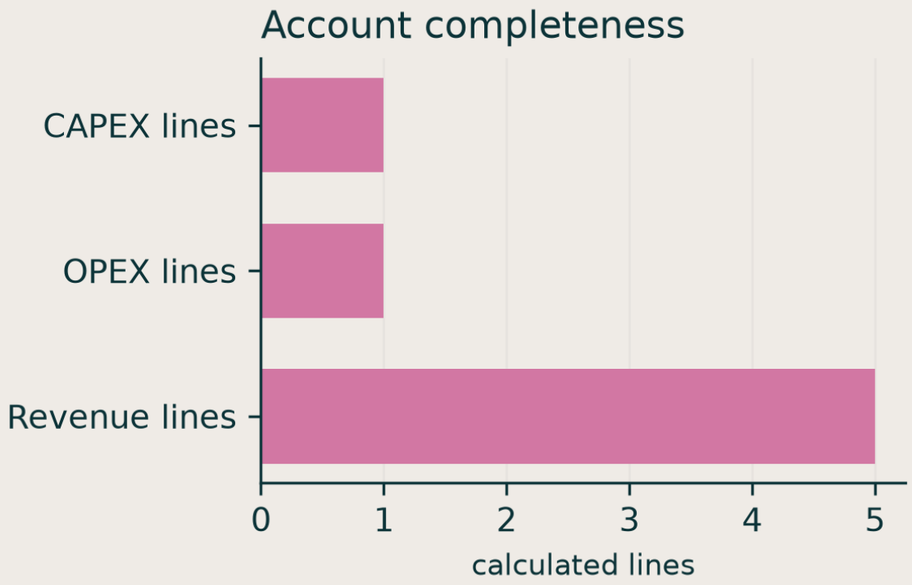
C	0 kg/year
N	0 kg/year
P	0 kg/year
K	0 kg/year

Numerical residuals

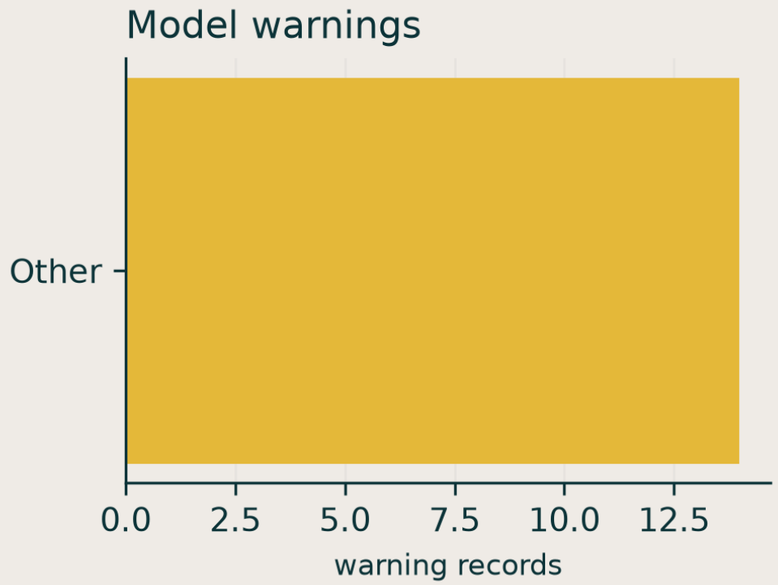
Parameter evidence



Account completeness



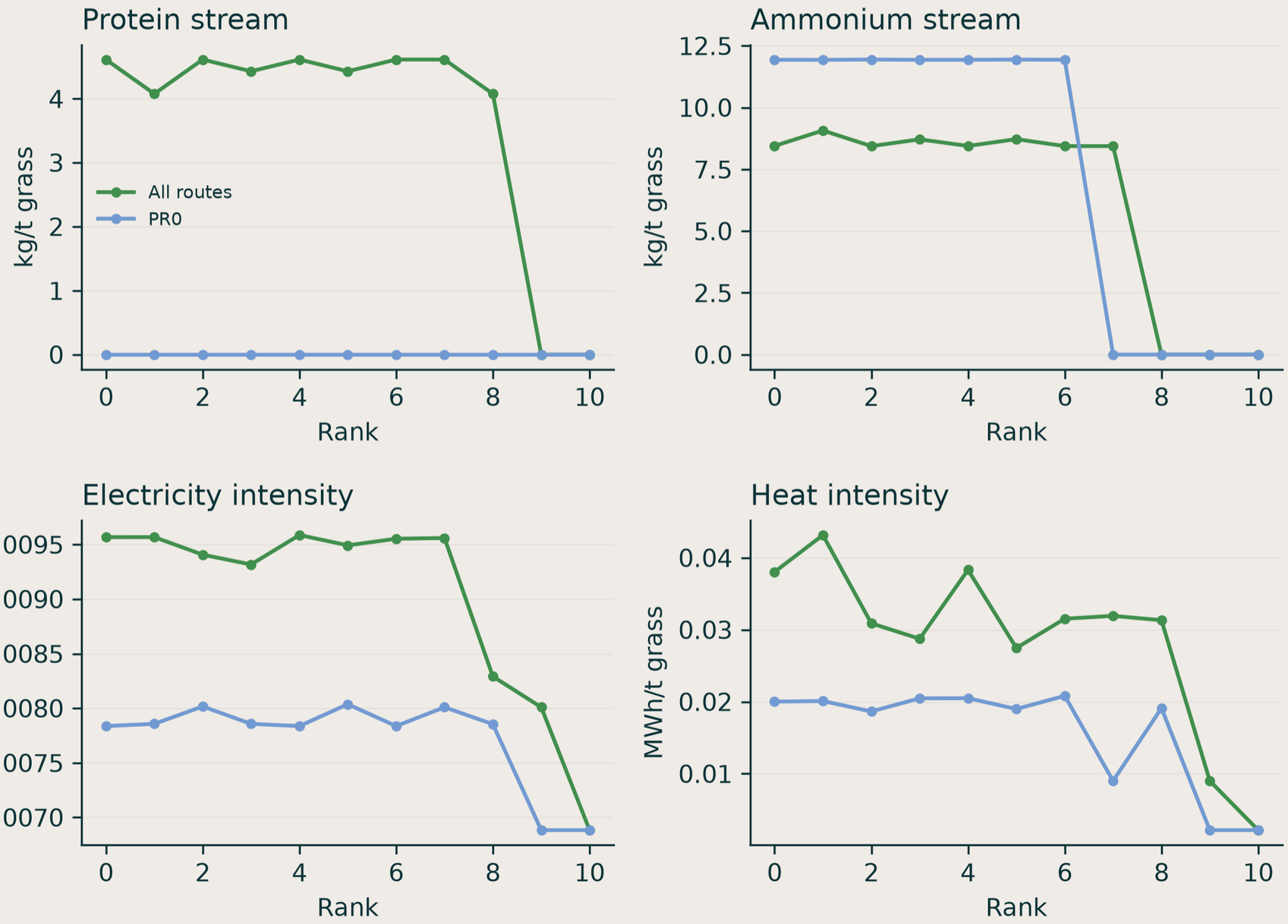
Model warnings



Numerical closure, evidence counts, calculated account lines and warnings

Scenario: F1 PR3 AD1 BG2 NR4 A reference; 0483640f1c194132b22e95fba0eaea1e; 1000 kg/h; 8760 h/year; zero savings/subsidy, benchmark product prices
Units: kg/year; records
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

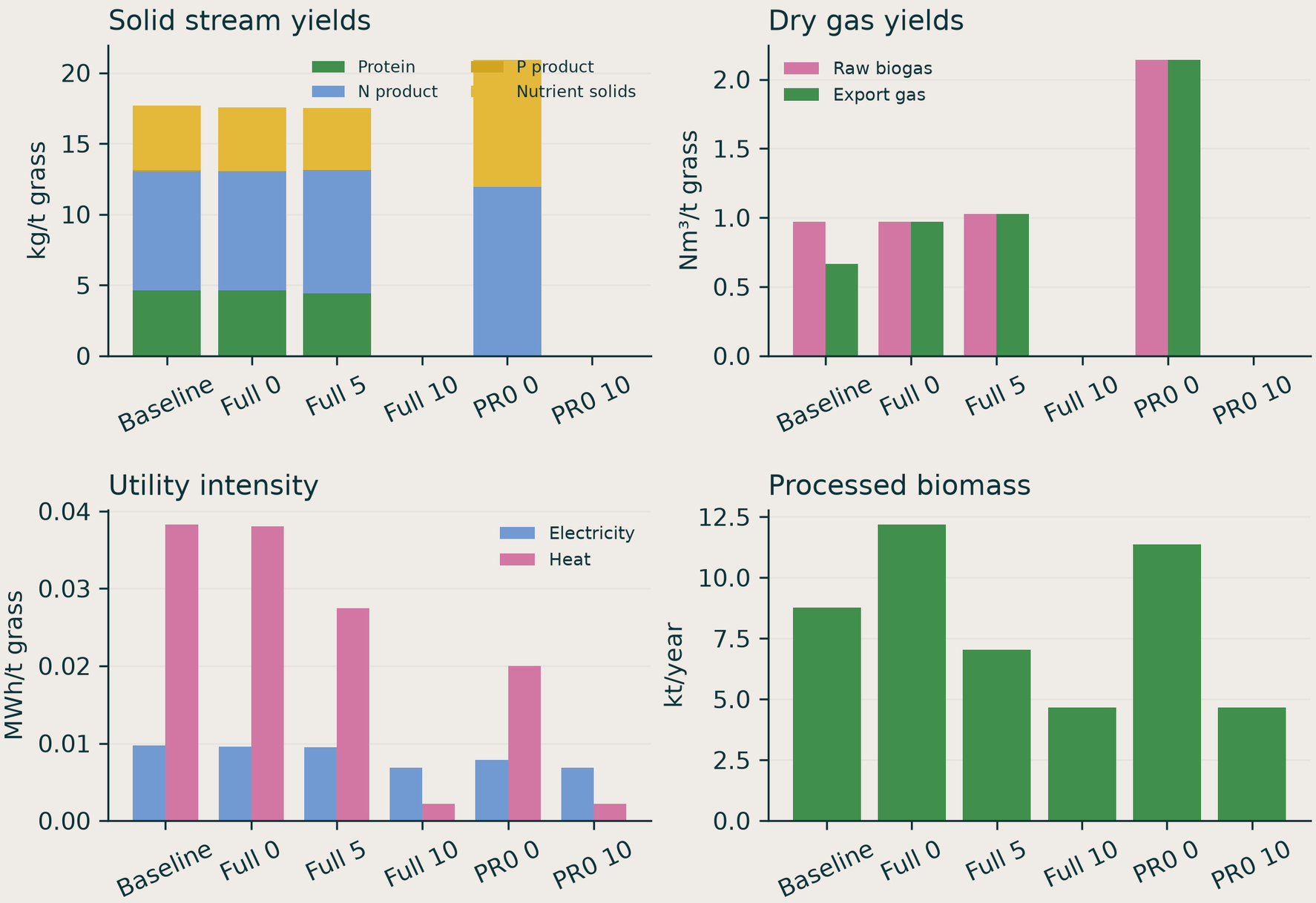
Process along the frontier



Product-stream and utility intensities across two independently selected F1 fronts

Scenario: F1 non-PEF saved BioSTEAM batch and selected exact reruns
Units: kg/t; MWh/t
Source: 22 exact selected non-PEF BioSTEAM reports

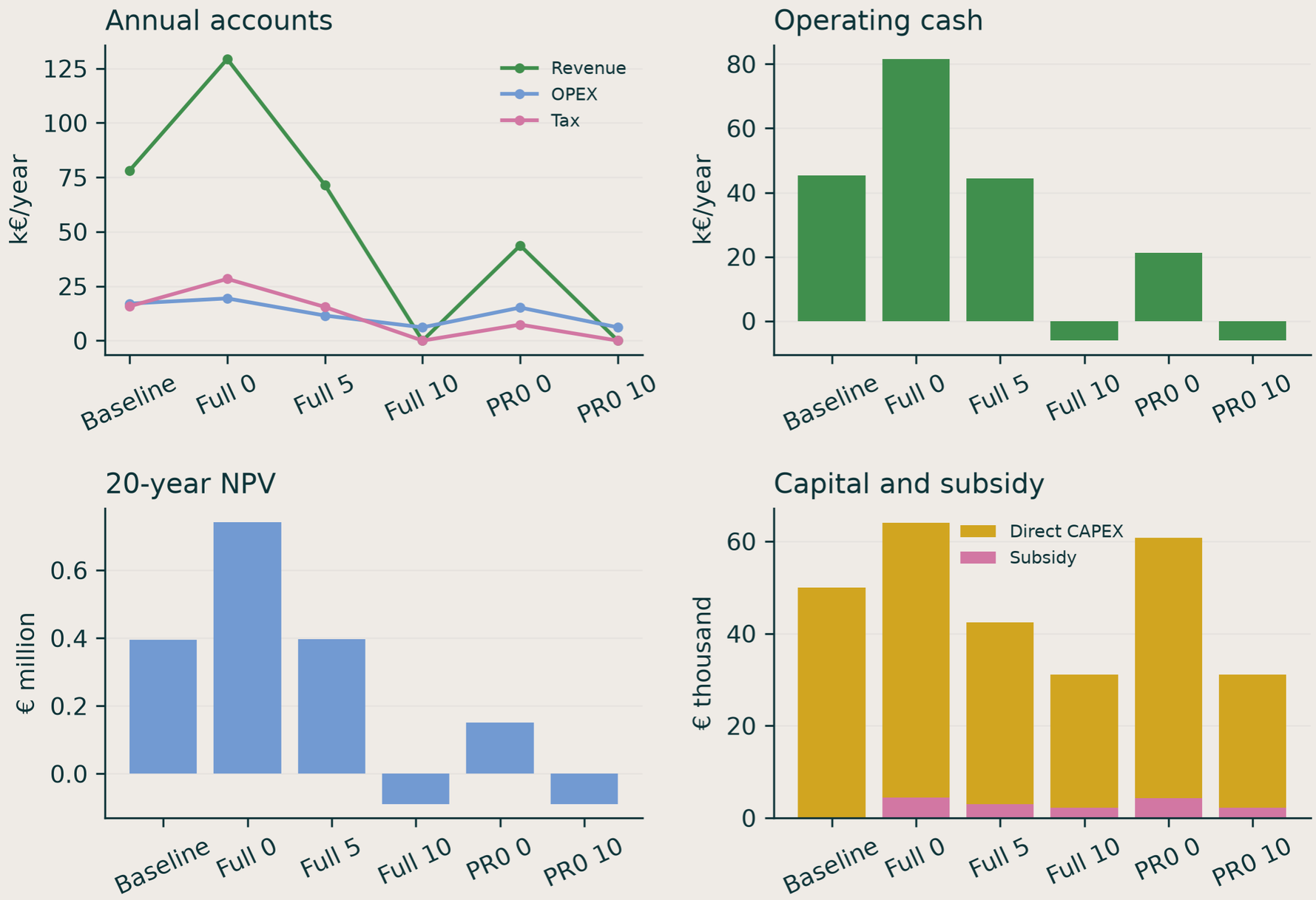
Technical route profiles



Feed-normalized solids, distinct raw/upgraded gas volumes, utility demand and annual feed for selected F1 routes

Scenario: F1 non-PEF saved BioSTEAM batch and selected exact reruns
Units: kg/t; dry Nm³/t; MWh/t; kt/year
Source: Exact F1 BioSTEAM reports and saved gas stream components

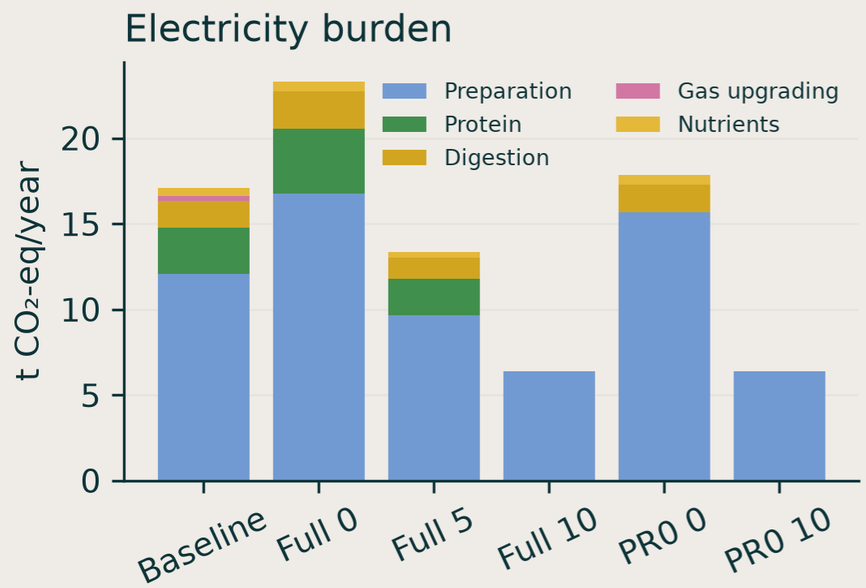
Economic route accounts

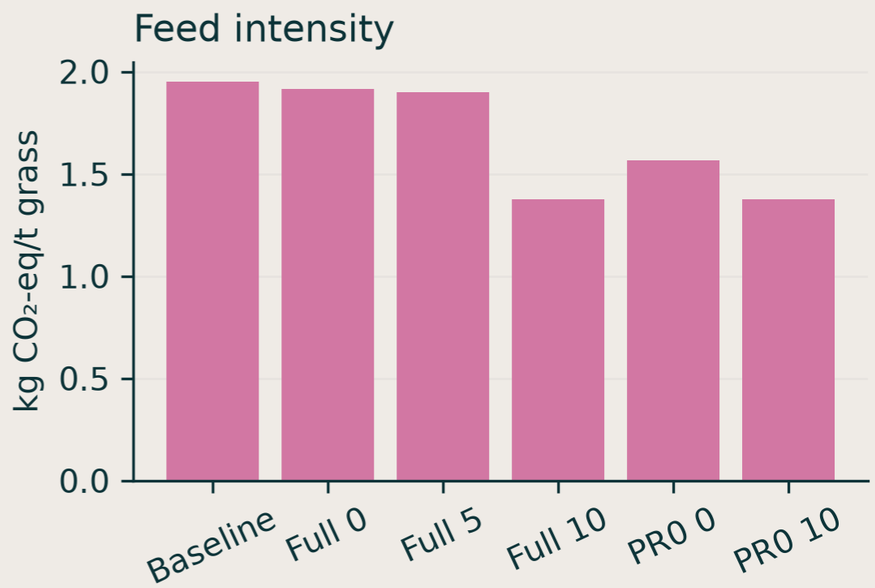


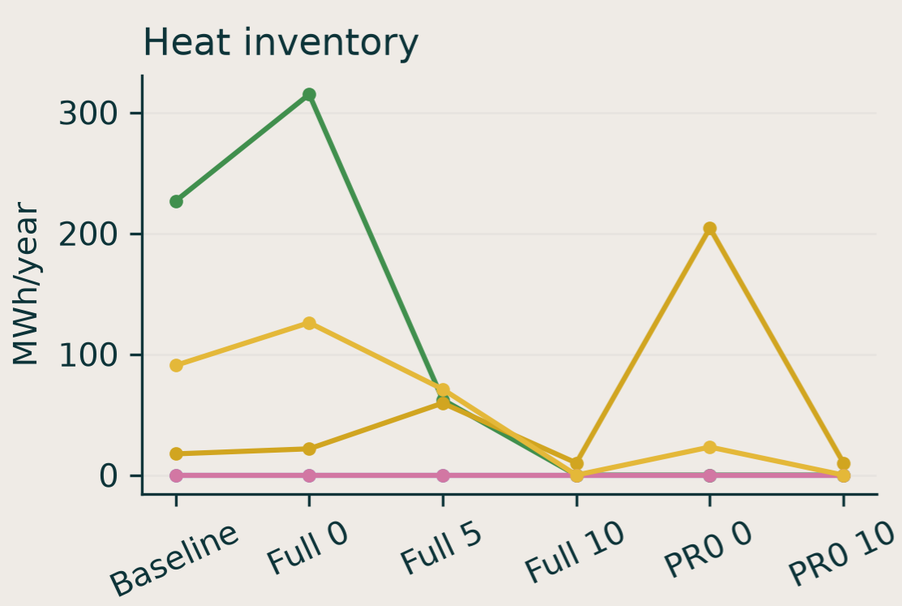
Partial BioSTEAM financial accounts and constant-account NPV for baseline and selected non-PEF candidates

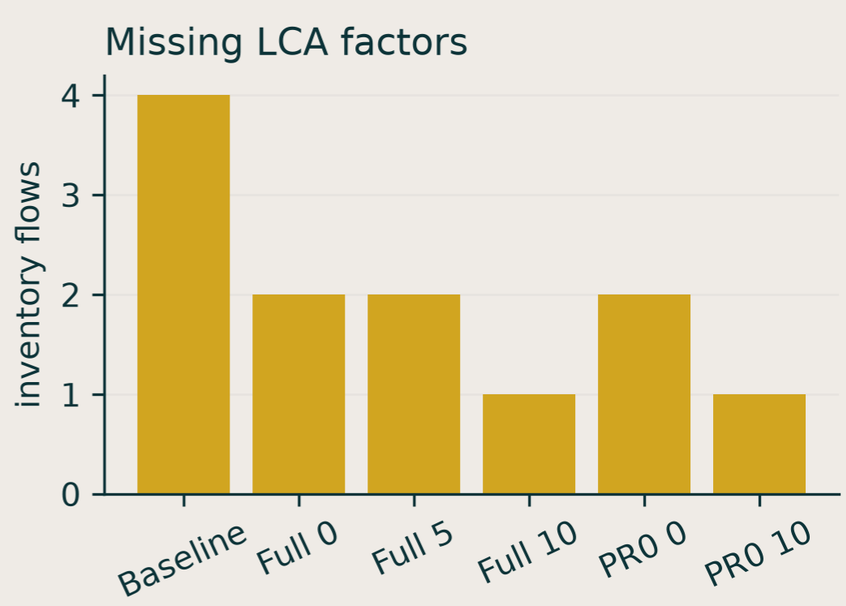
Scenario: F1 non-PEF saved BioSTEAM batch and selected exact reruns
Units: k€/year; € million; € thousand
Source: Baseline screening report; exact non-PEF Pareto reports

Attributional climate inventory



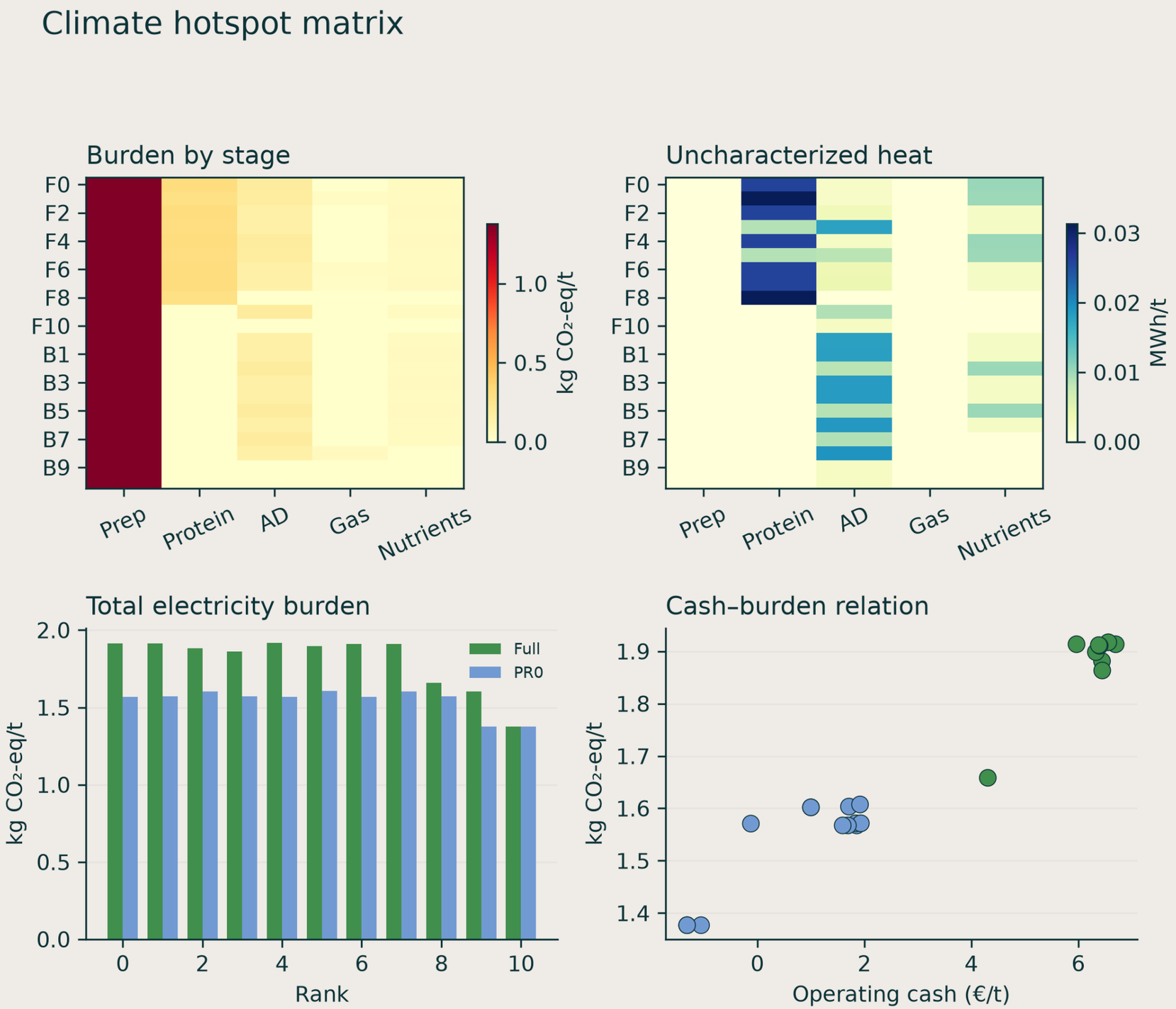






Electricity-only attributional burden and uncharacterized heat/inventory across selected non-PEF routes

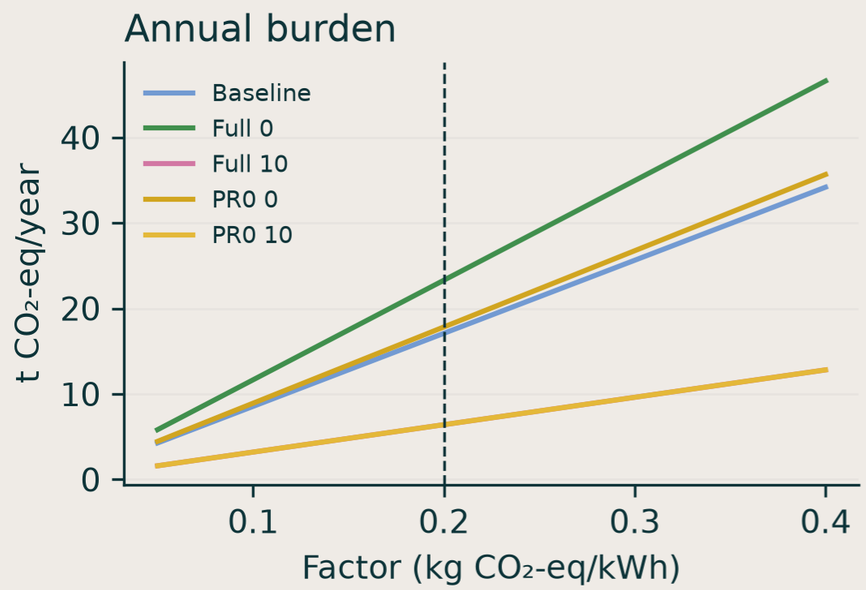
Scenario: F1 non-PEF saved BioSTEAM batch and selected exact reruns
Units: t CO_2-eq/year; kg CO_2-eq/t; MWh/year; inventory flows
Source: Exact BioSTEAM reports; annual process pathway; LCA inventory

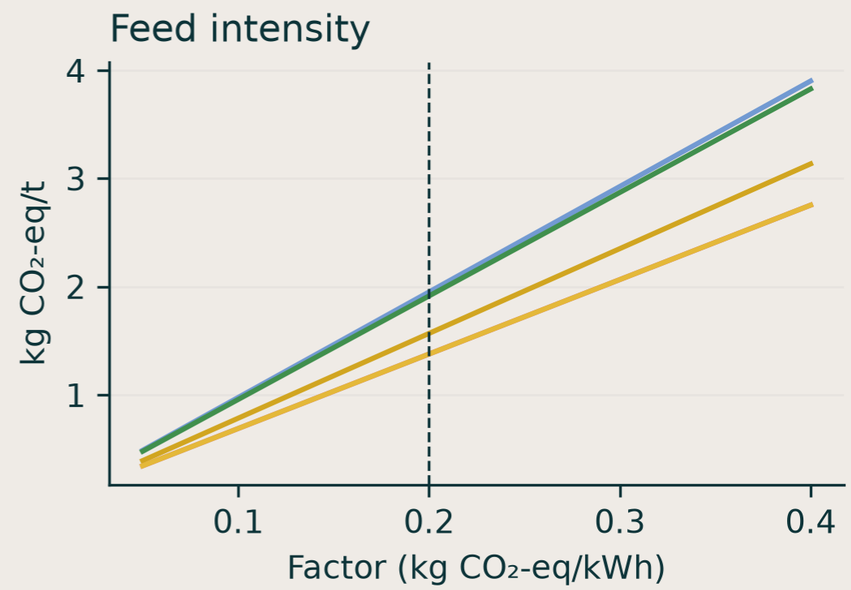


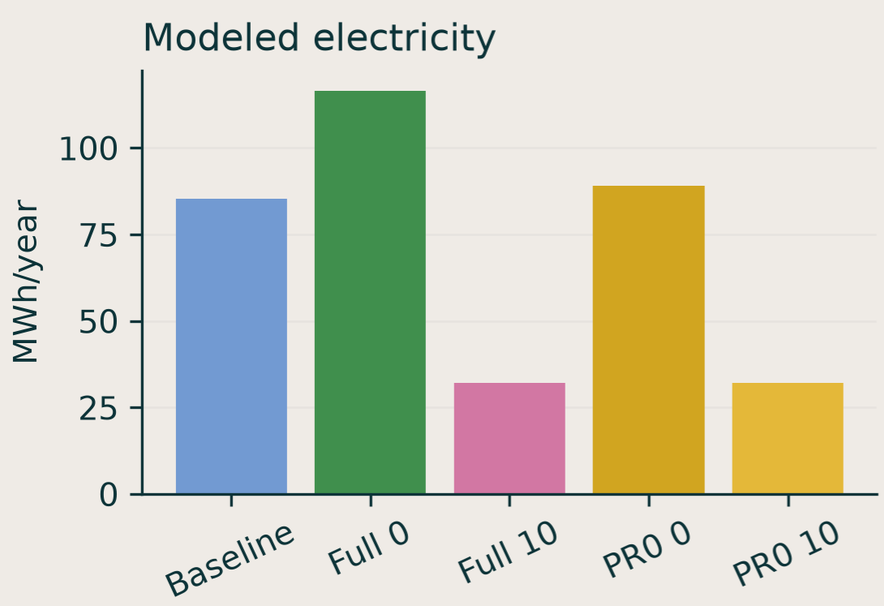
Stage attributional electricity burden and separately inventoried heat across 22 selected F1 routes

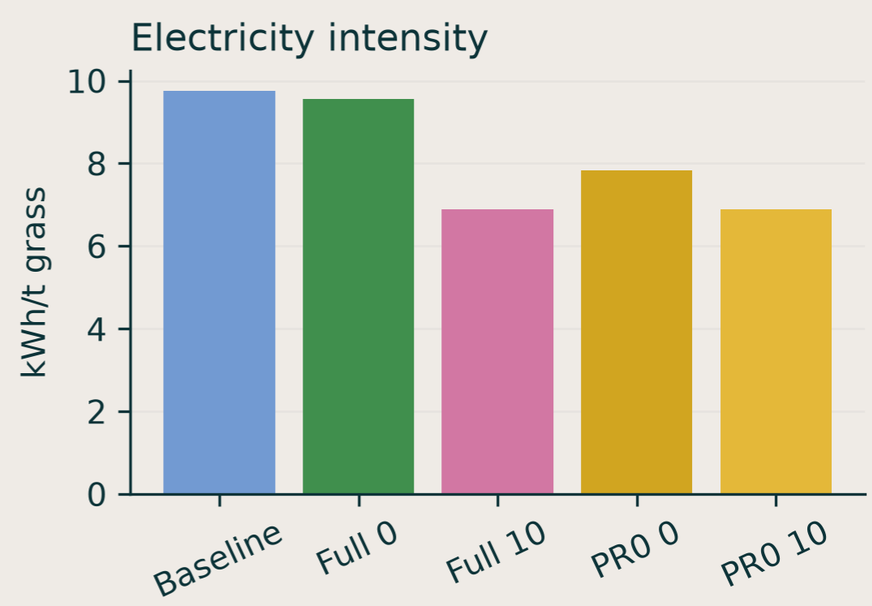
Scenario: F1 non-PEF saved BioSTEAM batch and selected exact reruns
Units: kg CO_2-eq/t; MWh/t; €/t
Source: 22 exact non-PEF BioSTEAM reports

Electricity factor sensitivity





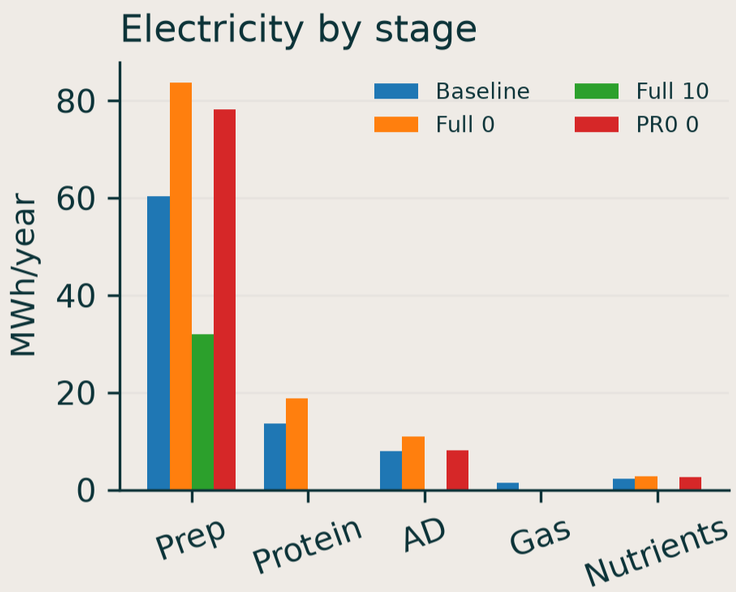


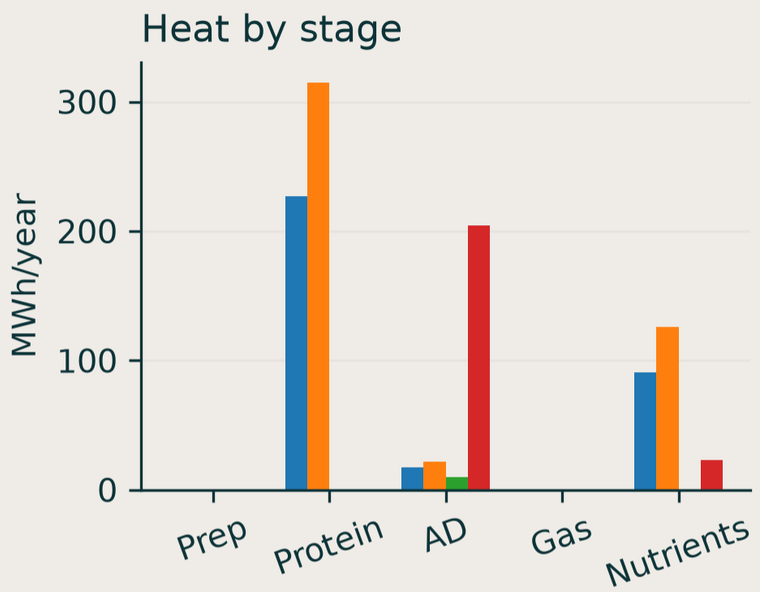


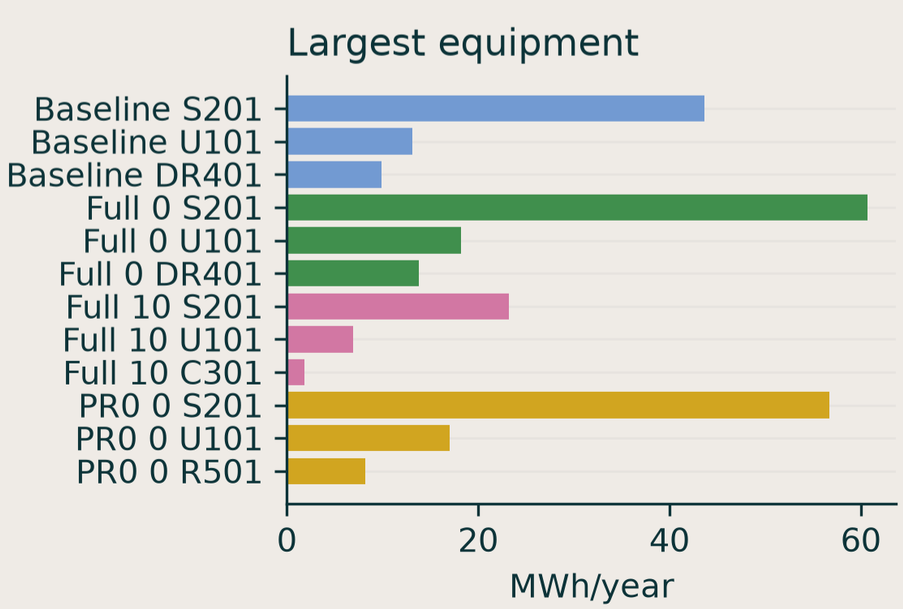
Deterministic attributional electricity factor scenarios on actual F1 utility inventories

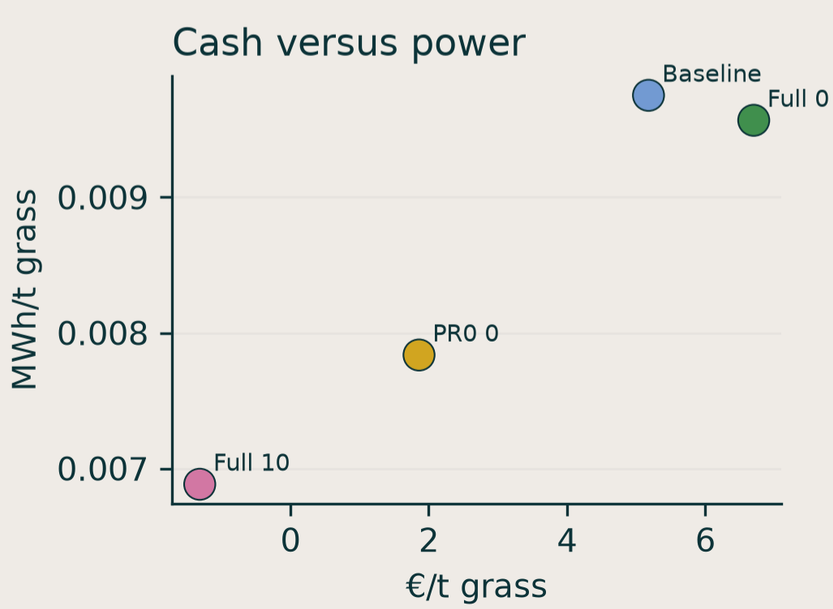
Scenario: F1 non-PEF saved BioSTEAM batch and selected exact reruns
Units: kg CO_2-eq/kWh; t CO_2-eq/year; kg CO_2-eq/t; MWh/year; kWh/t
Source: Exact non-PEF BioSTEAM utility inventories

Process hotspots





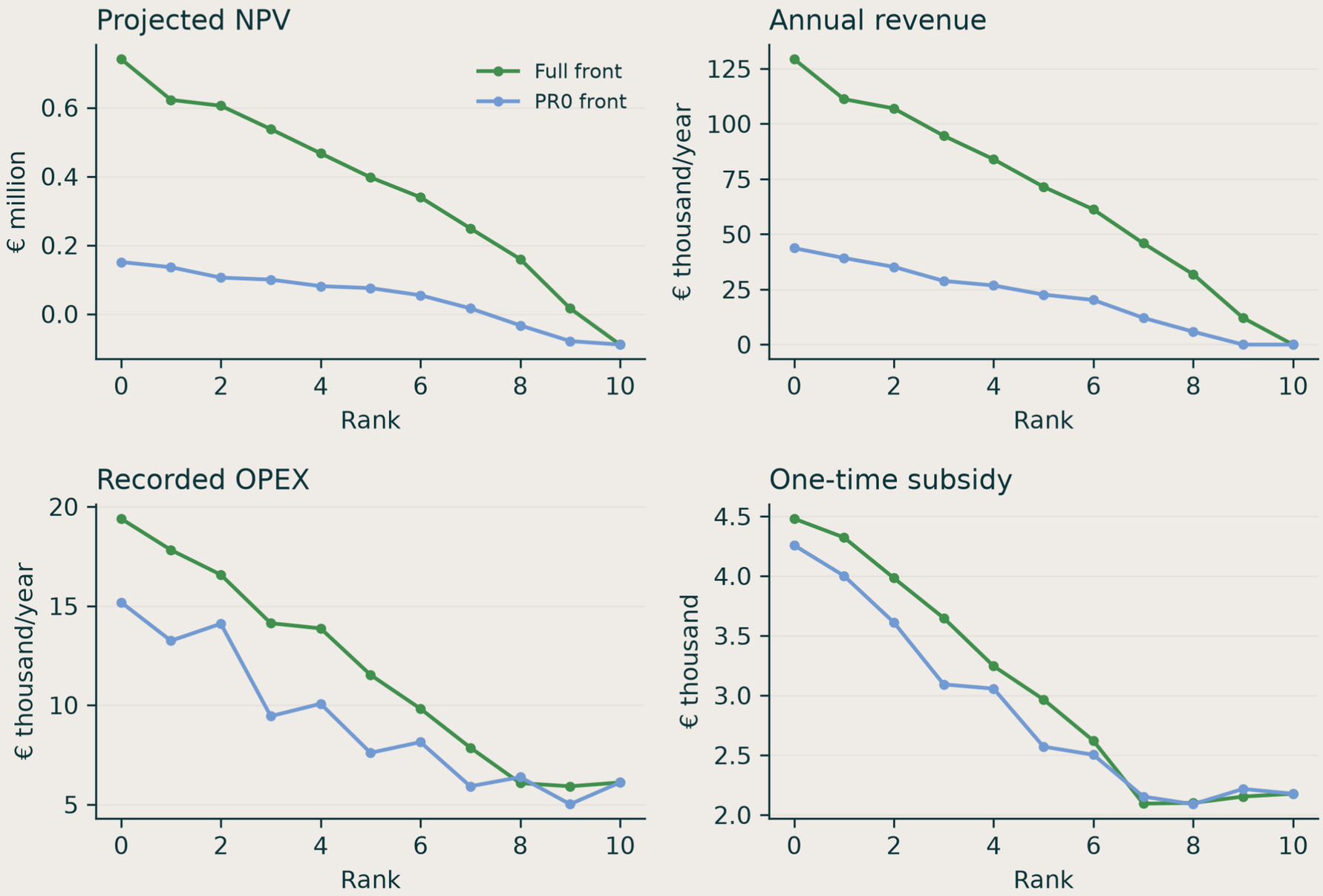




Stage and unit energy hotspots with feed-normalized economic relation

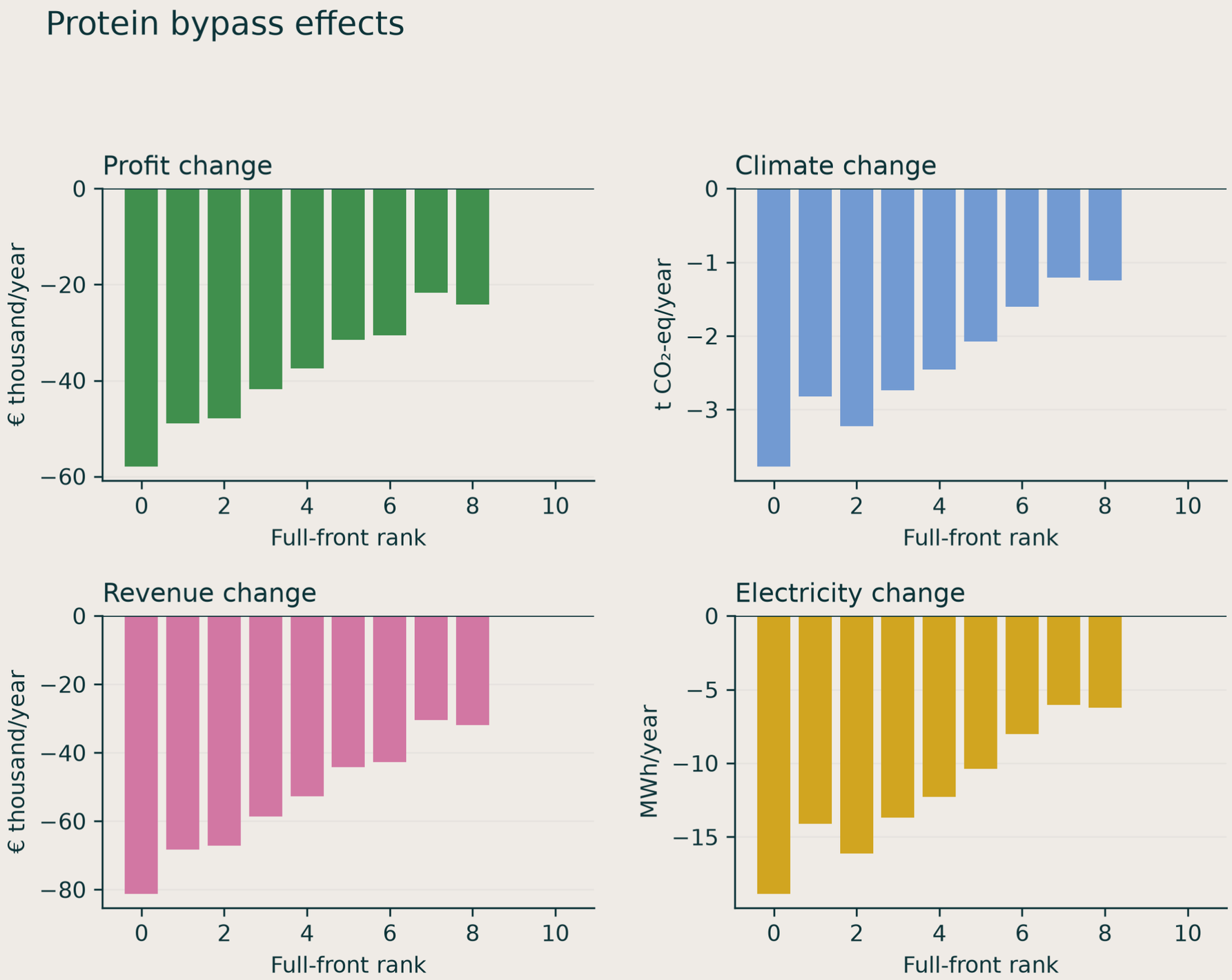
Scenario: F1 non-PEF saved BioSTEAM batch and selected exact reruns
Units: MWh/year; €/t; MWh/t
Source: Exact non-PEF BioSTEAM reports

Candidate financial comparison



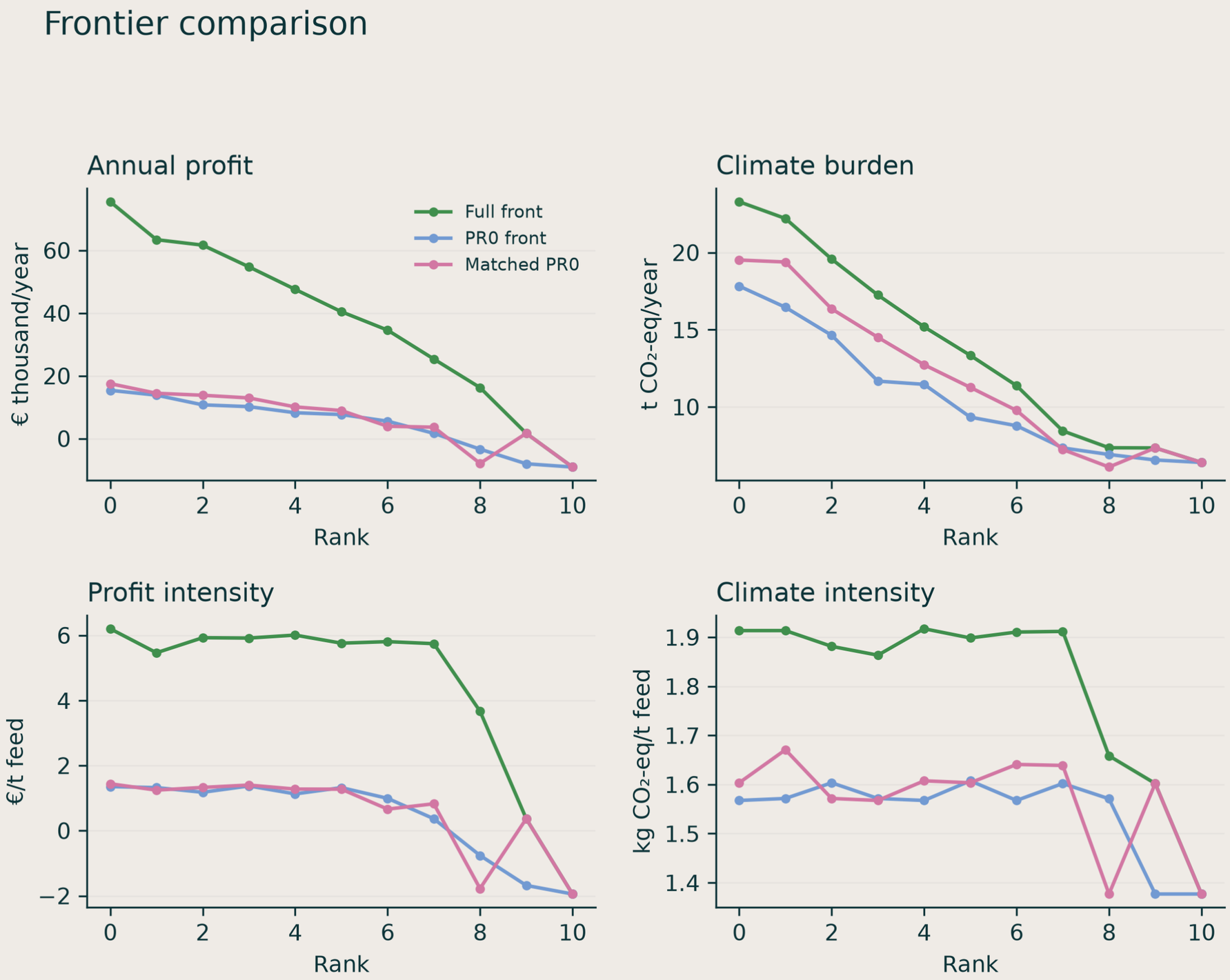
Projected NPV, revenue, recorded OPEX and subsidy across two independent frontiers

Scenario: Full and protein-bypass BioSTEAM screening fronts
Units: € million; € thousand/year; € thousand
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay



Paired PR0 minus original effects on profit, climate, revenue and electricity

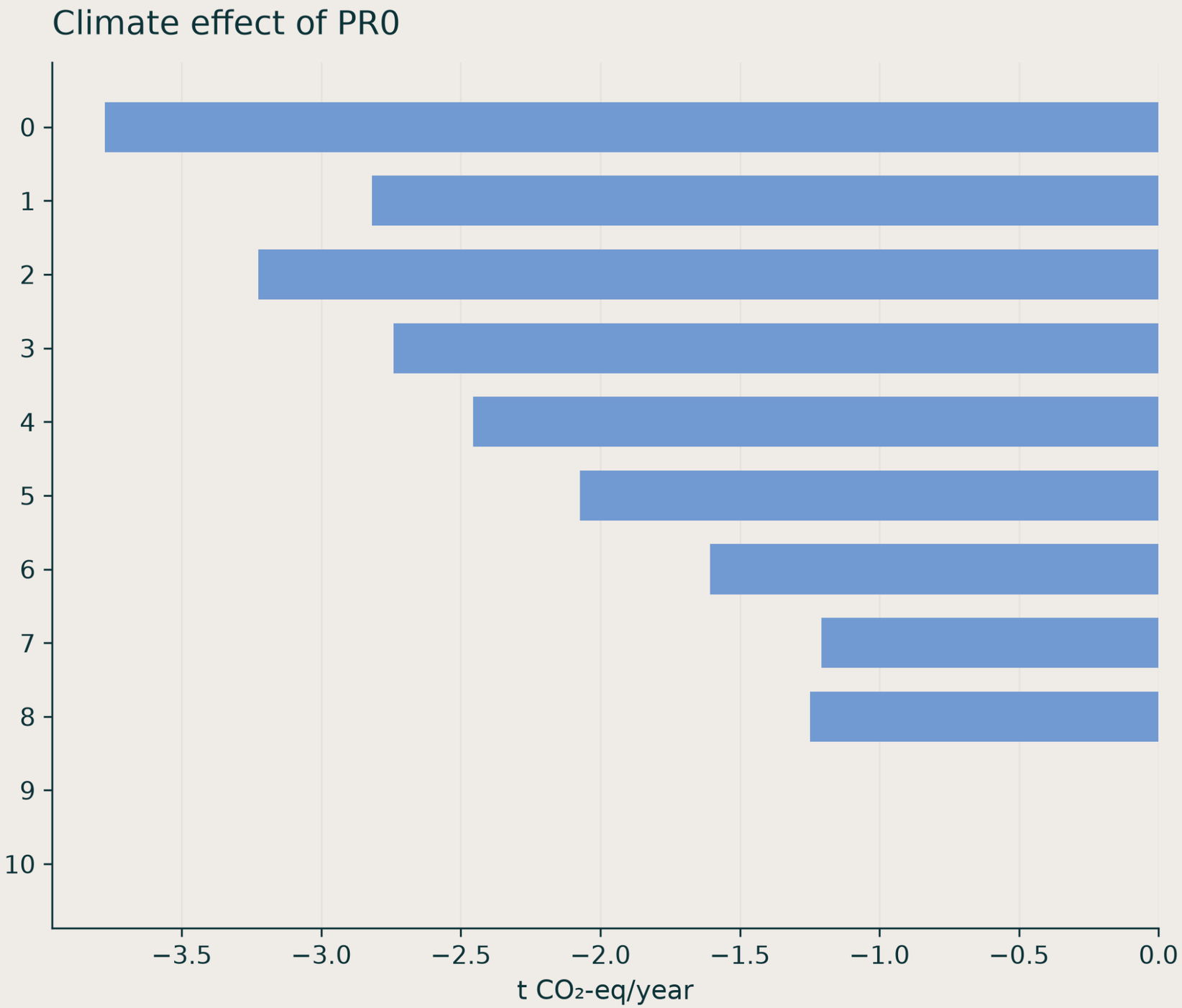
Scenario: 11 matched full-front protein-bypass reruns
Units: € thousand/year; t CO_2-eq/year; MWh/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay



Profit and partial climate burden across full, independent PR0 and matched PR0 series

Scenario: Full/PR0 fronts and matched protein-bypass reruns
Units: € thousand/year; t CO_2-eq/year; €/t; kg CO_2-eq/t
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

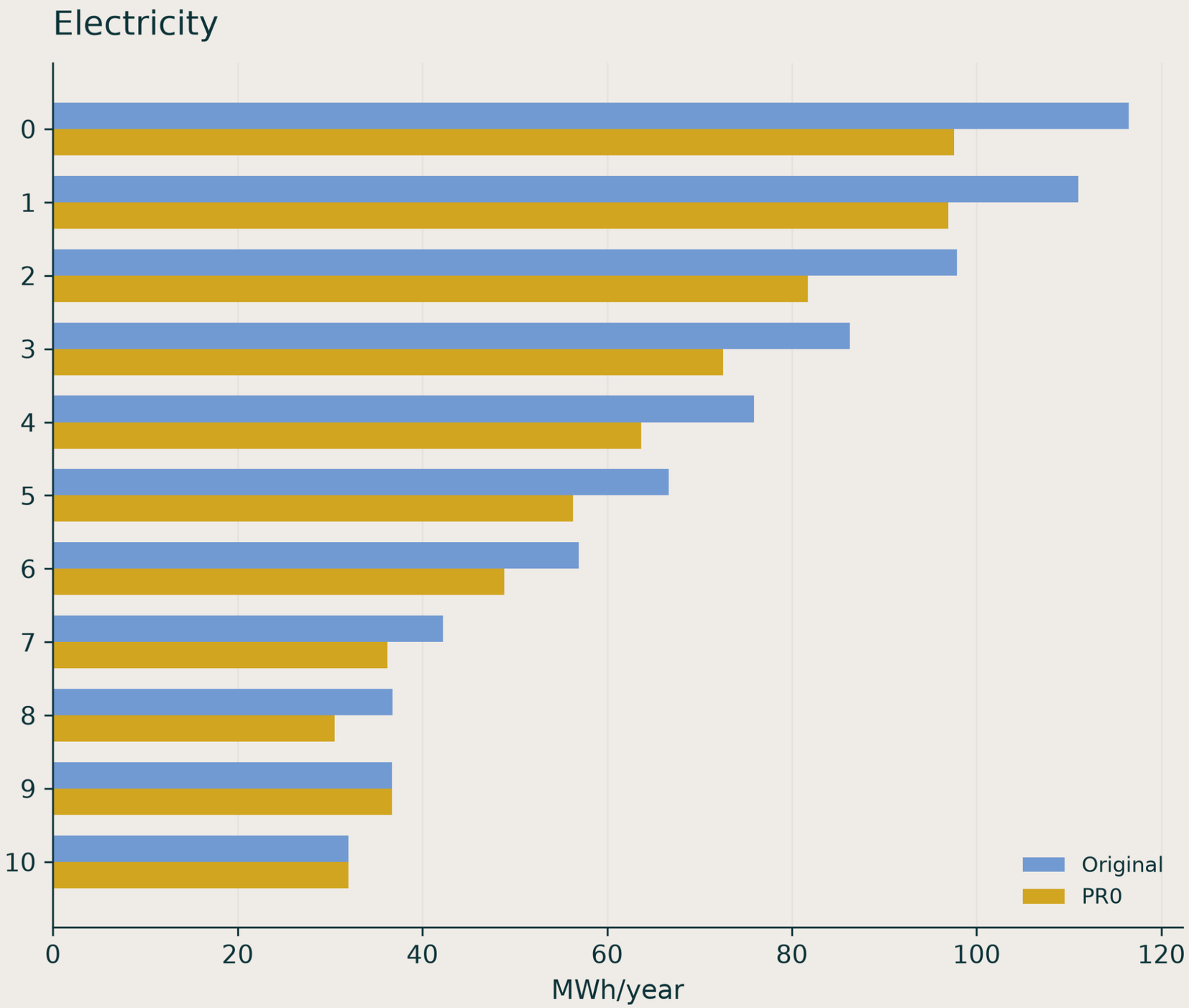
17. biosteam matched climate change



Change in partial attributional climate burden from bypassing protein

Scenario: Full-front ranks 0–10 rerun with PR0 only; same feed and candidate economics per pair
Units: t CO_2-eq/year
Source: Greenhub/results/actual_pareto_2000/decision_batches; biosteam_frontiers/rank_*.json

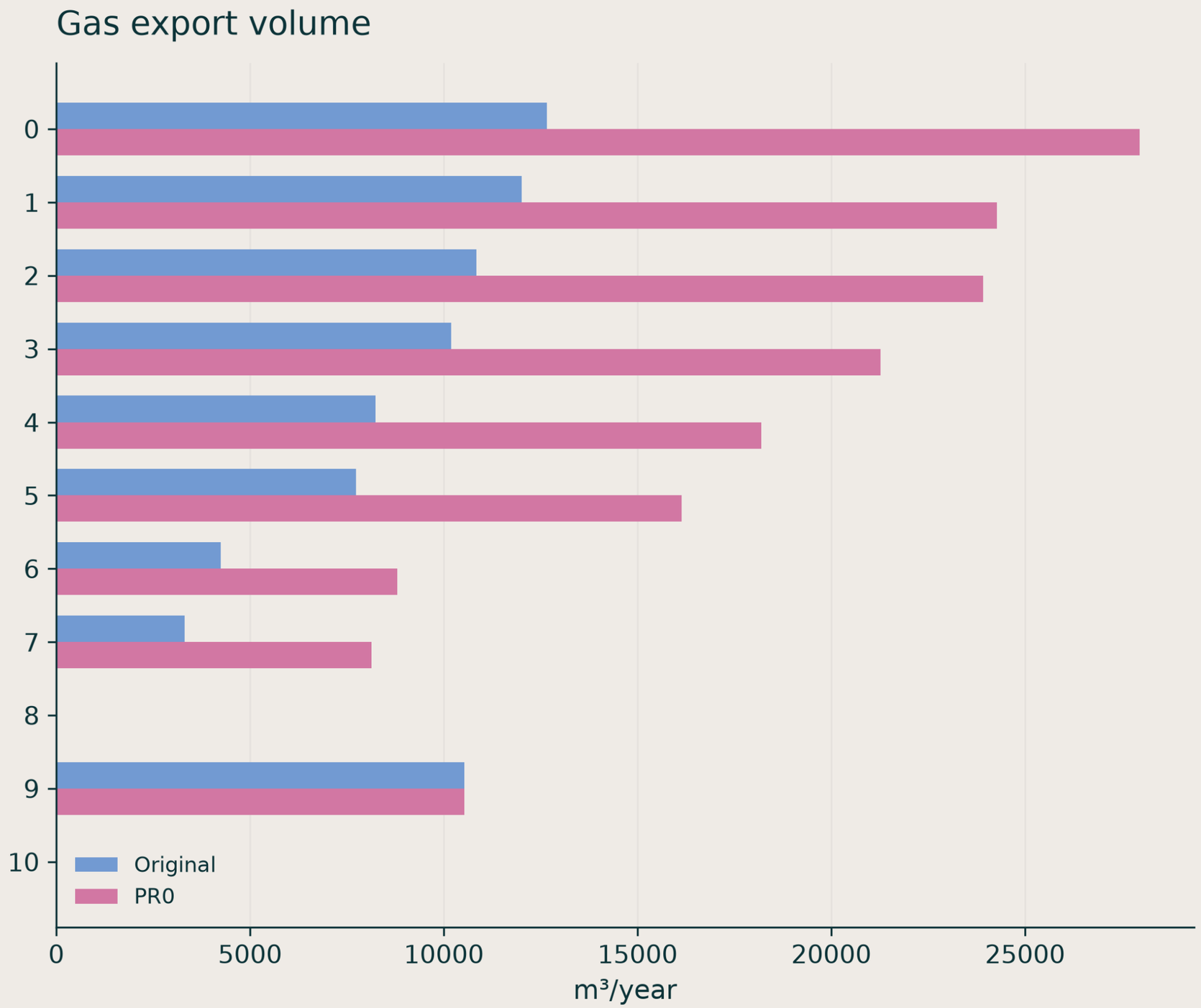
18. biosteam matched electricity comparison



Paired original versus protein-bypass electricity

Scenario: Full-front ranks 0–10 rerun with PR0 only; same feed and candidate economics per pair
Units: MWh/year
Source: Greenhub/results/actual_pareto_2000/decision_batches; biosteam_frontiers/rank_*.json

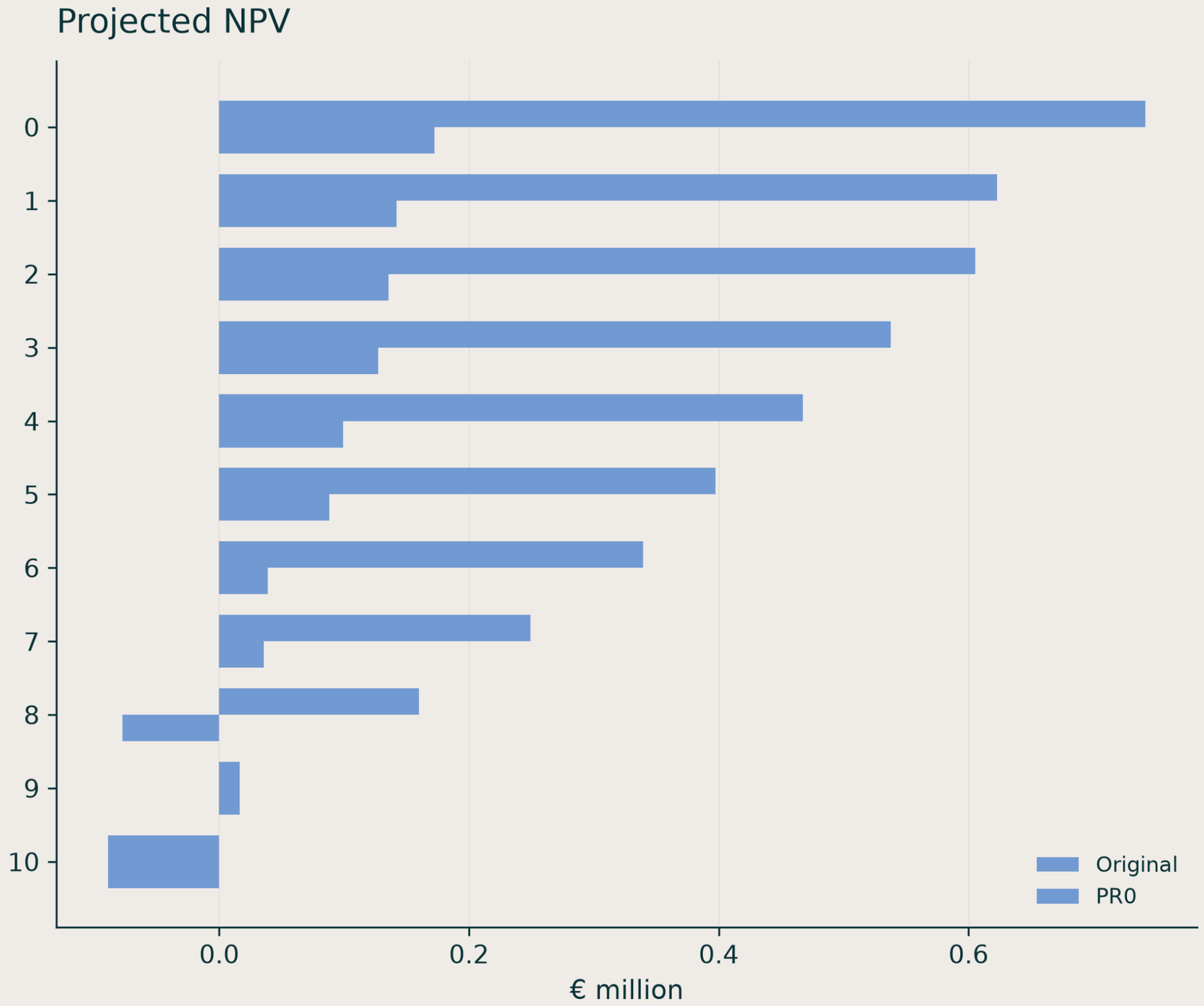
19. biosteam matched gas comparison



Paired original versus protein-bypass gas export volume

Scenario: Full-front ranks 0–10 rerun with PR0 only; same feed and candidate economics per pair
Units: m³/year
Source: Greenhub/results/actual_pareto_2000/decision_batches; biosteam_frontiers/rank_*.json

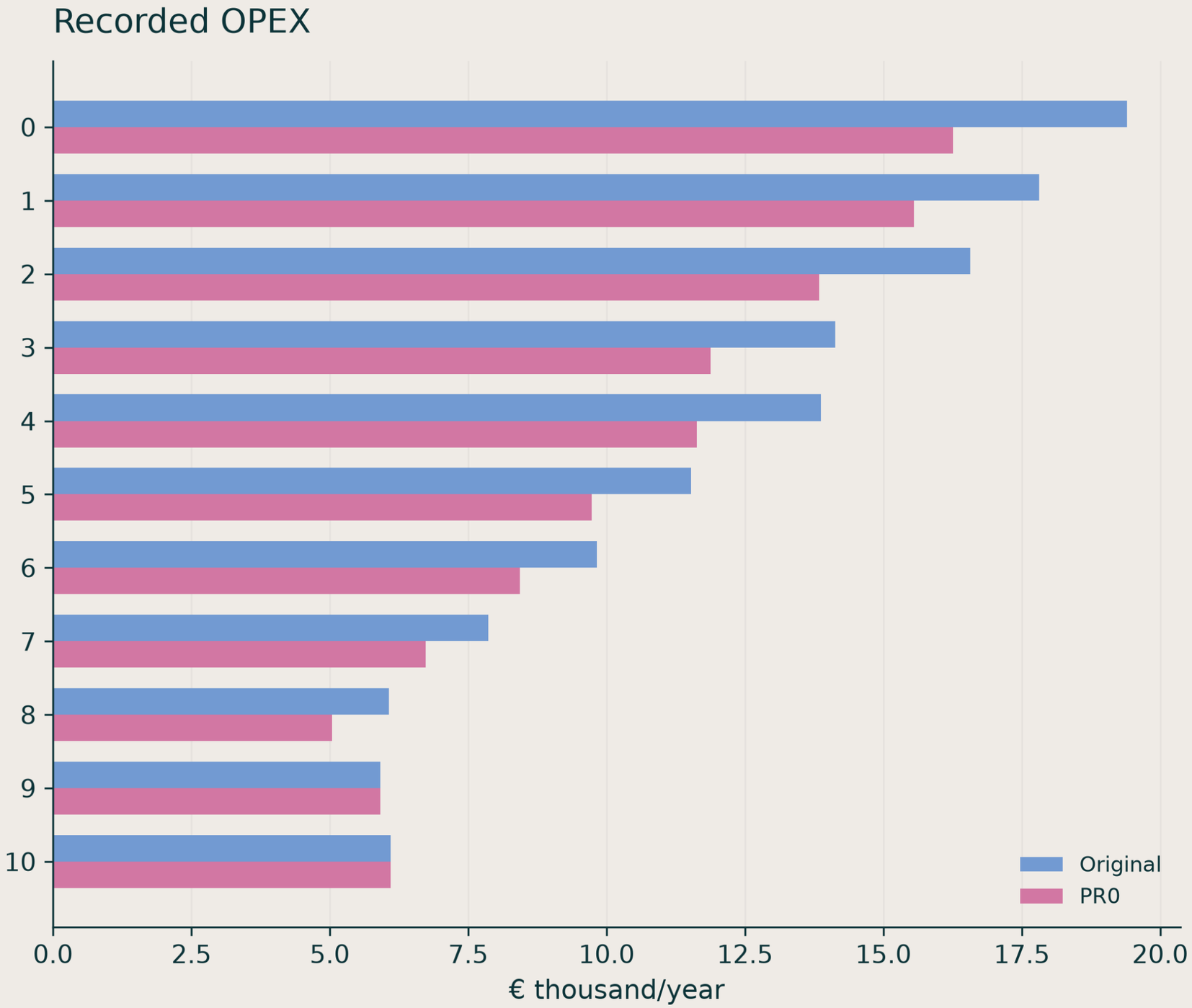
20. biosteam matched npv comparison



Paired original versus protein-bypass projected npv

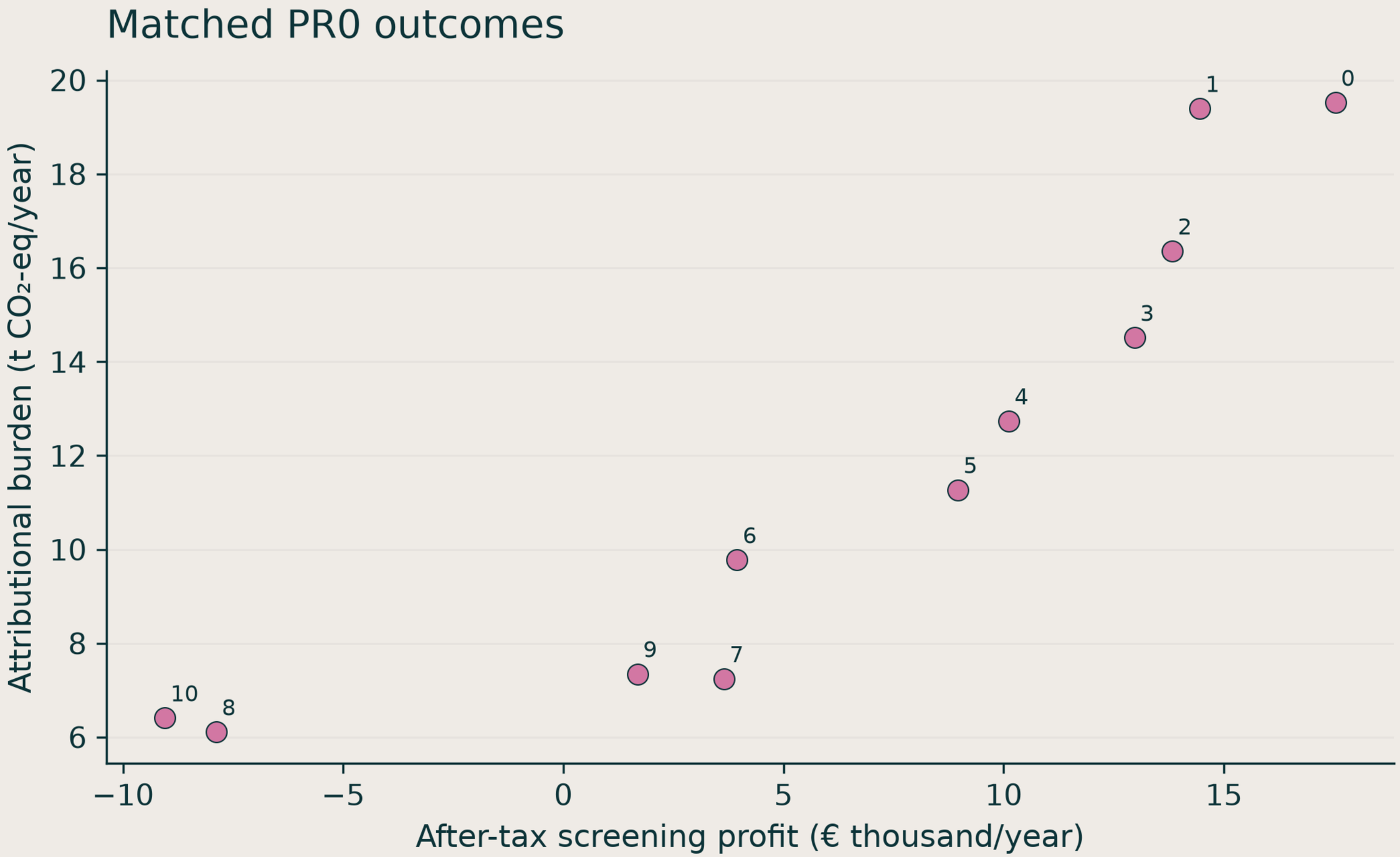
Scenario: Full-front ranks 0–10 rerun with PR0 only; same feed and candidate economics per pair
Units: € million
Source: Greenhub/results/actual_pareto_2000/decision_batches; biosteam_frontiers/rank_*.json

21. biosteam matched opex comparison



Paired original versus protein-bypass recorded opex

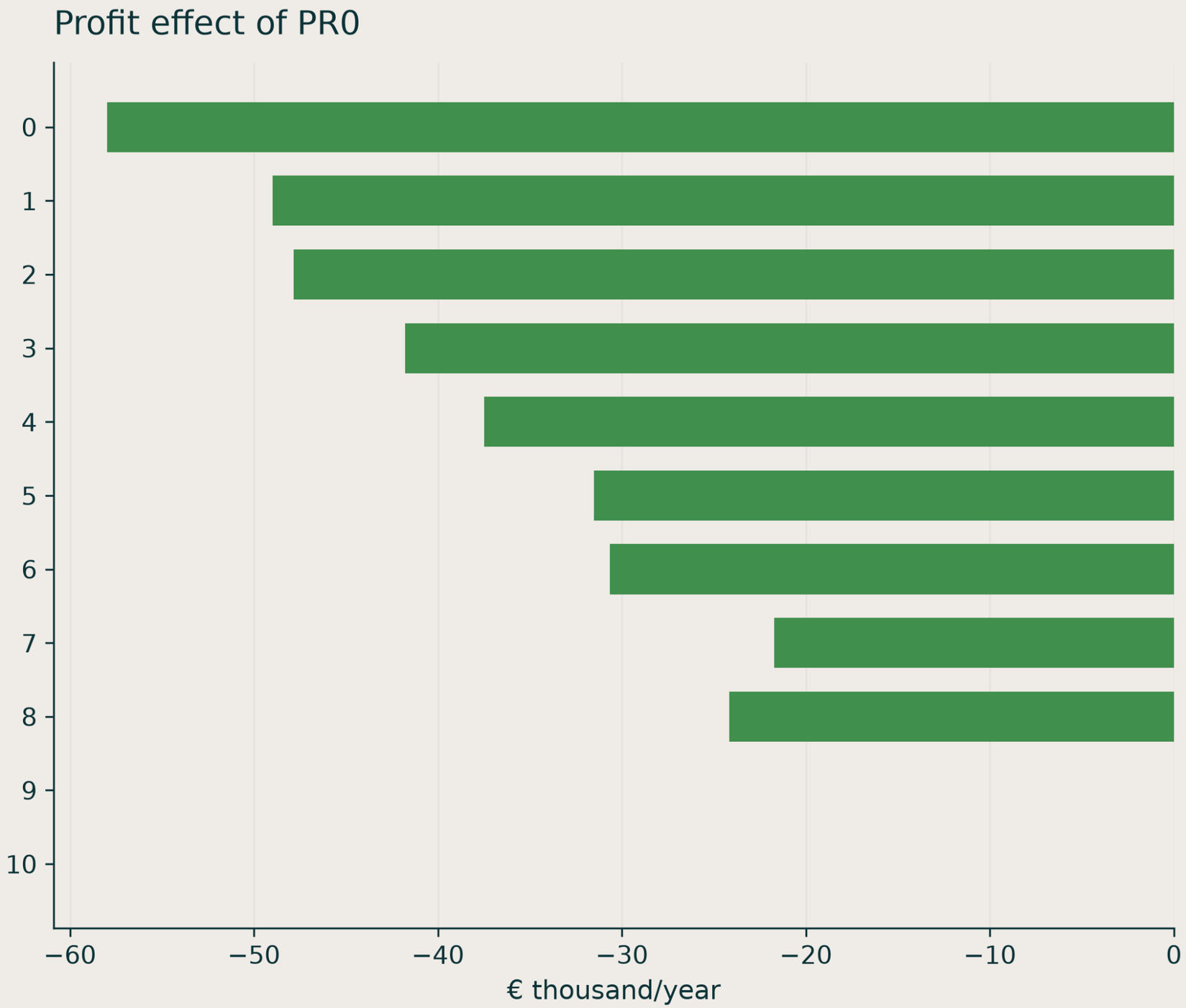
Scenario: Full-front ranks 0–10 rerun with PR0 only; same feed and candidate economics per pair
Units: € thousand/year
Source: Greenhub/results/actual_pareto_2000/decision_batches; biosteam_frontiers/rank_*.json



Matched protein-bypass outcomes retaining full-front rank labels

Scenario: Full-front ranks 0–10 rerun with PR0 only; same feed and candidate economics per pair
Units: € thousand/year; t CO_2-eq/year
Source: Greenhub/results/actual_pareto_2000/decision_batches; biosteam_frontiers/rank_*.json

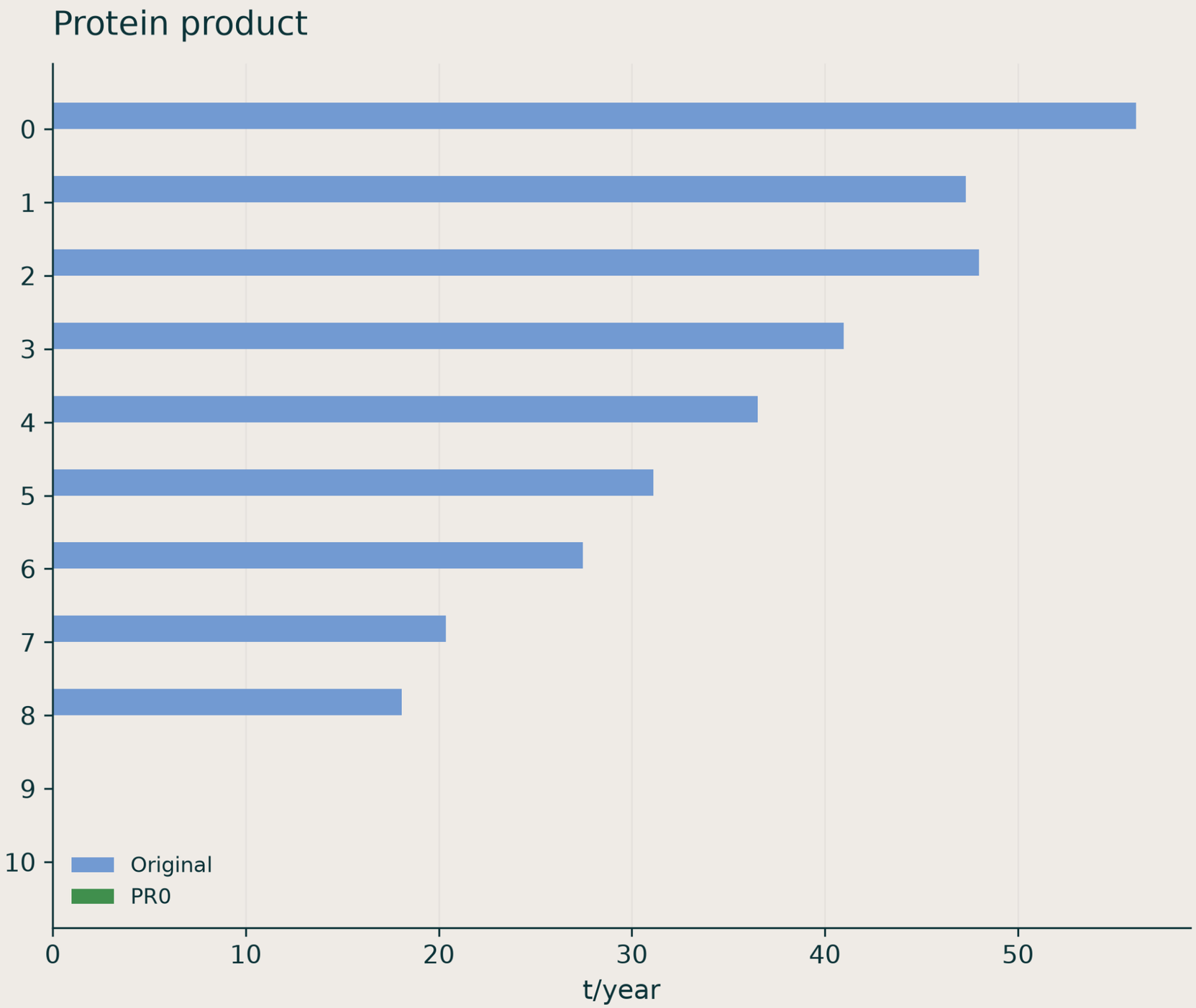
23. biosteam matched profit change



Change in after-tax screening profit from bypassing protein

Scenario: Full-front ranks 0–10 rerun with PR0 only; same feed and candidate economics per pair
Units: € thousand/year
Source: Greenhub/results/actual_pareto_2000/decision_batches; biosteam_frontiers/rank_*.json

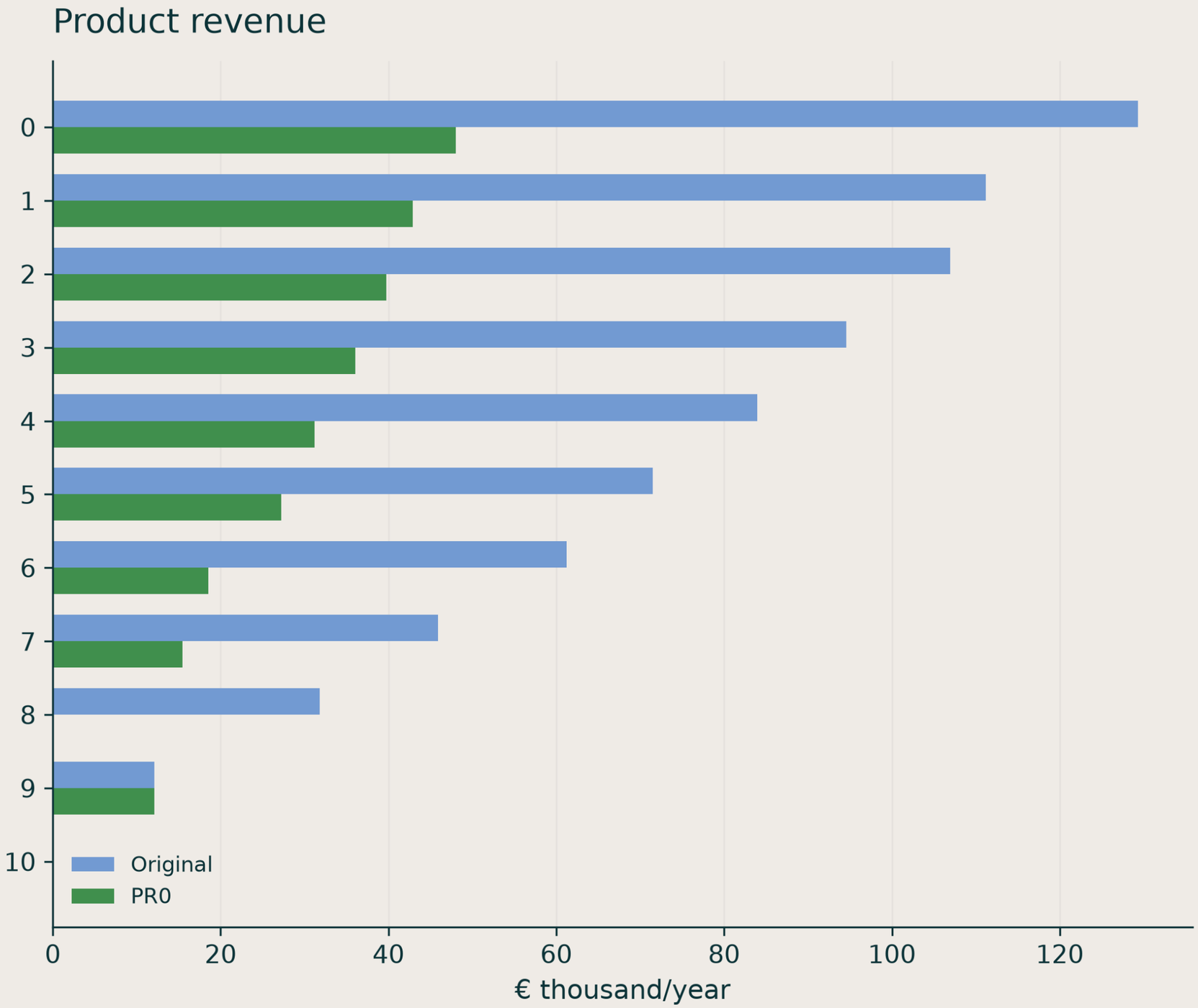
24. biosteam matched protein comparison



Paired original versus protein-bypass protein product

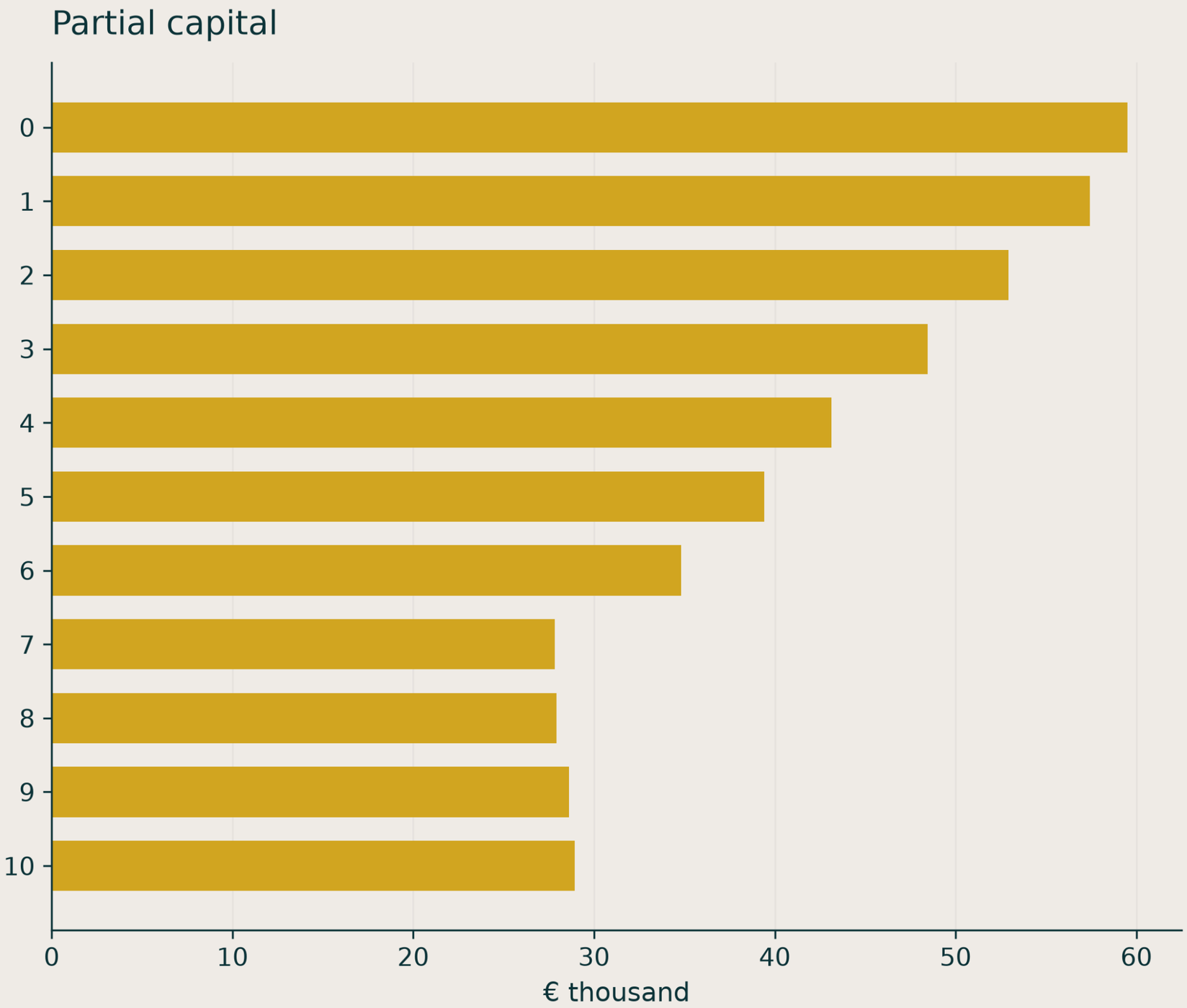
Scenario: Full-front ranks 0–10 rerun with PR0 only; same feed and candidate economics per pair
Units: t/year
Source: Greenhub/results/actual_pareto_2000/decision_batches; biosteam_frontiers/rank_*.json

25. biosteam matched revenue comparison



Paired original versus protein-bypass product revenue

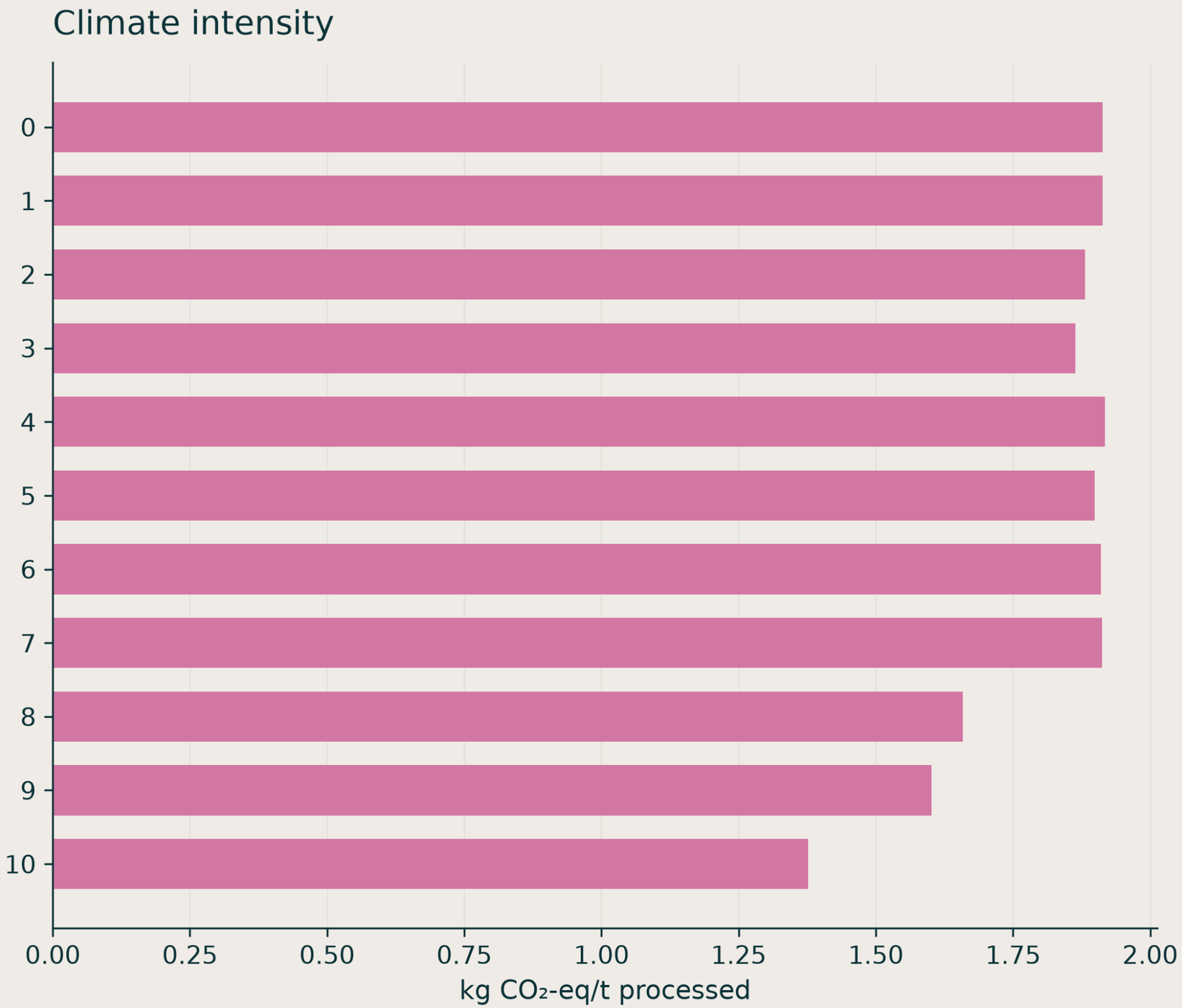
Scenario: Full-front ranks 0–10 rerun with PR0 only; same feed and candidate economics per pair
Units: € thousand/year
Source: Greenhub/results/actual_pareto_2000/decision_batches; biosteam_frontiers/rank_*.json



Partial capital among ranked candidates

Scenario: all backend batch; 2000 total simulations, seed 42; objective-eligible 1049
Units: € thousand
Source: Greenhub/results/actual_pareto_2000/decision_batches; biosteam_frontiers/rank_*.json

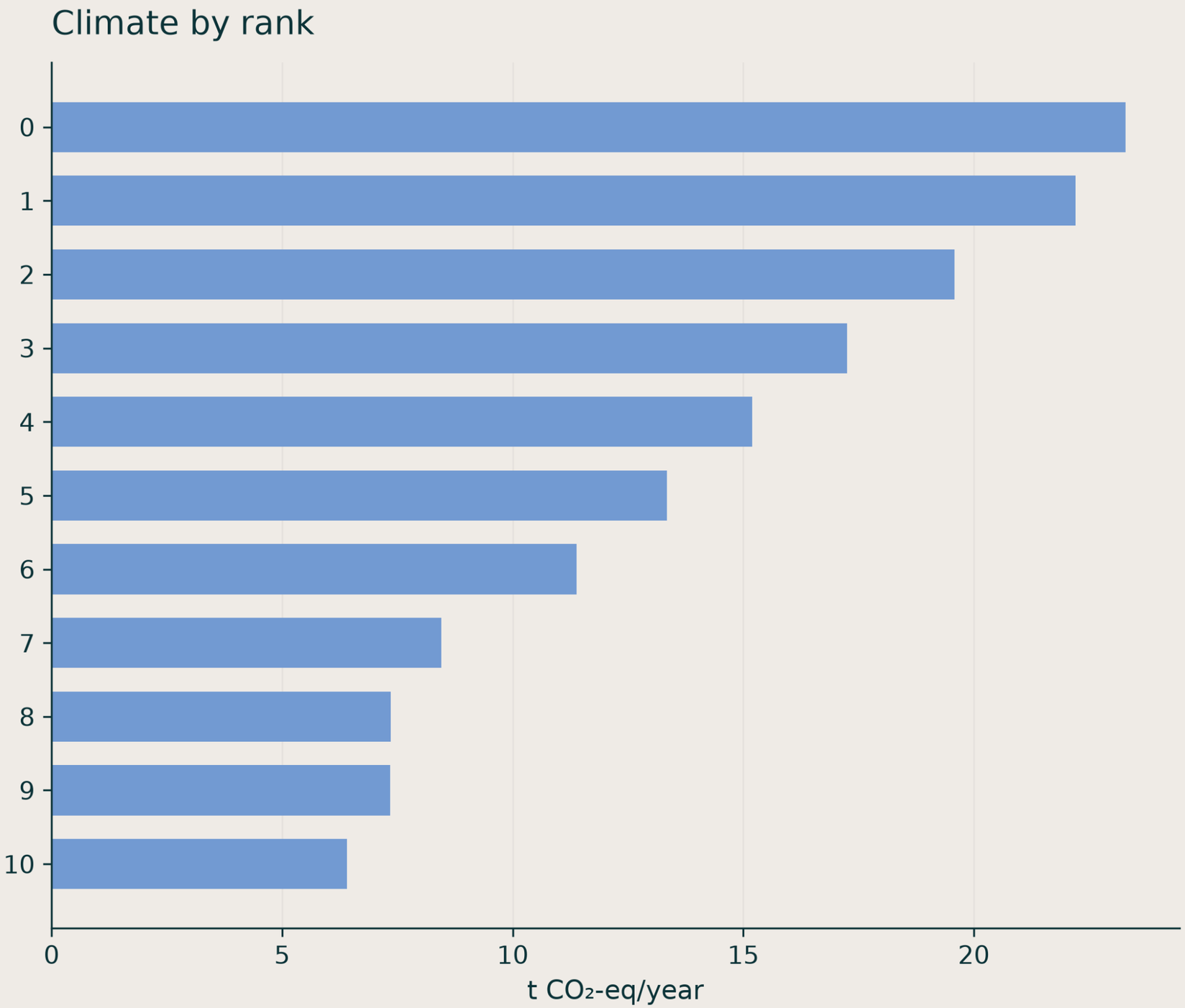
27. biosteam all climate intensity



Partial attributional climate burden per wet tonne

Scenario: all backend batch; 2000 total simulations, seed 42; objective-eligible 1049
Units: kg CO_2-eq/t processed
Source: Greenhub/results/actual_pareto_2000/decision_batches; biosteam_frontiers/rank_*.json

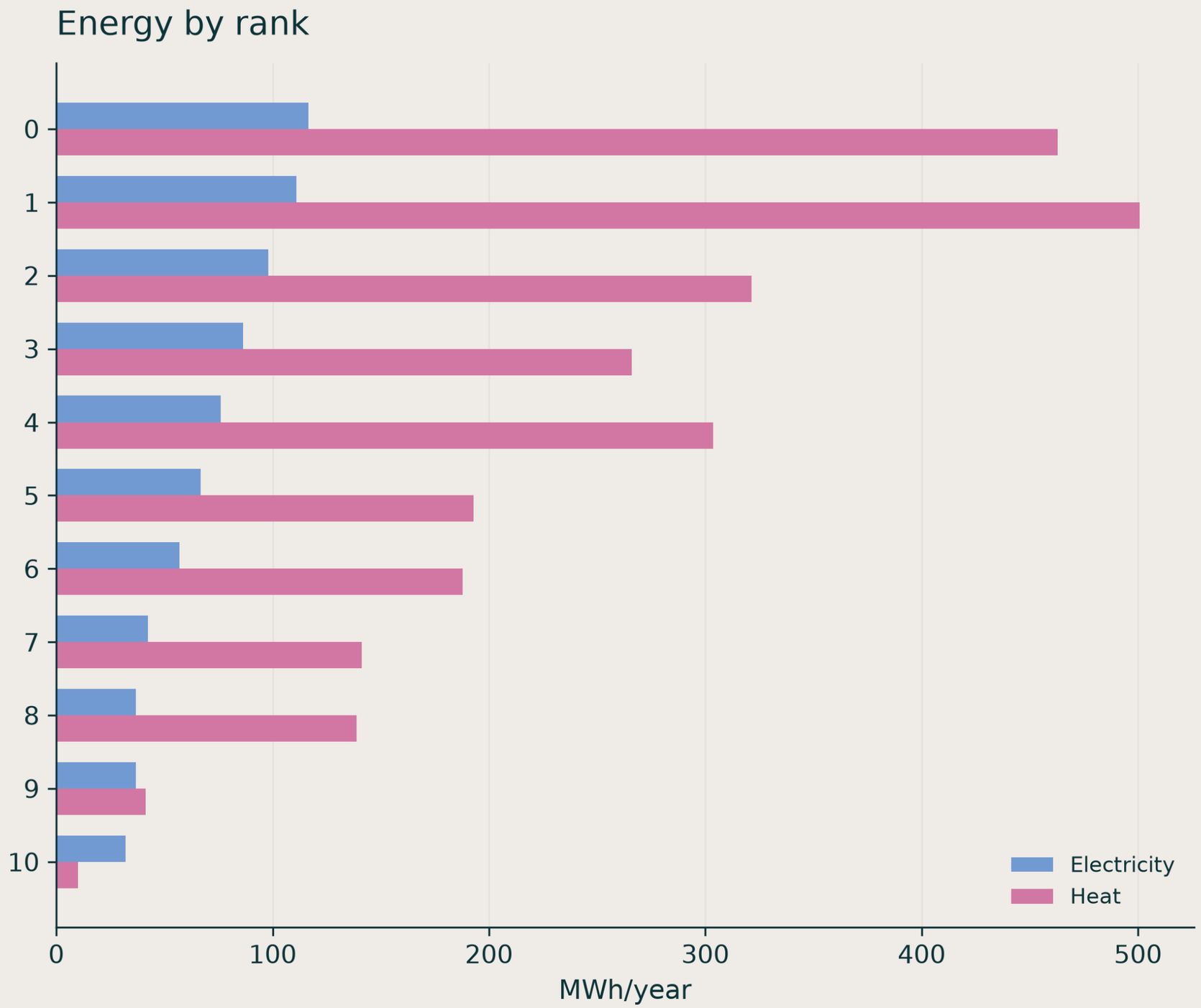
28. biosteam all climate ranks



Partial attributional electricity burden by ranked candidate

Scenario: all backend batch; 2000 total simulations, seed 42; objective-eligible 1049
Units: t CO_2-eq/year
Source: Greenhub/results/actual_pareto_2000/decision_batches; biosteam_frontiers/rank_*.json

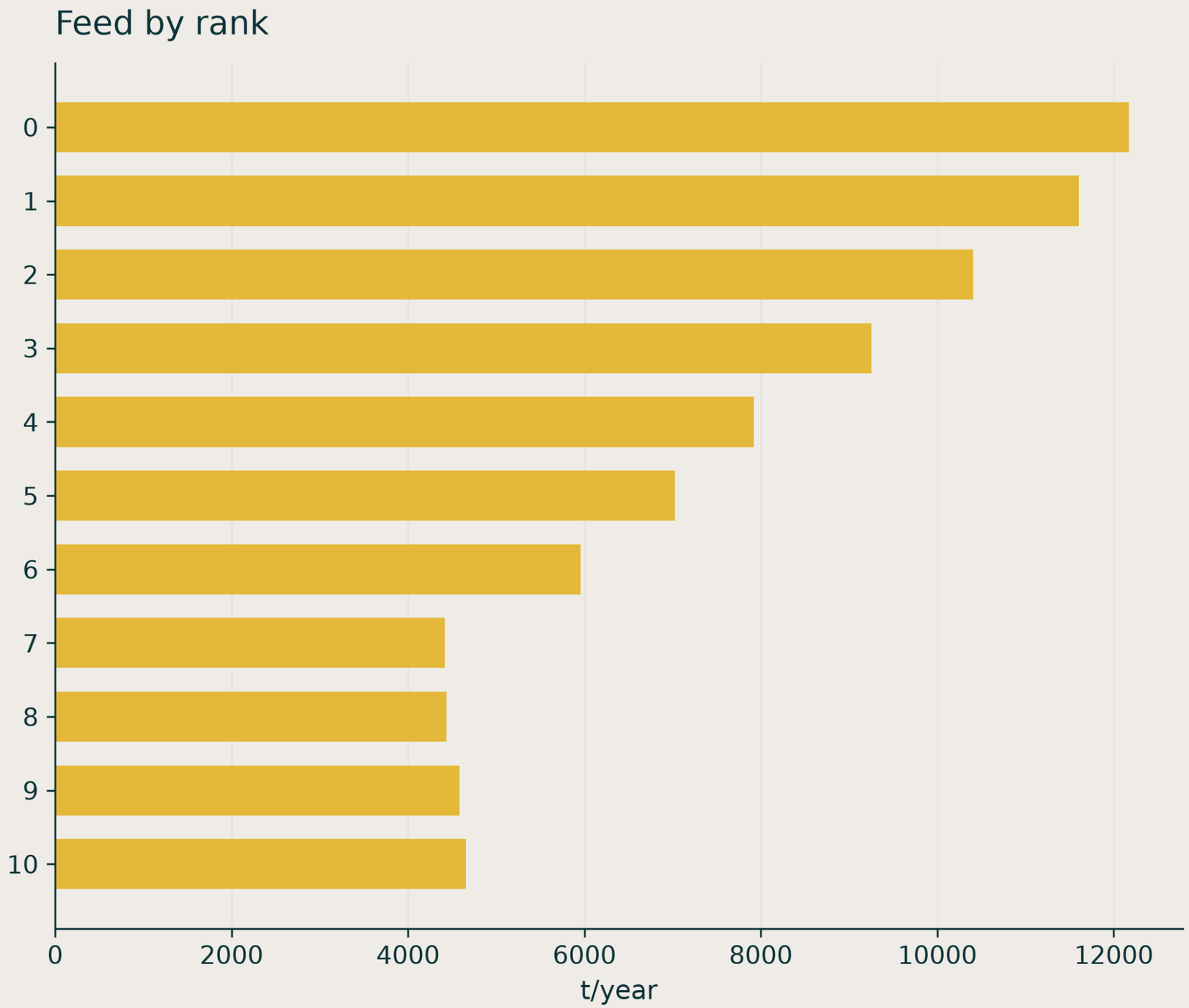
29. biosteam all energy ranks



Annual electricity and heating demand across ranked candidates

Scenario: all backend batch; 2000 total simulations, seed 42; objective-eligible 1049
Units: MWh/year
Source: Greenhub/results/actual_pareto_2000/decision_batches; biosteam_frontiers/rank_*.json

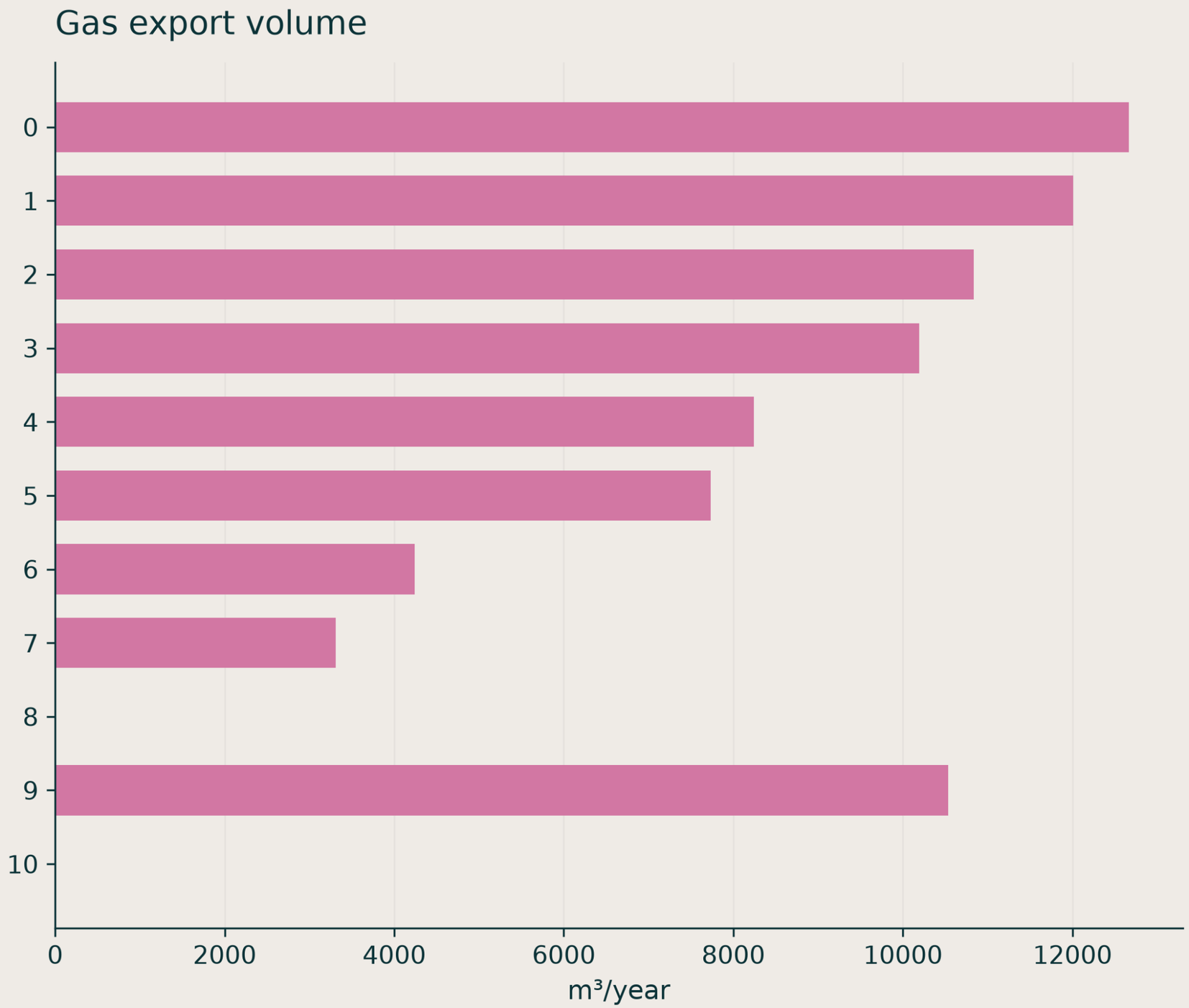
30. biosteam all feed ranks



Simulated annual wet feed across ranks

Scenario: all backend batch; 2000 total simulations, seed 42; objective-eligible 1049
Units: t/year
Source: Greenhub/results/actual_pareto_2000/decision_batches; biosteam_frontiers/rank_*.json

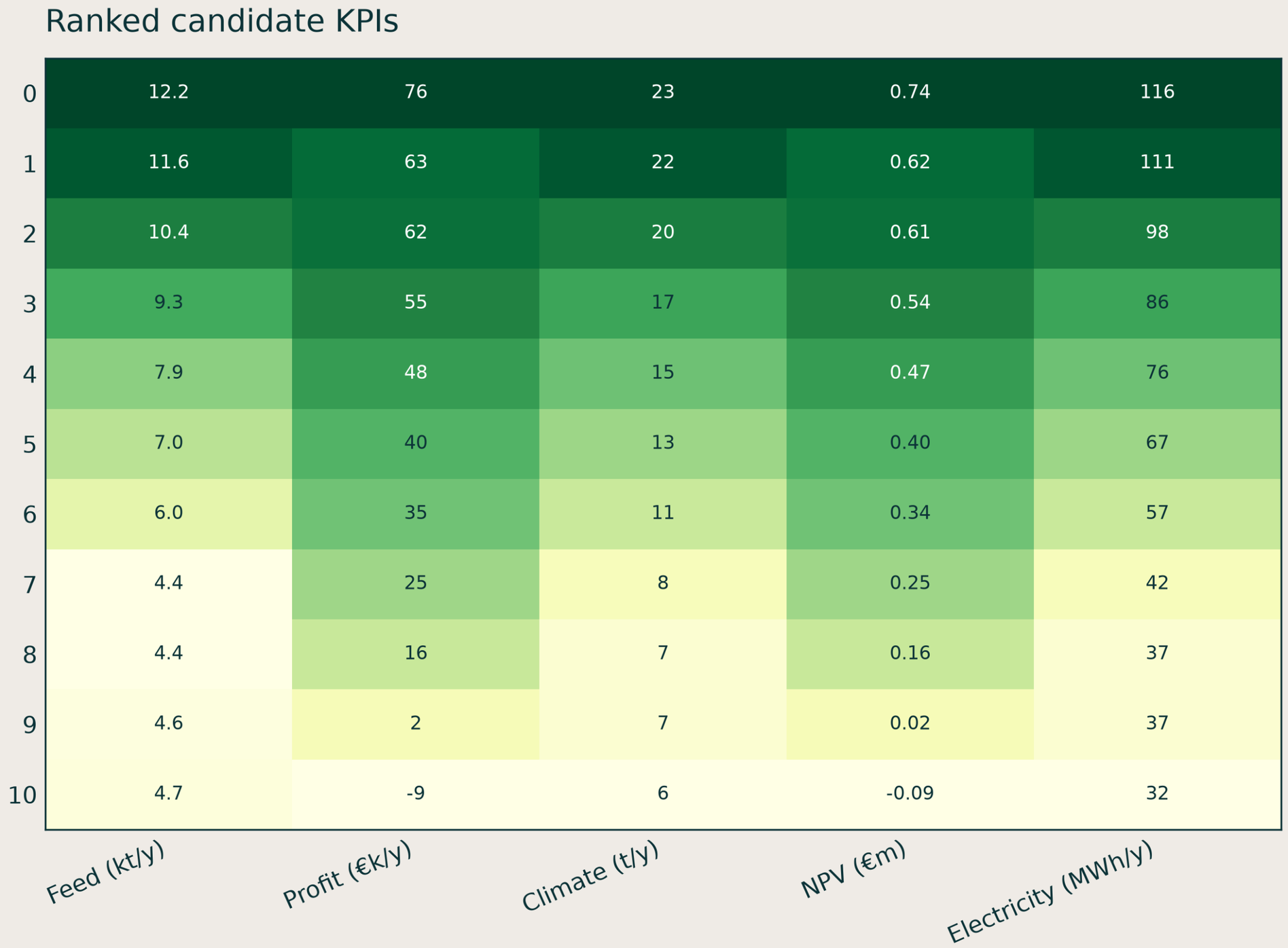
31. biosteam all gas ranks



Revenue-account dry gas export at 293.15 K; raw for BG0 and upgraded for selected BG routes

Scenario: all backend batch; 2000 total simulations, seed 42; objective-eligible 1049
Units: m³/year
Source: Greenhub/results/actual_pareto_2000/decision_batches; biosteam_frontiers/rank_*.json

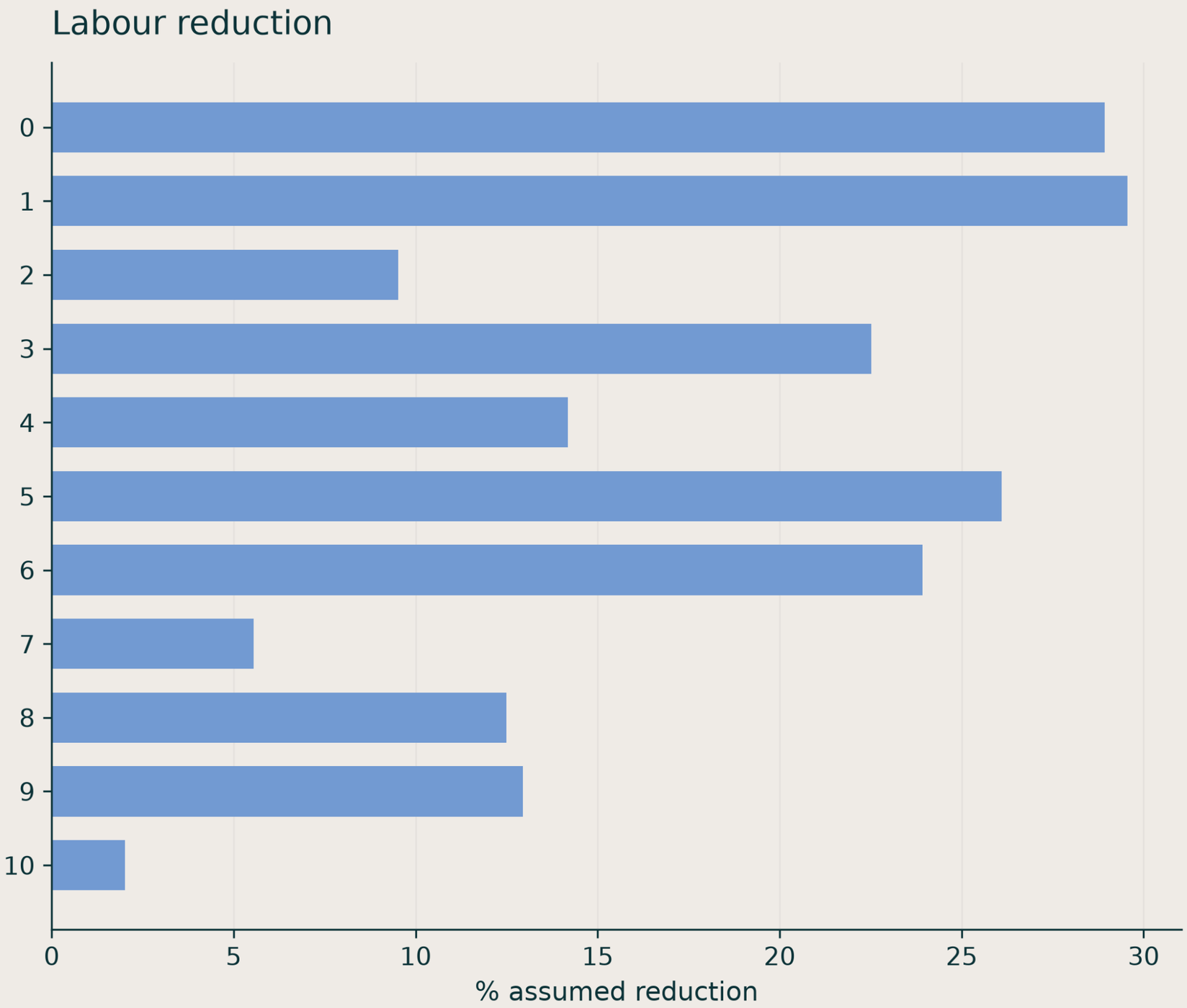
32. biosteam all kpi heatmap



Feed, screening finance, partial climate and electricity for ranked cases

Scenario: all backend batch; 2000 total simulations, seed 42; objective-eligible 1049
Units: Feed (kt/y); Profit (€k/y); Climate (t/y); NPV (€m); Electricity (MWh/y)
Source: Greenhub/results/actual_pareto_2000/decision_batches; biosteam_frontiers/rank_*.json

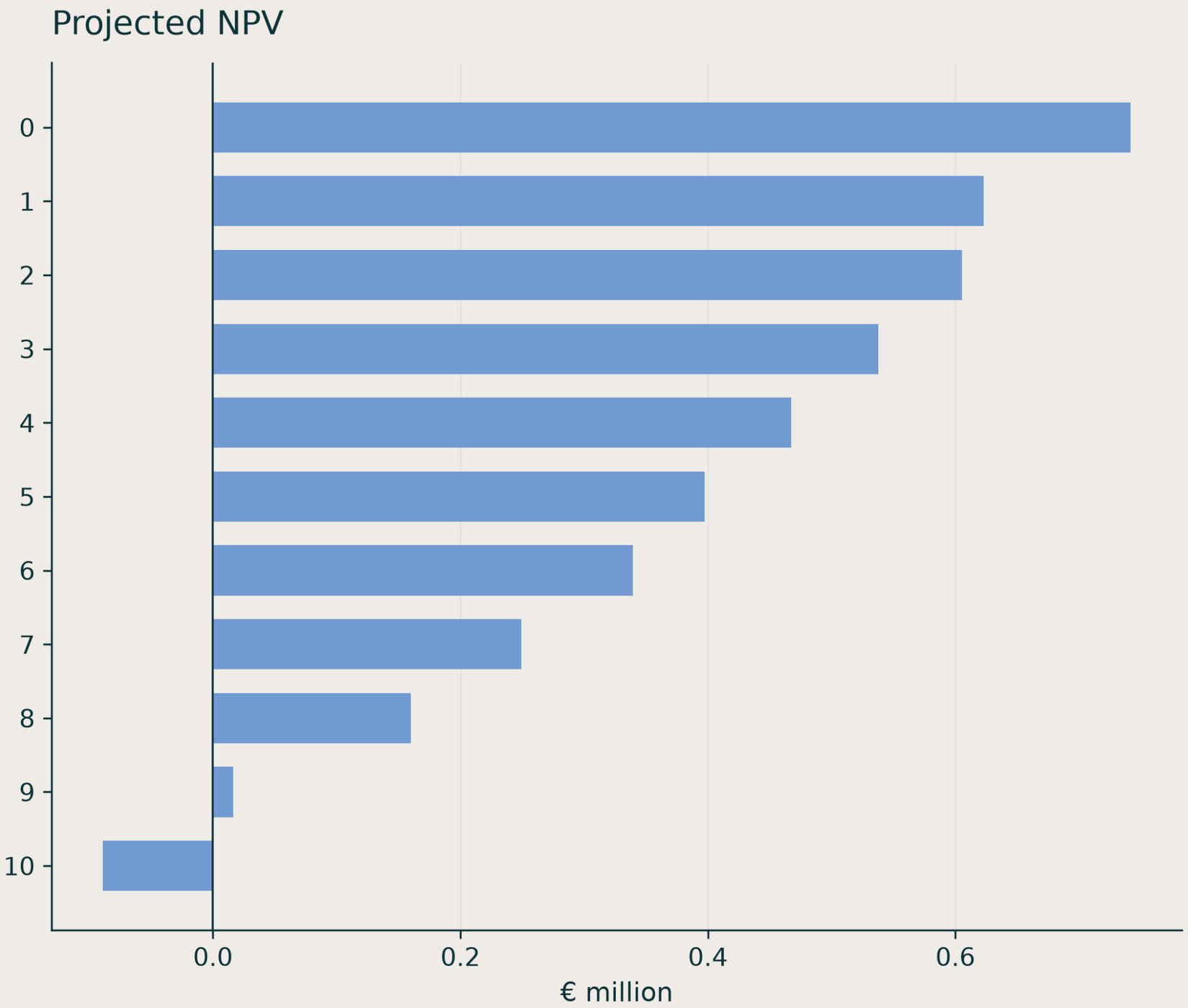
33. biosteam all labour ranks



Sampled candidate labour-reduction overlay

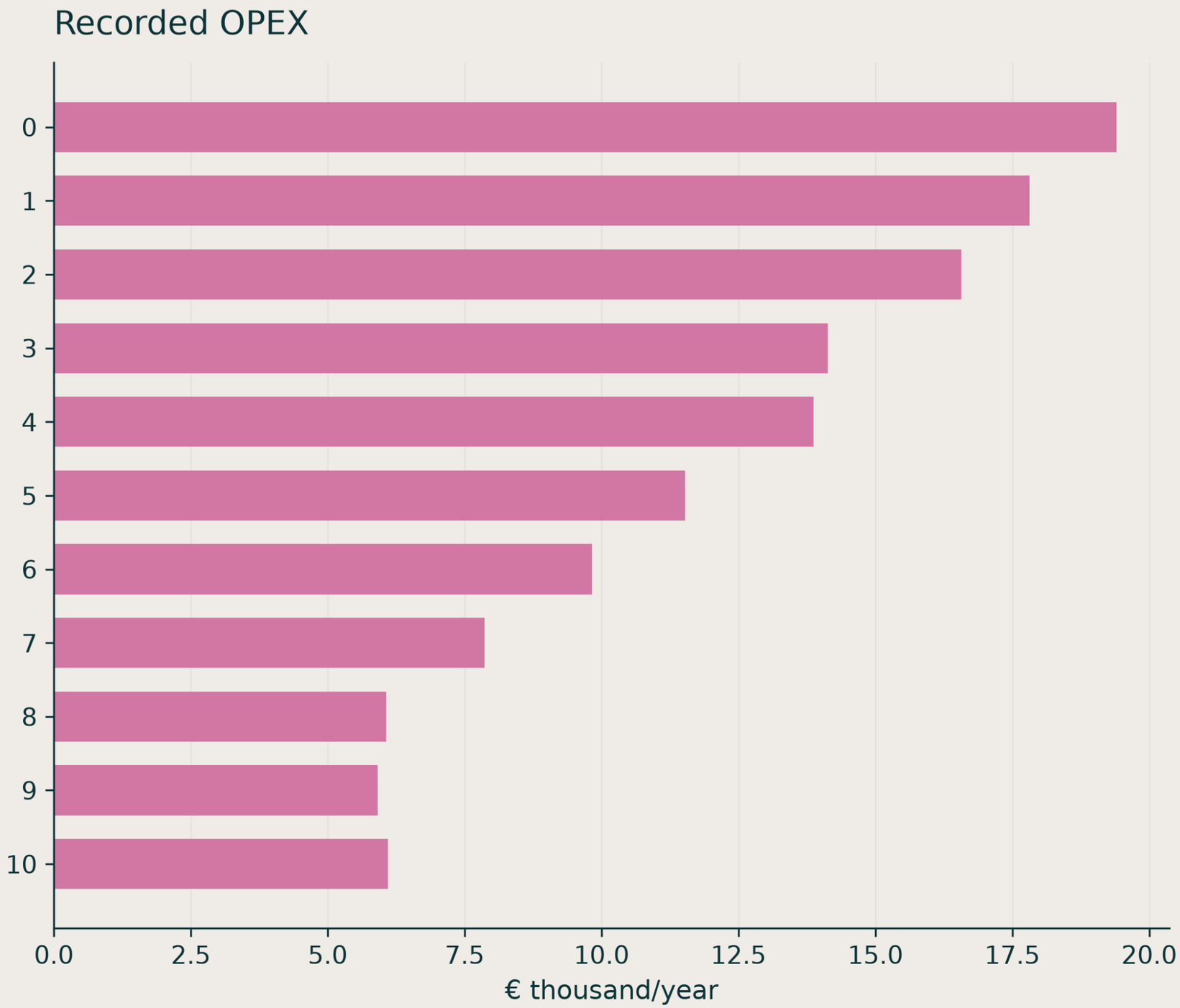
Scenario: all backend batch; 2000 total simulations, seed 42; objective-eligible 1049
Units: % assumed reduction
Source: Greenhub/results/actual_pareto_2000/decision_batches; biosteam_frontiers/rank_*.json

34. biosteam all npv ranks



20-year constant-account NPV across ranked candidates

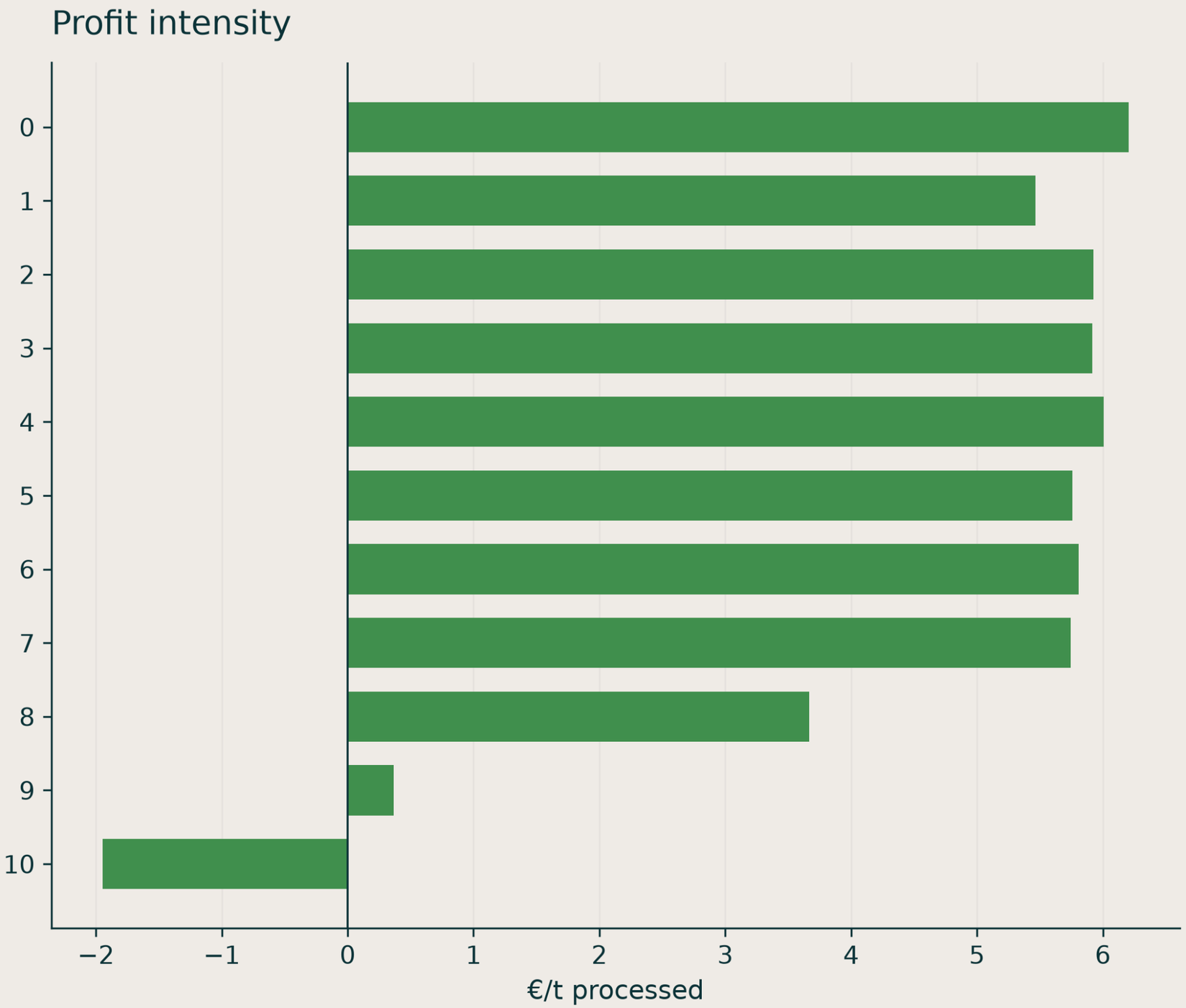
Scenario: all backend batch; 2000 total simulations, seed 42; objective-eligible 1049
Units: € million
Source: Greenhub/results/actual_pareto_2000/decision_batches; biosteam_frontiers/rank_*.json



Recorded OPEX among ranked candidates

Scenario: all backend batch; 2000 total simulations, seed 42; objective-eligible 1049
Units: € thousand/year
Source: Greenhub/results/actual_pareto_2000/decision_batches; biosteam_frontiers/rank_*.json

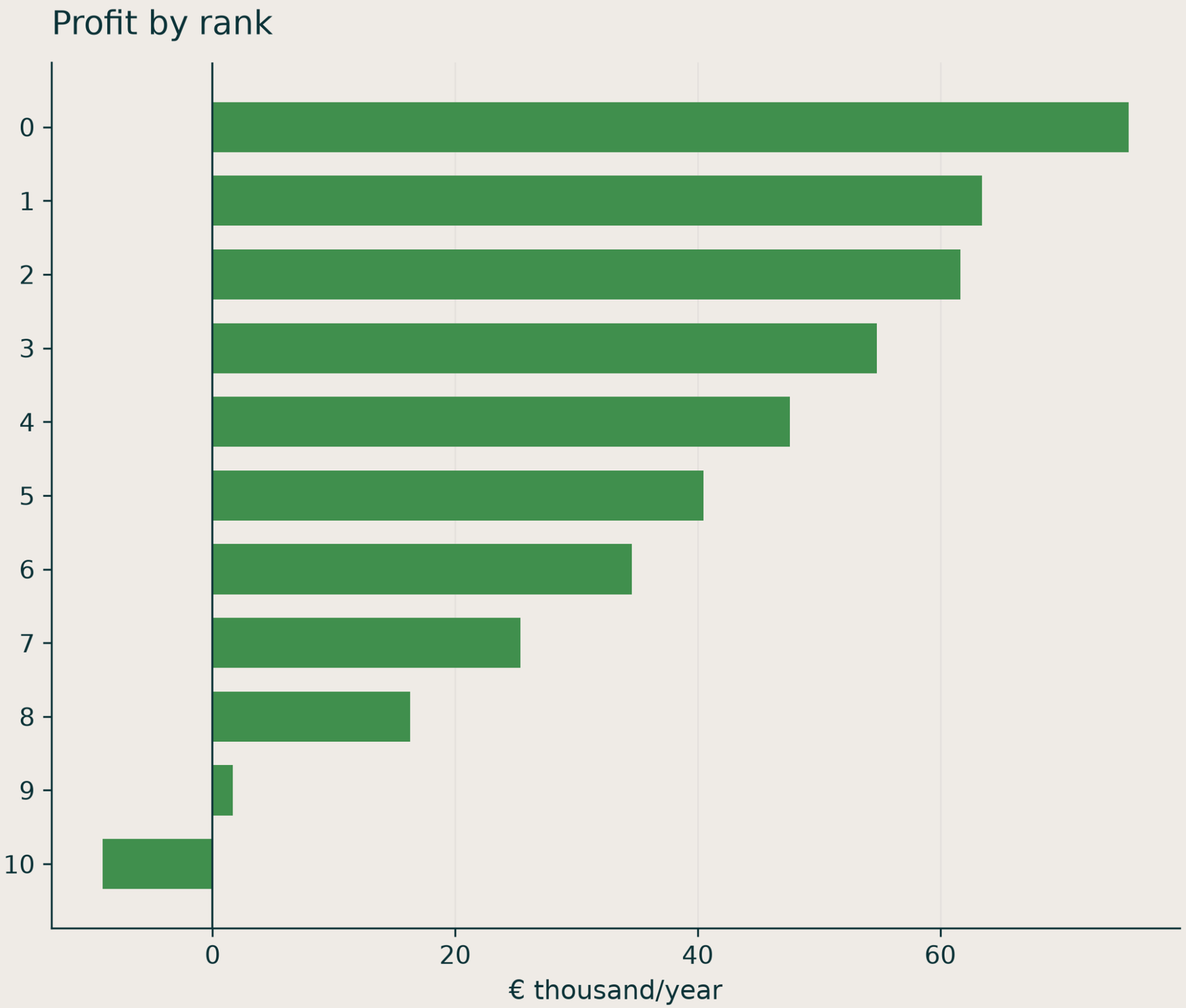
36. biosteam all profit intensity



Annualized screening profit normalized to simulated wet feed

Scenario: all backend batch; 2000 total simulations, seed 42; objective-eligible 1049
Units: €/t processed
Source: Greenhub/results/actual_pareto_2000/decision_batches; biosteam_frontiers/rank_*.json

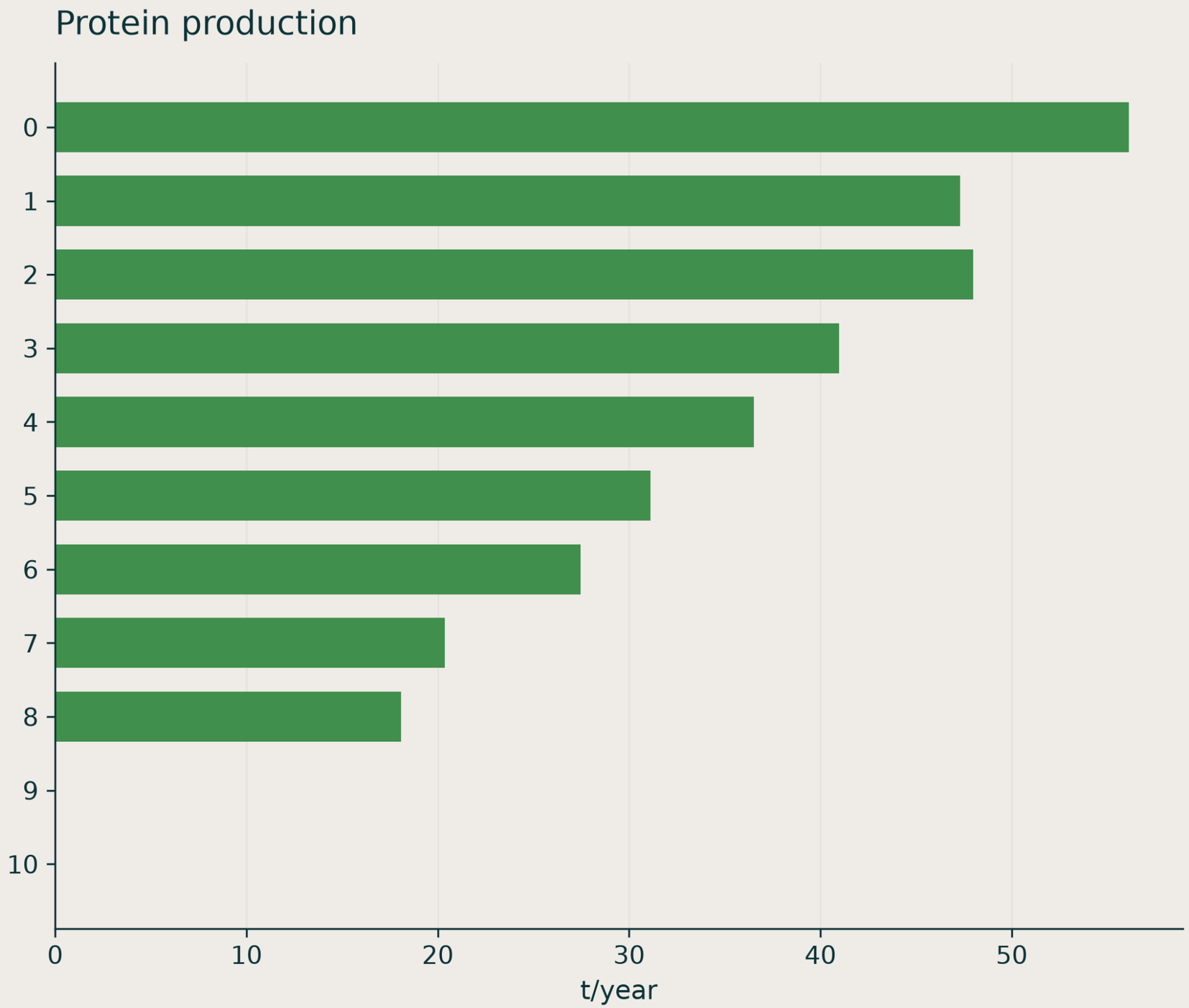
37. biosteam all profit ranks



Backend candidate-adjusted screening profit

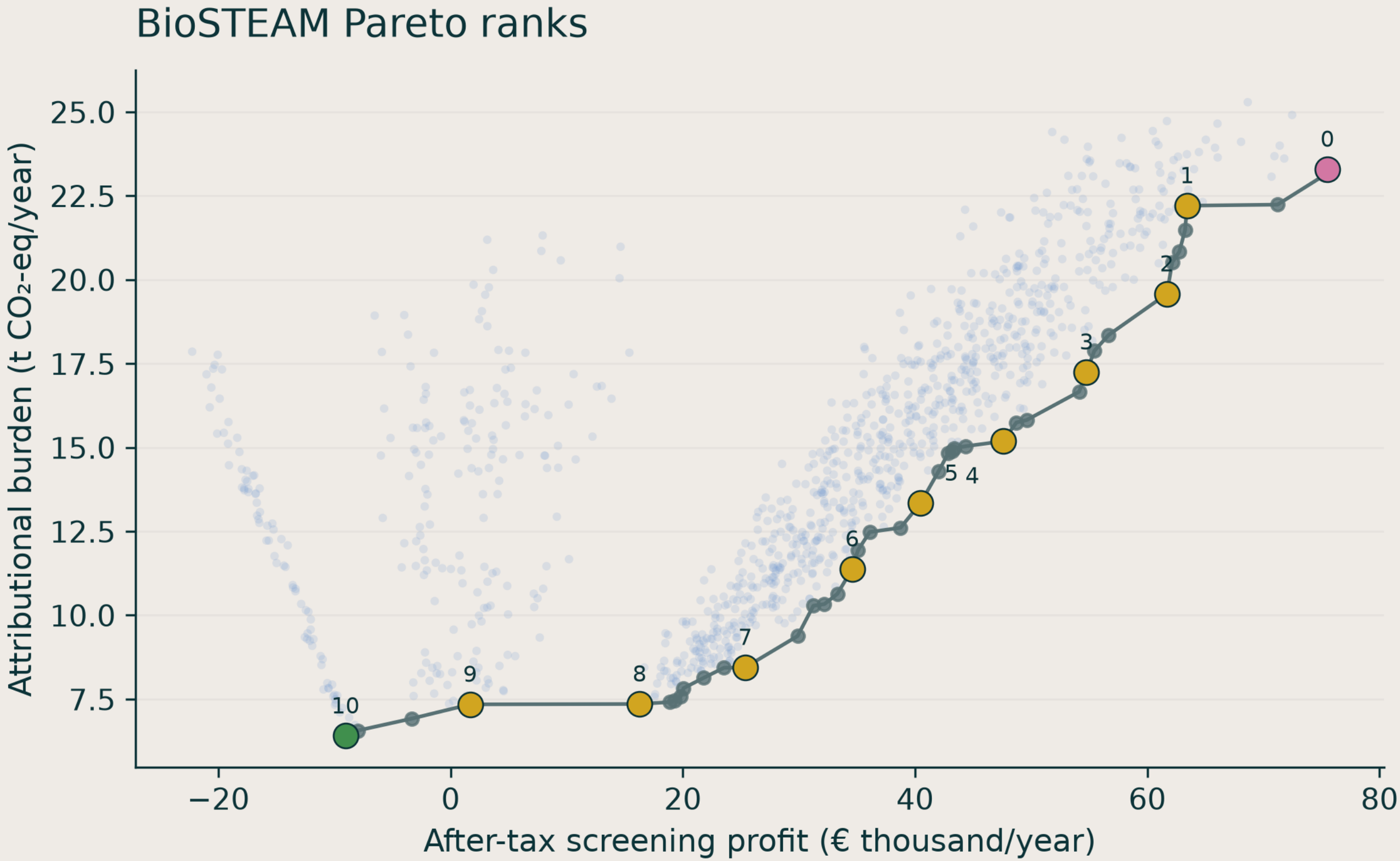
Scenario: all backend batch; 2000 total simulations, seed 42; objective-eligible 1049
Units: € thousand/year
Source: Greenhub/results/actual_pareto_2000/decision_batches; biosteam_frontiers/rank_*.json

38. biosteam all protein ranks



Modeled dry protein product by ranked case

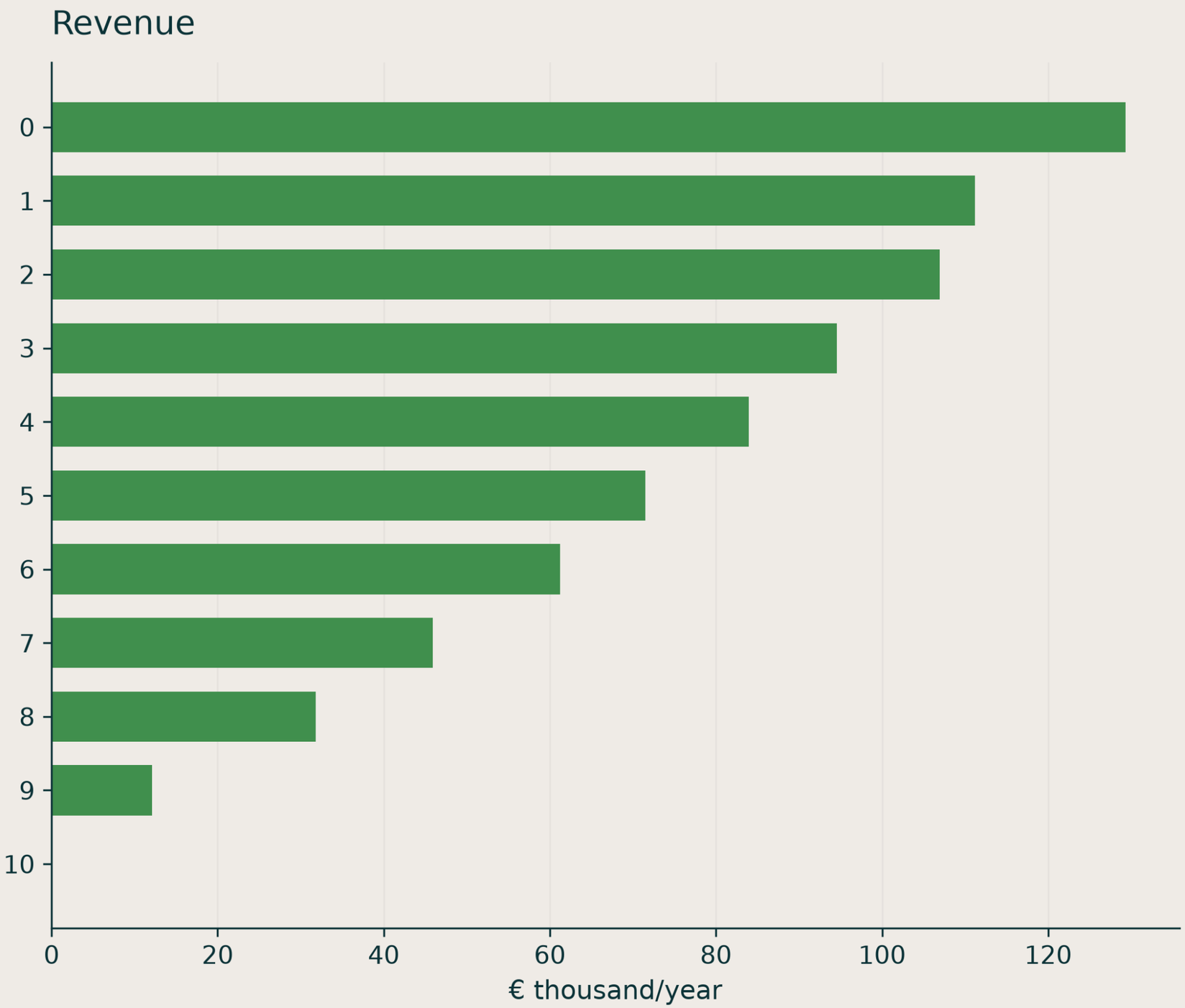
Scenario: all backend batch; 2000 total simulations, seed 42; objective-eligible 1049
Units: t/year
Source: Greenhub/results/actual_pareto_2000/decision_batches; biosteam_frontiers/rank_*.json



Backend-evaluated screening frontier with ranks 0–10

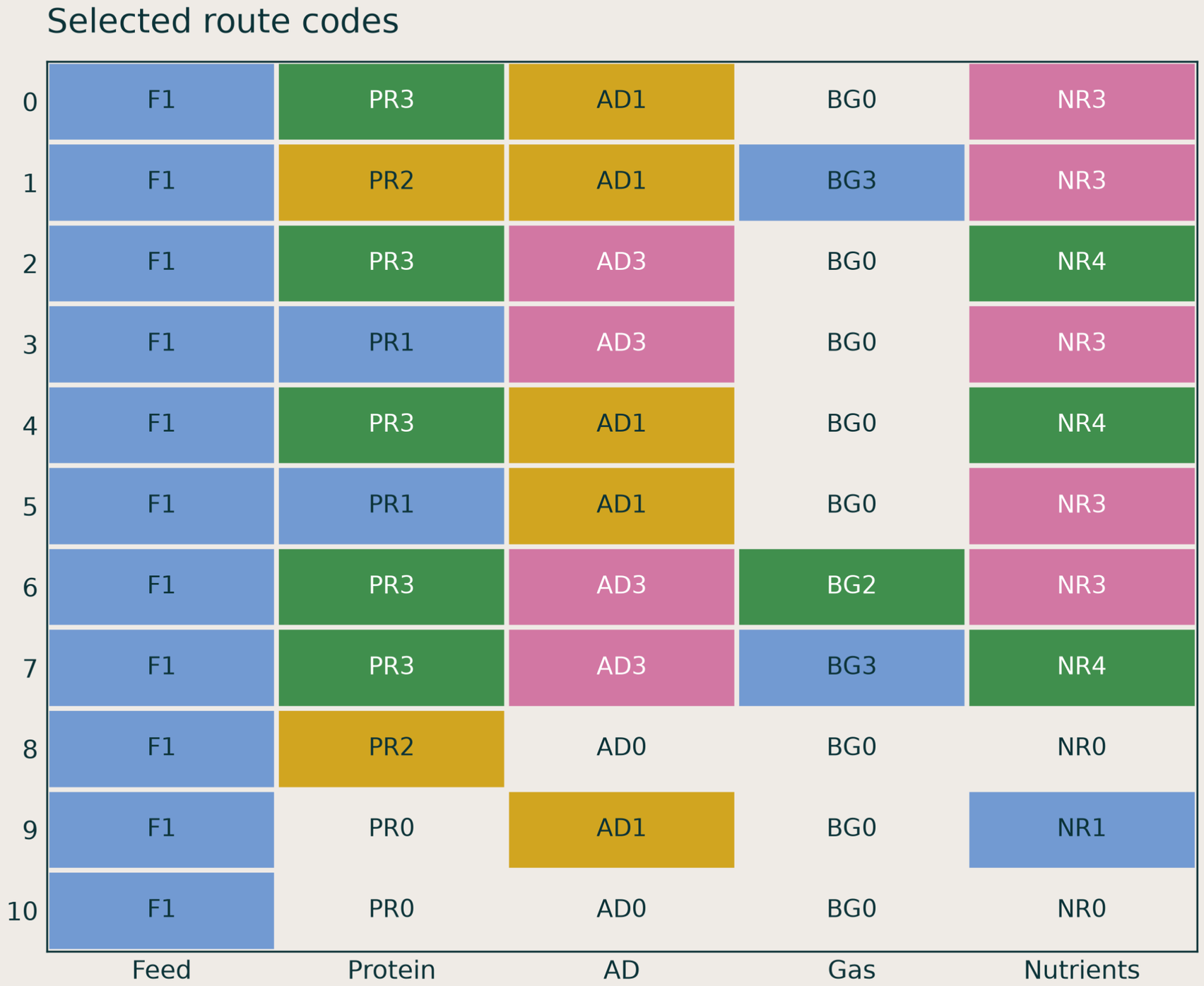
Scenario: all backend batch; 2000 total simulations, seed 42; objective-eligible 1049
Units: € thousand/year; t CO_2-eq/year
Source: Greenhub/results/actual_pareto_2000/decision_batches; biosteam_frontiers/rank_*.json

40. biosteam all revenue ranks



Revenue among ranked candidates

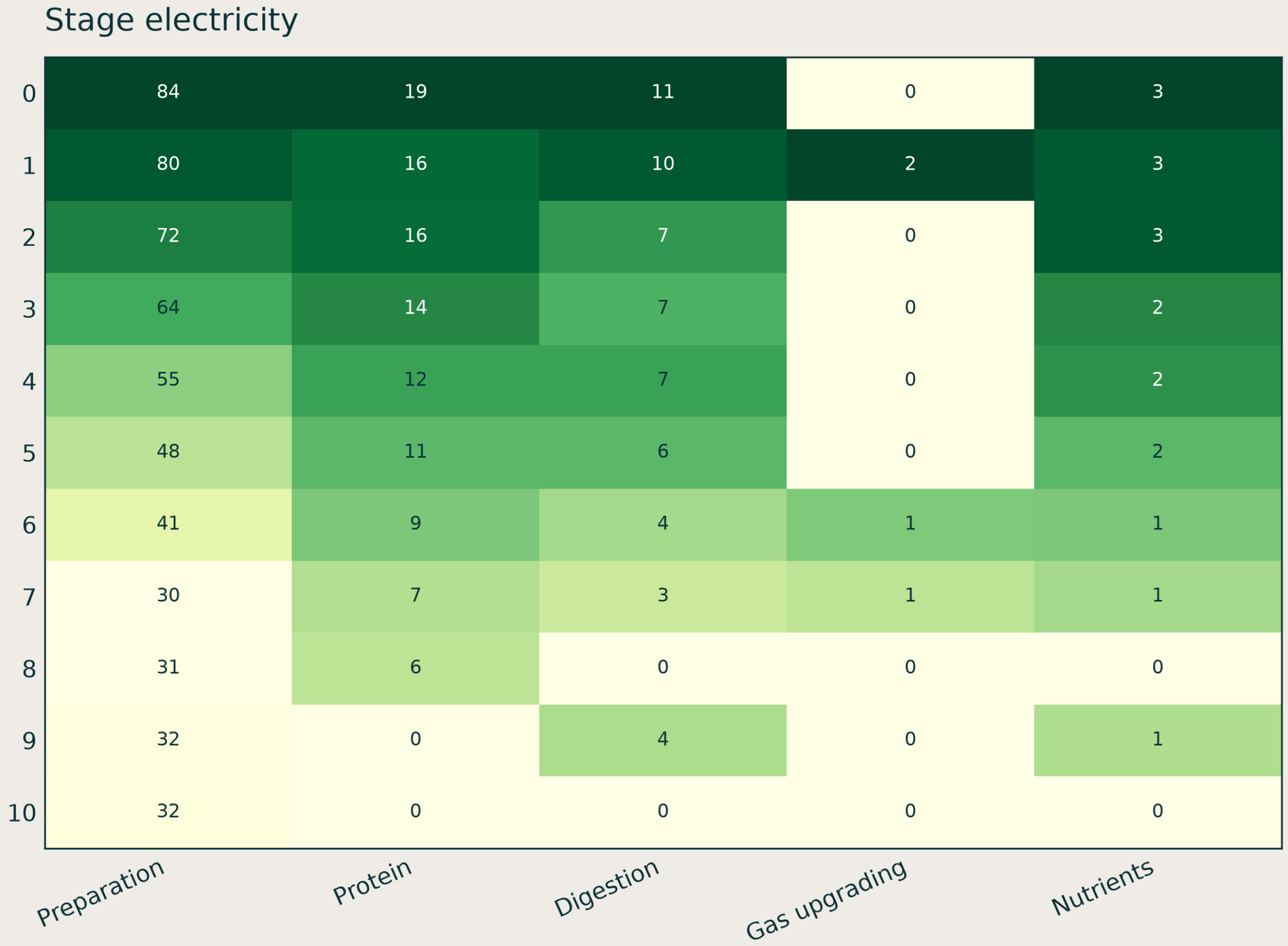
Scenario: all backend batch; 2000 total simulations, seed 42; objective-eligible 1049
Units: € thousand/year
Source: Greenhub/results/actual_pareto_2000/decision_batches; biosteam_frontiers/rank_*.json



Actual route configuration codes for ranked candidates

Scenario: all backend batch; 2000 total simulations, seed 42; objective-eligible 1049
Units: route code
Source: Greenhub/results/actual_pareto_2000/decision_batches; biosteam_frontiers/rank_*.json

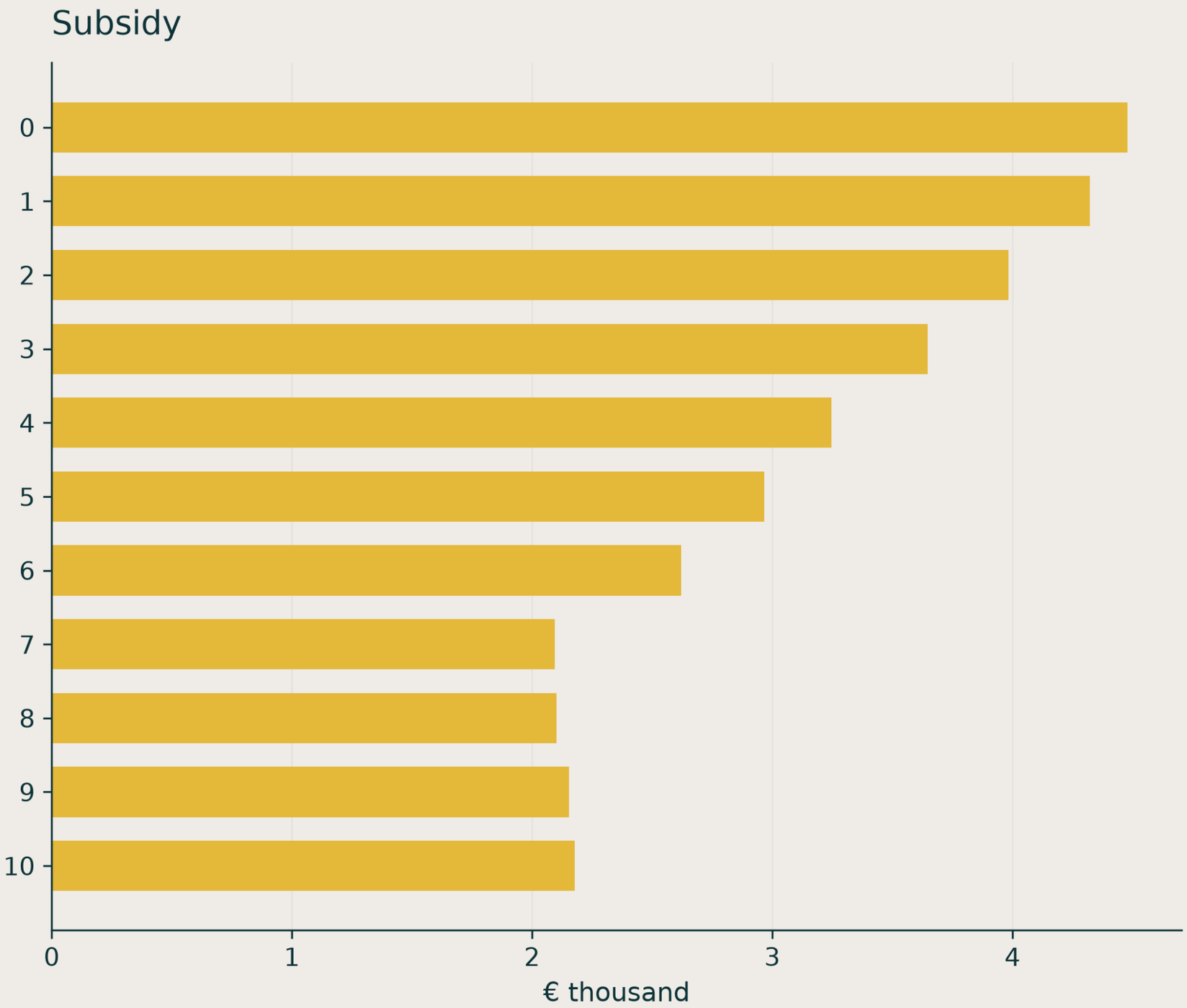
42. biosteam all stage energy heatmap



Simulated electricity by process stage across ranked candidates

Scenario: all backend batch; 2000 total simulations, seed 42; objective-eligible 1049
Units: MWh/year
Source: Greenhub/results/actual_pareto_2000/decision_batches; biosteam_frontiers/rank_*.json

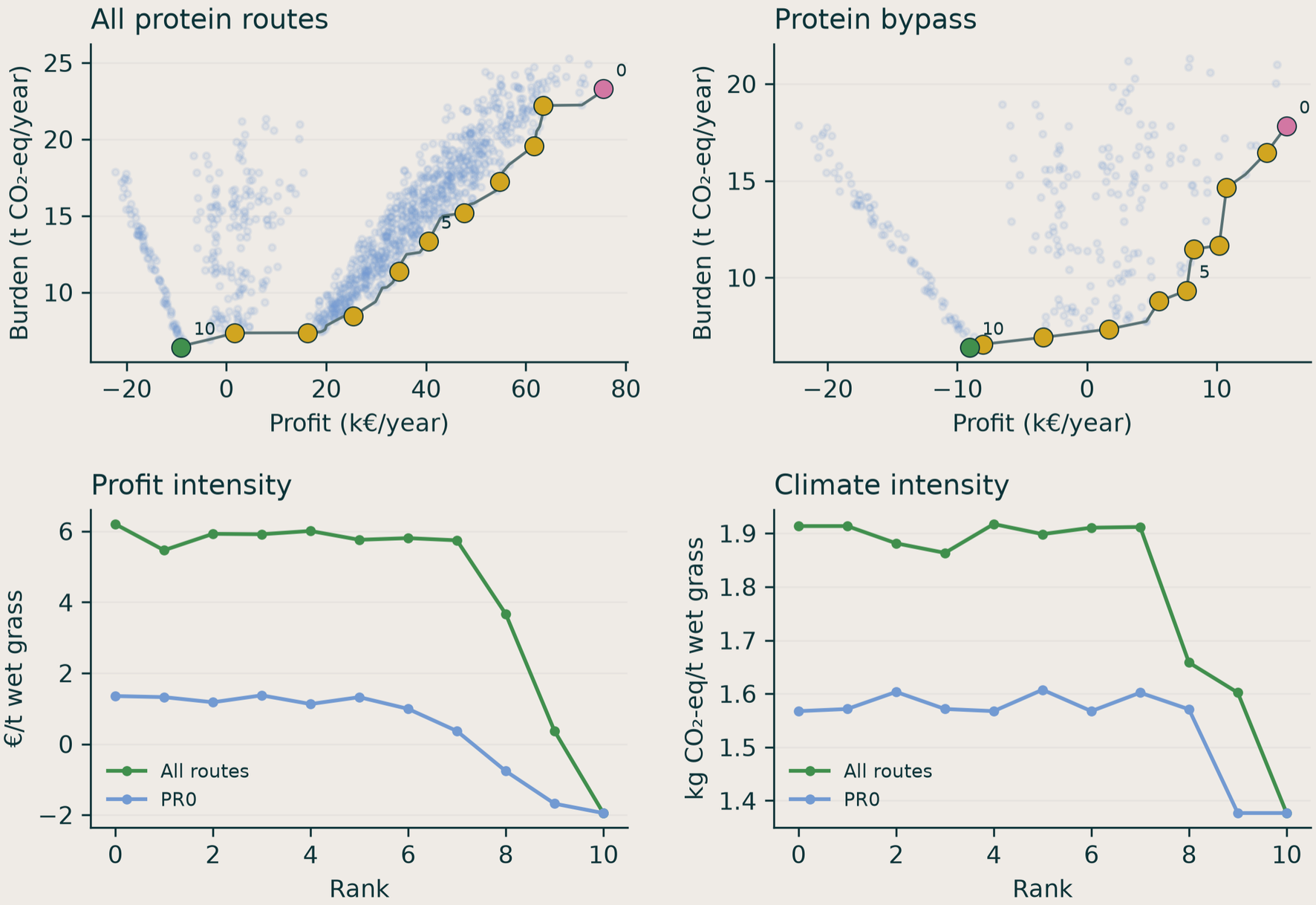
43. biosteam all subsidy ranks



Subsidy among ranked candidates

Scenario: all backend batch; 2000 total simulations, seed 42; objective-eligible 1049
Units: € thousand
Source: Greenhub/results/actual_pareto_2000/decision_batches; biosteam_frontiers/rank_*.json

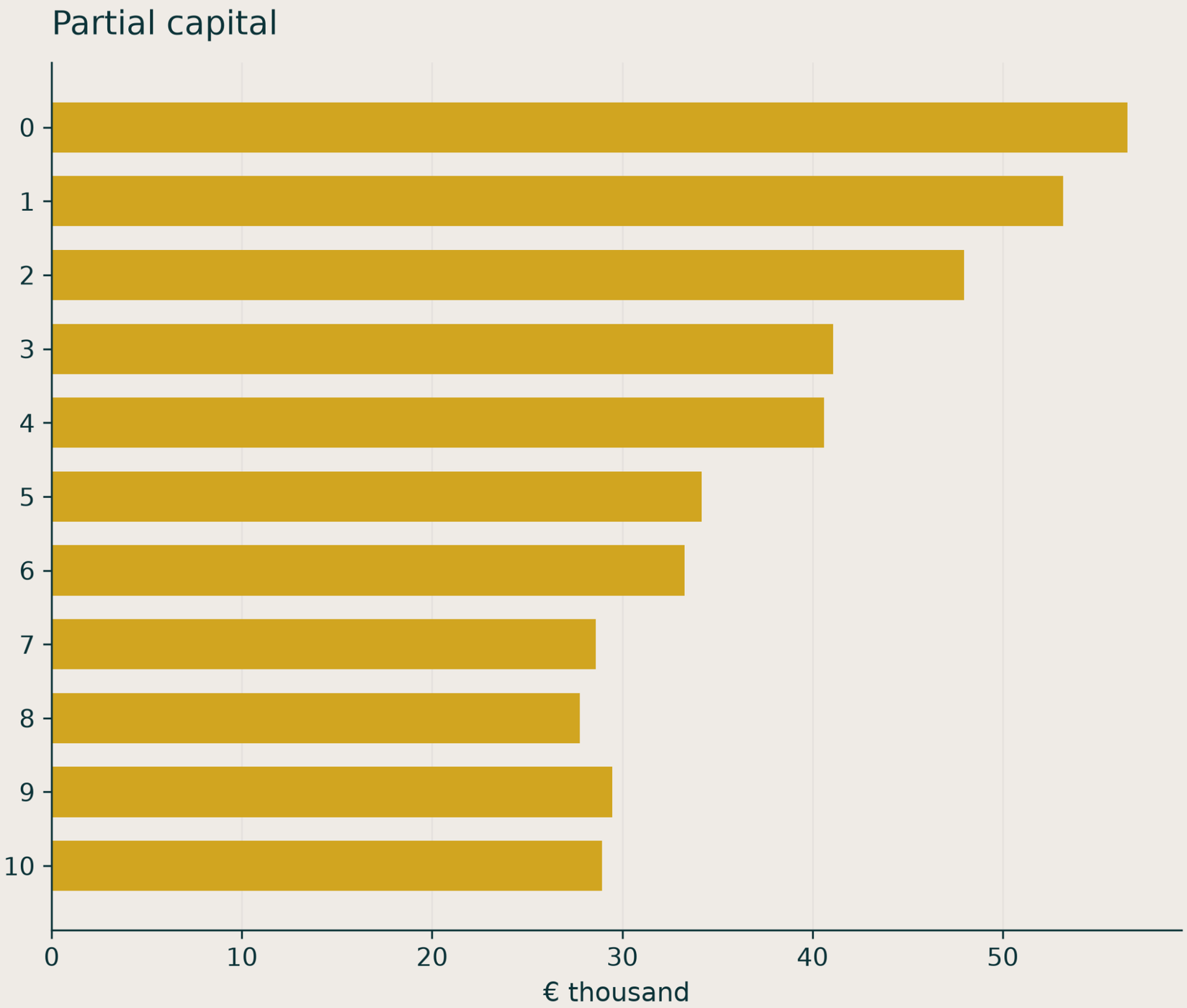
Non-PEF Pareto selection



Non-PEF screening fronts with selected ranks and feed-normalized objectives

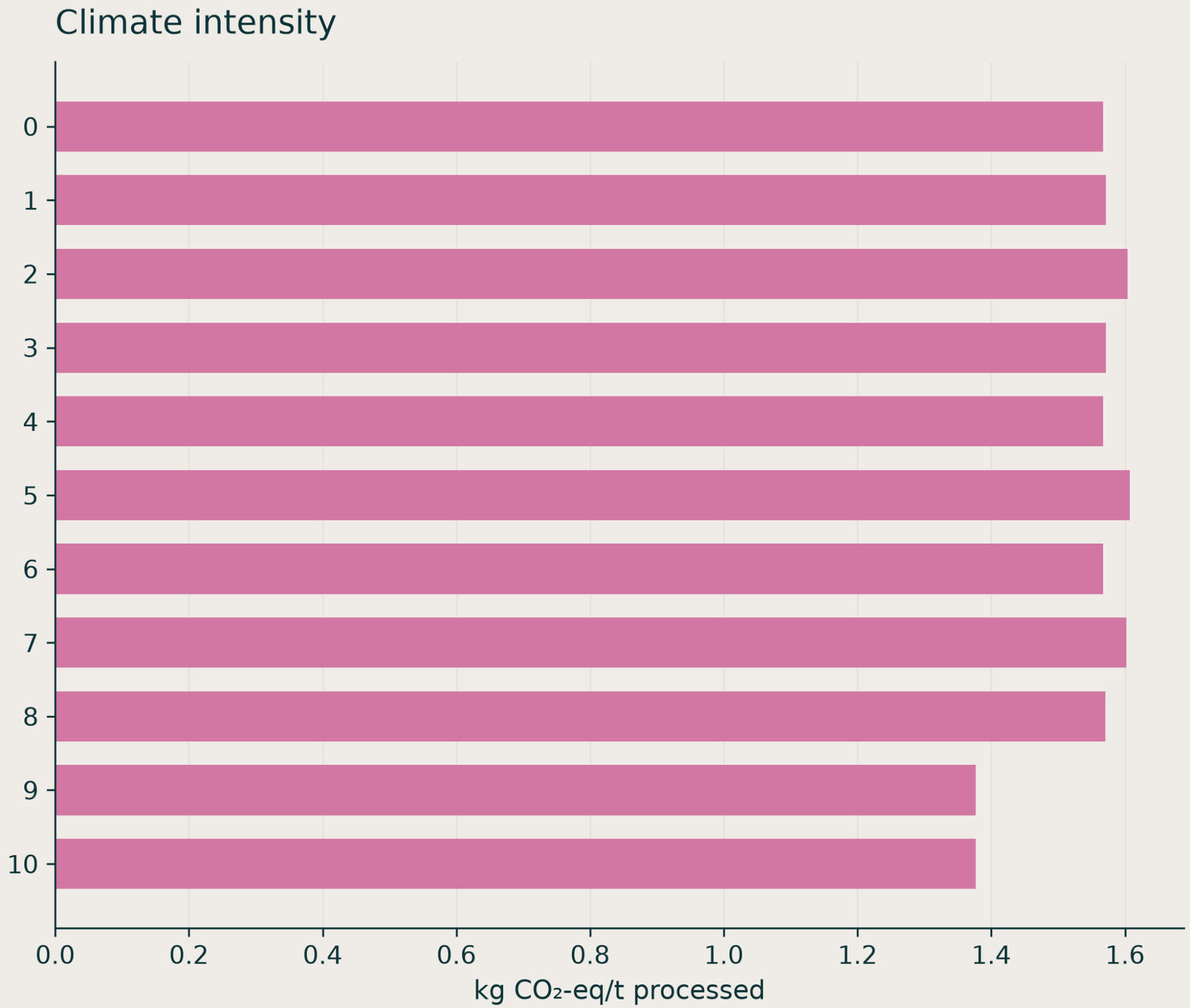
Scenario: F1 non-PEF saved BioSTEAM batch and selected exact reruns
Units: k€/year; t CO_2-eq/year; €/t; kg CO_2-eq/t
Source: C:\my\Spaak\Greenhub\Model\Greenhub\results\actual_pareto_2000\decision_batches\eed439a280ad488aa235a19b0efc32af.json

45. biosteam protein bypass capital ranks



Partial capital among ranked candidates

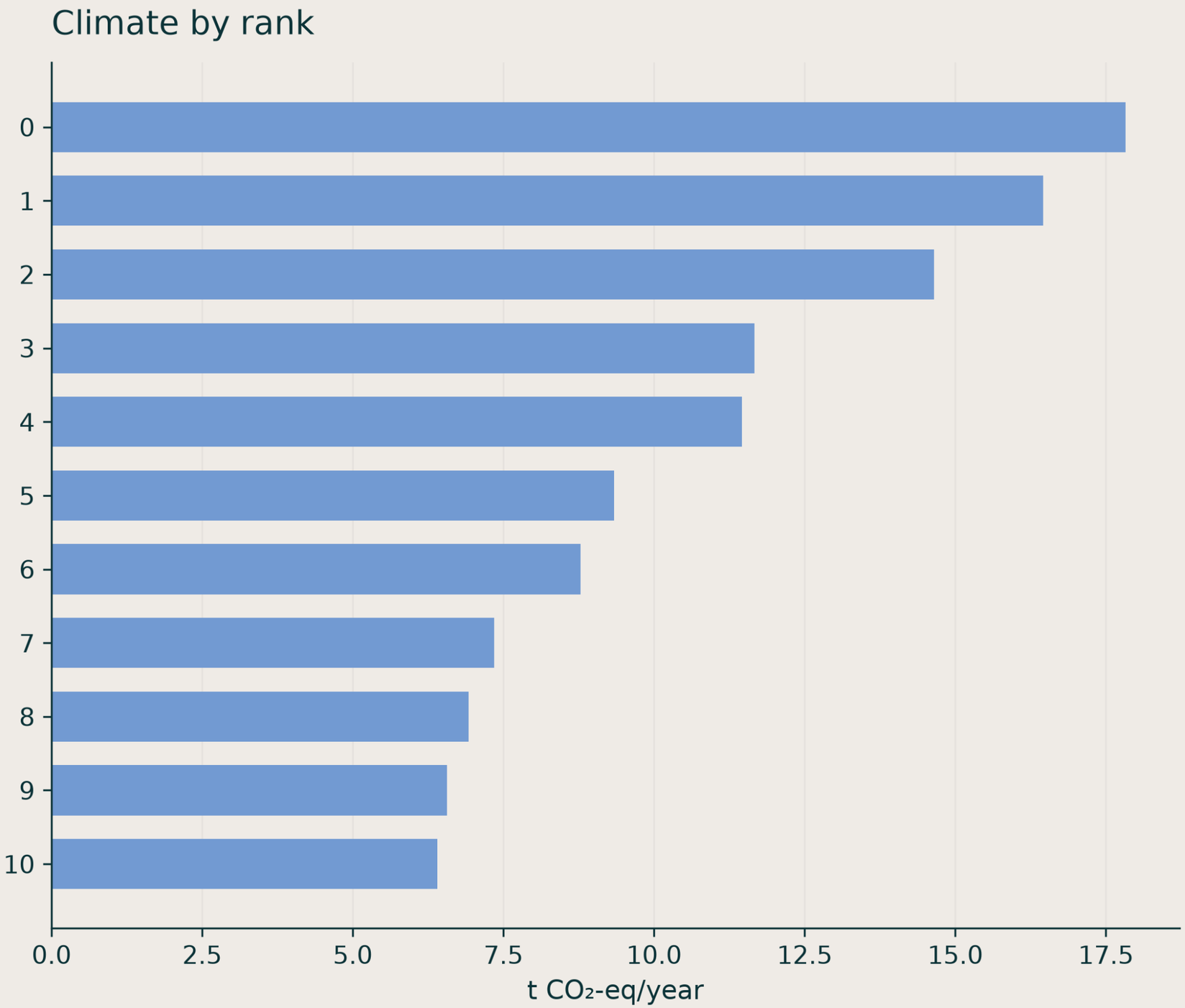
Scenario: protein_bypass backend batch; 2000 total simulations, seed 42; objective-eligible 247
Units: € thousand
Source: Greenhub/results/actual_pareto_2000/decision_batches; biosteam_frontiers/rank_*.json



Partial attributional climate burden per wet tonne

Scenario: protein_bypass backend batch; 2000 total simulations, seed 42; objective-eligible 247
Units: kg CO_2-eq/t processed
Source: Greenhub/results/actual_pareto_2000/decision_batches; biosteam_frontiers/rank_*.json

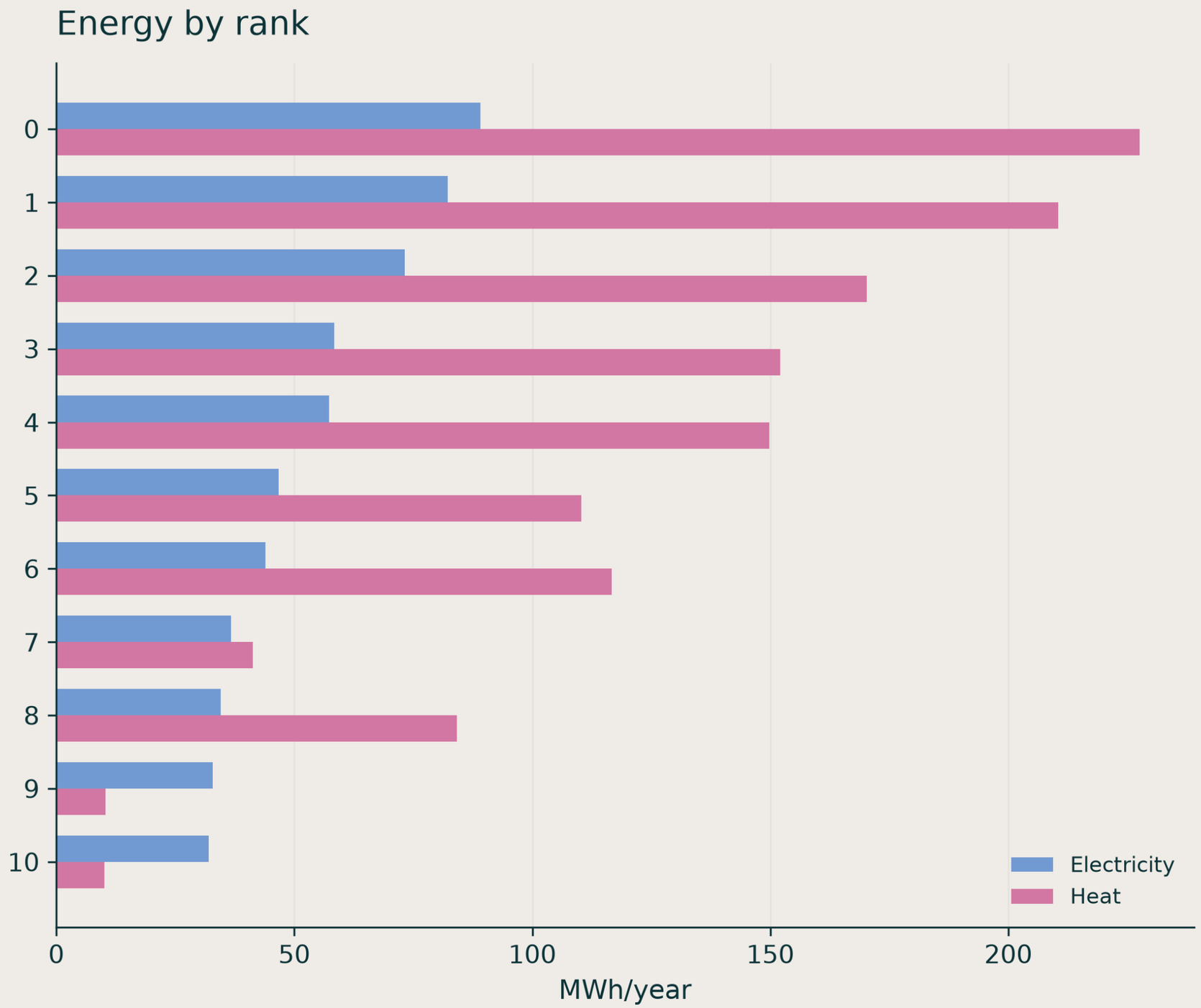
47. biosteam protein bypass climate ranks



Partial attributional electricity burden by ranked candidate

Scenario: protein_bypass backend batch; 2000 total simulations, seed 42; objective-eligible 247
Units: t CO_2-eq/year
Source: Greenhub/results/actual_pareto_2000/decision_batches; biosteam_frontiers/rank_*.json

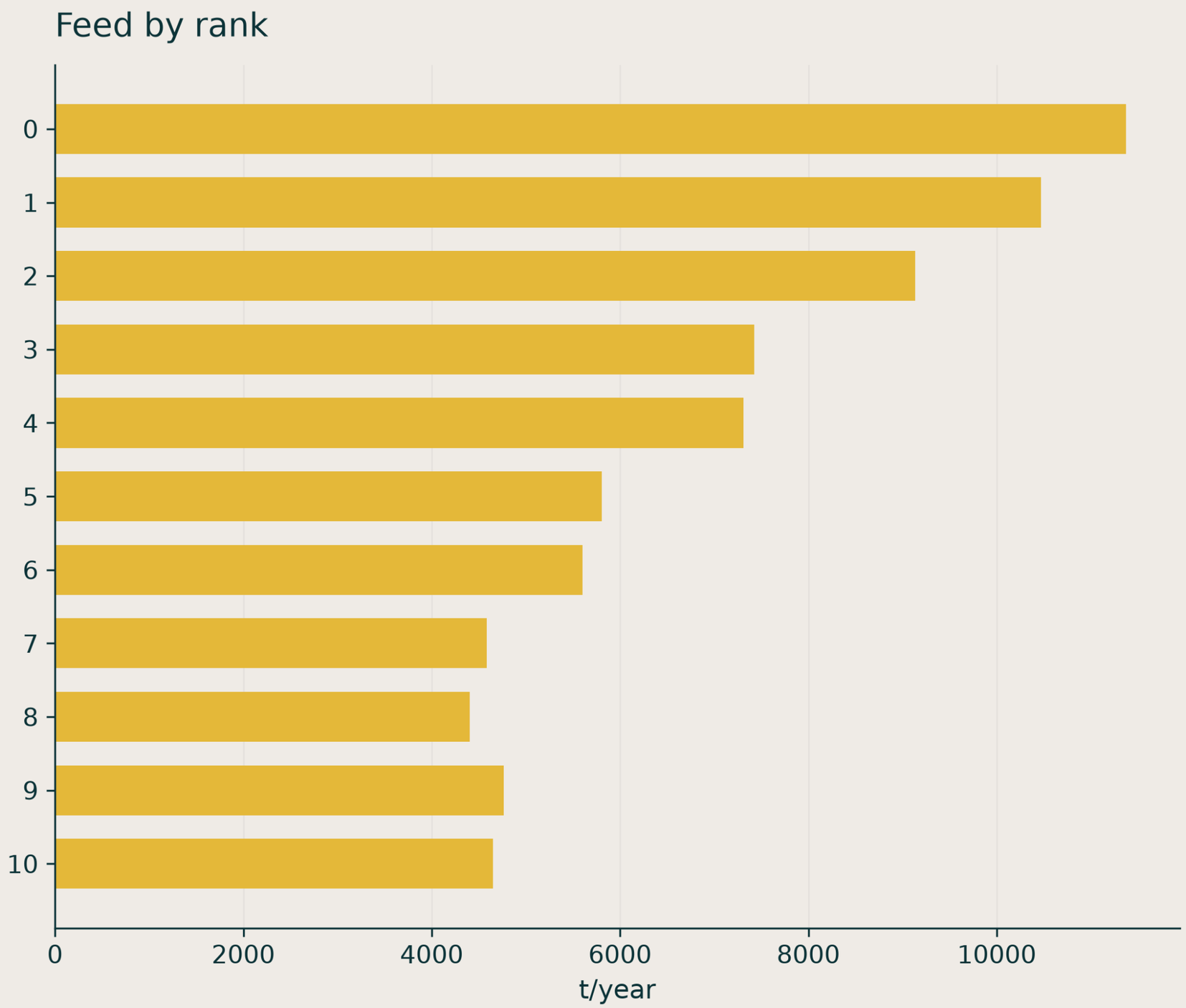
48. biosteam protein bypass energy ranks



Annual electricity and heating demand across ranked candidates

Scenario: protein_bypass backend batch; 2000 total simulations, seed 42; objective-eligible 247
Units: MWh/year
Source: Greenhub/results/actual_pareto_2000/decision_batches; biosteam_frontiers/rank_*.json

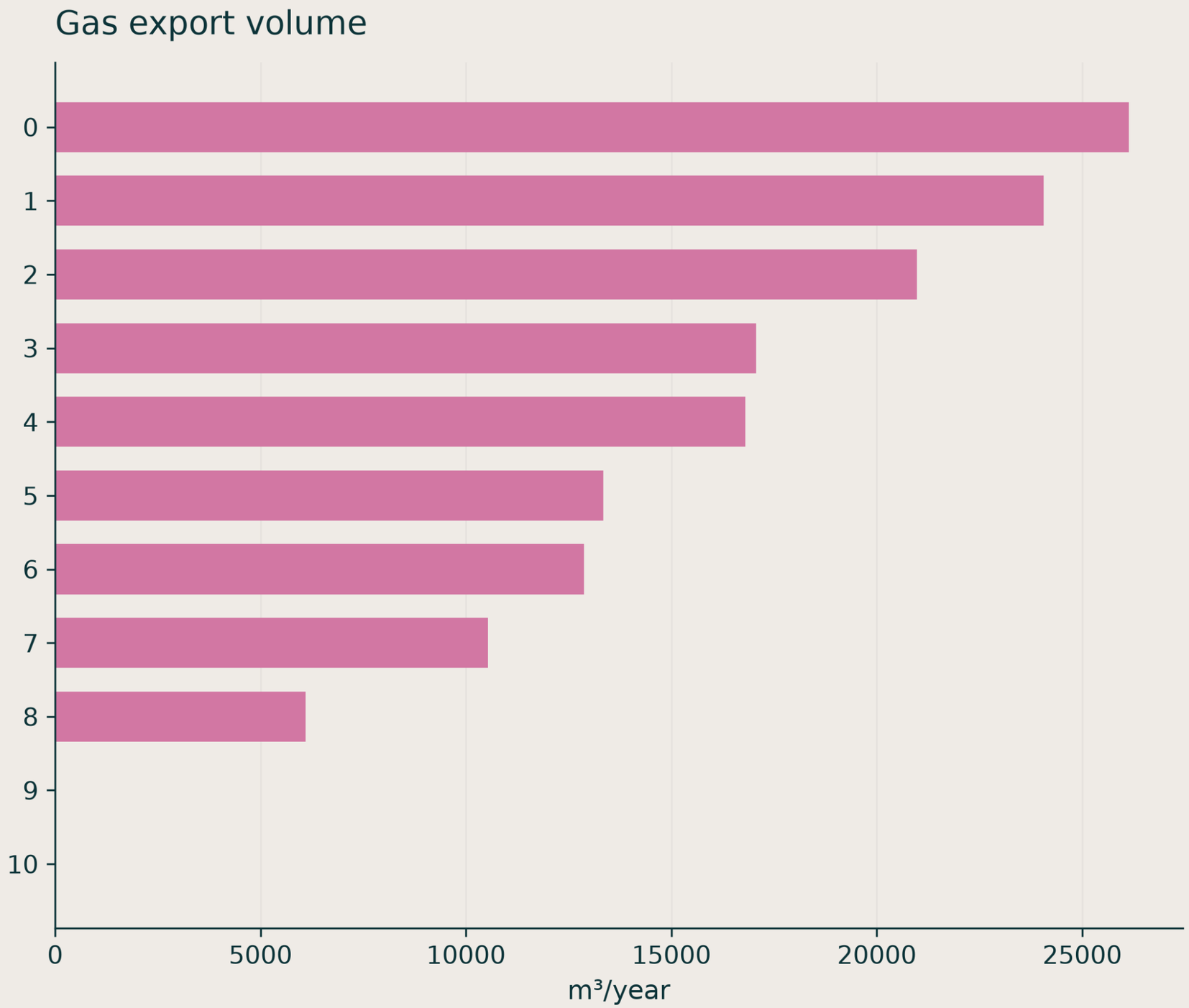
49. biosteam protein bypass feed ranks



Simulated annual wet feed across ranks

Scenario: protein_bypass backend batch; 2000 total simulations, seed 42; objective-eligible 247
Units: t/year
Source: Greenhub/results/actual_pareto_2000/decision_batches; biosteam_frontiers/rank_*.json

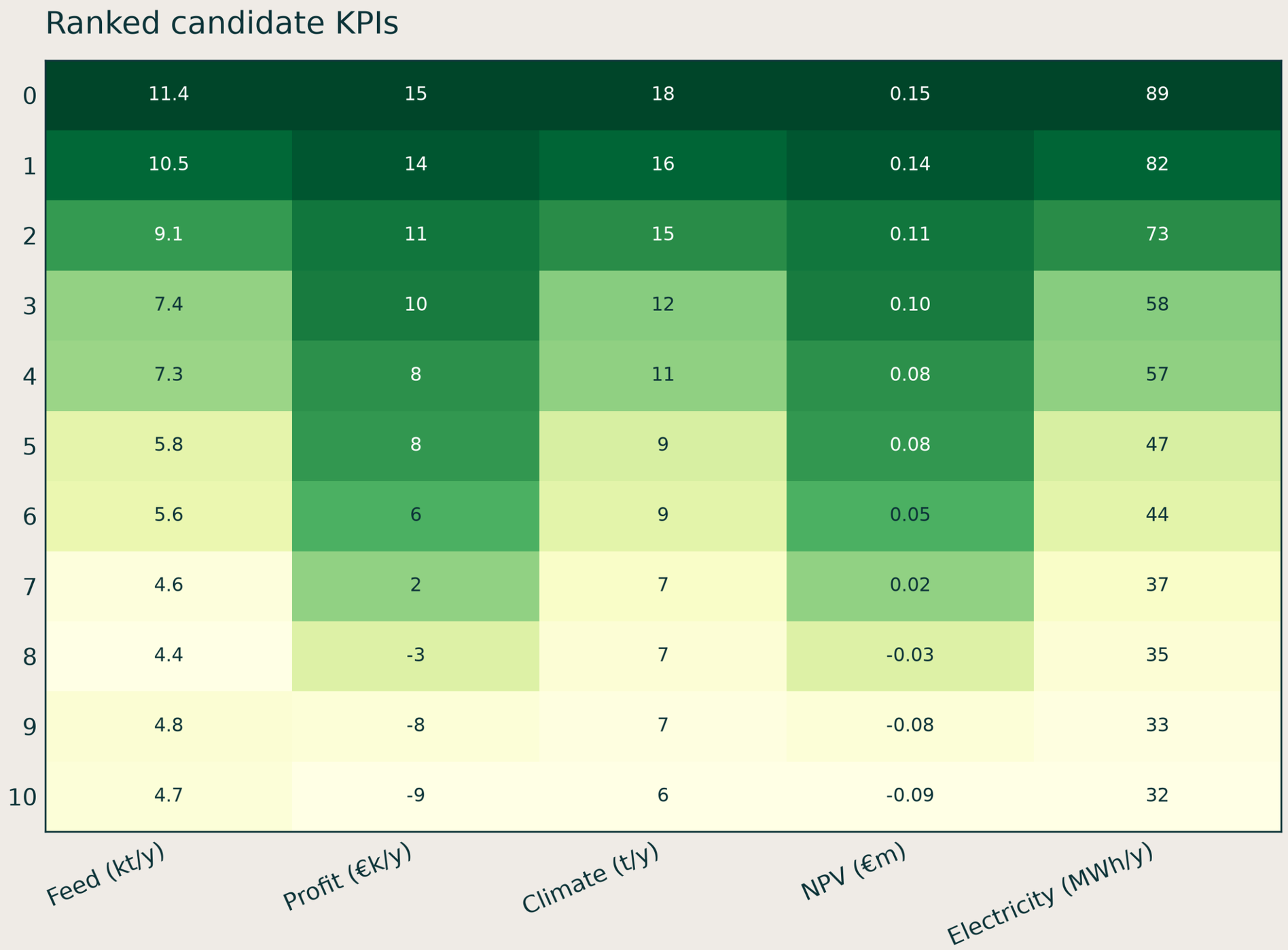
50. biosteam protein bypass gas ranks



Revenue-account dry gas export at 293.15 K; raw for BG0 and upgraded for selected BG routes

Scenario: protein_bypass backend batch; 2000 total simulations, seed 42; objective-eligible 247
Units: m³/year
Source: Greenhub/results/actual_pareto_2000/decision_batches; biosteam_frontiers/rank_*.json

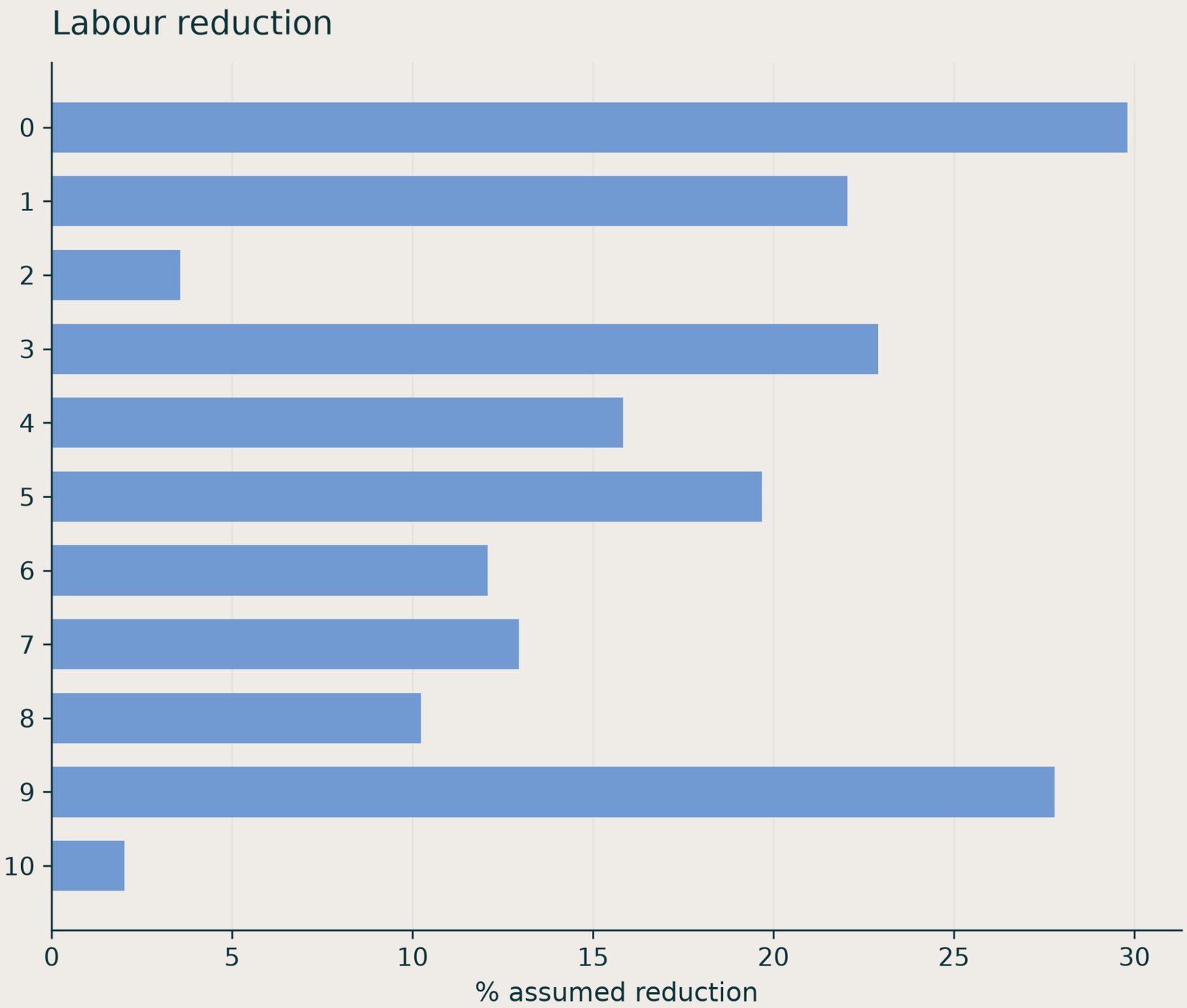
51. biosteam protein bypass kpi heatmap



Feed, screening finance, partial climate and electricity for ranked cases

Scenario: protein_bypass backend batch; 2000 total simulations, seed 42; objective-eligible 247
Units: Feed (kt/y); Profit (€k/y); Climate (t/y); NPV (€m); Electricity (MWh/y)
Source: Greenhub/results/actual_pareto_2000/decision_batches; biosteam_frontiers/rank_*.json

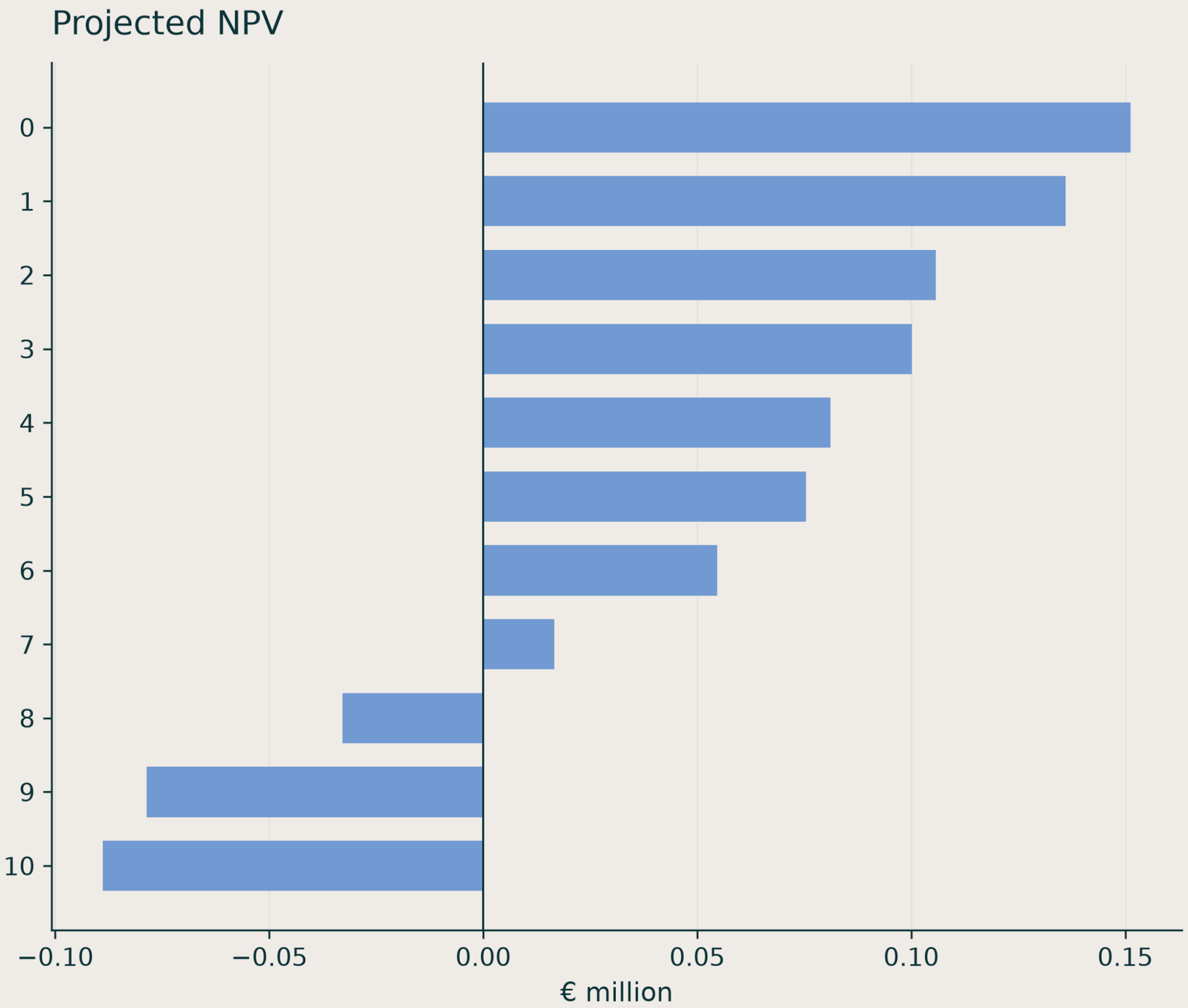
52. biosteam protein bypass labour ranks



Sampled candidate labour-reduction overlay

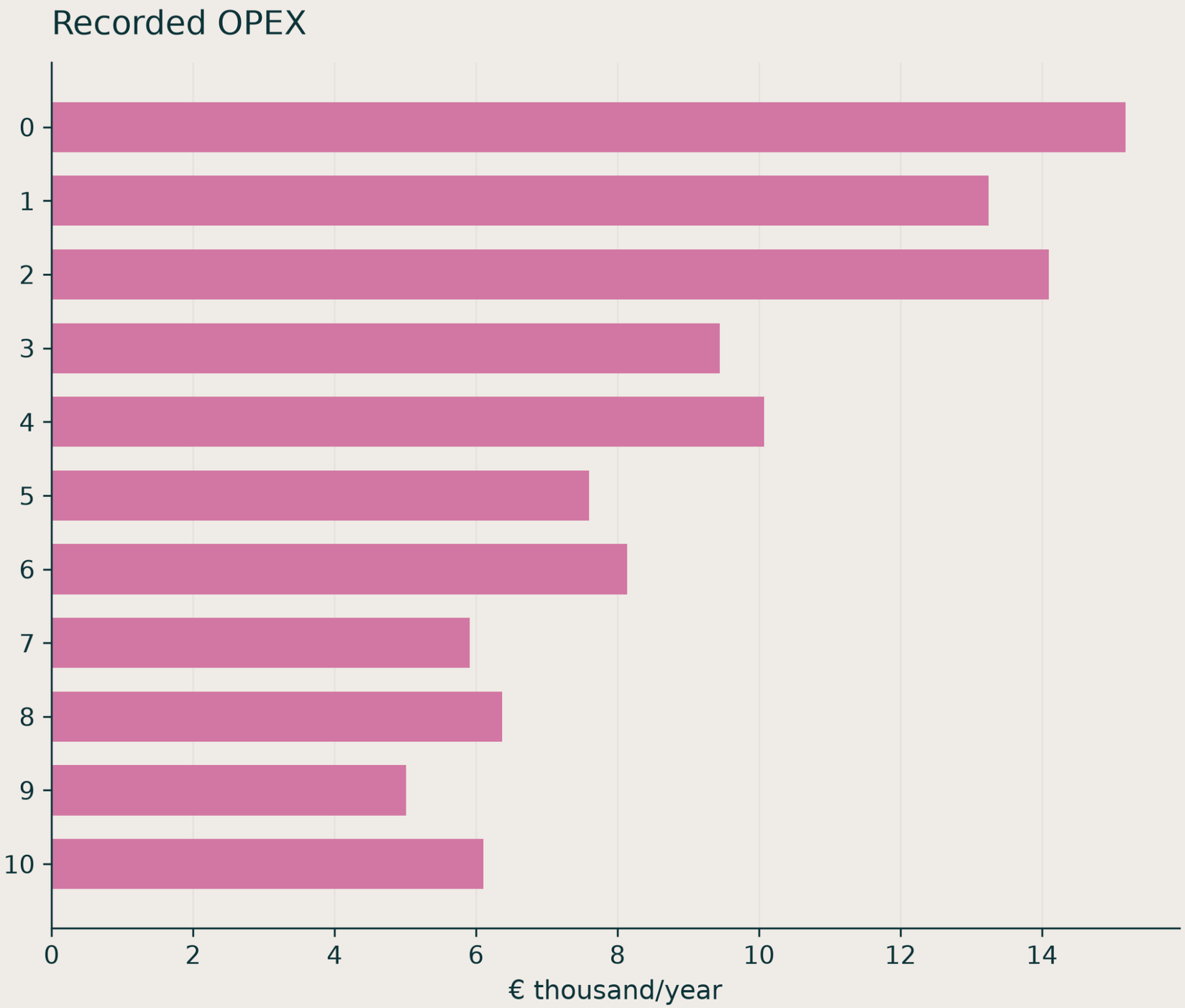
Scenario: protein_bypass backend batch; 2000 total simulations, seed 42; objective-eligible 247
Units: % assumed reduction
Source: Greenhub/results/actual_pareto_2000/decision_batches; biosteam_frontiers/rank_*.json

53. biosteam protein bypass npv ranks



20-year constant-account NPV across ranked candidates

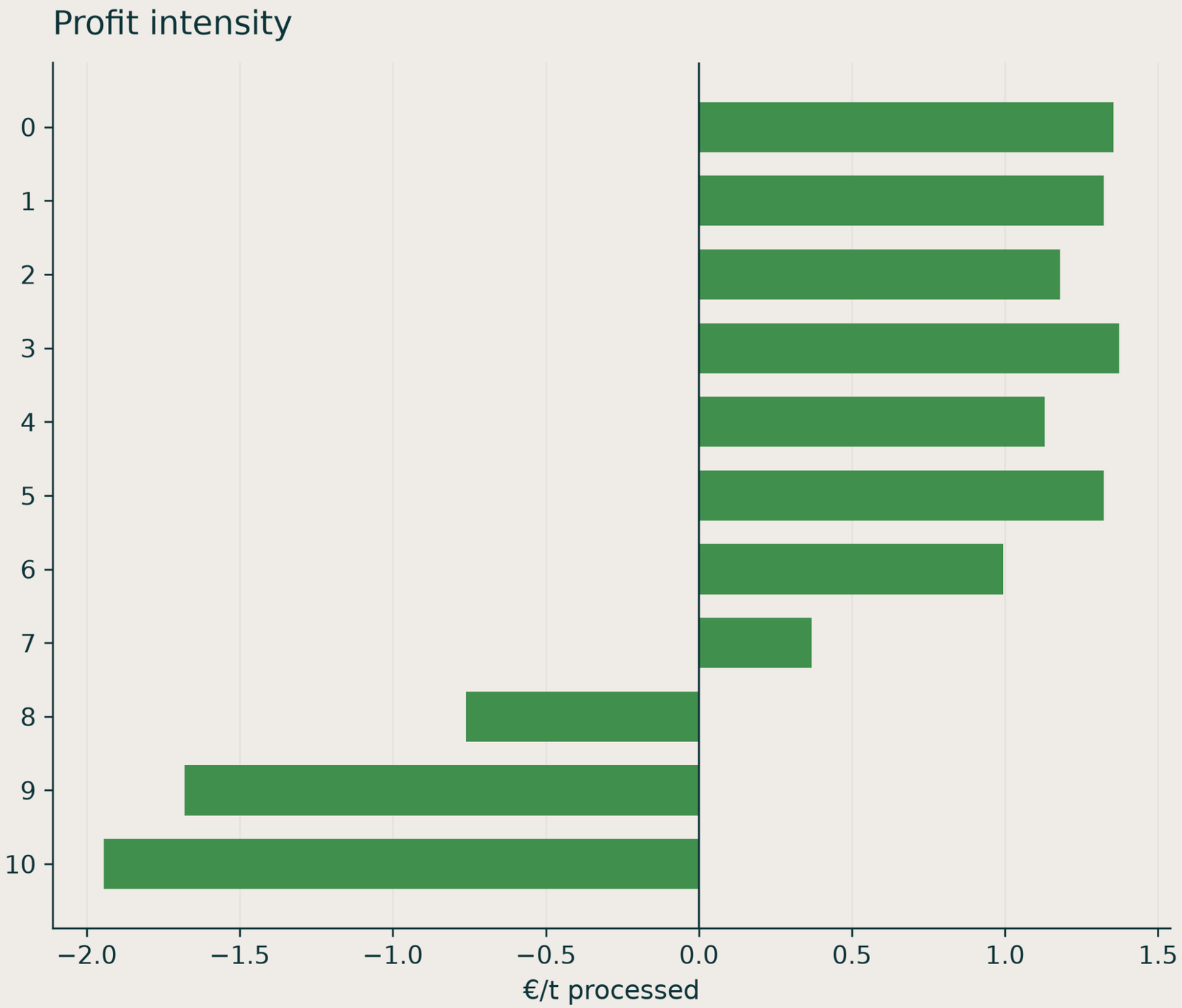
Scenario: protein_bypass backend batch; 2000 total simulations, seed 42; objective-eligible 247
Units: € million
Source: Greenhub/results/actual_pareto_2000/decision_batches; biosteam_frontiers/rank_*.json



Recorded OPEX among ranked candidates

Scenario: protein_bypass backend batch; 2000 total simulations, seed 42; objective-eligible 247
Units: € thousand/year
Source: Greenhub/results/actual_pareto_2000/decision_batches; biosteam_frontiers/rank_*.json

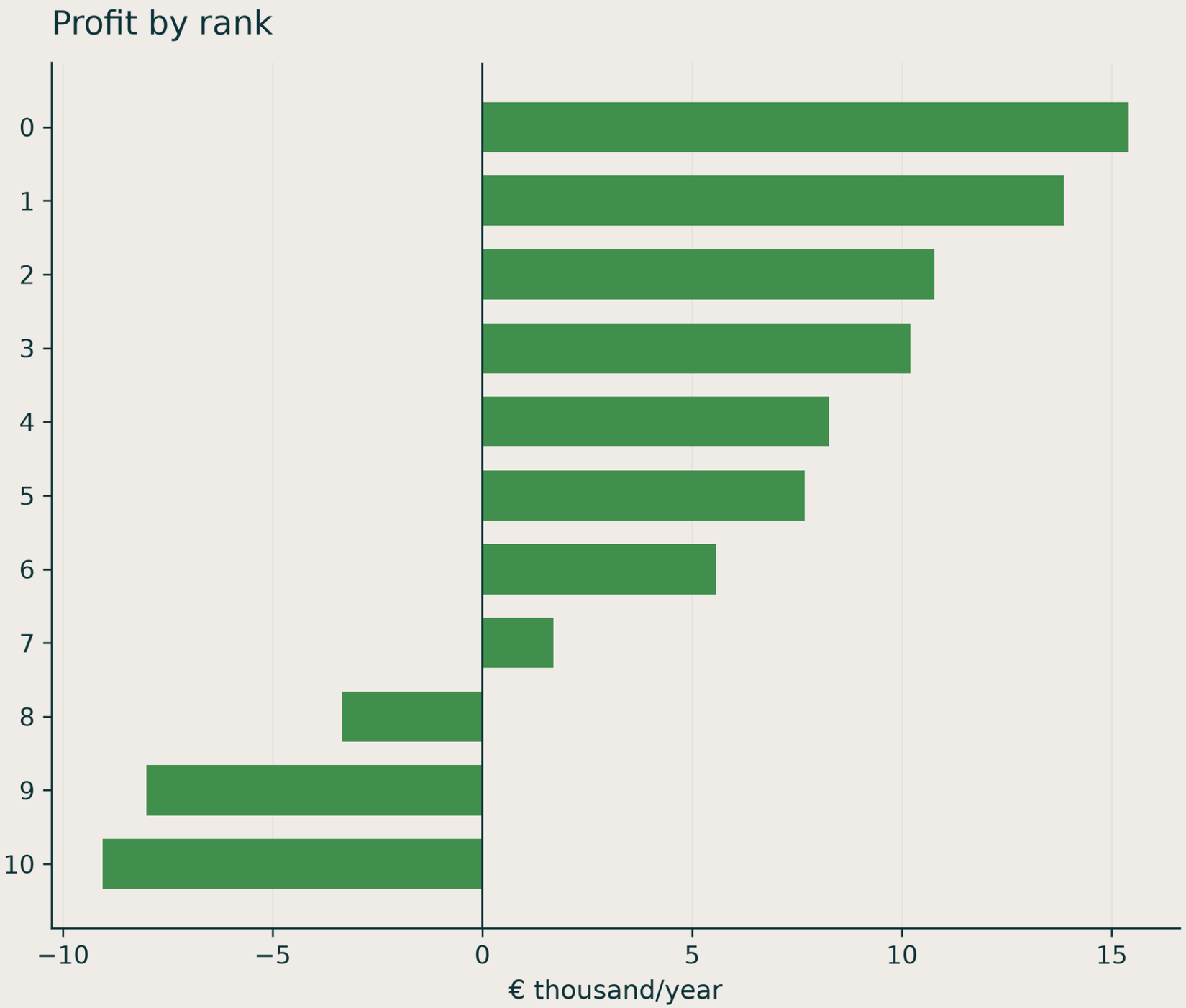
55. biosteam protein bypass profit intensity



Annualized screening profit normalized to simulated wet feed

Scenario: protein_bypass backend batch; 2000 total simulations, seed 42; objective-eligible 247
Units: €/t processed
Source: Greenhub/results/actual_pareto_2000/decision_batches; biosteam_frontiers/rank_*.json

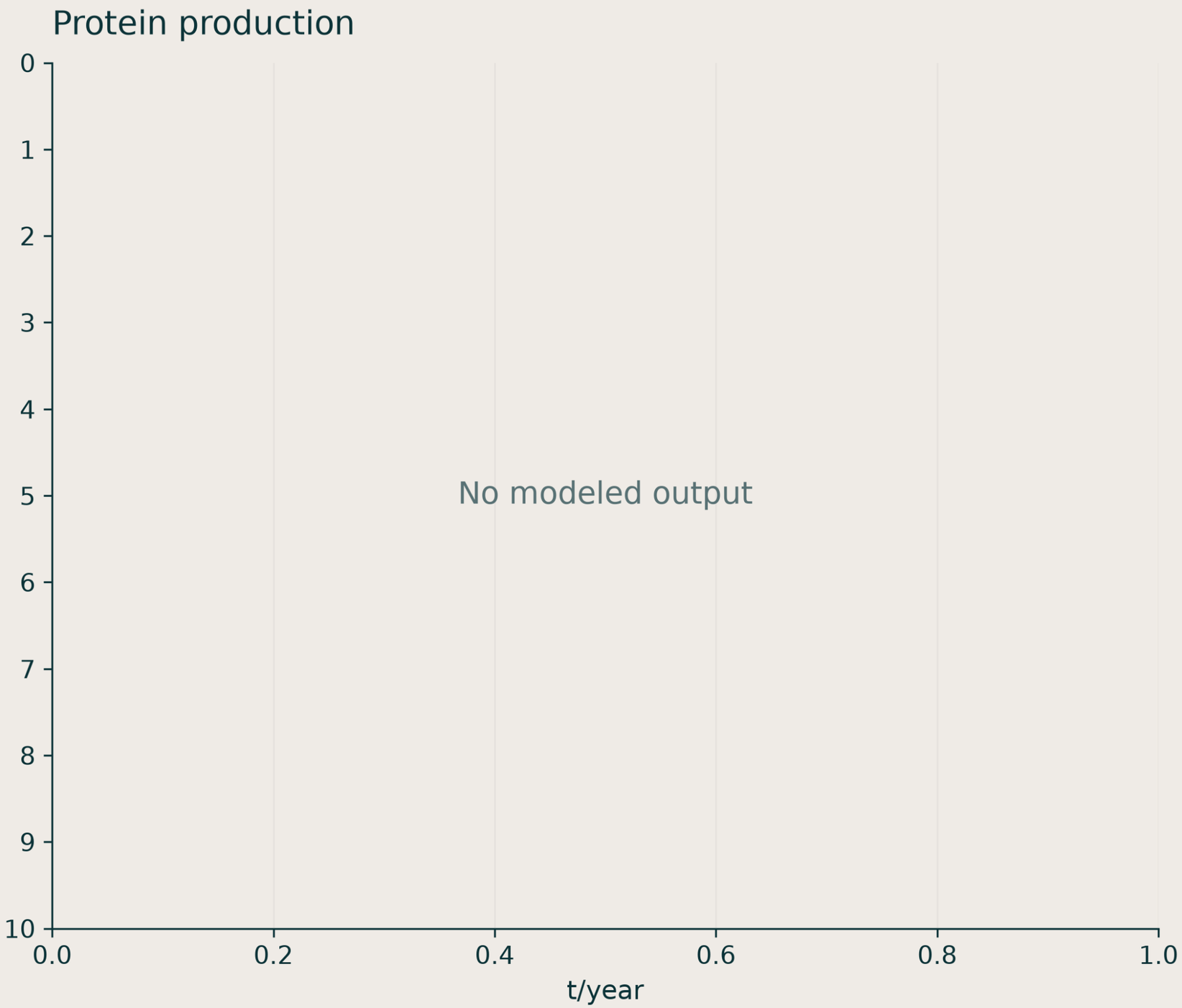
56. biosteam protein bypass profit ranks



Backend candidate-adjusted screening profit

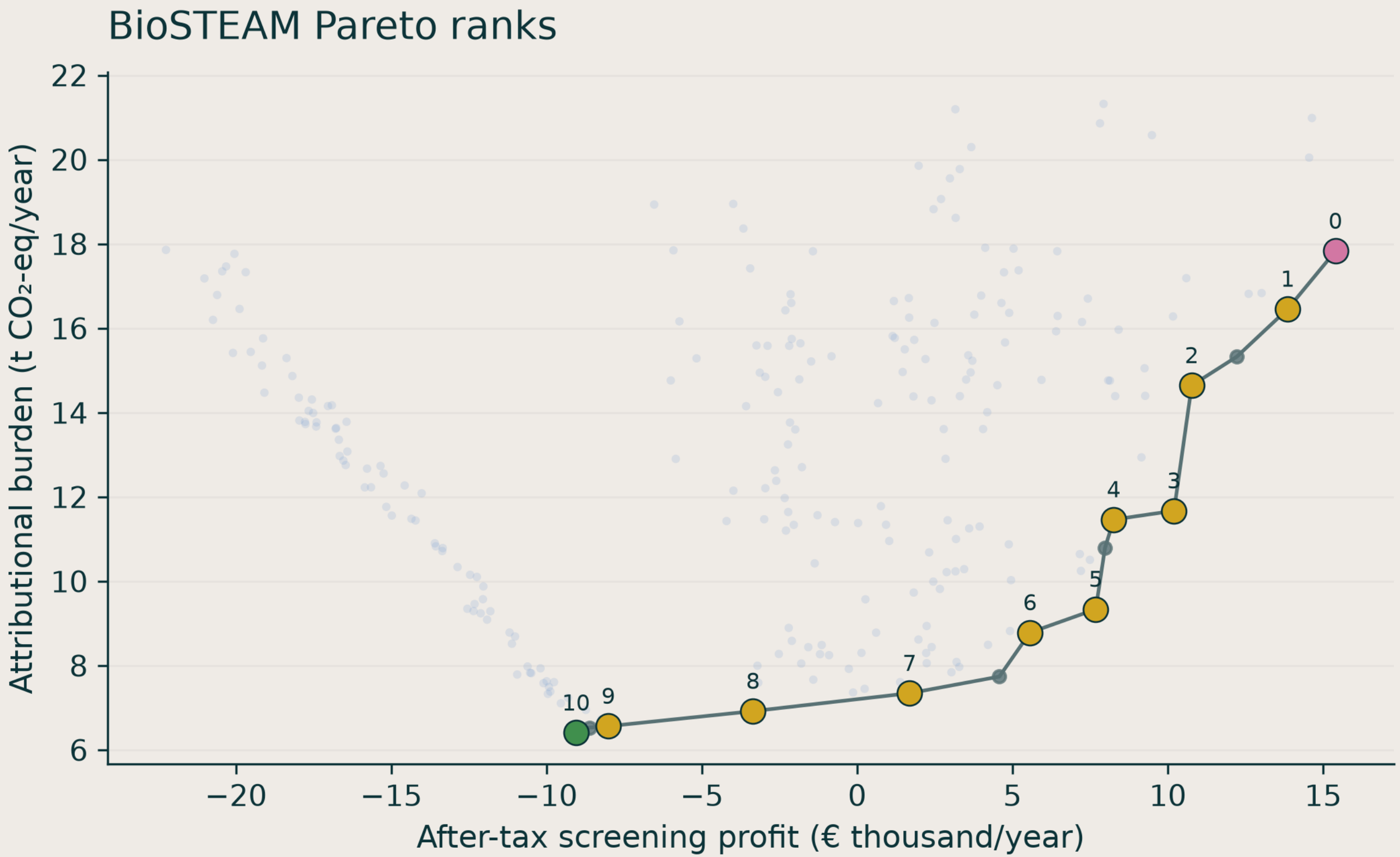
Scenario: protein_bypass backend batch; 2000 total simulations, seed 42; objective-eligible 247
Units: € thousand/year
Source: Greenhub/results/actual_pareto_2000/decision_batches; biosteam_frontiers/rank_*.json

57. biosteam protein bypass protein ranks



Modeled dry protein product by ranked case

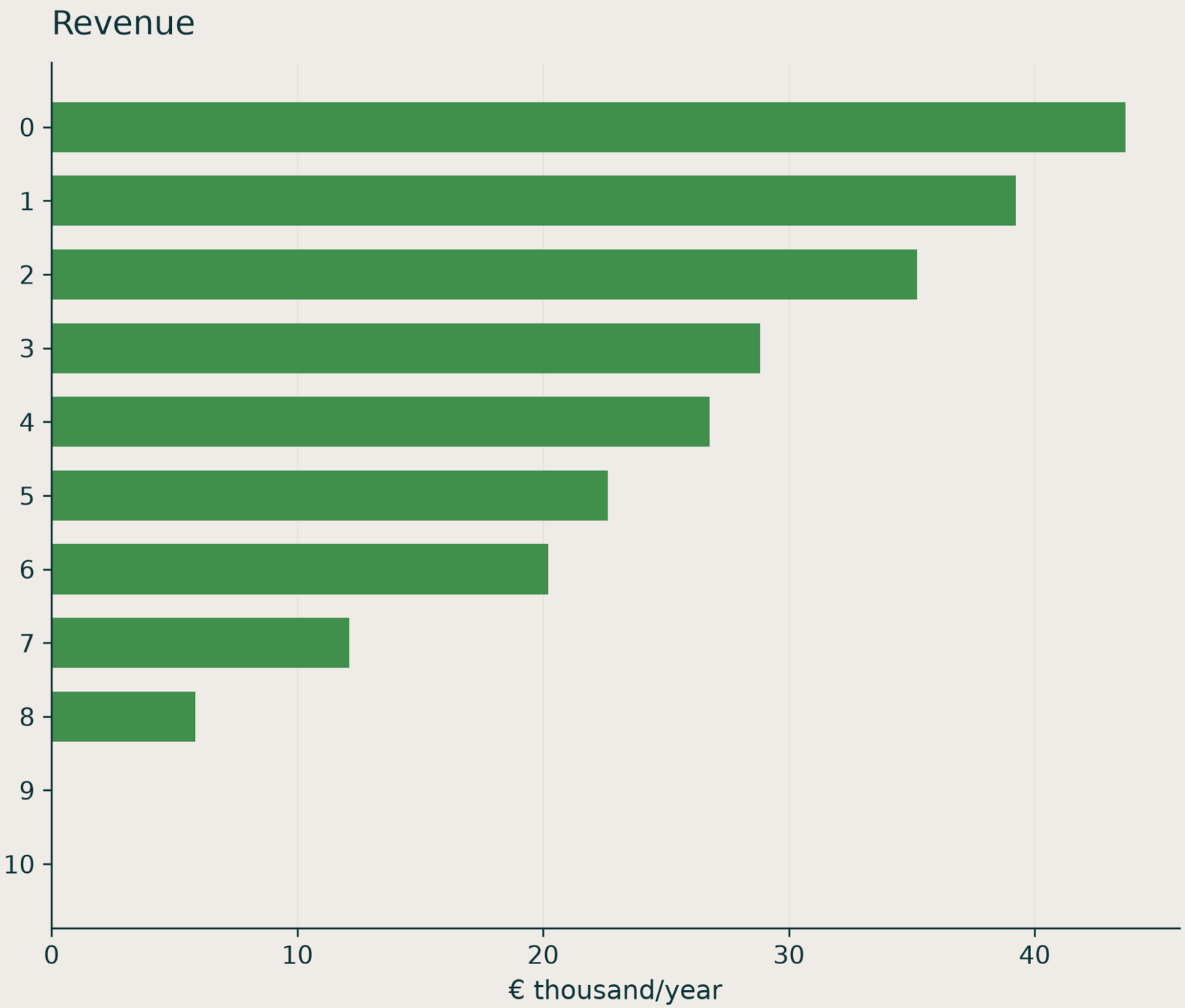
Scenario: protein_bypass backend batch; 2000 total simulations, seed 42; objective-eligible 247
Units: t/year
Source: Greenhub/results/actual_pareto_2000/decision_batches; biosteam_frontiers/rank_*.json



Backend-evaluated screening frontier with ranks 0–10

Scenario: protein_bypass backend batch; 2000 total simulations, seed 42; objective-eligible 247
Units: € thousand/year; t CO_2-eq/year
Source: Greenhub/results/actual_pareto_2000/decision_batches; biosteam_frontiers/rank_*.json

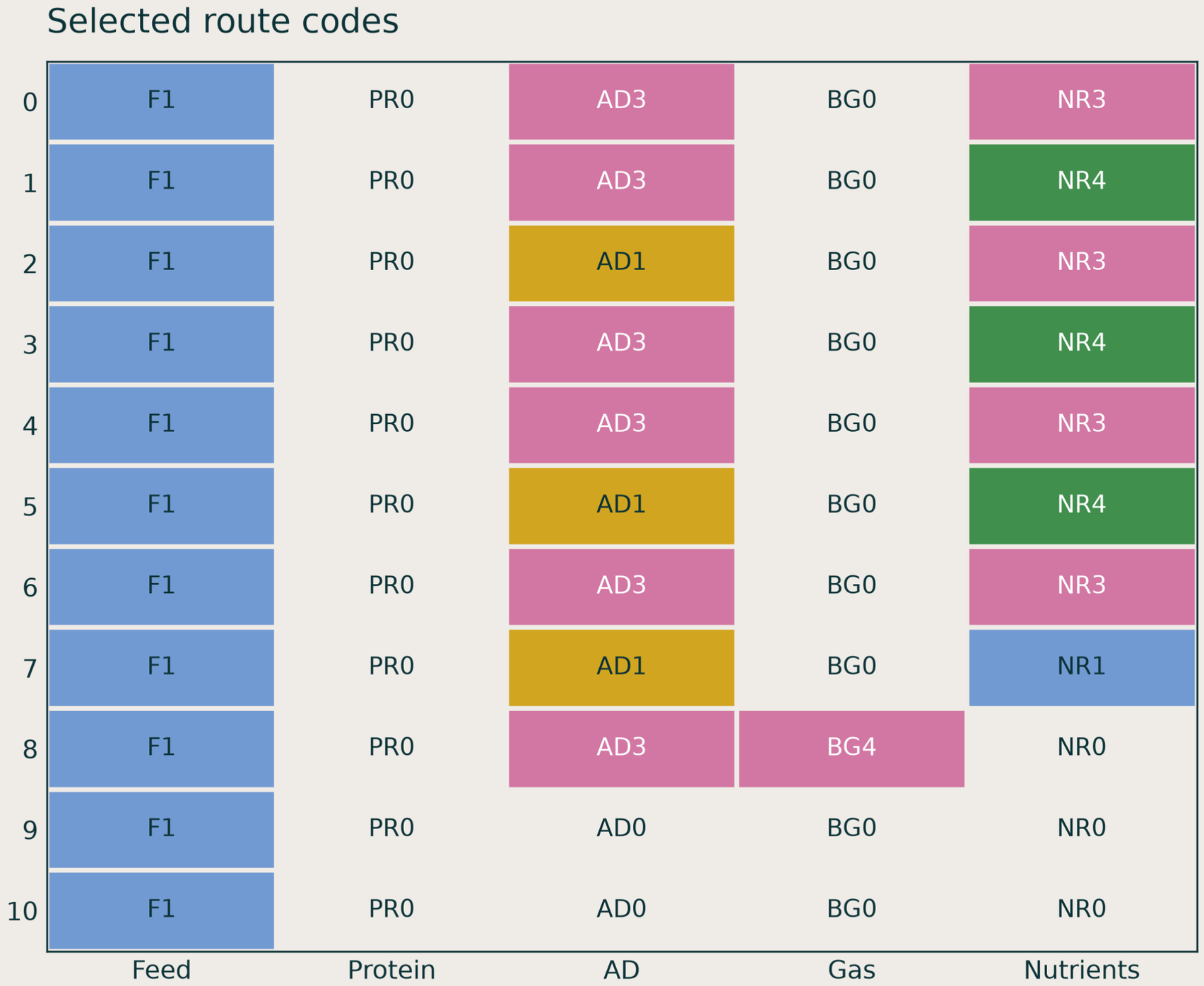
59. biosteam protein bypass revenue ranks



Revenue among ranked candidates

Scenario: protein_bypass backend batch; 2000 total simulations, seed 42; objective-eligible 247
Units: € thousand/year
Source: Greenhub/results/actual_pareto_2000/decision_batches; biosteam_frontiers/rank_*.json

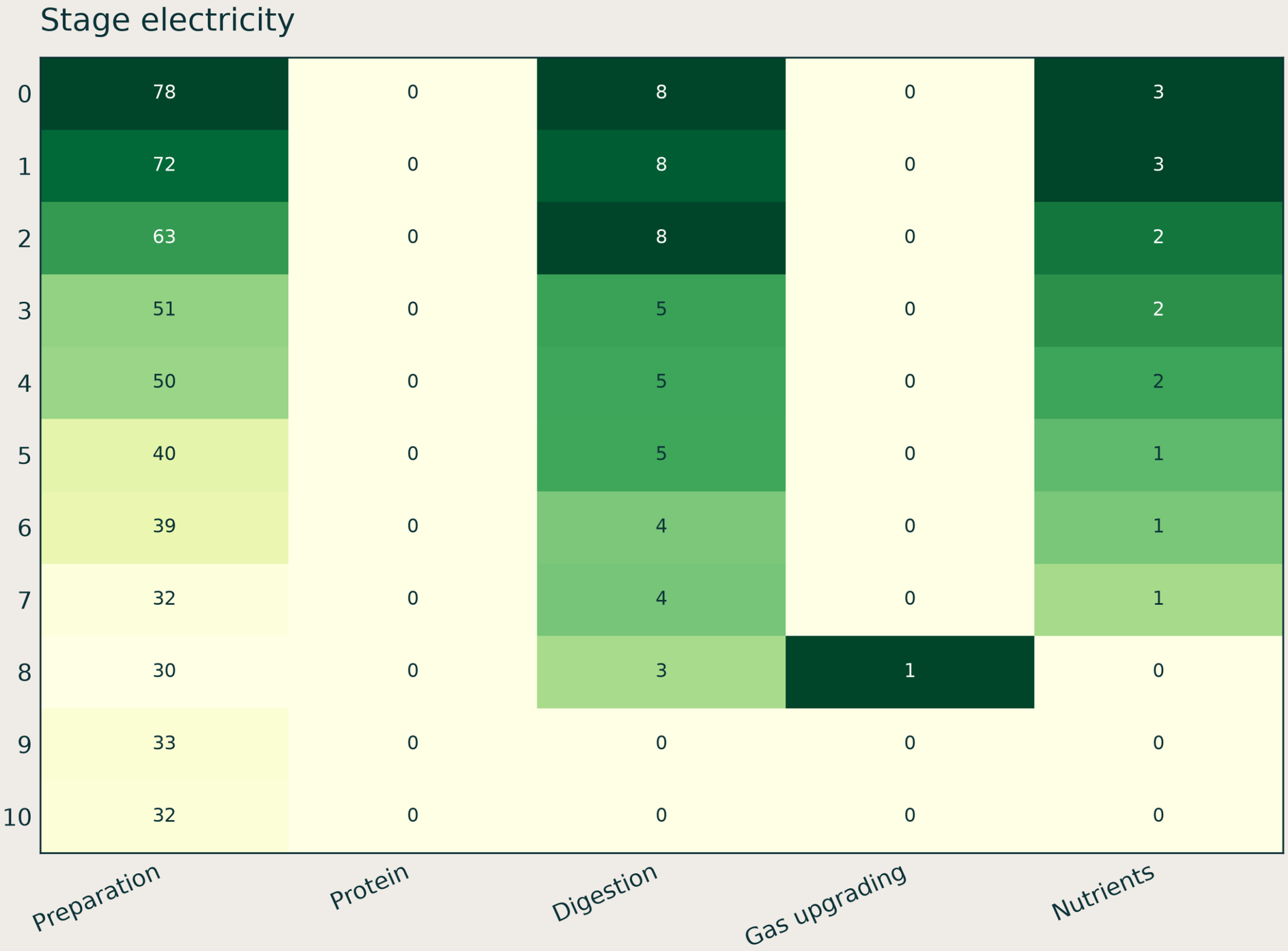
60. biosteam protein bypass route matrix



Actual route configuration codes for ranked candidates

Scenario: protein_bypass backend batch; 2000 total simulations, seed 42; objective-eligible 247
Units: route code
Source: Greenhub/results/actual_pareto_2000/decision_batches; biosteam_frontiers/rank_*.json

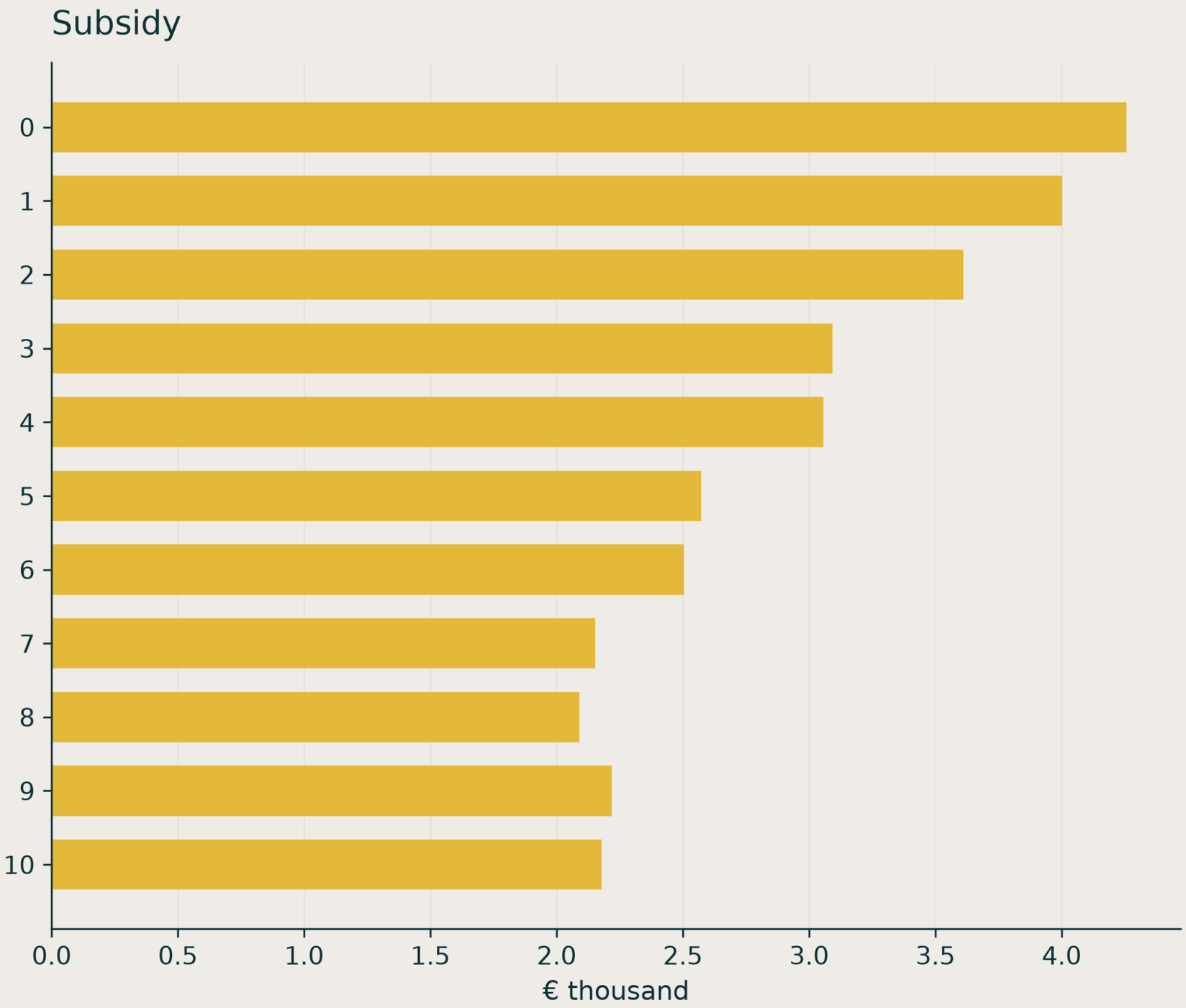
61. biosteam protein bypass stage energy heatmap



Simulated electricity by process stage across ranked candidates

Scenario: protein_bypass backend batch; 2000 total simulations, seed 42; objective-eligible 247
Units: MWh/year
Source: Greenhub/results/actual_pareto_2000/decision_batches; biosteam_frontiers/rank_*.json

62. biosteam protein bypass subsidy ranks



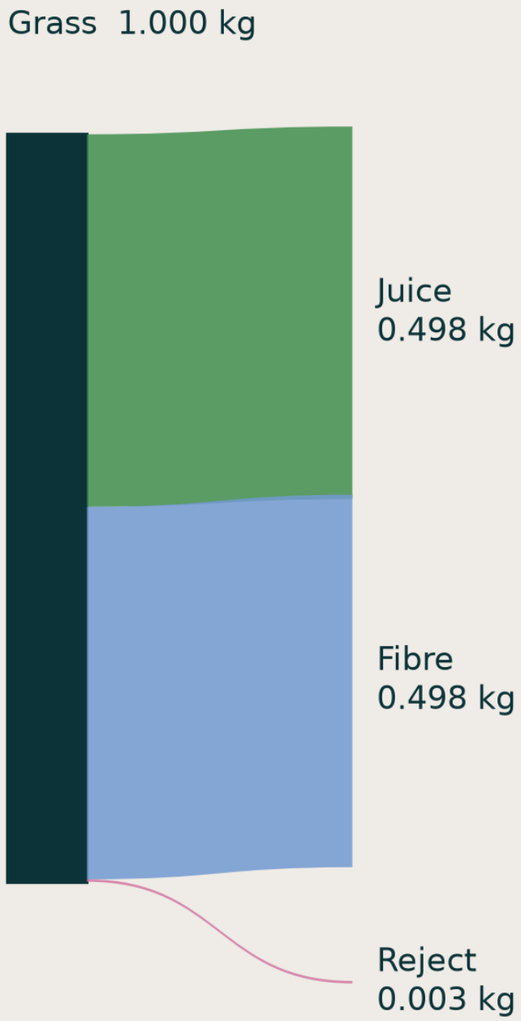
Subsidy among ranked candidates

Scenario: protein_bypass backend batch; 2000 total simulations, seed 42; objective-eligible 247
Units: € thousand
Source: Greenhub/results/actual_pareto_2000/decision_batches; biosteam_frontiers/rank_*.json

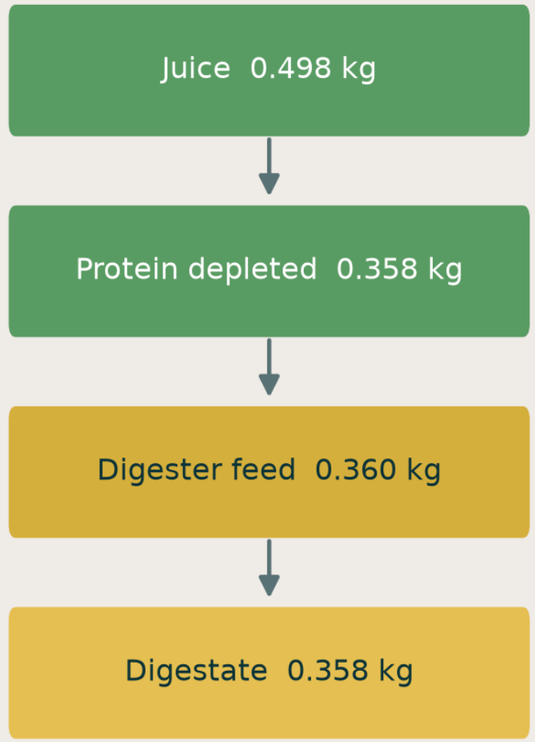
Grass to products

Protein ✓ · Biogas ✓ · N ✓ · P X · K X

Grass split



Juice pathway



Selected intermediate streams

Recovered outputs

Raw gas 0.0010 Nm³

Raw export 0.0010 Nm³

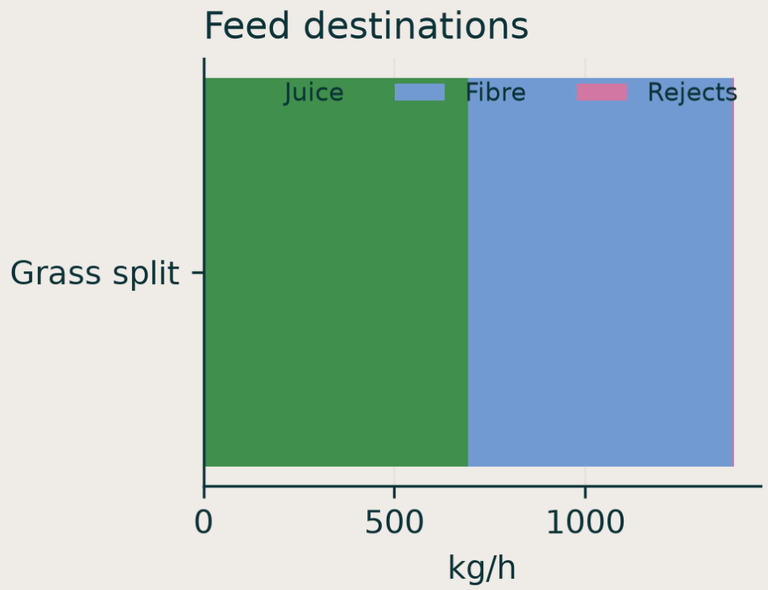
per kg wet grass

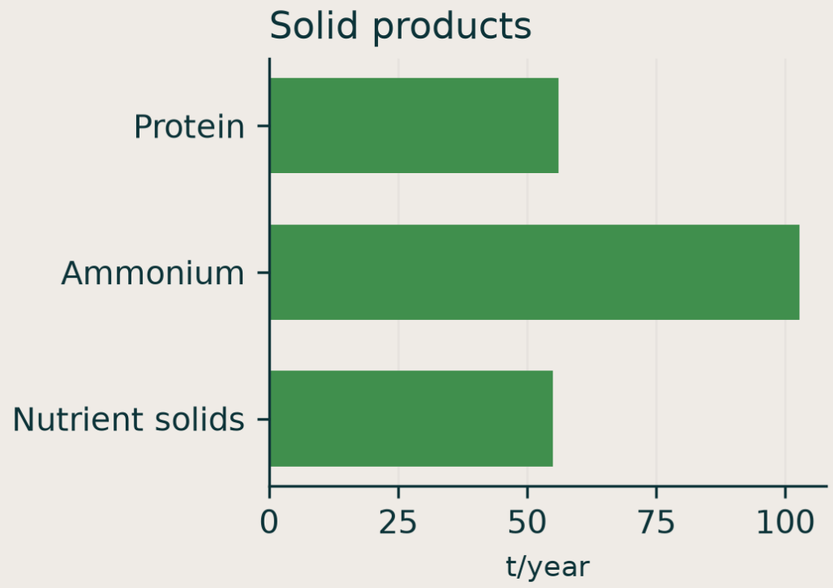
Mass-proportional grass split plus modeled downstream stream/product map

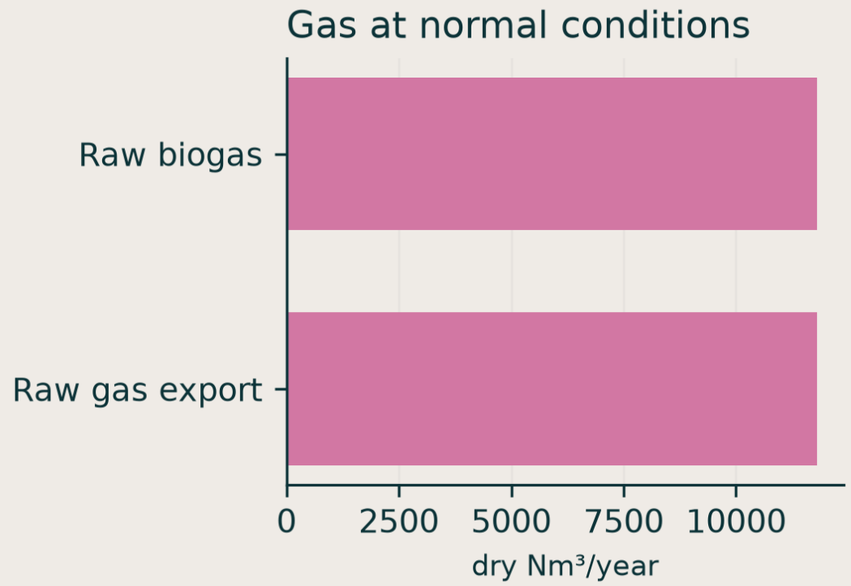
Scenario: all rank 0; 1bb3f182da164633a673cb57007ce90d; 1389.6 kg/h; 8760 h/year
Units: kg/kg wet grass; dry Nm³/kg wet grass
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

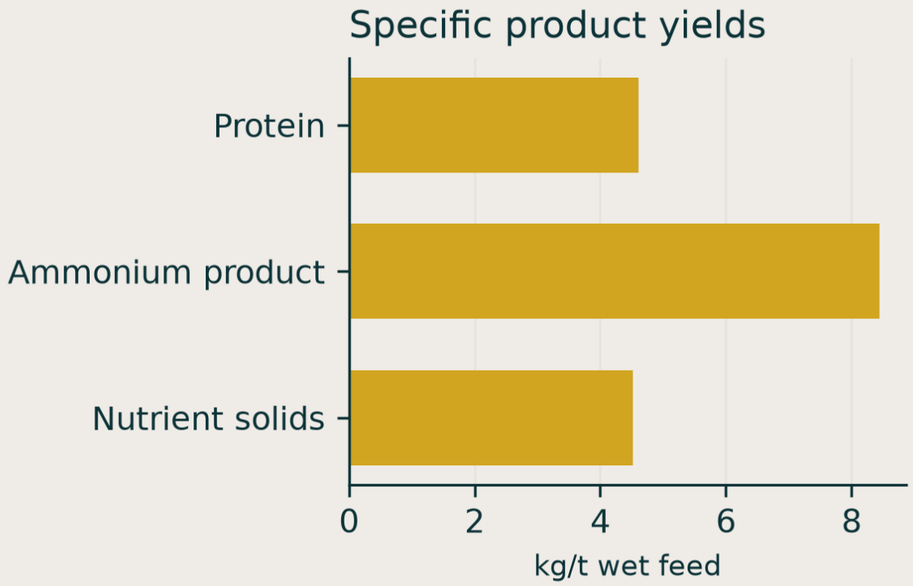
Material and products

Protein ✓ · Biogas ✓ · N ✓ · P X · K X



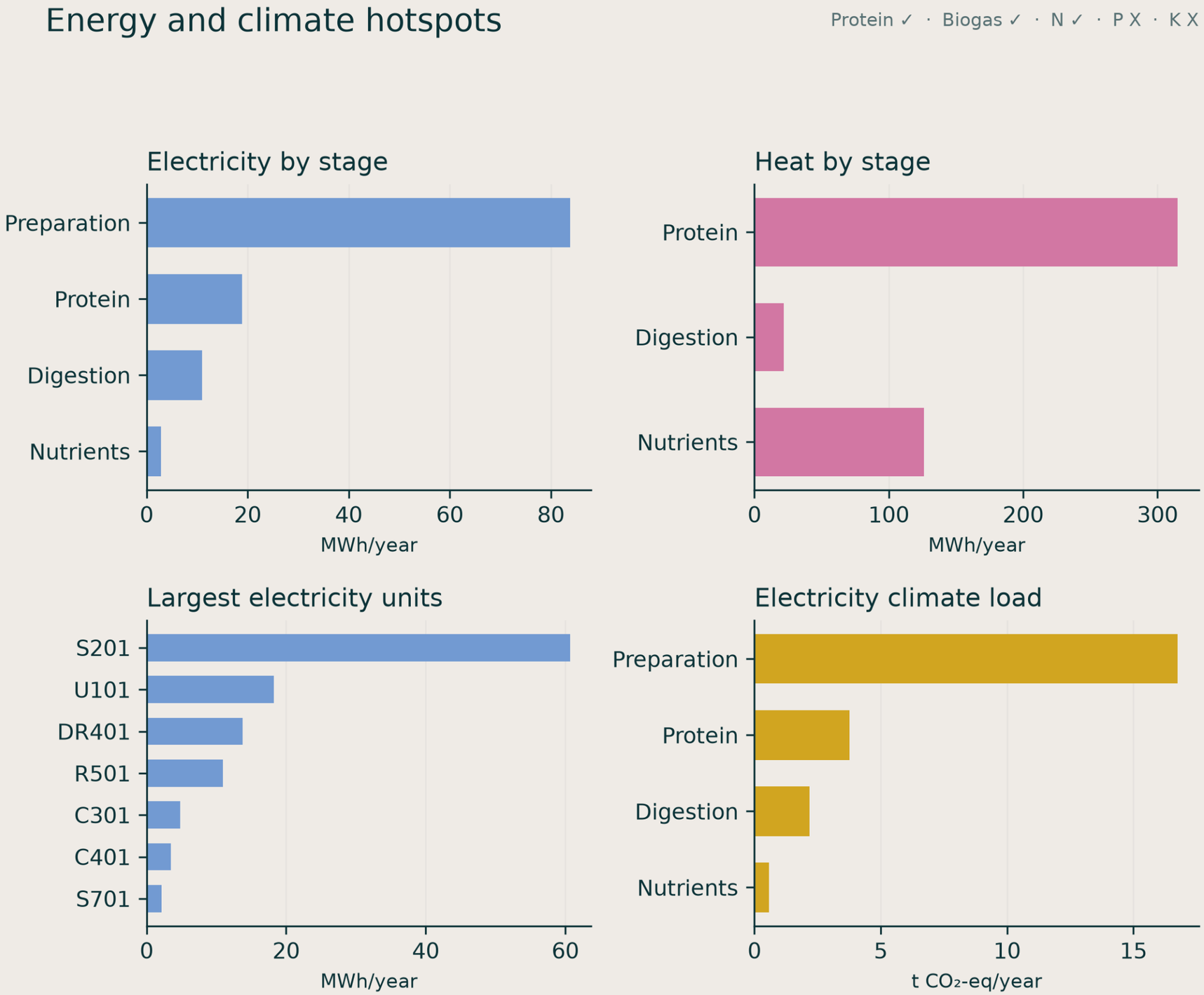






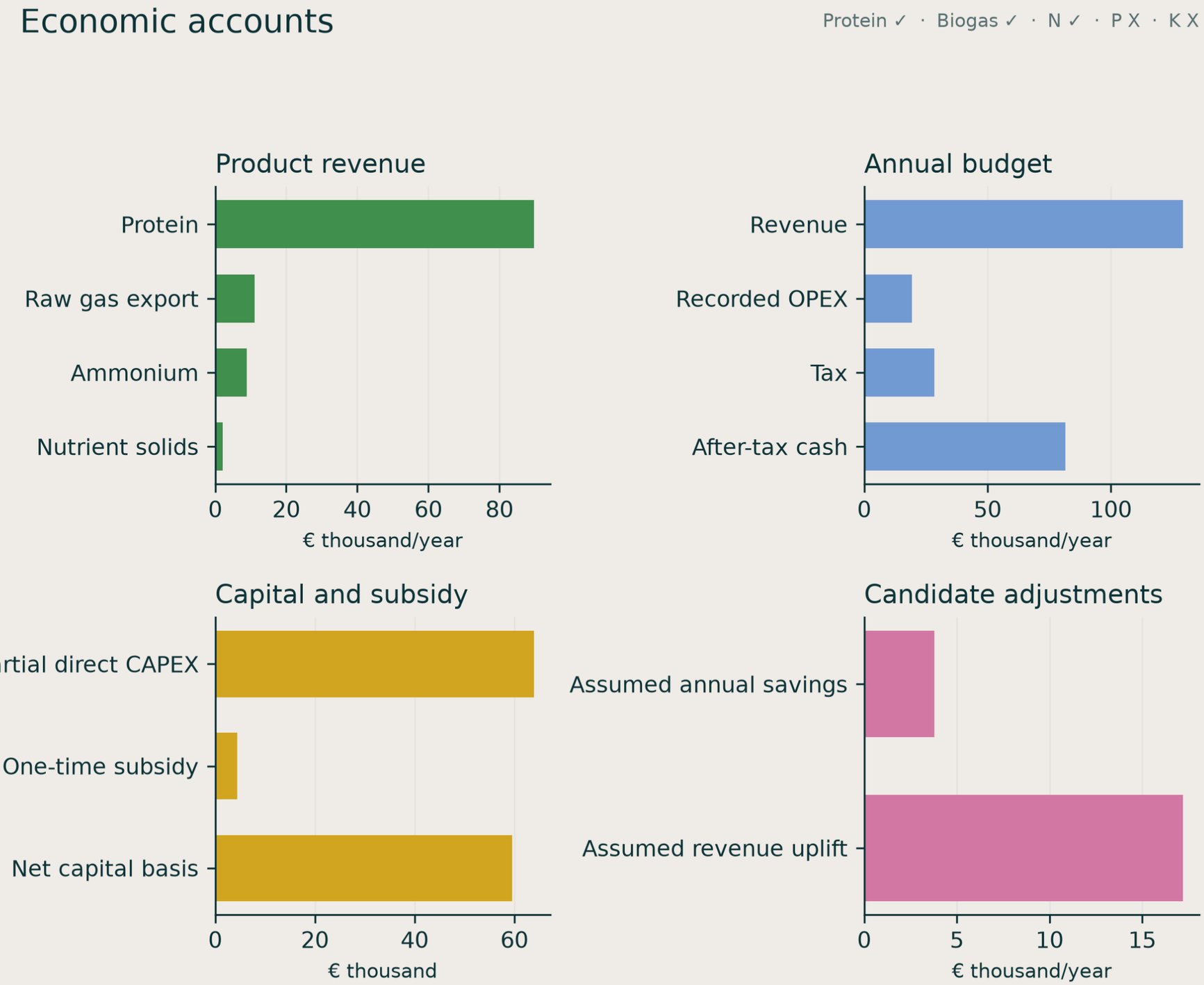
Front-end split, solid products, dry gas volumes and specific yields

Scenario: all rank 0; 1bb3f182da164633a673cb57007ce90d; 1389.6 kg/h; 8760 h/year
Units: kg/h; t/year; dry Nm³/year; kg/t feed
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay



Stage and unit utility hotspots and included climate load

Scenario: all rank 0; 1bb3f182da164633a673cb57007ce90d; 1389.6 kg/h; 8760 h/year
Units: MWh/year; t CO_2-eq/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

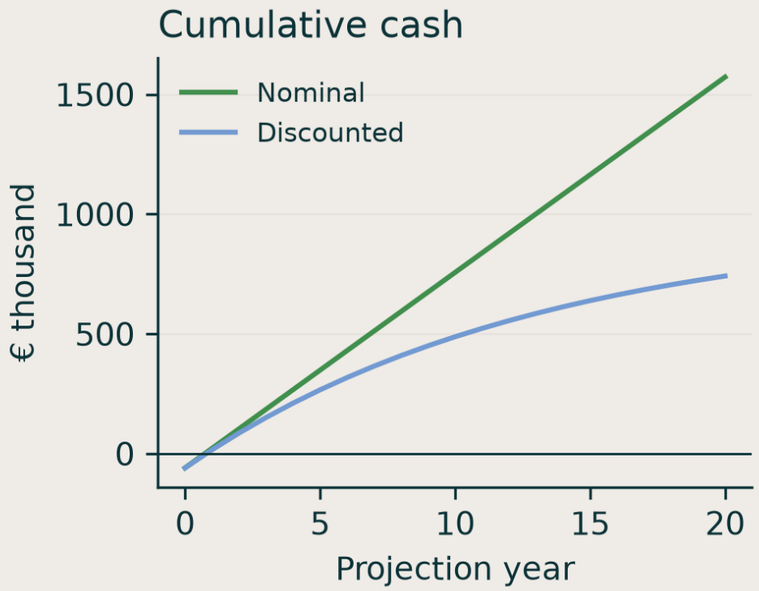


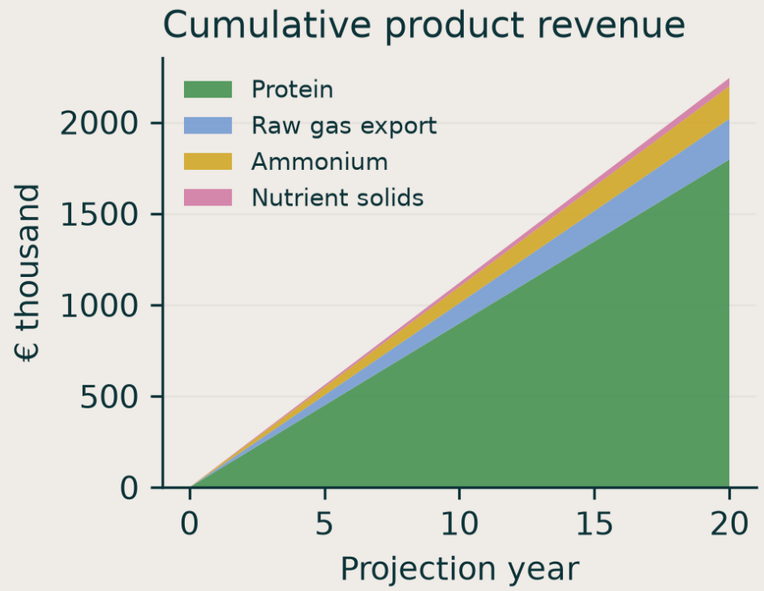
Product revenue, partial annual budget, capital subsidy and candidate overlays

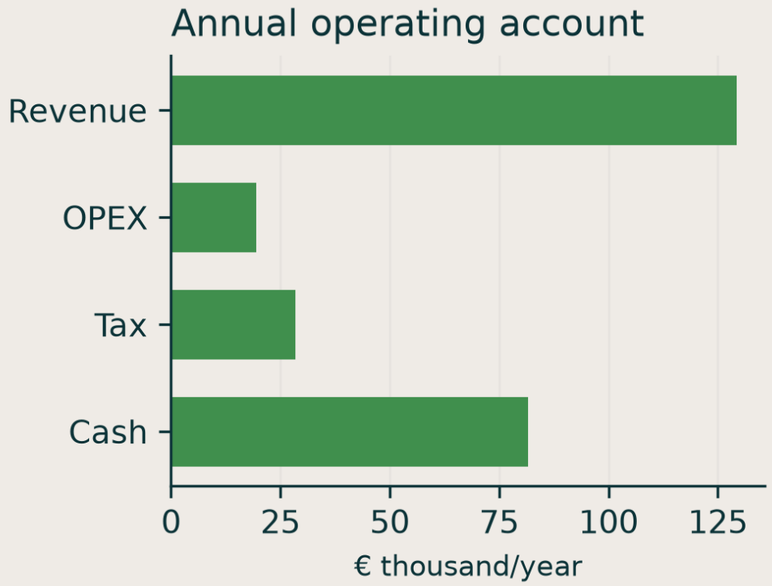
Scenario: all rank 0; 1bb3f182da164633a673cb57007ce90d; 1389.6 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

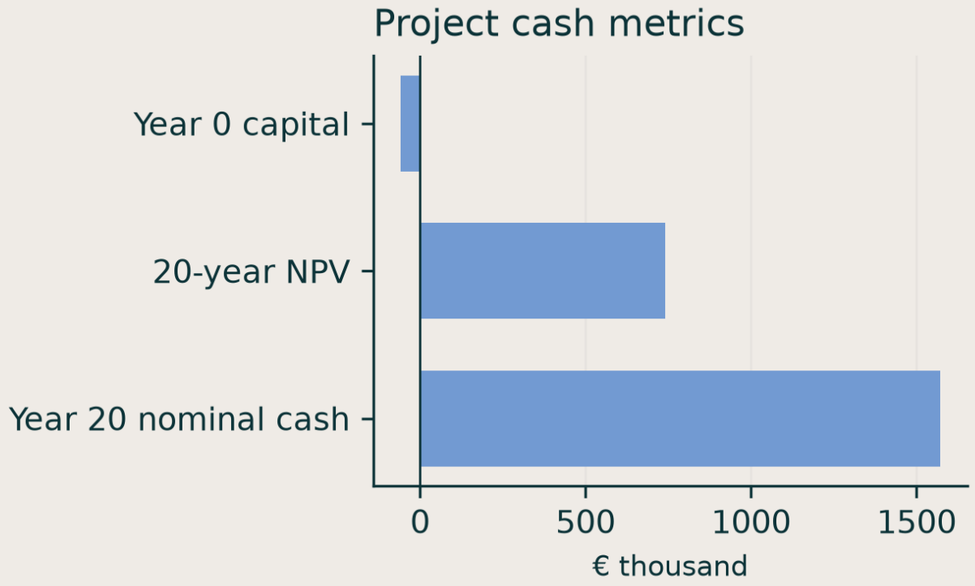
20-year financial projection

Protein ✓ · Biogas ✓ · N ✓ · P X · K X









Constant-account cash flow and cumulative product revenue over 20 years

Scenario: all rank 0; 1bb3f182da164633a673cb57007ce90d; 1389.6 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Balance and evidence

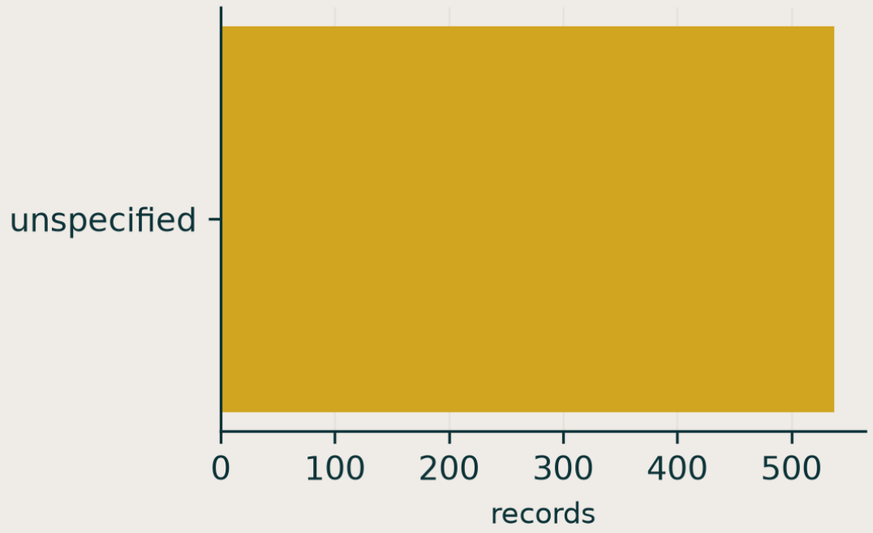
Protein ✓ · Biogas ✓ · N ✓ · P X · K X

Element closure

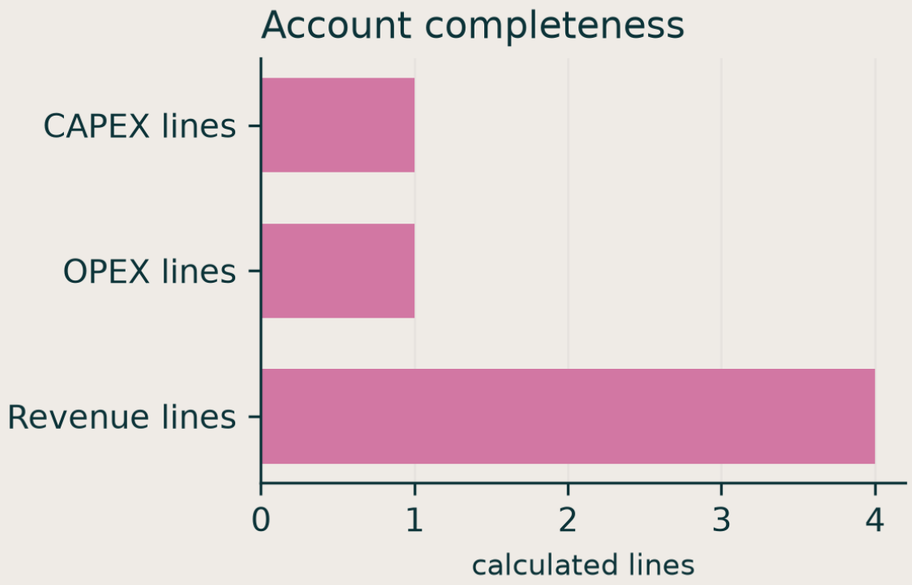
C	0 kg/year
N	0 kg/year
P	0 kg/year
K	0 kg/year

Numerical residuals

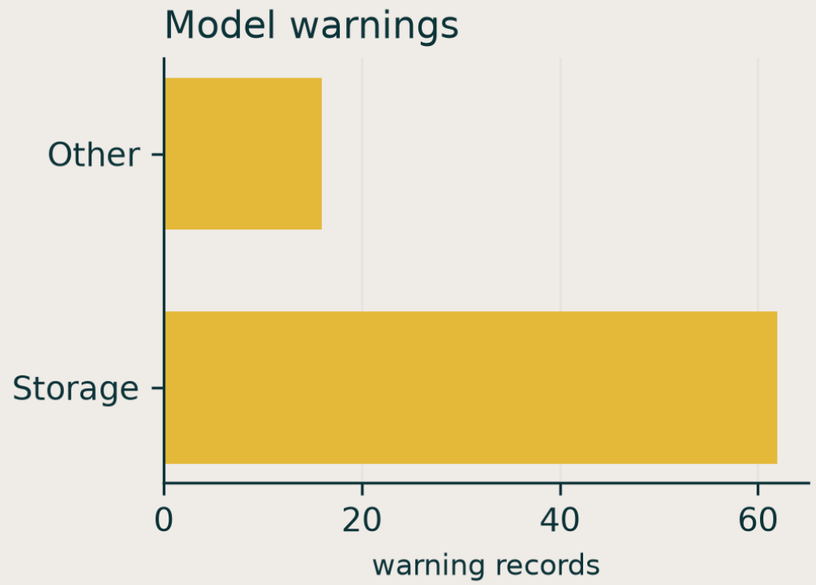
Parameter evidence



Account completeness



Model warnings



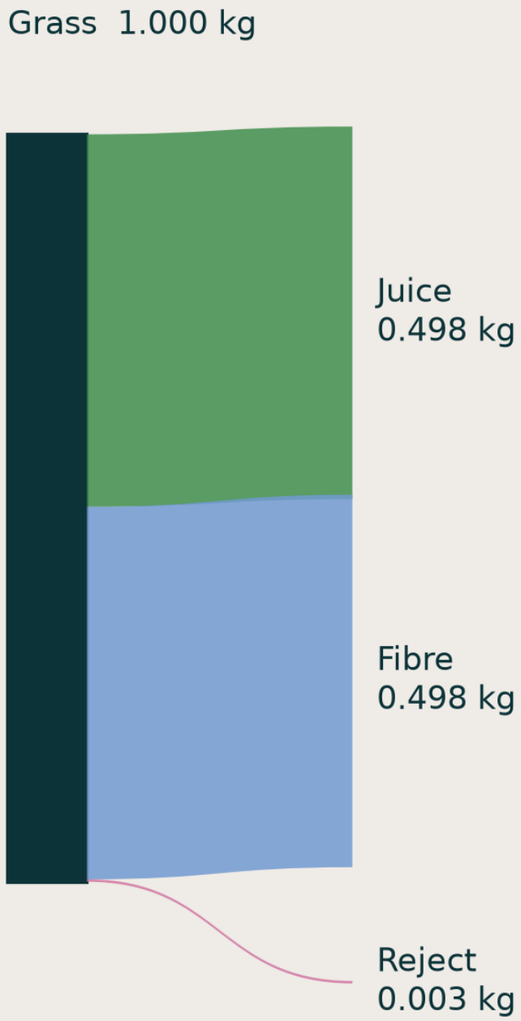
Numerical closure, evidence counts, calculated account lines and warnings

Scenario: all rank 0; 1bb3f182da164633a673cb57007ce90d; 1389.6 kg/h; 8760 h/year
Units: kg/year; records
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

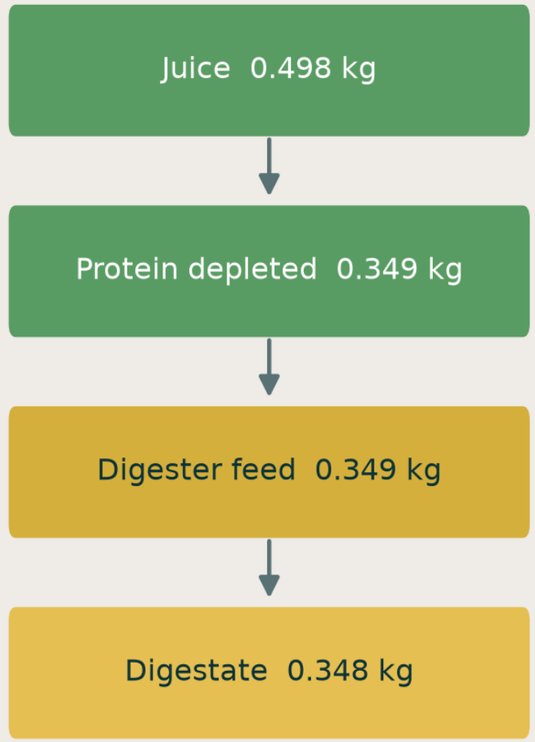
Grass to products

Protein ✓ · Biogas ✓ · N ✓ · P X · K X

Grass split



Juice pathway



Selected intermediate streams

Recovered outputs

Raw gas 0.0011 Nm³

Upgraded 0.0010 Nm³

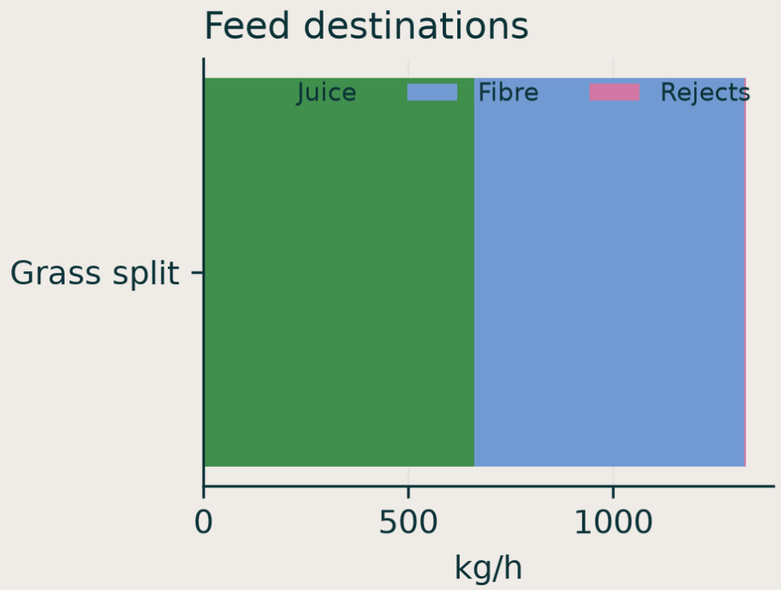
per kg wet grass

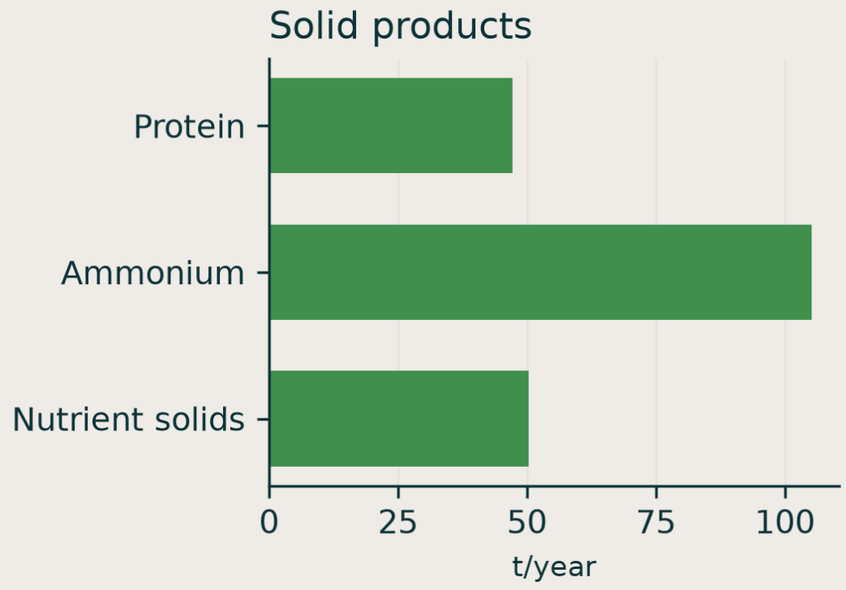
Mass-proportional grass split plus modeled downstream stream/product map

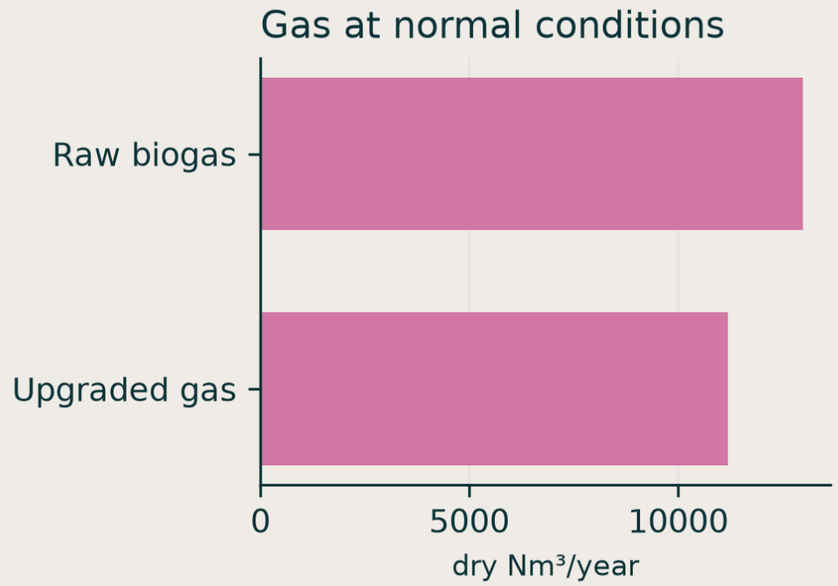
Scenario: all rank 1; 5bee2e26dc3c493494a3e59fb53be108; 1324.8 kg/h; 8760 h/year
Units: kg/kg wet grass; dry Nm³/kg wet grass
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

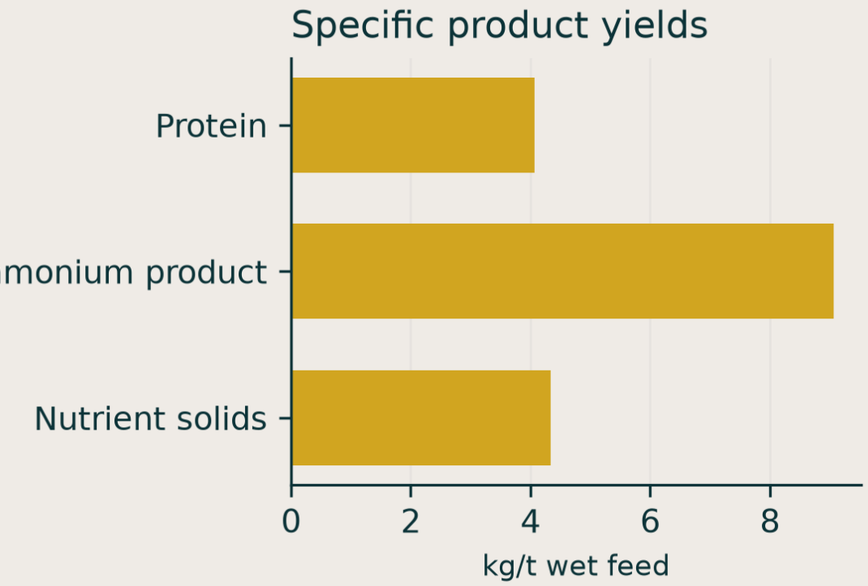
Material and products

Protein ✓ · Biogas ✓ · N ✓ · P X · K X







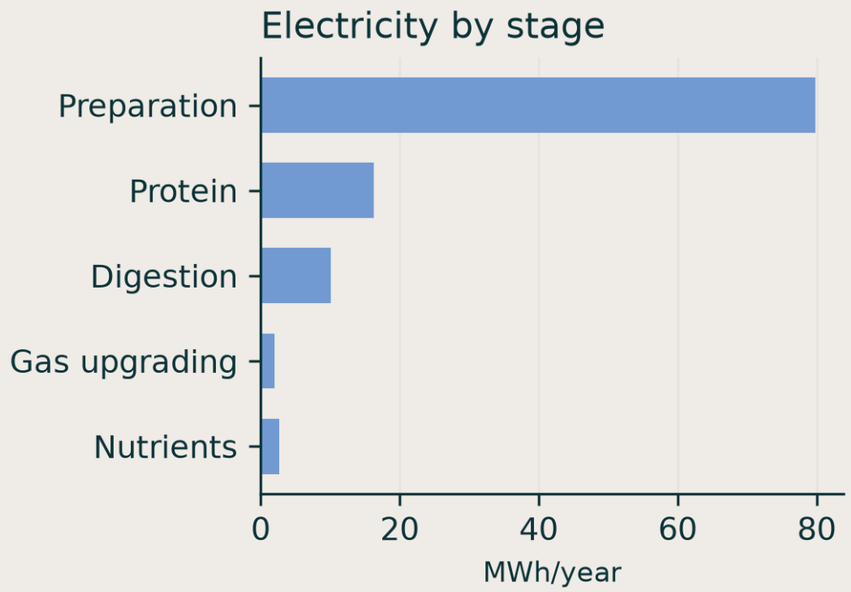


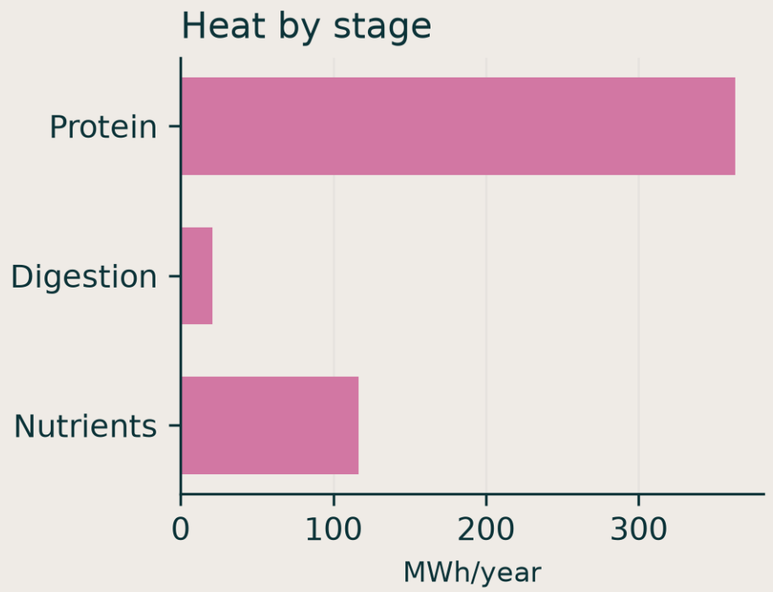
Front-end split, solid products, dry gas volumes and specific yields

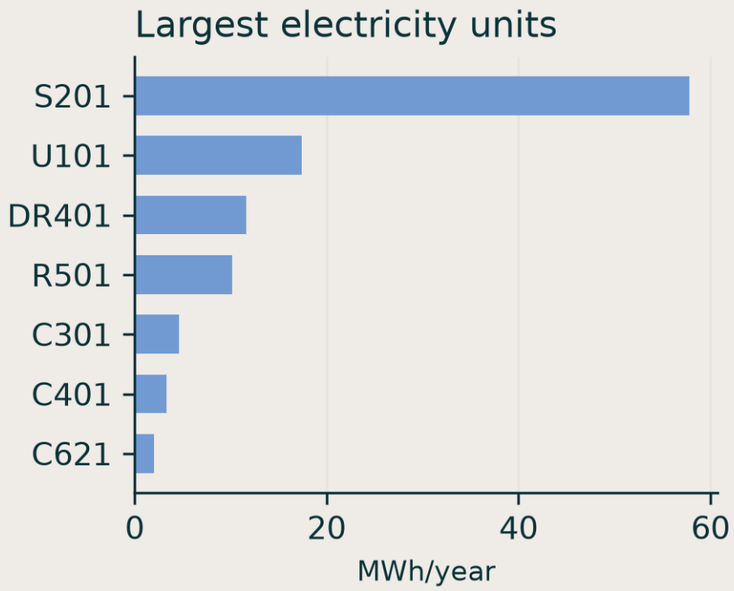
Scenario: all rank 1; 5bee2e26dc3c493494a3e59fb53be108; 1324.8 kg/h; 8760 h/year
Units: kg/h; t/year; dry Nm³/year; kg/t feed
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

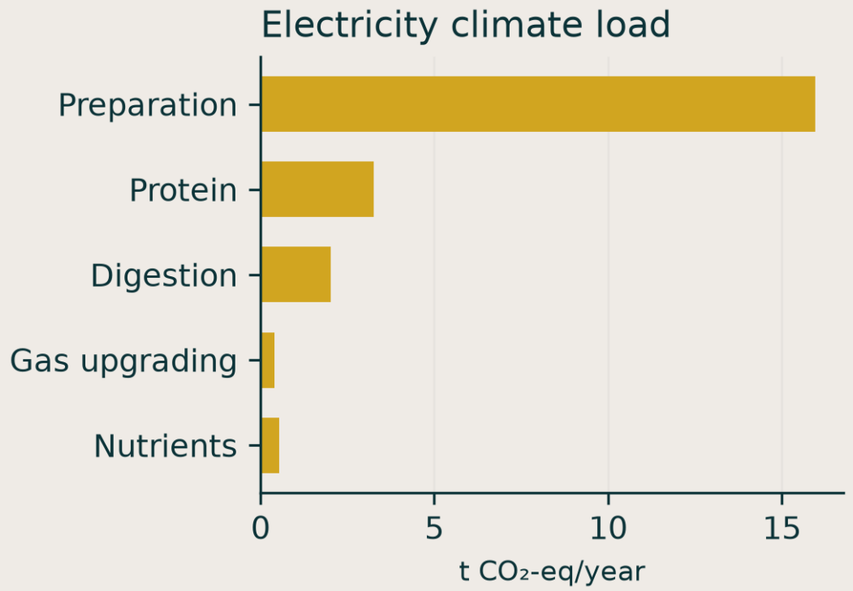
Energy and climate hotspots

Protein ✓ · Biogas ✓ · N ✓ · P X · K X



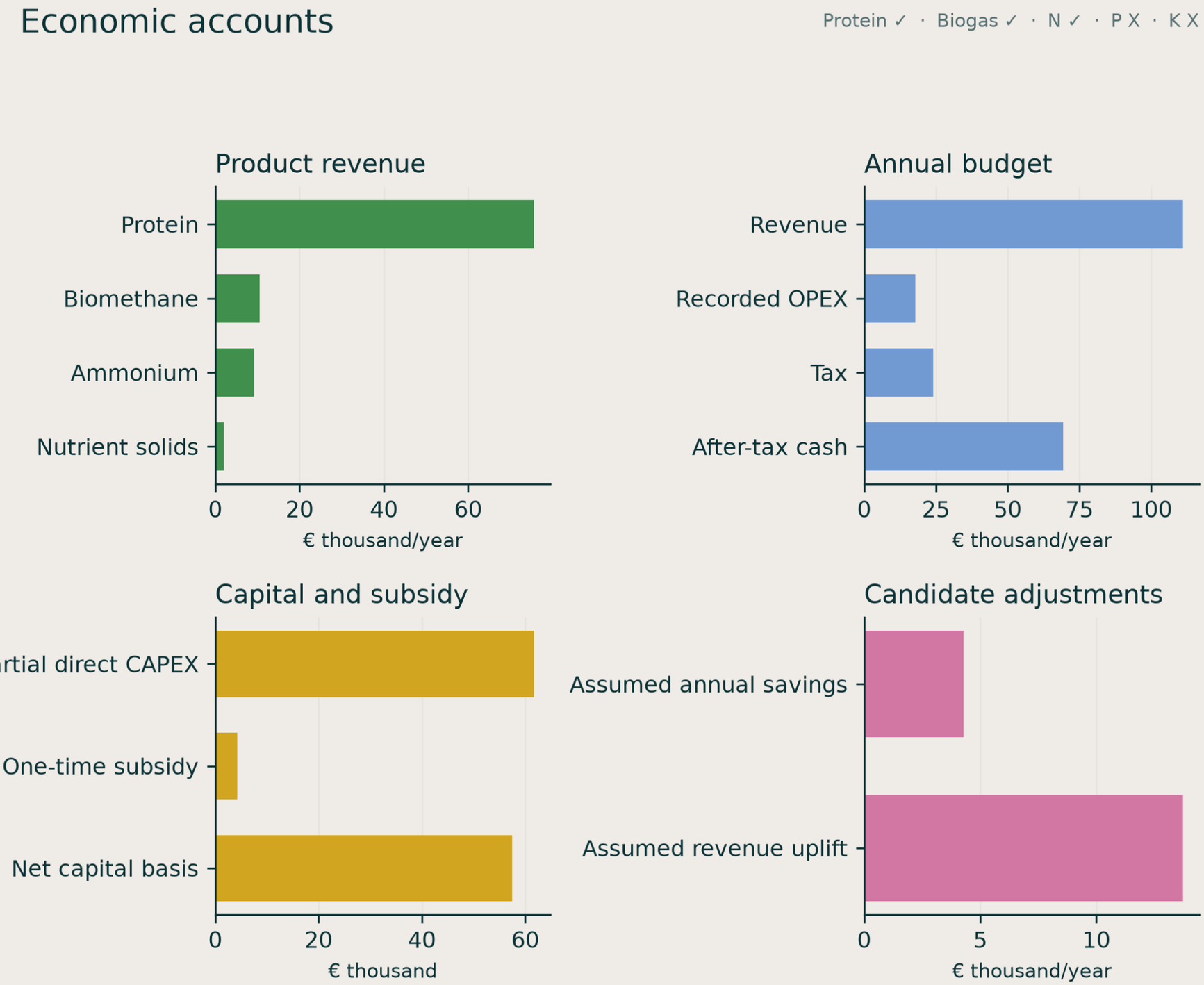






Stage and unit utility hotspots and included climate load

Scenario: all rank 1; 5bee2e26dc3c493494a3e59fb53be108; 1324.8 kg/h; 8760 h/year
Units: MWh/year; t CO_2-eq/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

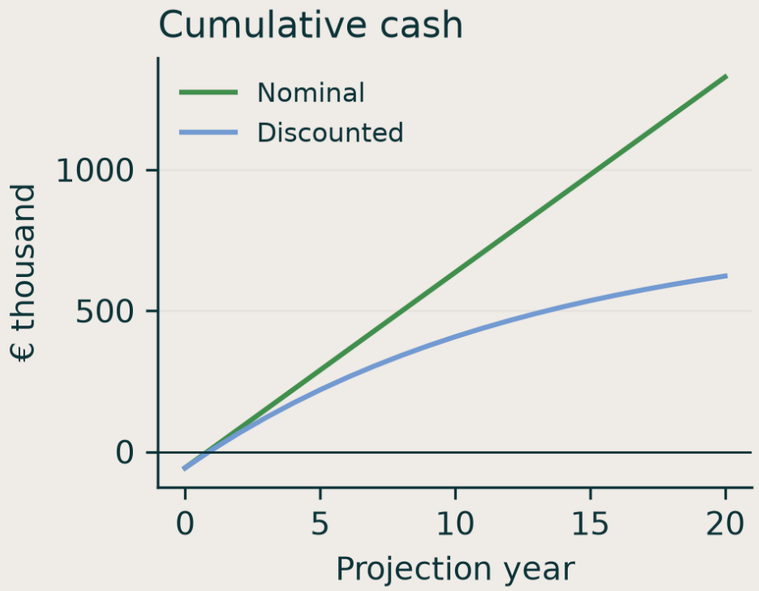


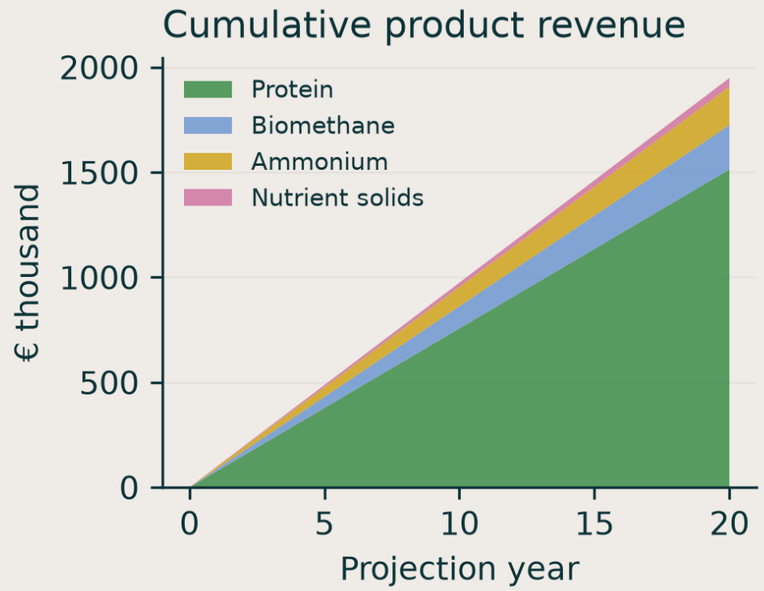
Product revenue, partial annual budget, capital subsidy and candidate overlays

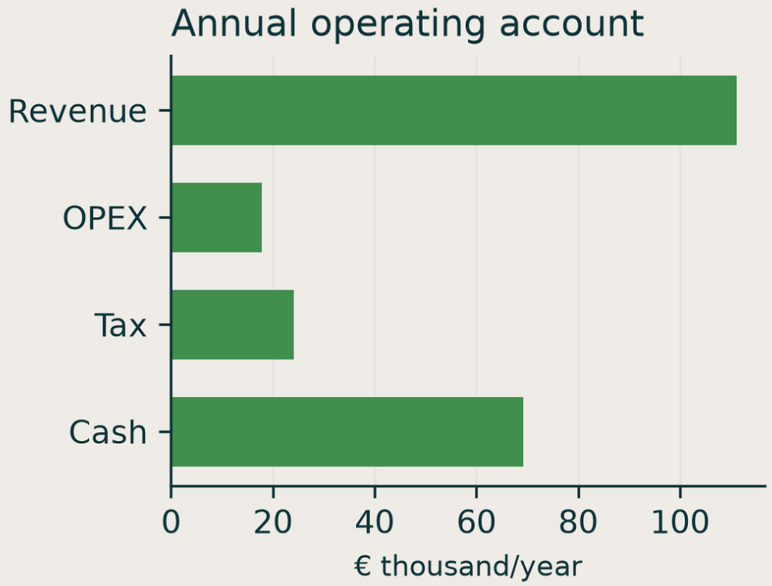
Scenario: all rank 1; 5bee2e26dc3c493494a3e59fb53be108; 1324.8 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

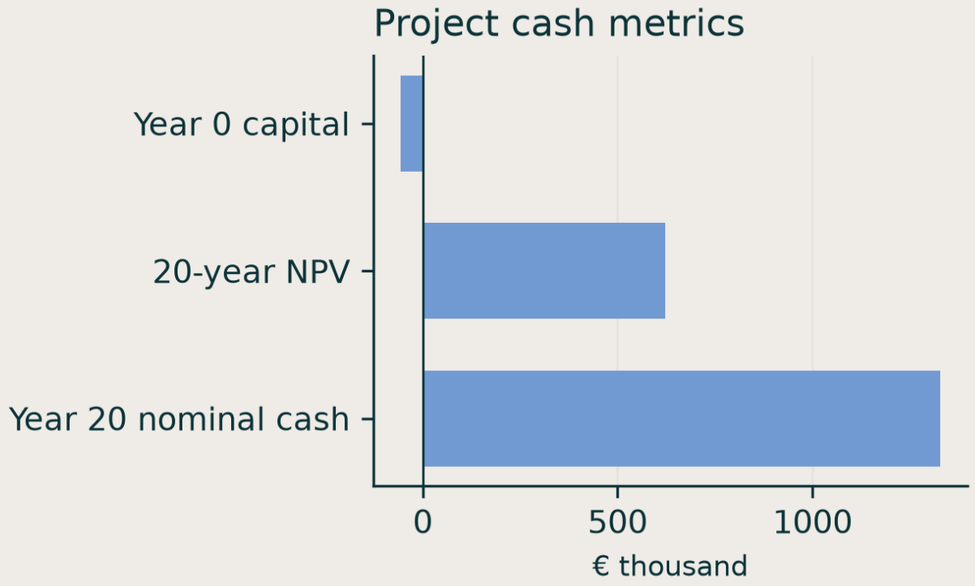
20-year financial projection

Protein ✓ · Biogas ✓ · N ✓ · P X · K X









Constant-account cash flow and cumulative product revenue over 20 years

Scenario: all rank 1; 5bee2e26dc3c493494a3e59fb53be108; 1324.8 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Balance and evidence

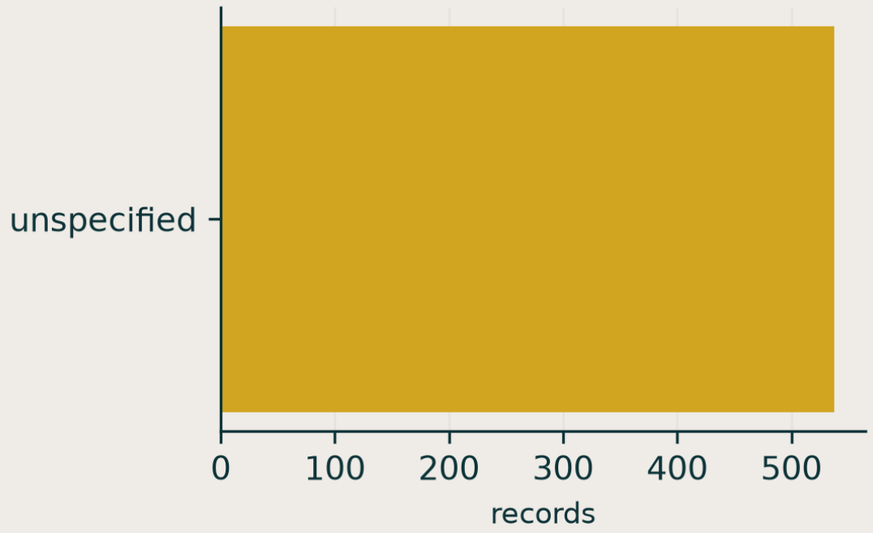
Protein ✓ · Biogas ✓ · N ✓ · P X · K X

Element closure

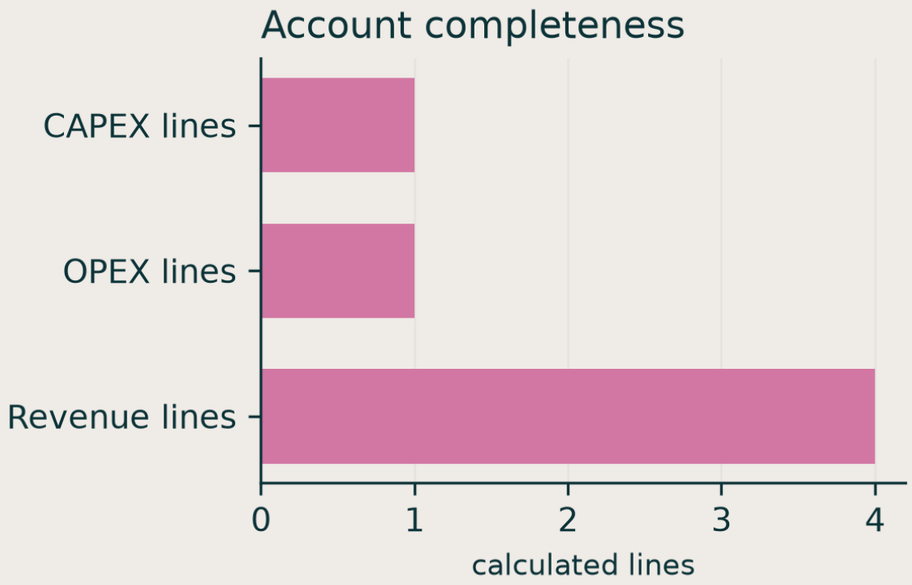
C	0 kg/year
N	0 kg/year
P	0 kg/year
K	0 kg/year

Numerical residuals

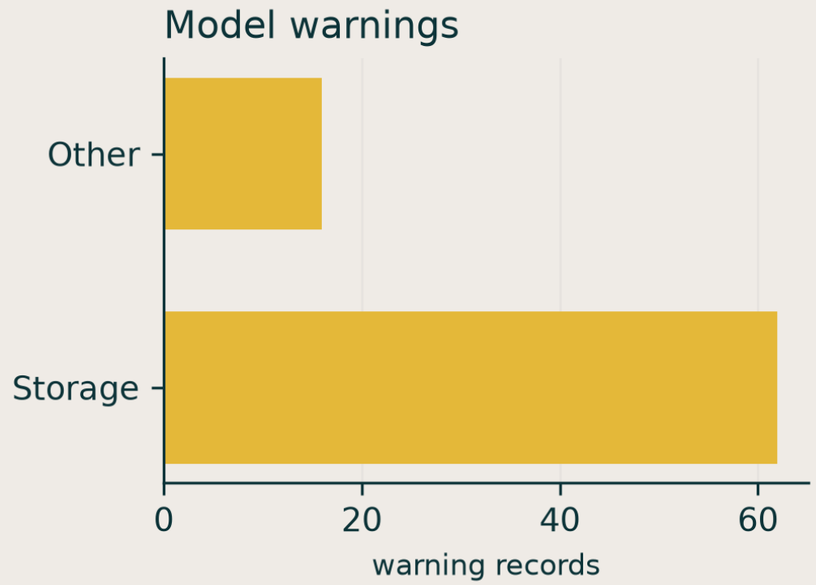
Parameter evidence



Account completeness



Model warnings



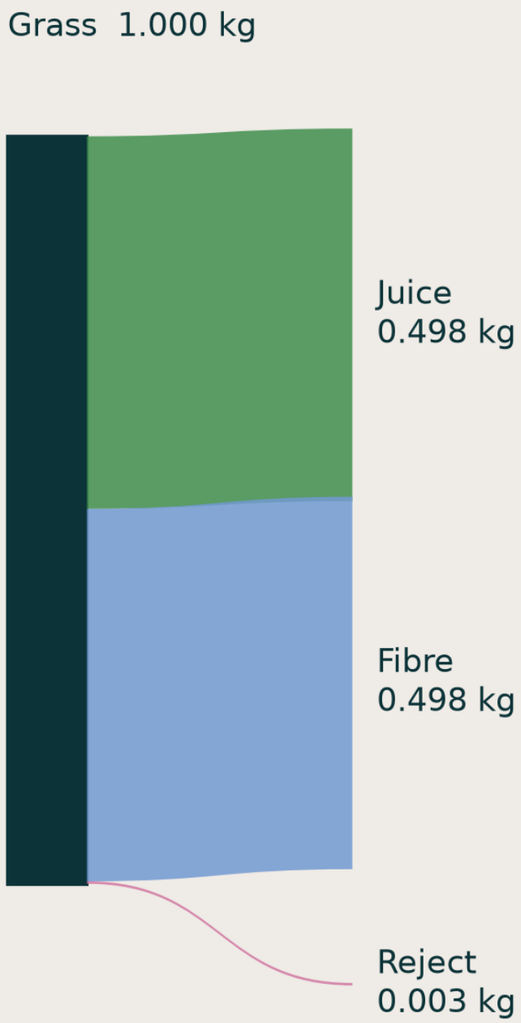
Numerical closure, evidence counts, calculated account lines and warnings

Scenario: all rank 1; 5bee2e26dc3c493494a3e59fb53be108; 1324.8 kg/h; 8760 h/year
Units: kg/year; records
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

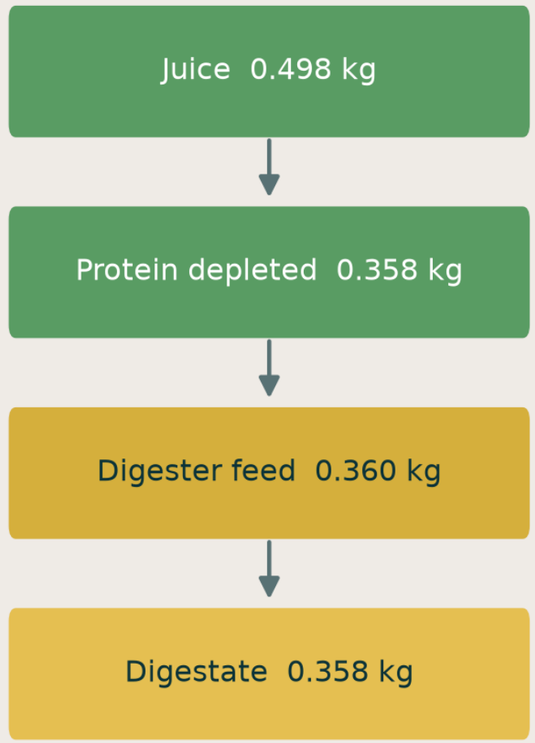
Grass to products

Protein ✓ · Biogas ✓ · N ✓ · P ✓ · K X

Grass split



Juice pathway



Selected intermediate streams

Recovered outputs

Protein 0.0046 kg

N product 0.0084 kg

P product 0.0001 kg

Nutrient solids 0.0045 kg

Raw gas 0.0010 Nm³

Raw export 0.0010 Nm³

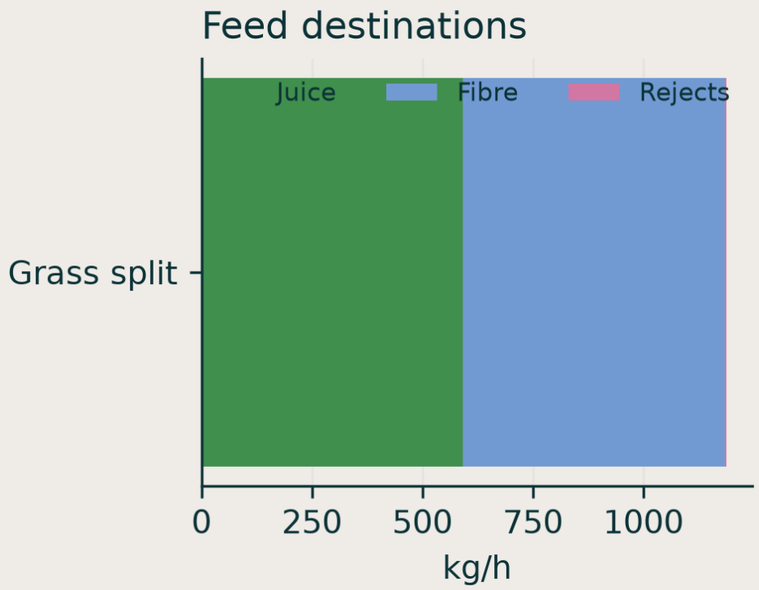
per kg wet grass

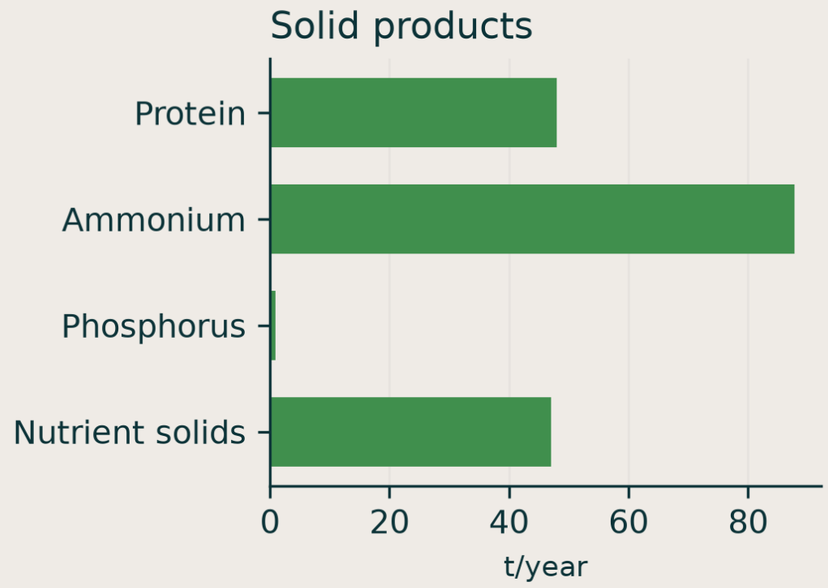
Mass-proportional grass split plus modeled downstream stream/product map

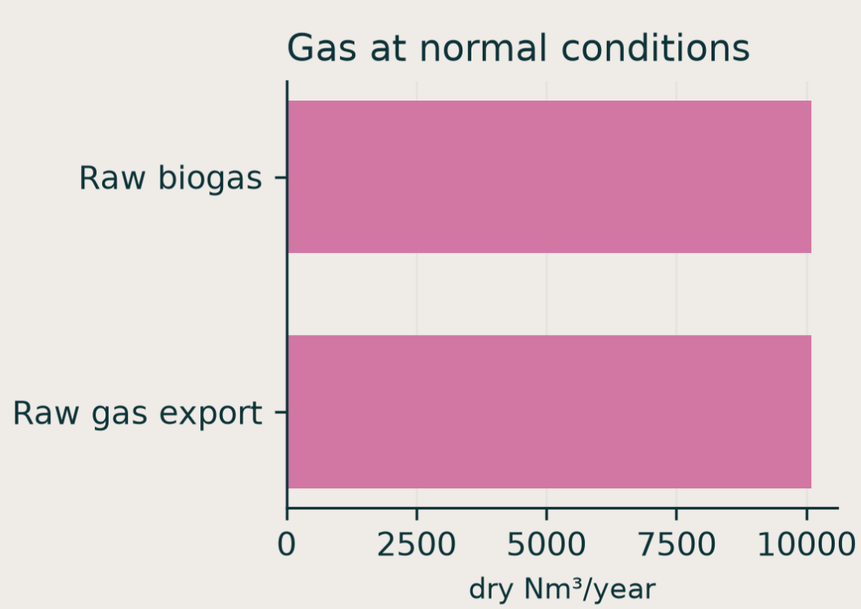
Scenario: all rank 2; 30068847a9d7468a9ad5cdec884bec5b; 1187.9 kg/h; 8760 h/year
Units: kg/kg wet grass; dry Nm³/kg wet grass
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

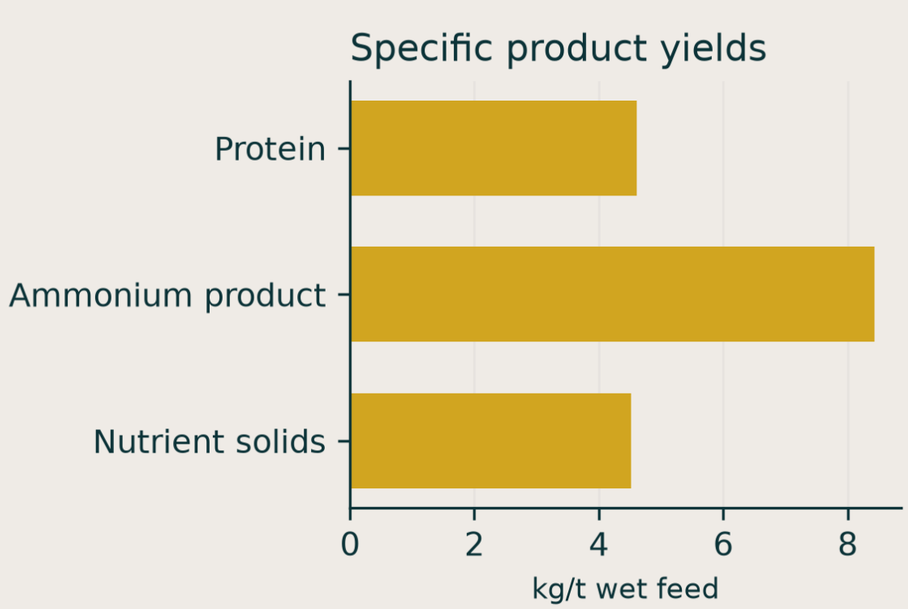
Material and products

Protein ✓ · Biogas ✓ · N ✓ · P ✓ · K X







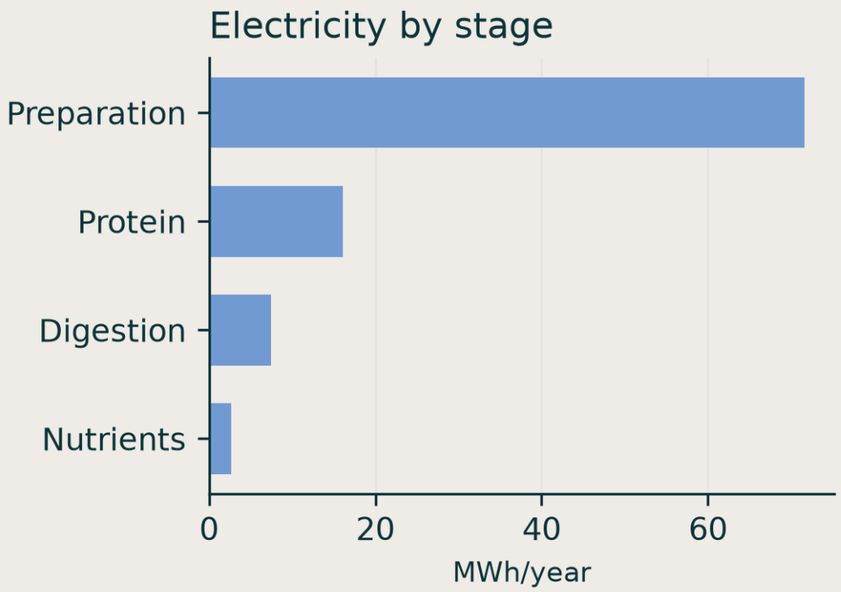


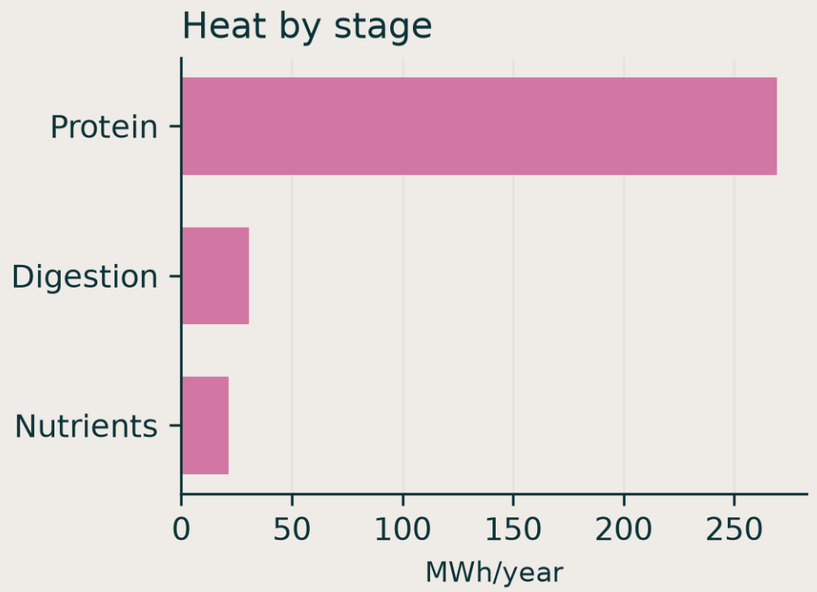
Front-end split, solid products, dry gas volumes and specific yields

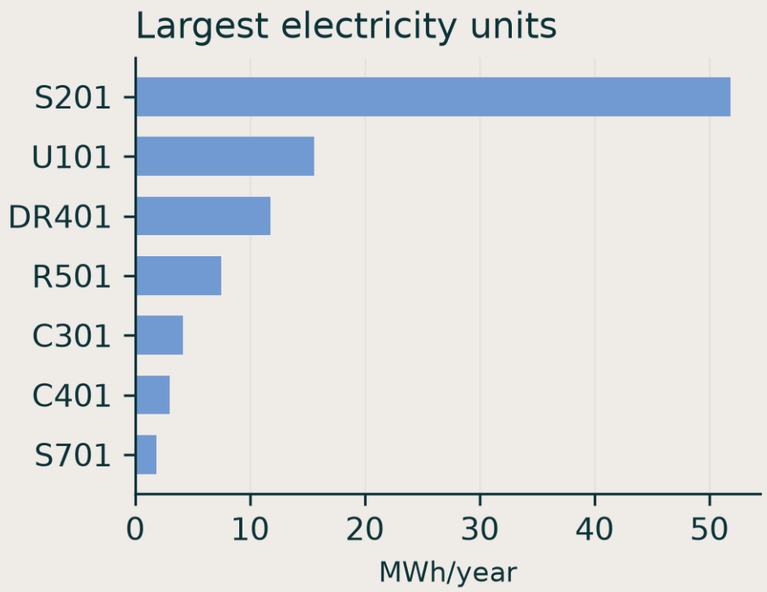
Scenario: all rank 2; 30068847a9d7468a9ad5cdec884bec5b; 1187.9 kg/h; 8760 h/year
Units: kg/h; t/year; dry Nm³/year; kg/t feed
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

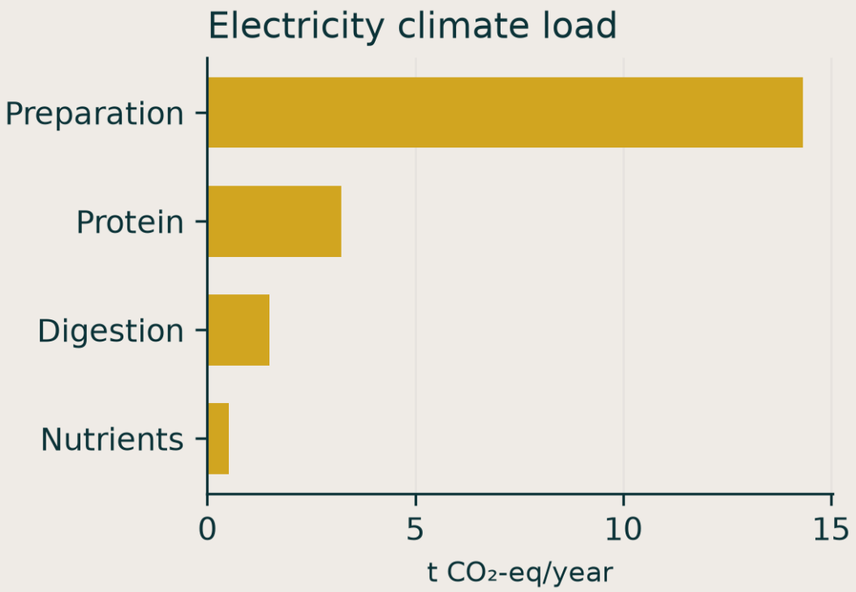
Energy and climate hotspots

Protein ✓ · Biogas ✓ · N ✓ · P ✓ · K X



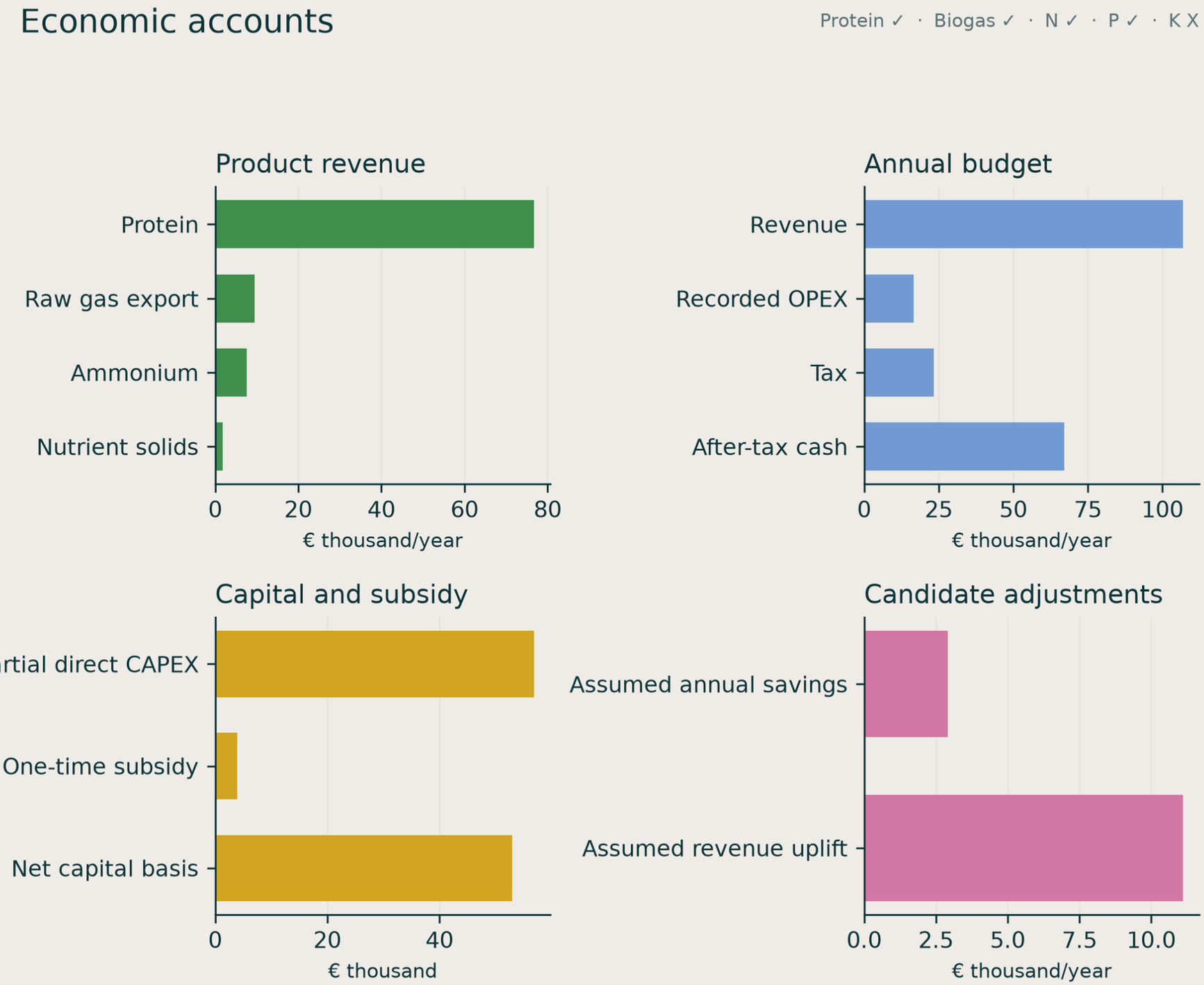






Stage and unit utility hotspots and included climate load

Scenario: all rank 2; 30068847a9d7468a9ad5cdec884bec5b; 1187.9 kg/h; 8760 h/year
Units: MWh/year; t CO_2-eq/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

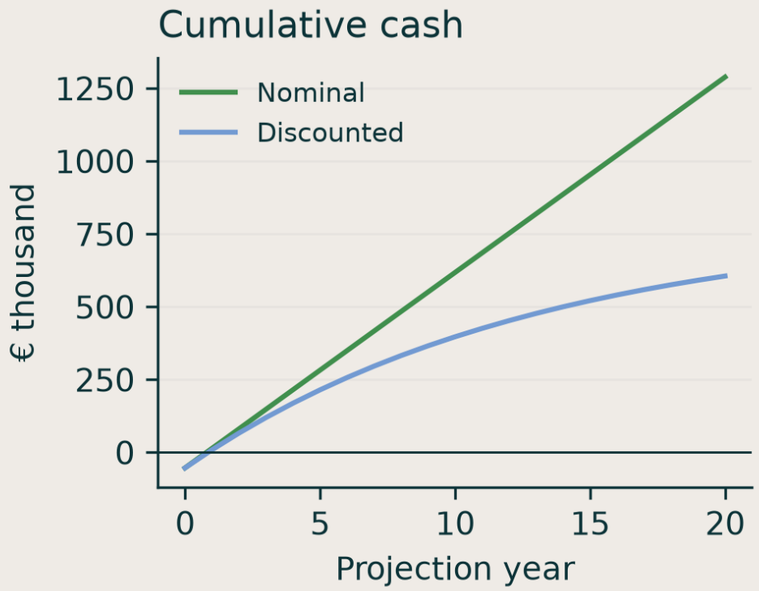


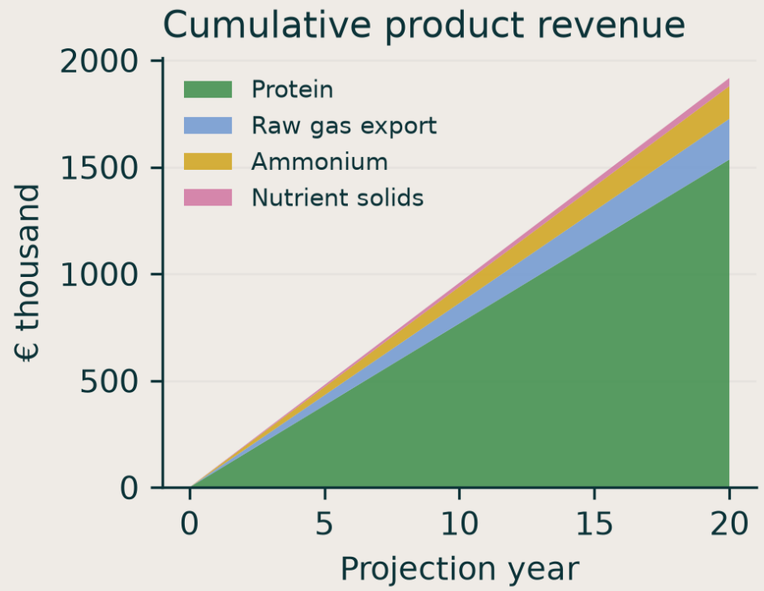
Product revenue, partial annual budget, capital subsidy and candidate overlays

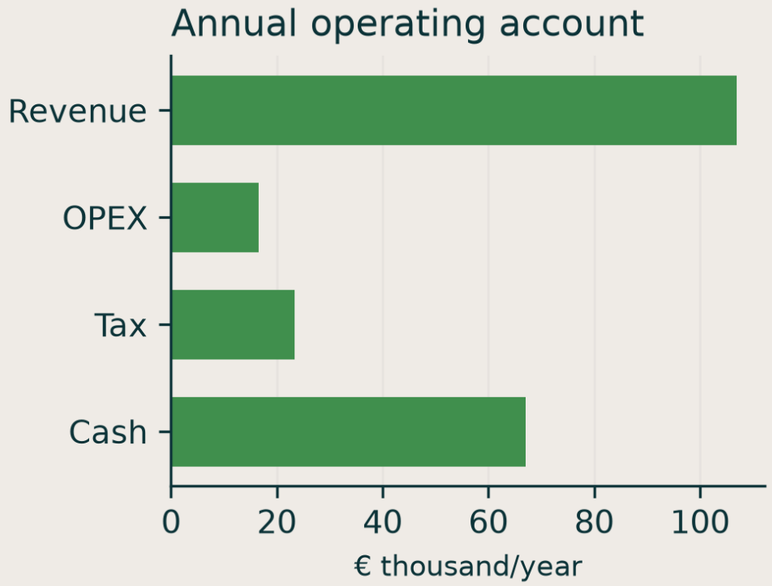
Scenario: all rank 2; 30068847a9d7468a9ad5cdec884bec5b; 1187.9 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

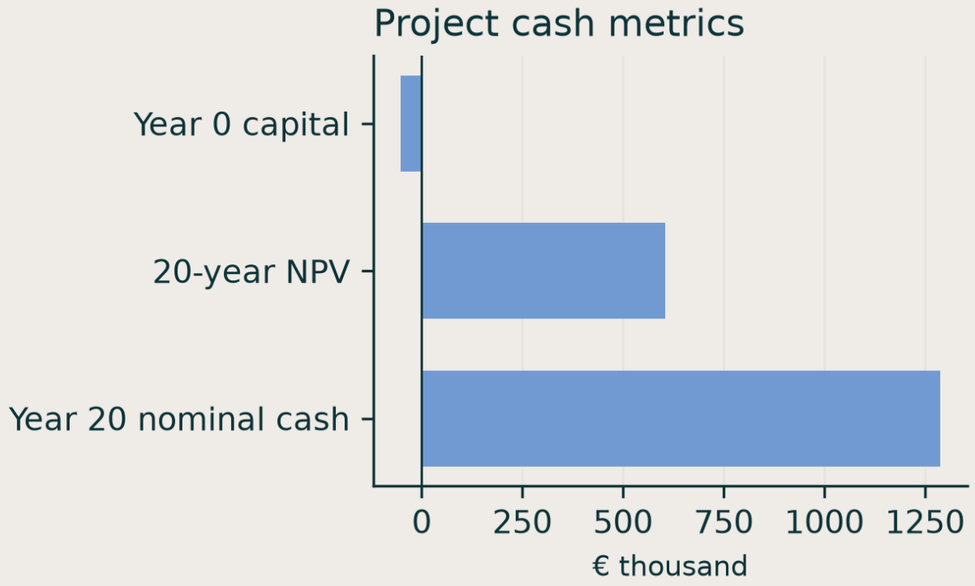
20-year financial projection

Protein ✓ · Biogas ✓ · N ✓ · P ✓ · K X









Constant-account cash flow and cumulative product revenue over 20 years

Scenario: all rank 2; 30068847a9d7468a9ad5cdec884bec5b; 1187.9 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Balance and evidence

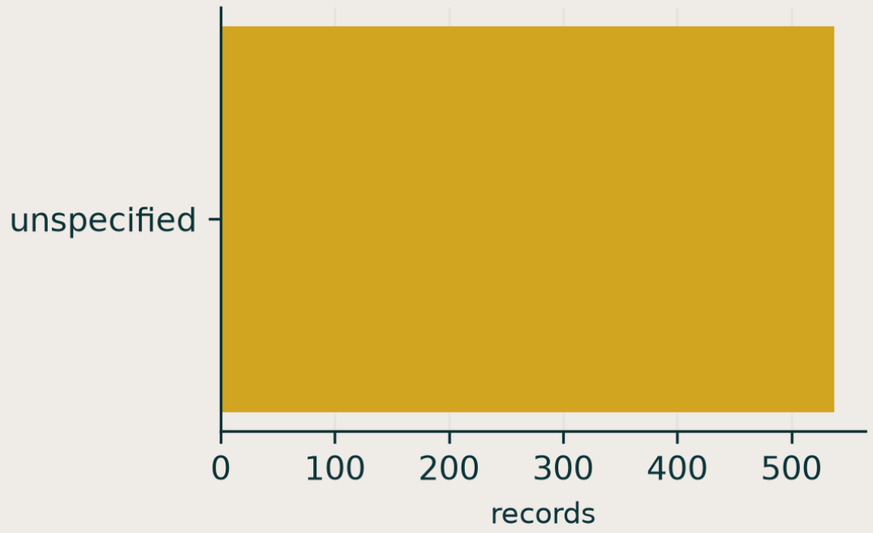
Protein ✓ · Biogas ✓ · N ✓ · P ✓ · K X

Element closure

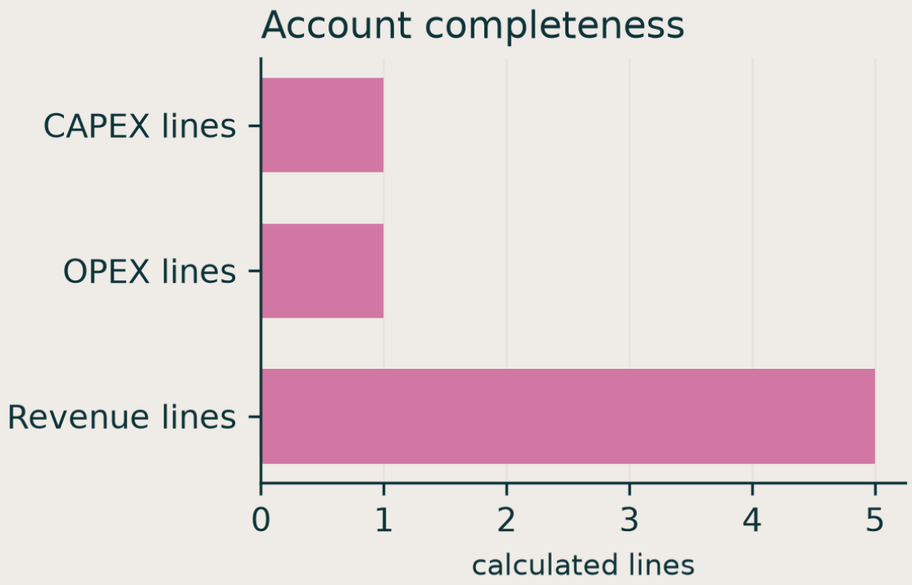
C	0 kg/year
N	0 kg/year
P	1.2e-13 kg/year
K	0 kg/year

Numerical residuals

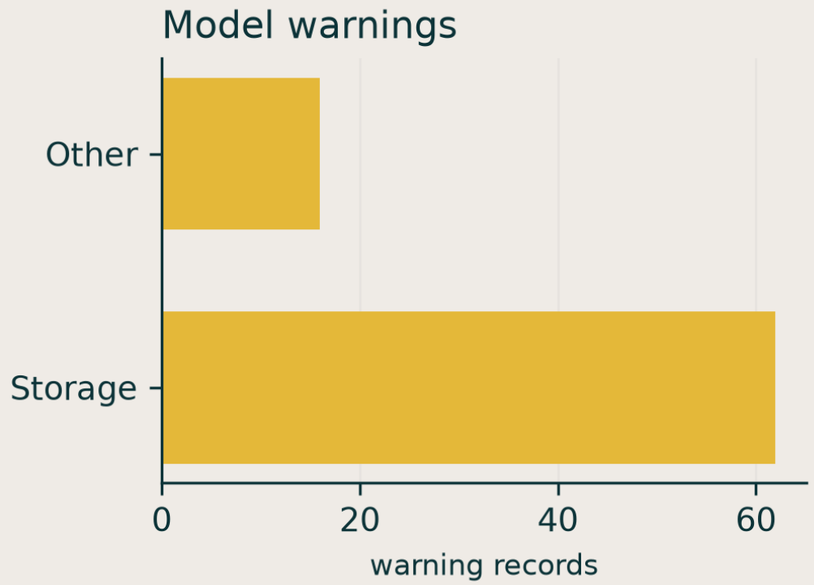
Parameter evidence



Account completeness



Model warnings



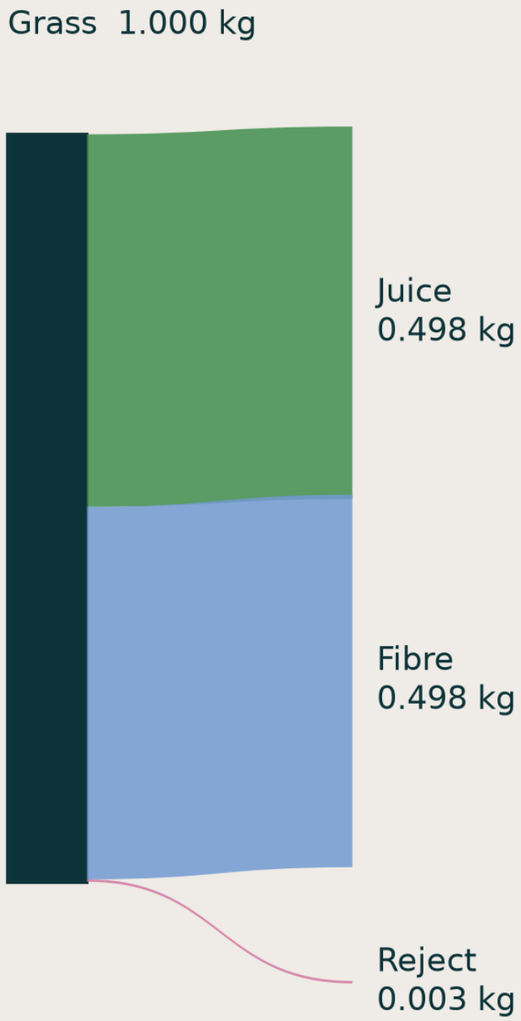
Numerical closure, evidence counts, calculated account lines and warnings

Scenario: all rank 2; 30068847a9d7468a9ad5cdec884bec5b; 1187.9 kg/h; 8760 h/year
Units: kg/year; records
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

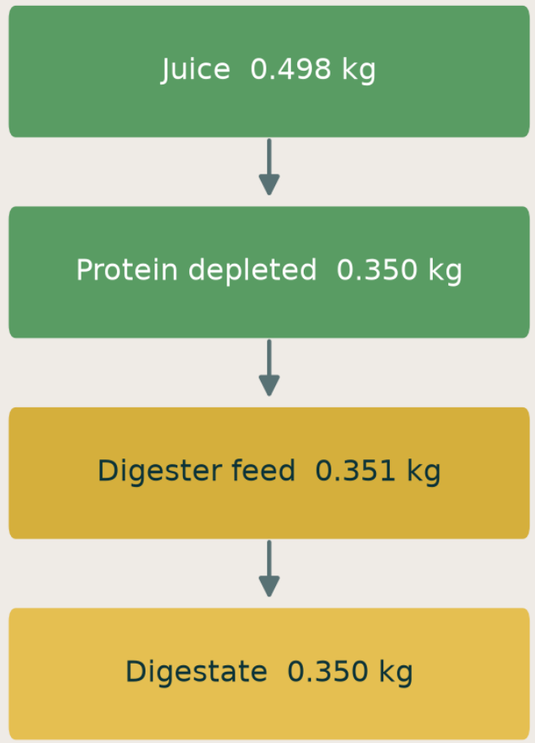
Grass to products

Protein ✓ · Biogas ✓ · N ✓ · P X · K X

Grass split



Juice pathway



Selected intermediate streams

Recovered outputs

Raw gas 0.0010 Nm³

Raw export 0.0010 Nm³

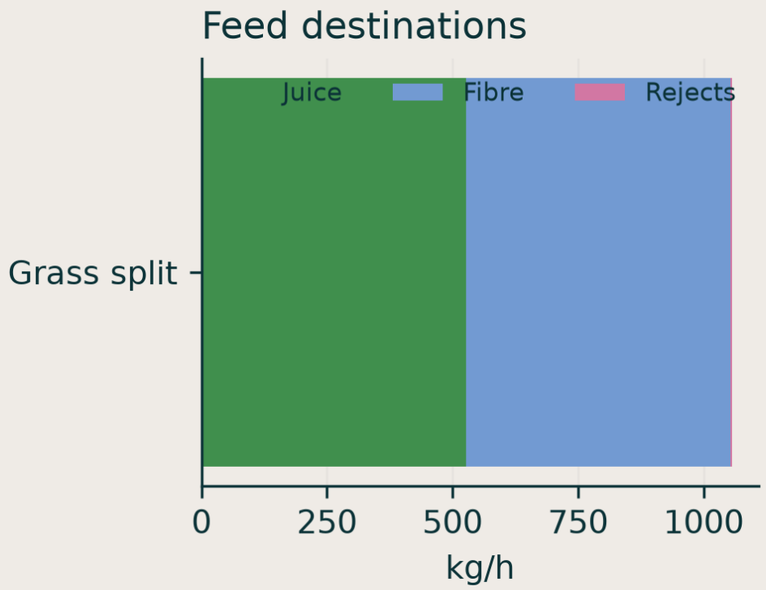
per kg wet grass

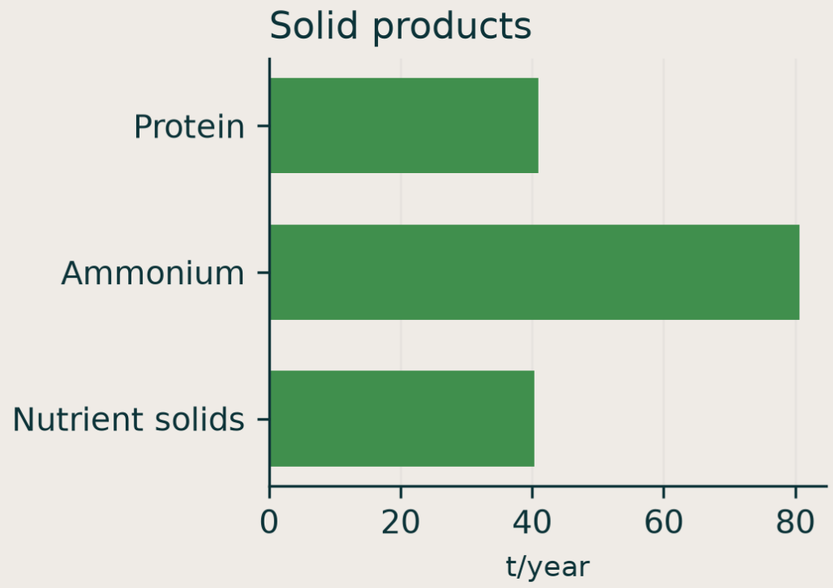
Mass-proportional grass split plus modeled downstream stream/product map

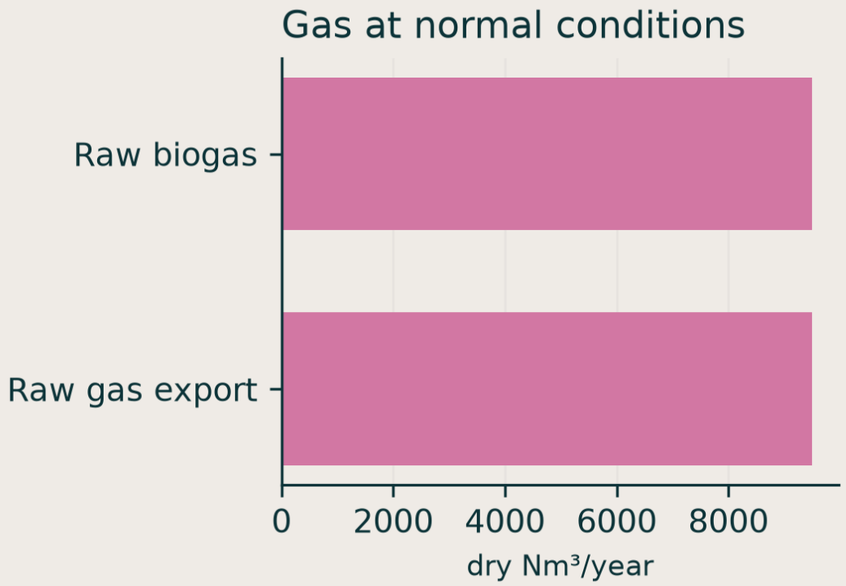
Scenario: all rank 3; e1229d22fd6a410383052c9dac19e6ee; 1458.6 kg/h; 8760 h/year
Units: kg/kg wet grass; dry Nm³/kg wet grass
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

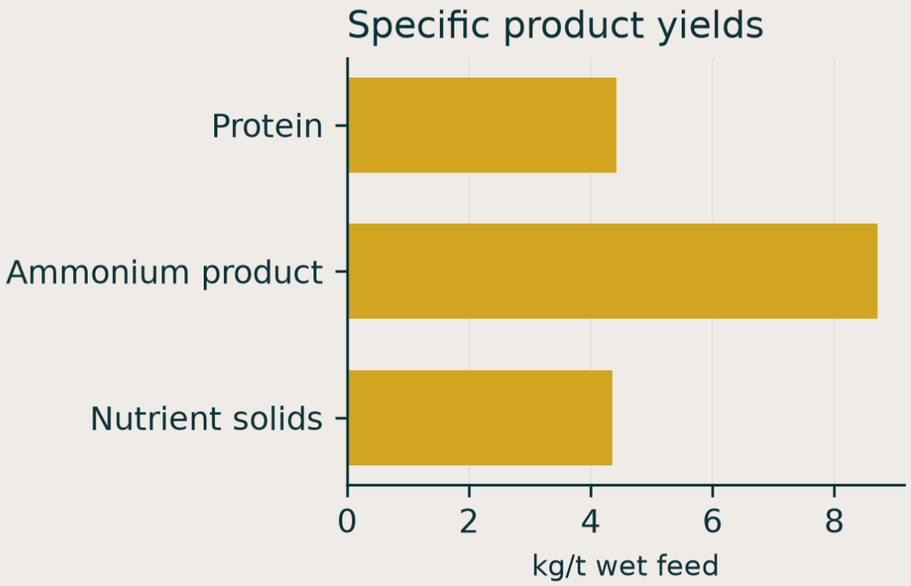
Material and products

Protein ✓ · Biogas ✓ · N ✓ · P X · K X







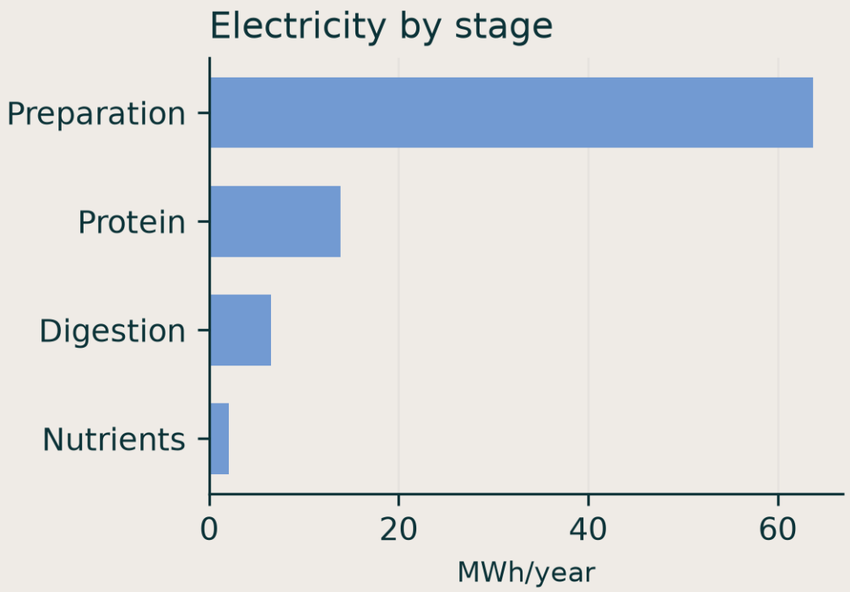


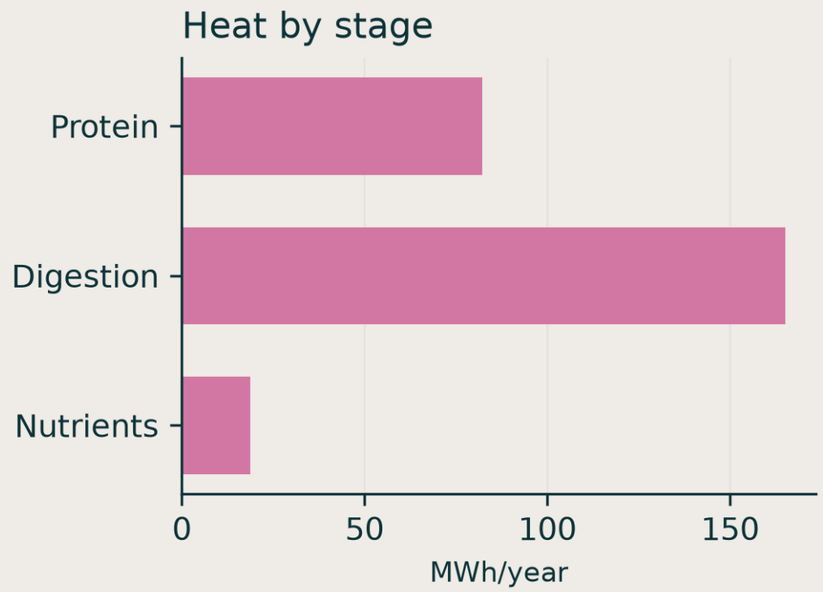
Front-end split, solid products, dry gas volumes and specific yields

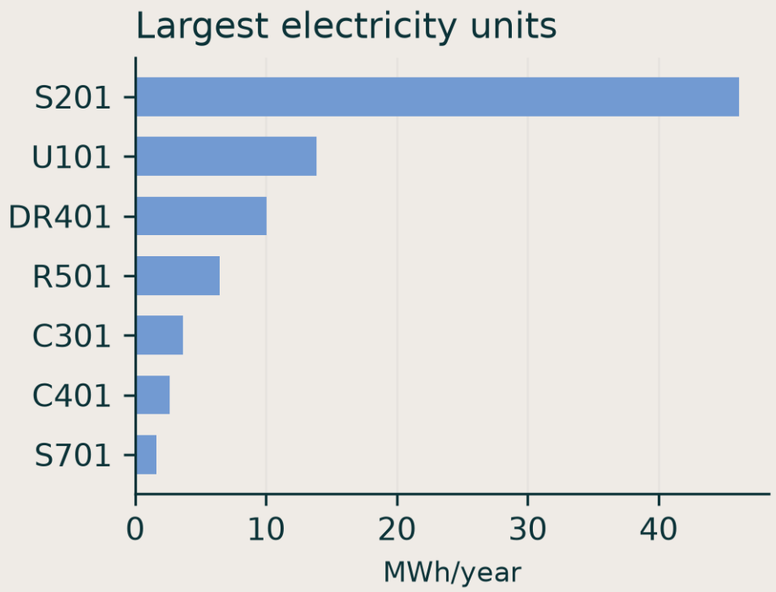
Scenario: all rank 3; e1229d22fd6a410383052c9dac19e6ee; 1458.6 kg/h; 8760 h/year
Units: kg/h; t/year; dry Nm³/year; kg/t feed
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

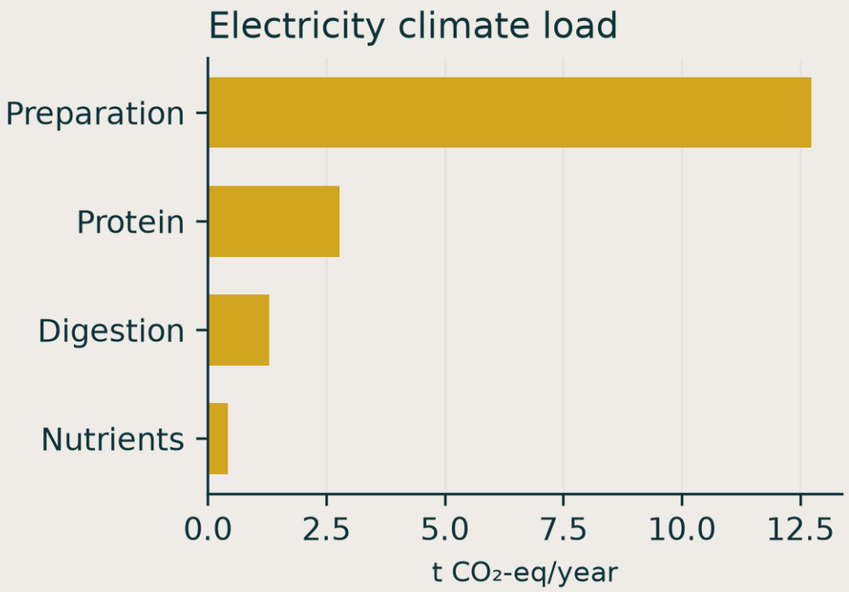
Energy and climate hotspots

Protein ✓ · Biogas ✓ · N ✓ · P X · K X



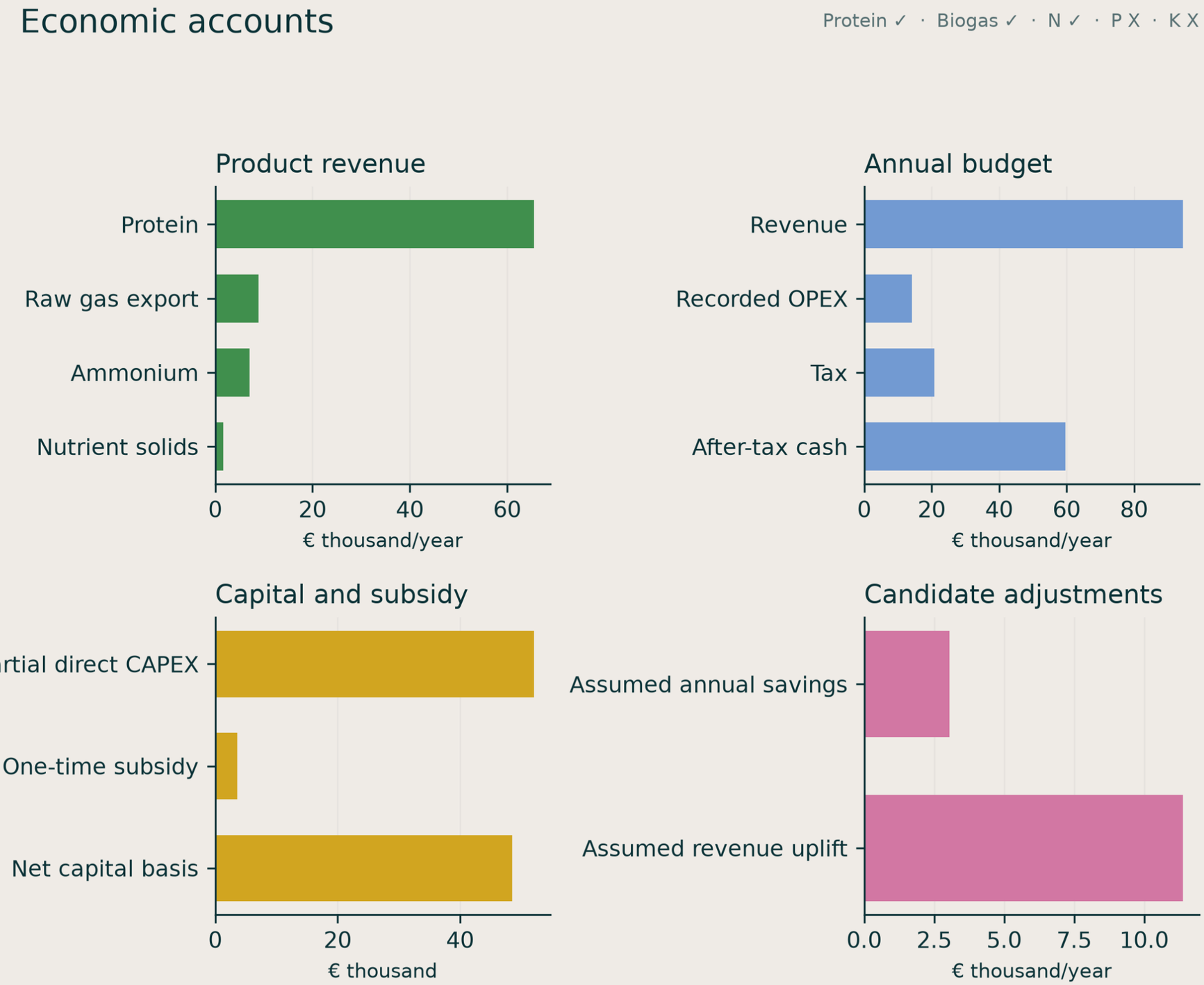






Stage and unit utility hotspots and included climate load

Scenario: all rank 3; e1229d22fd6a410383052c9dac19e6ee; 1458.6 kg/h; 8760 h/year
Units: MWh/year; t CO_2-eq/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

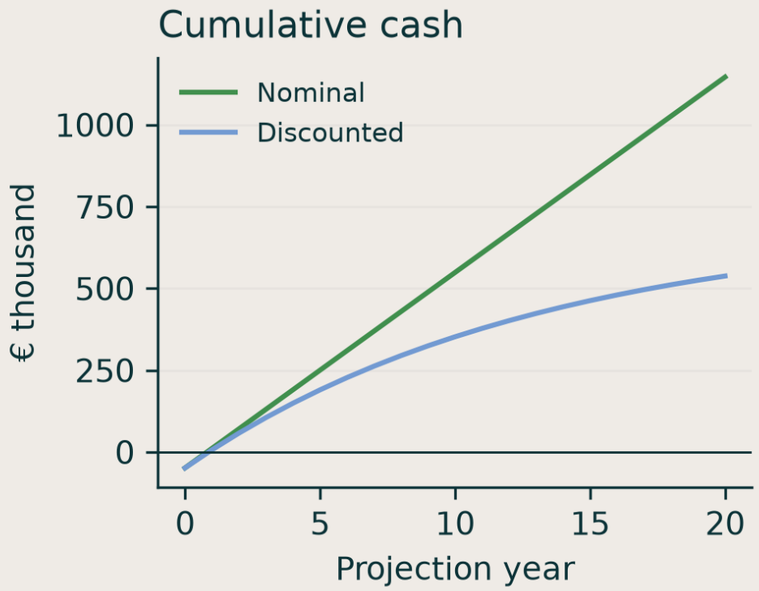


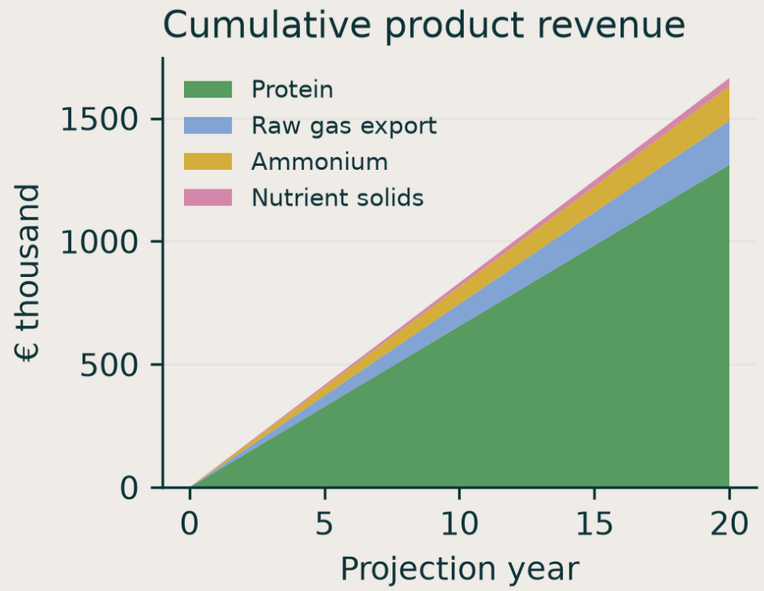
Product revenue, partial annual budget, capital subsidy and candidate overlays

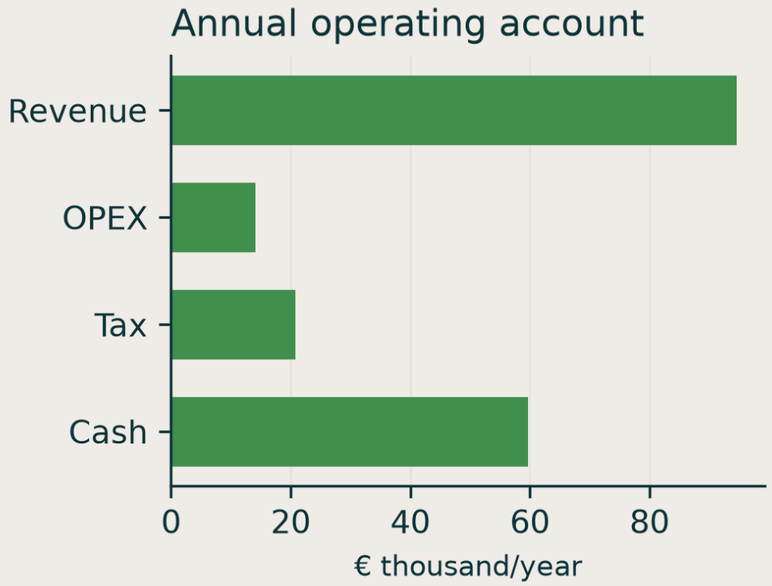
Scenario: all rank 3; e1229d22fd6a410383052c9dac19e6ee; 1458.6 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

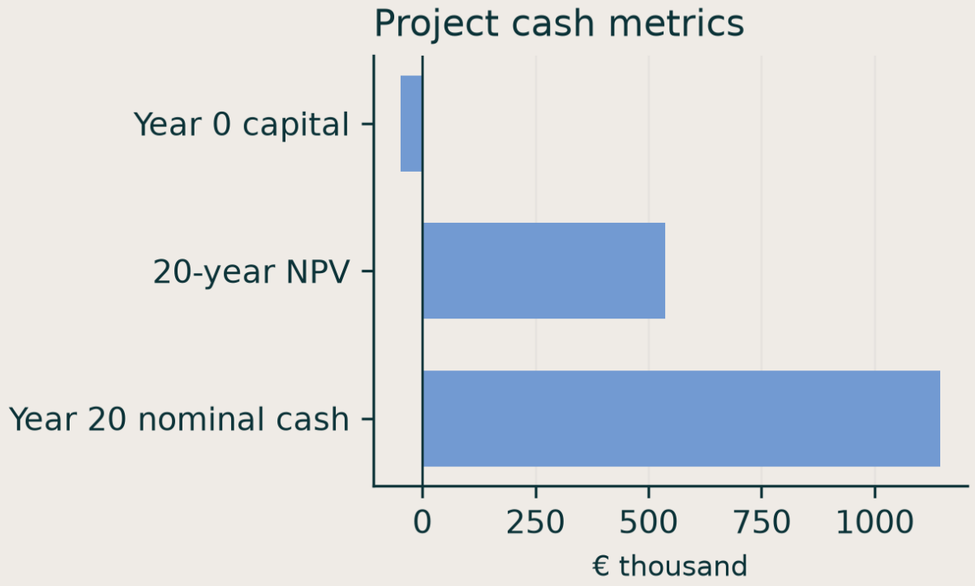
20-year financial projection

Protein ✓ · Biogas ✓ · N ✓ · P X · K X









Constant-account cash flow and cumulative product revenue over 20 years

Scenario: all rank 3; e1229d22fd6a410383052c9dac19e6ee; 1458.6 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Balance and evidence

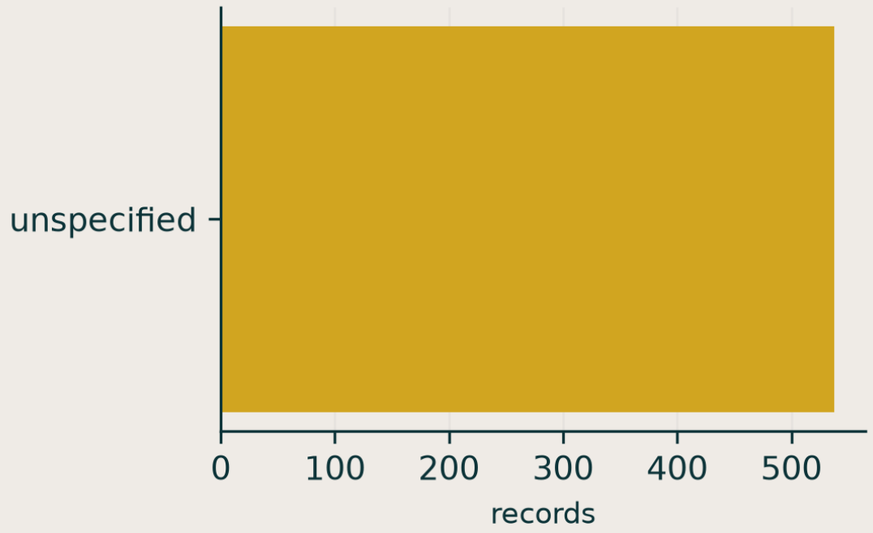
Protein ✓ · Biogas ✓ · N ✓ · P X · K X

Element closure

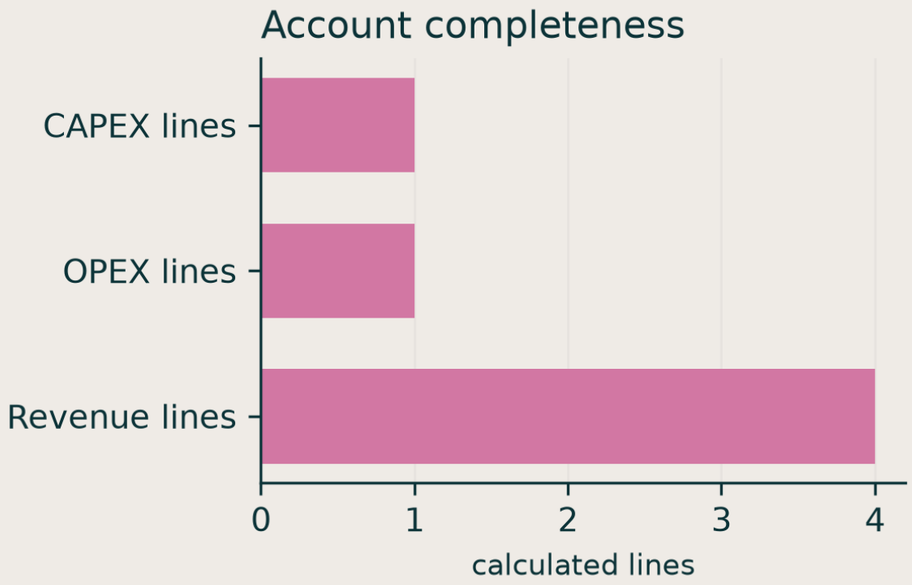
C	0 kg/year
N	0 kg/year
P	0 kg/year
K	0 kg/year

Numerical residuals

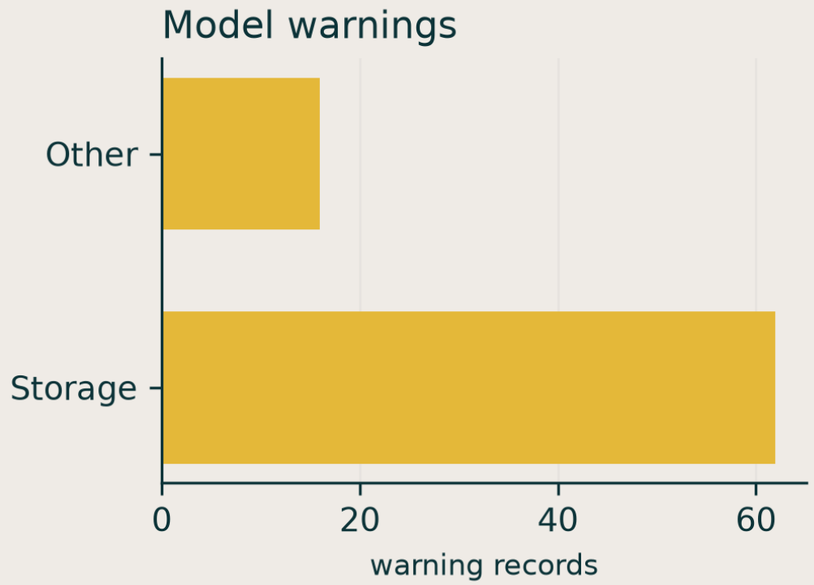
Parameter evidence



Account completeness



Model warnings



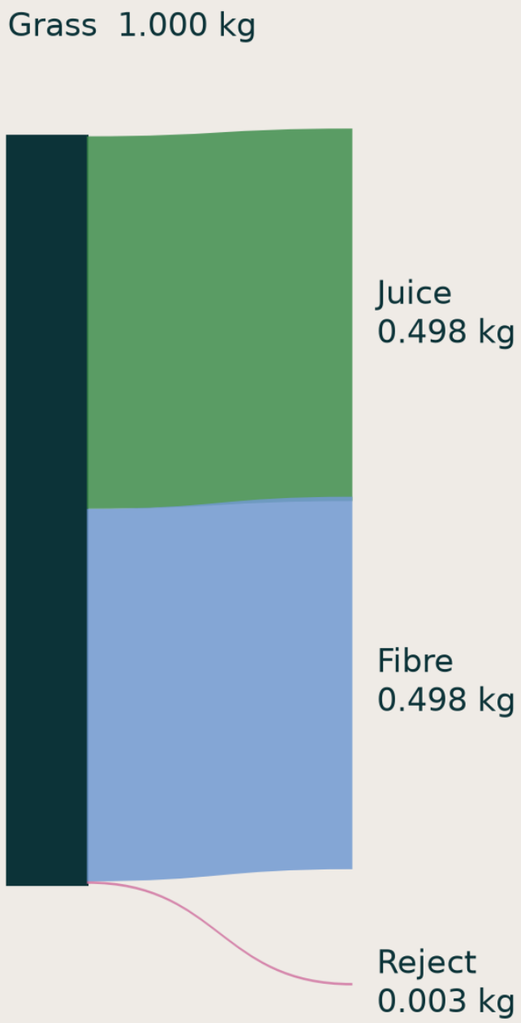
Numerical closure, evidence counts, calculated account lines and warnings

Scenario: all rank 3; e1229d22fd6a410383052c9dac19e6ee; 1458.6 kg/h; 8760 h/year
Units: kg/year; records
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

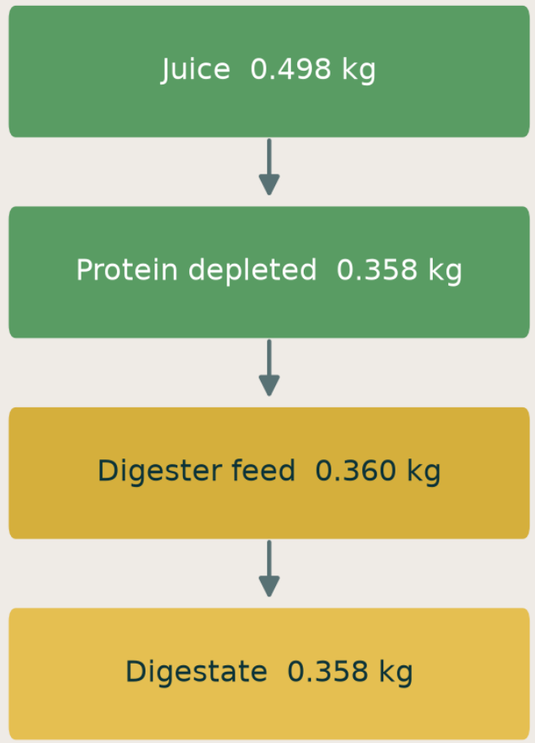
Grass to products

Protein ✓ · Biogas ✓ · N ✓ · P ✓ · K X

Grass split



Juice pathway



Selected intermediate streams

Recovered outputs

Raw gas 0.0010 Nm³

Raw export 0.0010 Nm³

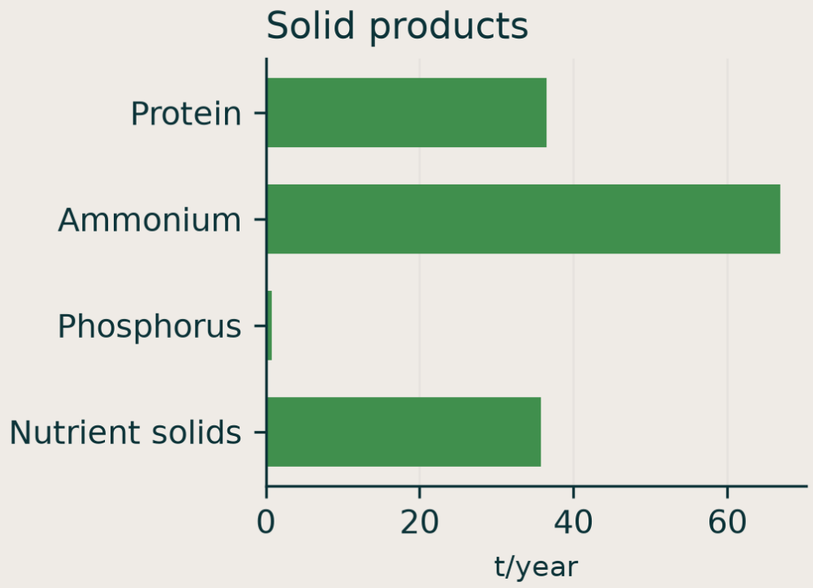
per kg wet grass

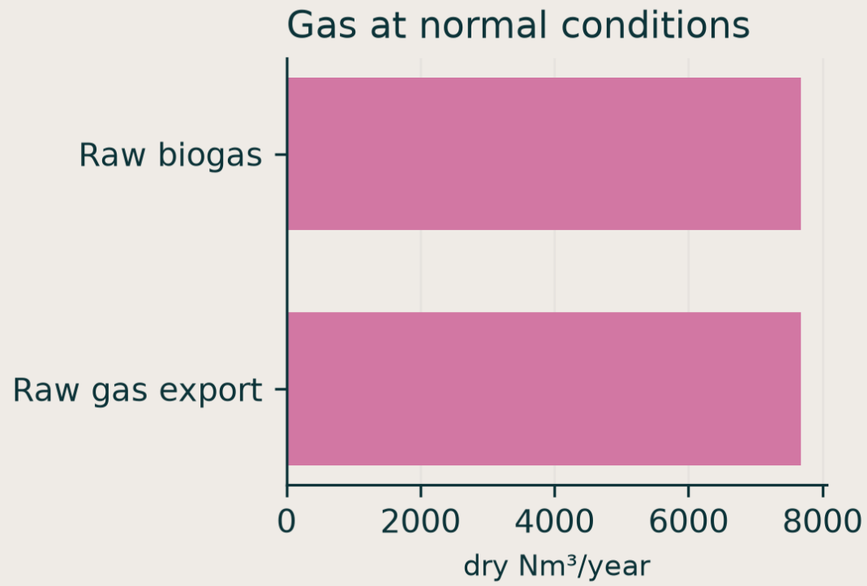
Mass-proportional grass split plus modeled downstream stream/product map

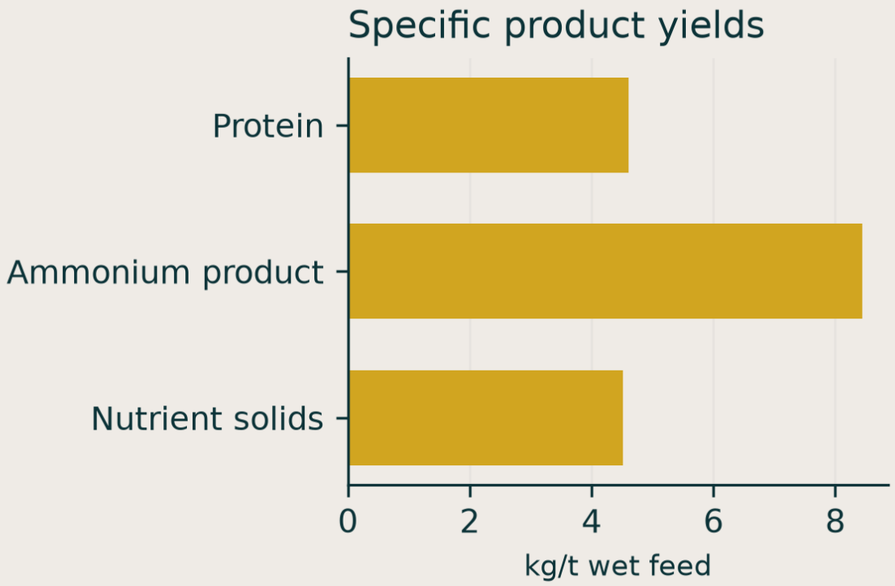
Scenario: all rank 4; 028ce1f6ff144c12be7f4f3cb4d82f98; 904.2 kg/h; 8760 h/year
Units: kg/kg wet grass; dry Nm³/kg wet grass
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Material and products

Protein ✓ · Biogas ✓ · N ✓ · P ✓ · K X





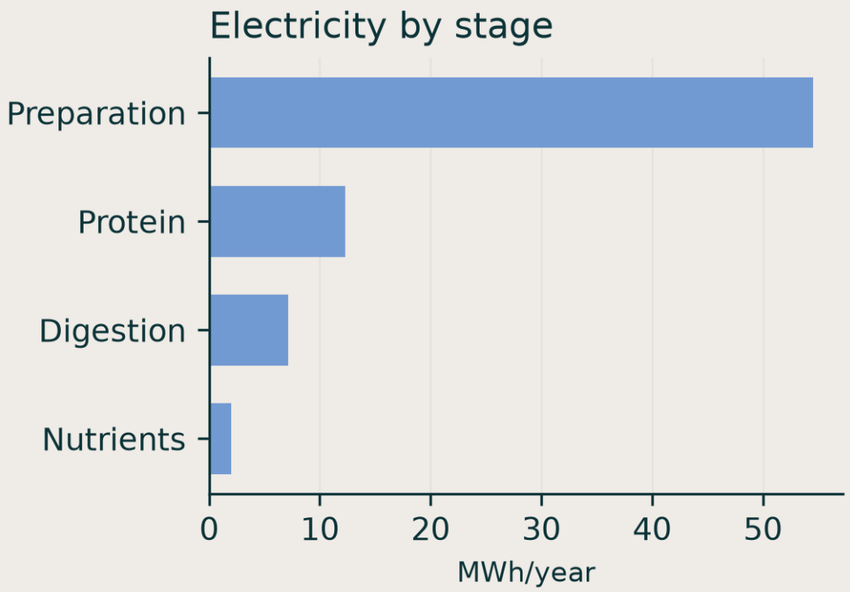


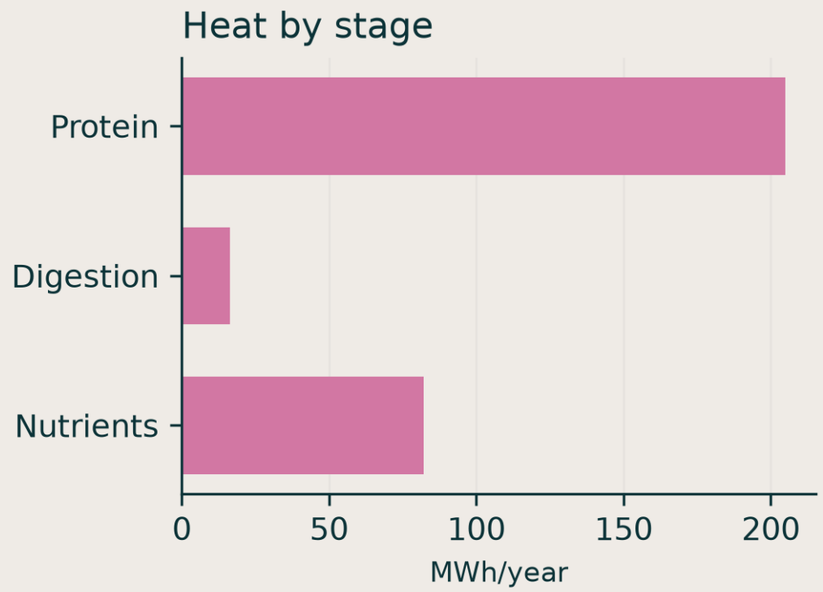
Front-end split, solid products, dry gas volumes and specific yields

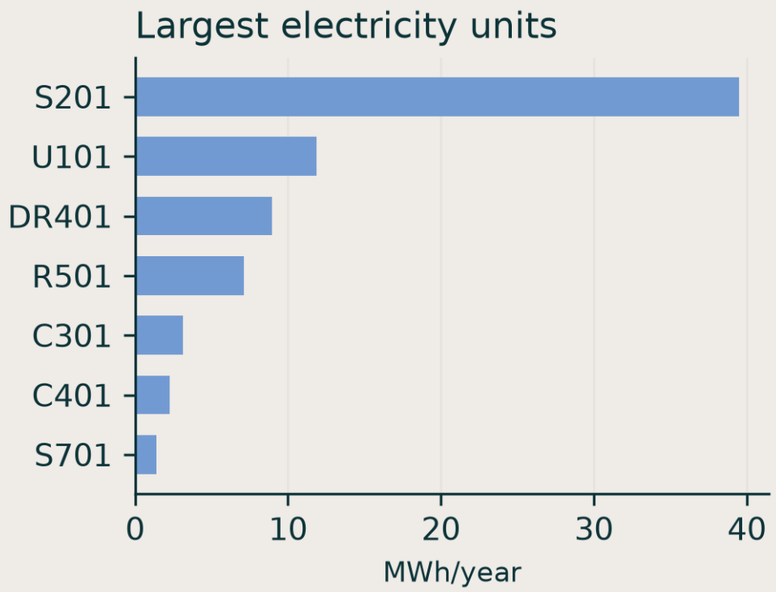
Scenario: all rank 4; 028ce1f6ff144c12be7f4f3cb4d82f98; 904.2 kg/h; 8760 h/year
Units: kg/h; t/year; dry Nm³/year; kg/t feed
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

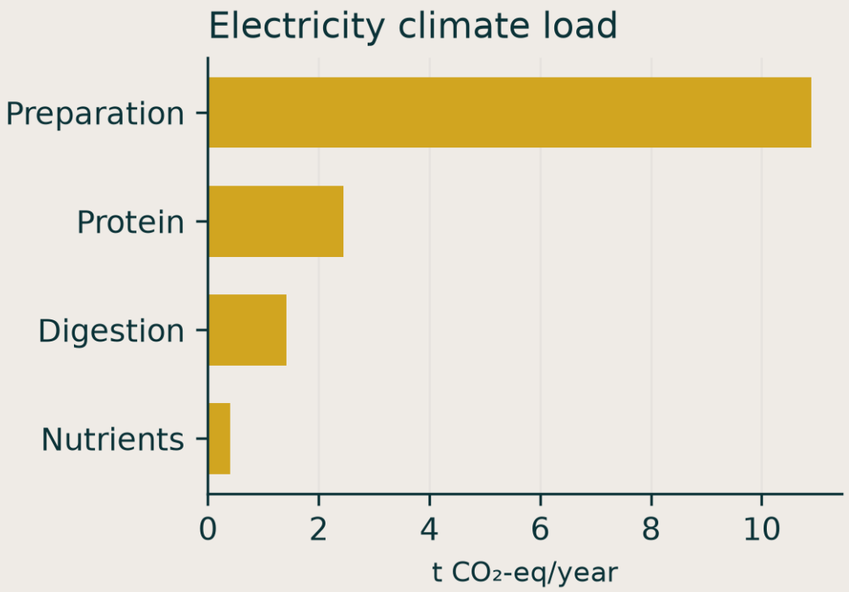
Energy and climate hotspots

Protein ✓ · Biogas ✓ · N ✓ · P ✓ · K X



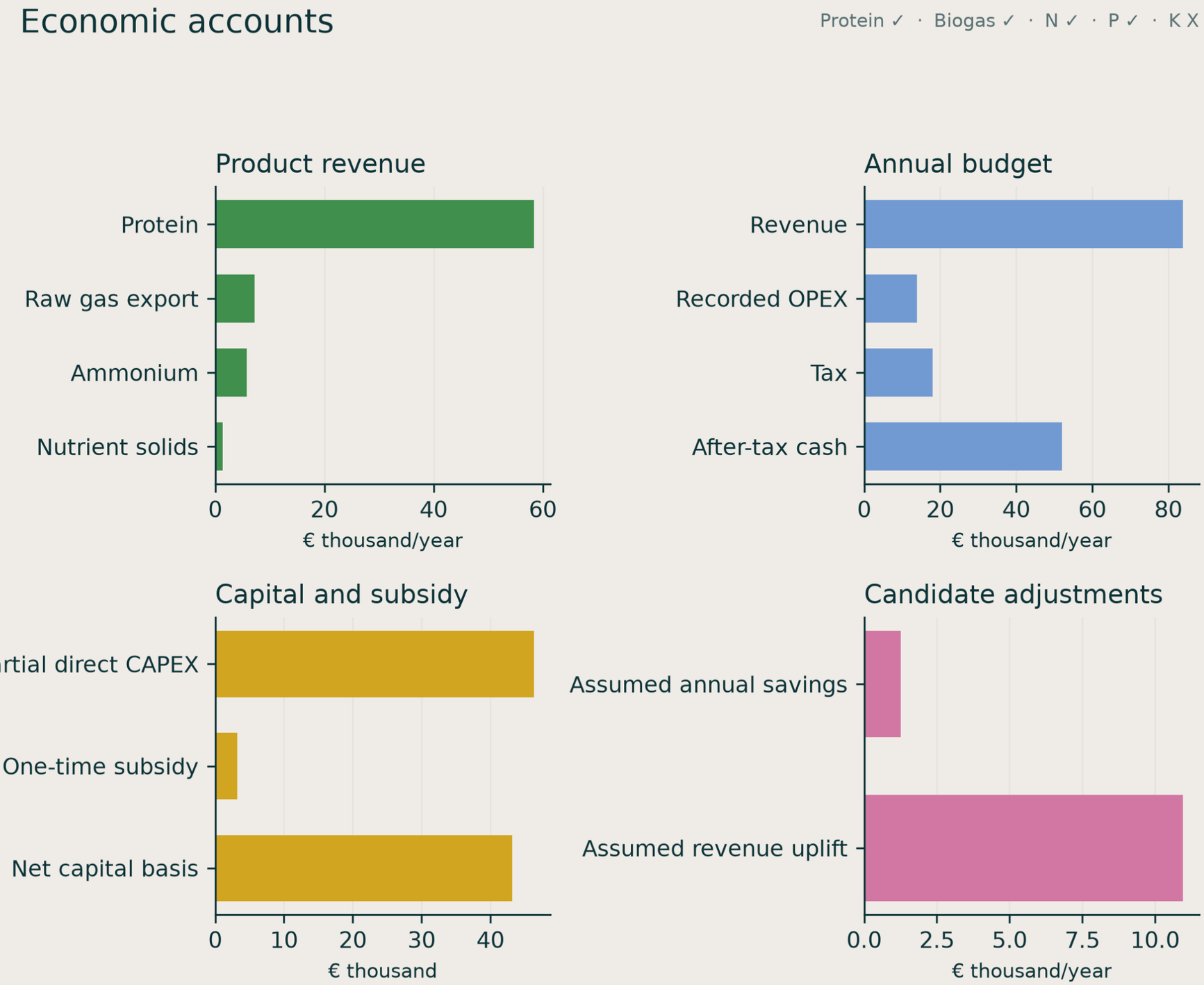






Stage and unit utility hotspots and included climate load

Scenario: all rank 4; 028ce1f6ff144c12be7f4f3cb4d82f98; 904.2 kg/h; 8760 h/year
Units: MWh/year; t CO_2-eq/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

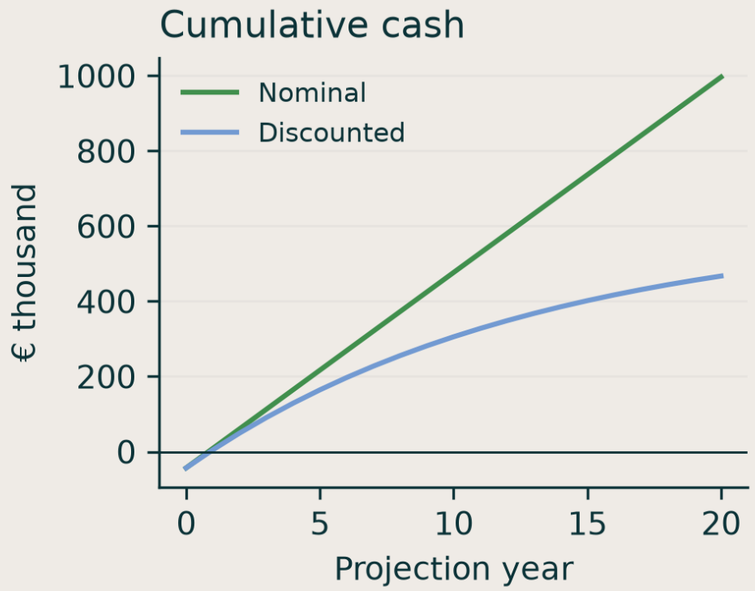


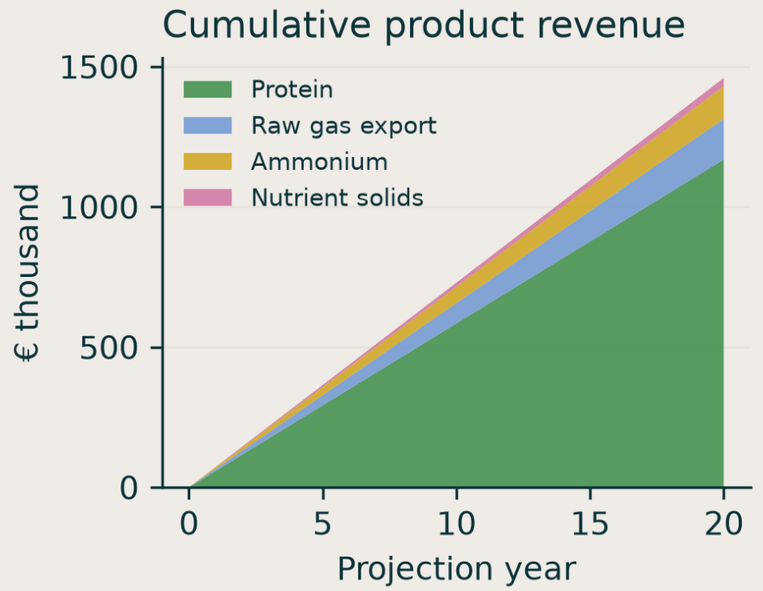
Product revenue, partial annual budget, capital subsidy and candidate overlays

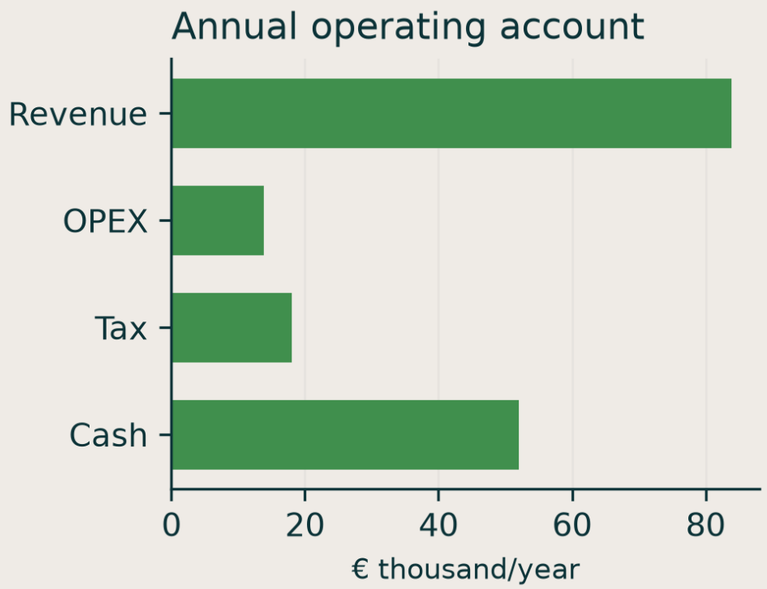
Scenario: all rank 4; 028ce1f6ff144c12be7f4f3cb4d82f98; 904.2 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

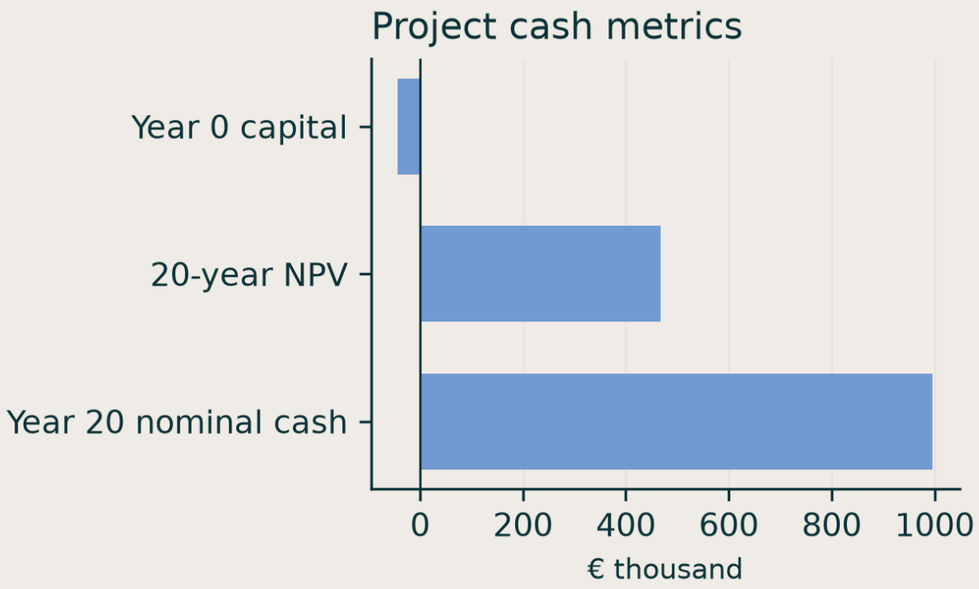
20-year financial projection

Protein ✓ · Biogas ✓ · N ✓ · P ✓ · K X









Constant-account cash flow and cumulative product revenue over 20 years

Scenario: all rank 4; 028ce1f6ff144c12be7f4f3cb4d82f98; 904.2 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Balance and evidence

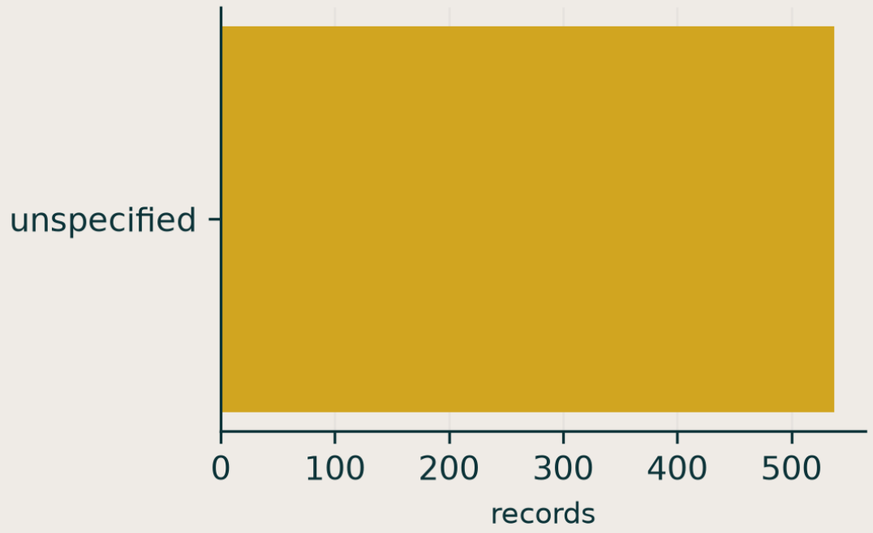
Protein ✓ · Biogas ✓ · N ✓ · P ✓ · K X

Element closure

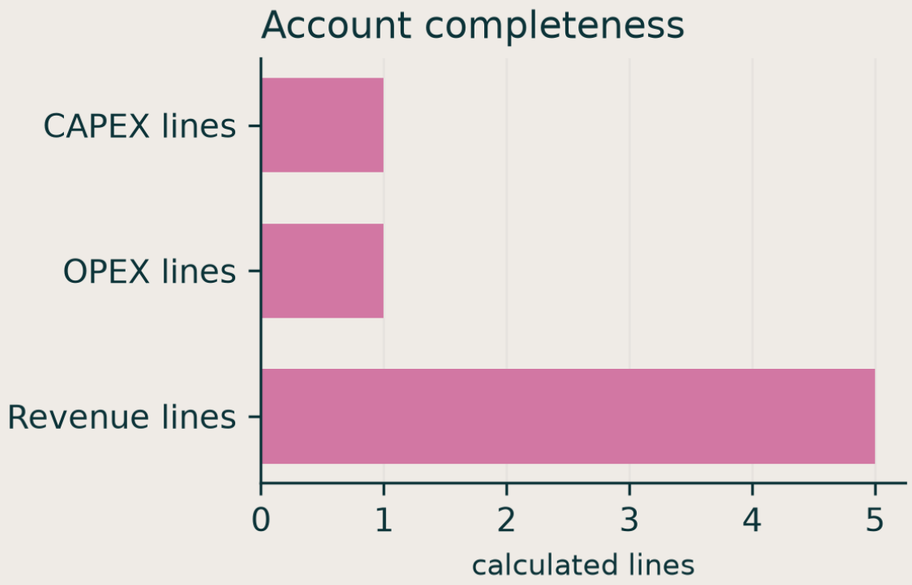
C	0 kg/year
N	0 kg/year
P	0 kg/year
K	0 kg/year

Numerical residuals

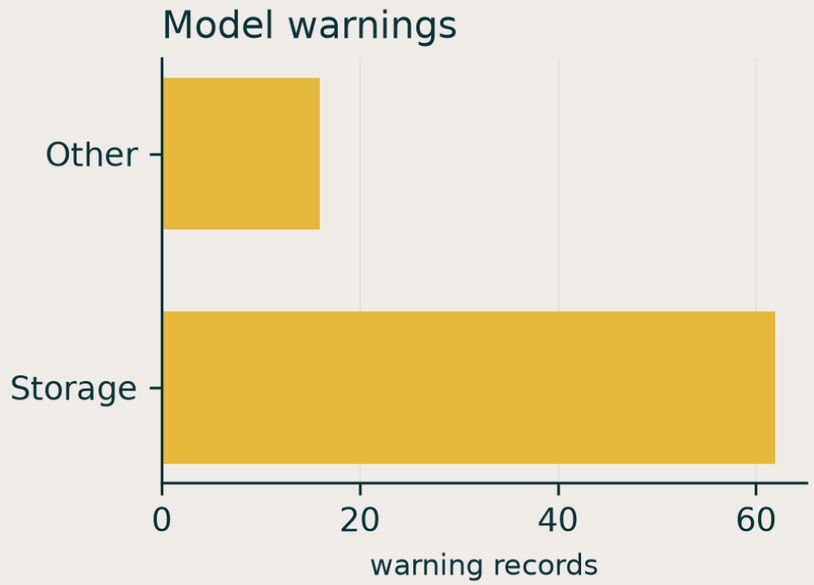
Parameter evidence



Account completeness



Model warnings



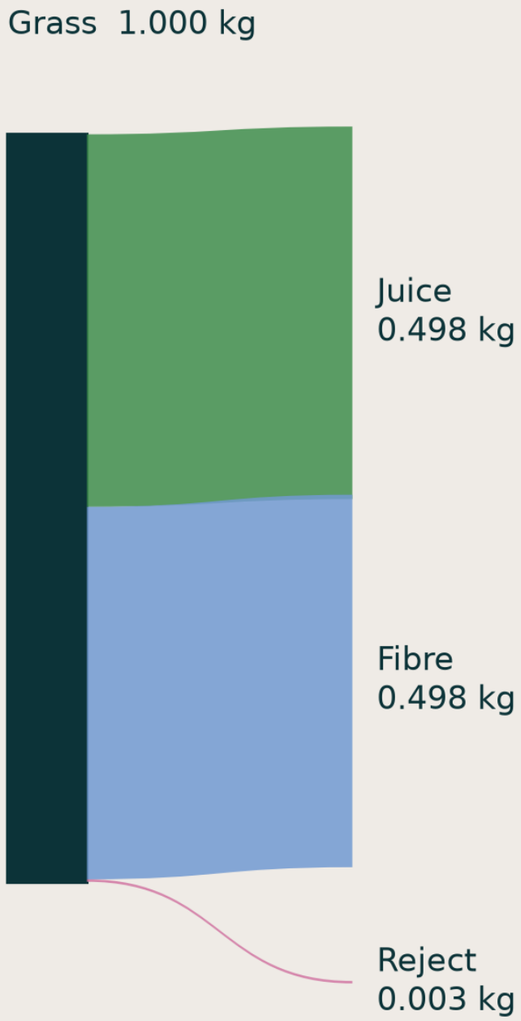
Numerical closure, evidence counts, calculated account lines and warnings

Scenario: all rank 4; 028ce1f6ff144c12be7f4f3cb4d82f98; 904.2 kg/h; 8760 h/year
Units: kg/year; records
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

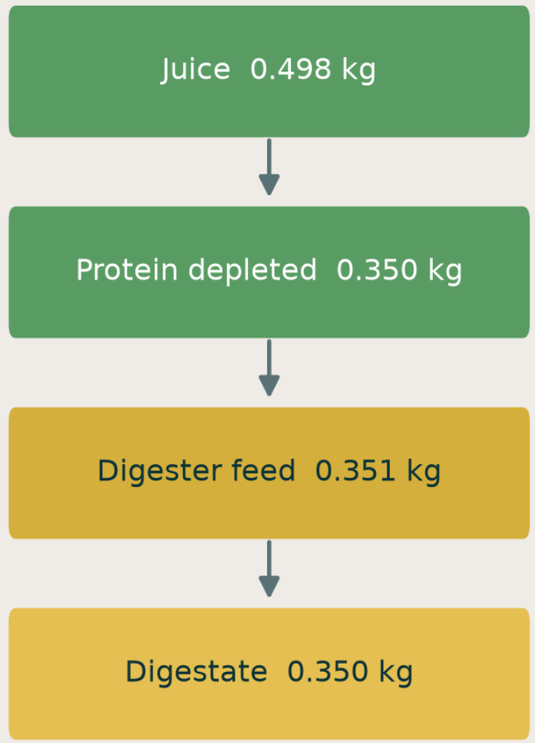
Grass to products

Protein ✓ · Biogas ✓ · N ✓ · P X · K X

Grass split



Juice pathway



Selected intermediate streams

Recovered outputs

Raw gas 0.0010 Nm³

Raw export 0.0010 Nm³

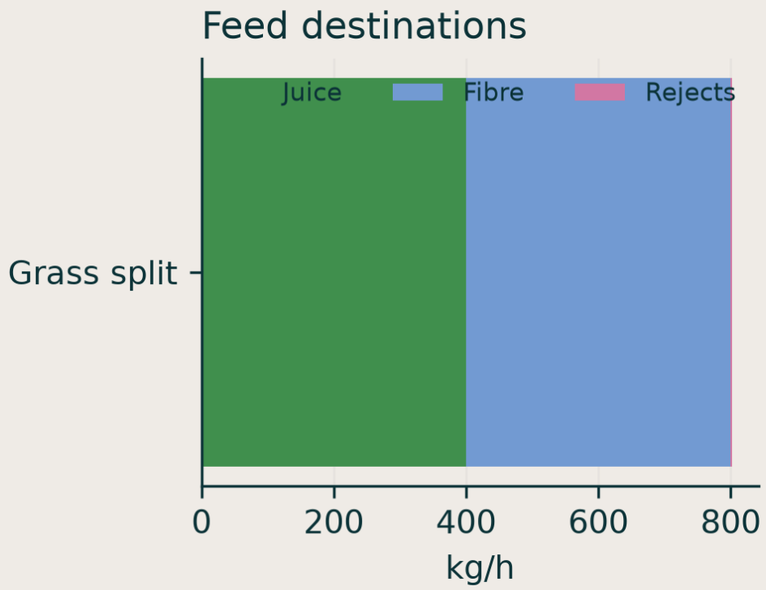
per kg wet grass

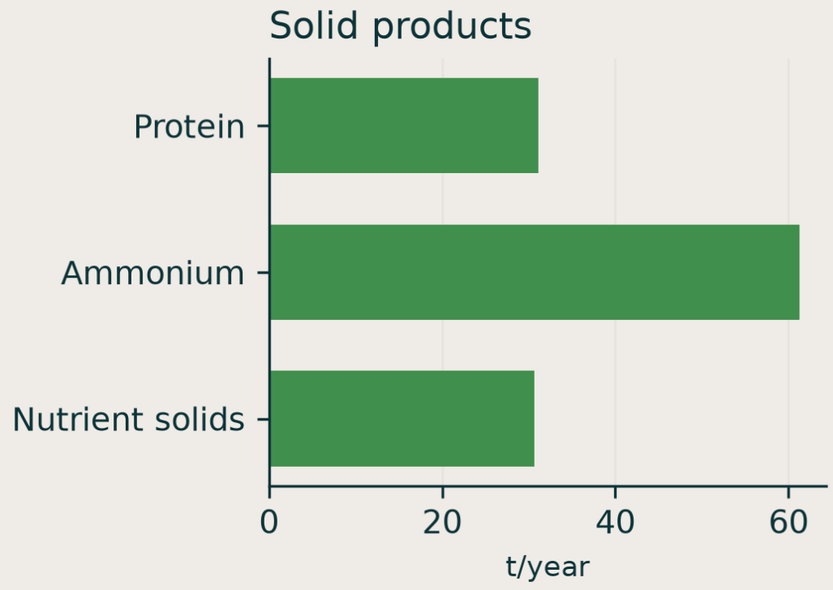
Mass-proportional grass split plus modeled downstream stream/product map

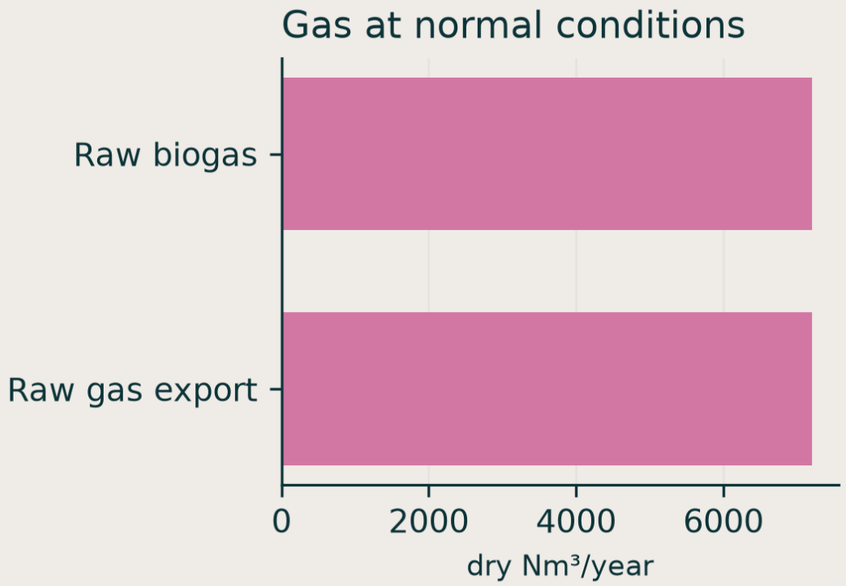
Scenario: all rank 5; 8088da95855d4bbaaa9599807db1ef50; 802.1 kg/h; 8760 h/year
Units: kg/kg wet grass; dry Nm³/kg wet grass
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

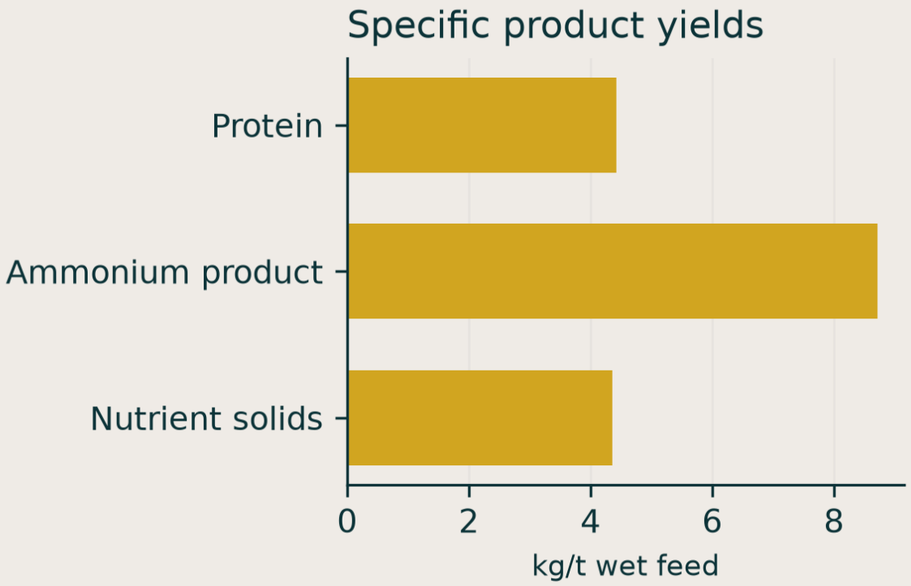
Material and products

Protein ✓ · Biogas ✓ · N ✓ · P X · K X







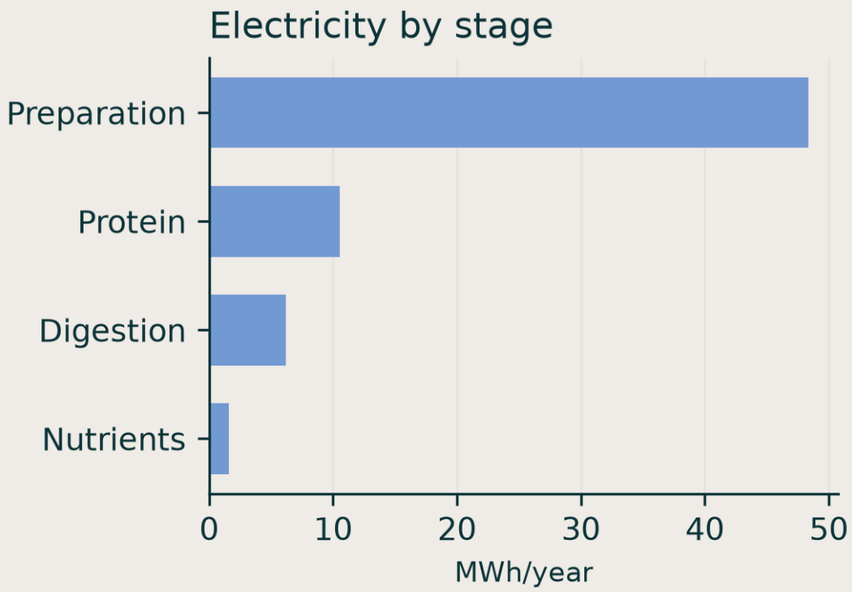


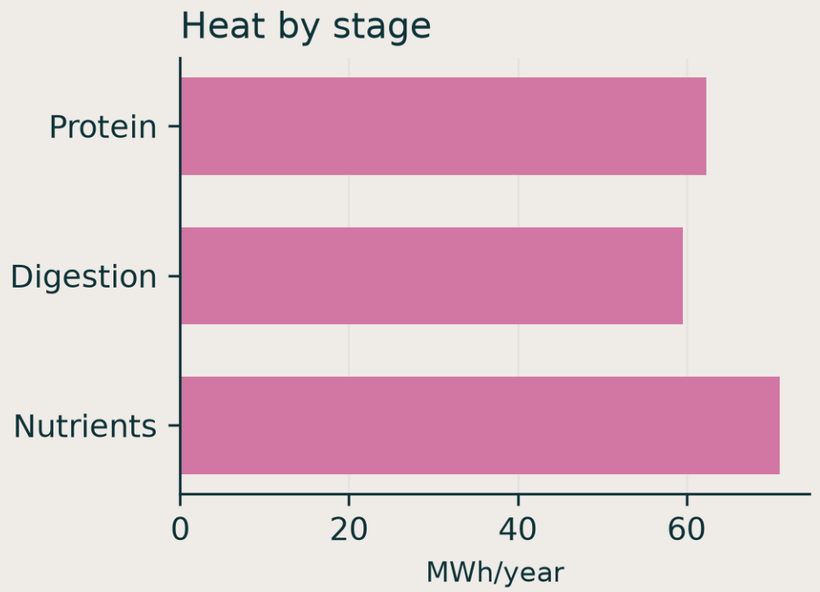
Front-end split, solid products, dry gas volumes and specific yields

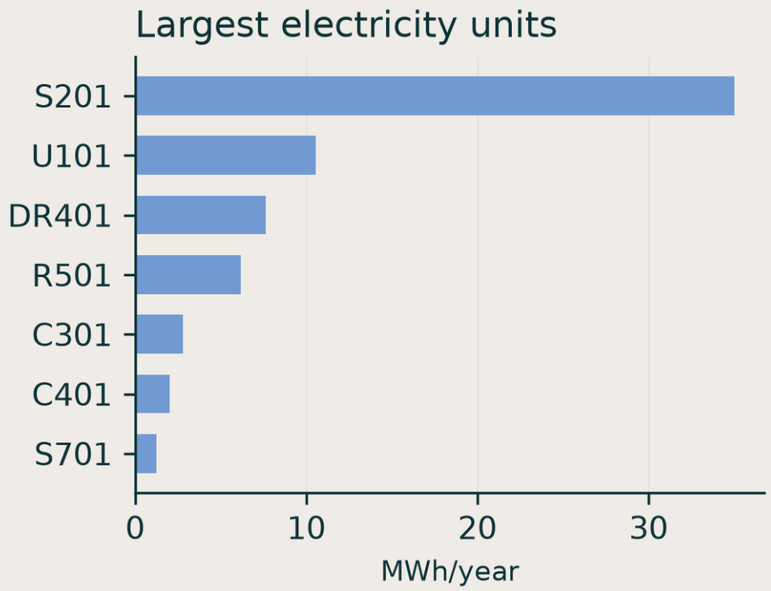
Scenario: all rank 5; 8088da95855d4bbaaa9599807db1ef50; 802.1 kg/h; 8760 h/year
Units: kg/h; t/year; dry Nm³/year; kg/t feed
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

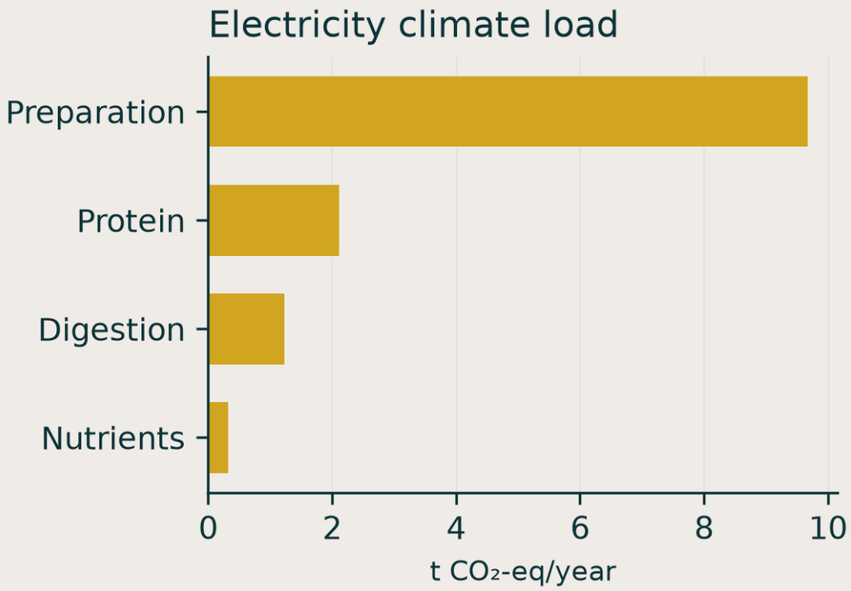
Energy and climate hotspots

Protein ✓ · Biogas ✓ · N ✓ · P X · K X



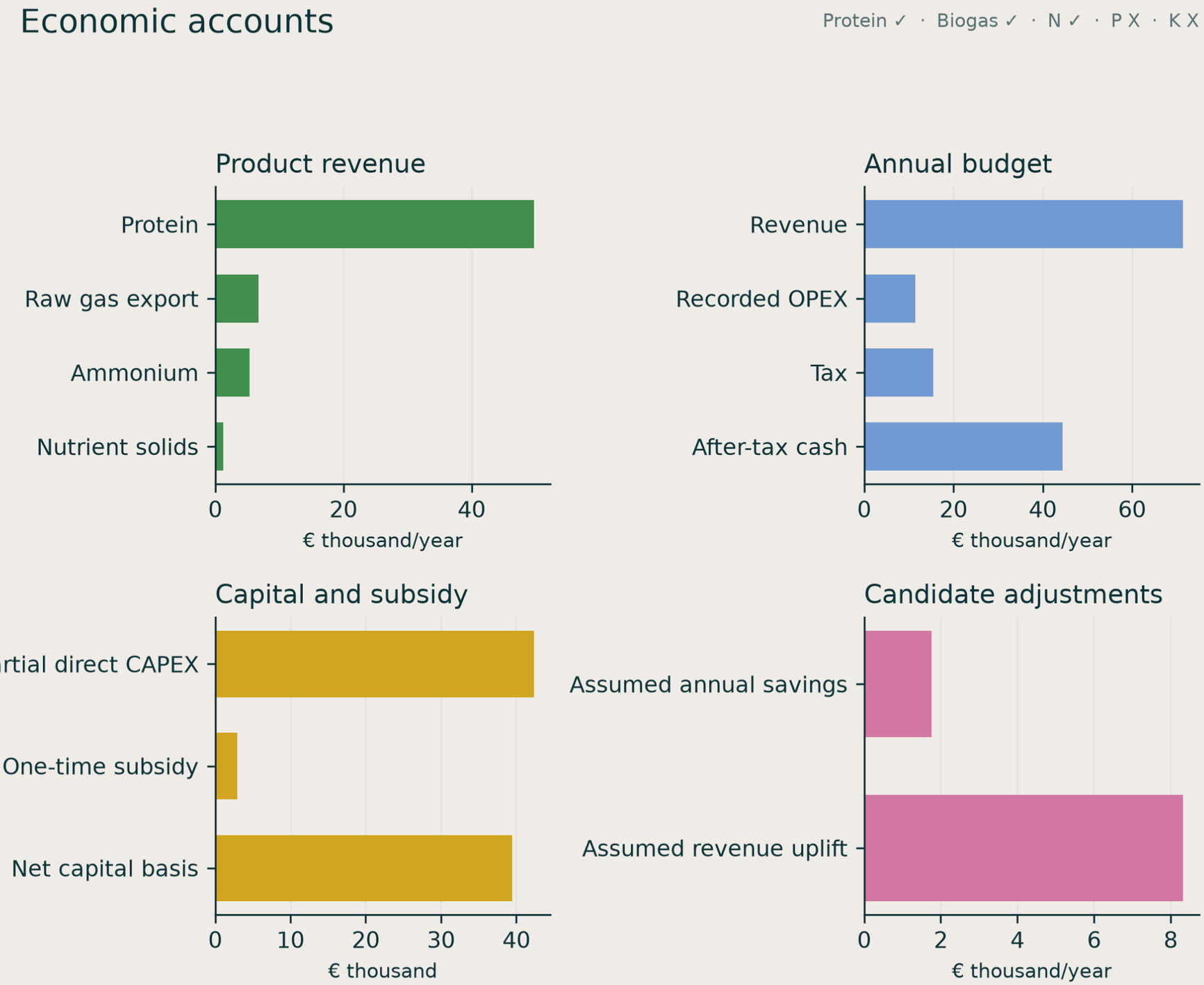






Stage and unit utility hotspots and included climate load

Scenario: all rank 5; 8088da95855d4bbaaa9599807db1ef50; 802.1 kg/h; 8760 h/year
Units: MWh/year; t CO_2-eq/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

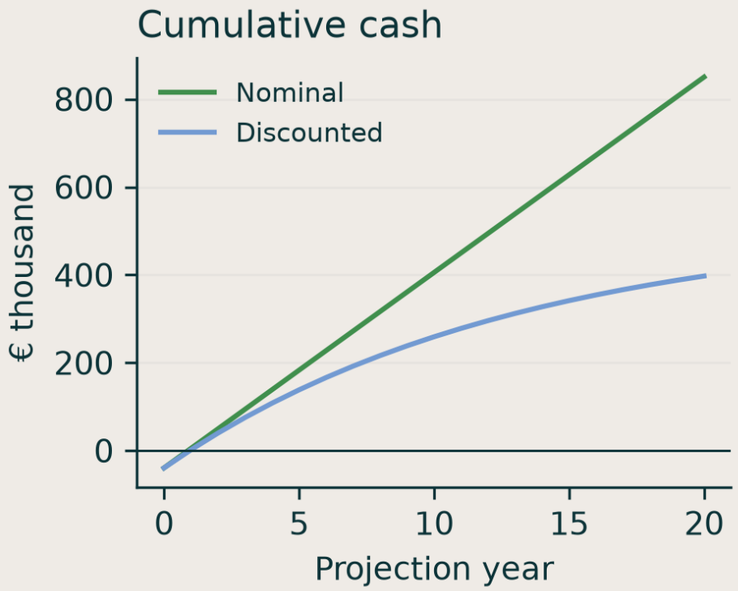


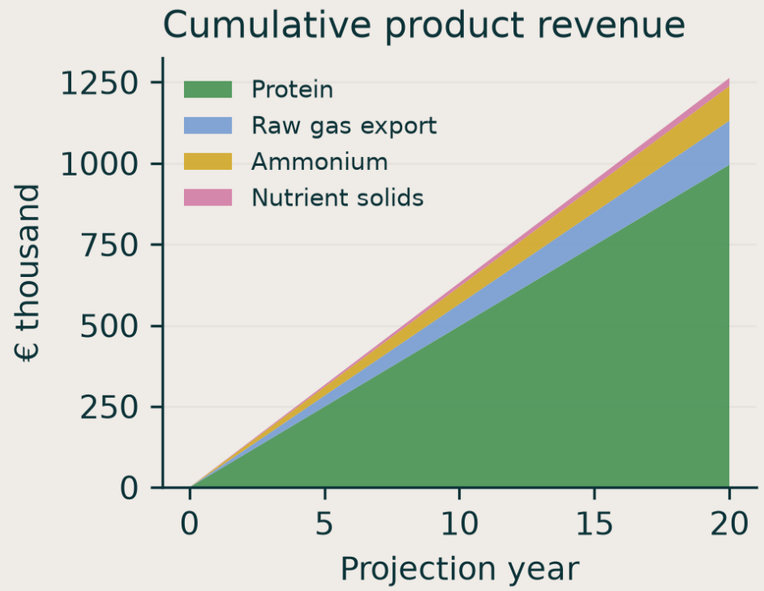
Product revenue, partial annual budget, capital subsidy and candidate overlays

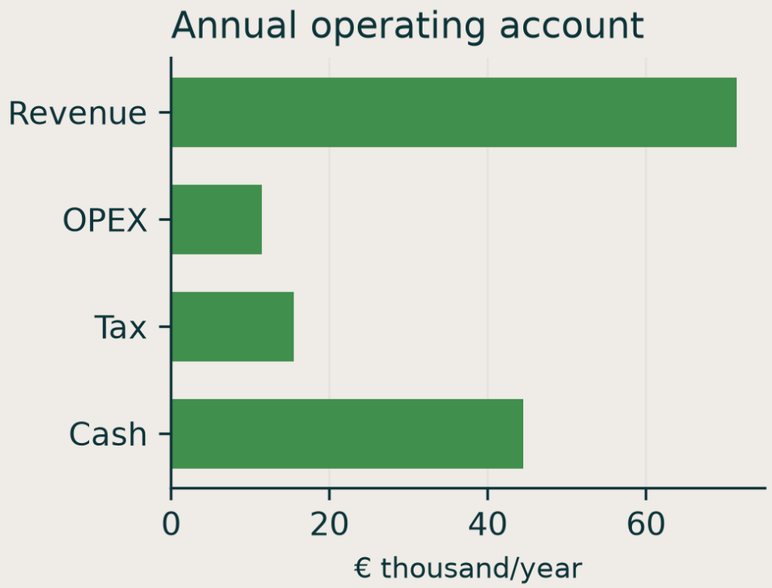
Scenario: all rank 5; 8088da95855d4bbaaa9599807db1ef50; 802.1 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

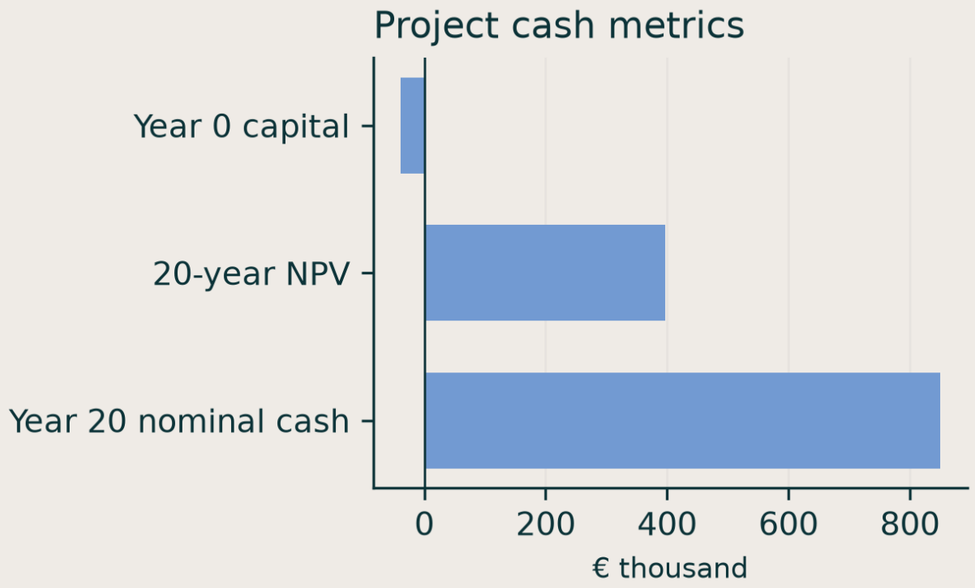
20-year financial projection

Protein ✓ · Biogas ✓ · N ✓ · P X · K X









Constant-account cash flow and cumulative product revenue over 20 years

Scenario: all rank 5; 8088da95855d4bbaaa9599807db1ef50; 802.1 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Balance and evidence

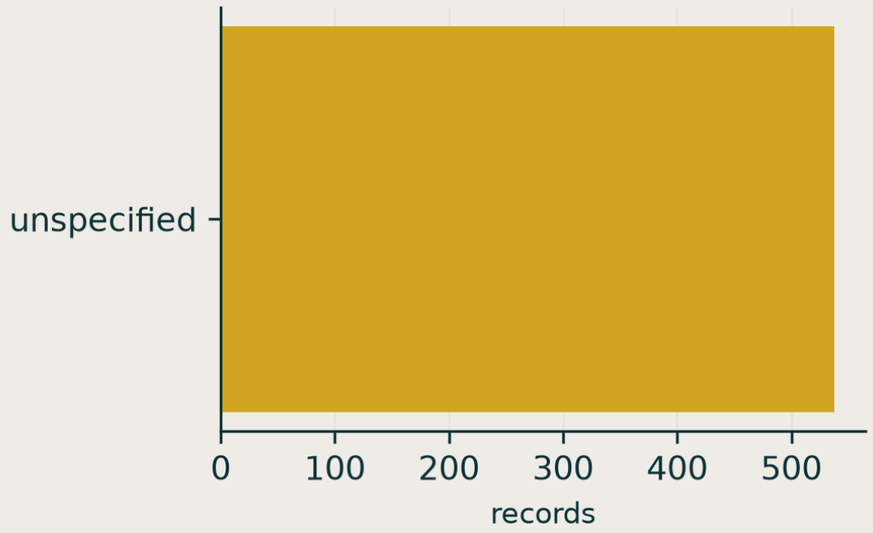
Protein ✓ · Biogas ✓ · N ✓ · P X · K X

Element closure

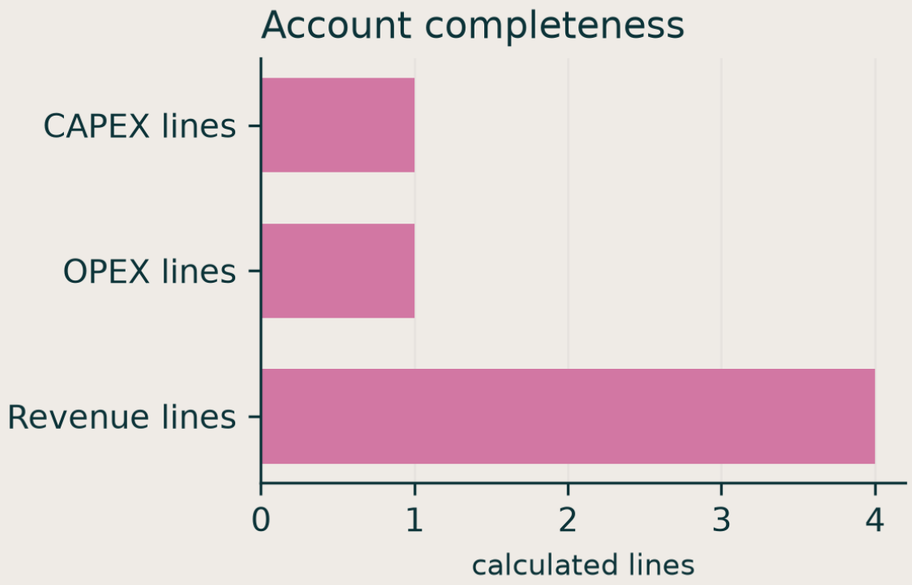
C	0 kg/year
N	0 kg/year
P	0 kg/year
K	0 kg/year

Numerical residuals

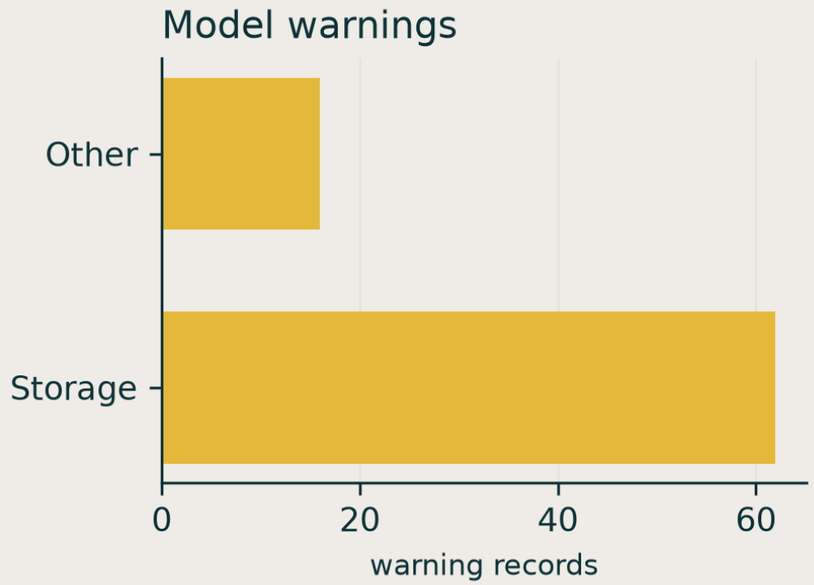
Parameter evidence



Account completeness



Model warnings



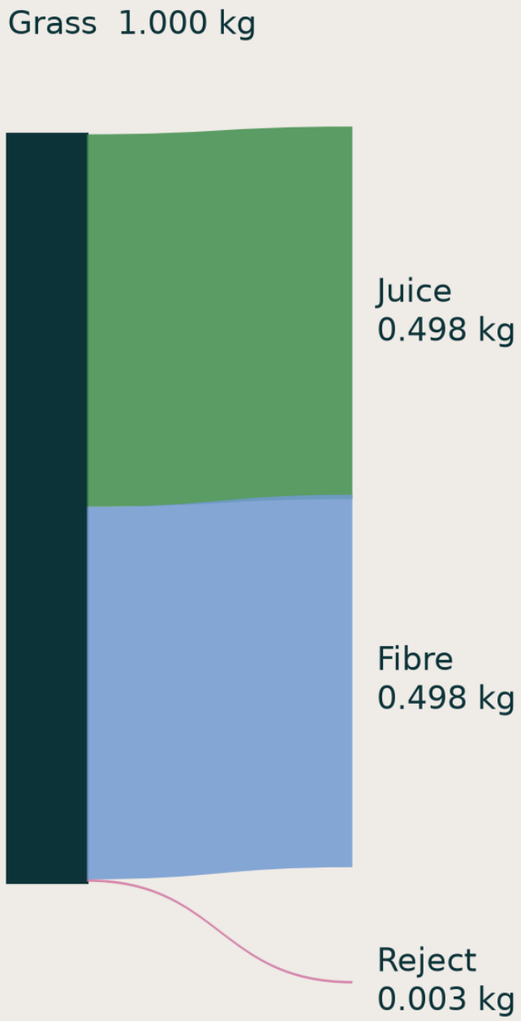
Numerical closure, evidence counts, calculated account lines and warnings

Scenario: all rank 5; 8088da95855d4bbaaa9599807db1ef50; 802.1 kg/h; 8760 h/year
Units: kg/year; records
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

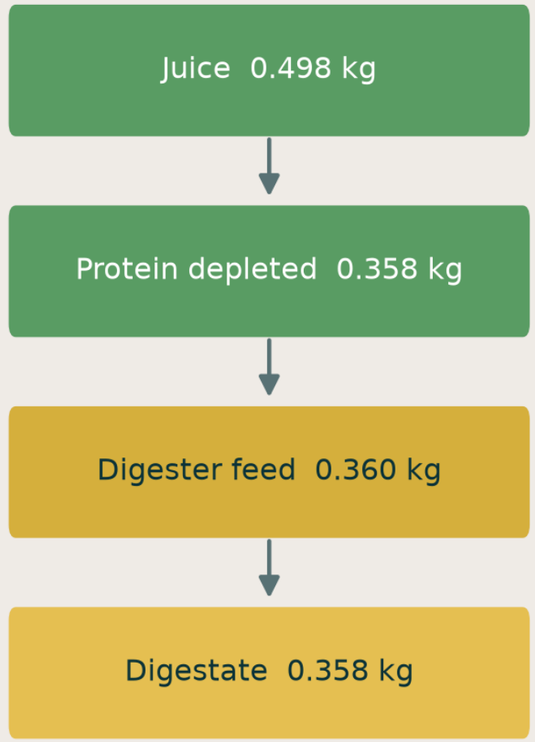
Grass to products

Protein ✓ · Biogas ✓ · N ✓ · P X · K X

Grass split



Juice pathway



Selected intermediate streams

Recovered outputs

Protein 0.0046 kg

N product 0.0084 kg

Nutrient solids 0.0045 kg

Raw gas 0.0010 Nm³

Upgraded 0.0007 Nm³

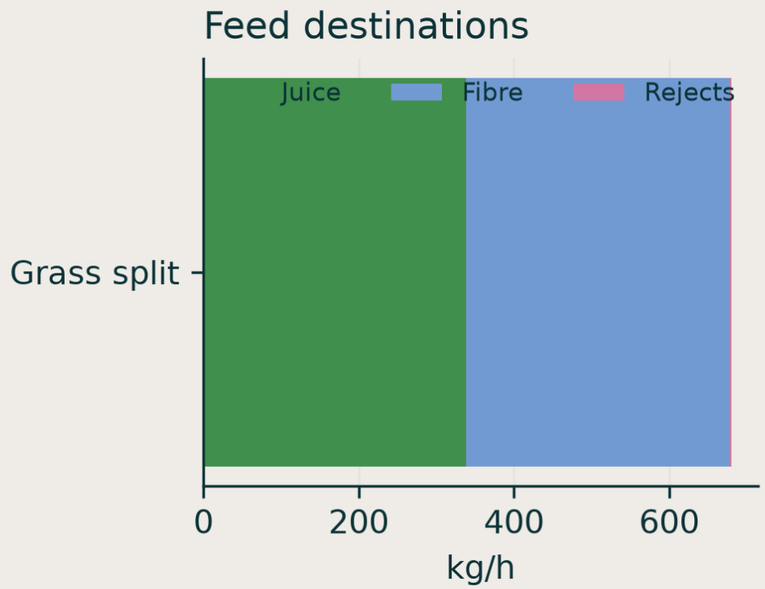
per kg wet grass

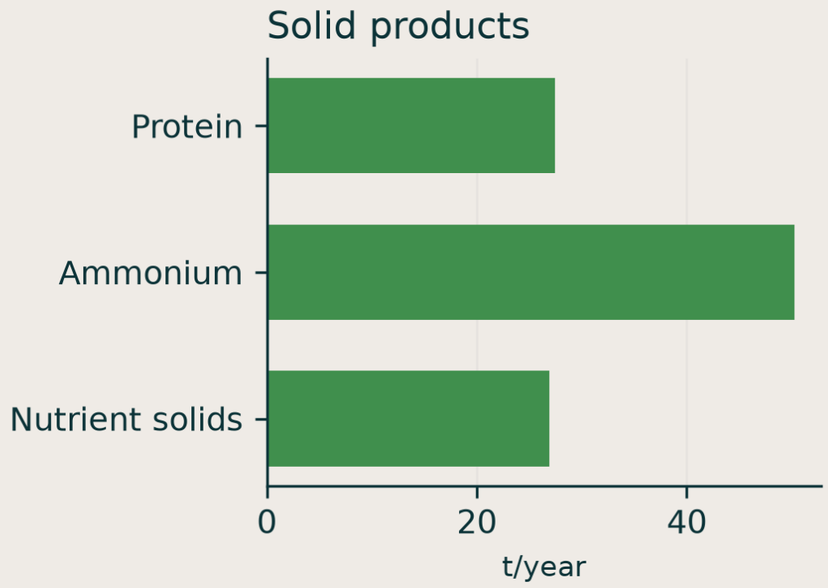
Mass-proportional grass split plus modeled downstream stream/product map

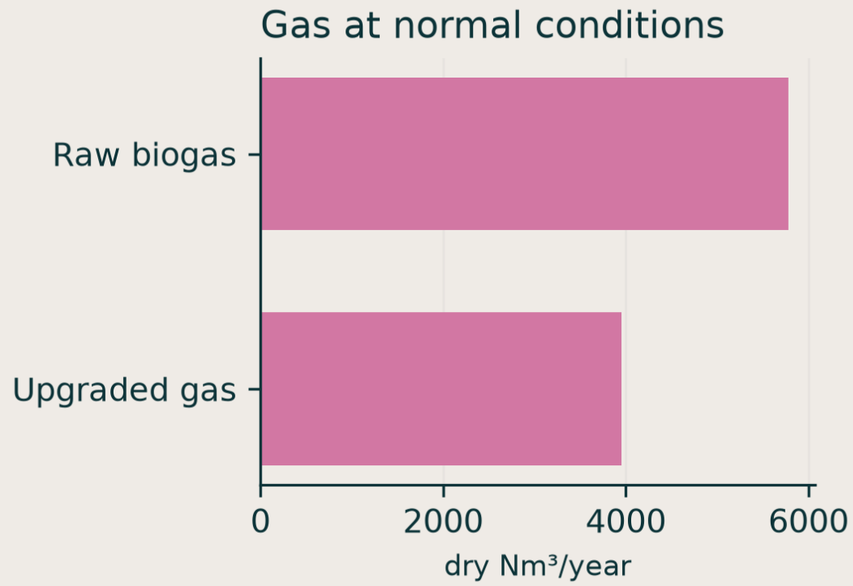
Scenario: all rank 6; 520944a3fae14541a889a13bda3bbd98; 679.9 kg/h; 8760 h/year
Units: kg/kg wet grass; dry Nm³/kg wet grass
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

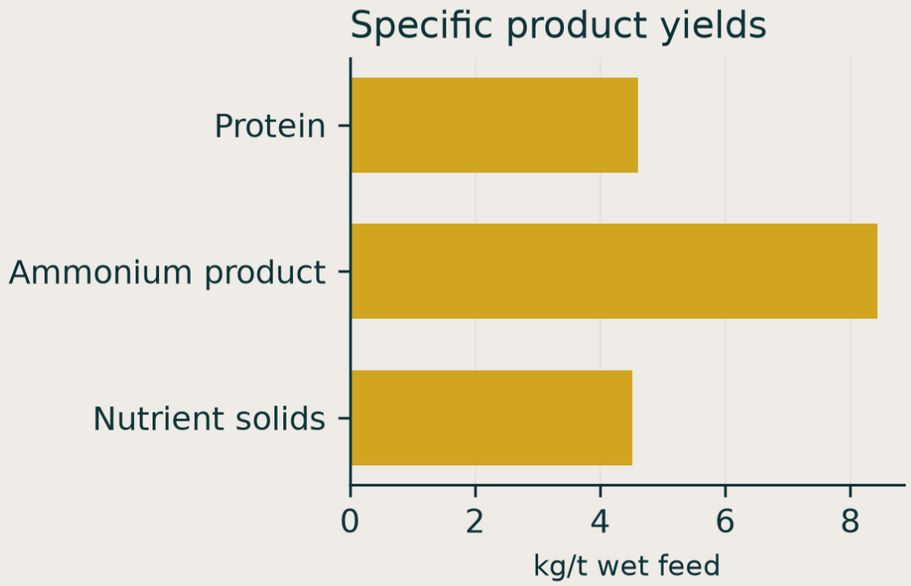
Material and products

Protein ✓ · Biogas ✓ · N ✓ · P X · K X







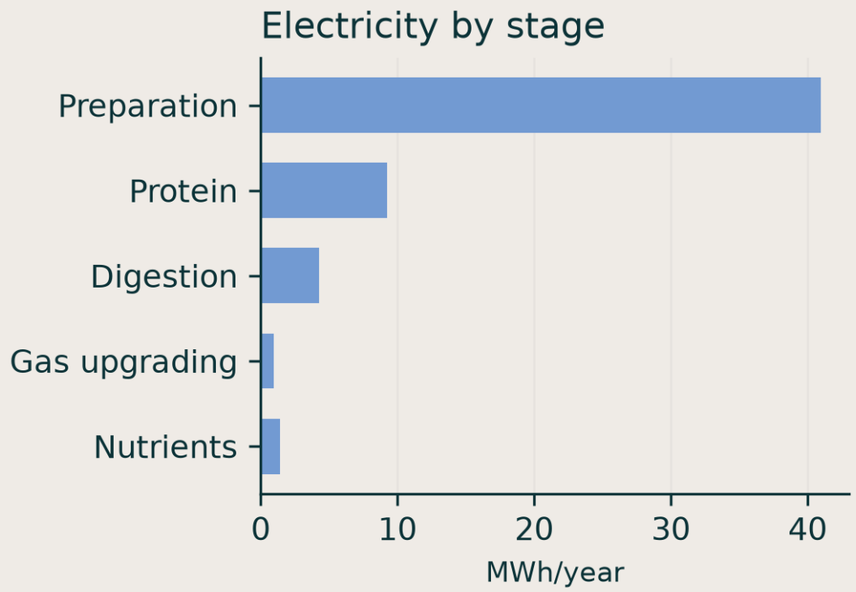


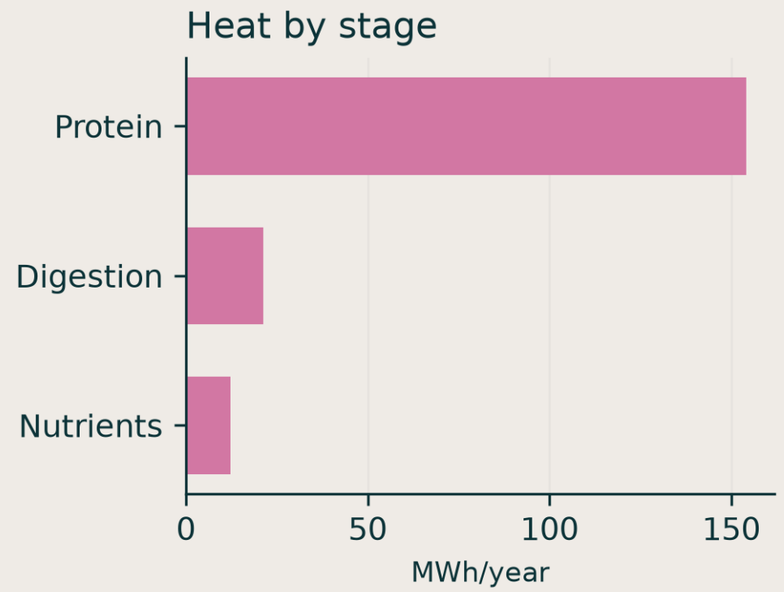
Front-end split, solid products, dry gas volumes and specific yields

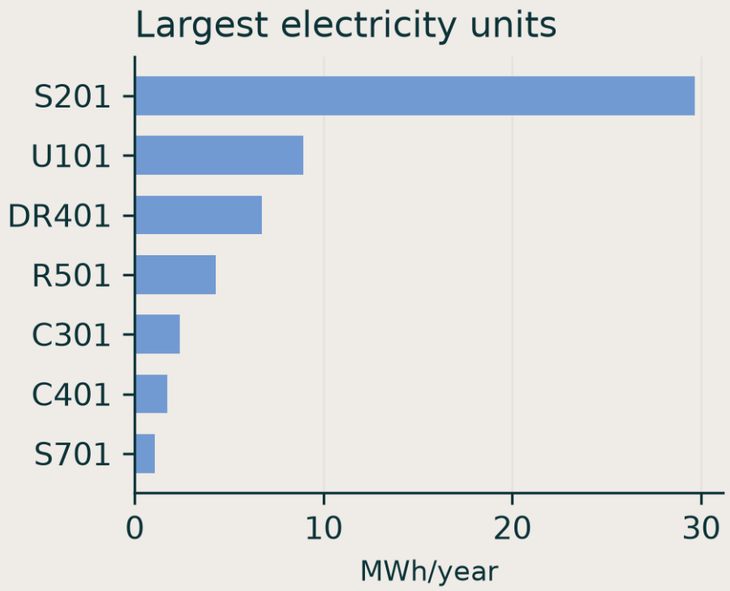
Scenario: all rank 6; 520944a3fae14541a889a13bda3bbd98; 679.9 kg/h; 8760 h/year
Units: kg/h; t/year; dry Nm³/year; kg/t feed
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

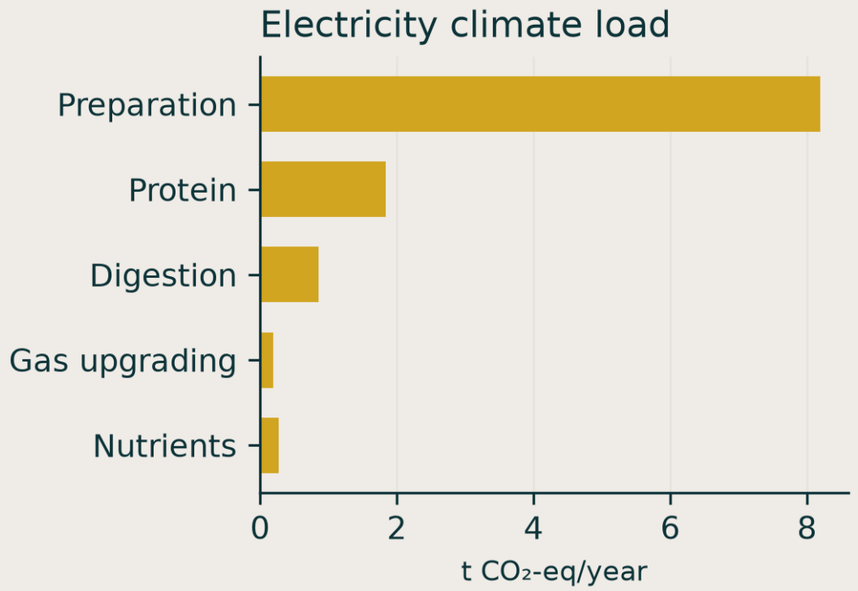
Energy and climate hotspots

Protein ✓ · Biogas ✓ · N ✓ · P X · K X



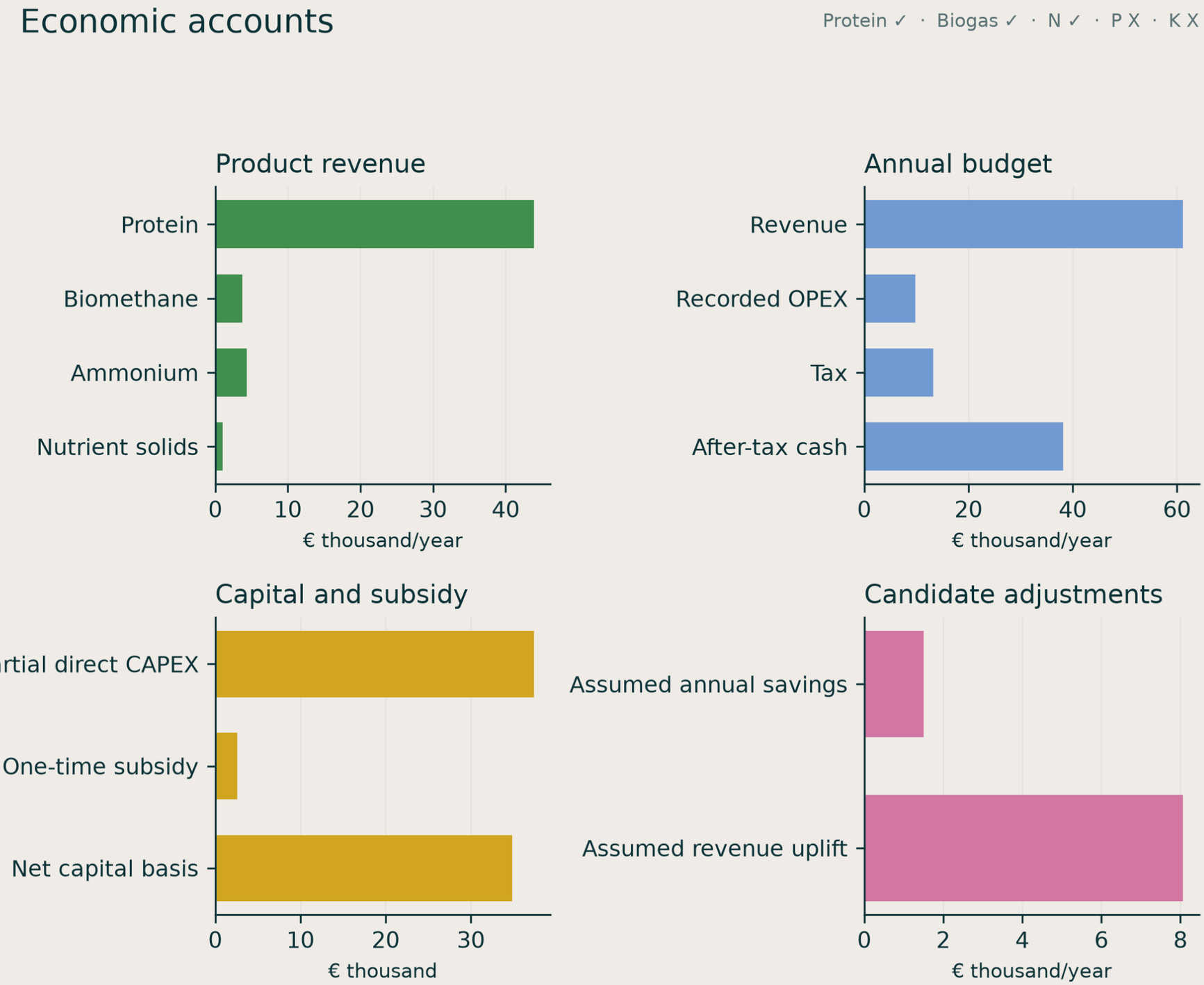






Stage and unit utility hotspots and included climate load

Scenario: all rank 6; 520944a3fae14541a889a13bda3bbd98; 679.9 kg/h; 8760 h/year
Units: MWh/year; t CO_2-eq/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

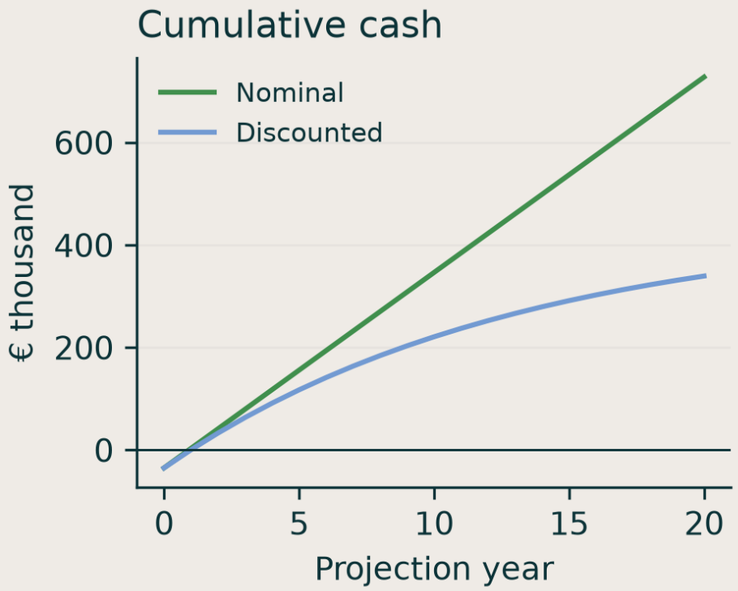


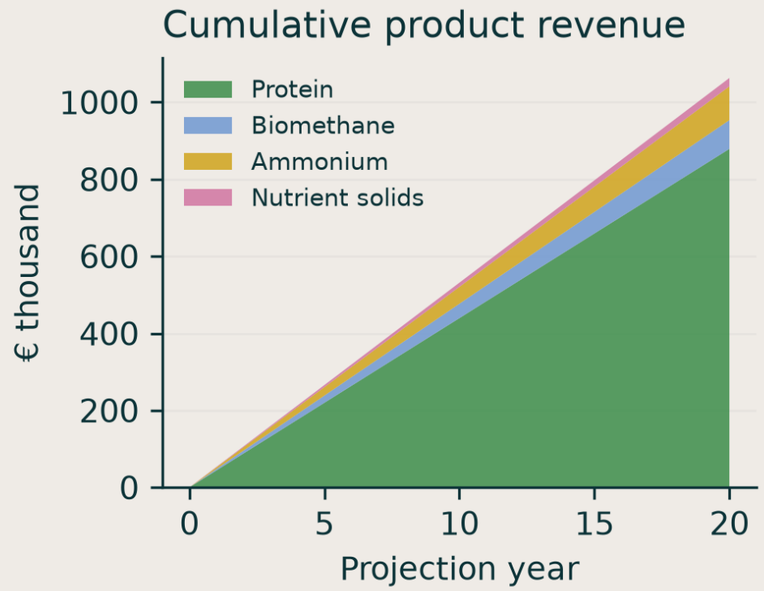
Product revenue, partial annual budget, capital subsidy and candidate overlays

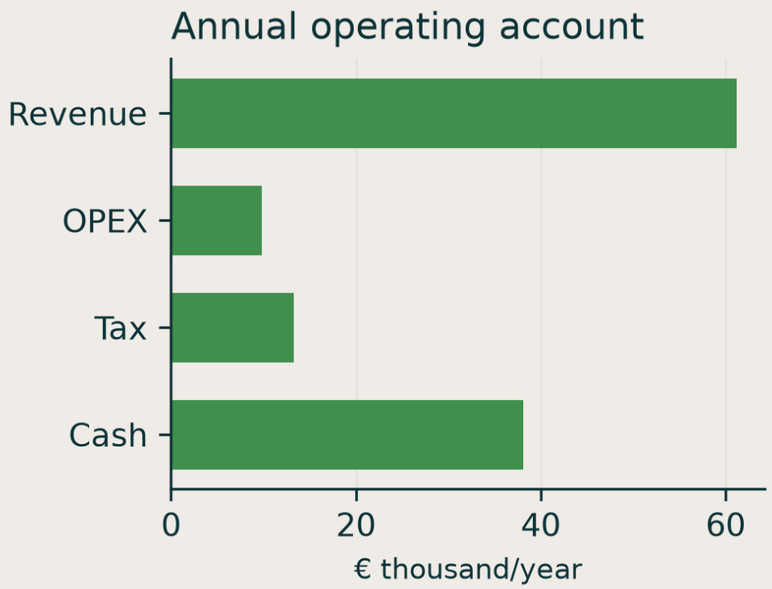
Scenario: all rank 6; 520944a3fae14541a889a13bda3bbd98; 679.9 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

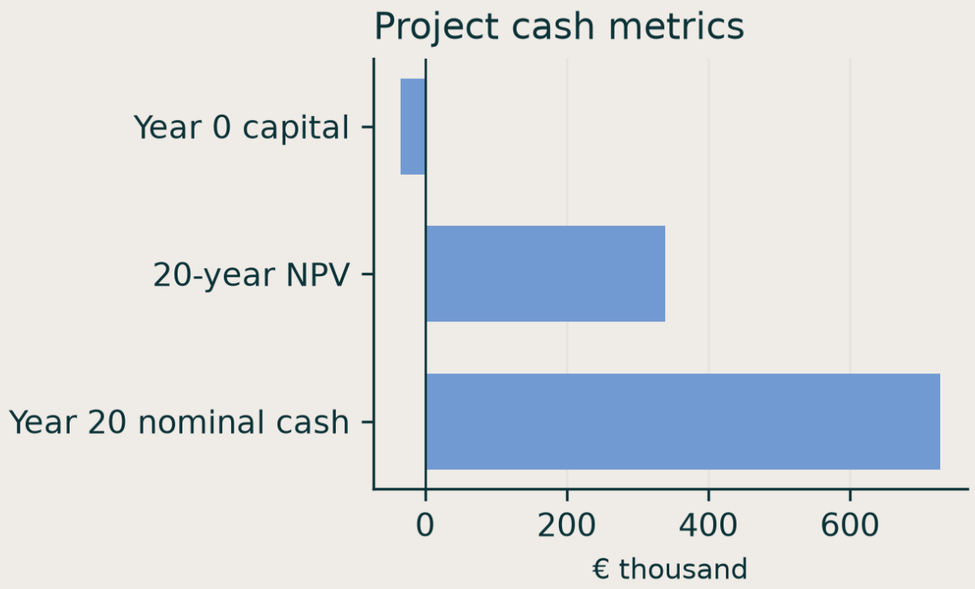
20-year financial projection

Protein ✓ · Biogas ✓ · N ✓ · P X · K X









Constant-account cash flow and cumulative product revenue over 20 years

Scenario: all rank 6; 520944a3fae14541a889a13bda3bbd98; 679.9 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Balance and evidence

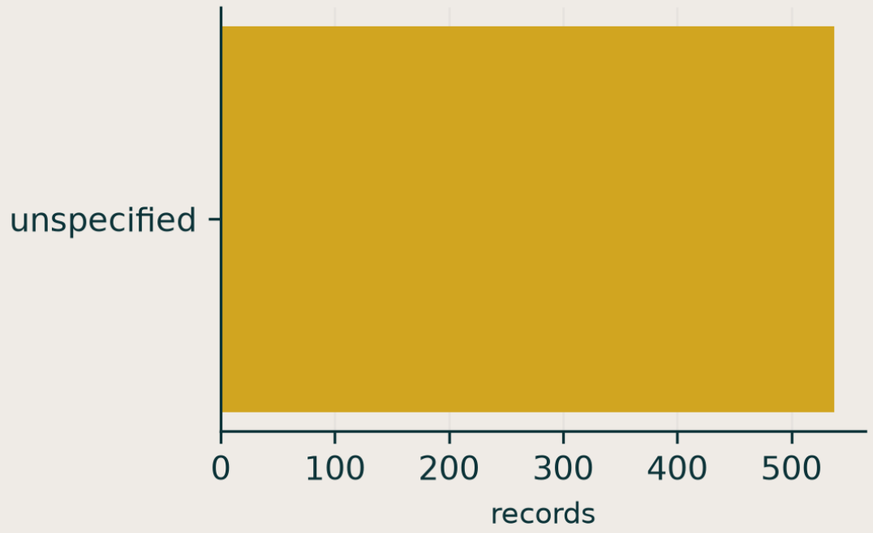
Protein ✓ · Biogas ✓ · N ✓ · P X · K X

Element closure

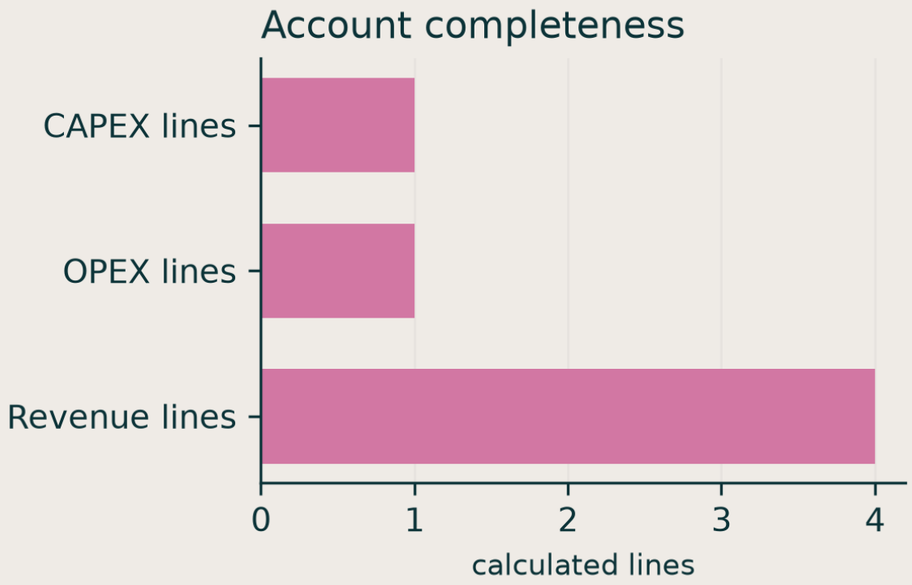
C	0 kg/year
N	0 kg/year
P	0 kg/year
K	0 kg/year

Numerical residuals

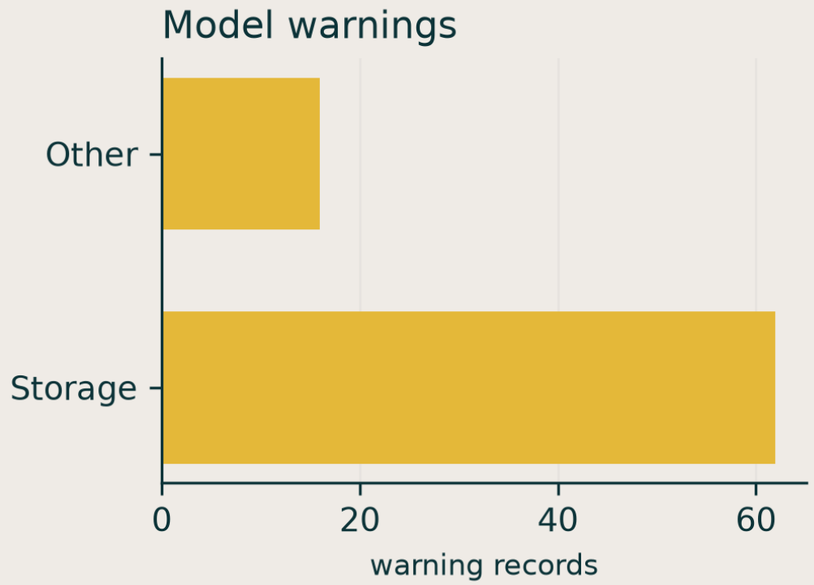
Parameter evidence



Account completeness



Model warnings



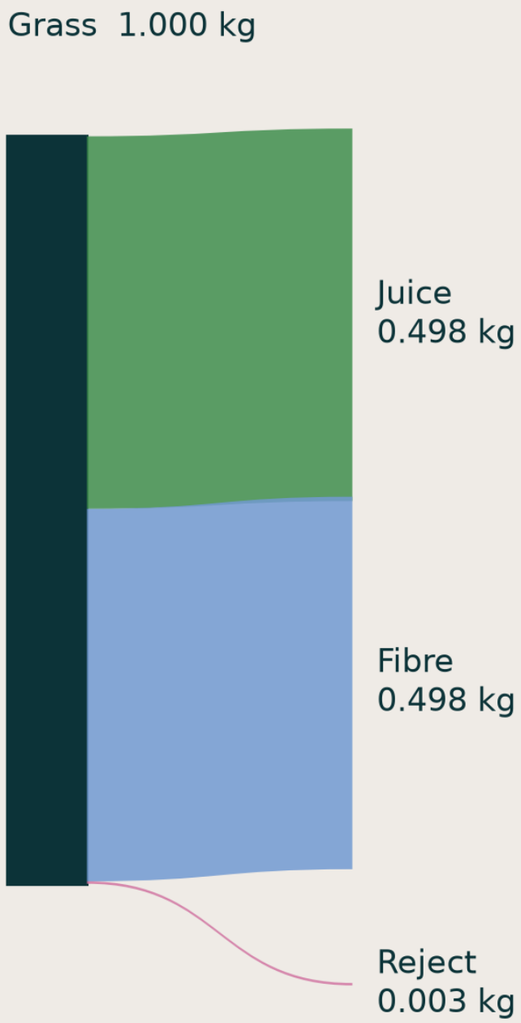
Numerical closure, evidence counts, calculated account lines and warnings

Scenario: all rank 6; 520944a3fae14541a889a13bda3bbd98; 679.9 kg/h; 8760 h/year
Units: kg/year; records
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

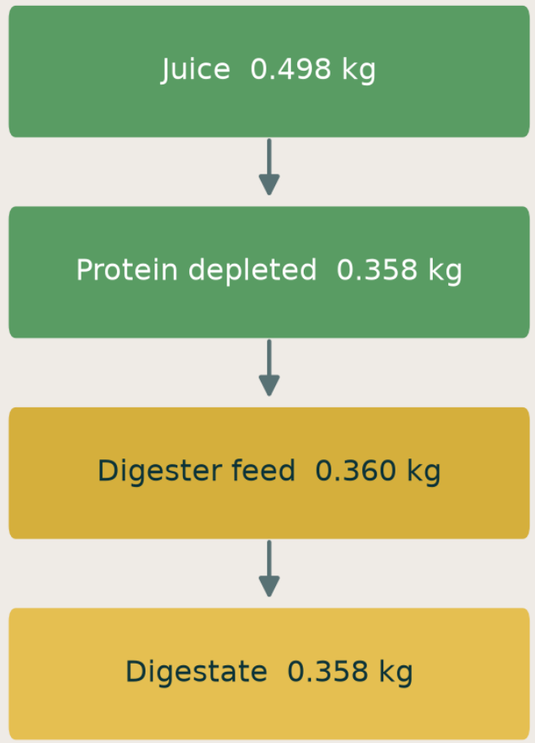
Grass to products

Protein ✓ · Biogas ✓ · N ✓ · P ✓ · K X

Grass split



Juice pathway



Selected intermediate streams

Recovered outputs

Raw gas 0.0010 Nm³

Upgraded 0.0007 Nm³

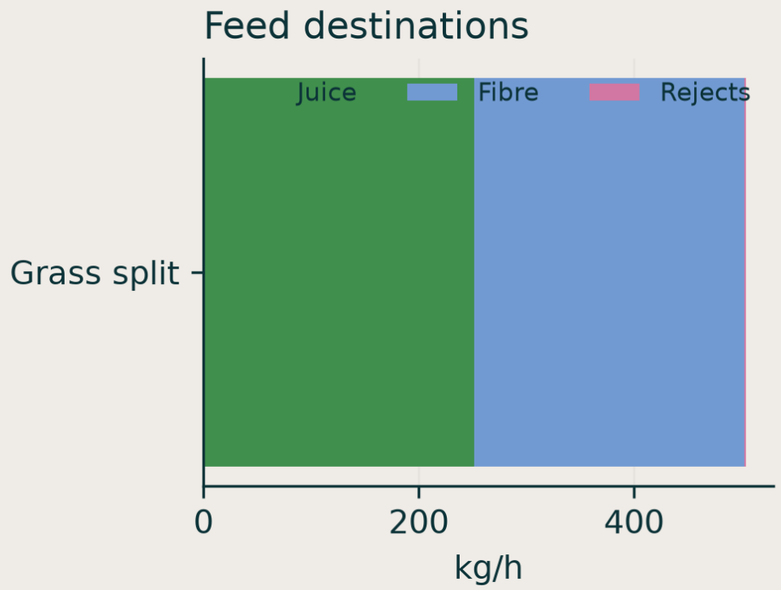
per kg wet grass

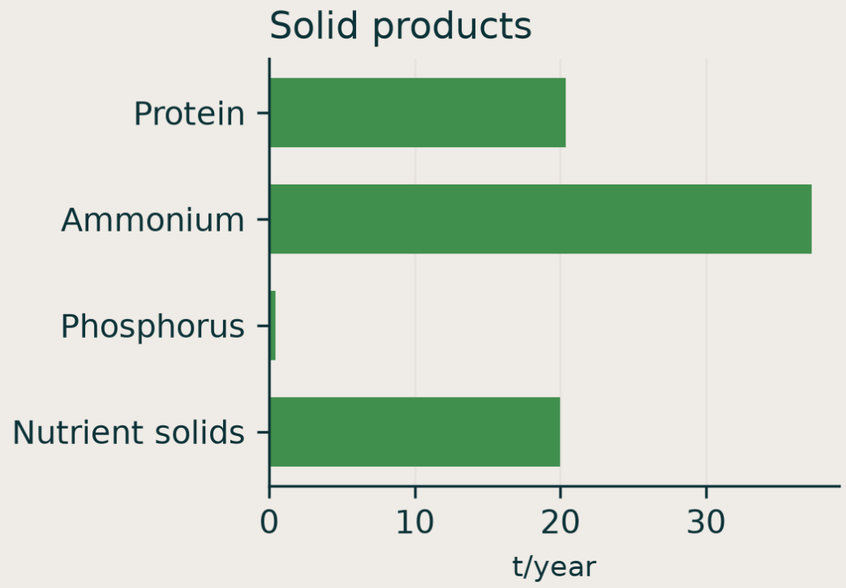
Mass-proportional grass split plus modeled downstream stream/product map

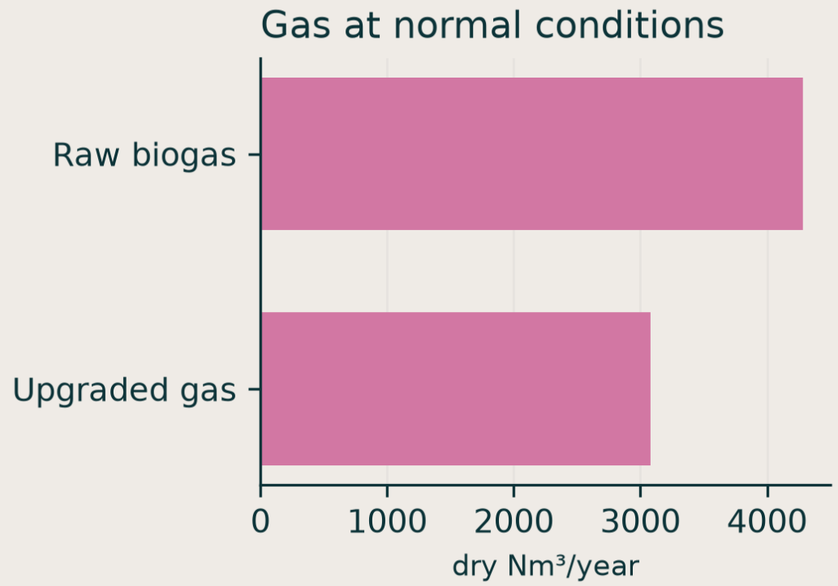
Scenario: all rank 7; 990eb9c13db24eea9696f0ee5d2dbdec; 504.3 kg/h; 8760 h/year
Units: kg/kg wet grass; dry Nm³/kg wet grass
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

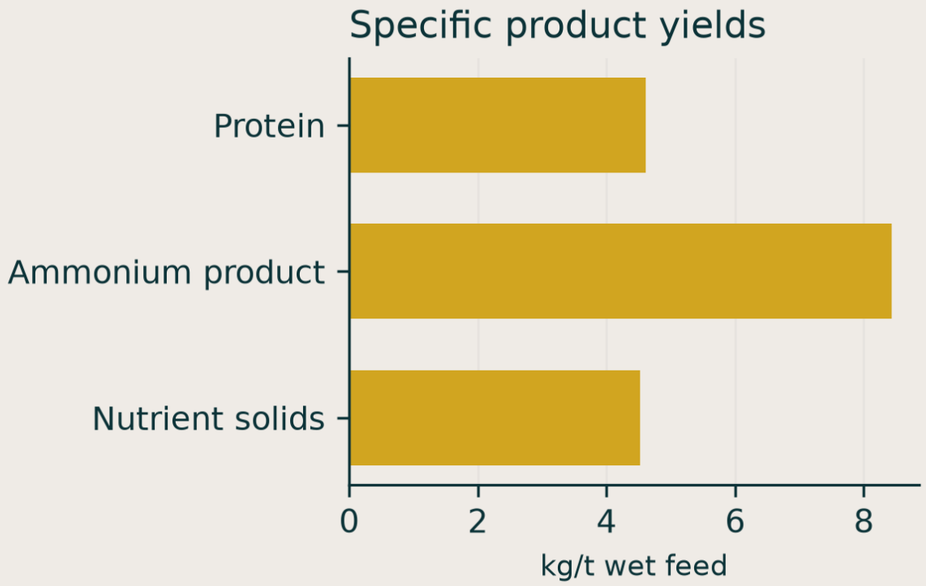
Material and products

Protein ✓ · Biogas ✓ · N ✓ · P ✓ · K X







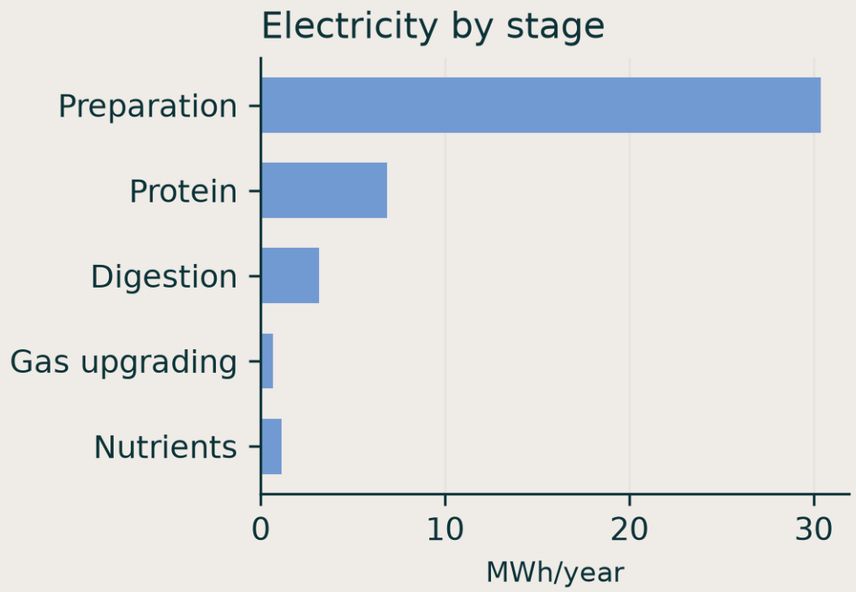


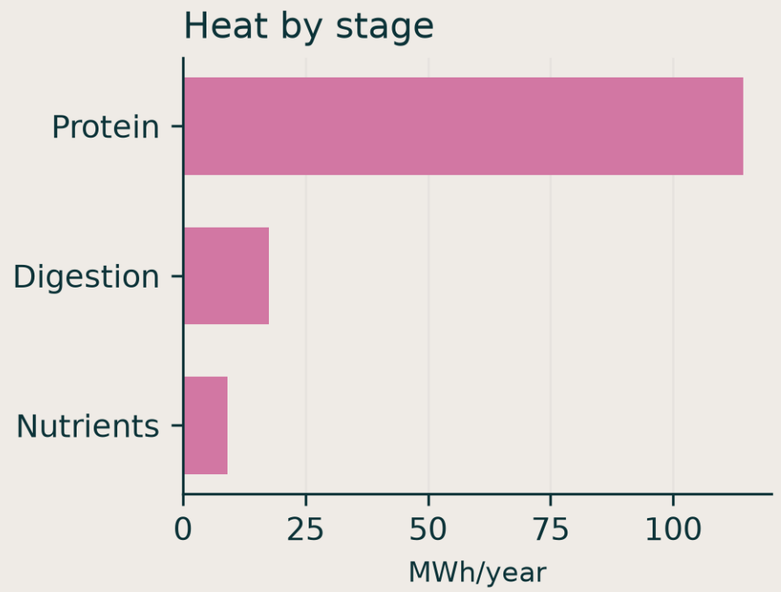
Front-end split, solid products, dry gas volumes and specific yields

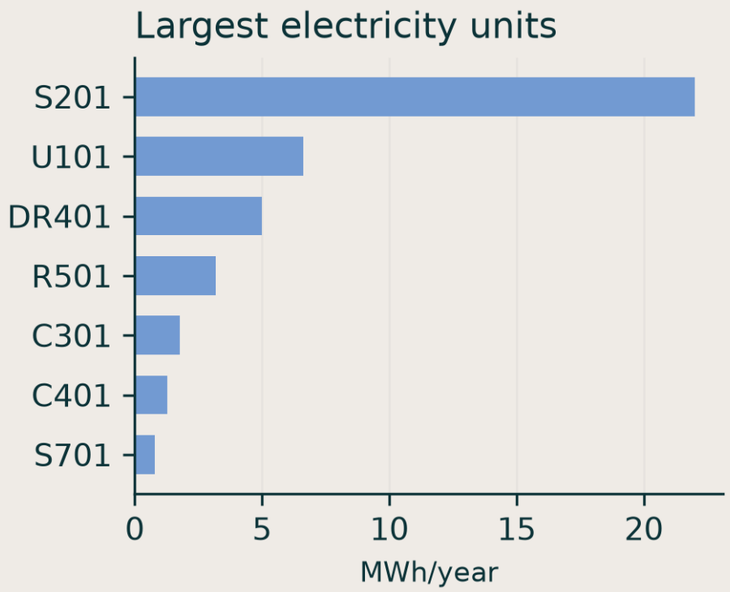
Scenario: all rank 7; 990eb9c13db24eea9696f0ee5d2dbdec; 504.3 kg/h; 8760 h/year
Units: kg/h; t/year; dry Nm³/year; kg/t feed
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

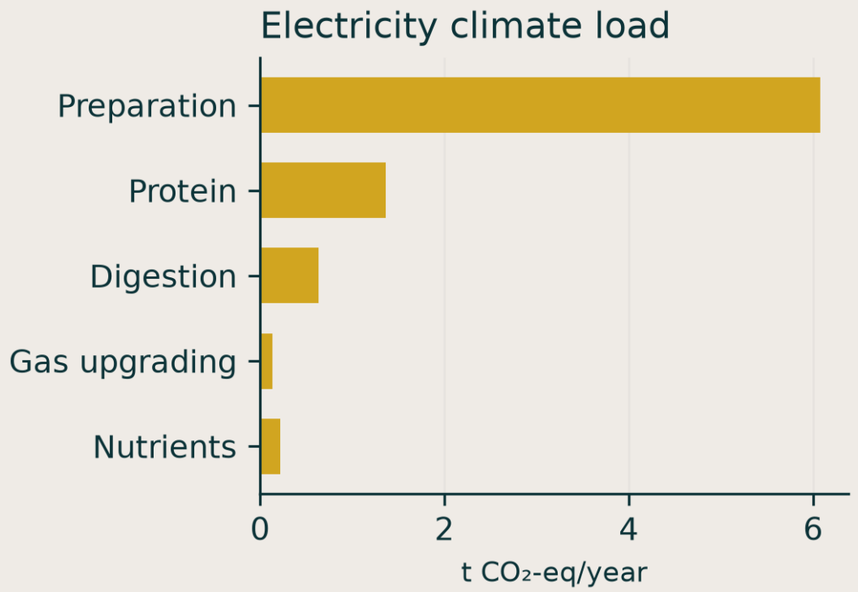
Energy and climate hotspots

Protein ✓ · Biogas ✓ · N ✓ · P ✓ · K X



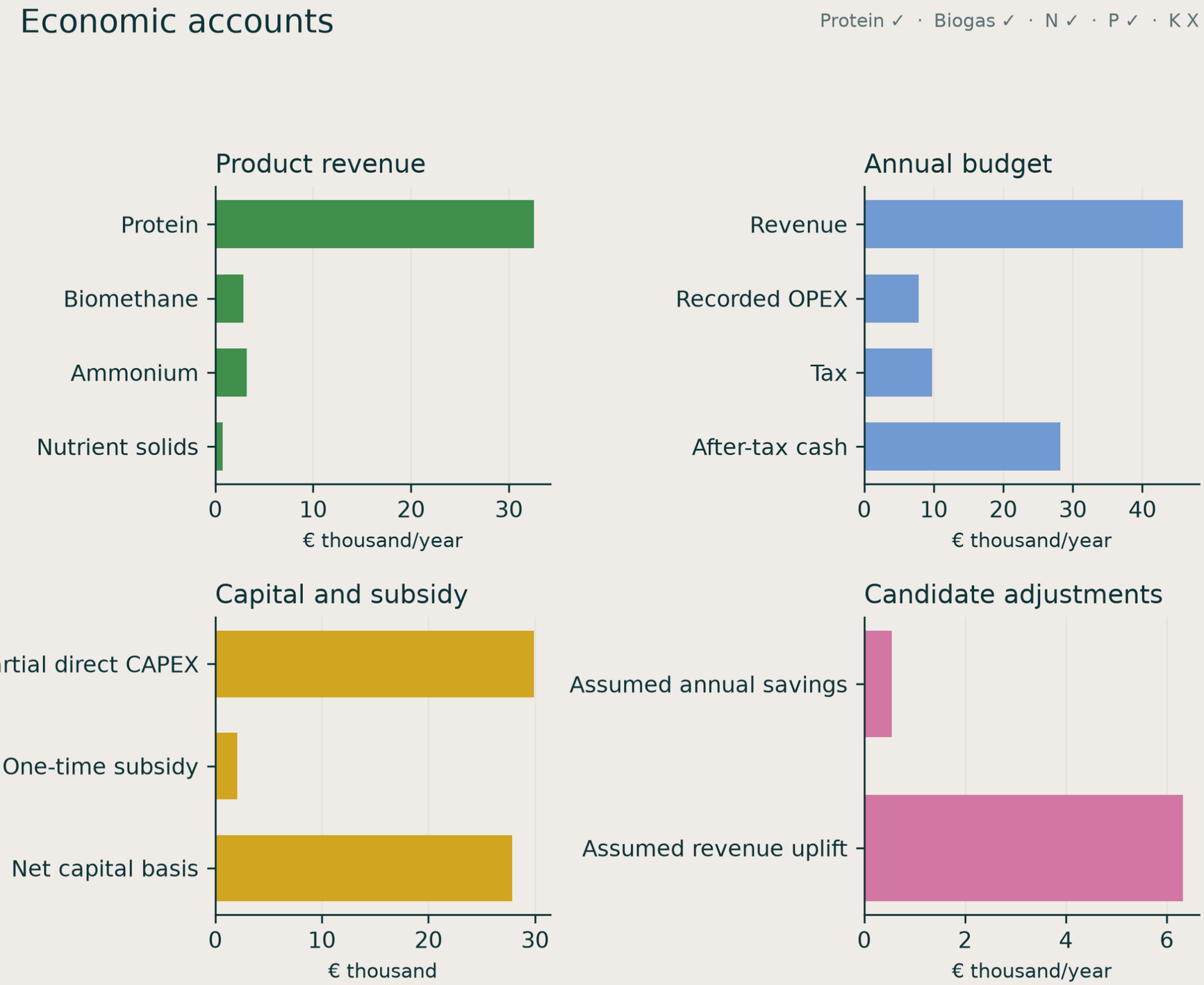






Stage and unit utility hotspots and included climate load

Scenario: all rank 7; 990eb9c13db24eea9696f0ee5d2dbdec; 504.3 kg/h; 8760 h/year
Units: MWh/year; t CO_2-eq/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

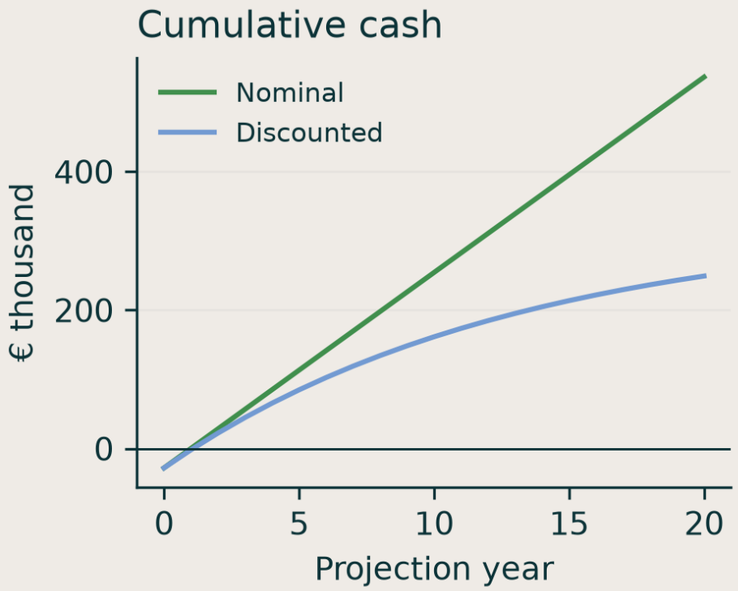


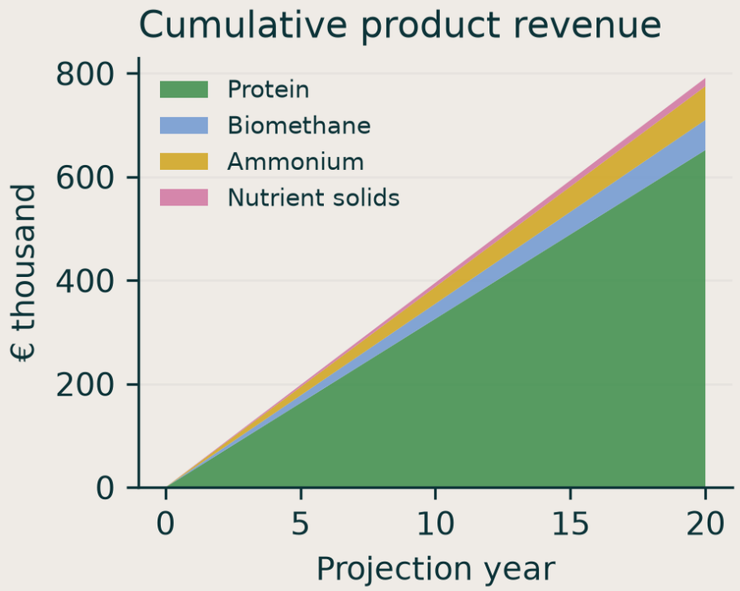
Product revenue, partial annual budget, capital subsidy and candidate overlays

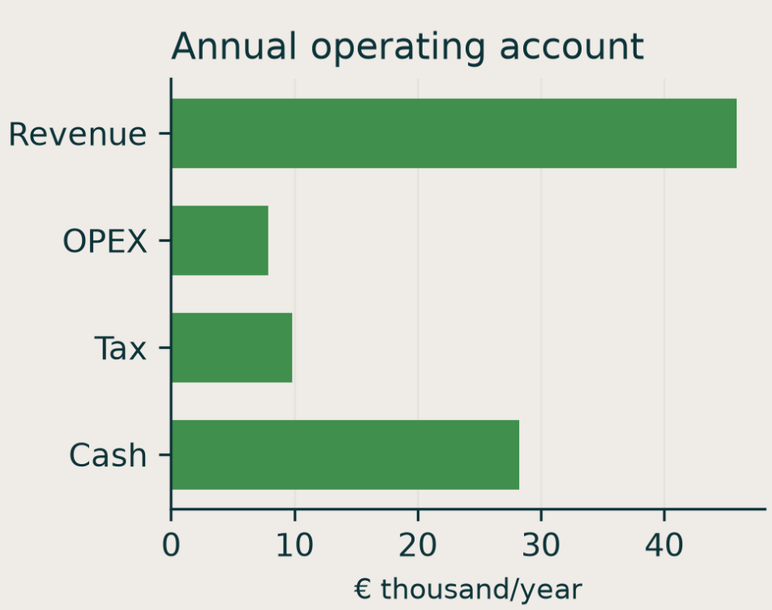
Scenario: all rank 7; 990eb9c13db24eea9696f0ee5d2dbdec; 504.3 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

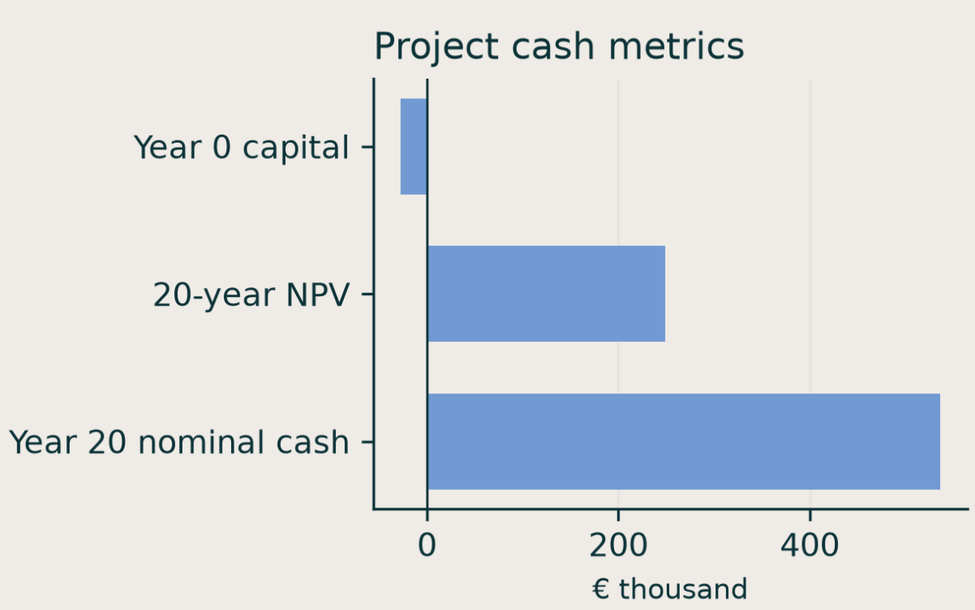
20-year financial projection

Protein ✓ · Biogas ✓ · N ✓ · P ✓ · K X









Constant-account cash flow and cumulative product revenue over 20 years

Scenario: all rank 7; 990eb9c13db24eea9696f0ee5d2dbdec; 504.3 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Balance and evidence

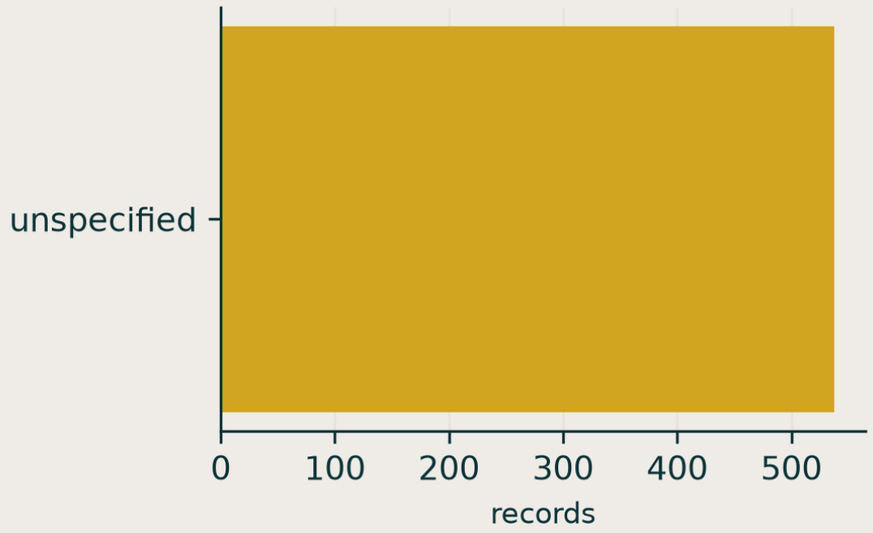
Protein ✓ · Biogas ✓ · N ✓ · P ✓ · K X

Element closure

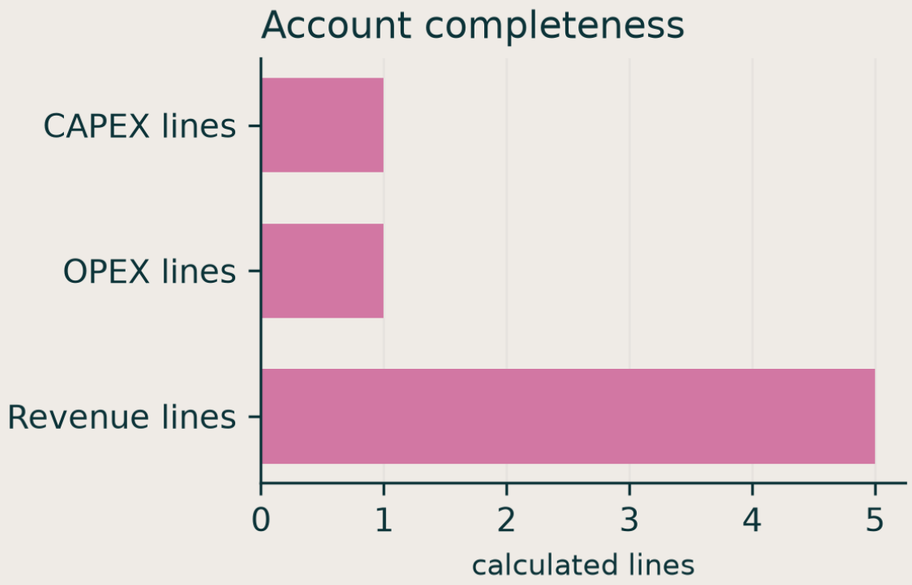
C	0 kg/year
N	0 kg/year
P	0 kg/year
K	2.4e-13 kg/year

Numerical residuals

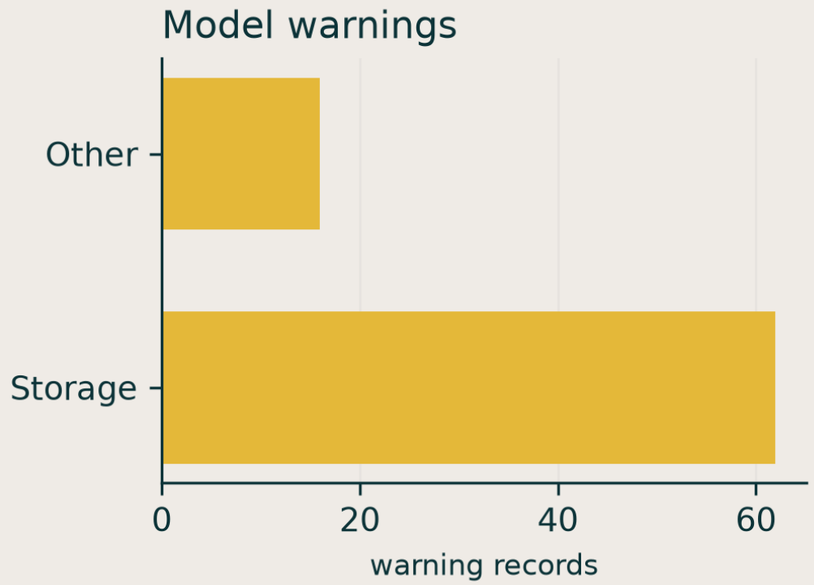
Parameter evidence



Account completeness



Model warnings



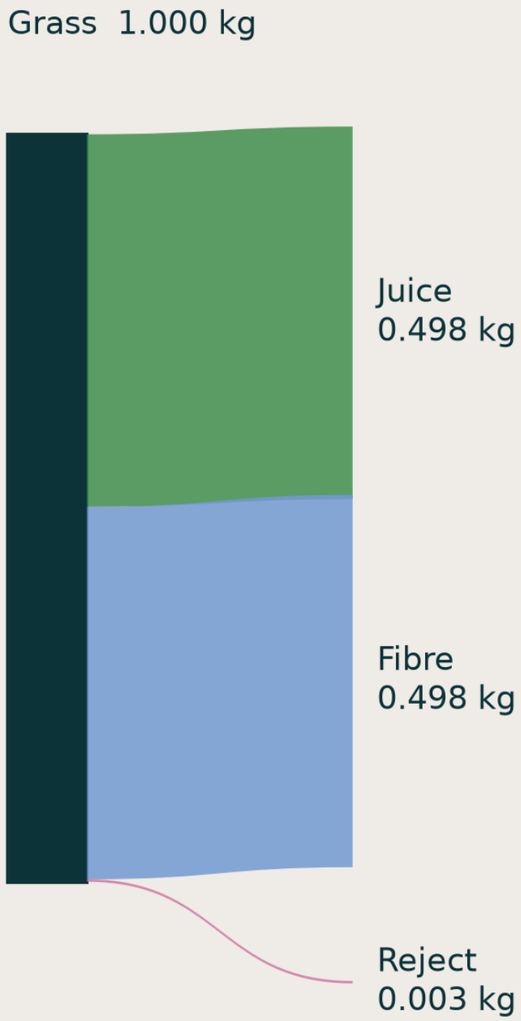
Numerical closure, evidence counts, calculated account lines and warnings

Scenario: all rank 7; 990eb9c13db24eea9696f0ee5d2dbdec; 504.3 kg/h; 8760 h/year
Units: kg/year; records
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

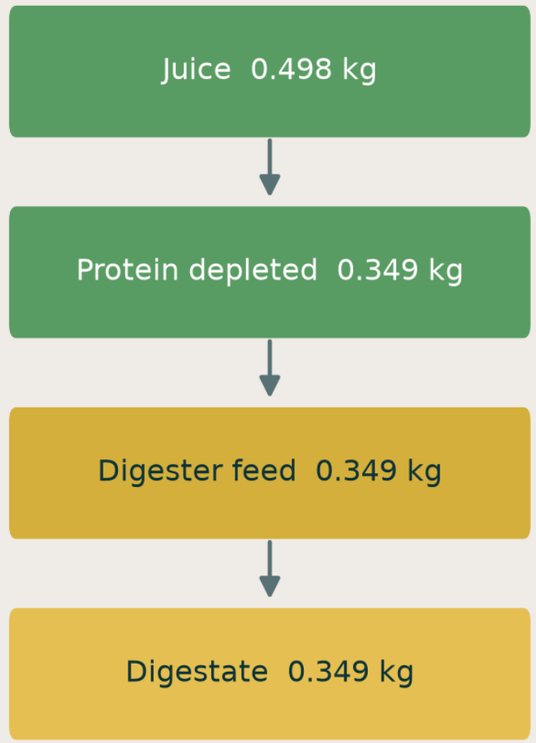
Grass to products

Protein ✓ · Biogas X · N X · P X · K X

Grass split

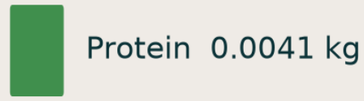


Juice pathway



Selected intermediate streams

Recovered outputs



per kg wet grass

Mass-proportional grass split plus modeled downstream stream/product map

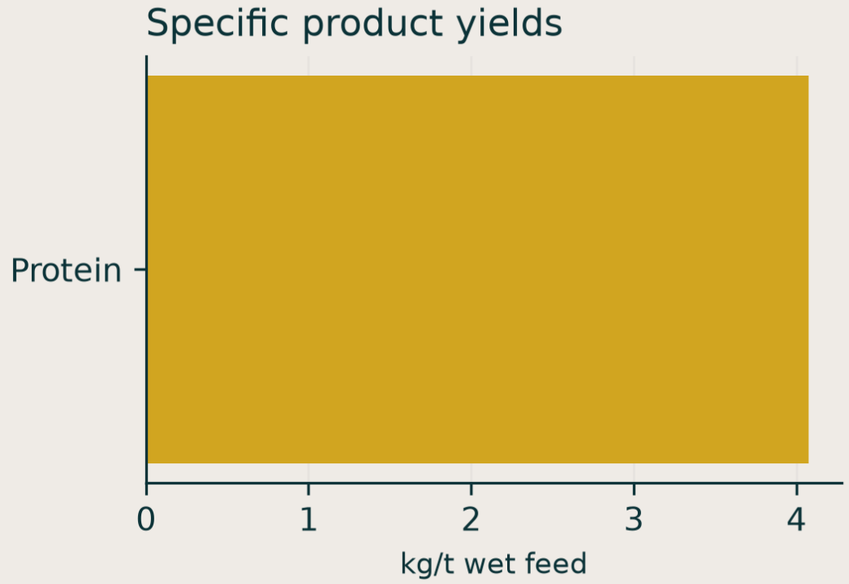
Scenario: all rank 8; 30302c69493c42b29b36b846fd77725d; 506.3 kg/h; 8760 h/year
Units: kg/kg wet grass; dry Nm³/kg wet grass
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Material and products

Protein ✓ · Biogas X · N X · P X · K X

Gas at normal conditions

No modeled output

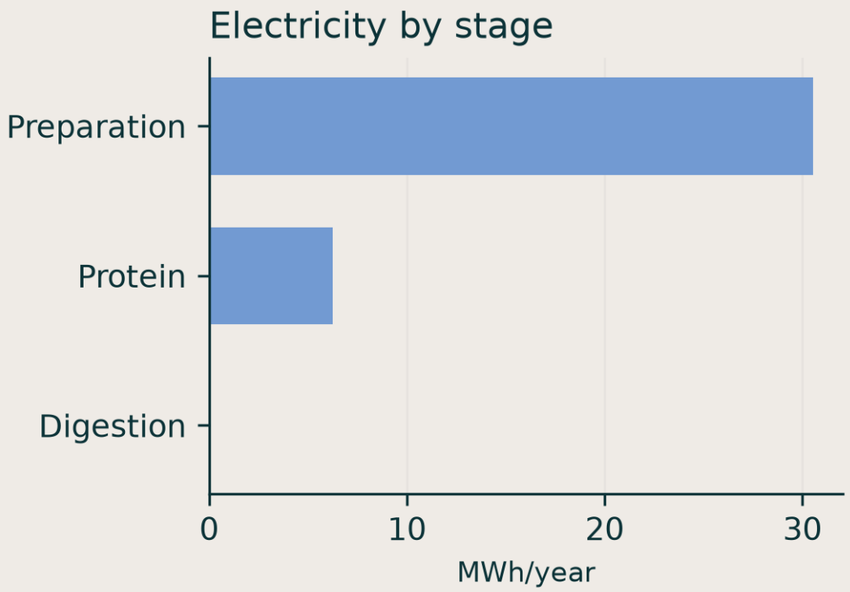


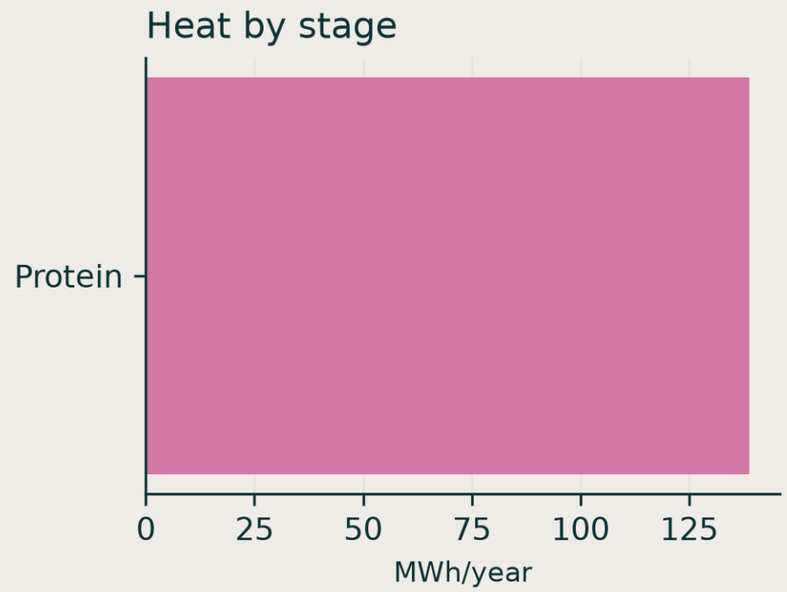
Front-end split, solid products, dry gas volumes and specific yields

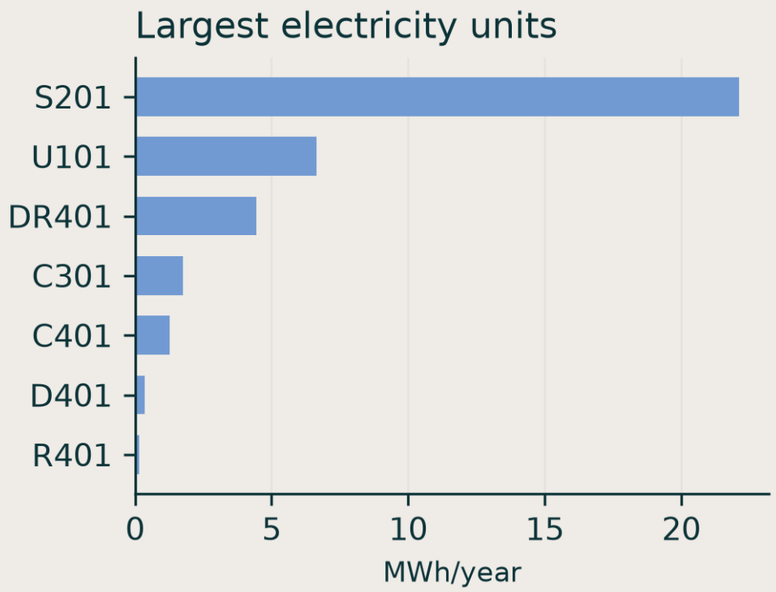
Scenario: all rank 8; 30302c69493c42b29b36b846fd77725d; 506.3 kg/h; 8760 h/year
Units: kg/h; t/year; dry Nm³/year; kg/t feed
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

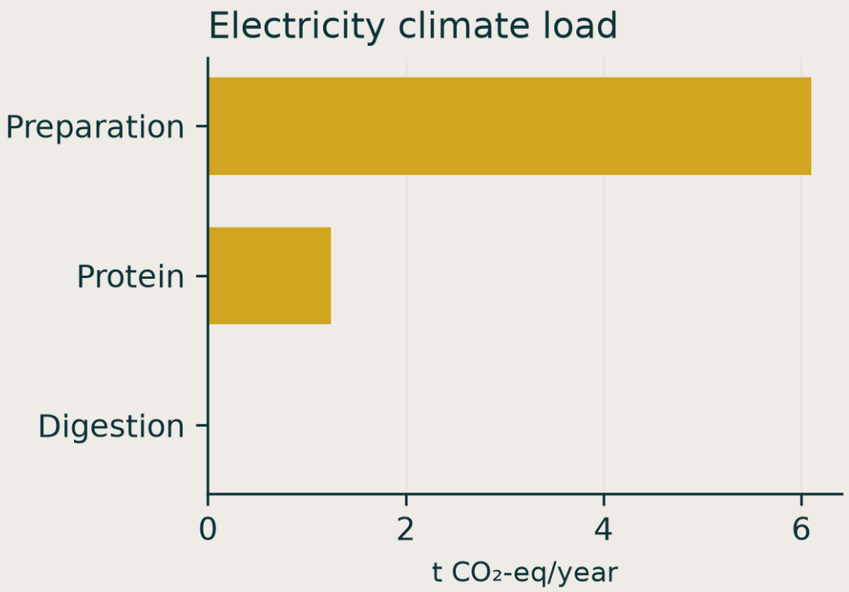
Energy and climate hotspots

Protein ✓ · Biogas X · N X · P X · K X







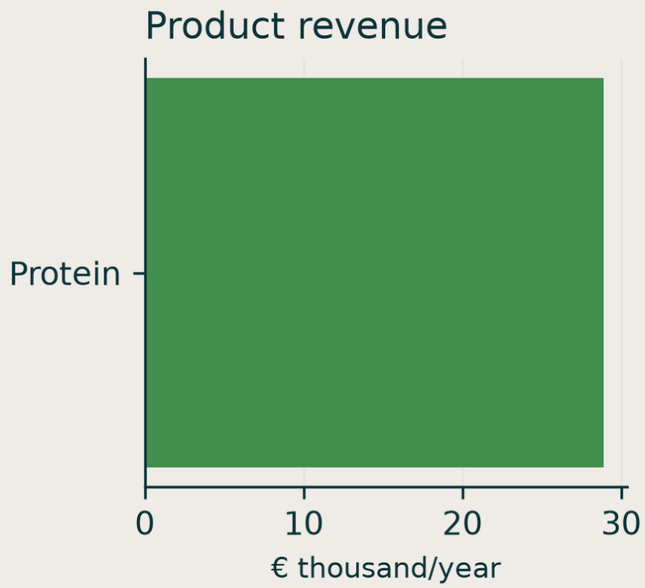


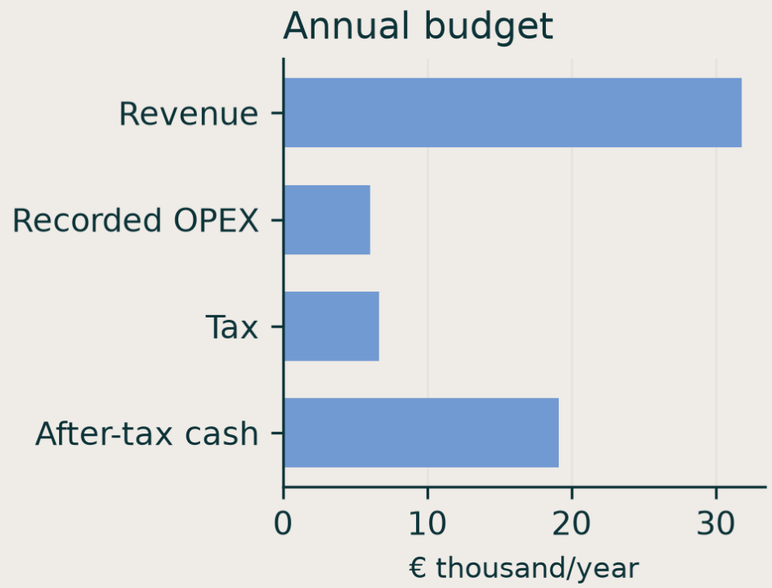
Stage and unit utility hotspots and included climate load

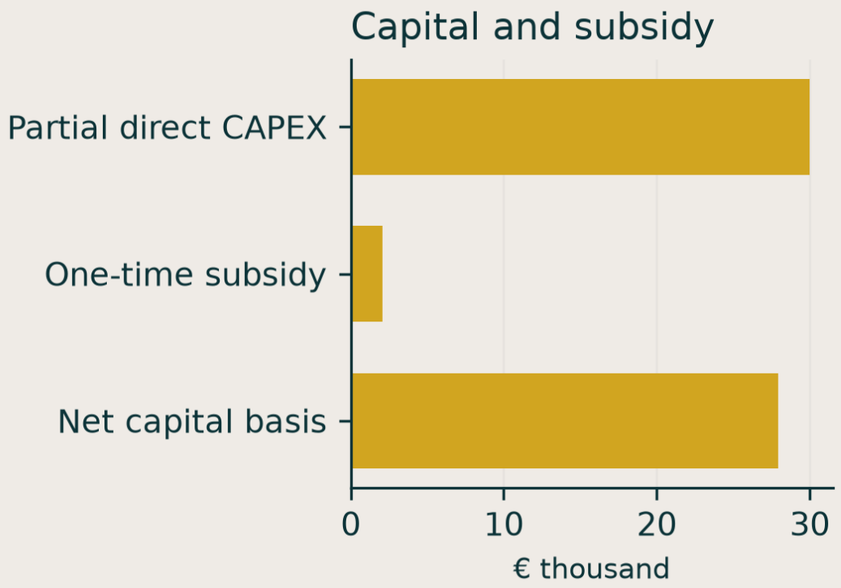
Scenario: all rank 8; 30302c69493c42b29b36b846fd77725d; 506.3 kg/h; 8760 h/year
Units: MWh/year; t CO_2-eq/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

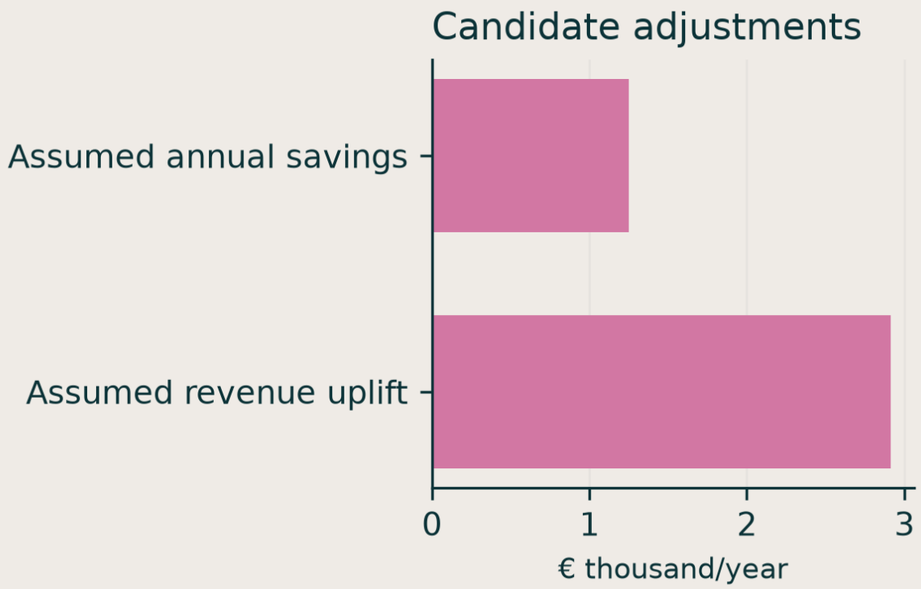
Economic accounts

Protein ✓ · Biogas X · N X · P X · K X







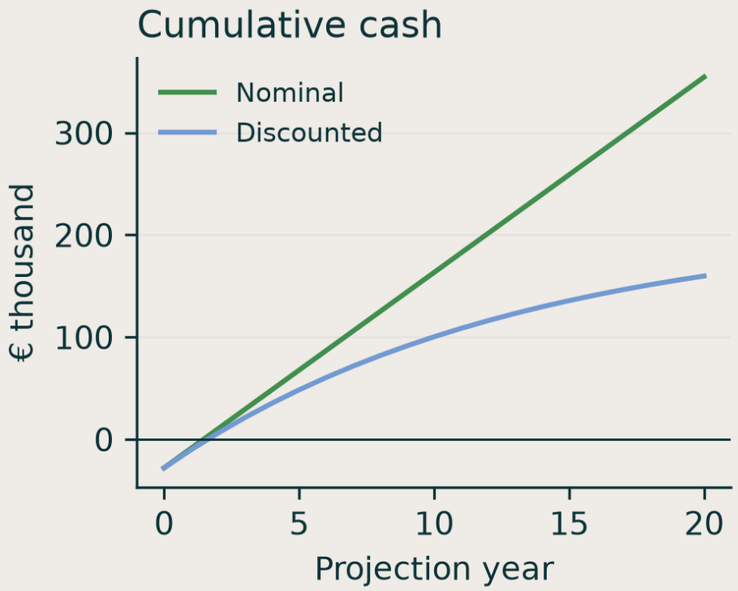


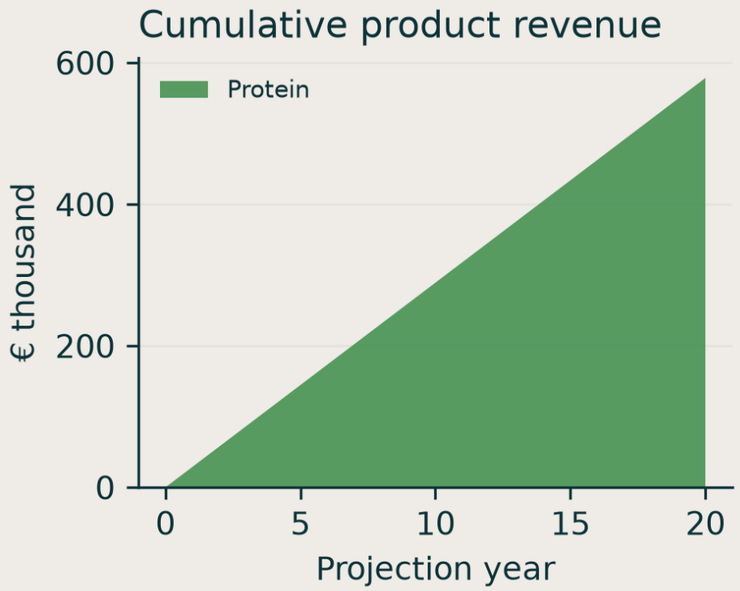
Product revenue, partial annual budget, capital subsidy and candidate overlays

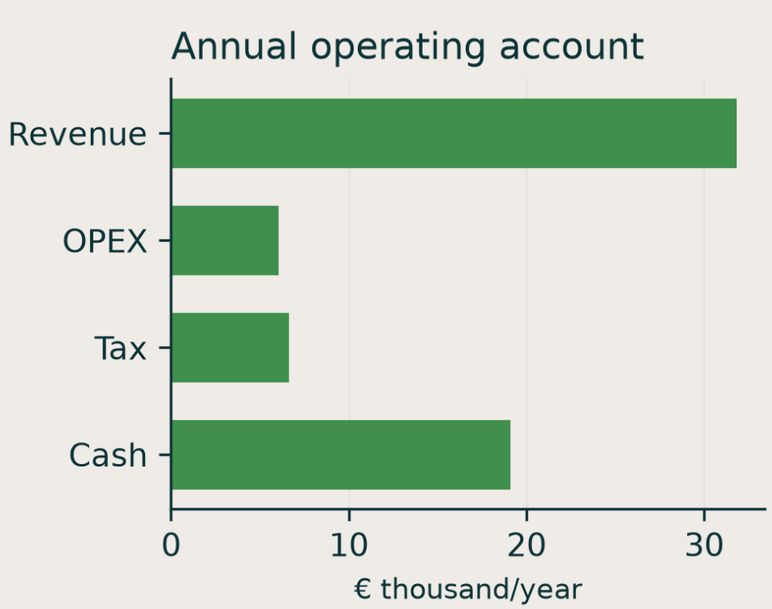
Scenario: all rank 8; 30302c69493c42b29b36b846fd77725d; 506.3 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

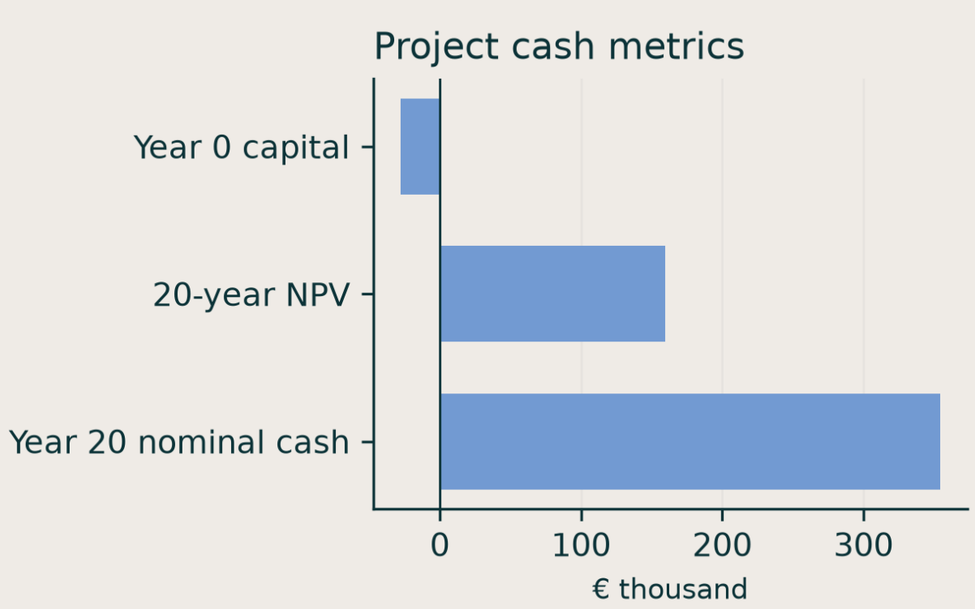
20-year financial projection

Protein ✓ · Biogas X · N X · P X · K X









Constant-account cash flow and cumulative product revenue over 20 years

Scenario: all rank 8; 30302c69493c42b29b36b846fd77725d; 506.3 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Balance and evidence

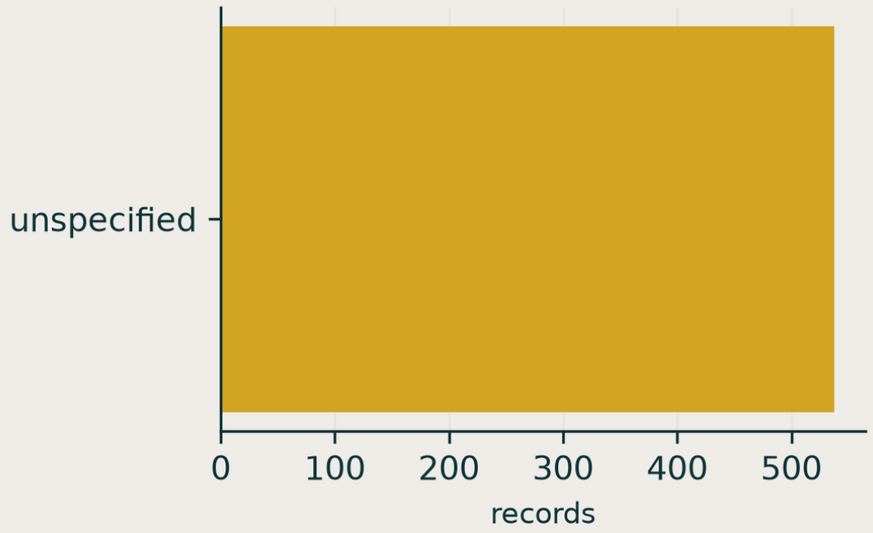
Protein ✓ · Biogas X · N X · P X · K X

Element closure

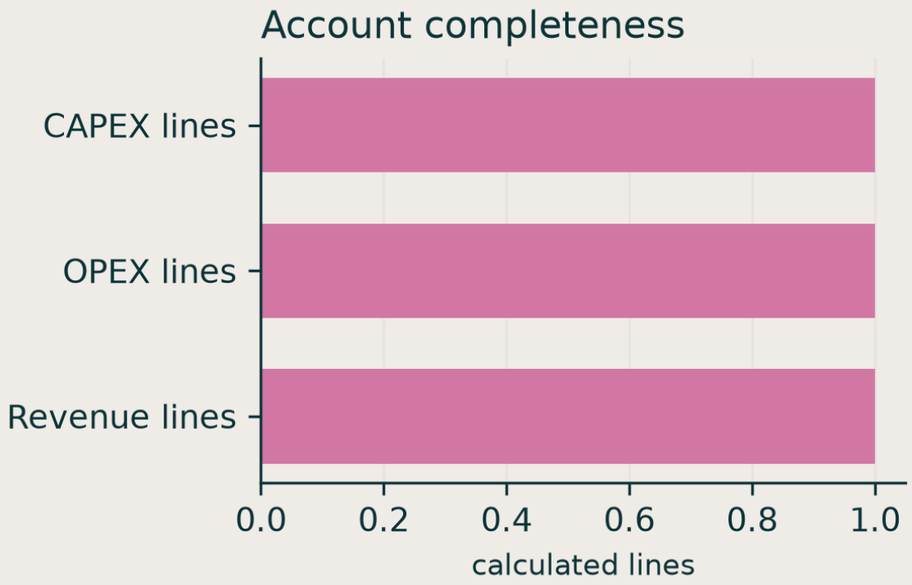
C	0 kg/year
N	0 kg/year
P	0 kg/year
K	0 kg/year

Numerical residuals

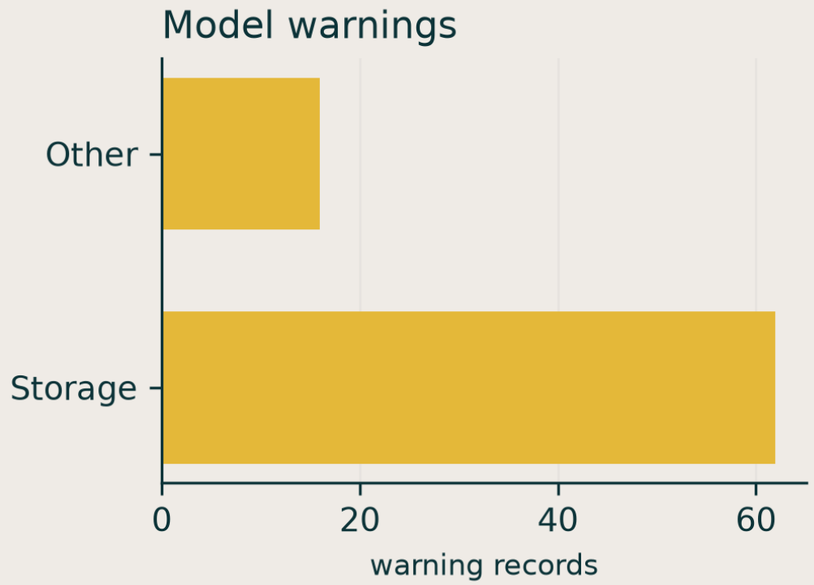
Parameter evidence



Account completeness



Model warnings



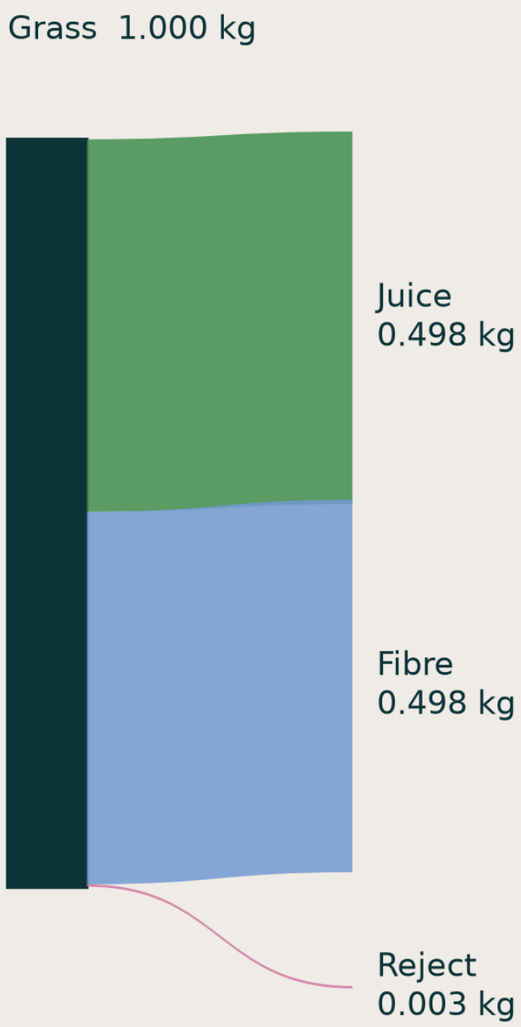
Numerical closure, evidence counts, calculated account lines and warnings

Scenario: all rank 8; 30302c69493c42b29b36b846fd77725d; 506.3 kg/h; 8760 h/year
Units: kg/year; records
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

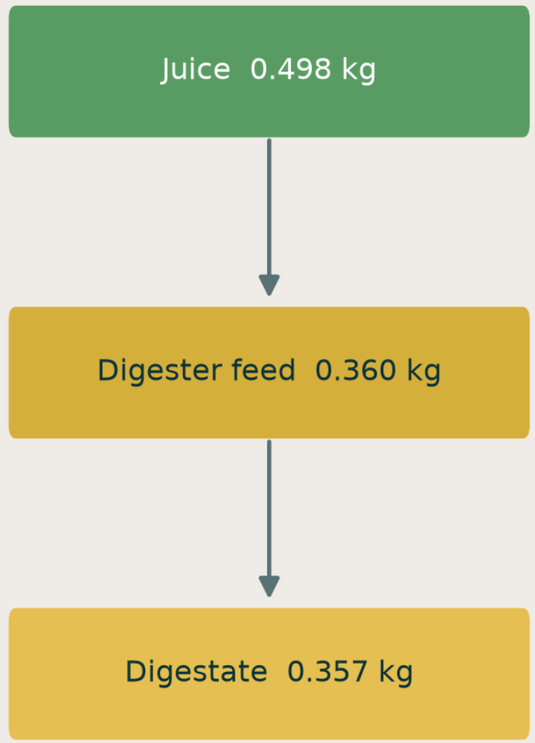
Grass to products

Protein X · Biogas ✓ · N X · P X · K X

Grass split



Juice pathway



Selected intermediate streams

Recovered outputs

Nutrient solids 0.0090 kg

Raw gas 0.0021 Nm³

Raw export 0.0021 Nm³

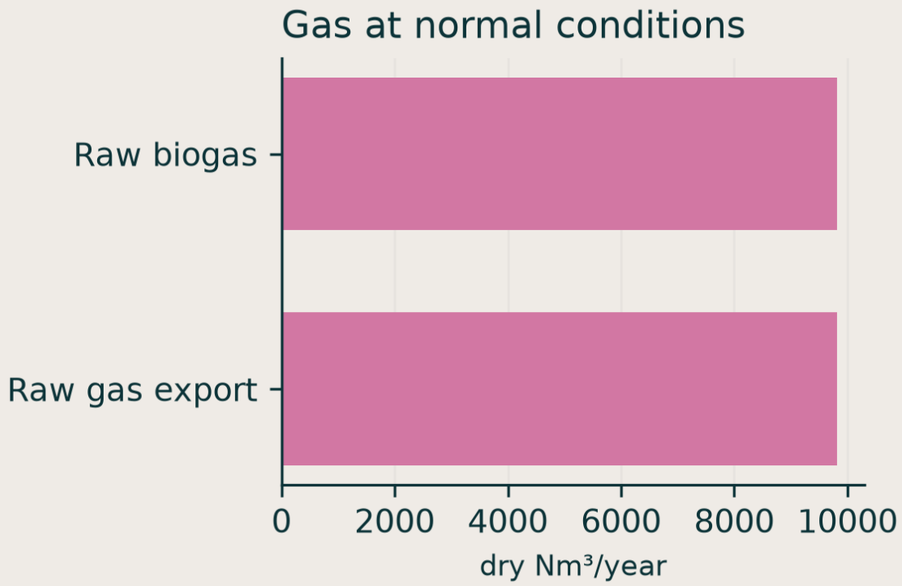
per kg wet grass

Mass-proportional grass split plus modeled downstream stream/product map

Scenario: all rank 9; 98b8467e23fc45c8a0136c61f4149274; 523.4 kg/h; 8760 h/year
Units: kg/kg wet grass; dry Nm³/kg wet grass
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Material and products

Protein X · Biogas ✓ · N X · P X · K X

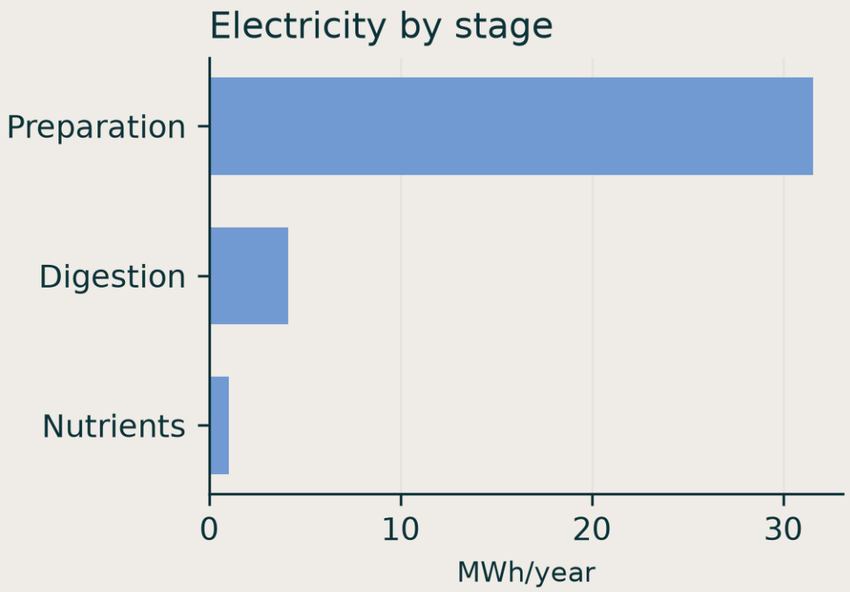


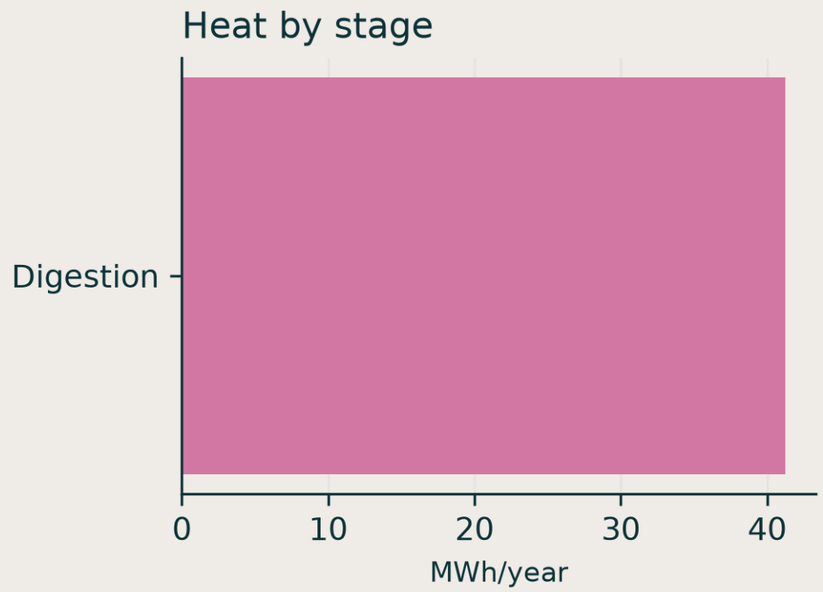
Front-end split, solid products, dry gas volumes and specific yields

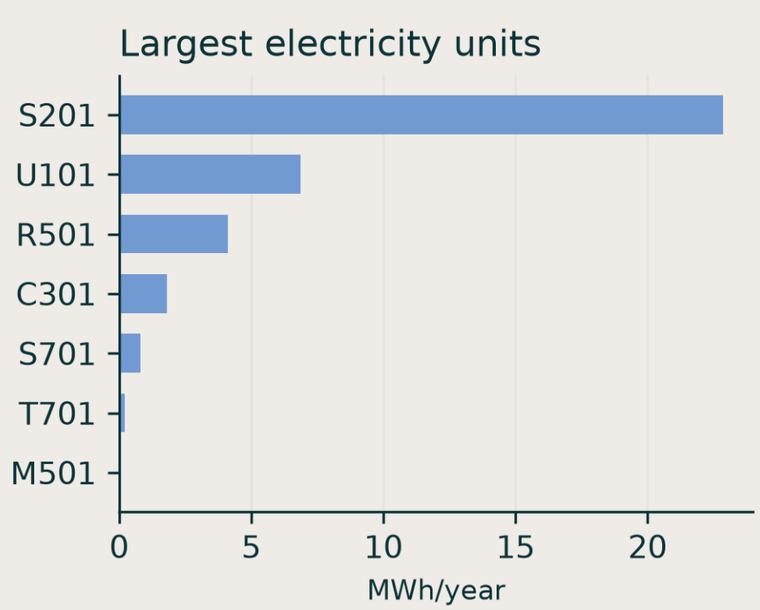
Scenario: all rank 9; 98b8467e23fc45c8a0136c61f4149274; 523.4 kg/h; 8760 h/year
Units: kg/h; t/year; dry Nm³/year; kg/t feed
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

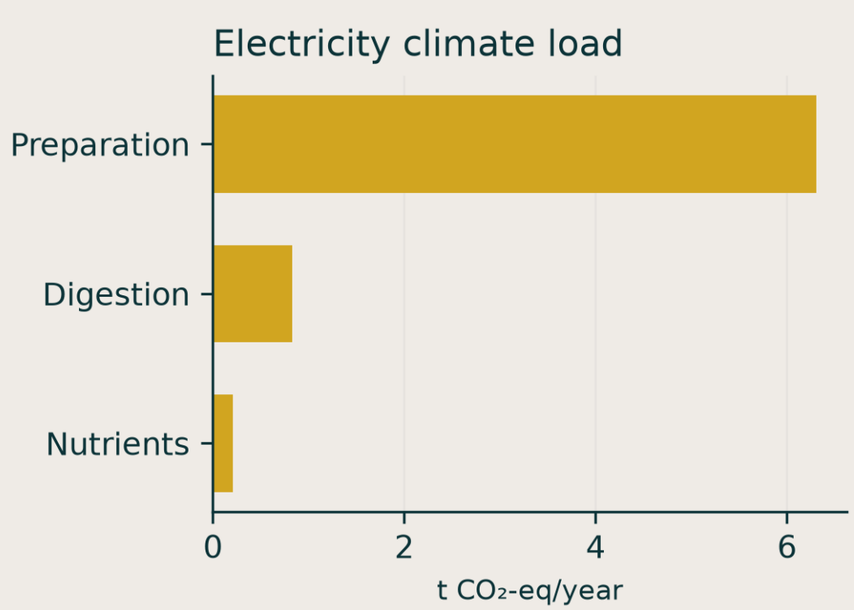
Energy and climate hotspots

Protein X · Biogas ✓ · N X · P X · K X







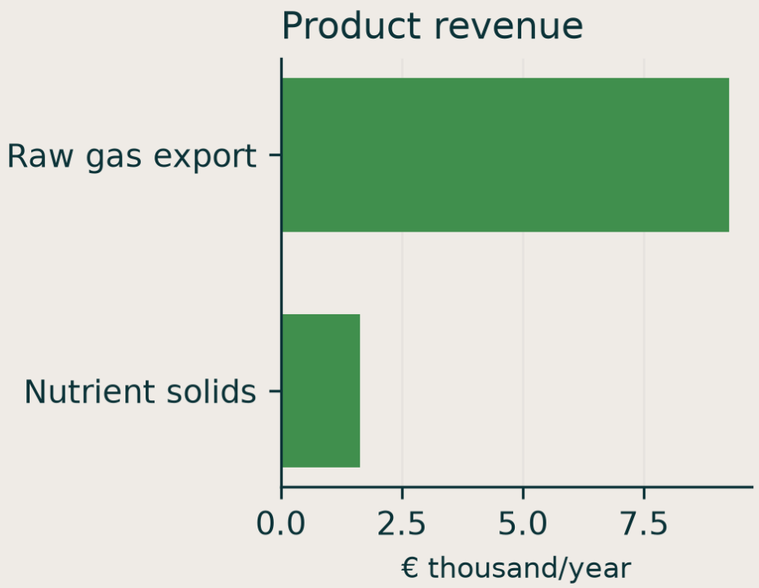


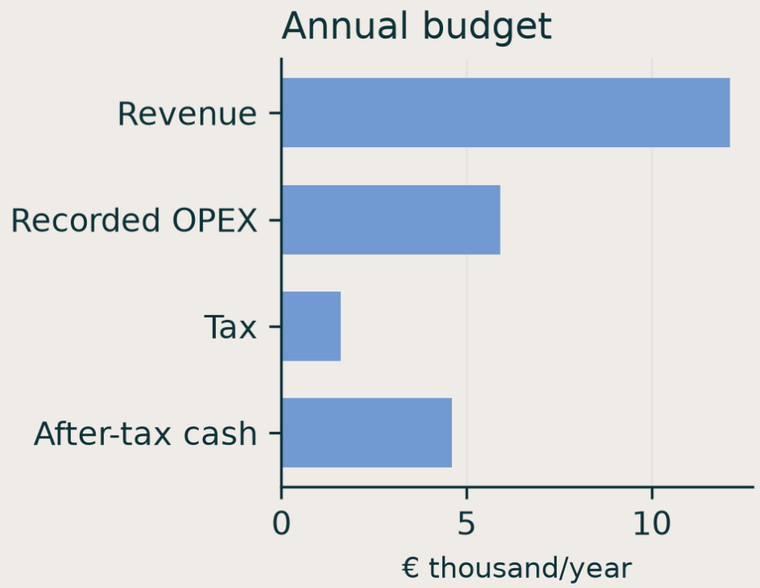
Stage and unit utility hotspots and included climate load

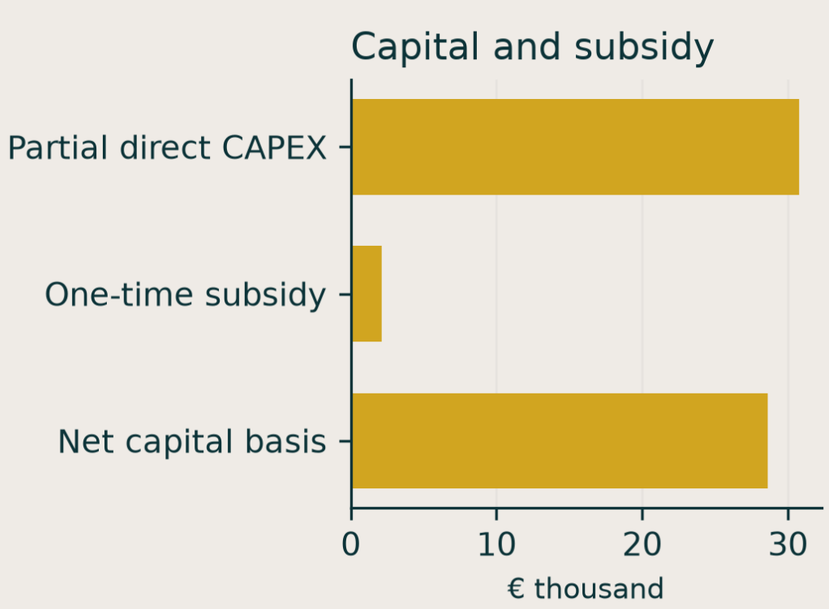
Scenario: all rank 9; 98b8467e23fc45c8a0136c61f4149274; 523.4 kg/h; 8760 h/year
Units: MWh/year; t CO_2-eq/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

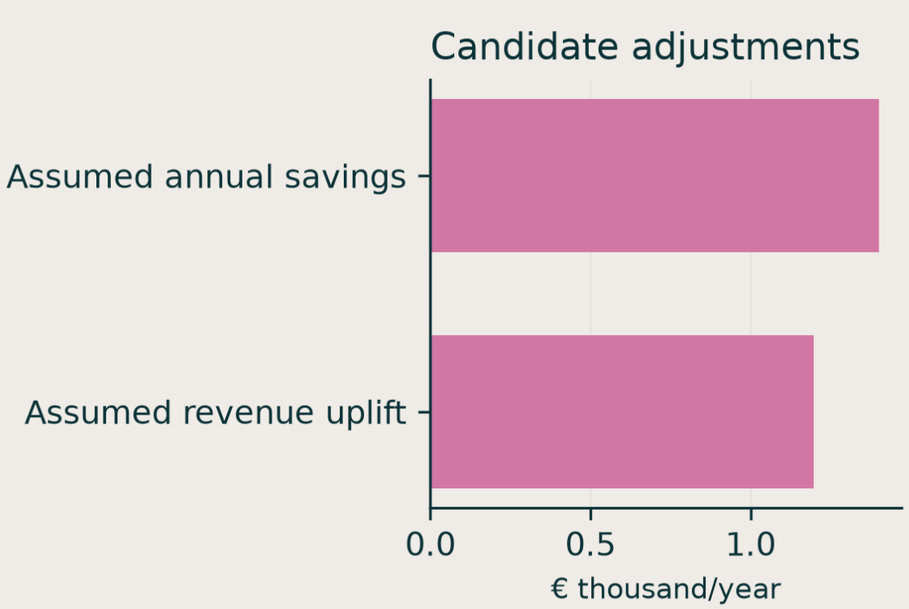
Economic accounts

Protein X · Biogas ✓ · N X · P X · K X







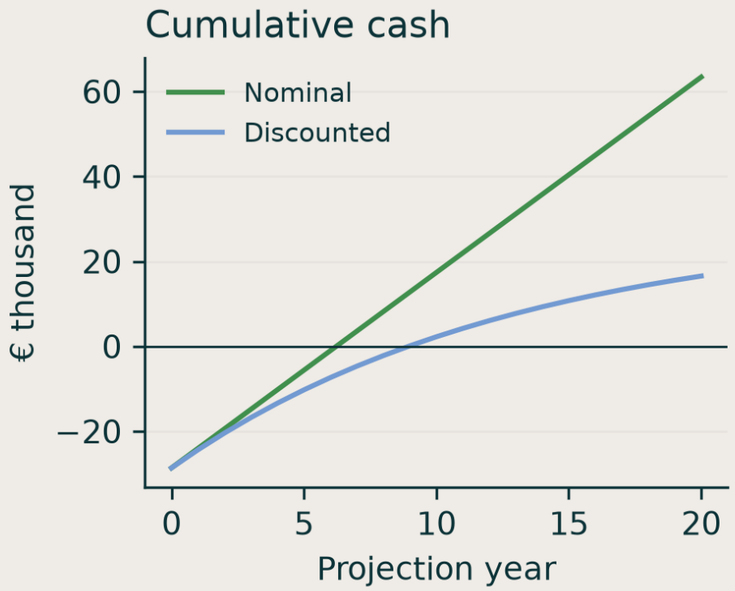


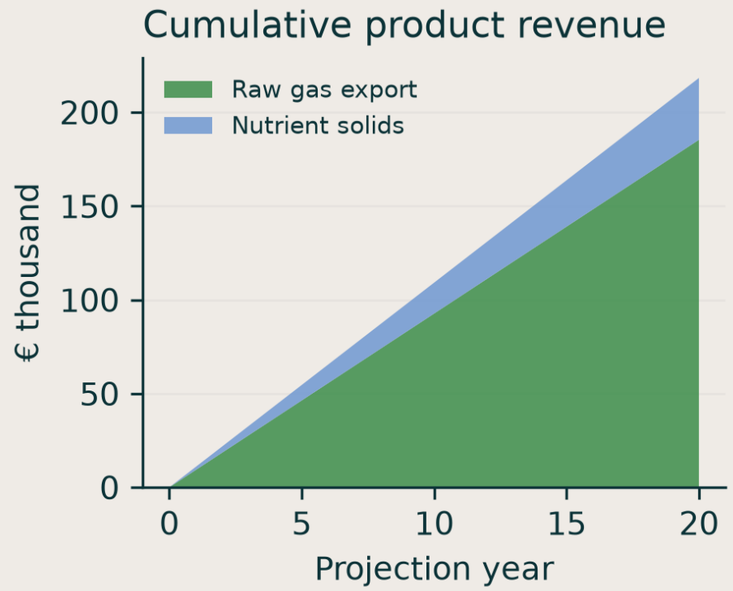
Product revenue, partial annual budget, capital subsidy and candidate overlays

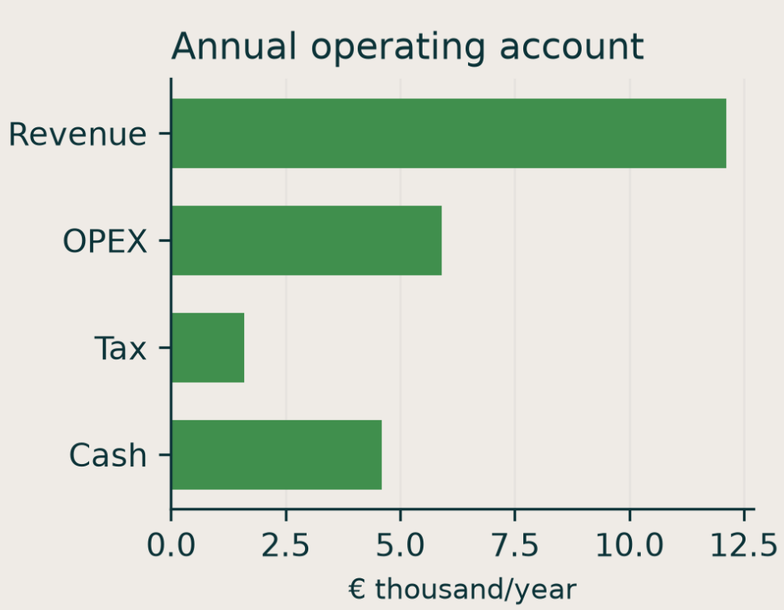
Scenario: all rank 9; 98b8467e23fc45c8a0136c61f4149274; 523.4 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

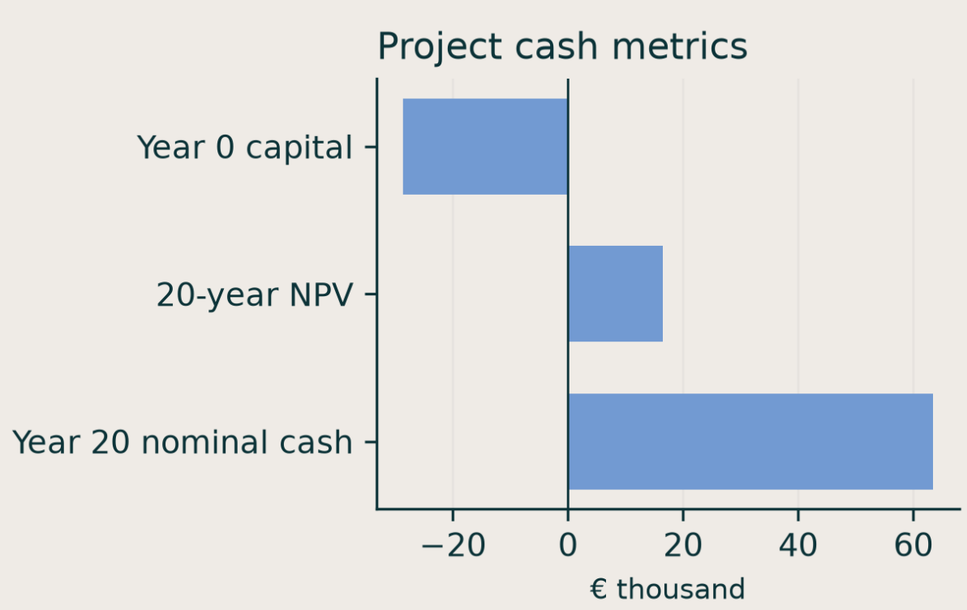
20-year financial projection

Protein X · Biogas ✓ · N X · P X · K X









Constant-account cash flow and cumulative product revenue over 20 years

Scenario: all rank 9; 98b8467e23fc45c8a0136c61f4149274; 523.4 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Balance and evidence

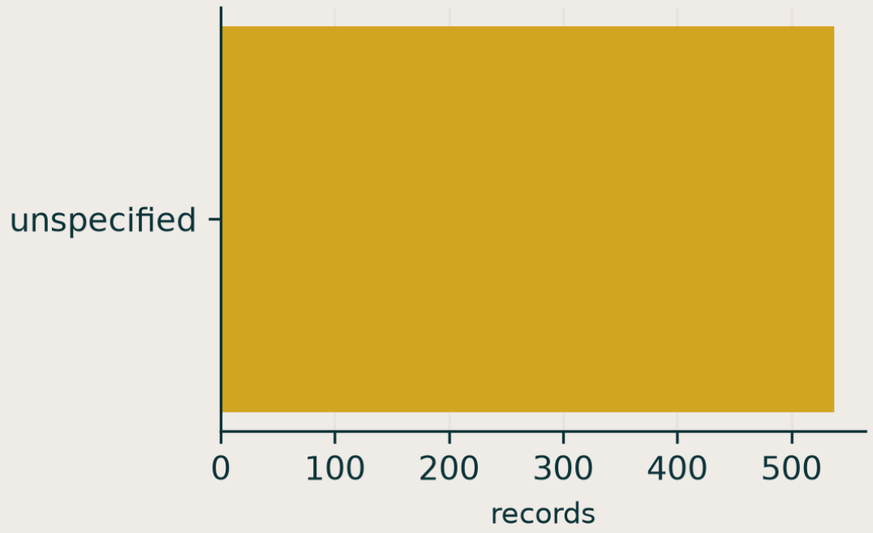
Protein X · Biogas ✓ · N X · P X · K X

Element closure

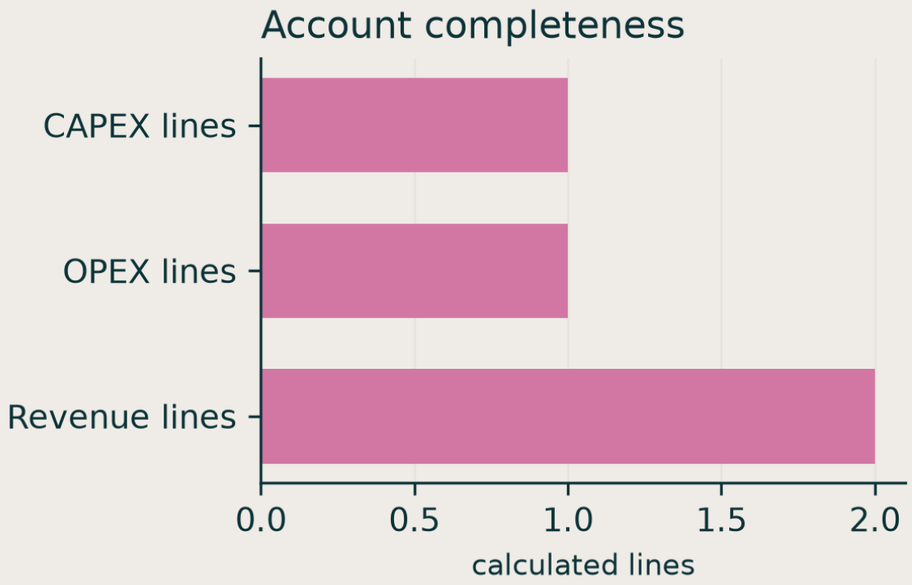
C	1.9e-12 kg/year
N	0 kg/year
P	0 kg/year
K	2.4e-13 kg/year

Numerical residuals

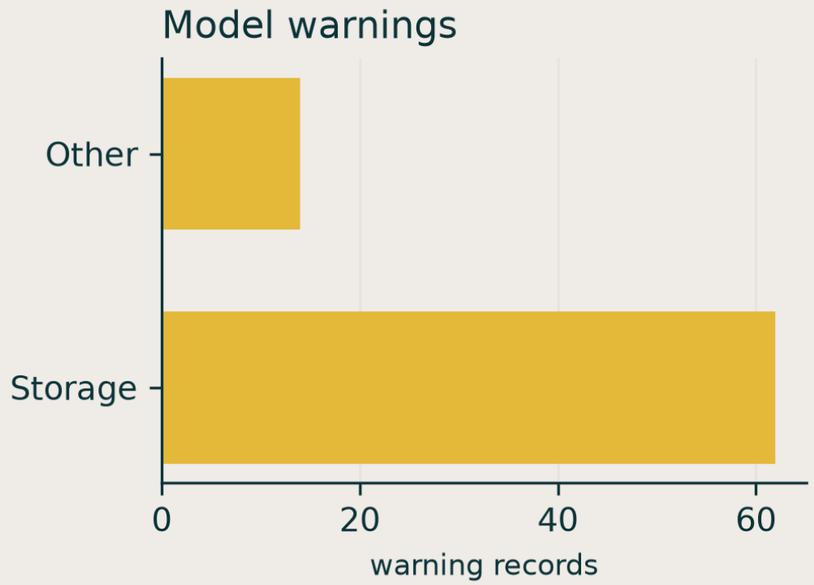
Parameter evidence



Account completeness



Model warnings



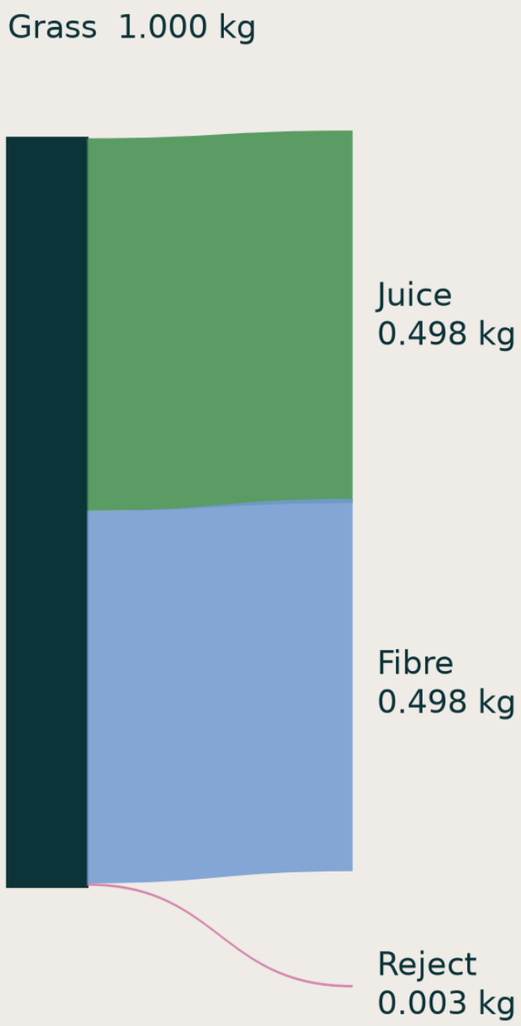
Numerical closure, evidence counts, calculated account lines and warnings

Scenario: all rank 9; 98b8467e23fc45c8a0136c61f4149274; 523.4 kg/h; 8760 h/year
Units: kg/year; records
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

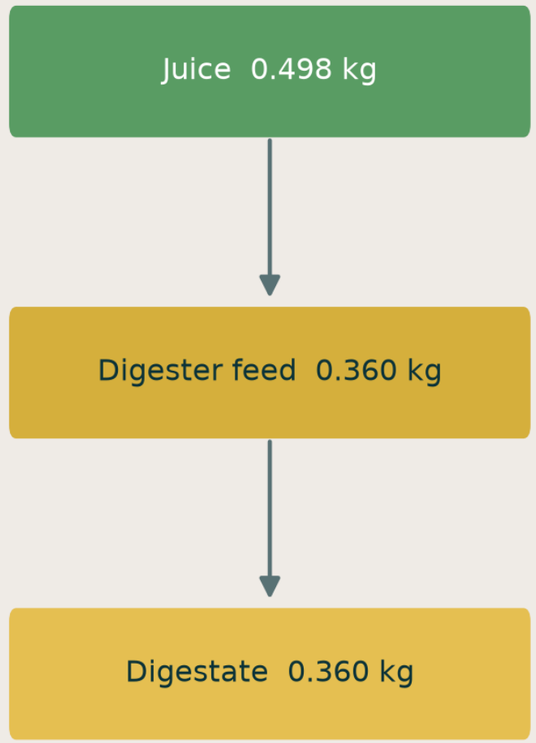
Grass to products

Protein X · Biogas X · N X · P X · K X

Grass split



Juice pathway



Selected intermediate streams

Recovered outputs

per kg wet grass

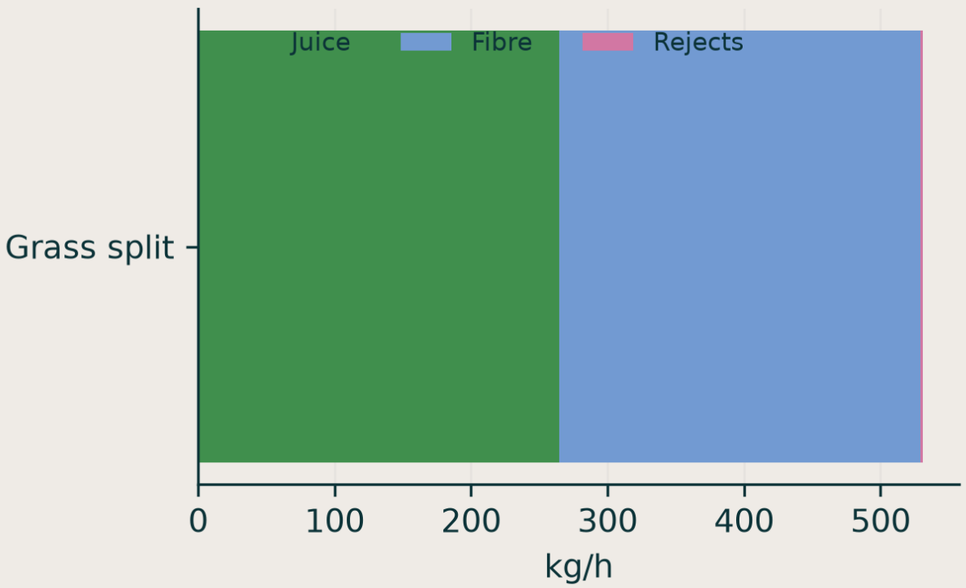
Mass-proportional grass split plus modeled downstream stream/product map

Scenario: all rank 10; 108b8cc293534e359644d2d9a95b1b08; 531.1 kg/h; 8760 h/year
Units: kg/kg wet grass; dry Nm³/kg wet grass
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Material and products

Protein X · Biogas X · N X · P X · K X

Feed destinations



Solid products

No modeled output

Gas at normal conditions

No modeled output

Specific product yields

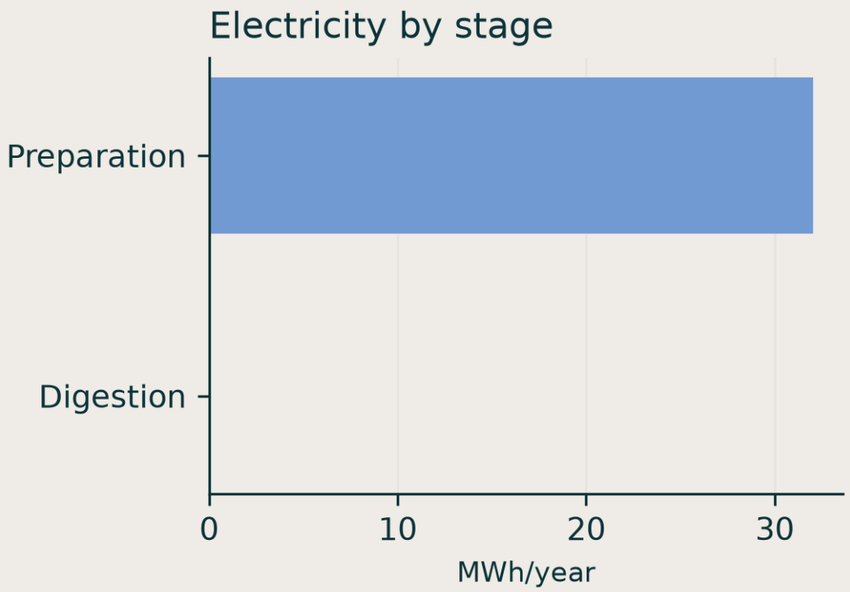
No modeled output

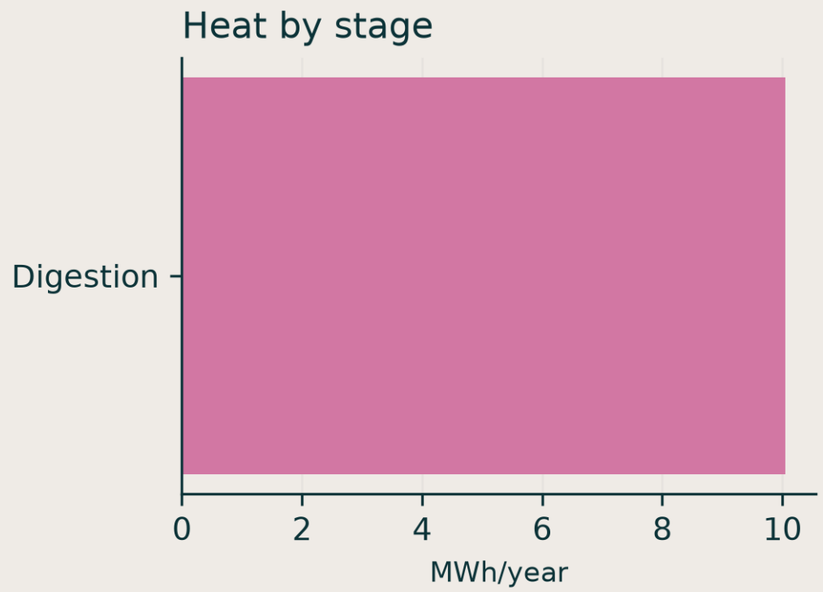
Front-end split, solid products, dry gas volumes and specific yields

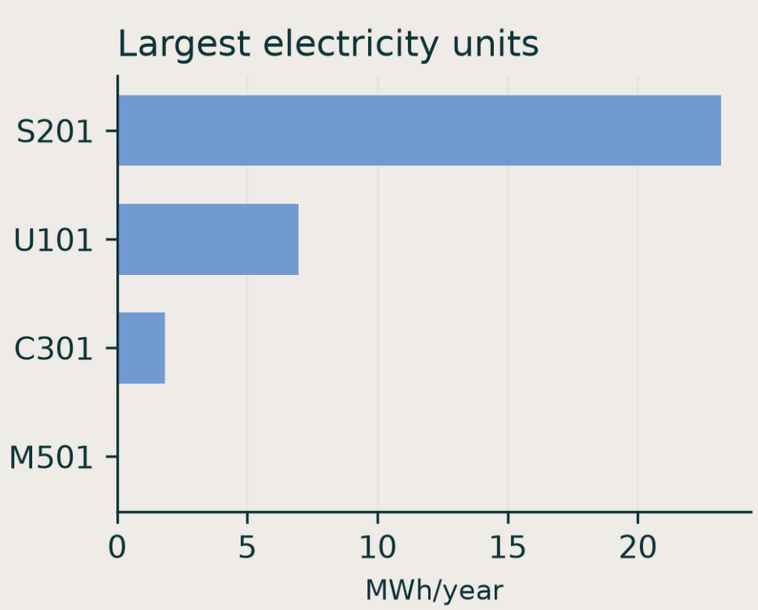
Scenario: all rank 10; 108b8cc293534e359644d2d9a95b1b08; 531.1 kg/h; 8760 h/year
Units: kg/h; t/year; dry Nm³/year; kg/t feed
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

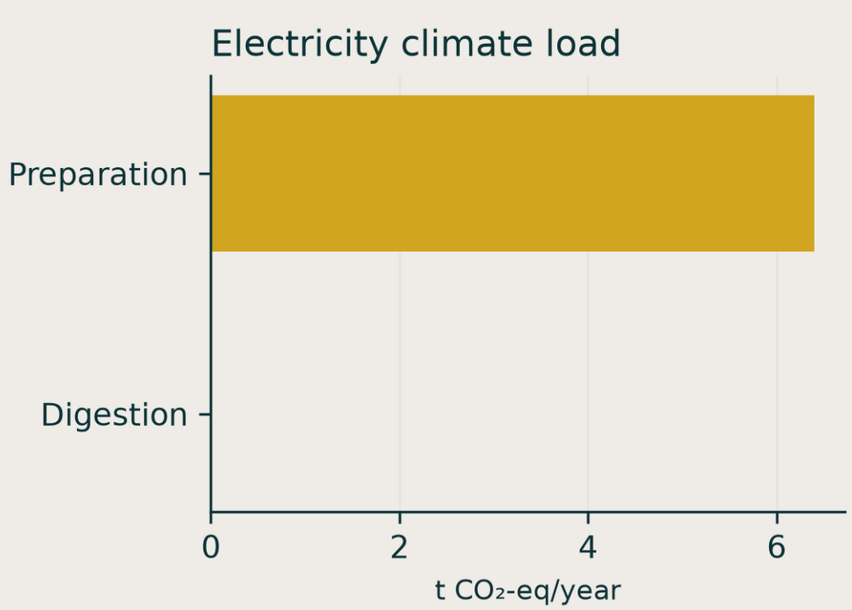
Energy and climate hotspots

Protein X · Biogas X · N X · P X · K X









Stage and unit utility hotspots and included climate load

Scenario: all rank 10; 108b8cc293534e359644d2d9a95b1b08; 531.1 kg/h; 8760 h/year
Units: MWh/year; t CO_2-eq/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

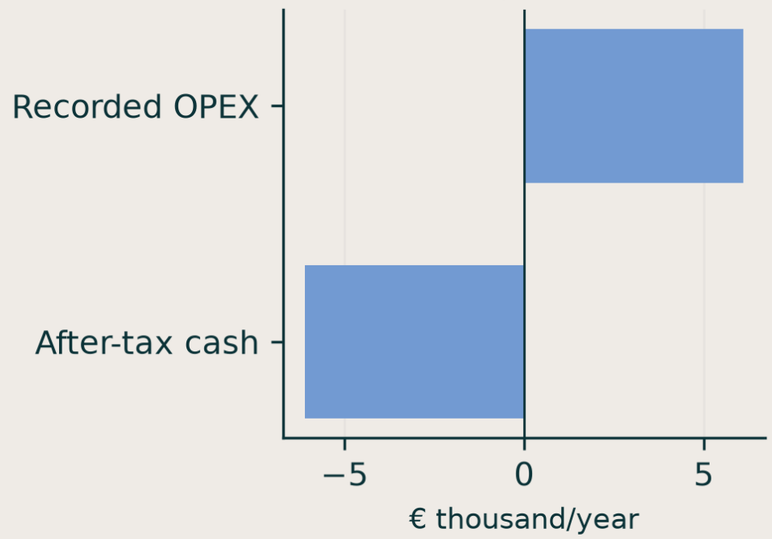
Economic accounts

Protein X · Biogas X · N X · P X · K X

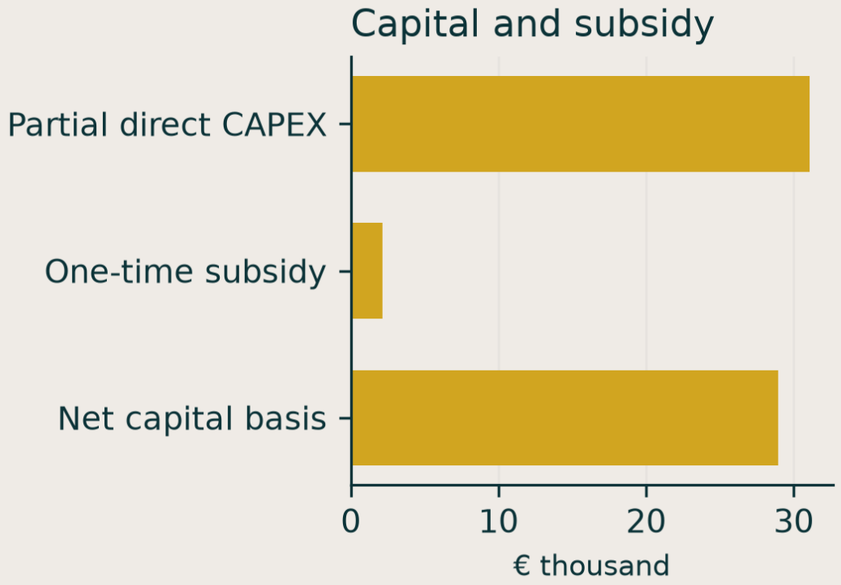
Product revenue

No modeled output

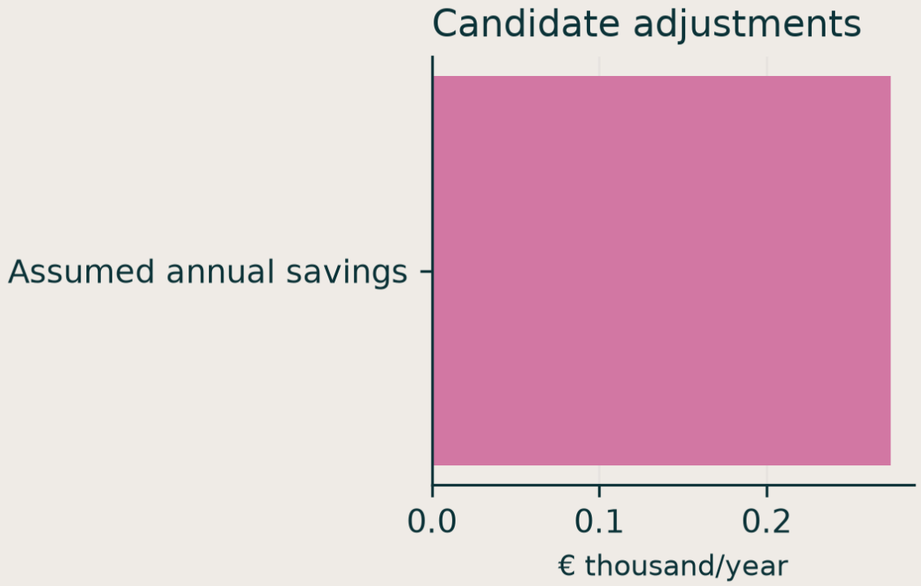
Annual budget



Capital and subsidy



Candidate adjustments

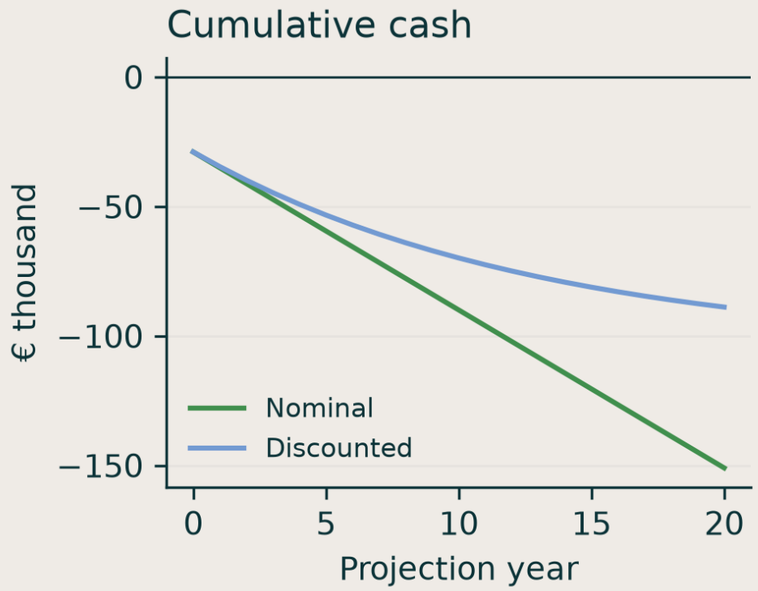


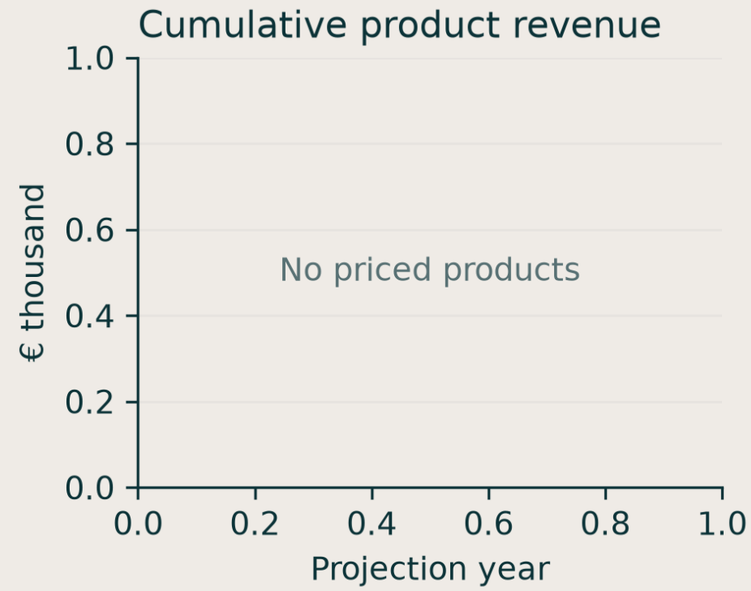
Product revenue, partial annual budget, capital subsidy and candidate overlays

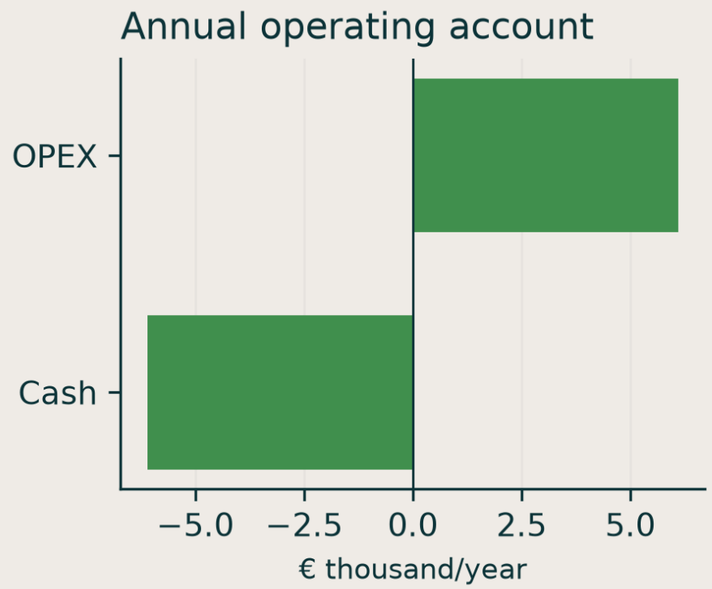
Scenario: all rank 10; 108b8cc293534e359644d2d9a95b1b08; 531.1 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

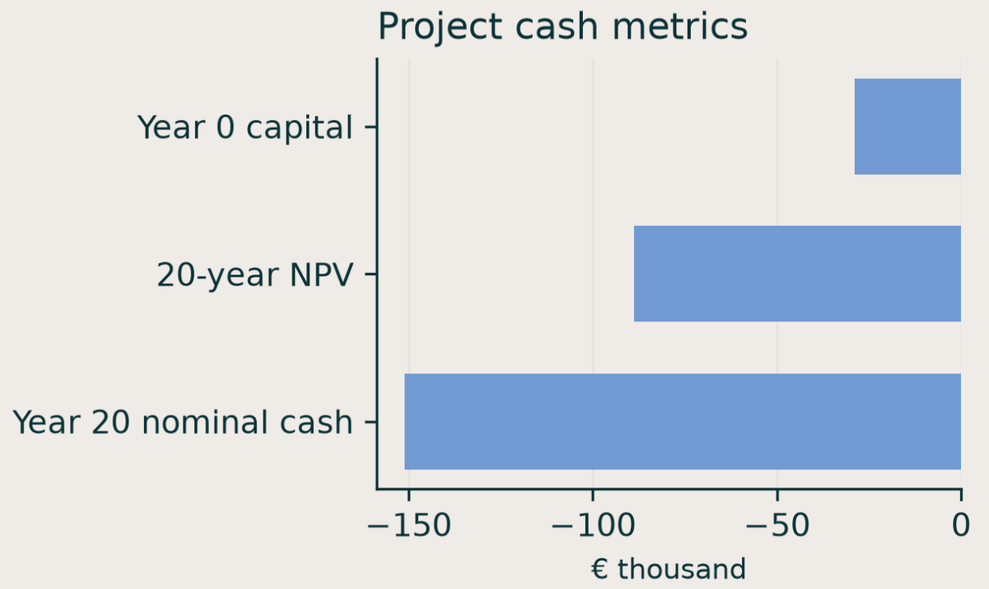
20-year financial projection

Protein X · Biogas X · N X · P X · K X









Constant-account cash flow and cumulative product revenue over 20 years

Scenario: all rank 10; 108b8cc293534e359644d2d9a95b1b08; 531.1 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Balance and evidence

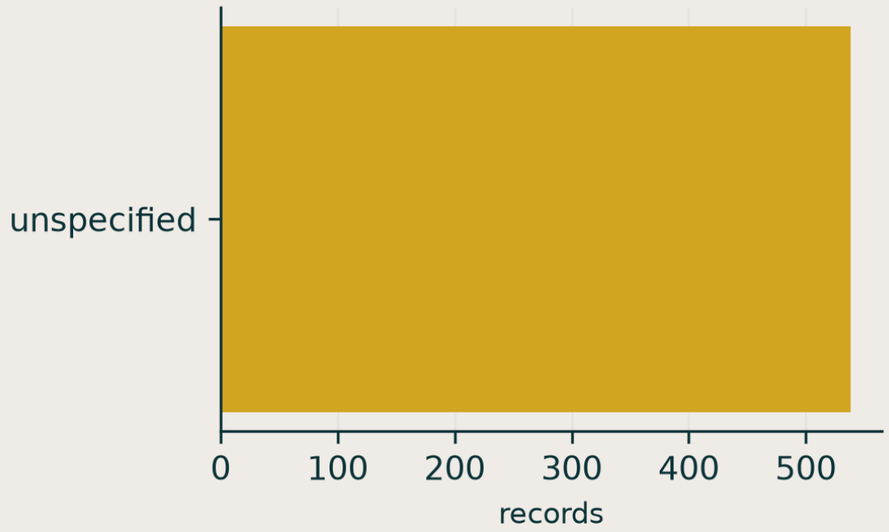
Protein X · Biogas X · N X · P X · K X

Element closure

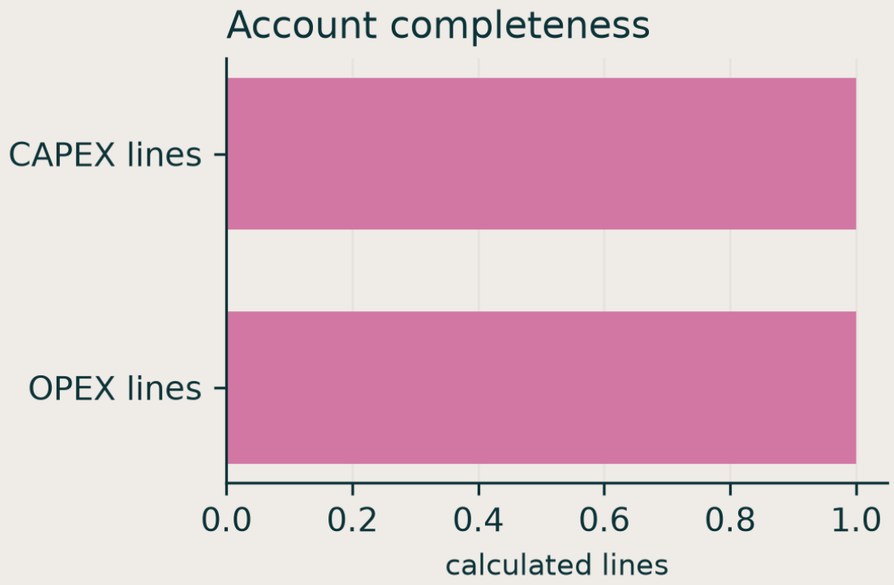
C	0 kg/year
N	0 kg/year
P	0 kg/year
K	0 kg/year

Numerical residuals

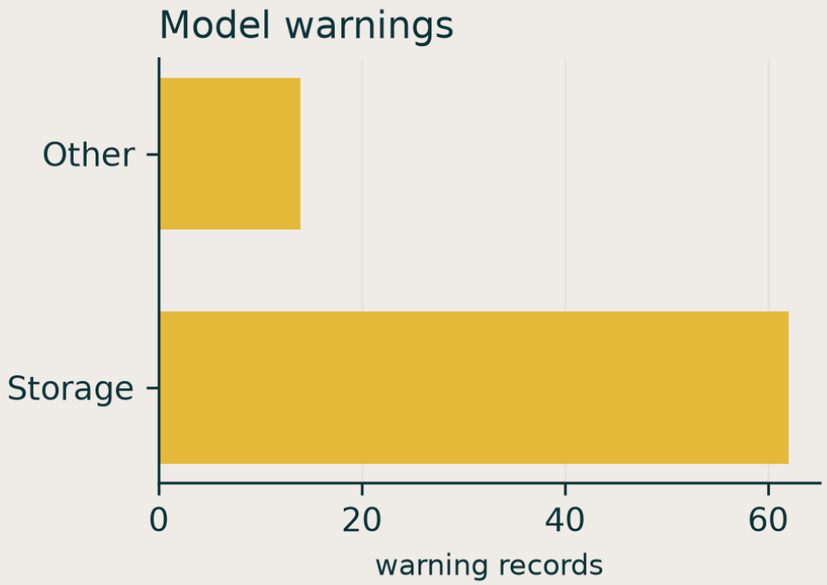
Parameter evidence



Account completeness



Model warnings



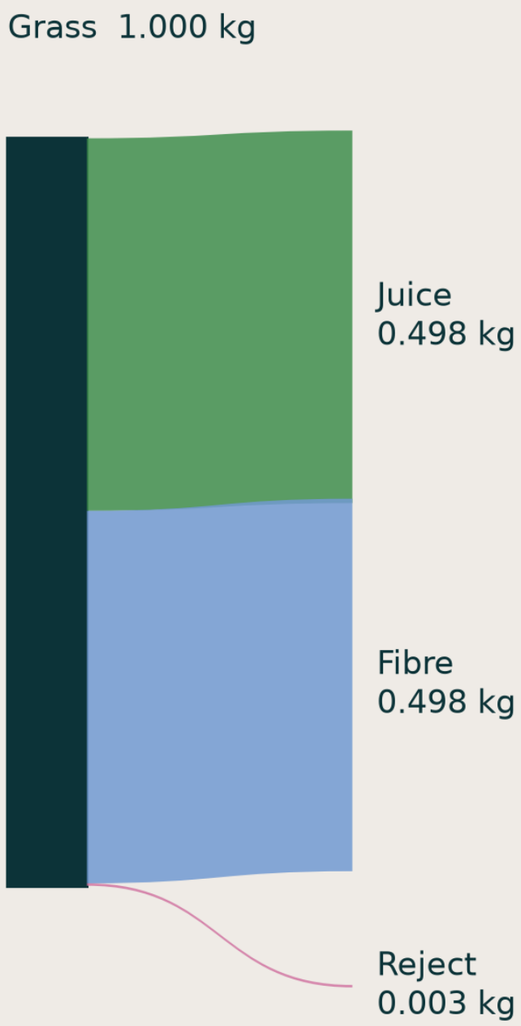
Numerical closure, evidence counts, calculated account lines and warnings

Scenario: all rank 10; 108b8cc293534e359644d2d9a95b1b08; 531.1 kg/h; 8760 h/year
Units: kg/year; records
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

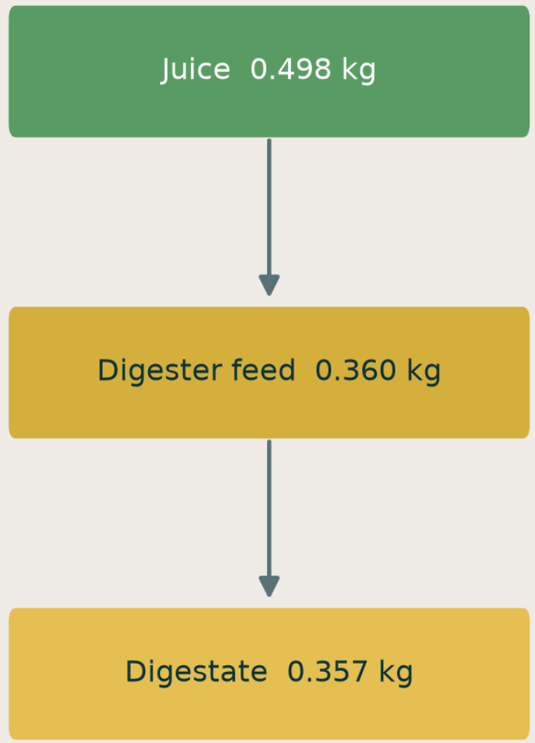
Grass to products

Protein X · Biogas ✓ · N ✓ · P X · K X

Grass split

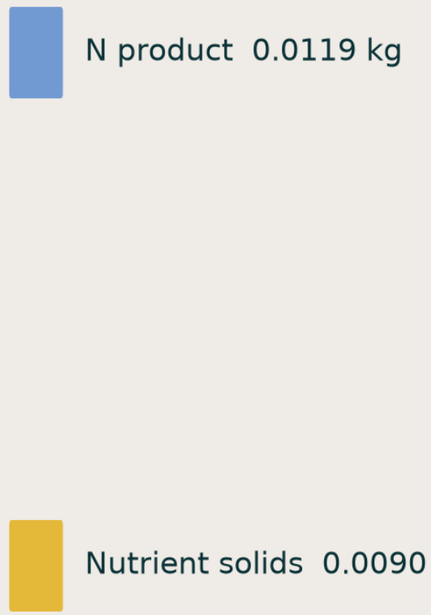


Juice pathway



Selected intermediate streams

Recovered outputs



Raw gas 0.0021 Nm³

Raw export 0.0021 Nm³

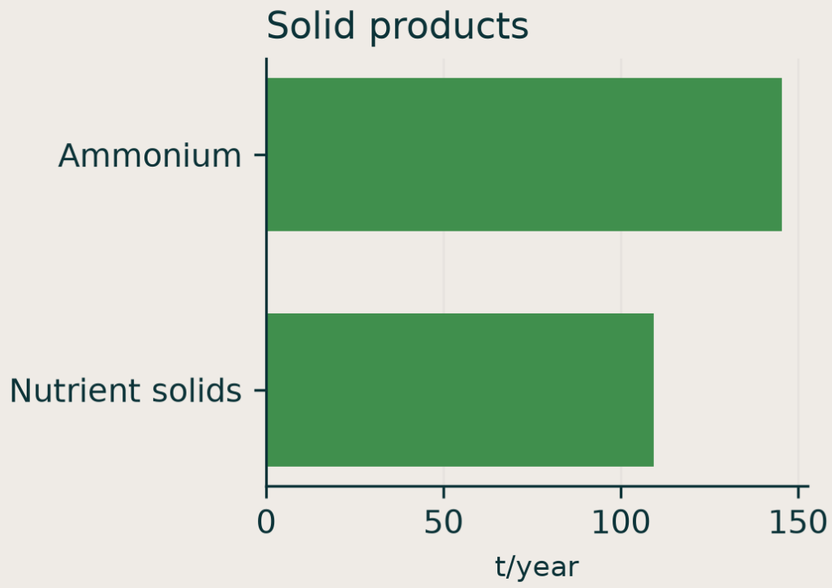
per kg wet grass

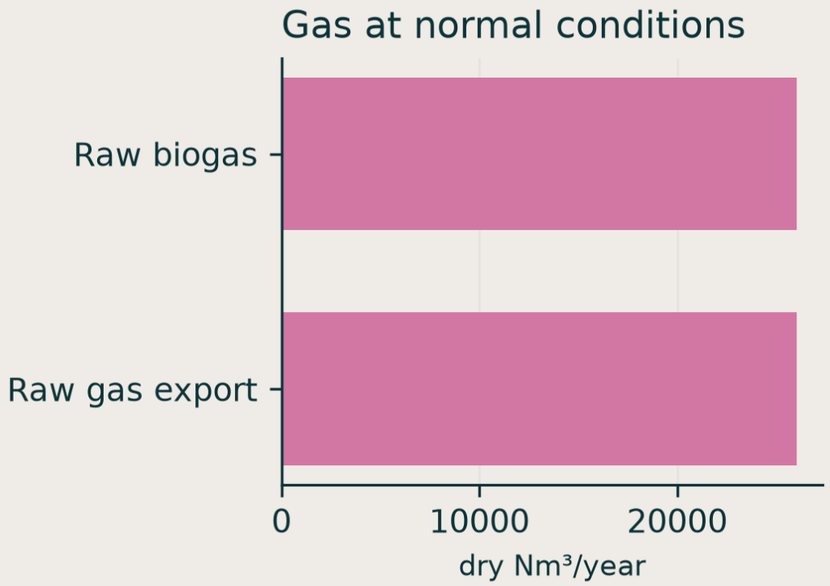
Mass-proportional grass split plus modeled downstream stream/product map

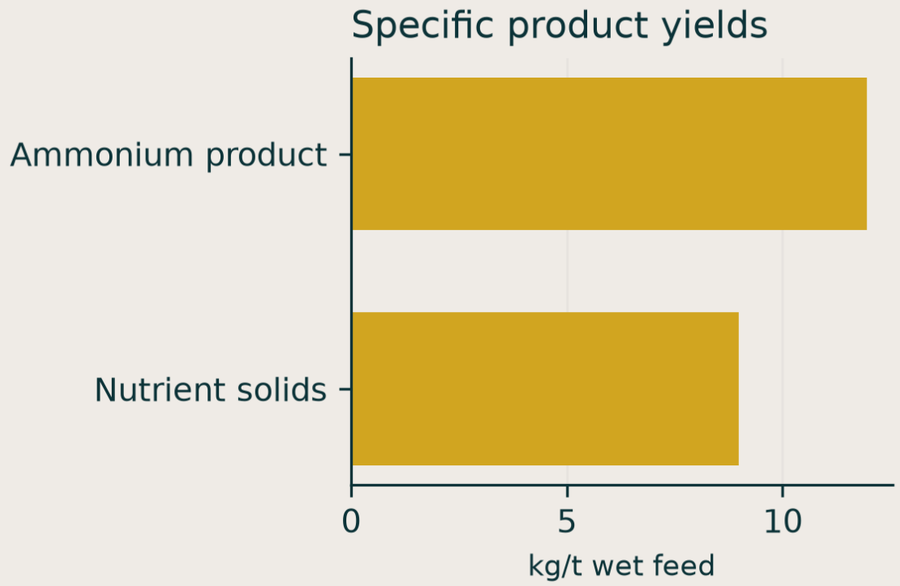
Scenario: matched_protein_bypass rank 0; 72e8ca02a5f84e1ab22d09f6eef68c9a; 1389.6 kg/h; 8760 h/year
Units: kg/kg wet grass; dry Nm³/kg wet grass
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Material and products

Protein X · Biogas ✓ · N ✓ · P X · K X





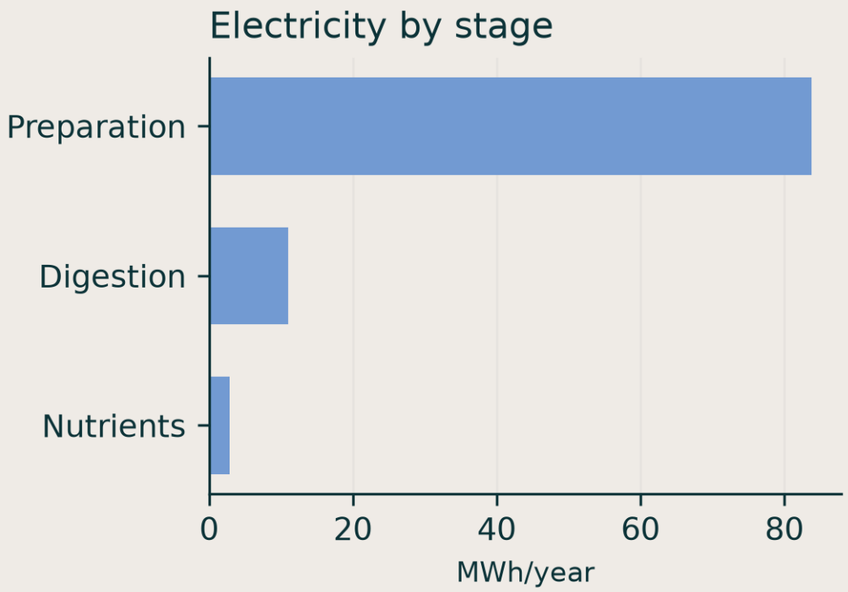


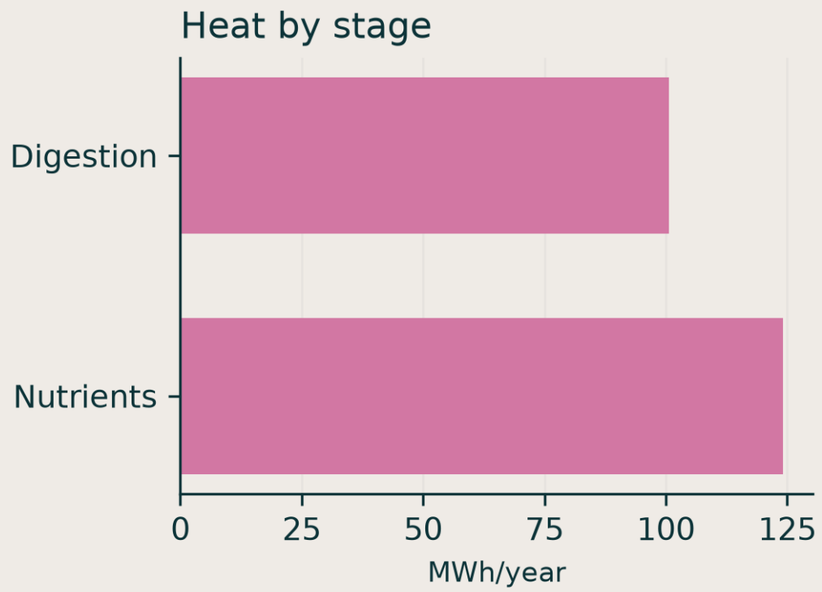
Front-end split, solid products, dry gas volumes and specific yields

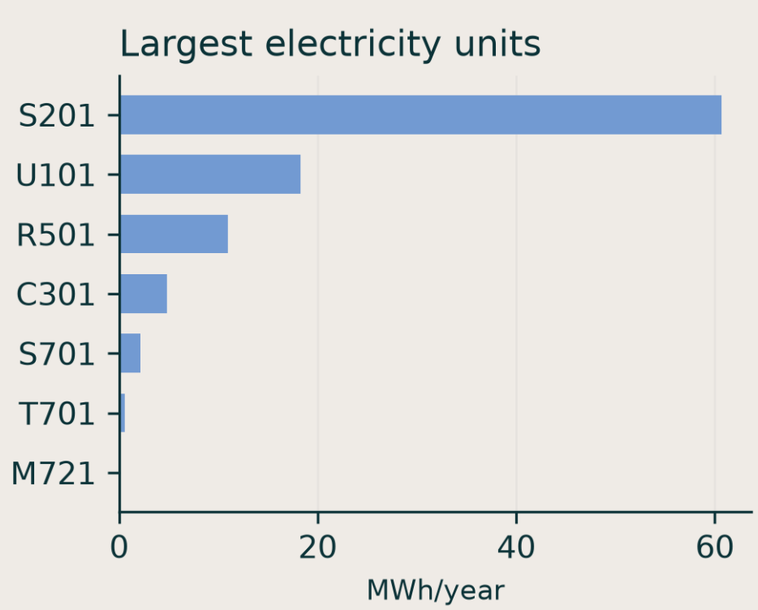
Scenario: matched_protein_bypass rank 0; 72e8ca02a5f84e1ab22d09f6eef68c9a; 1389.6 kg/h; 8760 h/year
Units: kg/h; t/year; dry Nm³/year; kg/t feed
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

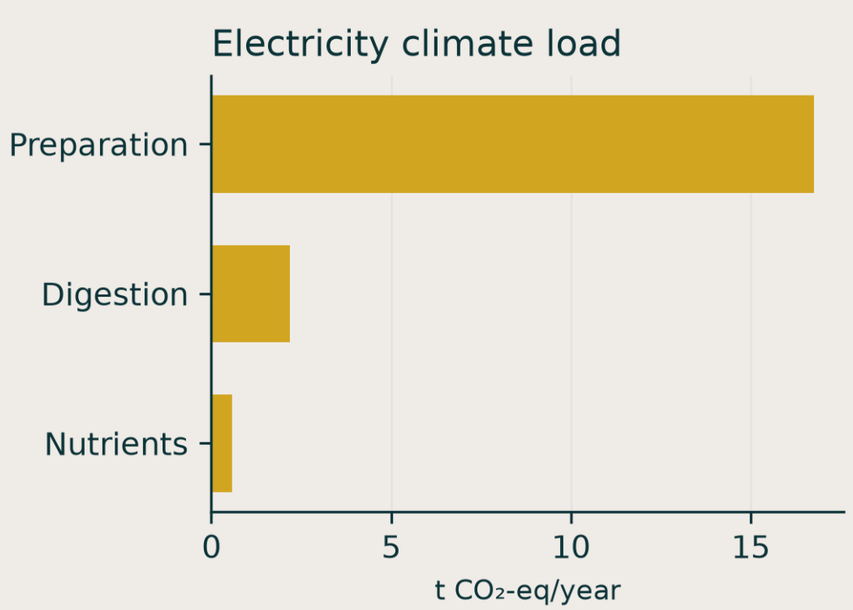
Energy and climate hotspots

Protein X · Biogas ✓ · N ✓ · P X · K X







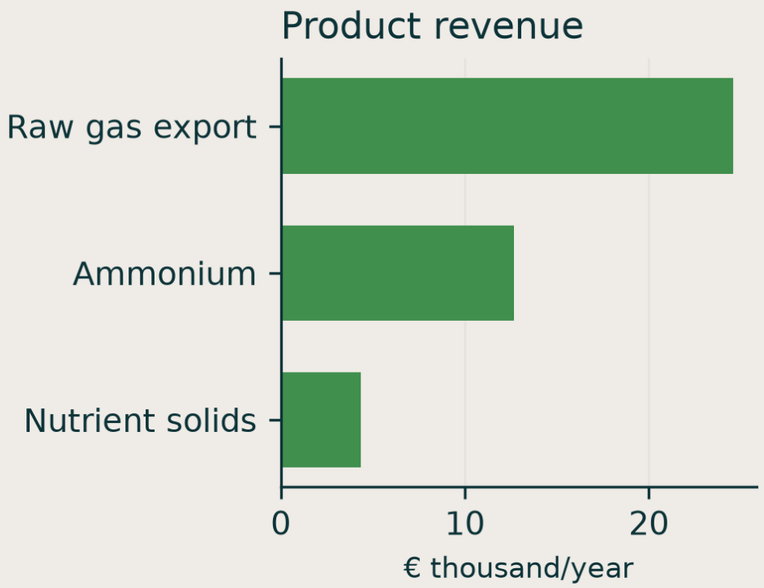


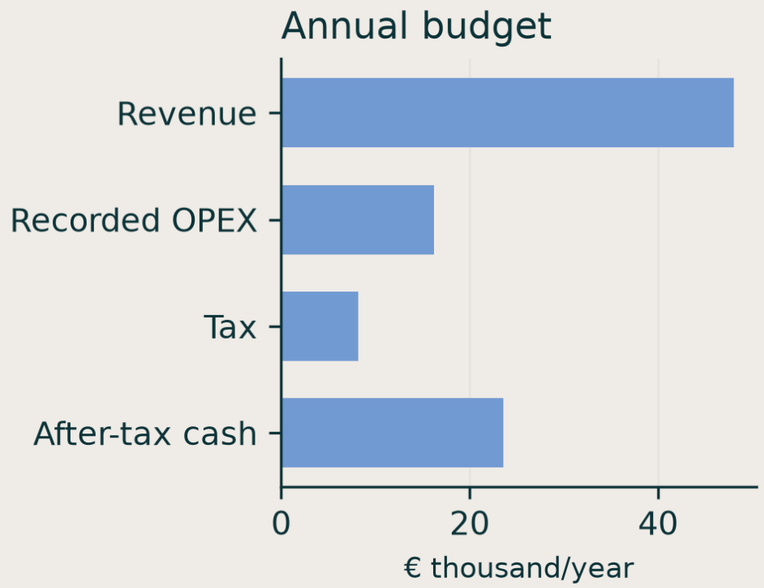
Stage and unit utility hotspots and included climate load

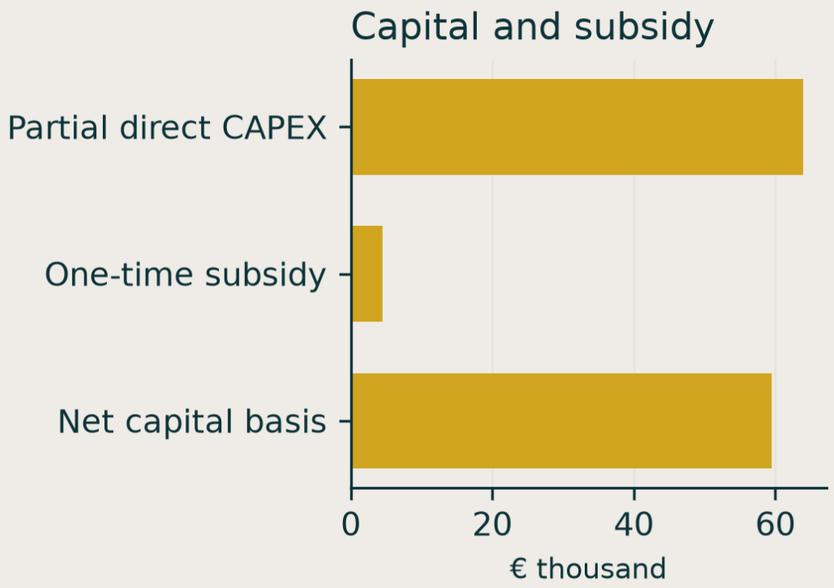
Scenario: matched_protein_bypass rank 0; 72e8ca02a5f84e1ab22d09f6eef68c9a; 1389.6 kg/h; 8760 h/year
Units: MWh/year; t CO_2-eq/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

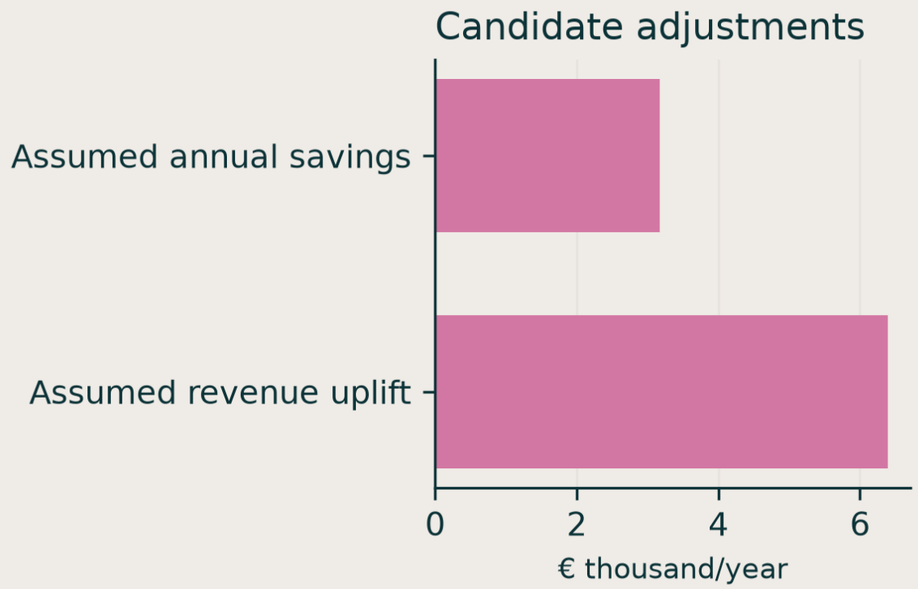
Economic accounts

Protein X · Biogas ✓ · N ✓ · P X · K X







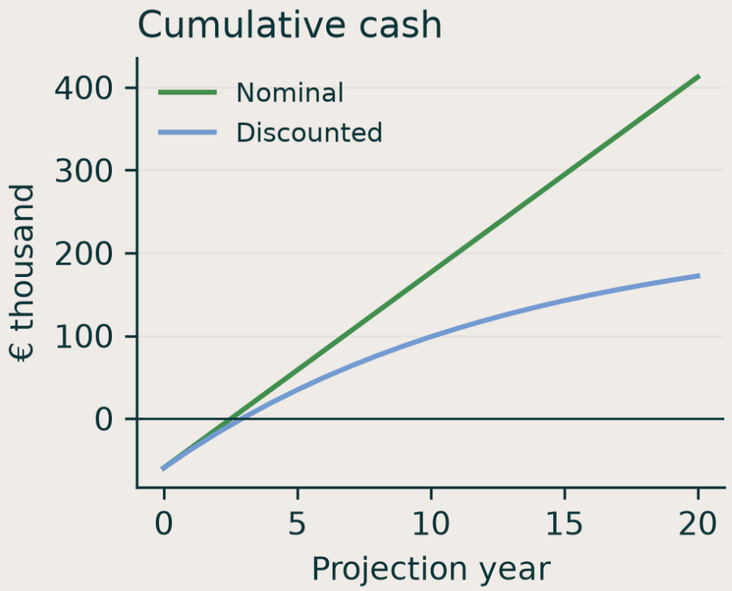


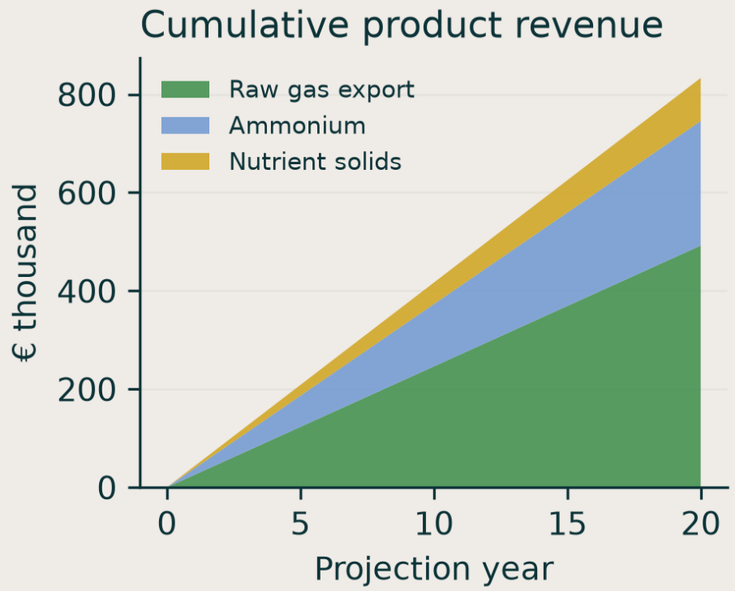
Product revenue, partial annual budget, capital subsidy and candidate overlays

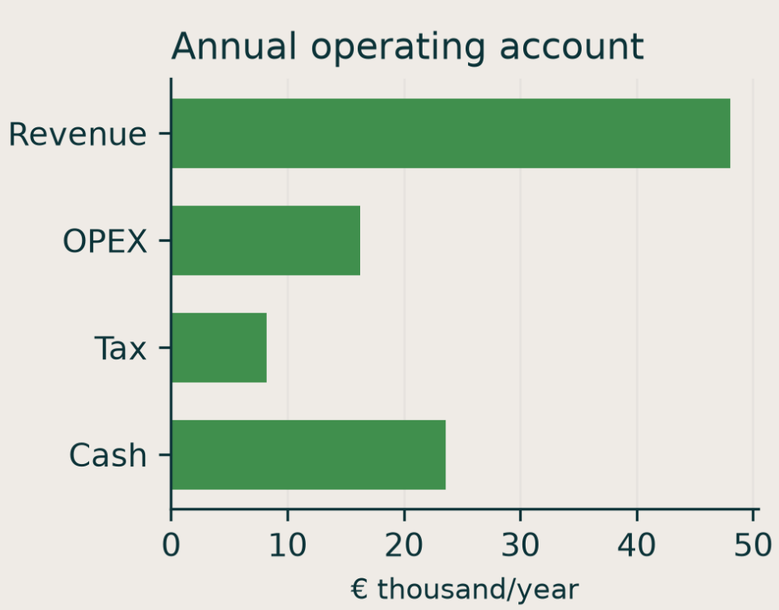
Scenario: matched_protein_bypass rank 0; 72e8ca02a5f84e1ab22d09f6eef68c9a; 1389.6 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

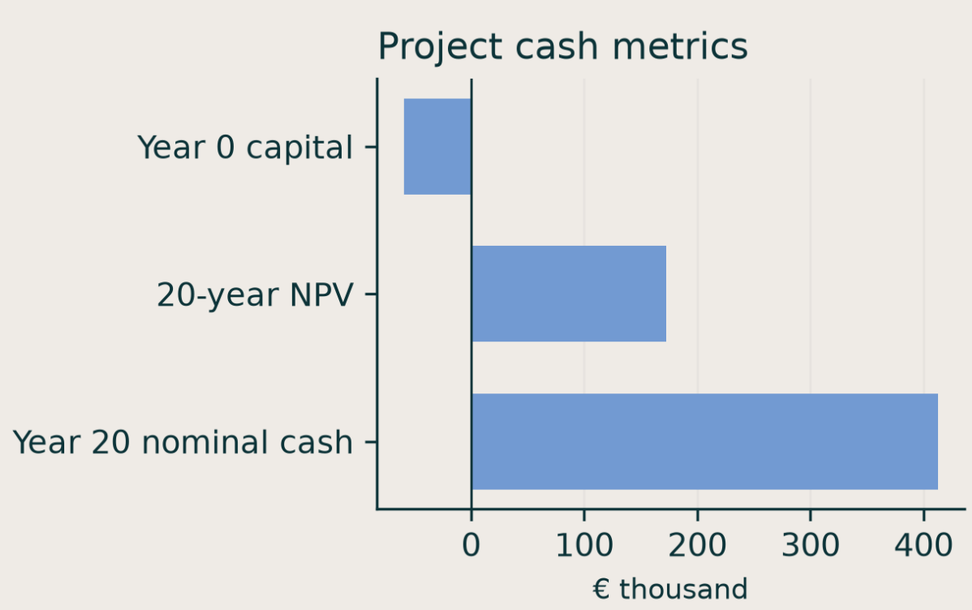
20-year financial projection

Protein X · Biogas ✓ · N ✓ · P X · K X









Constant-account cash flow and cumulative product revenue over 20 years

Scenario: matched_protein_bypass rank 0; 72e8ca02a5f84e1ab22d09f6eef68c9a; 1389.6 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Balance and evidence

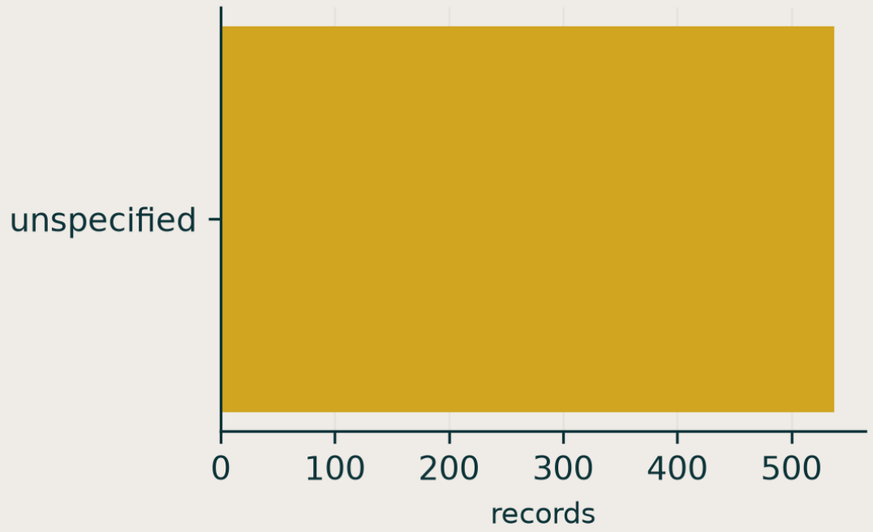
Protein X · Biogas ✓ · N ✓ · P X · K X

Element closure

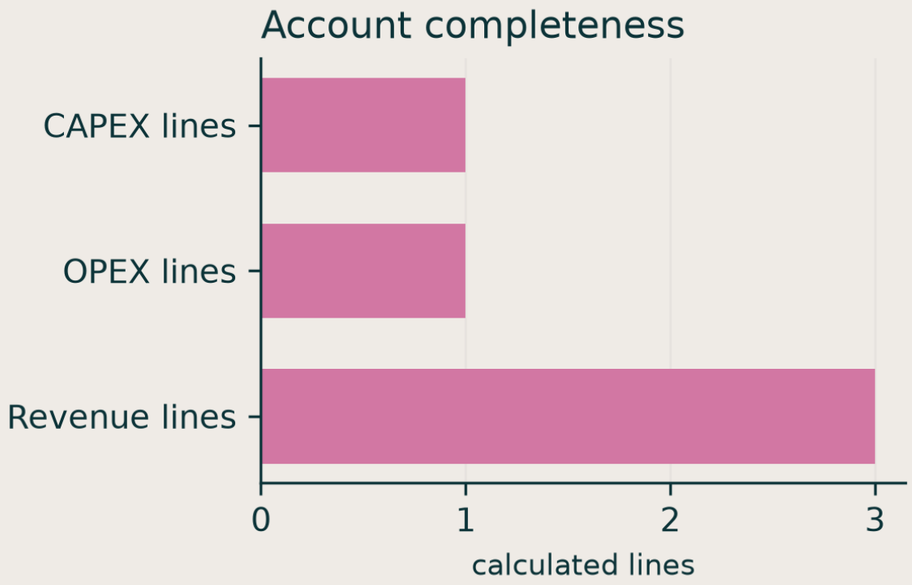
C	0 kg/year
N	0 kg/year
P	0 kg/year
K	0 kg/year

Numerical residuals

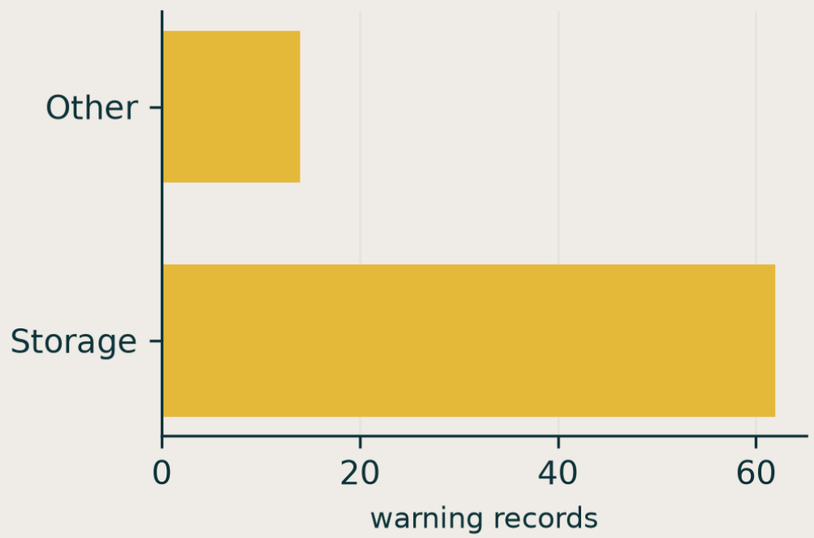
Parameter evidence



Account completeness



Model warnings



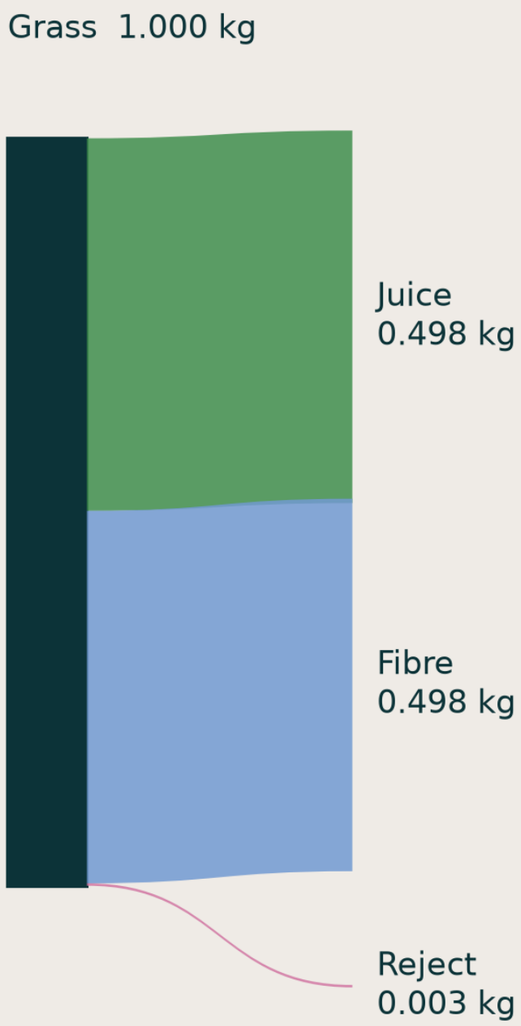
Numerical closure, evidence counts, calculated account lines and warnings

Scenario: matched_protein_bypass rank 0; 72e8ca02a5f84e1ab22d09f6eef68c9a; 1389.6 kg/h; 8760 h/year
Units: kg/year; records
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

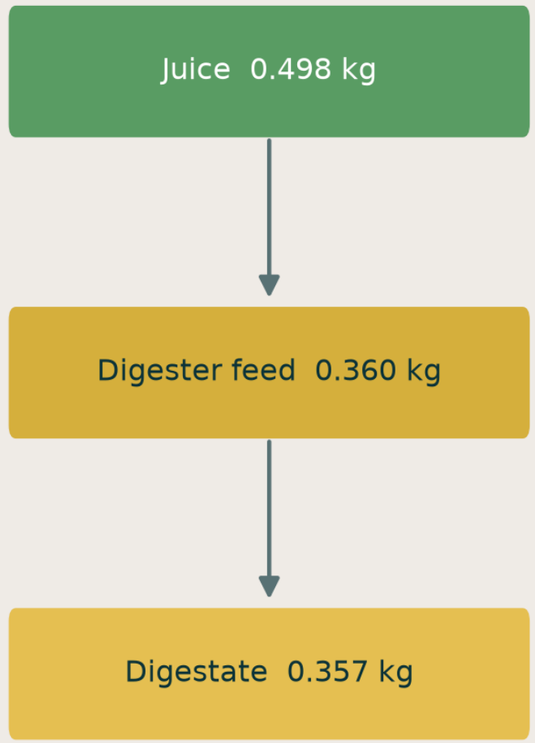
Grass to products

Protein X · Biogas ✓ · N ✓ · P X · K X

Grass split

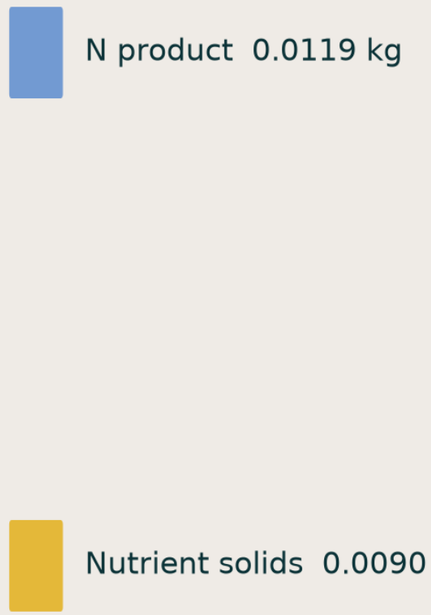


Juice pathway



Selected intermediate streams

Recovered outputs



Raw gas 0.0021 Nm³

Upgraded 0.0019 Nm³

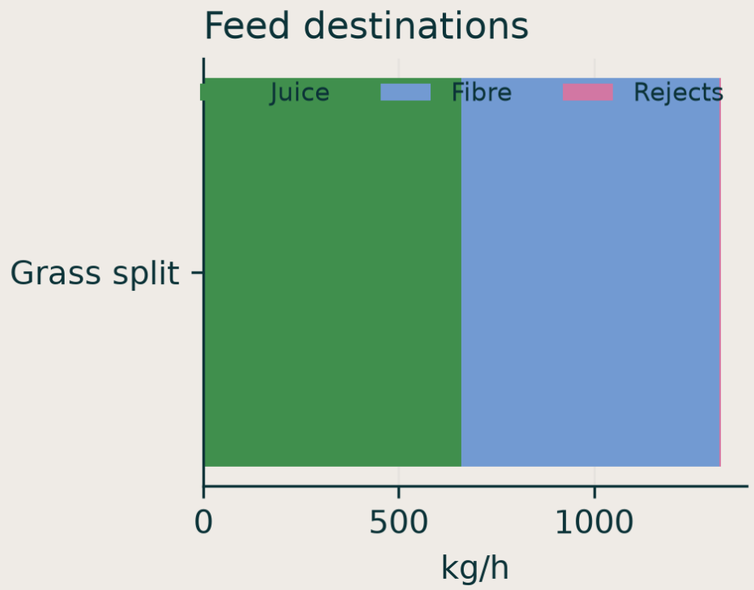
per kg wet grass

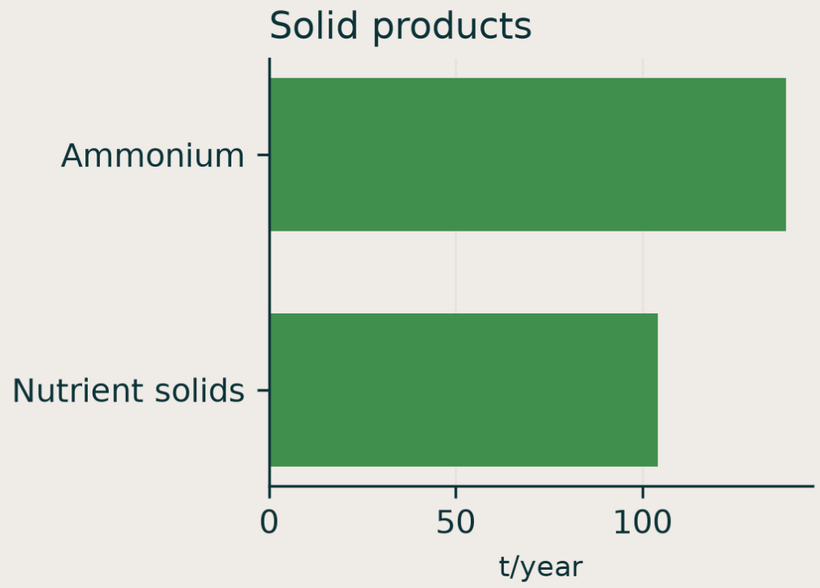
Mass-proportional grass split plus modeled downstream stream/product map

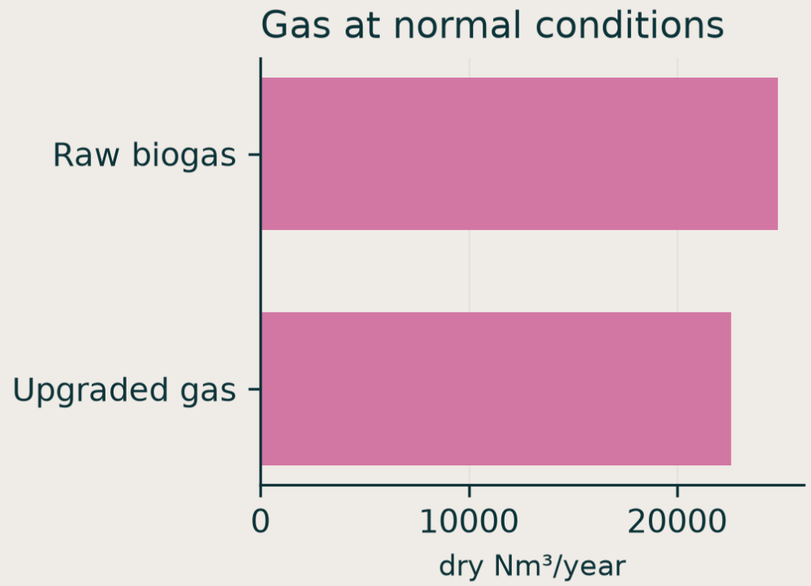
Scenario: matched_protein_bypass rank 1; 9b68969ce02b4f5c94b6ff0f59b53447; 1324.8 kg/h; 8760 h/year
Units: kg/kg wet grass; dry Nm³/kg wet grass
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

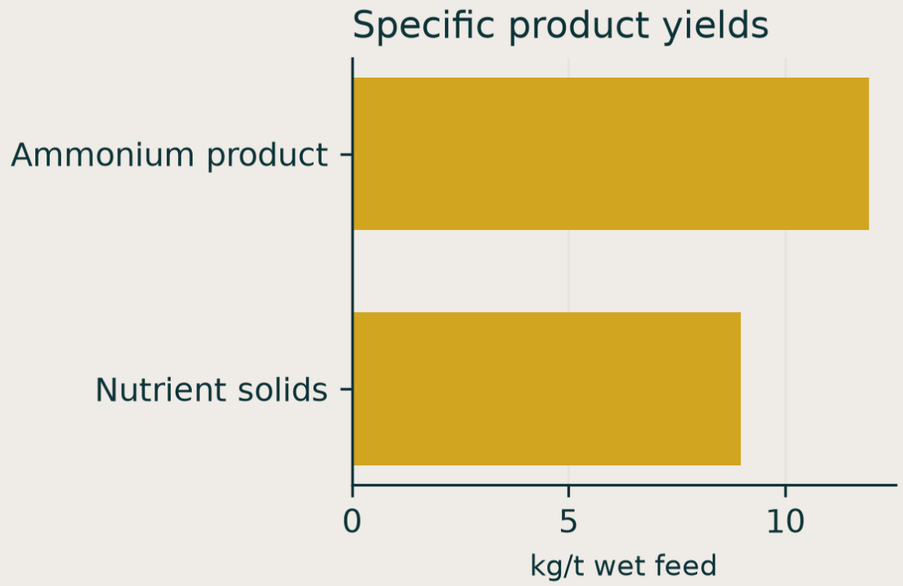
Material and products

Protein X · Biogas ✓ · N ✓ · P X · K X







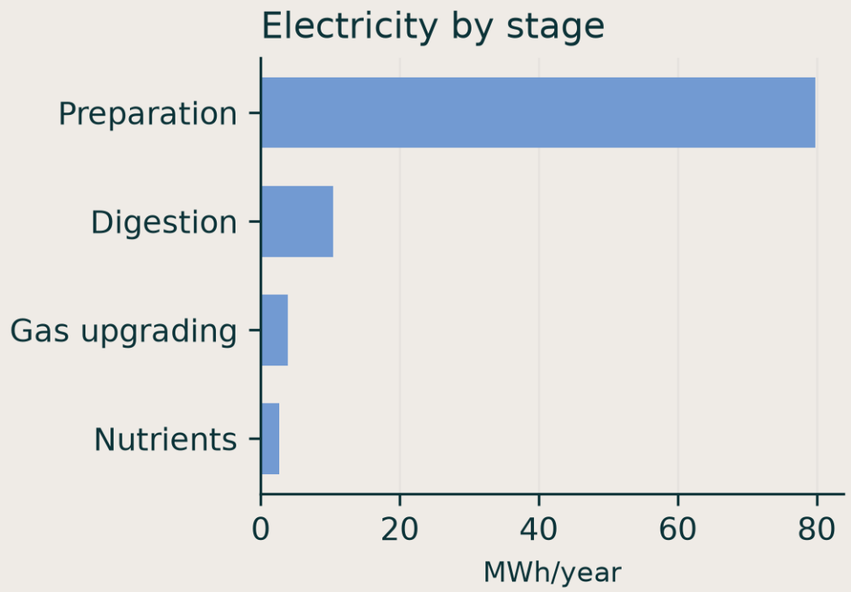


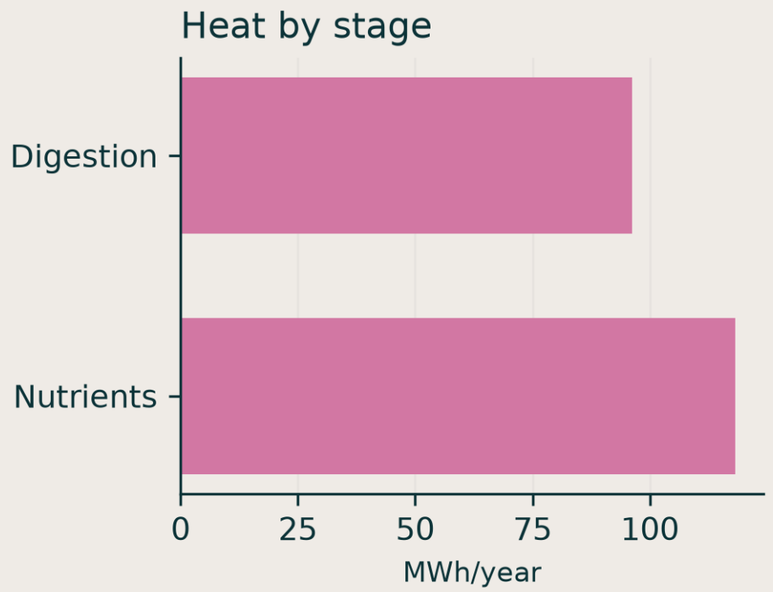
Front-end split, solid products, dry gas volumes and specific yields

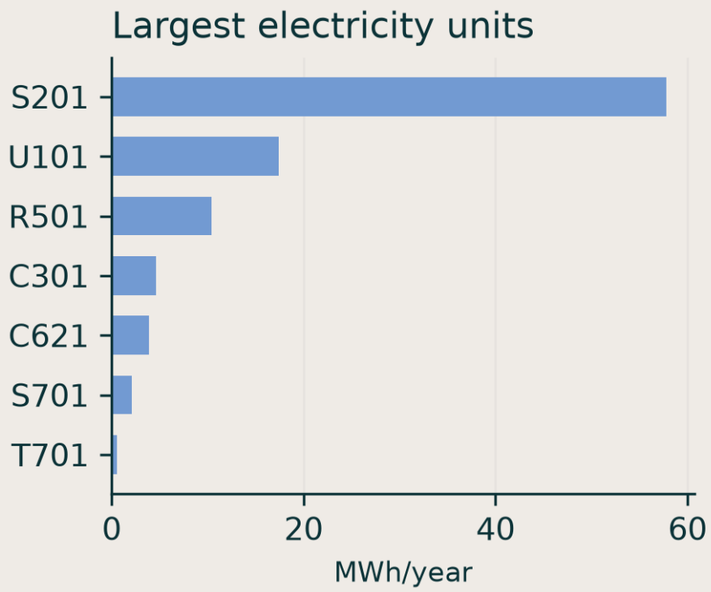
Scenario: matched_protein_bypass rank 1; 9b68969ce02b4f5c94b6ff0f59b53447; 1324.8 kg/h; 8760 h/year
Units: kg/h; t/year; dry Nm³/year; kg/t feed
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

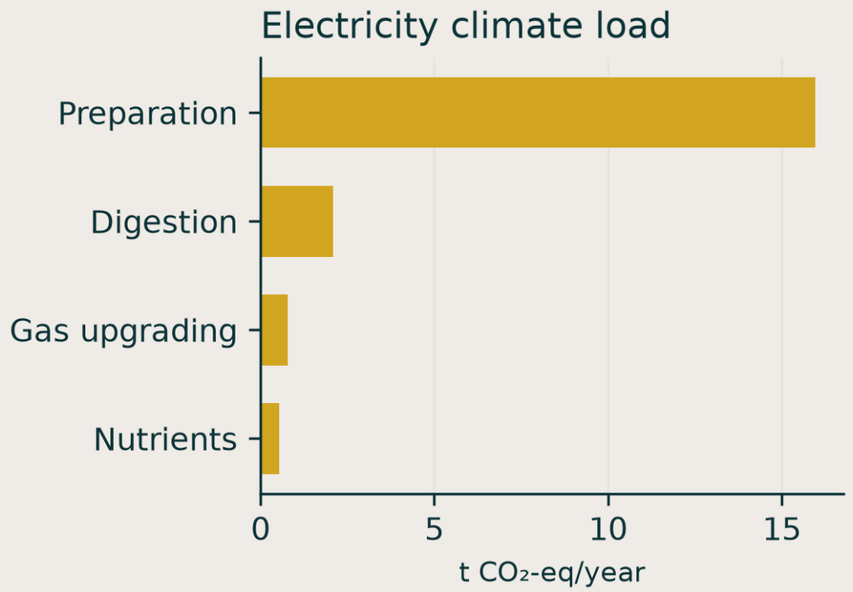
Energy and climate hotspots

Protein X · Biogas ✓ · N ✓ · P X · K X







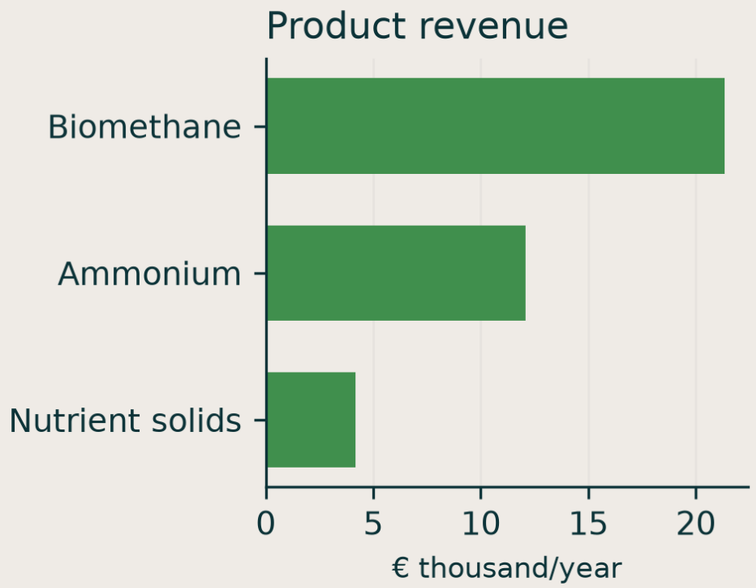


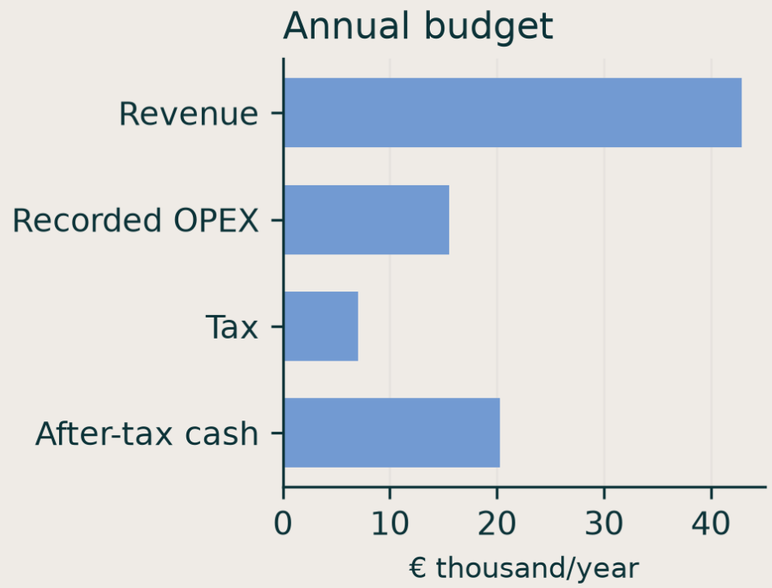
Stage and unit utility hotspots and included climate load

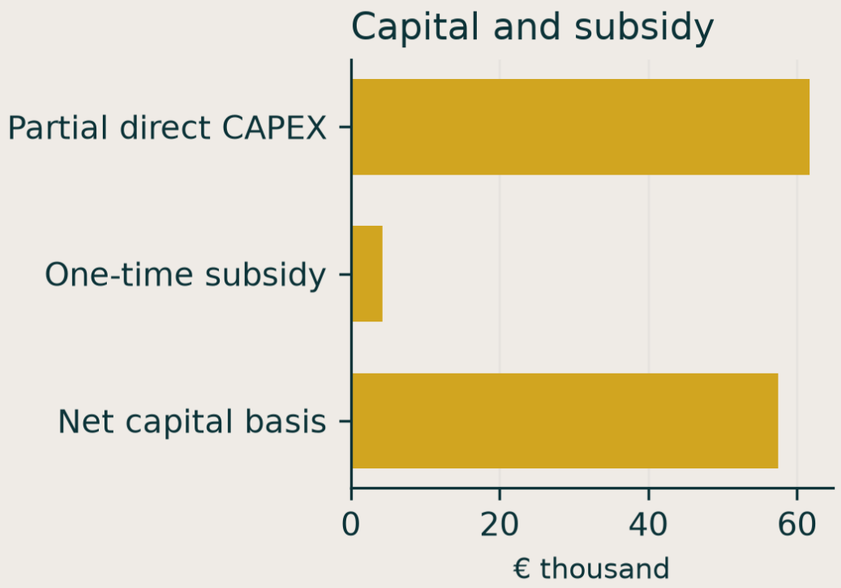
Scenario: matched_protein_bypass rank 1; 9b68969ce02b4f5c94b6ff0f59b53447; 1324.8 kg/h; 8760 h/year
Units: MWh/year; t CO_2-eq/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

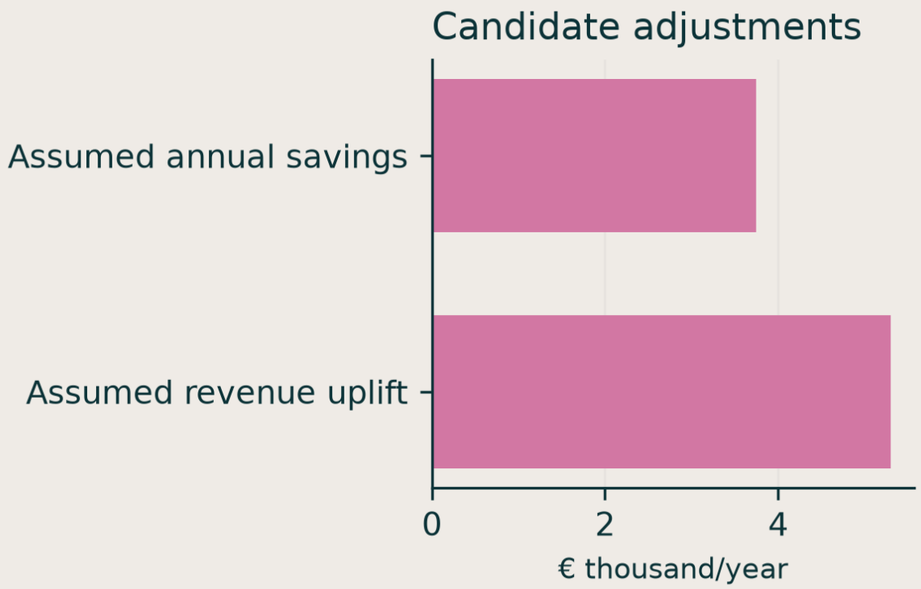
Economic accounts

Protein X · Biogas ✓ · N ✓ · P X · K X







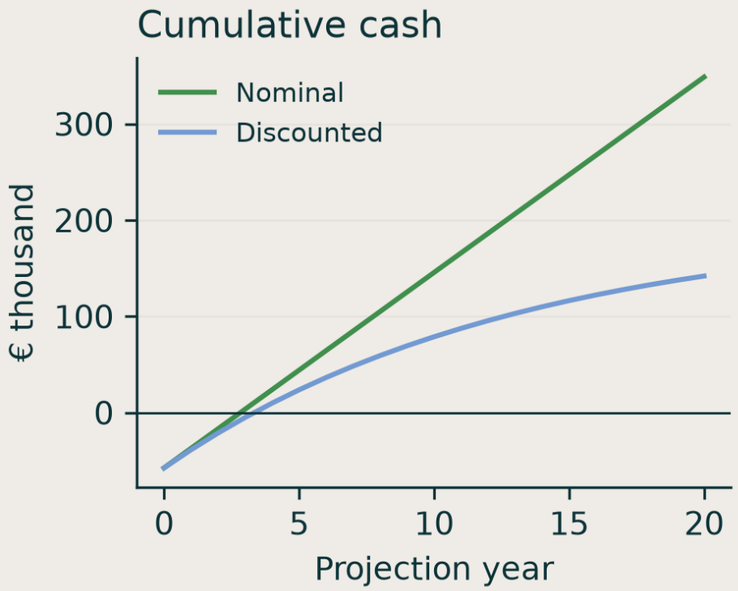


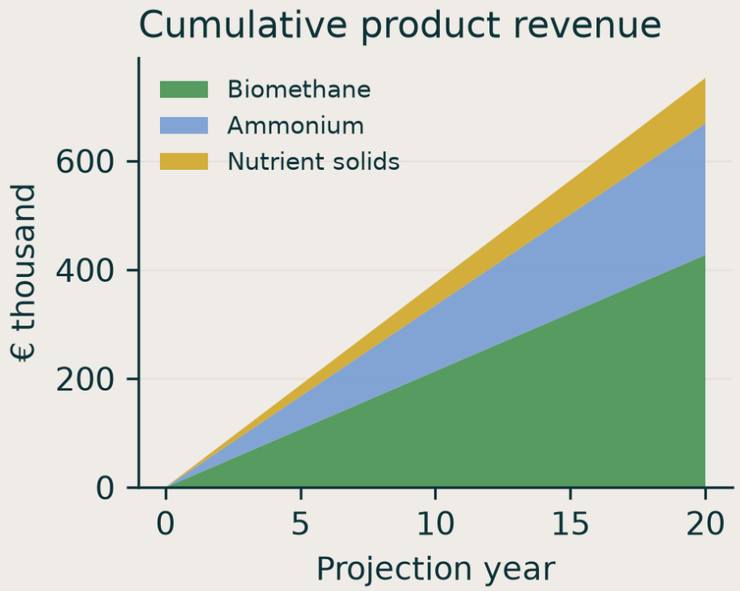
Product revenue, partial annual budget, capital subsidy and candidate overlays

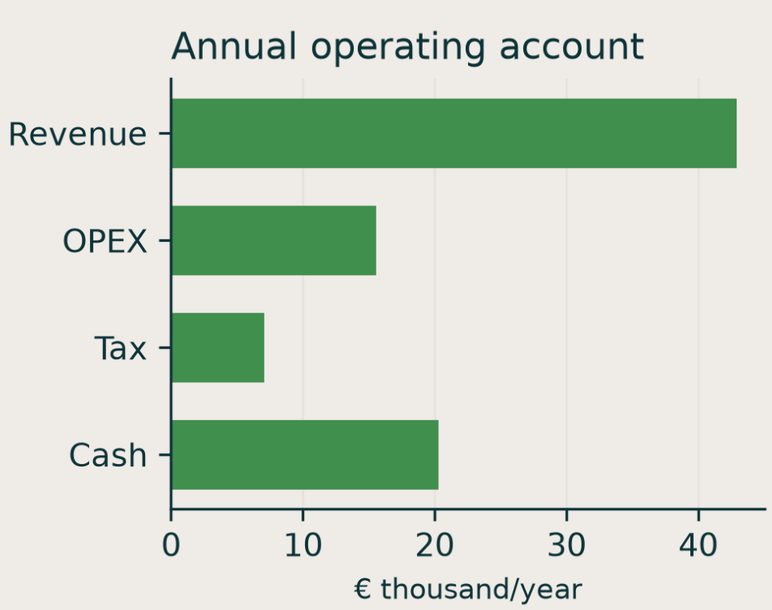
Scenario: matched_protein_bypass rank 1; 9b68969ce02b4f5c94b6ff0f59b53447; 1324.8 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

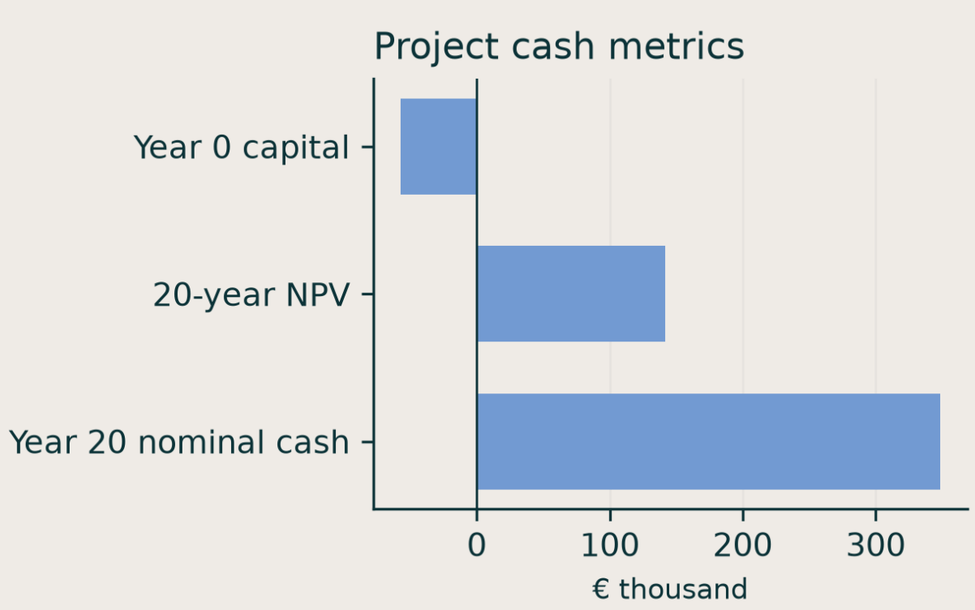
20-year financial projection

Protein X · Biogas ✓ · N ✓ · P X · K X









Constant-account cash flow and cumulative product revenue over 20 years

Scenario: matched_protein_bypass rank 1; 9b68969ce02b4f5c94b6ff0f59b53447; 1324.8 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Balance and evidence

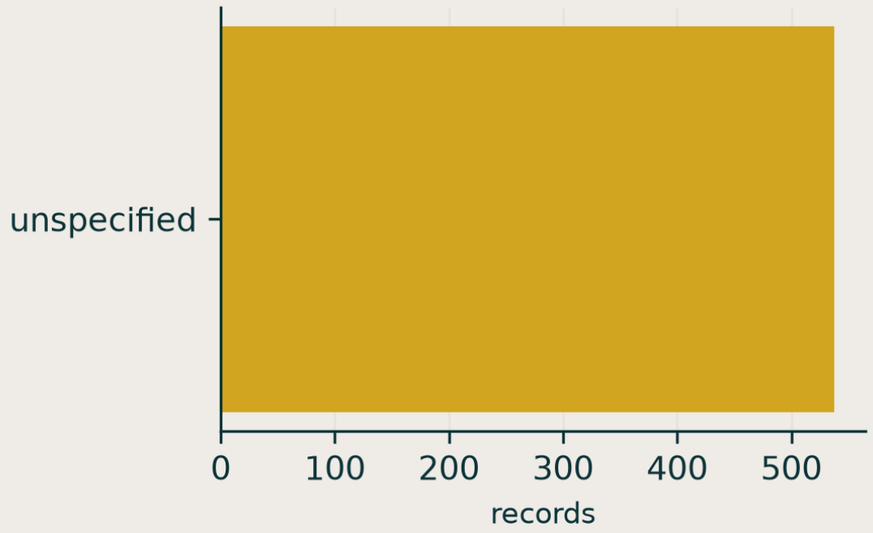
Protein X · Biogas ✓ · N ✓ · P X · K X

Element closure

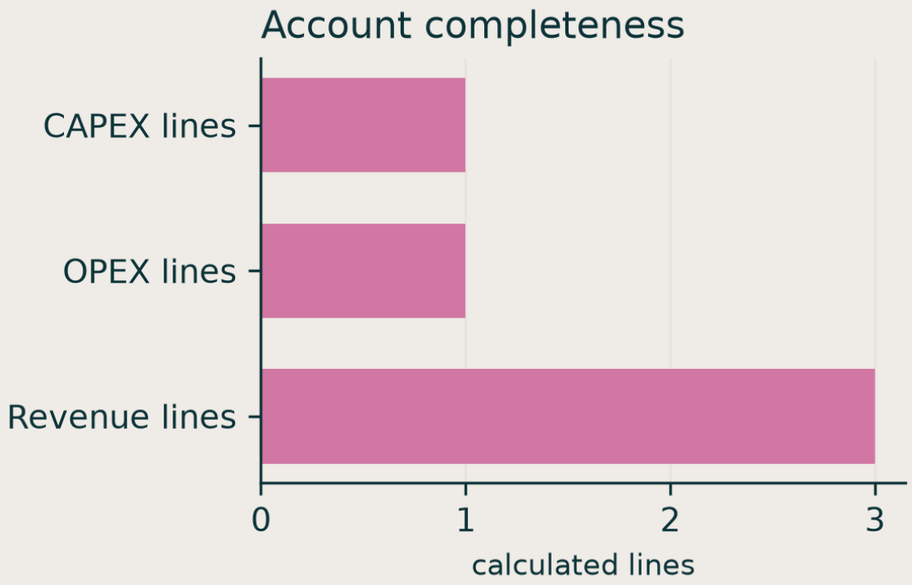
C	0 kg/year
N	0 kg/year
P	0 kg/year
K	0 kg/year

Numerical residuals

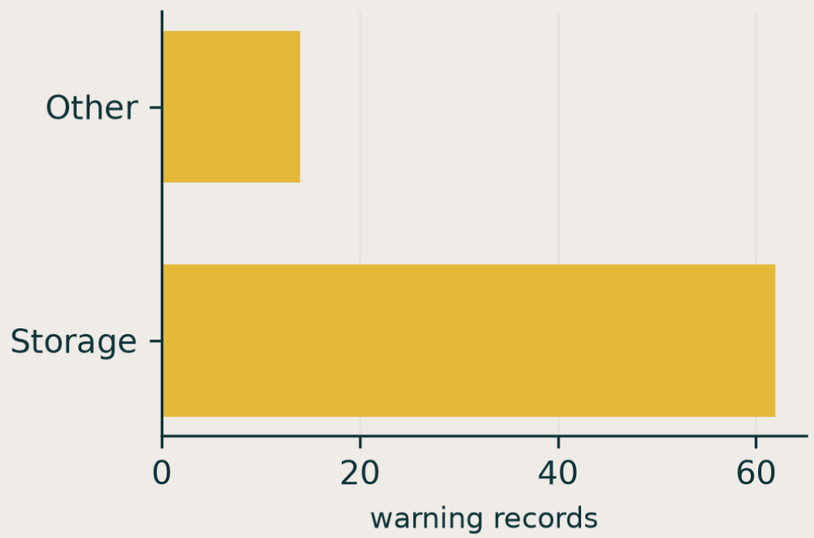
Parameter evidence



Account completeness



Model warnings



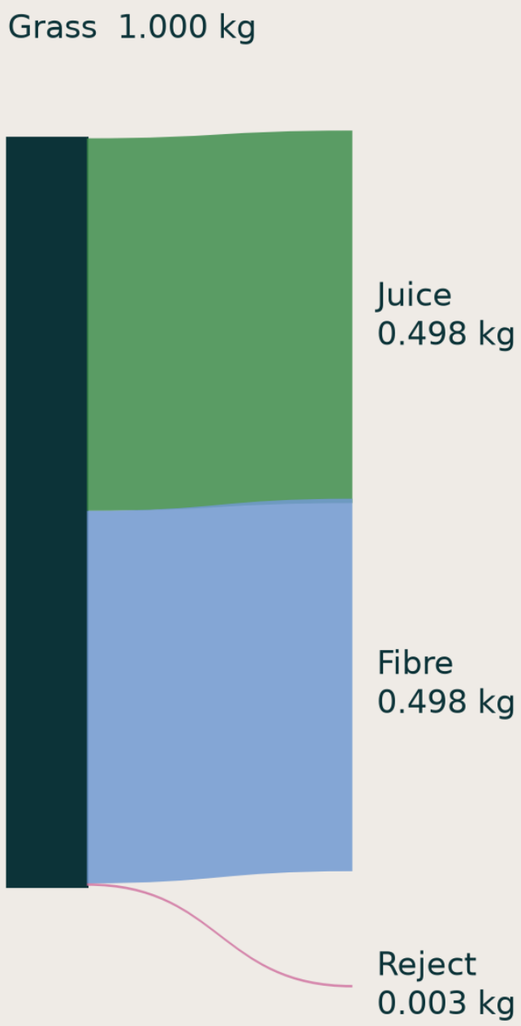
Numerical closure, evidence counts, calculated account lines and warnings

Scenario: matched_protein_bypass rank 1; 9b68969ce02b4f5c94b6ff0f59b53447; 1324.8 kg/h; 8760 h/year
Units: kg/year; records
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

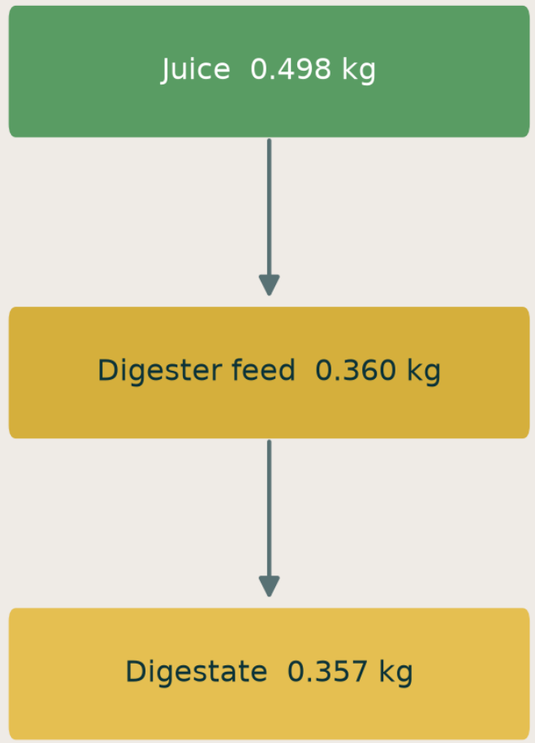
Grass to products

Protein X · Biogas ✓ · N ✓ · P ✓ · K X

Grass split



Juice pathway



Selected intermediate streams

Recovered outputs

Raw gas 0.0021 Nm³

Raw export 0.0021 Nm³

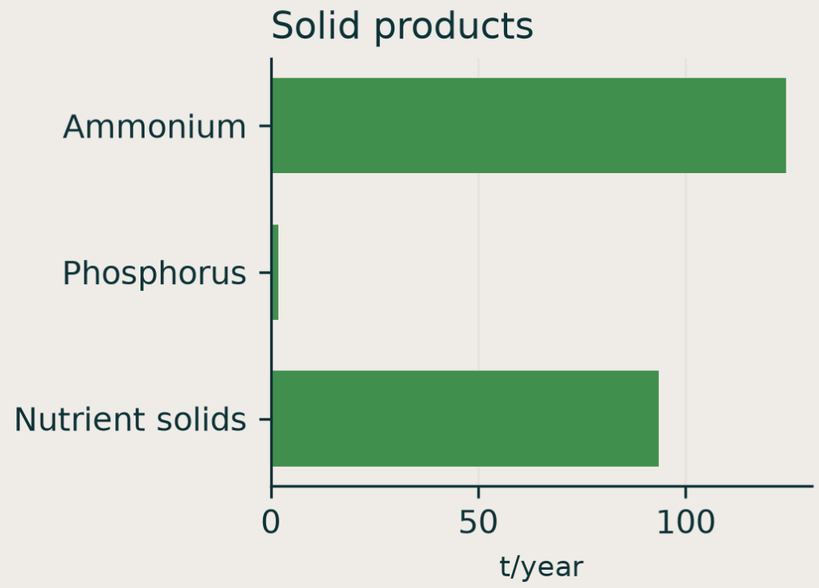
per kg wet grass

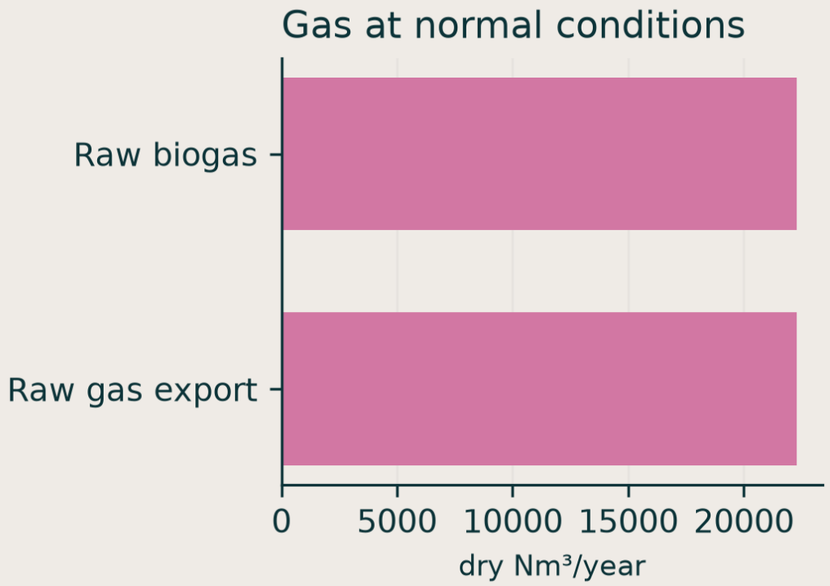
Mass-proportional grass split plus modeled downstream stream/product map

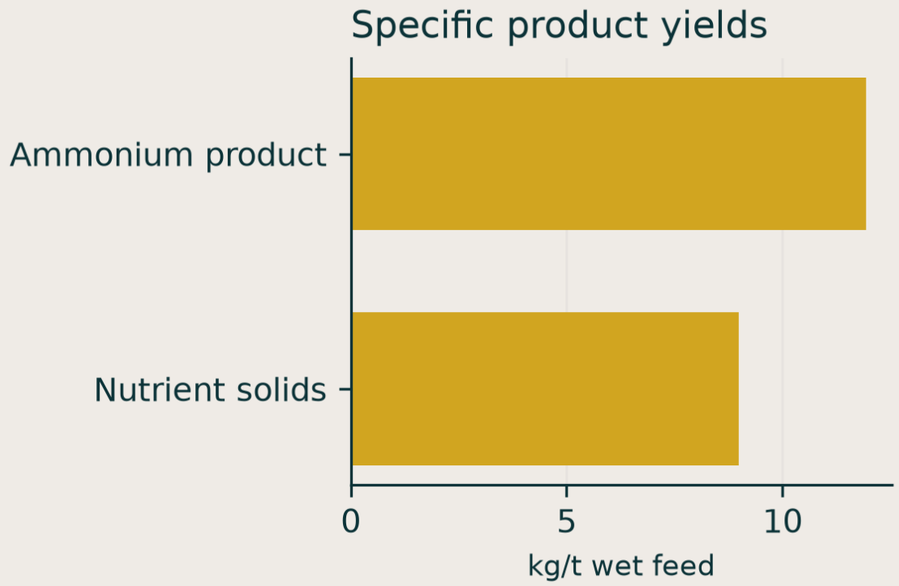
Scenario: matched_protein_bypass rank 2; fc4dc441b0034f7fab5083eae8e8a0f3; 1187.9 kg/h; 8760 h/year
Units: kg/kg wet grass; dry Nm³/kg wet grass
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Material and products

Protein X · Biogas ✓ · N ✓ · P ✓ · K X





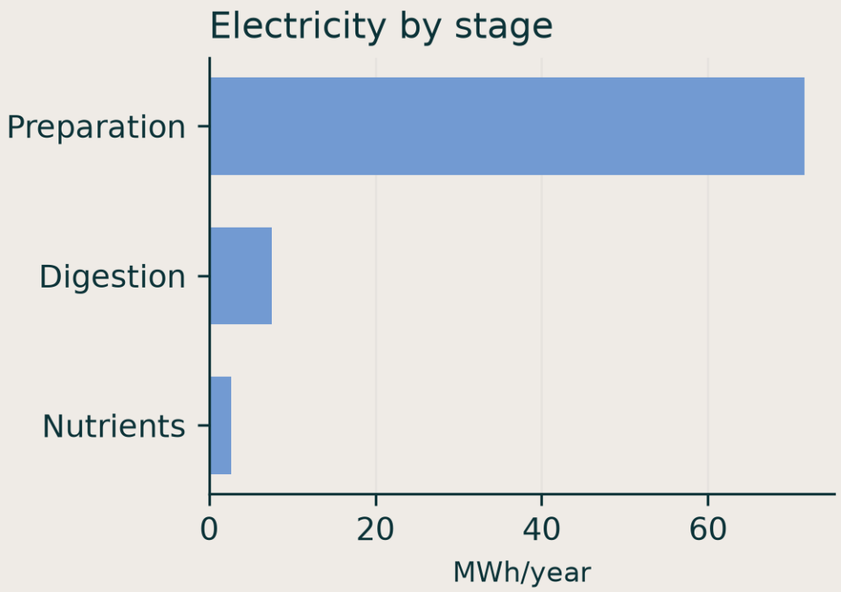


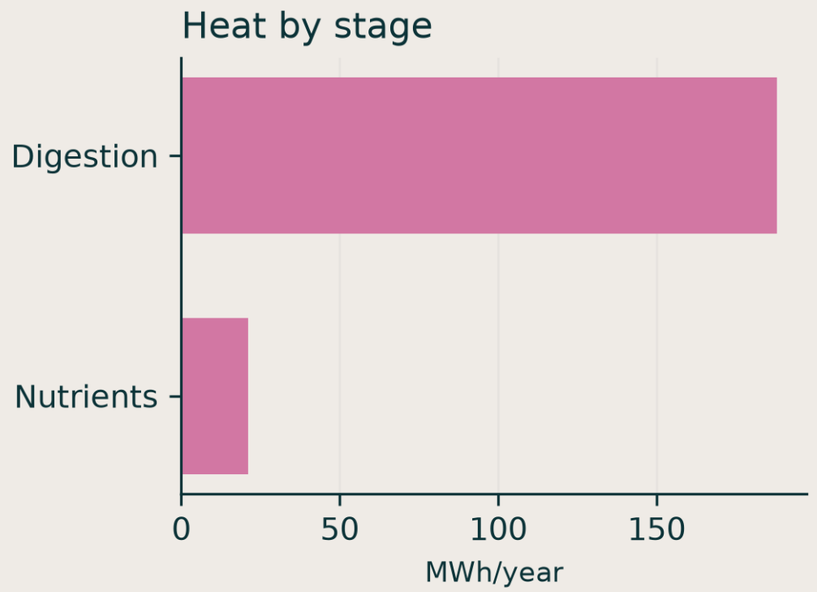
Front-end split, solid products, dry gas volumes and specific yields

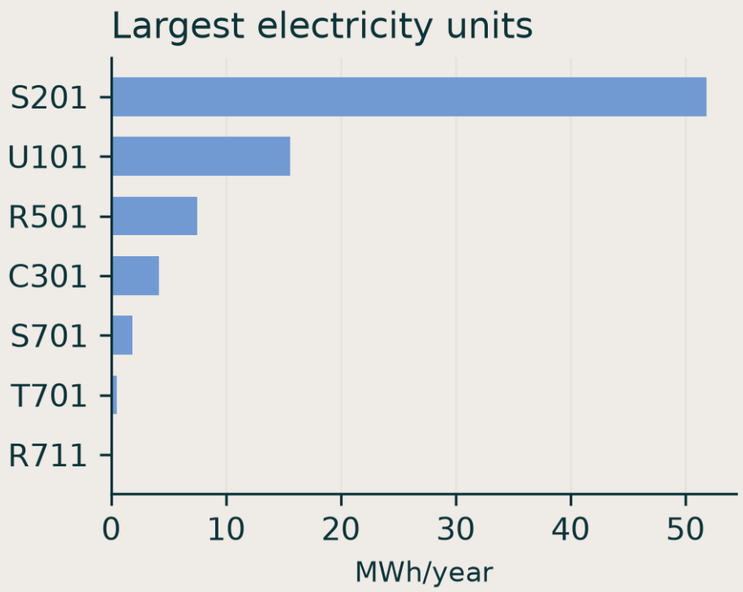
Scenario: matched_protein_bypass rank 2; fc4dc441b0034f7fab5083eae8e8a0f3; 1187.9 kg/h; 8760 h/year
Units: kg/h; t/year; dry Nm³/year; kg/t feed
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

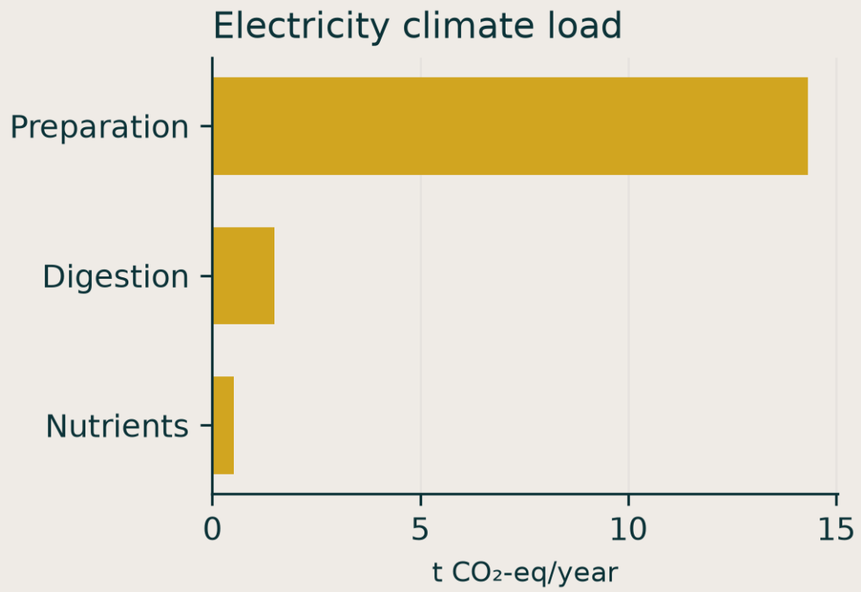
Energy and climate hotspots

Protein X · Biogas ✓ · N ✓ · P ✓ · K X



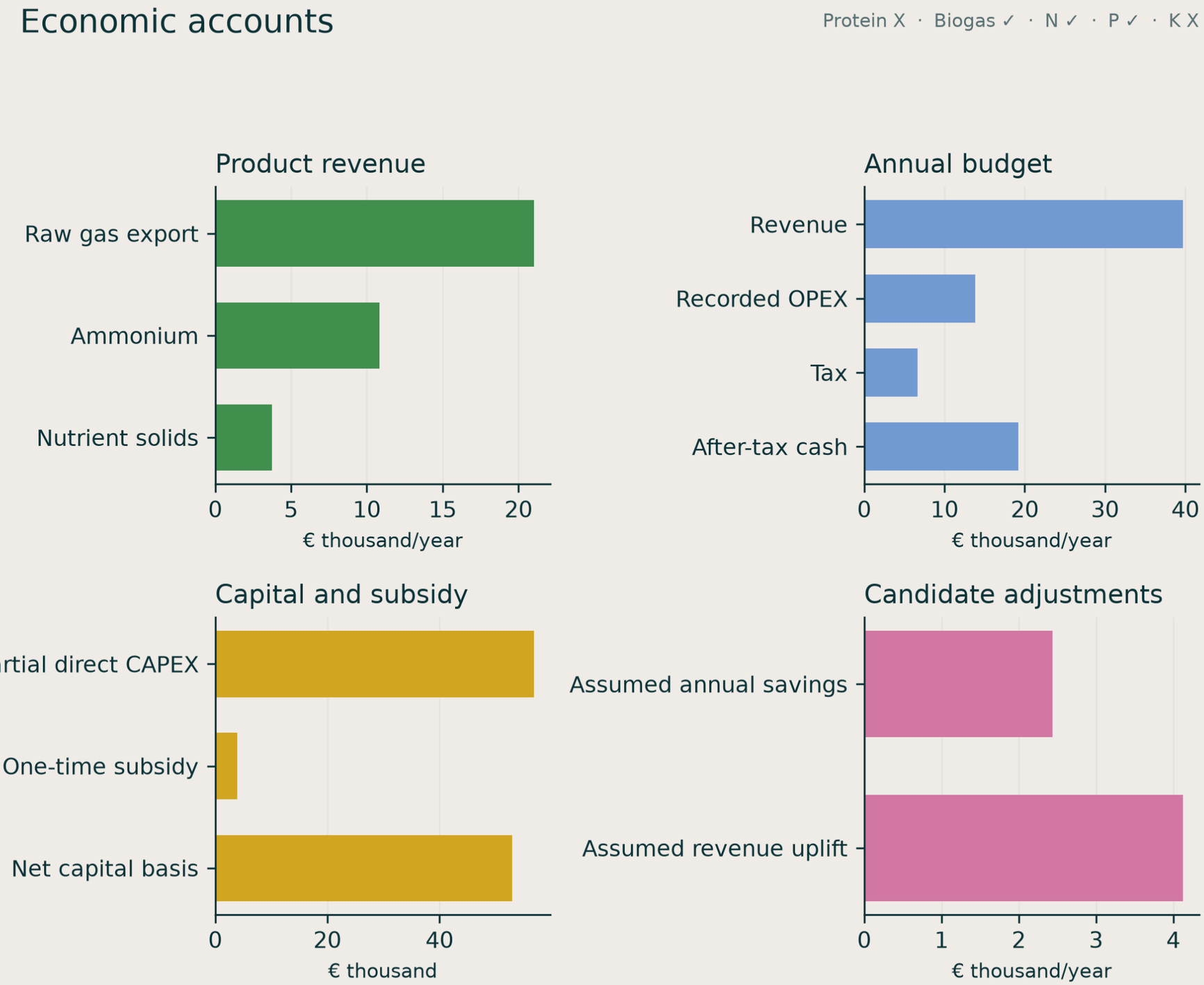






Stage and unit utility hotspots and included climate load

Scenario: matched_protein_bypass rank 2; fc4dc441b0034f7fab5083eae8e8a0f3; 1187.9 kg/h; 8760 h/year
Units: MWh/year; t CO_2-eq/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

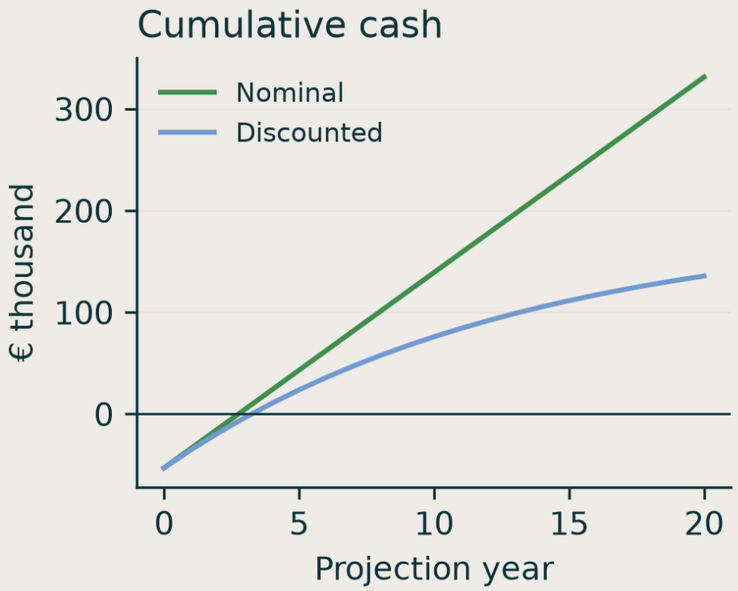


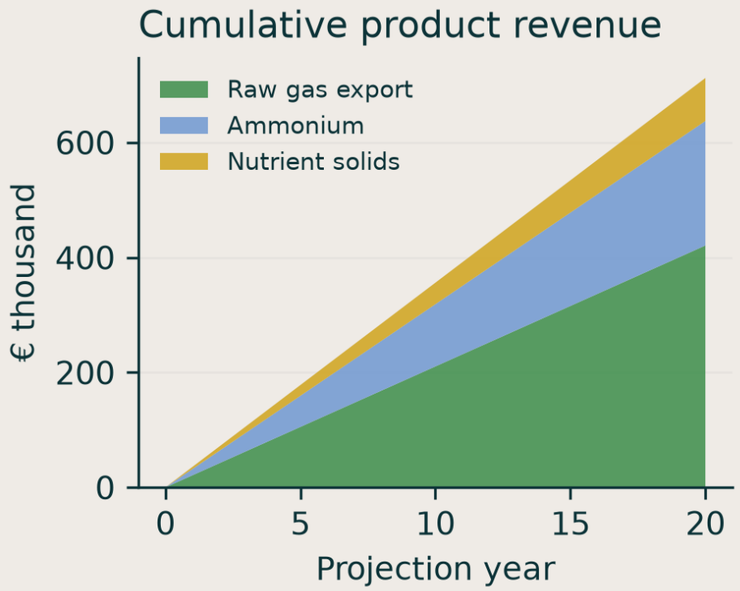
Product revenue, partial annual budget, capital subsidy and candidate overlays

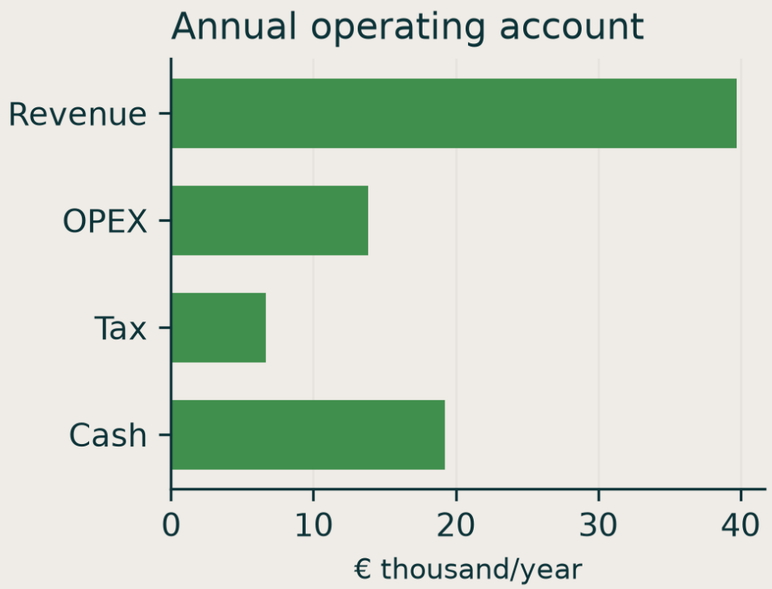
Scenario: matched_protein_bypass rank 2; fc4dc441b0034f7fab5083eae8e8a0f3; 1187.9 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

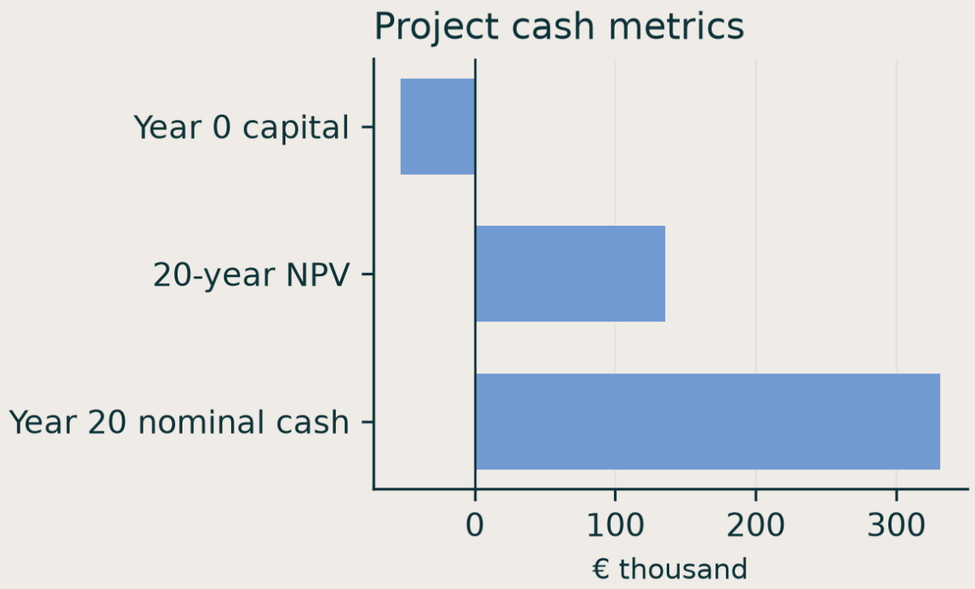
20-year financial projection

Protein X · Biogas ✓ · N ✓ · P ✓ · K X









Constant-account cash flow and cumulative product revenue over 20 years

Scenario: matched_protein_bypass rank 2; fc4dc441b0034f7fab5083eae8e8a0f3; 1187.9 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Balance and evidence

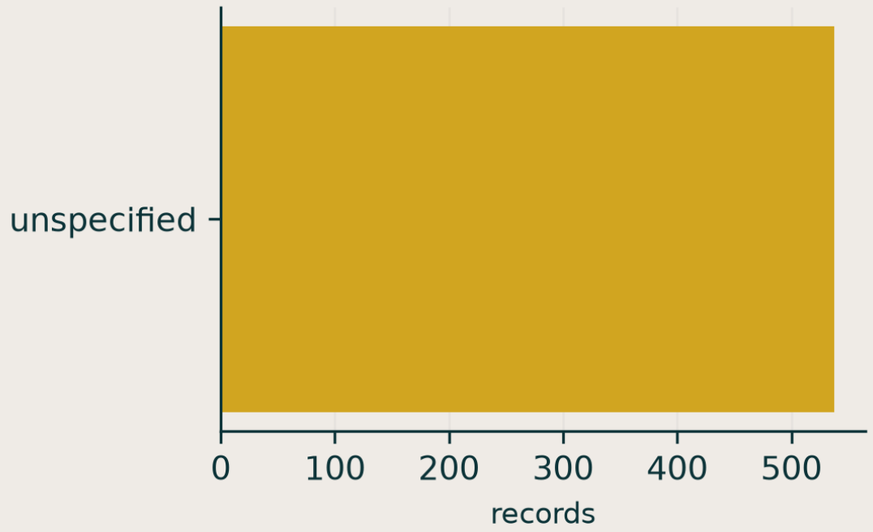
Protein X · Biogas ✓ · N ✓ · P ✓ · K X

Element closure

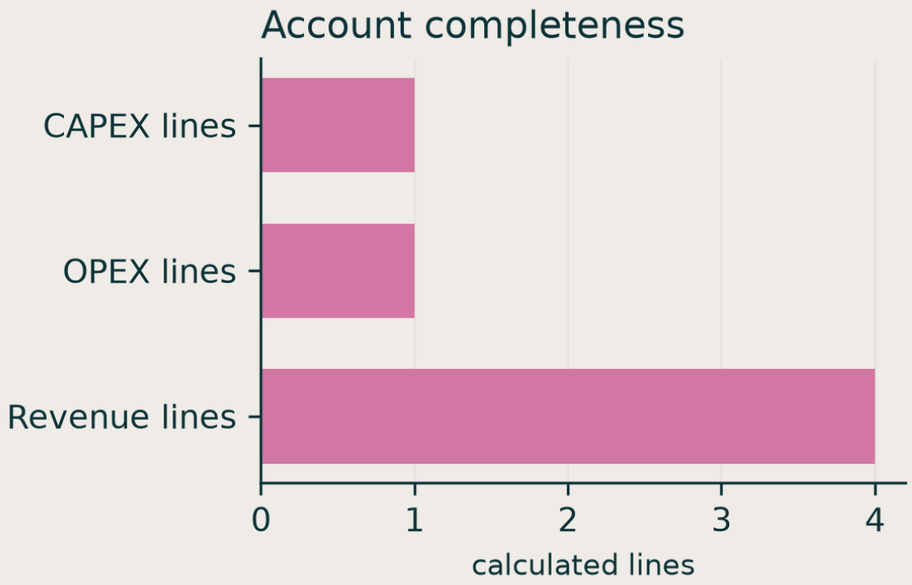
C	0 kg/year
N	0 kg/year
P	0 kg/year
K	0 kg/year

Numerical residuals

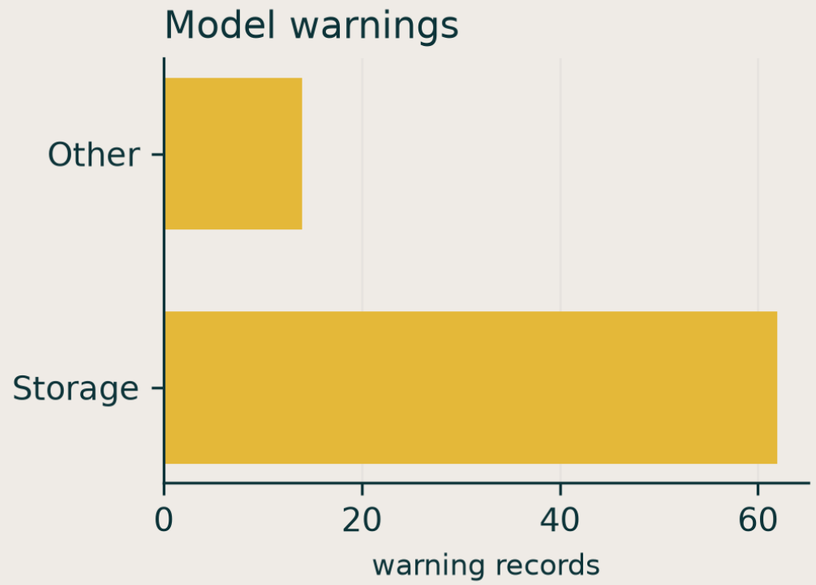
Parameter evidence



Account completeness



Model warnings



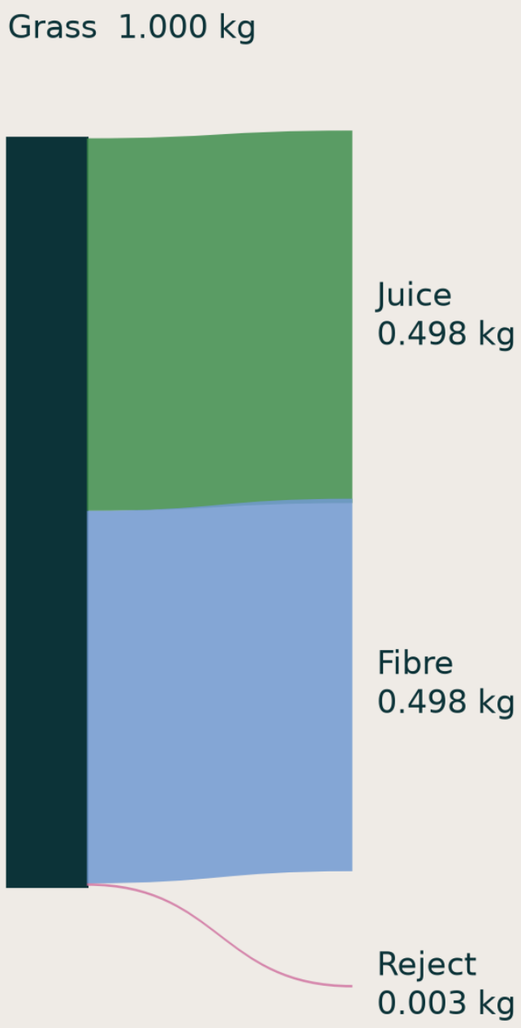
Numerical closure, evidence counts, calculated account lines and warnings

Scenario: matched_protein_bypass rank 2; fc4dc441b0034f7fab5083eae8e8a0f3; 1187.9 kg/h; 8760 h/year
Units: kg/year; records
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

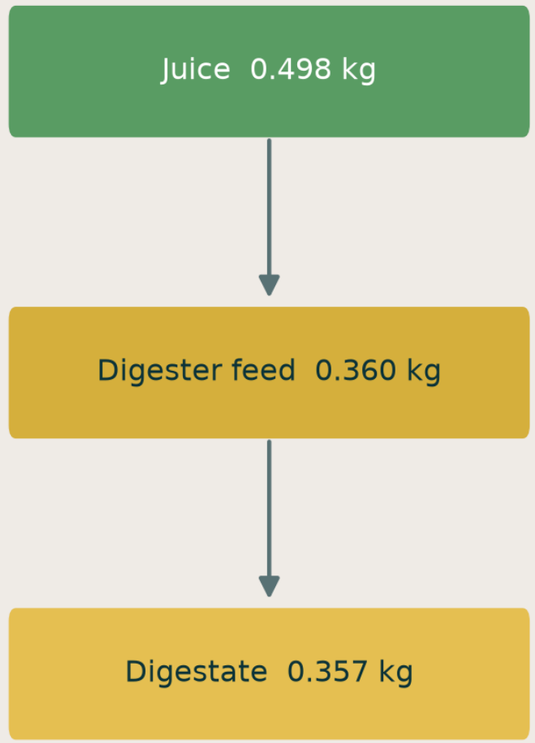
Grass to products

Protein X · Biogas ✓ · N ✓ · P X · K X

Grass split

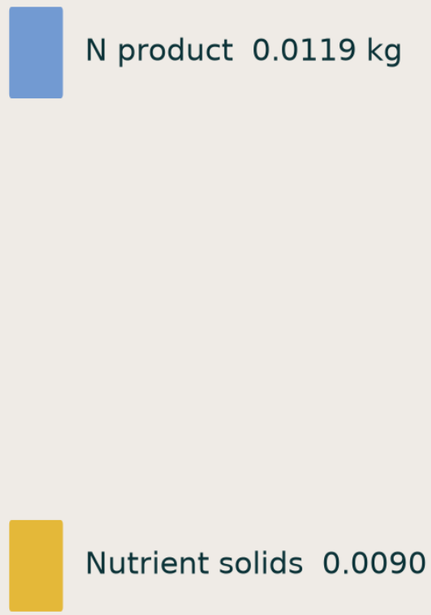


Juice pathway



Selected intermediate streams

Recovered outputs



Raw gas 0.0021 Nm³

Raw export 0.0021 Nm³

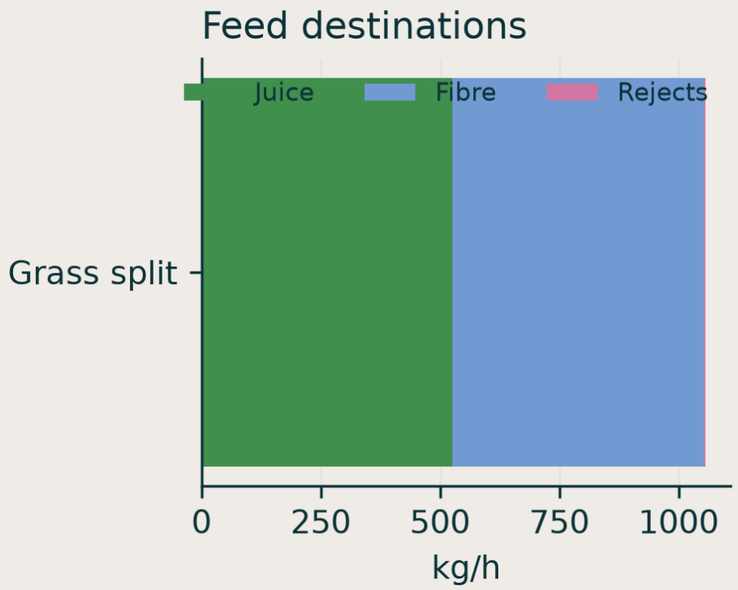
per kg wet grass

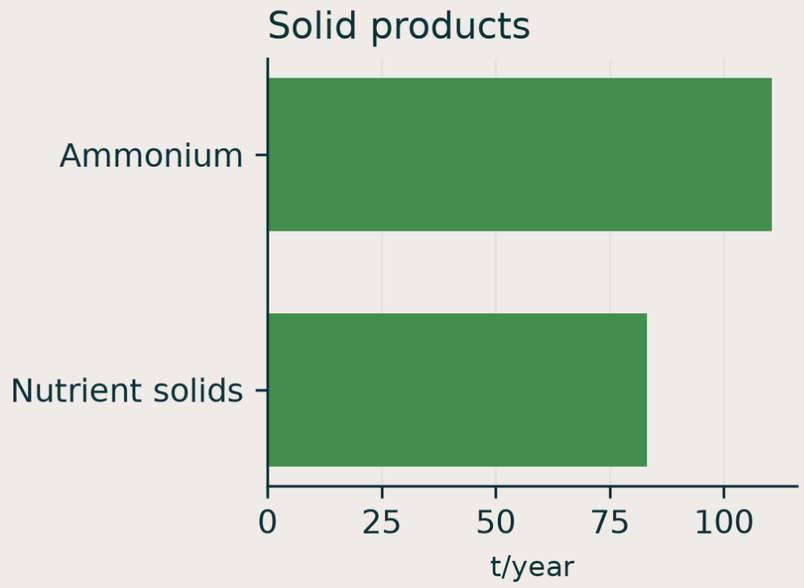
Mass-proportional grass split plus modeled downstream stream/product map

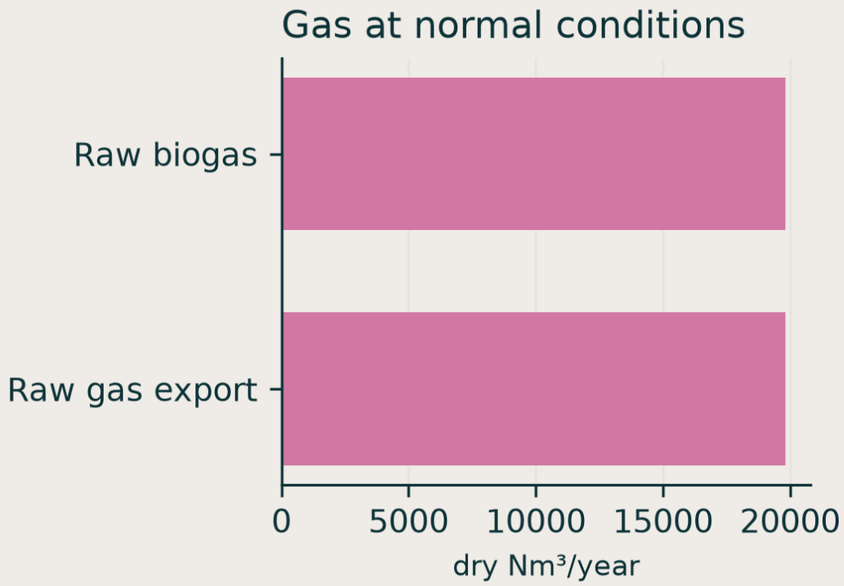
Scenario: matched_protein_bypass rank 3; 267beb9505fe47e99c877f19b4f9588a; 1458.6 kg/h; 8760 h/year
Units: kg/kg wet grass; dry Nm³/kg wet grass
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

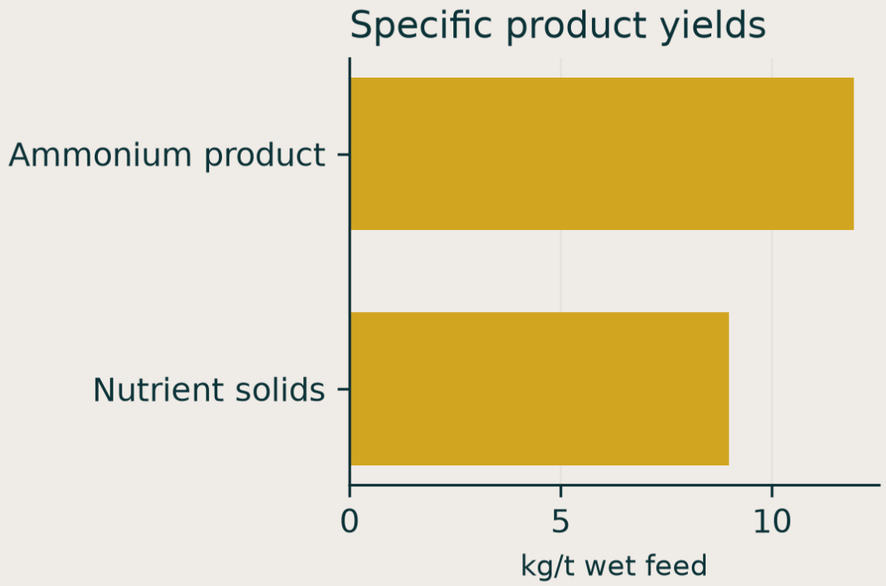
Material and products

Protein X · Biogas ✓ · N ✓ · P X · K X







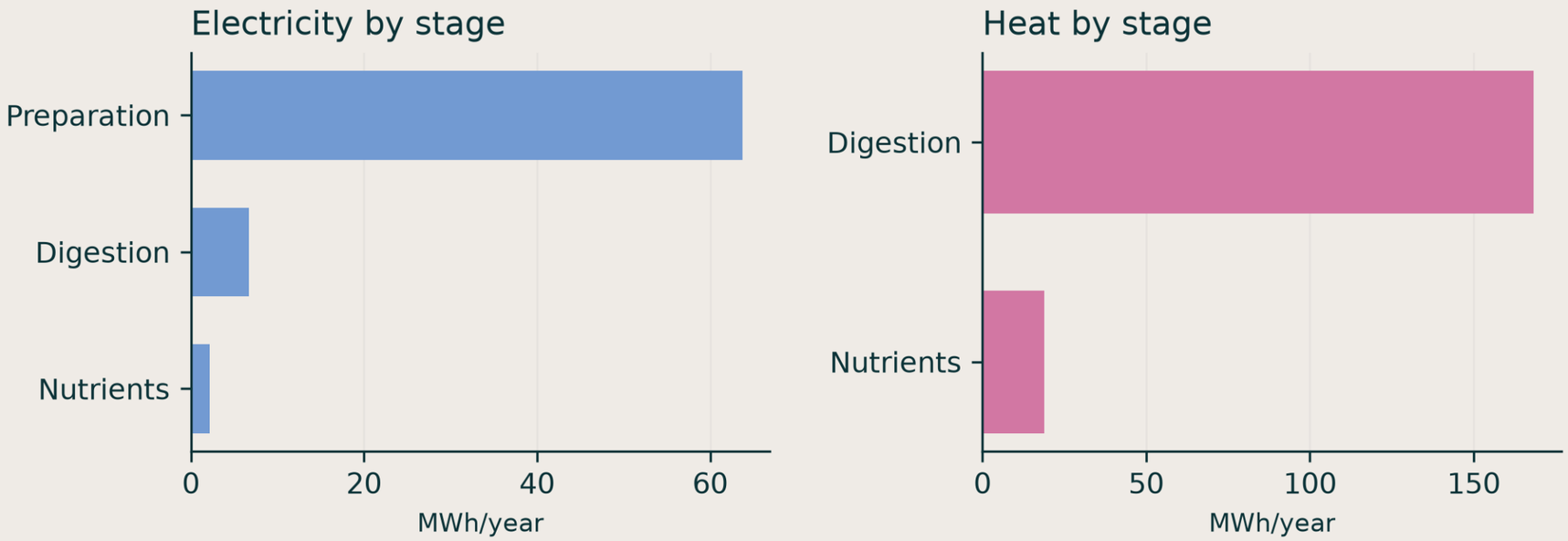


Front-end split, solid products, dry gas volumes and specific yields

Scenario: matched_protein_bypass rank 3; 267beb9505fe47e99c877f19b4f9588a; 1458.6 kg/h; 8760 h/year
Units: kg/h; t/year; dry Nm³/year; kg/t feed
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Energy and climate hotspots

Protein X · Biogas ✓ · N ✓ · P X · K X

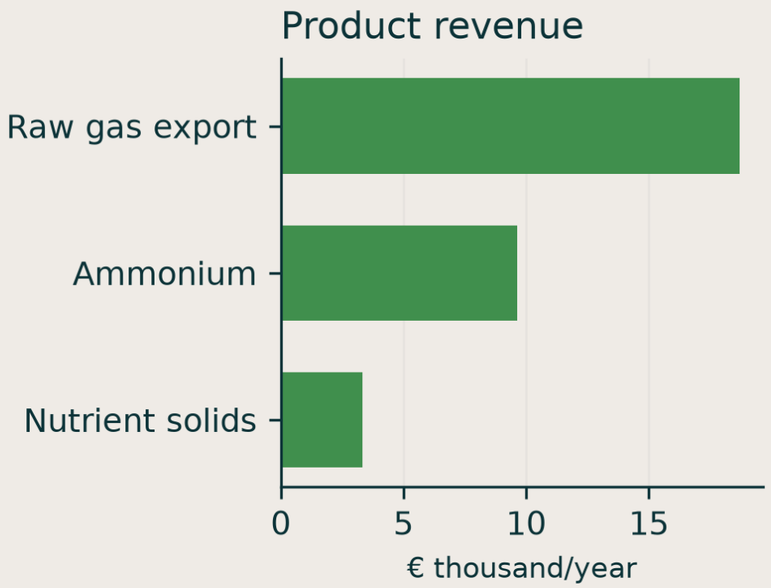


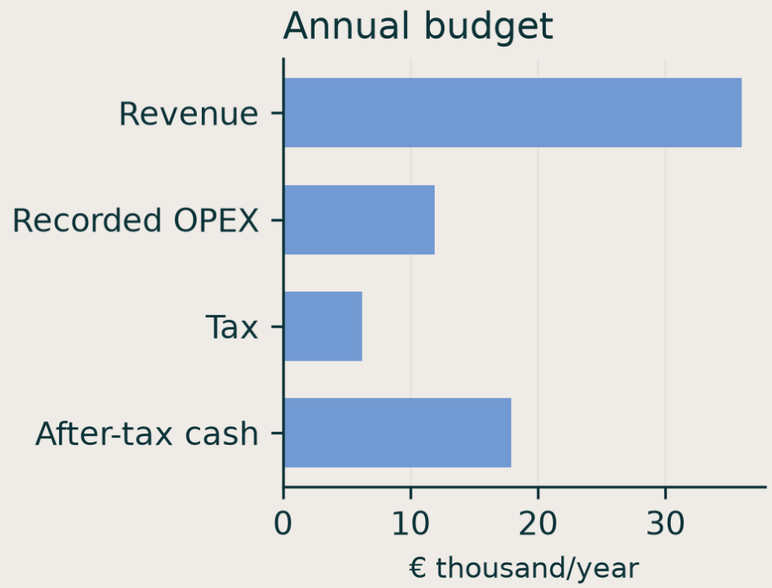
Stage and unit utility hotspots and included climate load

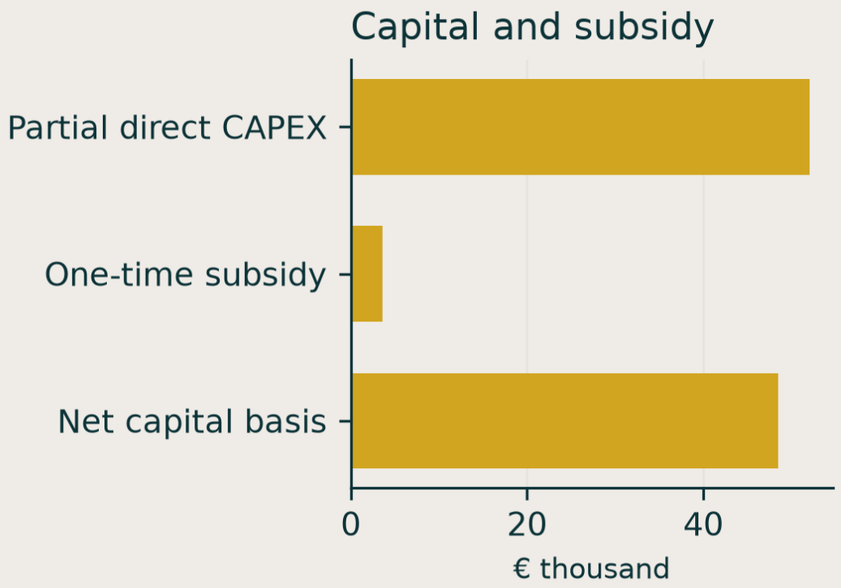
Scenario: matched_protein_bypass rank 3; 267beb9505fe47e99c877f19b4f9588a; 1458.6 kg/h; 8760 h/year
Units: MWh/year; t CO_2-eq/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

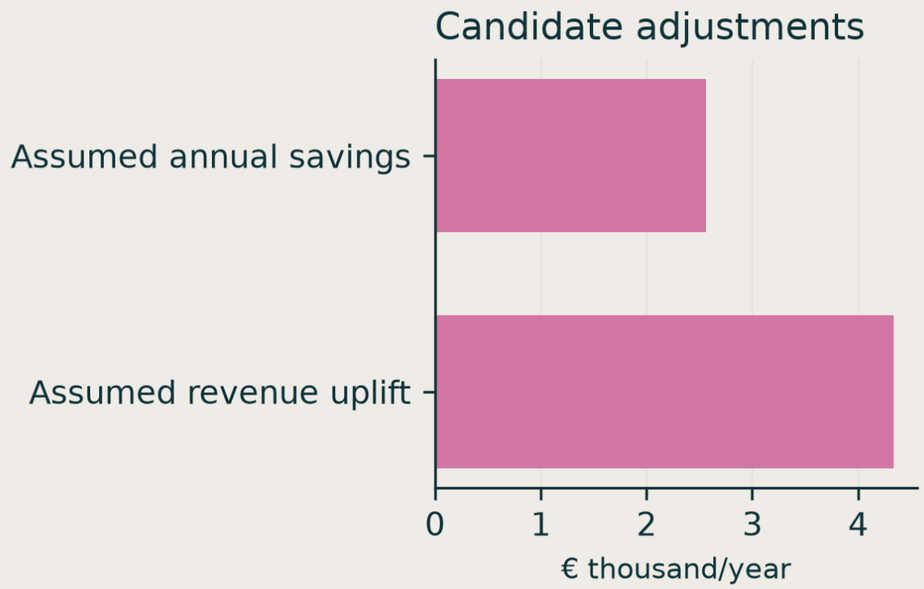
Economic accounts

Protein X · Biogas ✓ · N ✓ · P X · K X







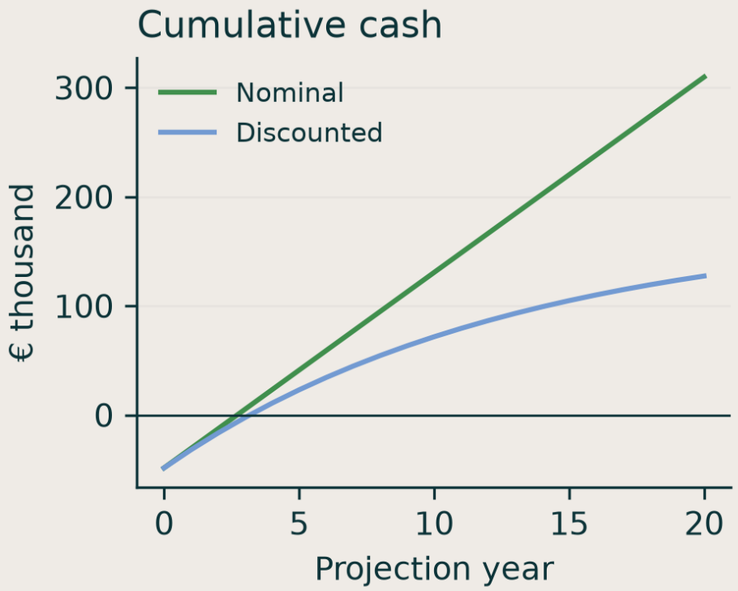


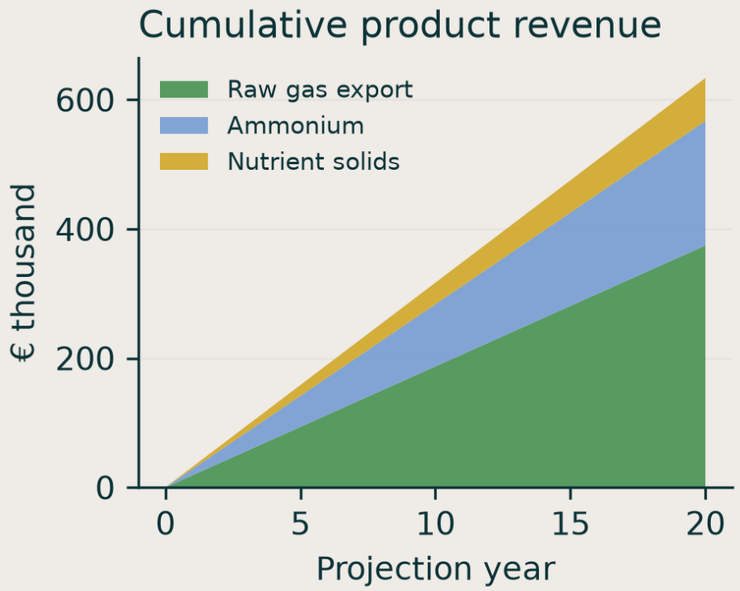
Product revenue, partial annual budget, capital subsidy and candidate overlays

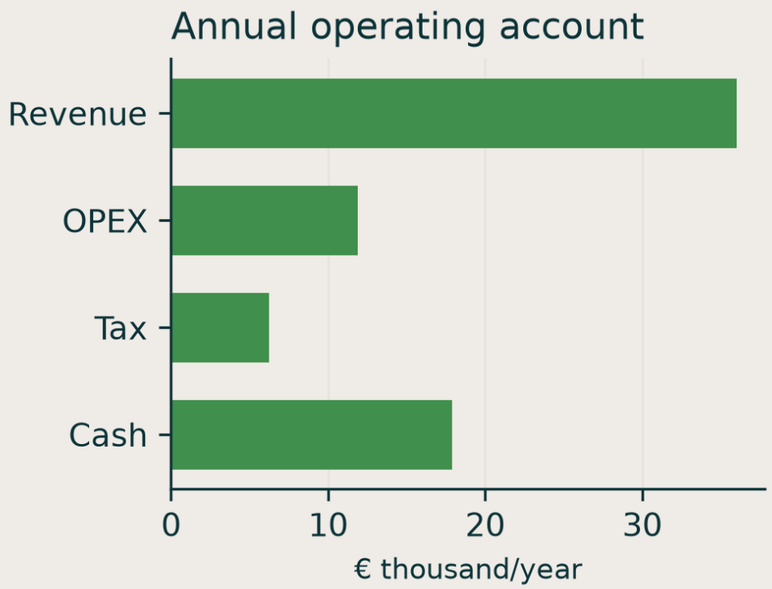
Scenario: matched_protein_bypass rank 3; 267beb9505fe47e99c877f19b4f9588a; 1458.6 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

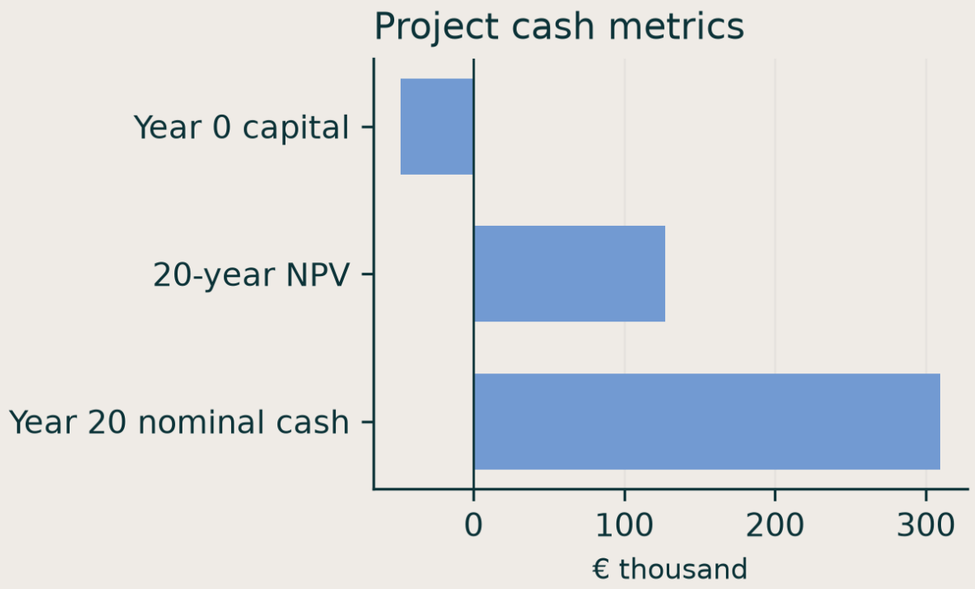
20-year financial projection

Protein X · Biogas ✓ · N ✓ · P X · K X









Constant-account cash flow and cumulative product revenue over 20 years

Scenario: matched_protein_bypass rank 3; 267beb9505fe47e99c877f19b4f9588a; 1458.6 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Balance and evidence

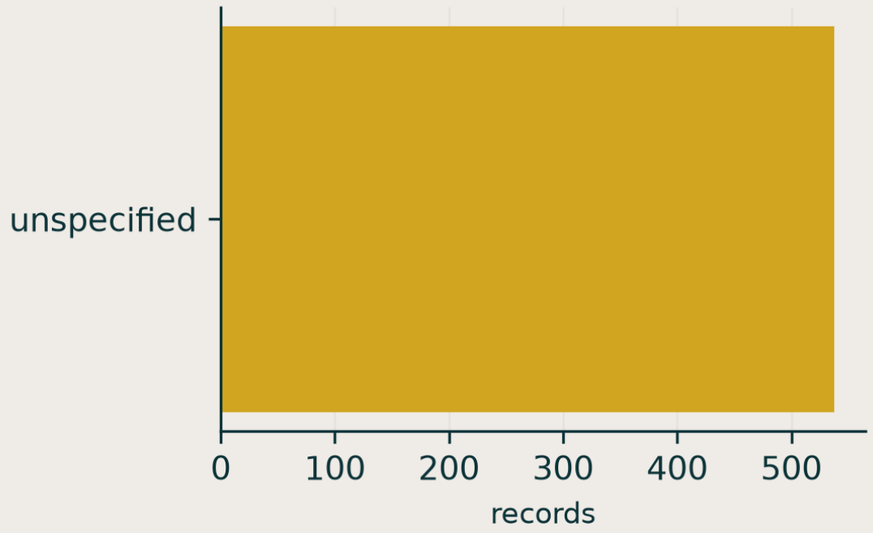
Protein X · Biogas ✓ · N ✓ · P X · K X

Element closure

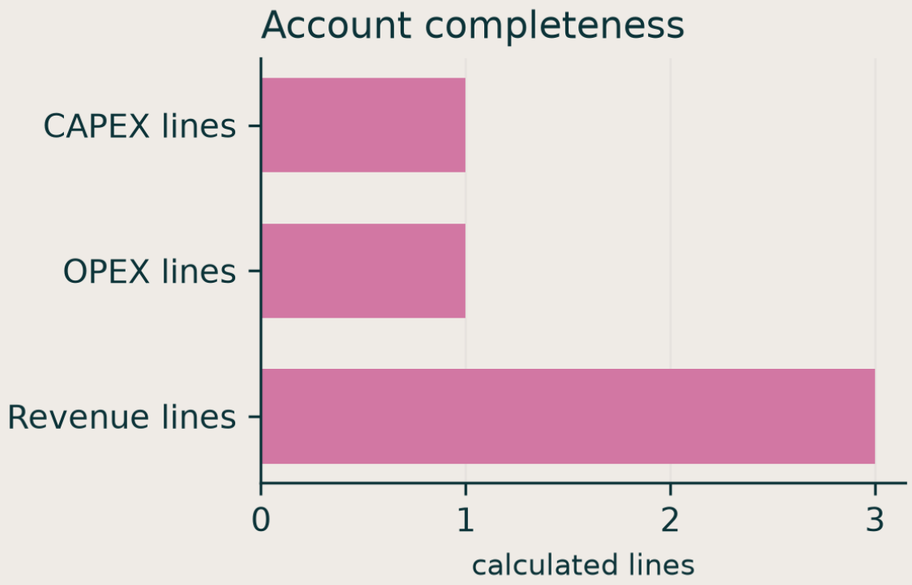
C	0 kg/year
N	0 kg/year
P	0 kg/year
K	0 kg/year

Numerical residuals

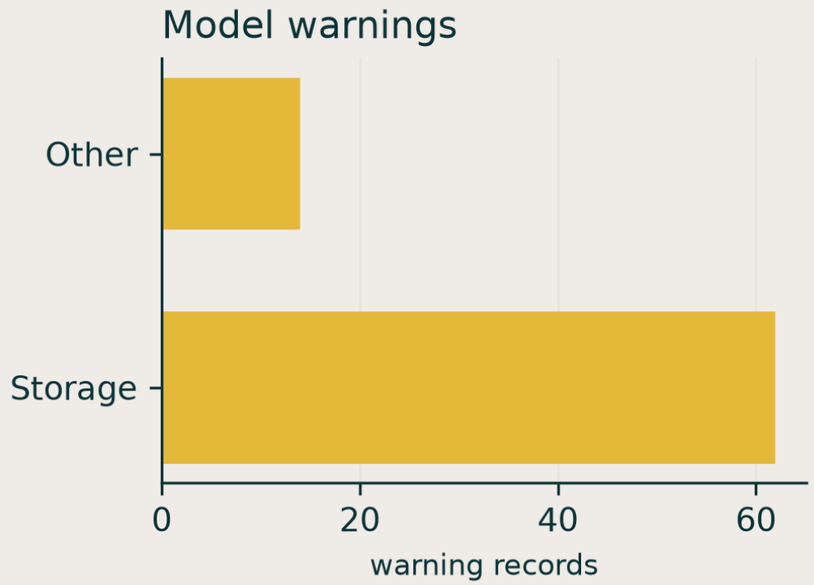
Parameter evidence



Account completeness



Model warnings



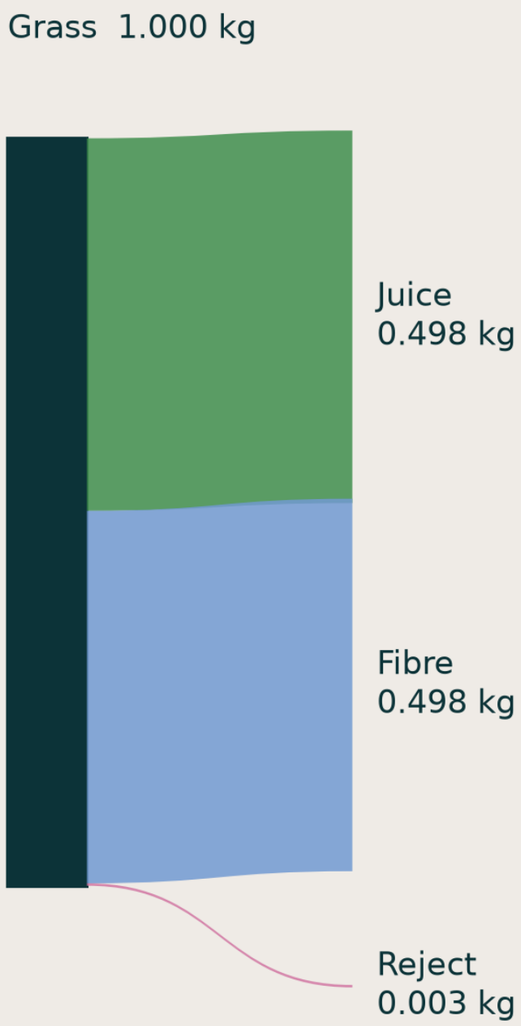
Numerical closure, evidence counts, calculated account lines and warnings

Scenario: matched_protein_bypass rank 3; 267beb9505fe47e99c877f19b4f9588a; 1458.6 kg/h; 8760 h/year
Units: kg/year; records
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

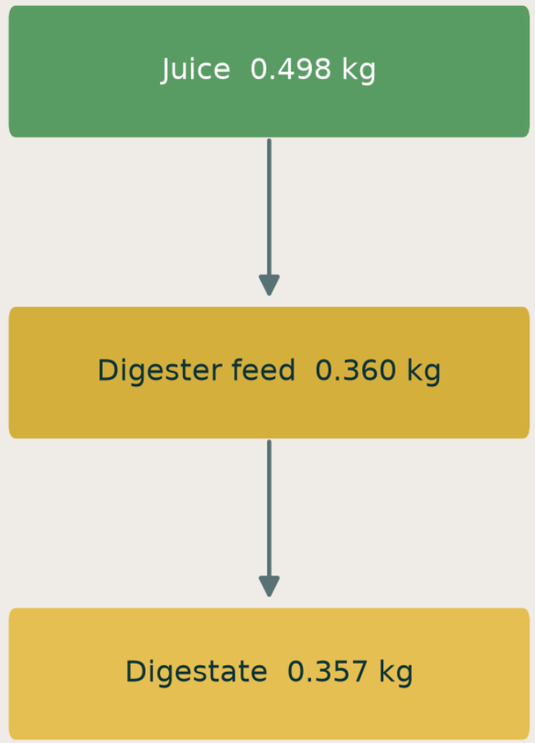
Grass to products

Protein X · Biogas ✓ · N ✓ · P ✓ · K X

Grass split



Juice pathway



Selected intermediate streams

Recovered outputs

Raw gas 0.0021 Nm³

Raw export 0.0021 Nm³

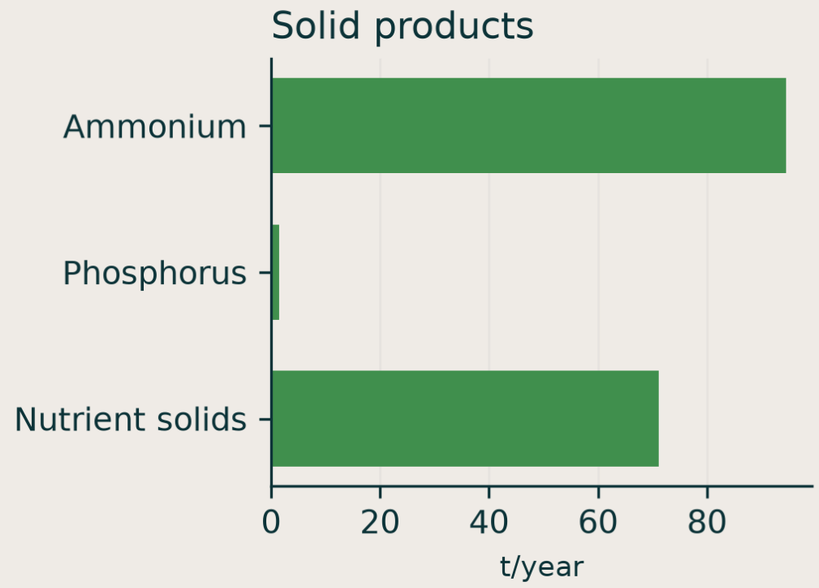
per kg wet grass

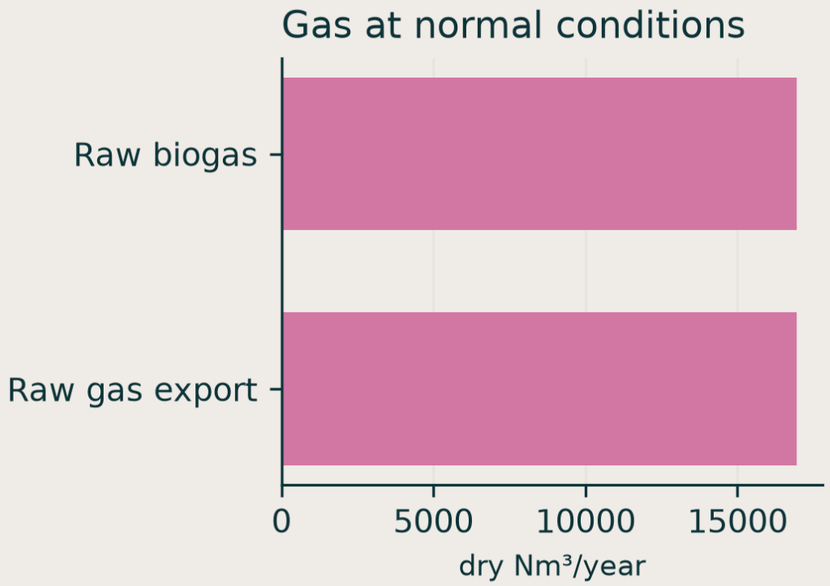
Mass-proportional grass split plus modeled downstream stream/product map

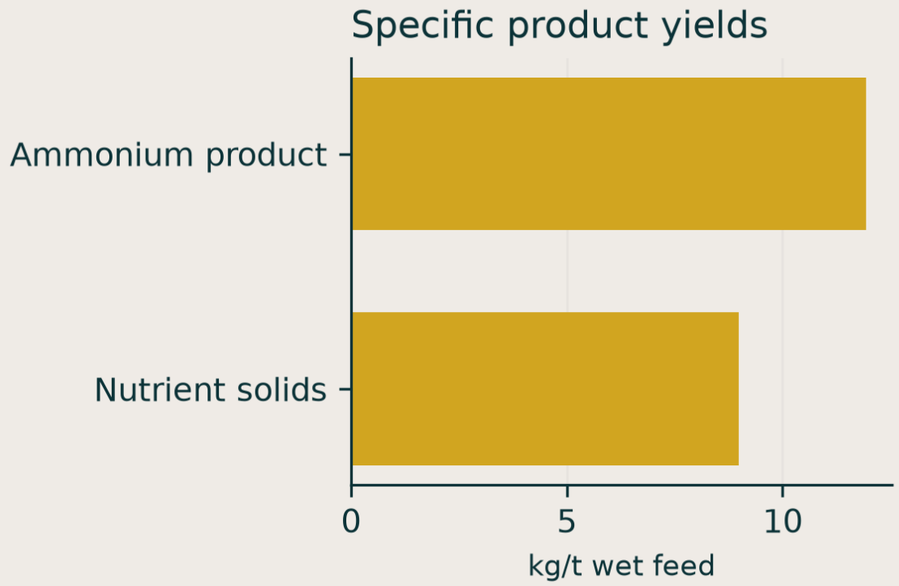
Scenario: matched_protein_bypass rank 4; ba18c74cc7304fcba1a2c3977bb4bf2c; 904.2 kg/h; 8760 h/year
Units: kg/kg wet grass; dry Nm³/kg wet grass
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Material and products

Protein X · Biogas ✓ · N ✓ · P ✓ · K X





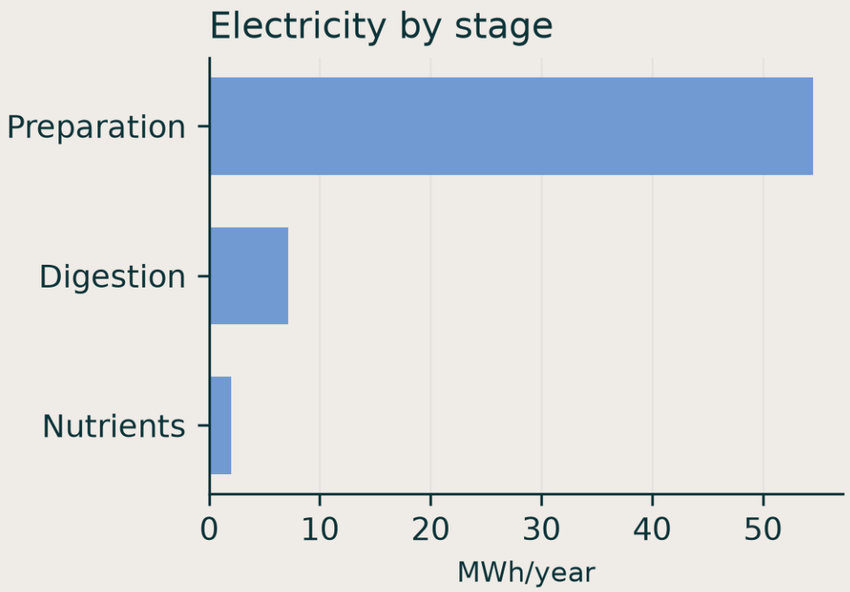


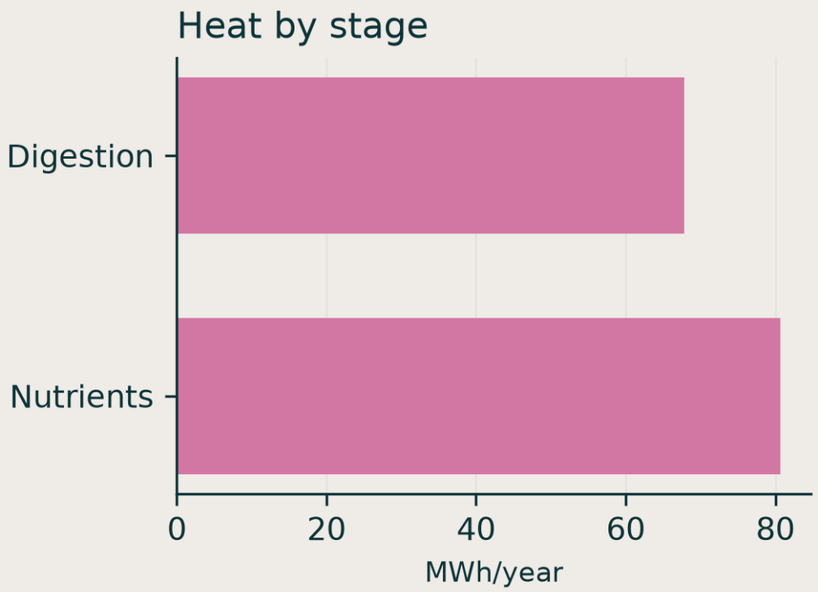
Front-end split, solid products, dry gas volumes and specific yields

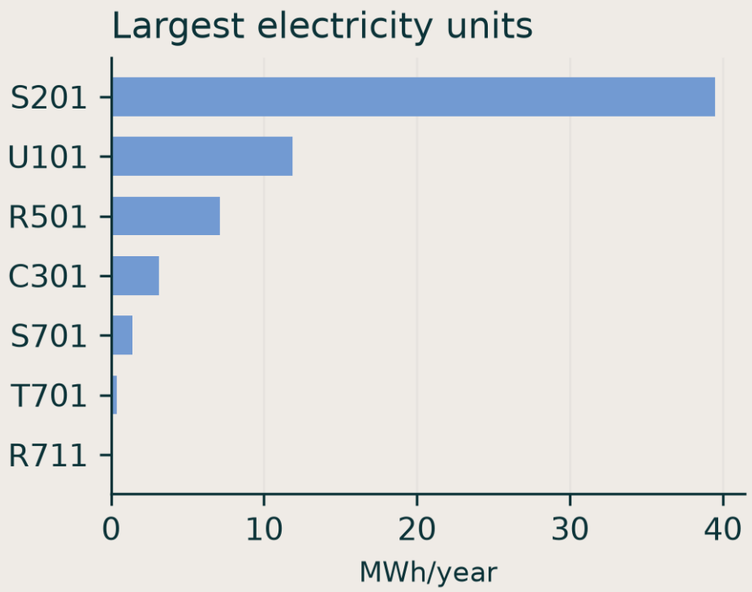
Scenario: matched_protein_bypass rank 4; ba18c74cc7304fcba1a2c3977bb4bf2c; 904.2 kg/h; 8760 h/year
Units: kg/h; t/year; dry Nm³/year; kg/t feed
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

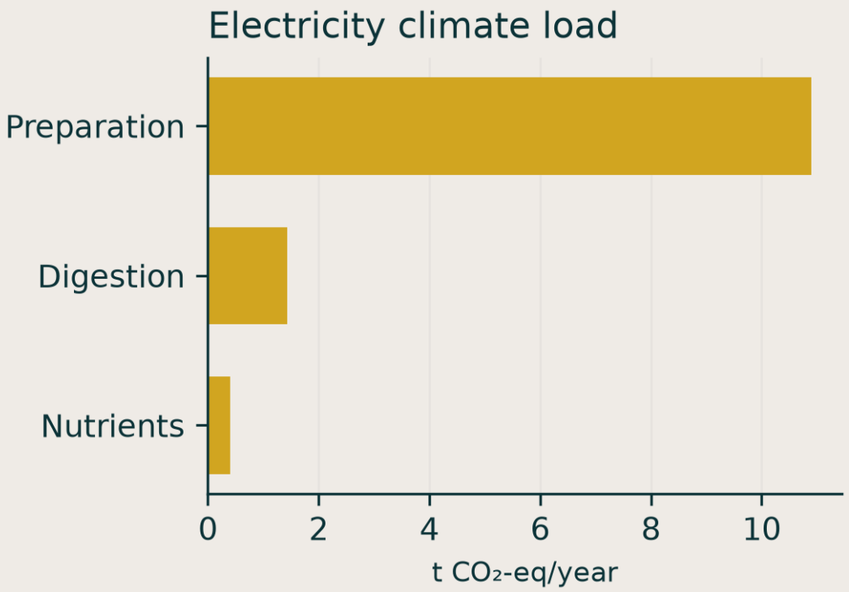
Energy and climate hotspots

Protein X · Biogas ✓ · N ✓ · P ✓ · K X







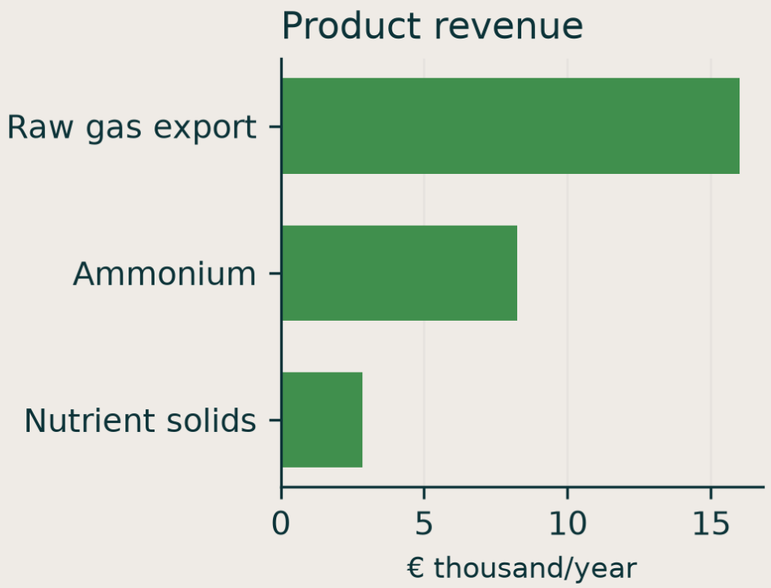


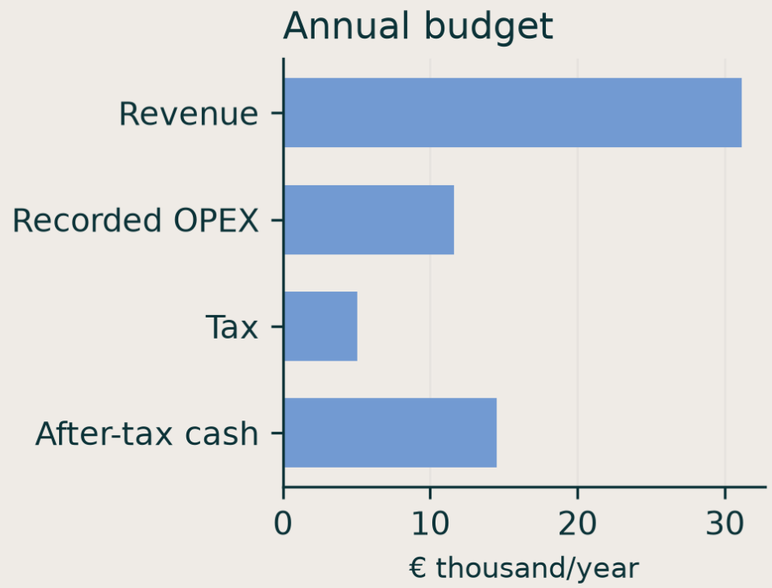
Stage and unit utility hotspots and included climate load

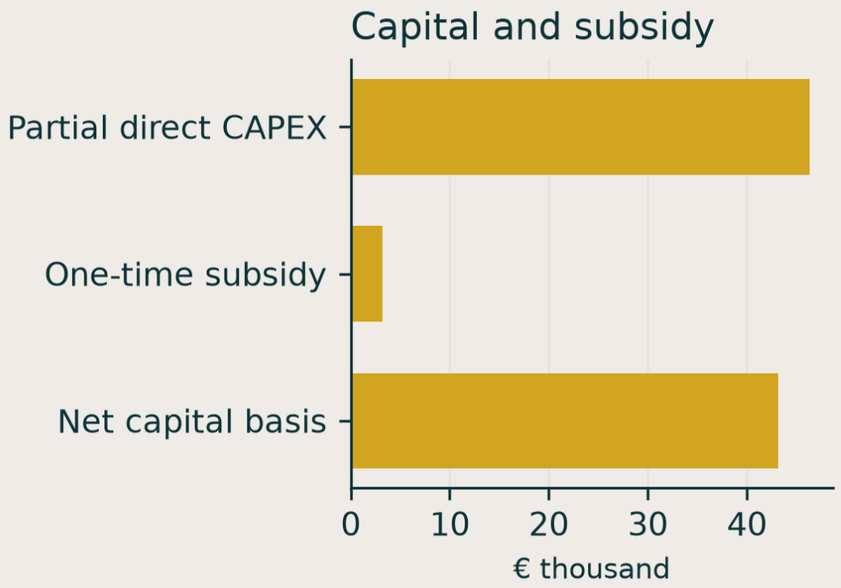
Scenario: matched_protein_bypass rank 4; ba18c74cc7304fcba1a2c3977bb4bf2c; 904.2 kg/h; 8760 h/year
Units: MWh/year; t CO_2-eq/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

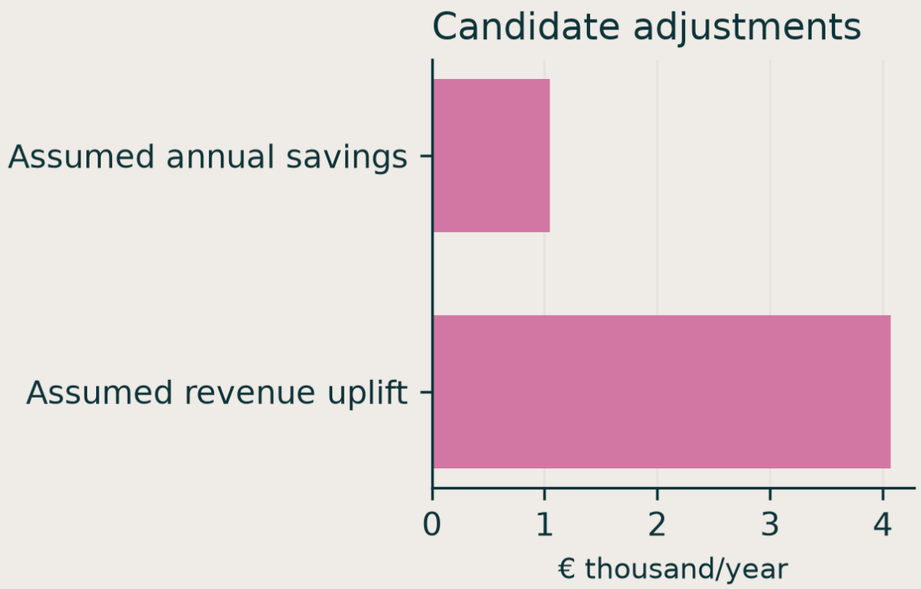
Economic accounts

Protein X · Biogas ✓ · N ✓ · P ✓ · K X







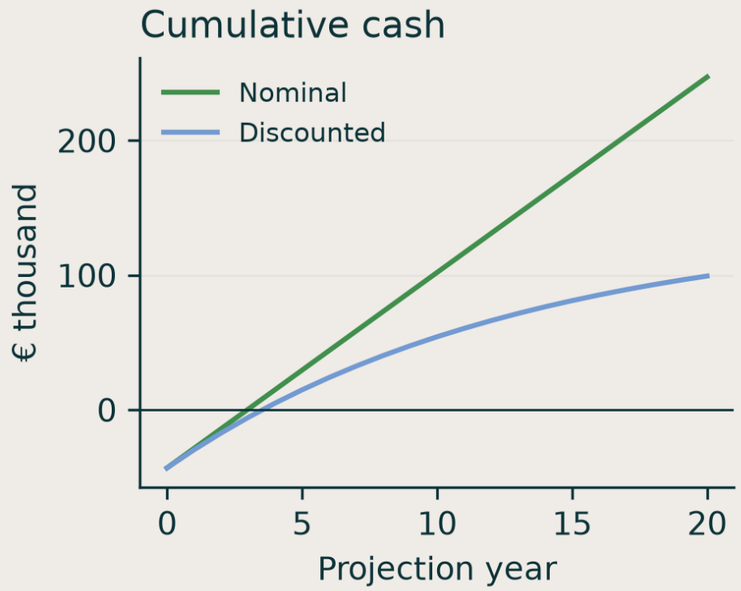


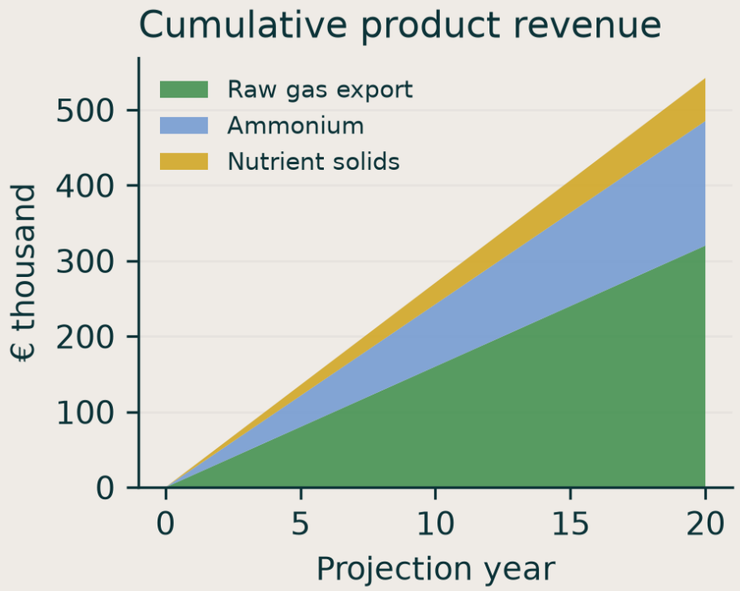
Product revenue, partial annual budget, capital subsidy and candidate overlays

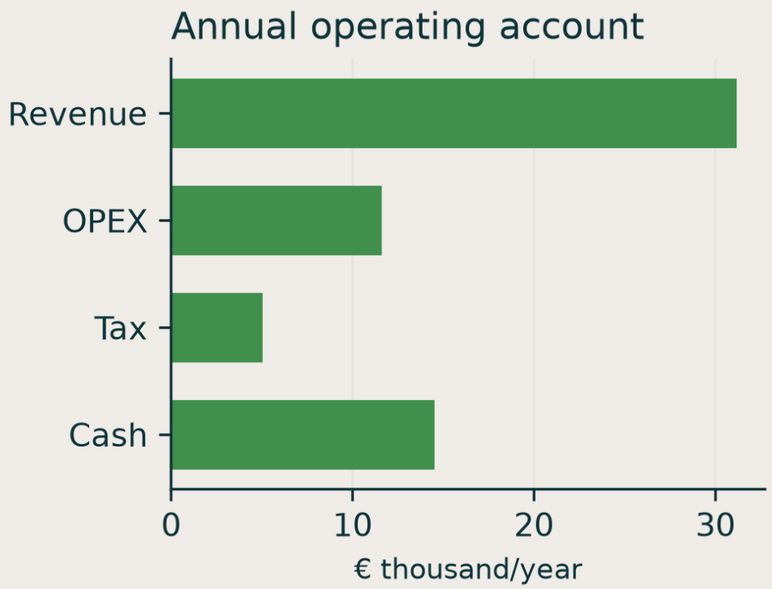
Scenario: matched_protein_bypass rank 4; ba18c74cc7304fcba1a2c3977bb4bf2c; 904.2 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

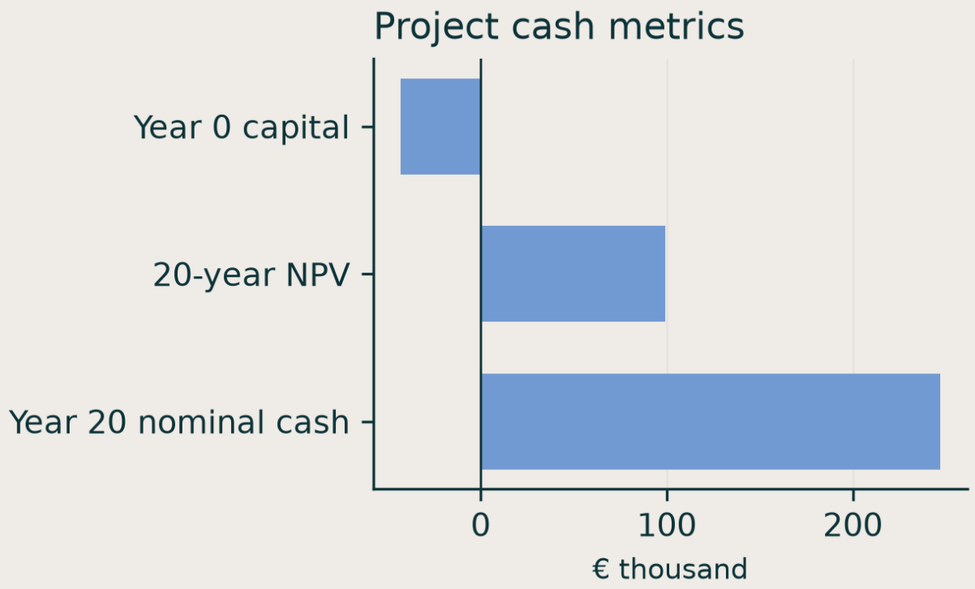
20-year financial projection

Protein X · Biogas ✓ · N ✓ · P ✓ · K X









Constant-account cash flow and cumulative product revenue over 20 years

Scenario: matched_protein_bypass rank 4; ba18c74cc7304fcba1a2c3977bb4bf2c; 904.2 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Balance and evidence

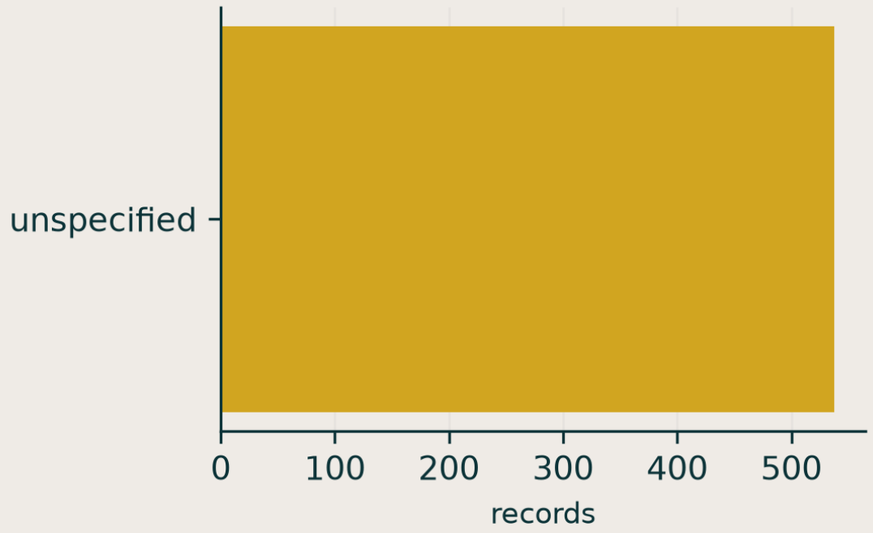
Protein X · Biogas ✓ · N ✓ · P ✓ · K X

Element closure

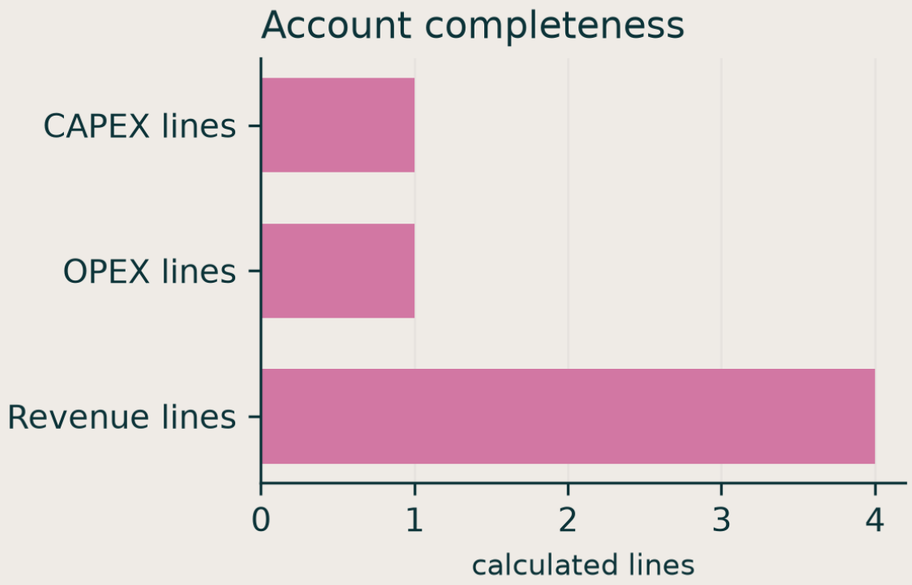
C	0 kg/year
N	0 kg/year
P	0 kg/year
K	0 kg/year

Numerical residuals

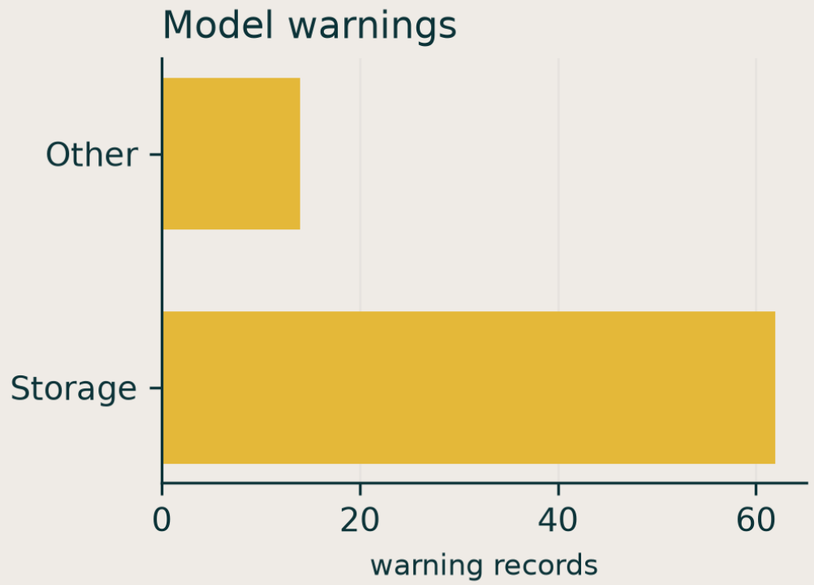
Parameter evidence



Account completeness



Model warnings



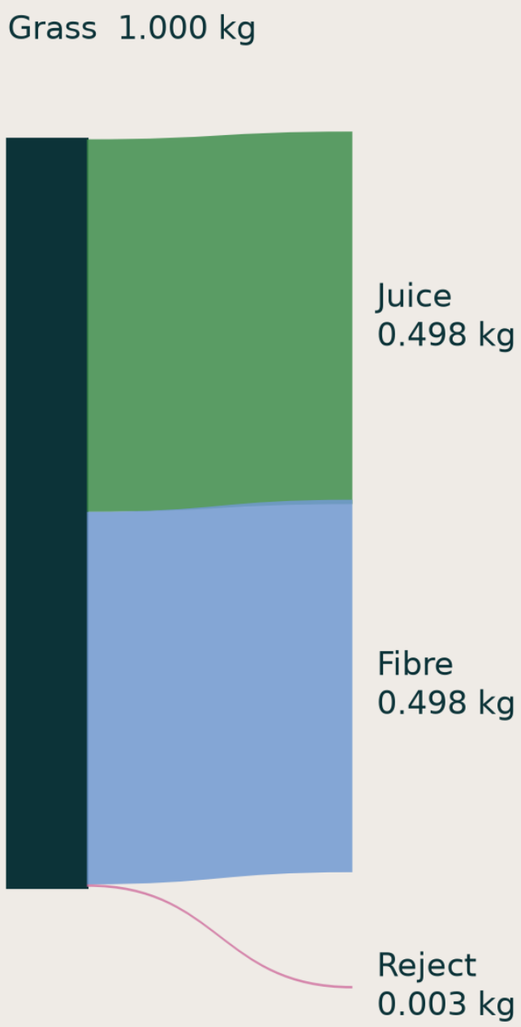
Numerical closure, evidence counts, calculated account lines and warnings

Scenario: matched_protein_bypass rank 4; ba18c74cc7304fcba1a2c3977bb4bf2c; 904.2 kg/h; 8760 h/year
Units: kg/year; records
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

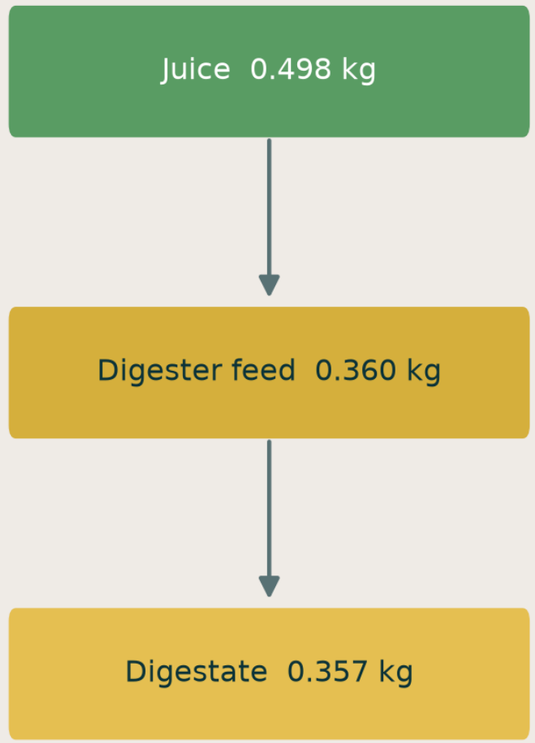
Grass to products

Protein X · Biogas ✓ · N ✓ · P X · K X

Grass split



Juice pathway



Selected intermediate streams

Recovered outputs

N product 0.0119 kg

Nutrient solids 0.0090 kg

Raw gas 0.0021 Nm³

Raw export 0.0021 Nm³

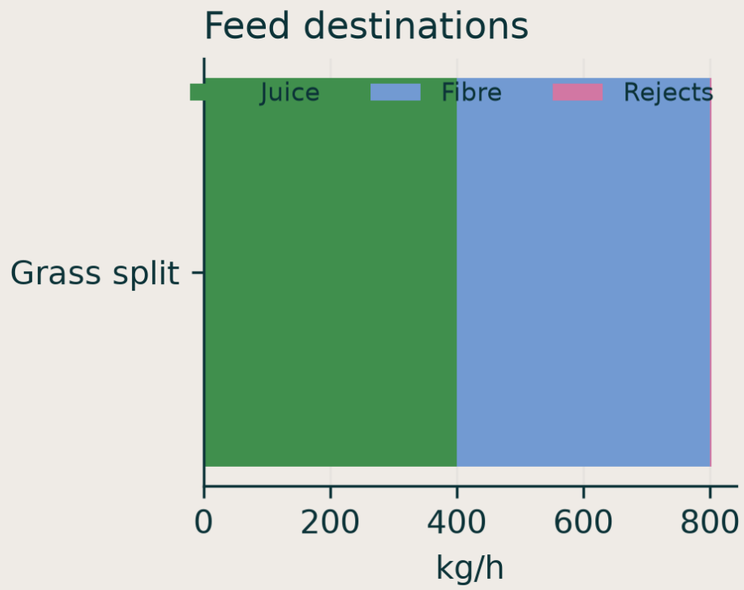
per kg wet grass

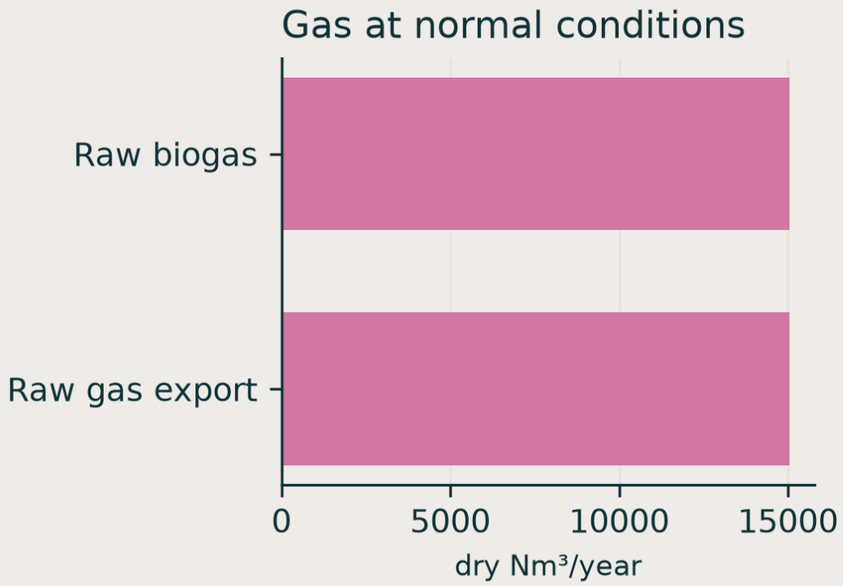
Mass-proportional grass split plus modeled downstream stream/product map

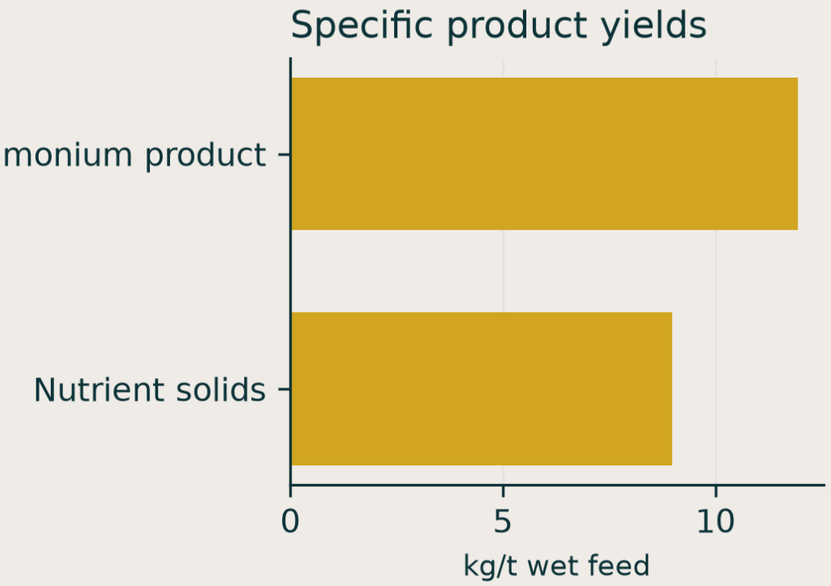
Scenario: matched_protein_bypass rank 5; ec49a089e1e44856adc154029124e8d3; 802.1 kg/h; 8760 h/year
Units: kg/kg wet grass; dry Nm³/kg wet grass
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Material and products

Protein X · Biogas ✓ · N ✓ · P X · K X





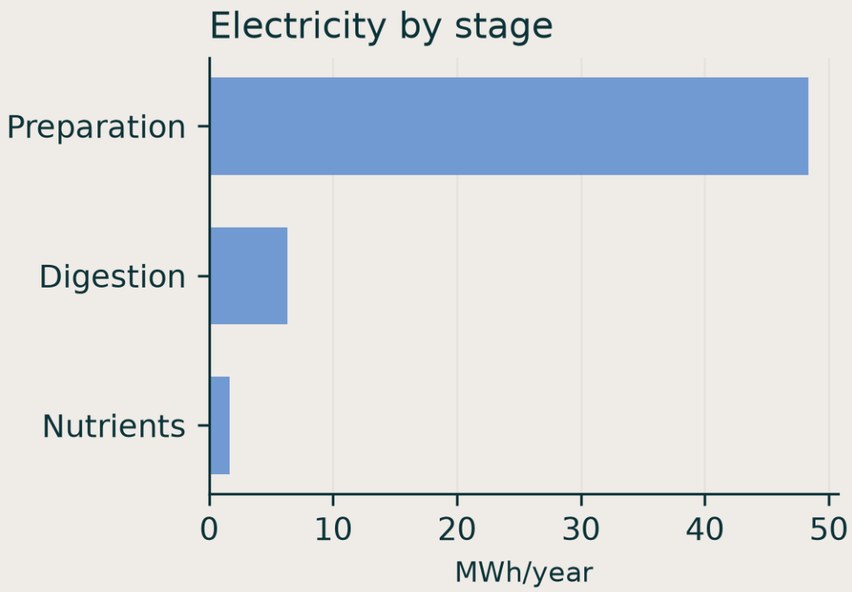


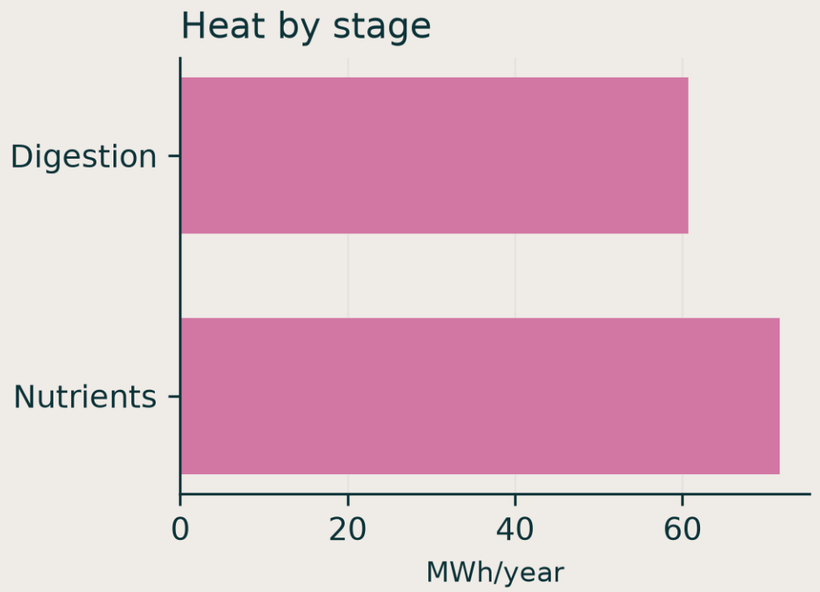
Front-end split, solid products, dry gas volumes and specific yields

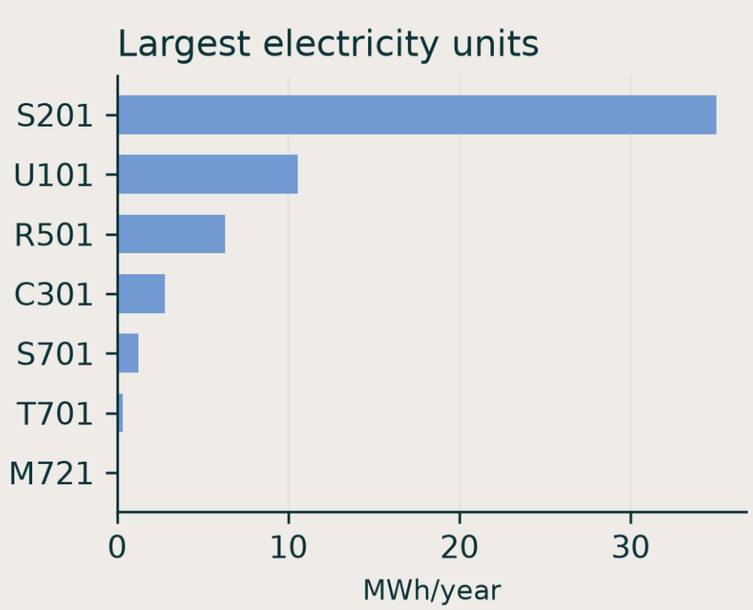
Scenario: matched_protein_bypass rank 5; ec49a089e1e44856adc154029124e8d3; 802.1 kg/h; 8760 h/year
Units: kg/h; t/year; dry Nm³/year; kg/t feed
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

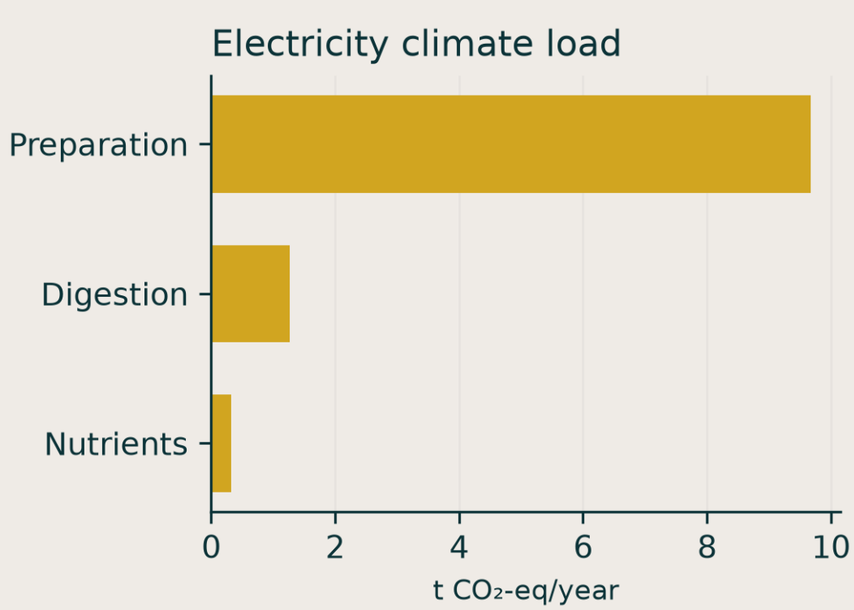
Energy and climate hotspots

Protein X · Biogas ✓ · N ✓ · P X · K X







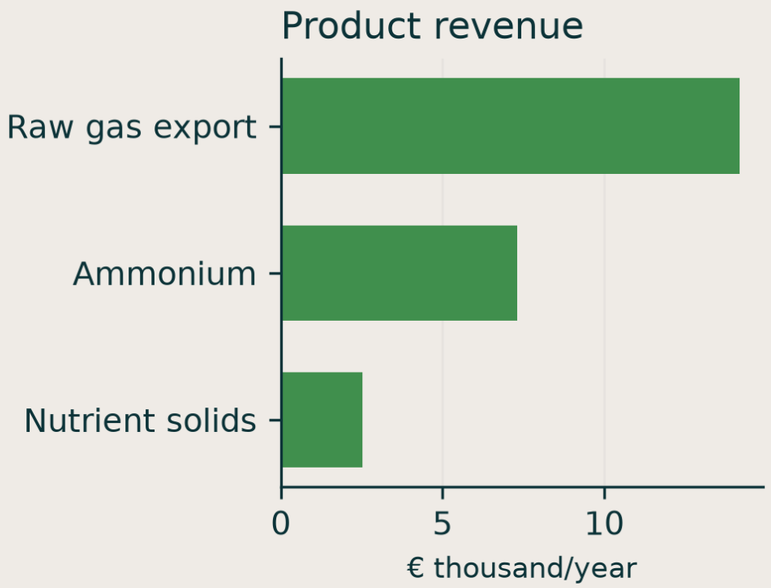


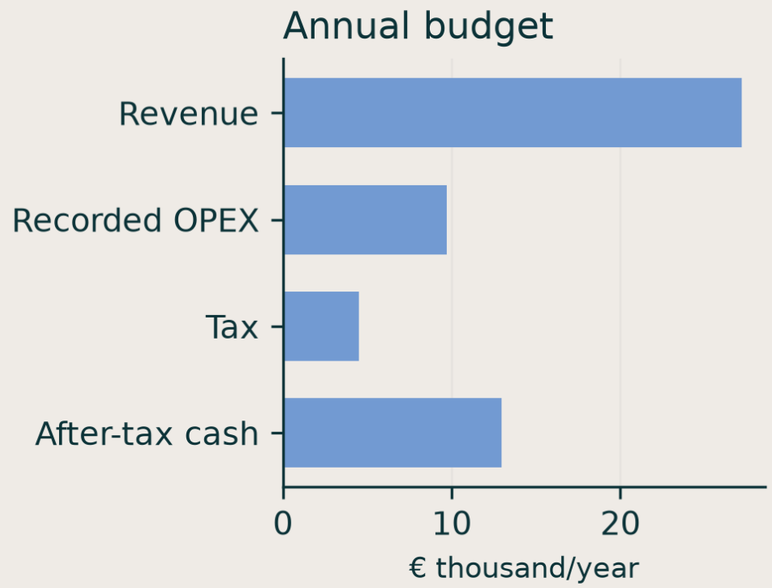
Stage and unit utility hotspots and included climate load

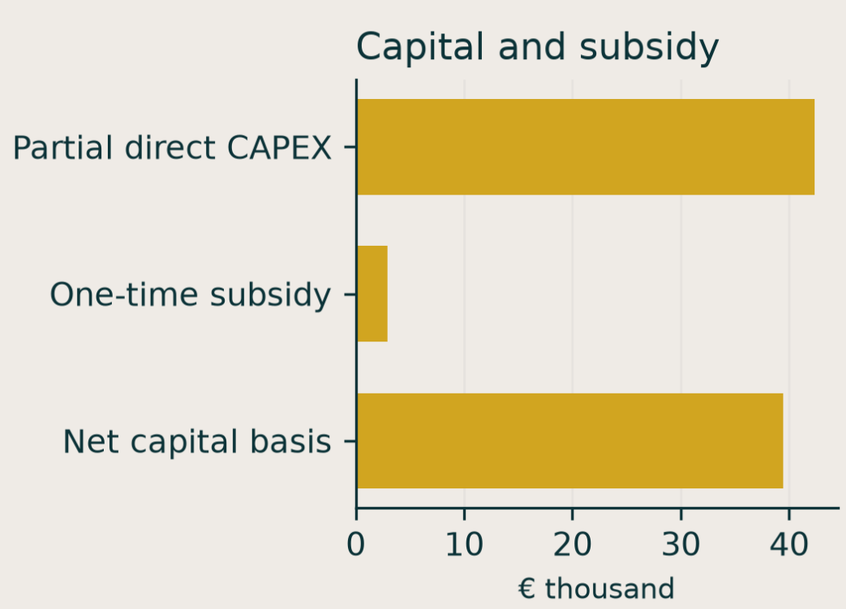
Scenario: matched_protein_bypass rank 5; ec49a089e1e44856adc154029124e8d3; 802.1 kg/h; 8760 h/year
Units: MWh/year; t CO_2-eq/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

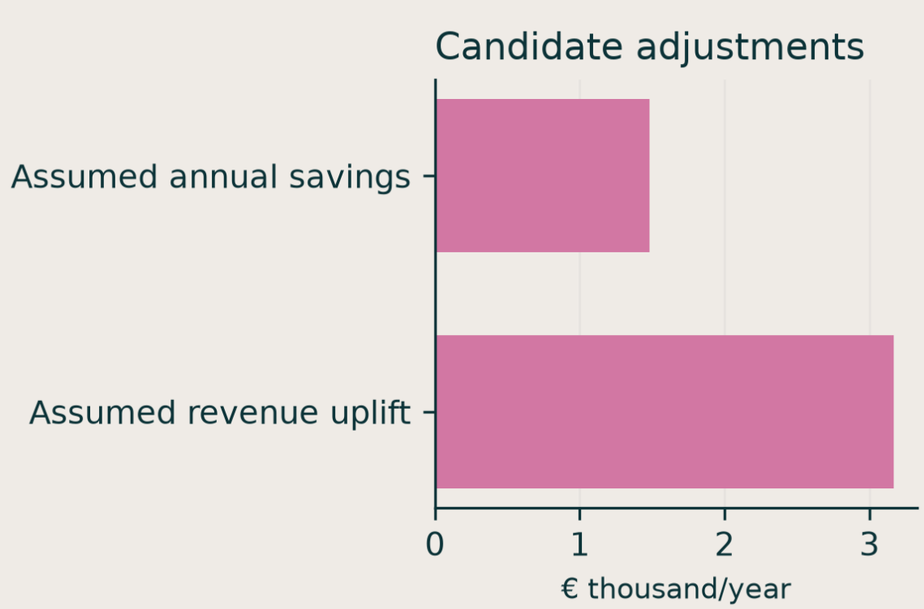
Economic accounts

Protein X · Biogas ✓ · N ✓ · P X · K X







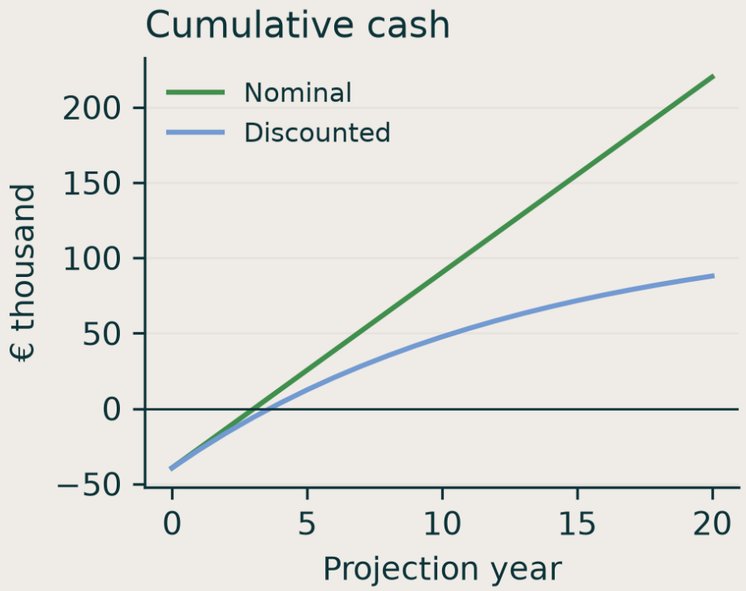


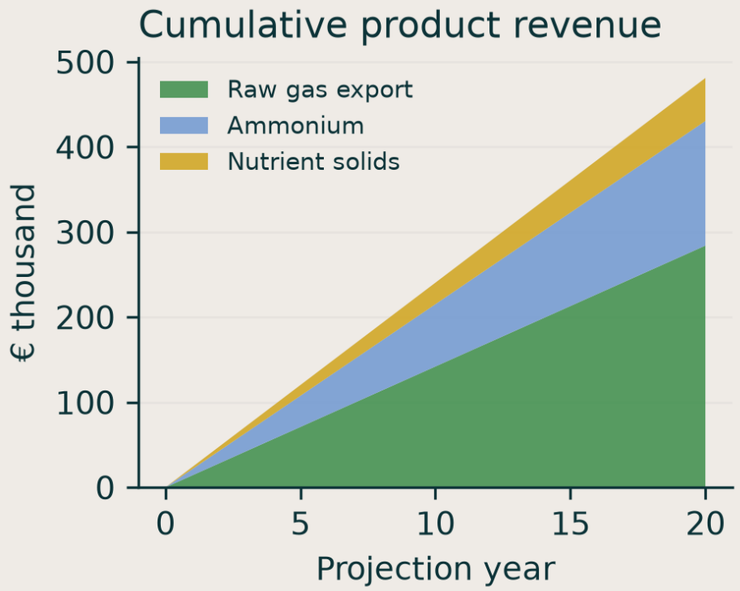
Product revenue, partial annual budget, capital subsidy and candidate overlays

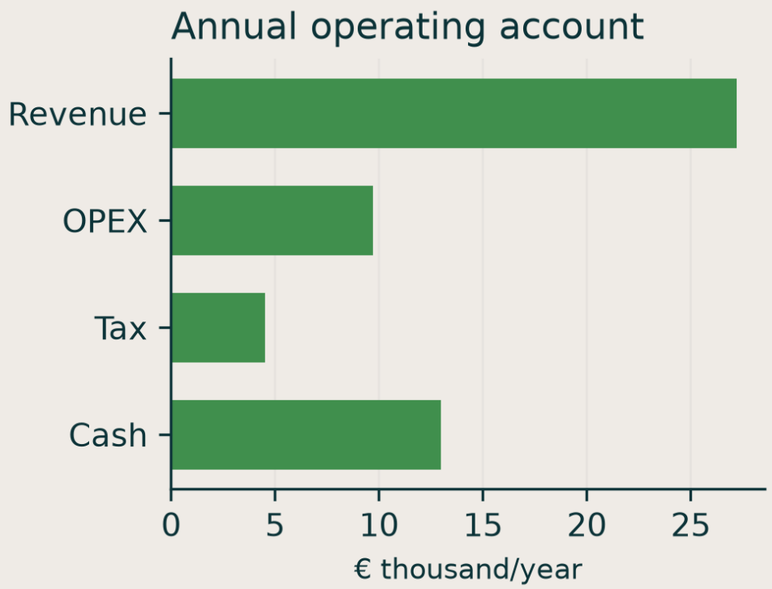
Scenario: matched_protein_bypass rank 5; ec49a089e1e44856adc154029124e8d3; 802.1 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

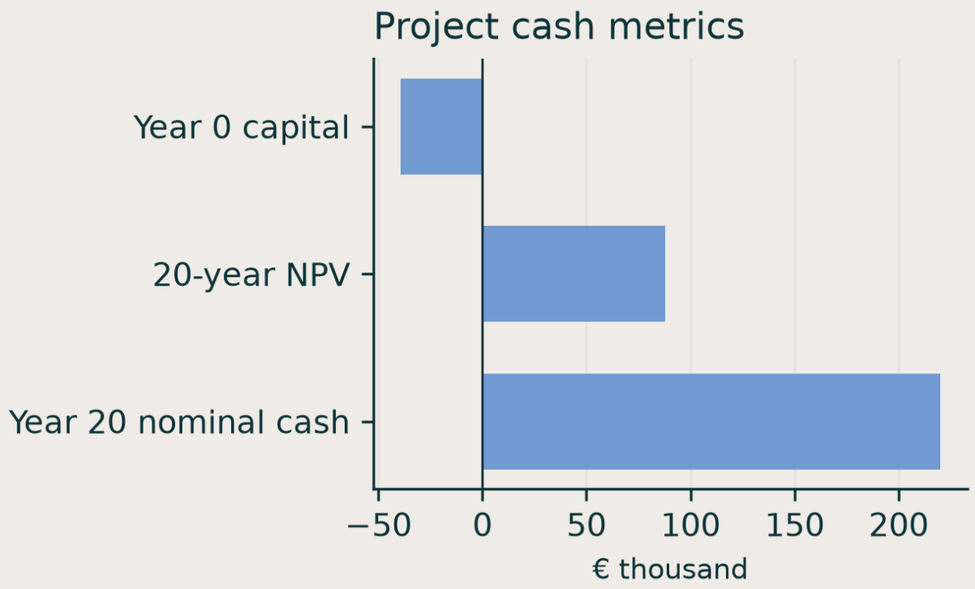
20-year financial projection

Protein X · Biogas ✓ · N ✓ · P X · K X









Constant-account cash flow and cumulative product revenue over 20 years

Scenario: matched_protein_bypass rank 5; ec49a089e1e44856adc154029124e8d3; 802.1 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Balance and evidence

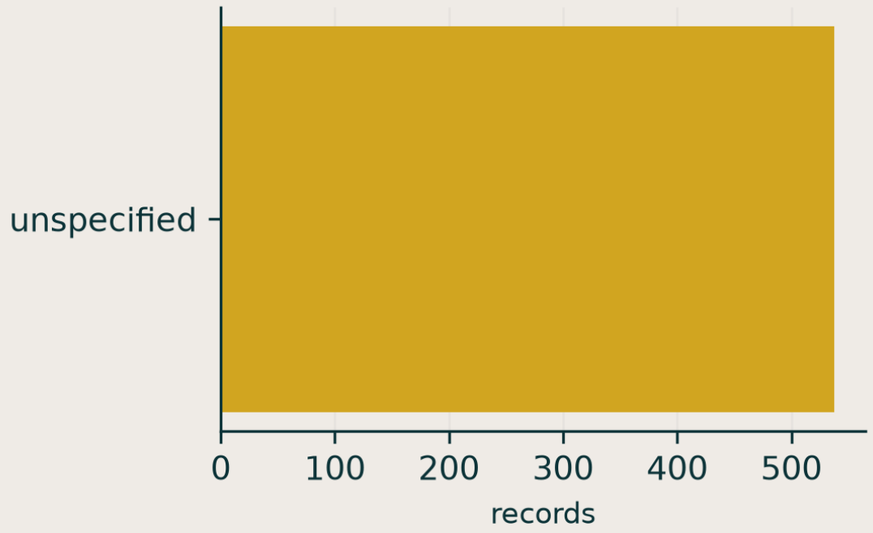
Protein X · Biogas ✓ · N ✓ · P X · K X

Element closure

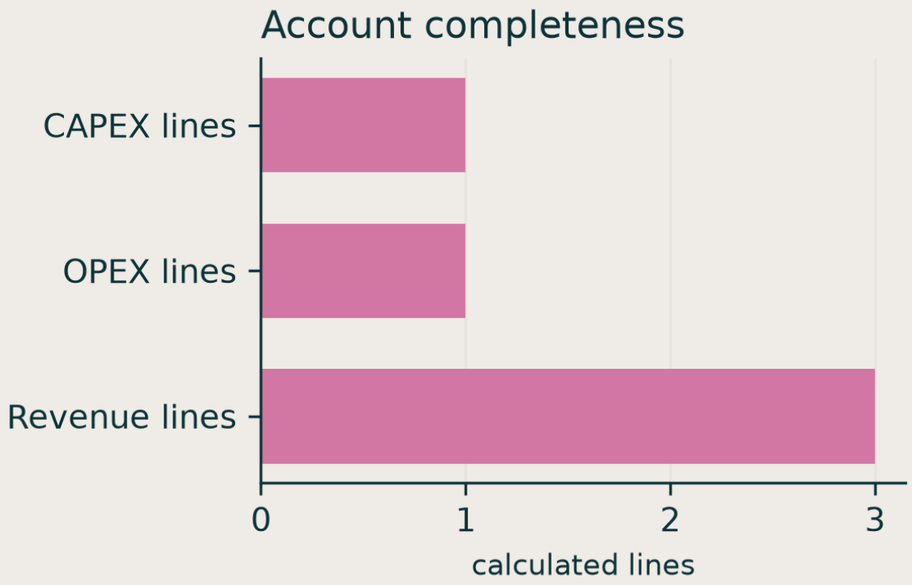
C	0 kg/year
N	0 kg/year
P	0 kg/year
K	0 kg/year

Numerical residuals

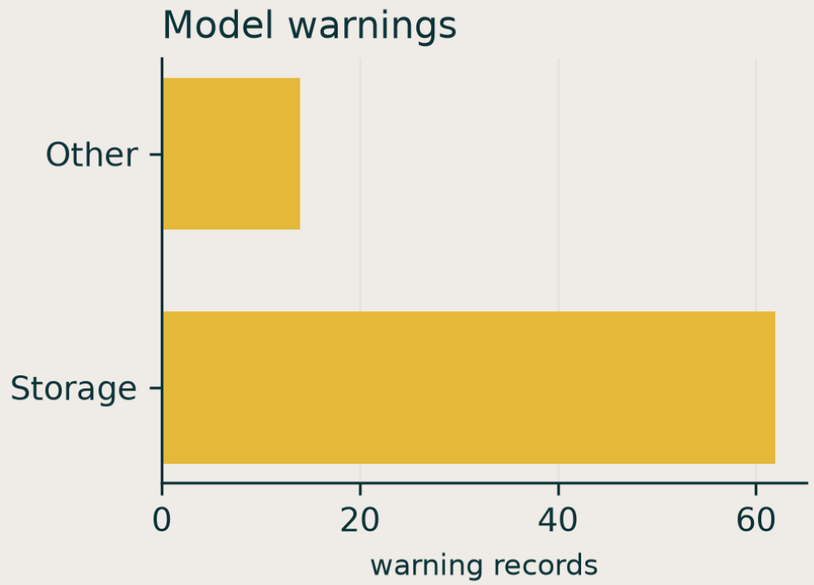
Parameter evidence



Account completeness



Model warnings



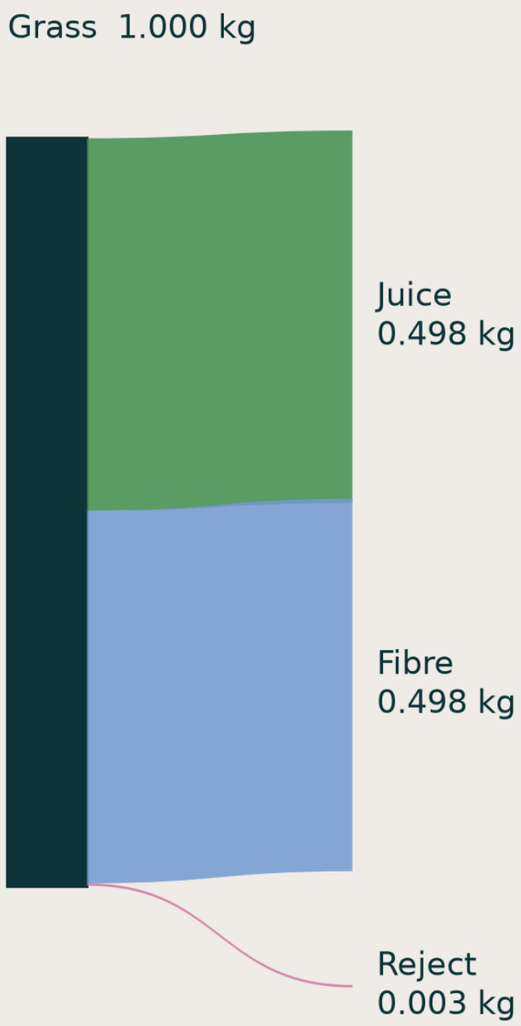
Numerical closure, evidence counts, calculated account lines and warnings

Scenario: matched_protein_bypass rank 5; ec49a089e1e44856adc154029124e8d3; 802.1 kg/h; 8760 h/year
Units: kg/year; records
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

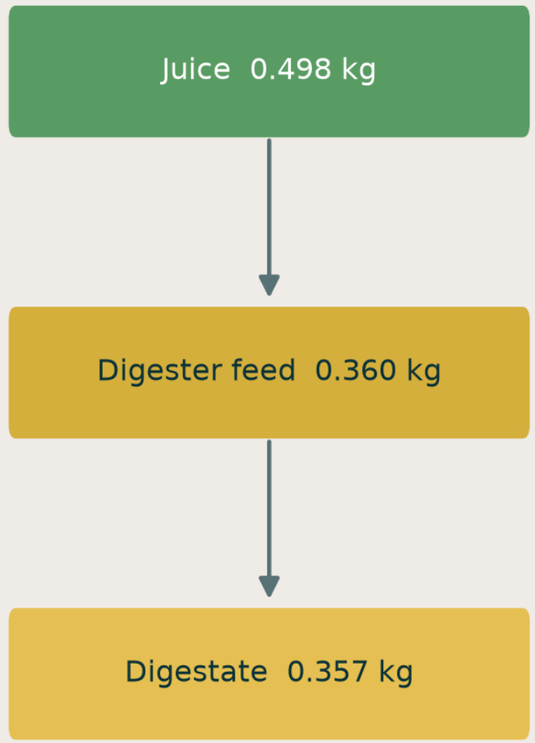
Grass to products

Protein X · Biogas ✓ · N ✓ · P X · K X

Grass split

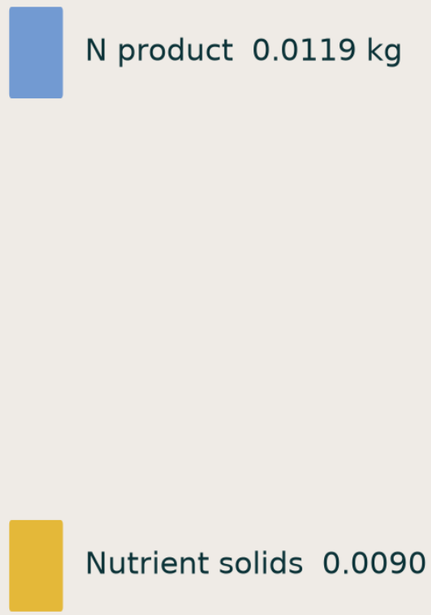


Juice pathway



Selected intermediate streams

Recovered outputs



Raw gas 0.0021 Nm³

Upgraded 0.0014 Nm³

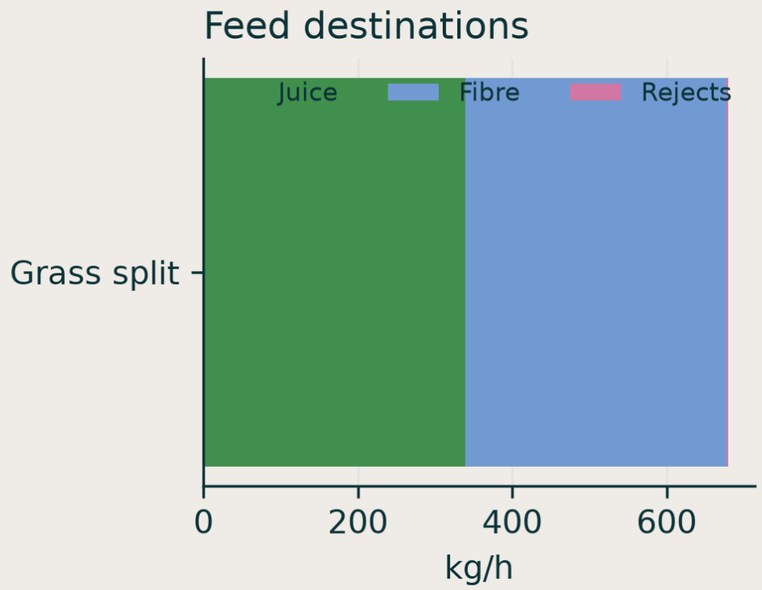
per kg wet grass

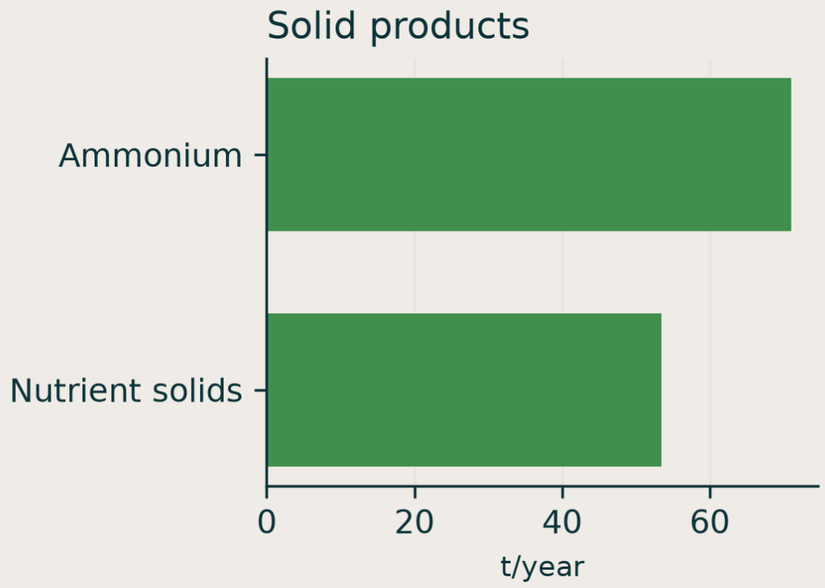
Mass-proportional grass split plus modeled downstream stream/product map

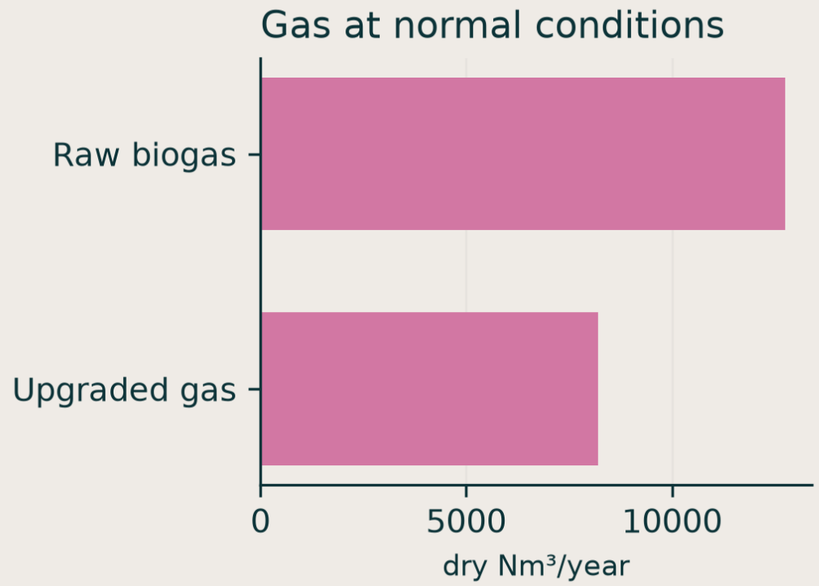
Scenario: matched_protein_bypass rank 6; 0f8fc768eeb941e0b1ebd9a216bfbe93; 679.9 kg/h; 8760 h/year
Units: kg/kg wet grass; dry Nm³/kg wet grass
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

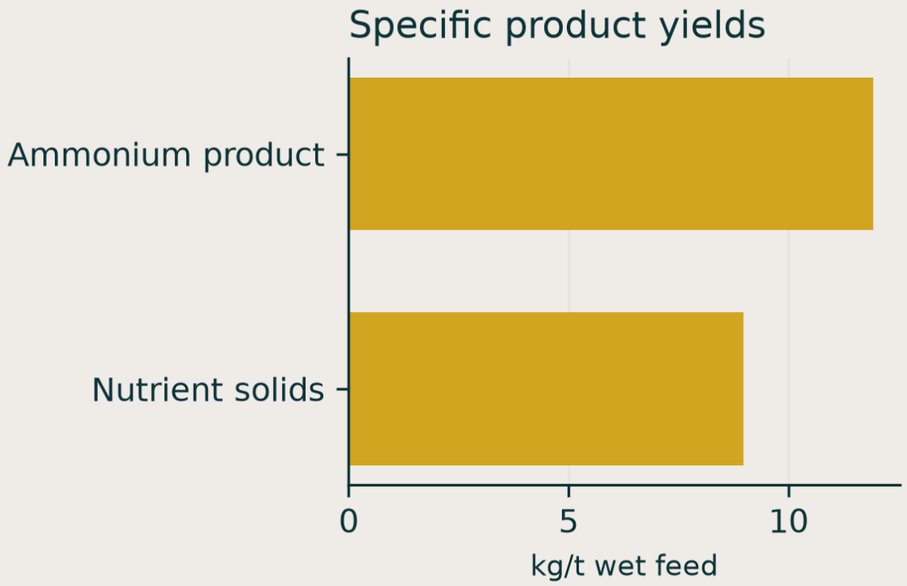
Material and products

Protein X · Biogas ✓ · N ✓ · P X · K X







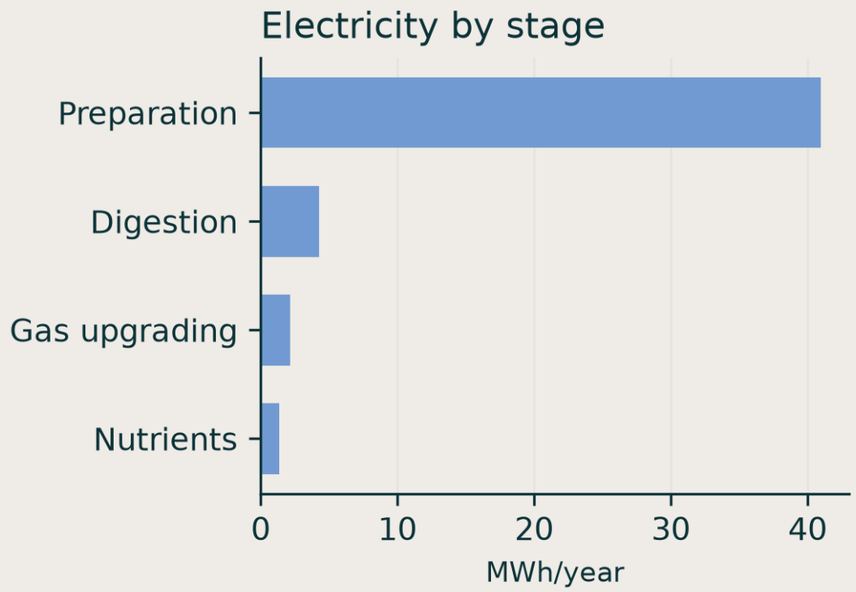


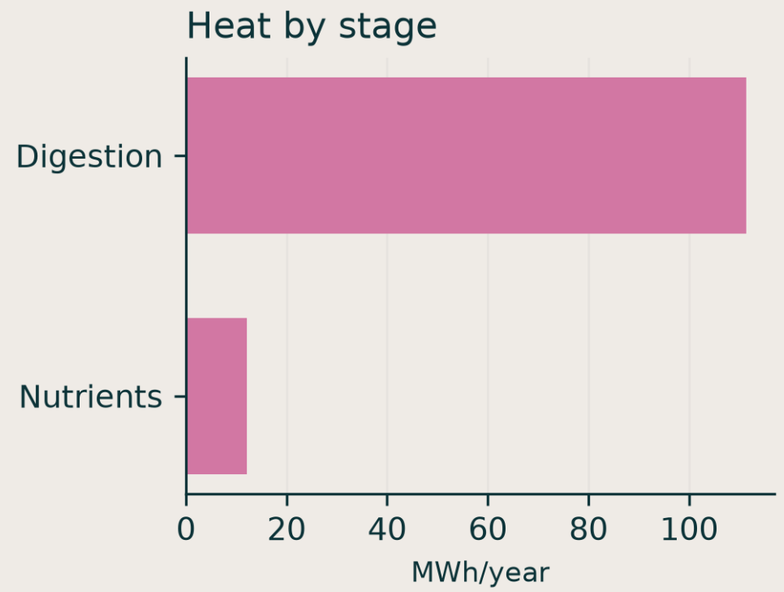
Front-end split, solid products, dry gas volumes and specific yields

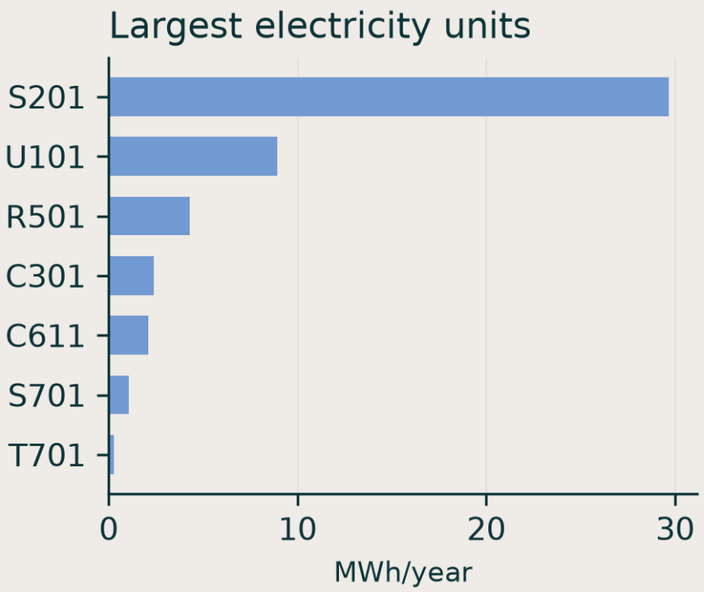
Scenario: matched_protein_bypass rank 6; 0f8fc768eeb941e0b1ebd9a216bfbe93; 679.9 kg/h; 8760 h/year
Units: kg/h; t/year; dry Nm³/year; kg/t feed
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

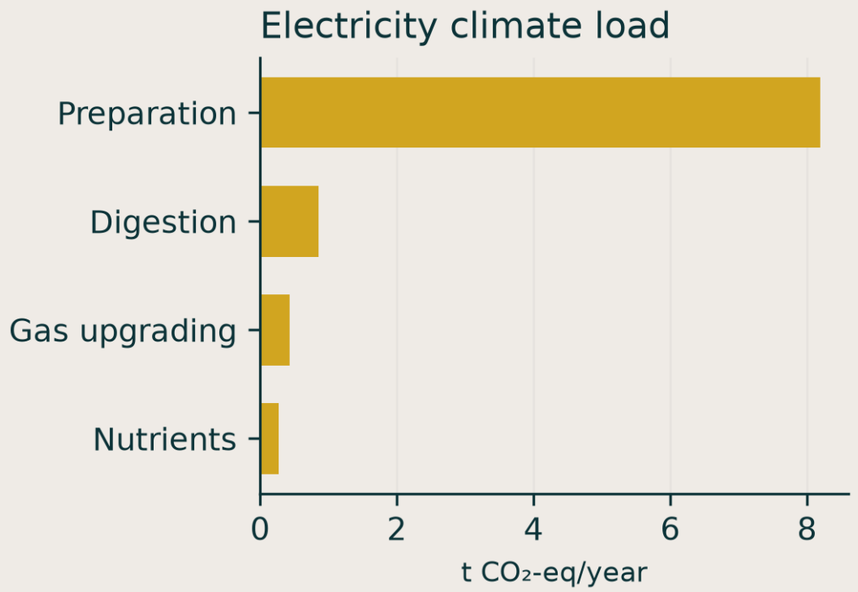
Energy and climate hotspots

Protein X · Biogas ✓ · N ✓ · P X · K X







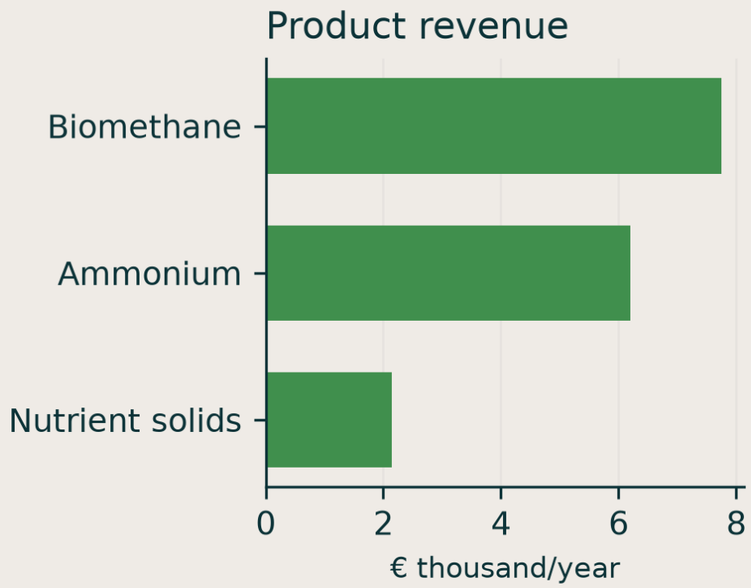


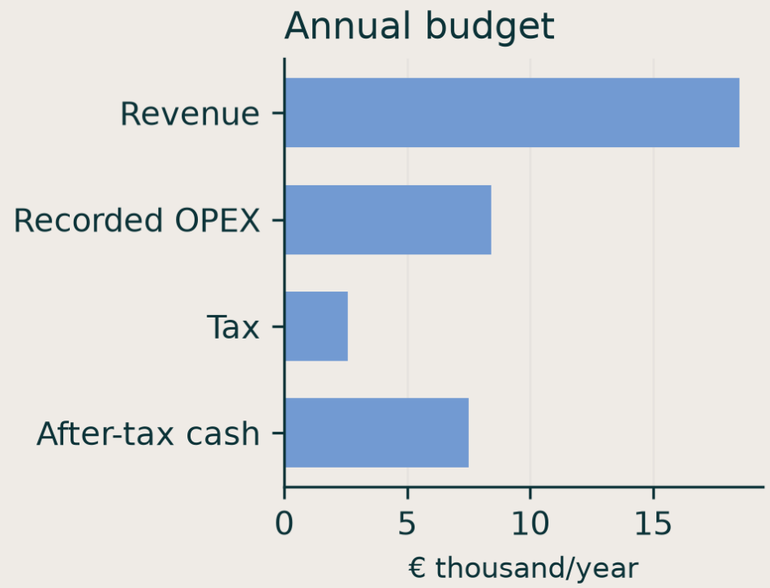
Stage and unit utility hotspots and included climate load

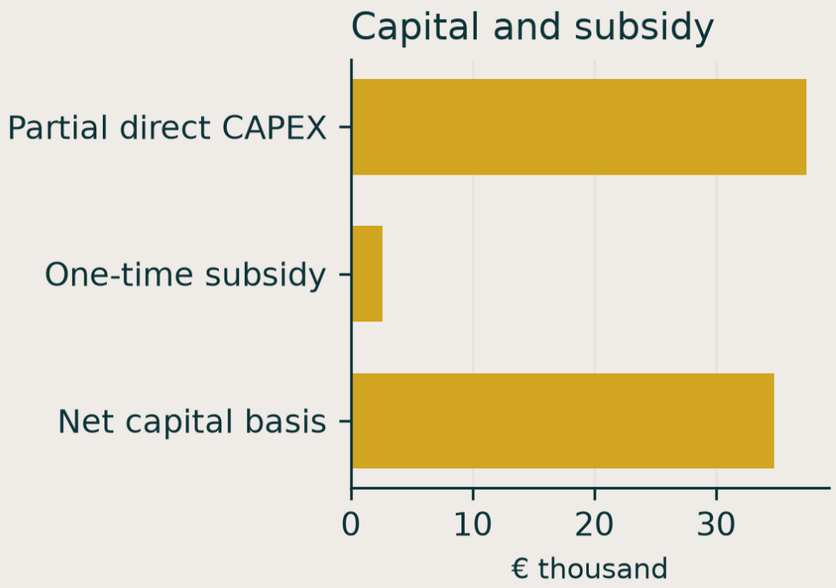
Scenario: matched_protein_bypass rank 6; 0f8fc768eeb941e0b1ebd9a216bfbe93; 679.9 kg/h; 8760 h/year
Units: MWh/year; t CO_2-eq/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

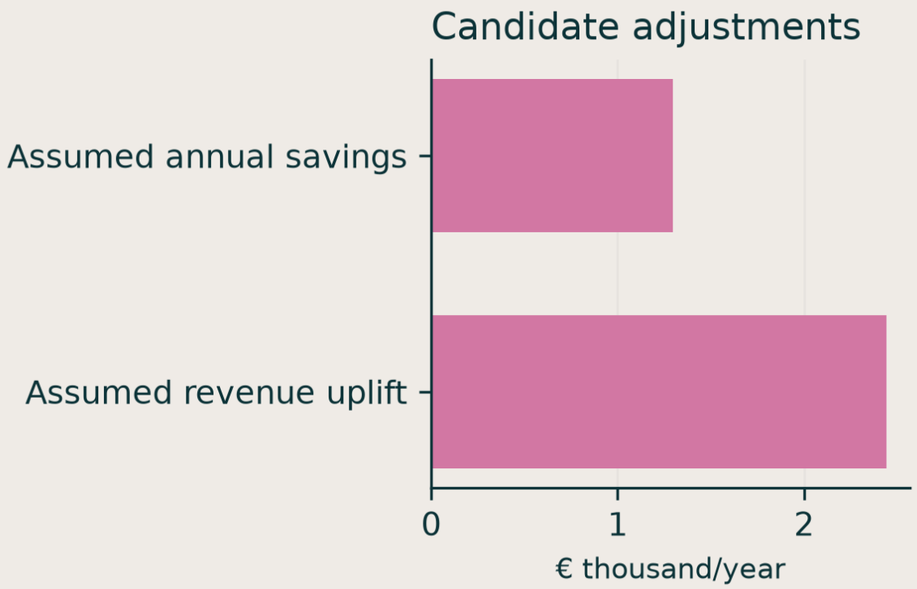
Economic accounts

Protein X · Biogas ✓ · N ✓ · P X · K X







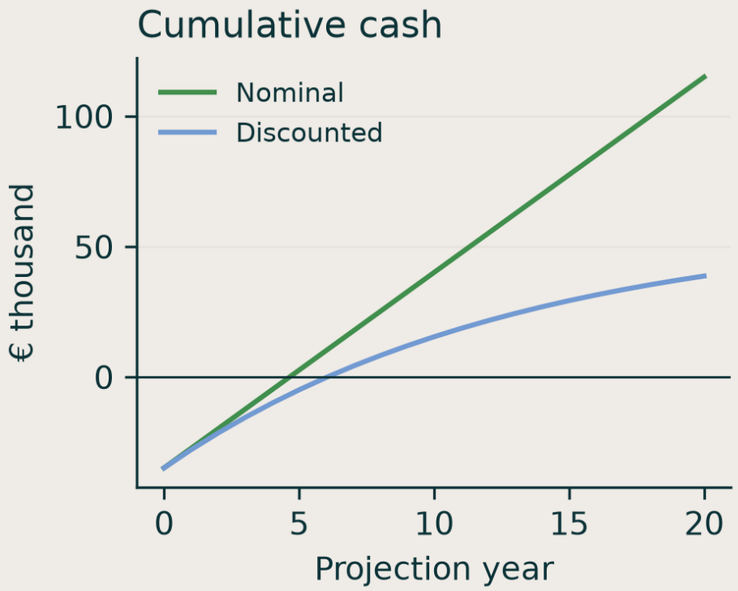


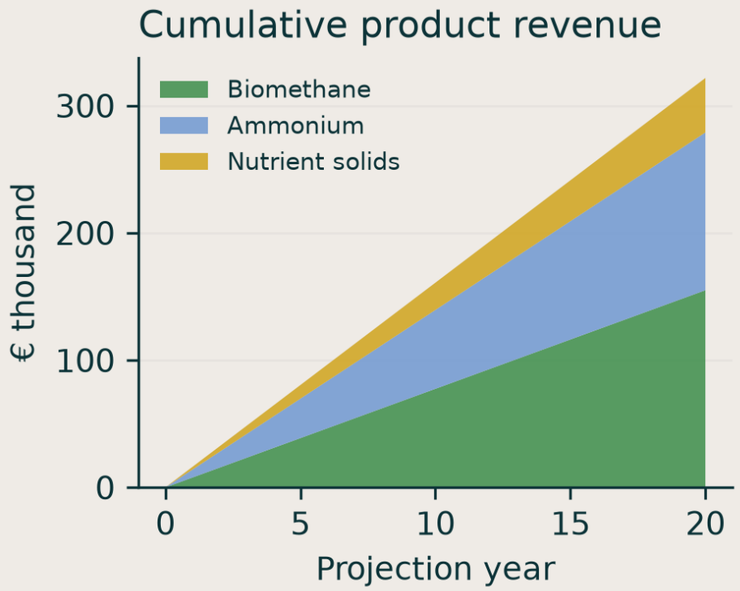
Product revenue, partial annual budget, capital subsidy and candidate overlays

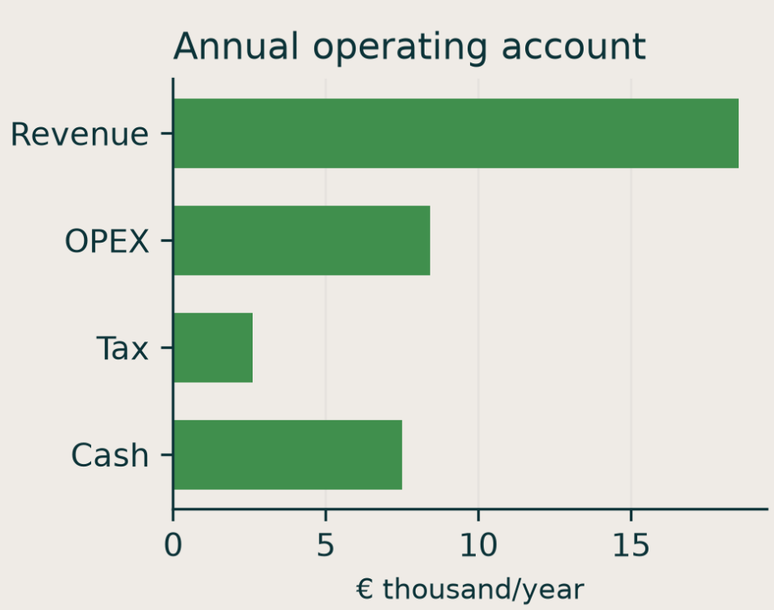
Scenario: matched_protein_bypass rank 6; 0f8fc768eeb941e0b1ebd9a216bfbe93; 679.9 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

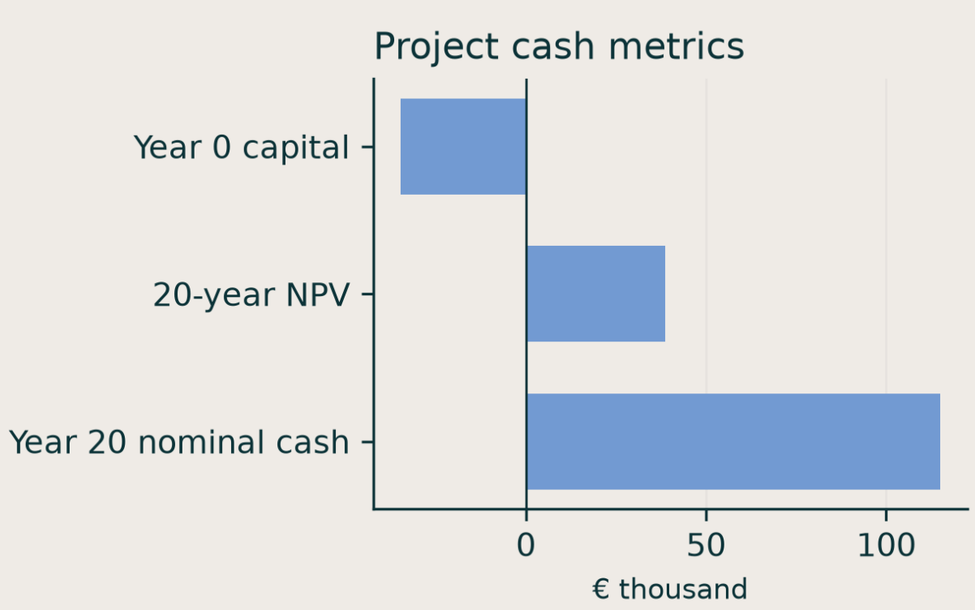
20-year financial projection

Protein X · Biogas ✓ · N ✓ · P X · K X









Constant-account cash flow and cumulative product revenue over 20 years

Scenario: matched_protein_bypass rank 6; 0f8fc768eeb941e0b1ebd9a216bfbe93; 679.9 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Balance and evidence

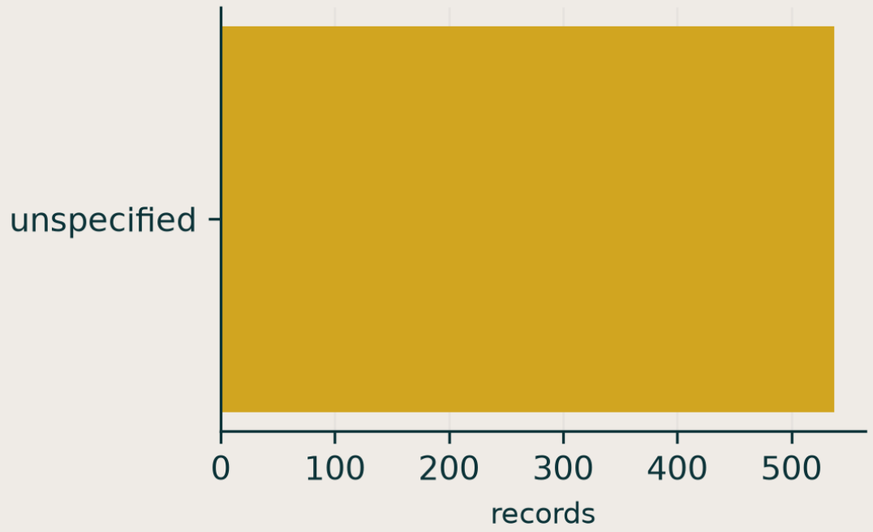
Protein X · Biogas ✓ · N ✓ · P X · K X

Element closure

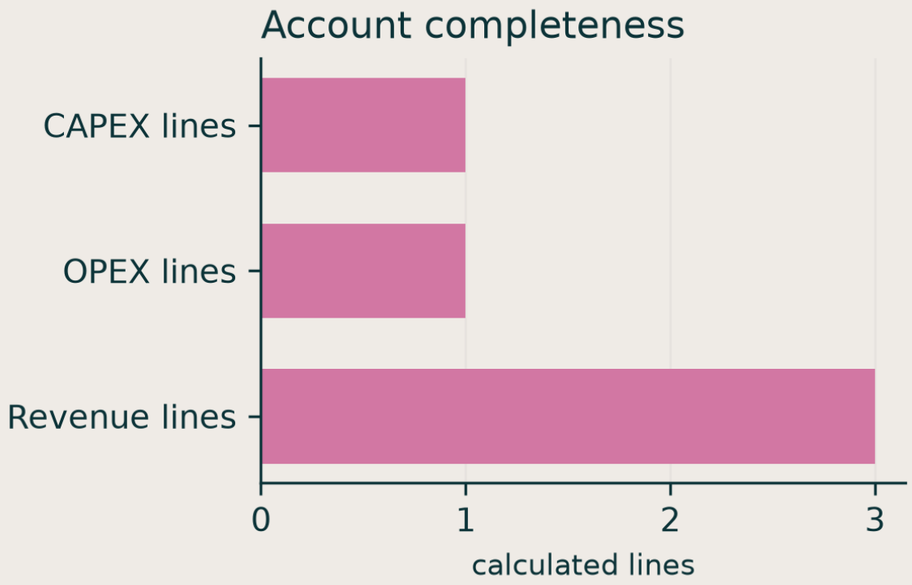
C	0 kg/year
N	0 kg/year
P	0 kg/year
K	0 kg/year

Numerical residuals

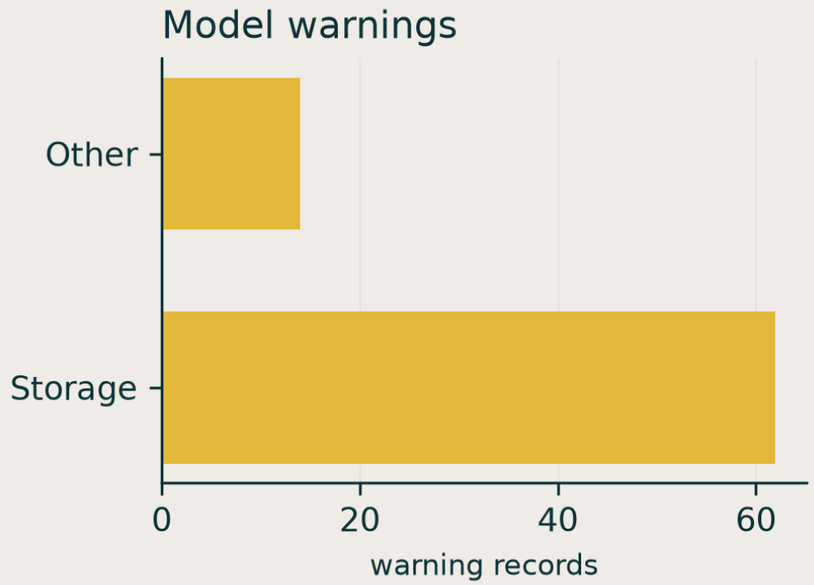
Parameter evidence



Account completeness



Model warnings



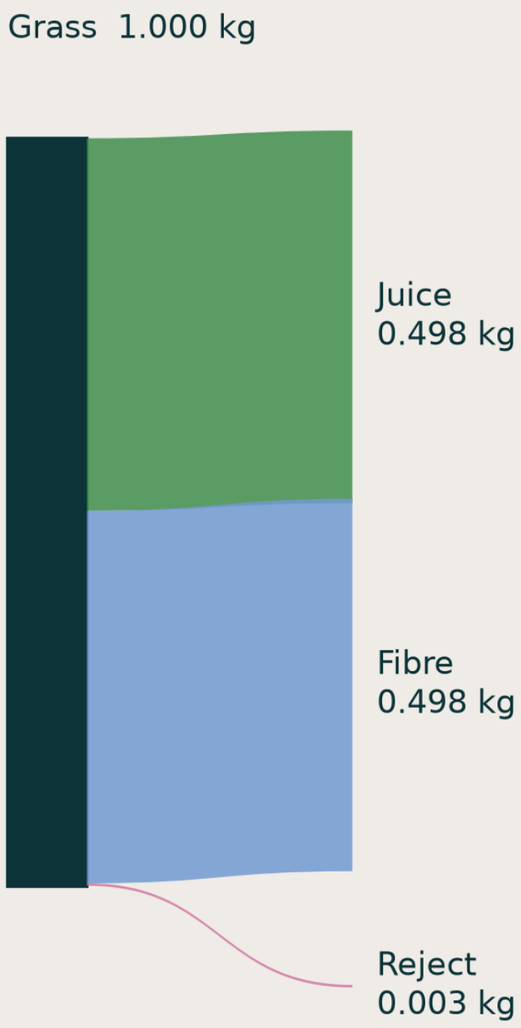
Numerical closure, evidence counts, calculated account lines and warnings

Scenario: matched_protein_bypass rank 6; 0f8fc768eeb941e0b1ebd9a216bfbe93; 679.9 kg/h; 8760 h/year
Units: kg/year; records
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

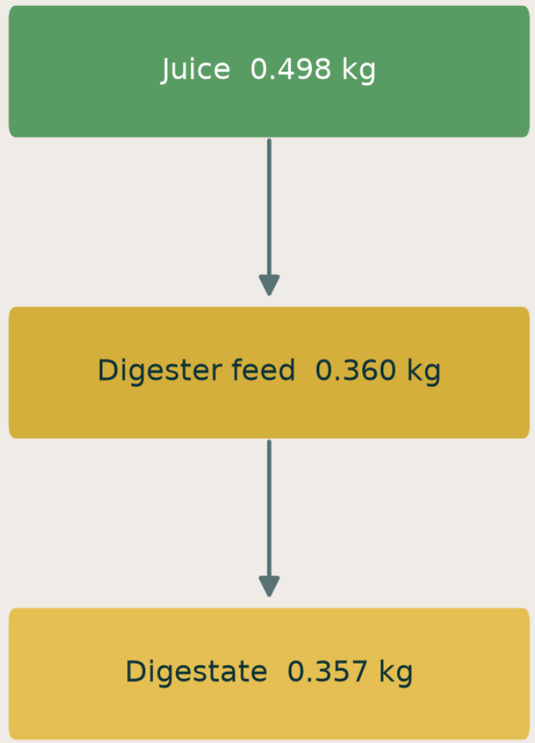
Grass to products

Protein X · Biogas ✓ · N ✓ · P ✓ · K X

Grass split



Juice pathway



Selected intermediate streams

Recovered outputs

Raw gas 0.0021 Nm³

Upgraded 0.0017 Nm³

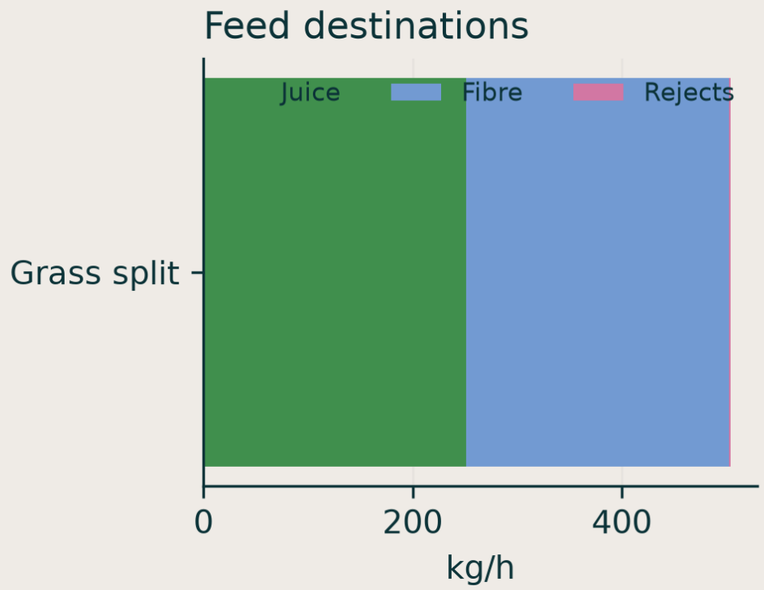
per kg wet grass

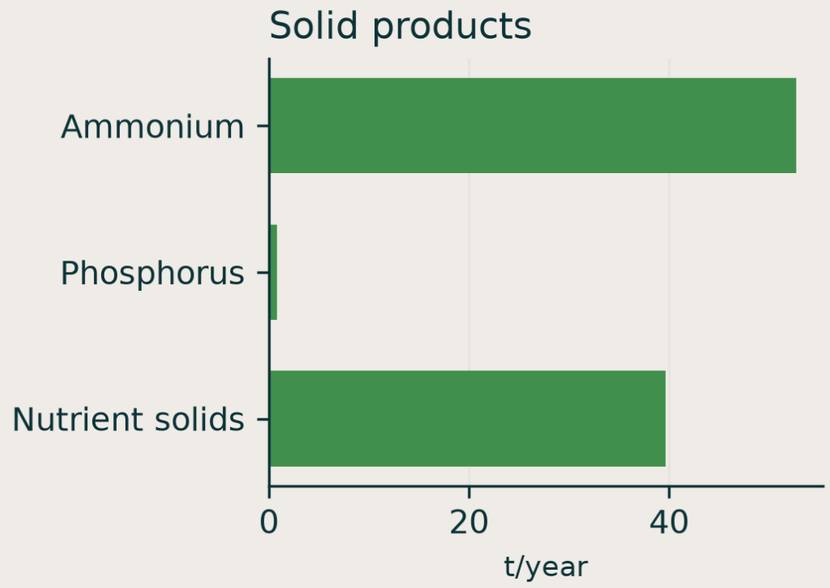
Mass-proportional grass split plus modeled downstream stream/product map

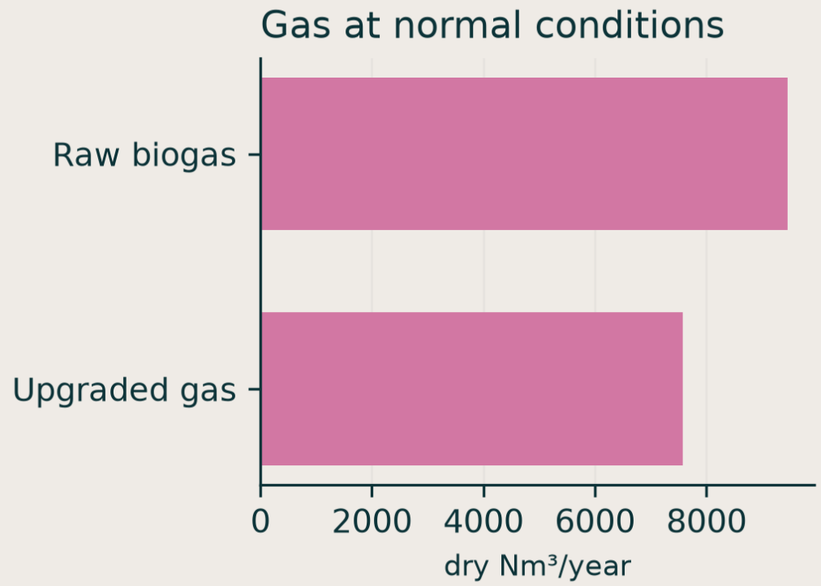
Scenario: matched_protein_bypass rank 7; ab67fb691fc940ceb37c54dd707d8501; 504.3 kg/h; 8760 h/year
Units: kg/kg wet grass; dry Nm³/kg wet grass
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

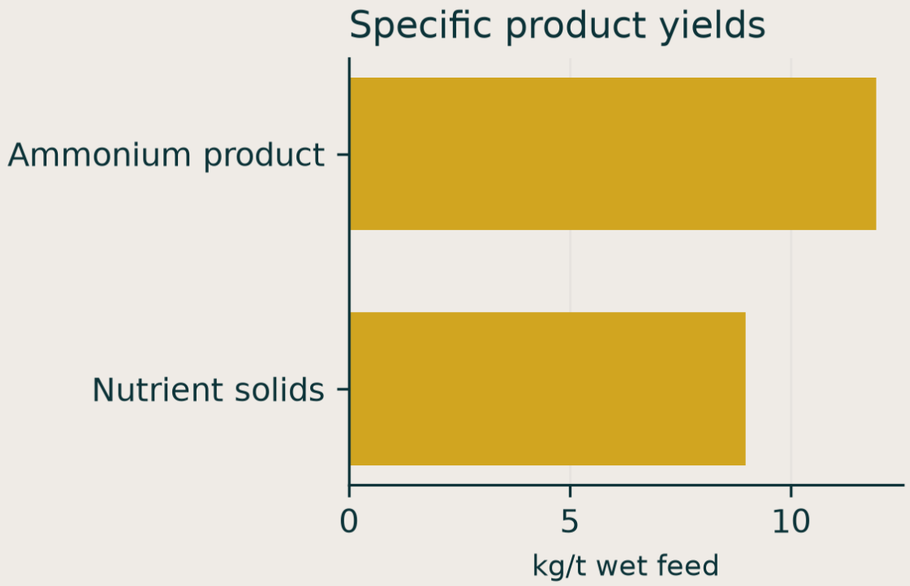
Material and products

Protein X · Biogas ✓ · N ✓ · P ✓ · K X







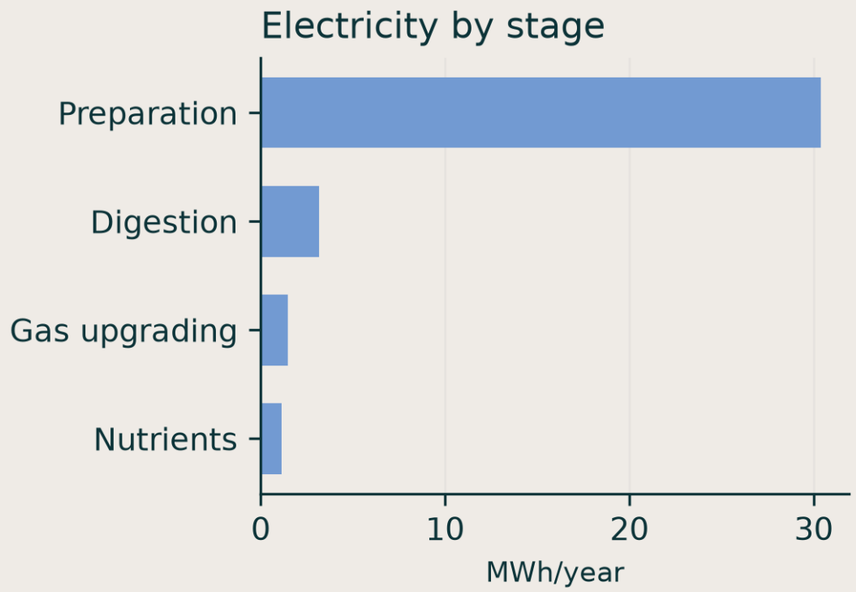


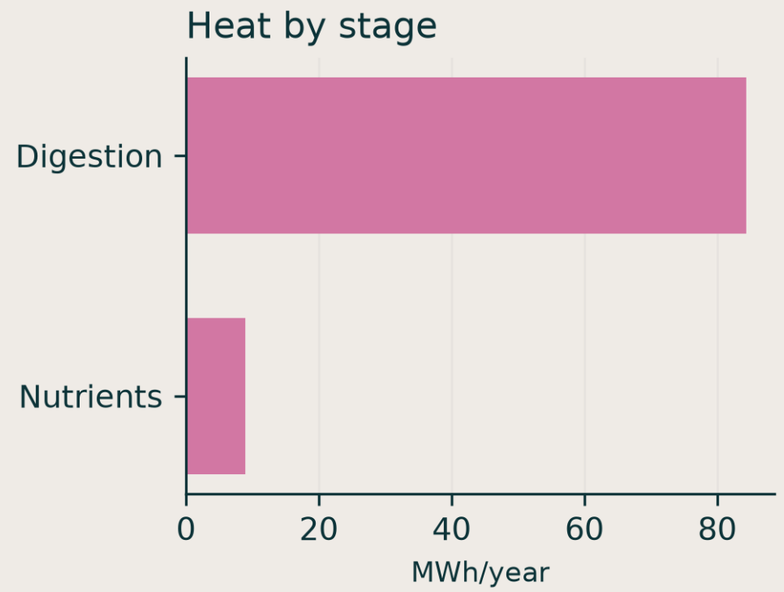
Front-end split, solid products, dry gas volumes and specific yields

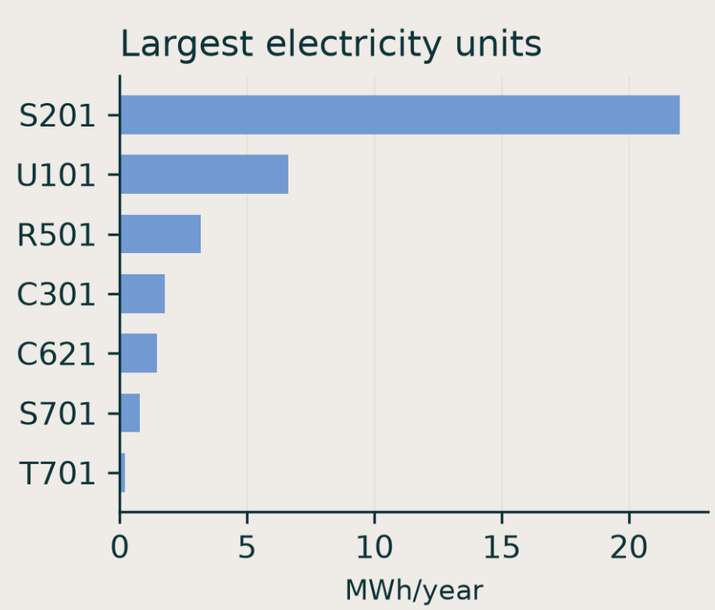
Scenario: matched_protein_bypass rank 7; ab67fb691fc940ceb37c54dd707d8501; 504.3 kg/h; 8760 h/year
Units: kg/h; t/year; dry Nm³/year; kg/t feed
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

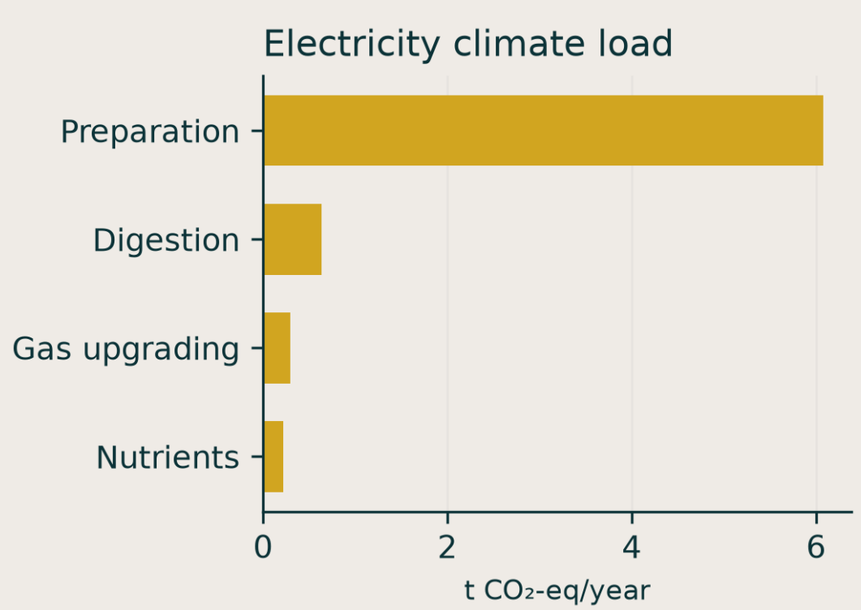
Energy and climate hotspots

Protein X · Biogas ✓ · N ✓ · P ✓ · K X







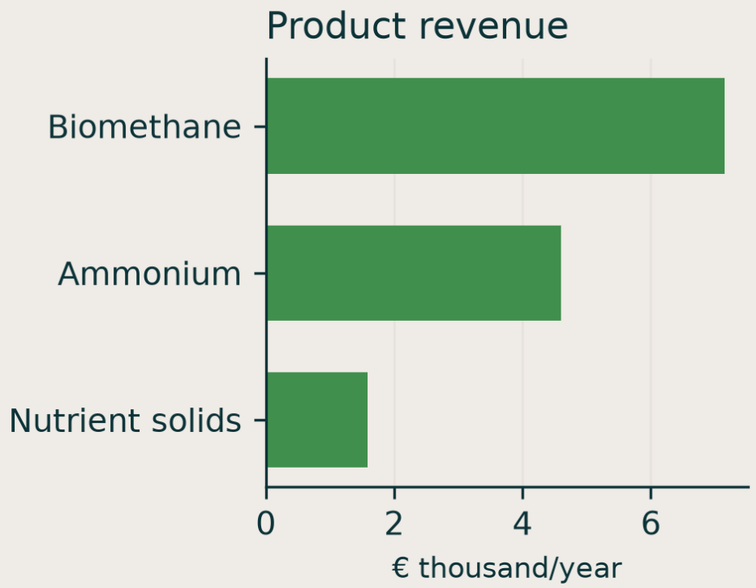


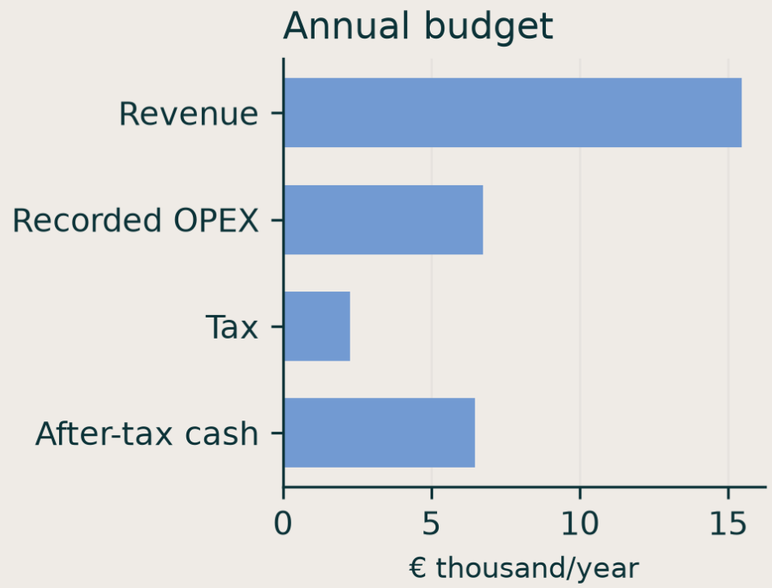
Stage and unit utility hotspots and included climate load

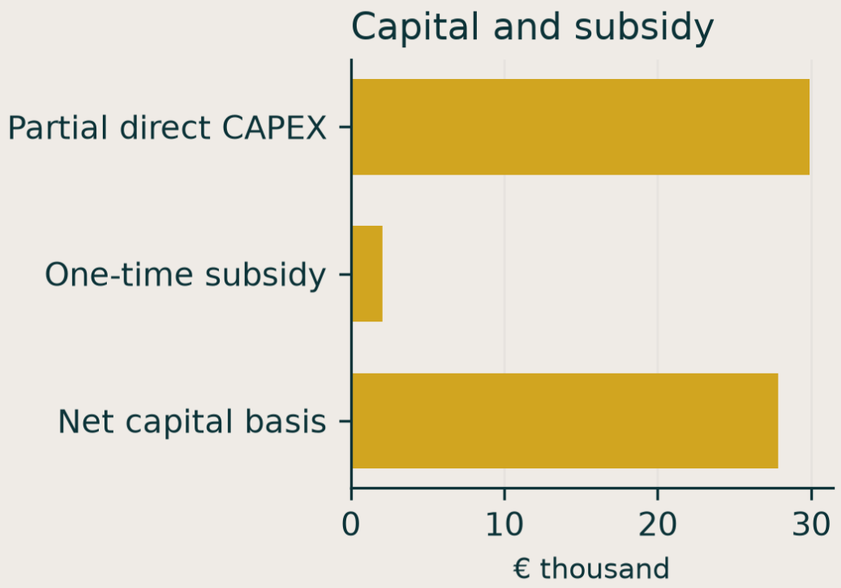
Scenario: matched_protein_bypass rank 7; ab67fb691fc940ceb37c54dd707d8501; 504.3 kg/h; 8760 h/year
Units: MWh/year; t CO_2-eq/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

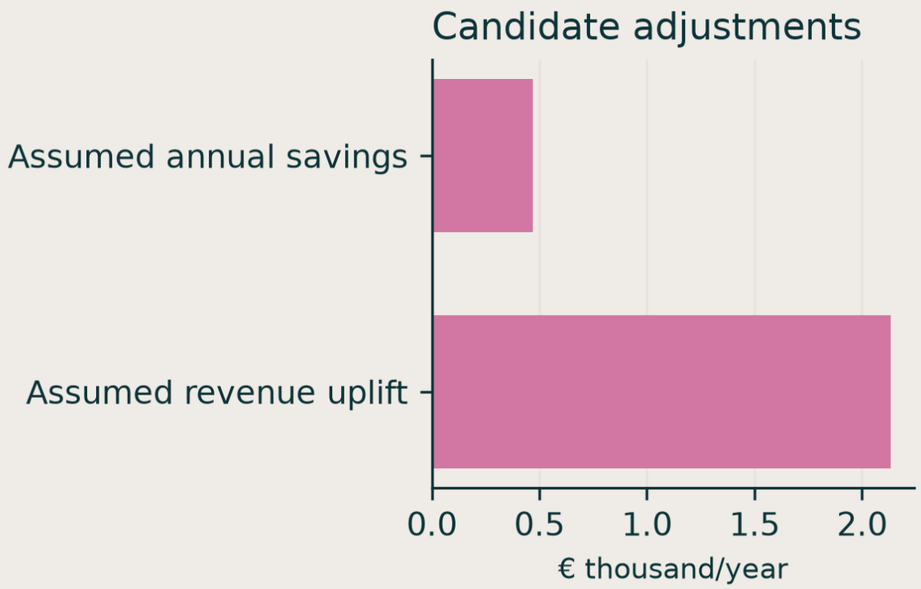
Economic accounts

Protein X · Biogas ✓ · N ✓ · P ✓ · K X







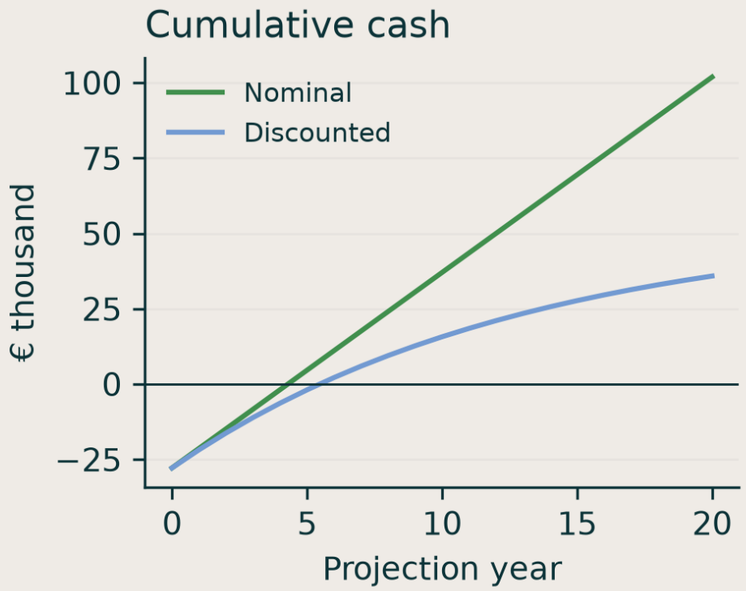


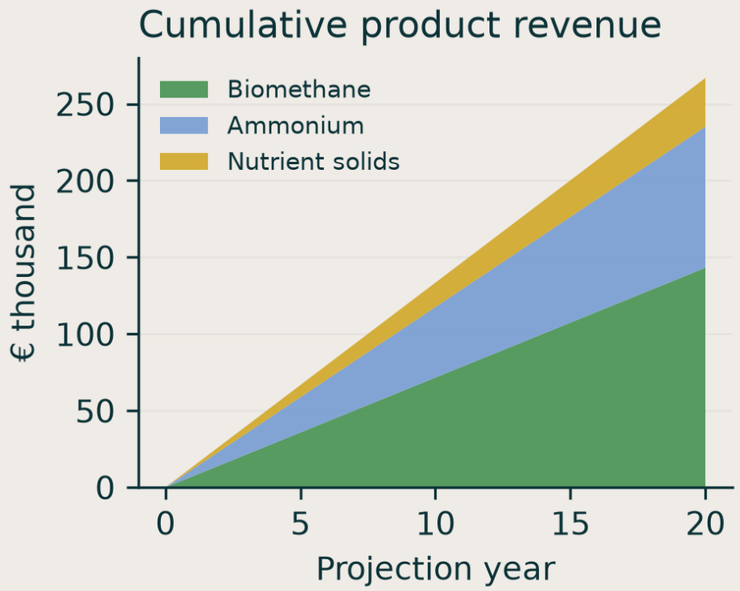
Product revenue, partial annual budget, capital subsidy and candidate overlays

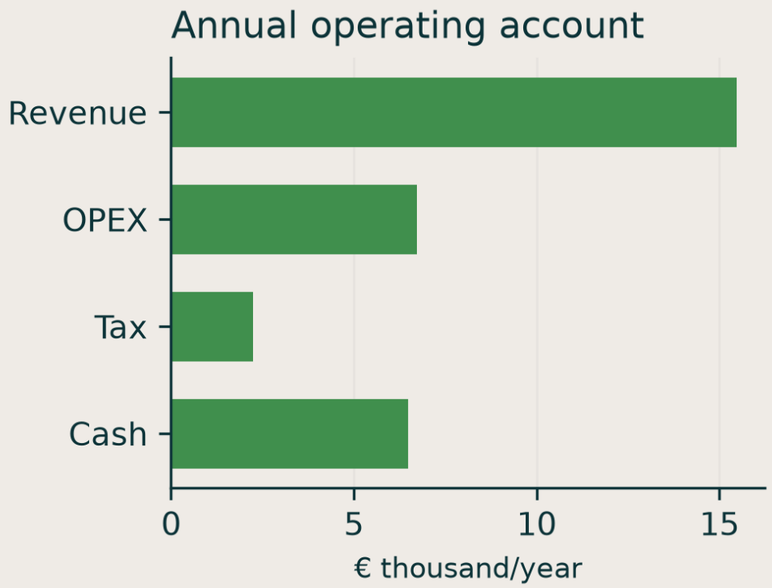
Scenario: matched_protein_bypass rank 7; ab67fb691fc940ceb37c54dd707d8501; 504.3 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

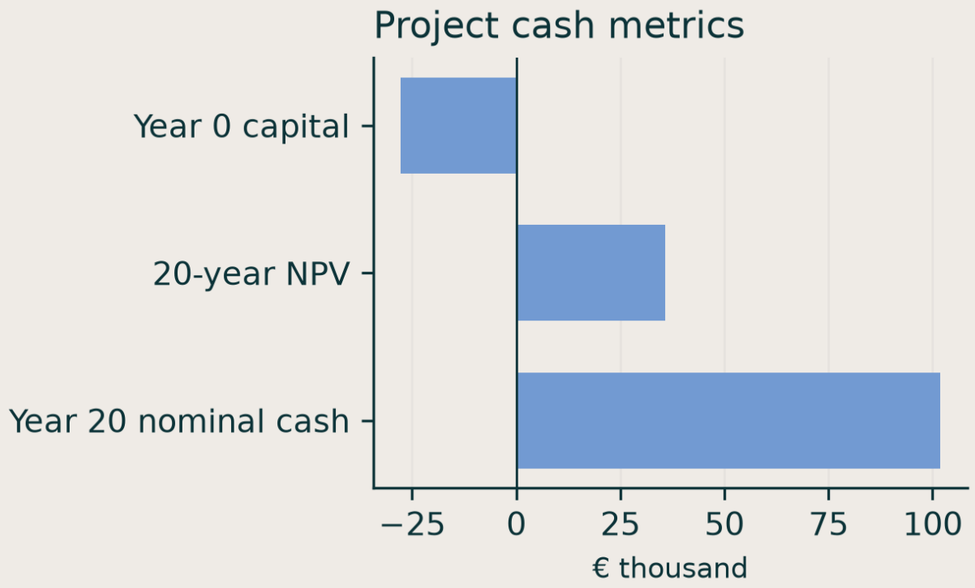
20-year financial projection

Protein X · Biogas ✓ · N ✓ · P ✓ · K X









Constant-account cash flow and cumulative product revenue over 20 years

Scenario: matched_protein_bypass rank 7; ab67fb691fc940ceb37c54dd707d8501; 504.3 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Balance and evidence

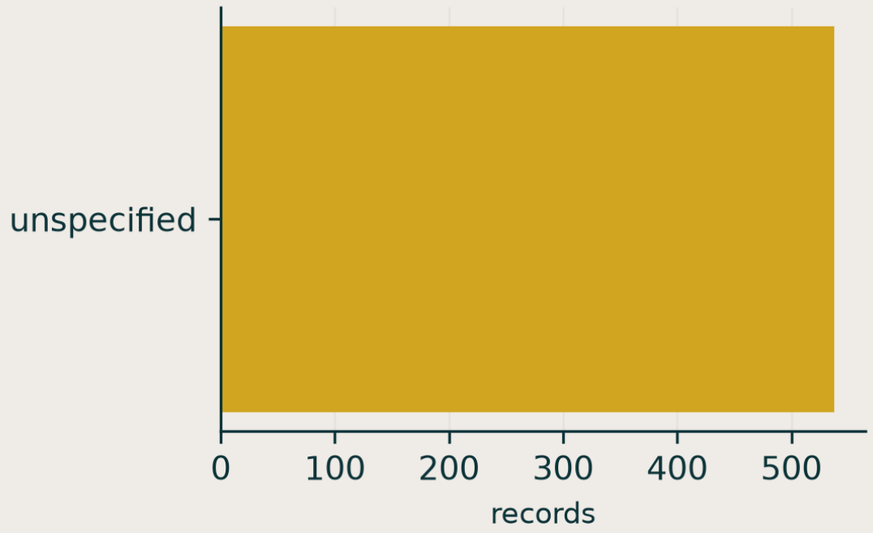
Protein X · Biogas ✓ · N ✓ · P ✓ · K X

Element closure

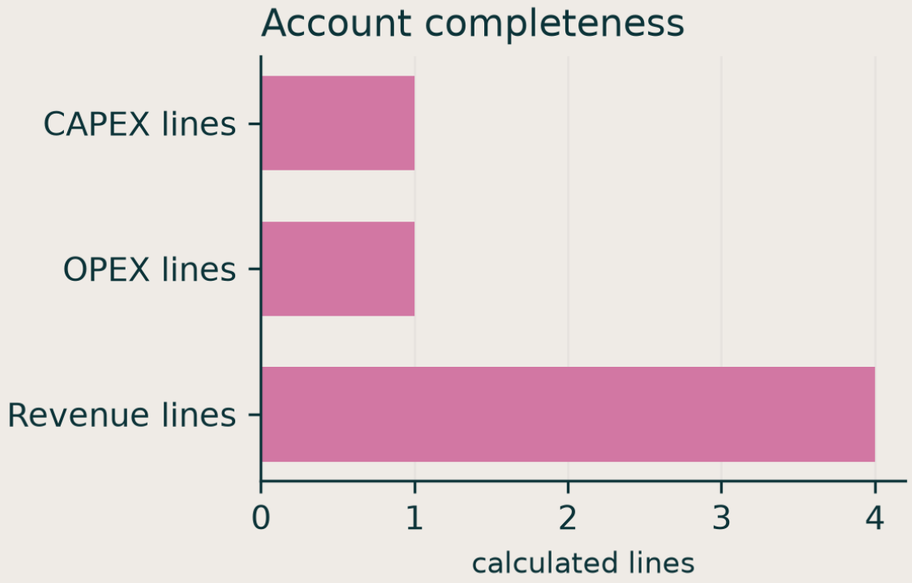
C	0 kg/year
N	0 kg/year
P	0 kg/year
K	2.4e-13 kg/year

Numerical residuals

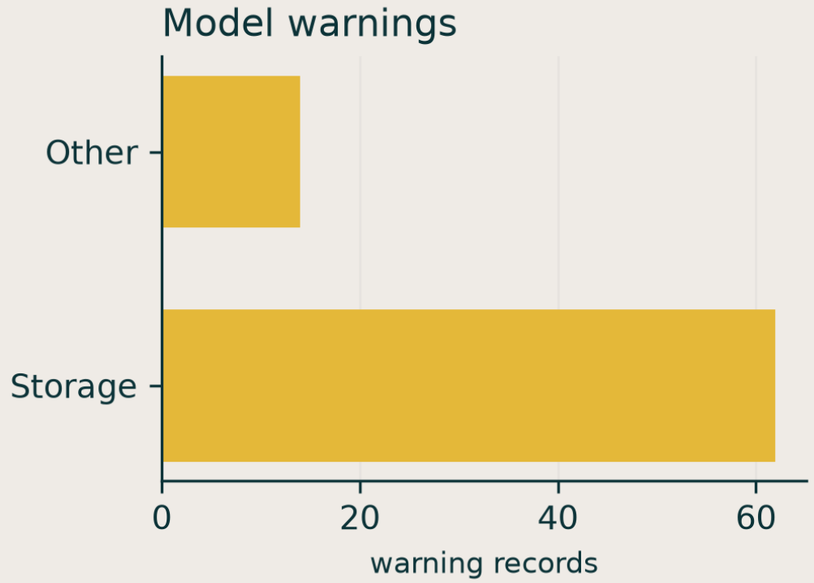
Parameter evidence



Account completeness



Model warnings



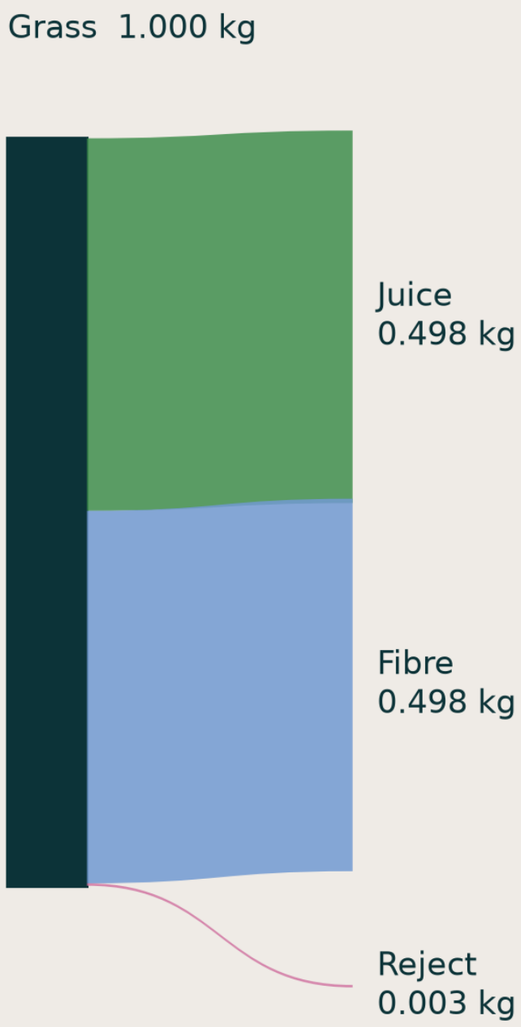
Numerical closure, evidence counts, calculated account lines and warnings

Scenario: matched_protein_bypass rank 7; ab67fb691fc940ceb37c54dd707d8501; 504.3 kg/h; 8760 h/year
Units: kg/year; records
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

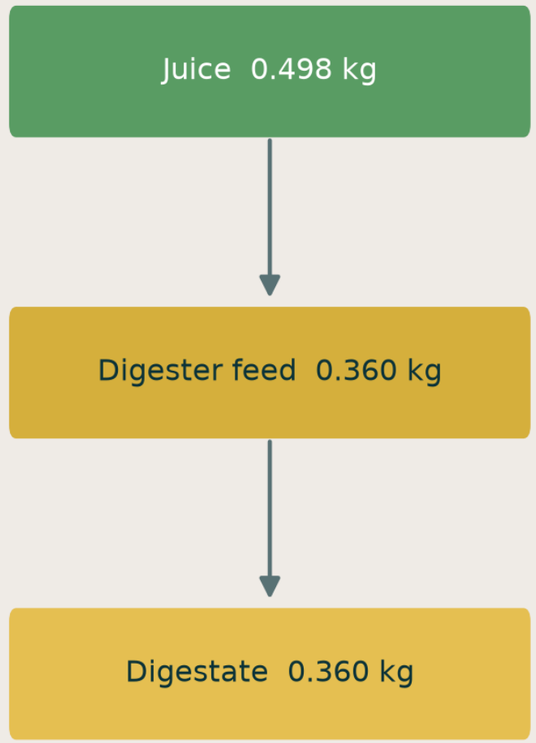
Grass to products

Protein X · Biogas X · N X · P X · K X

Grass split



Juice pathway



Selected intermediate streams

Recovered outputs

per kg wet grass

Mass-proportional grass split plus modeled downstream stream/product map

Scenario: matched_protein_bypass rank 8; 280e23dac98c416b887fdcac52b99f40; 506.3 kg/h; 8760 h/year
Units: kg/kg wet grass; dry Nm³/kg wet grass
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Material and products

Protein X · Biogas X · N X · P X · K X

Feed destinations

Solid products

No modeled output

Gas at normal conditions

No modeled output

Specific product yields

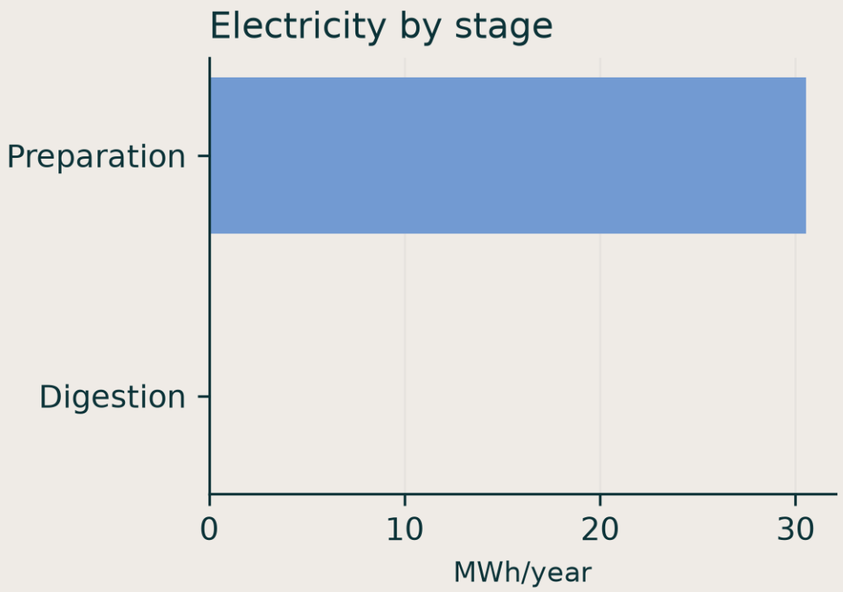
No modeled output

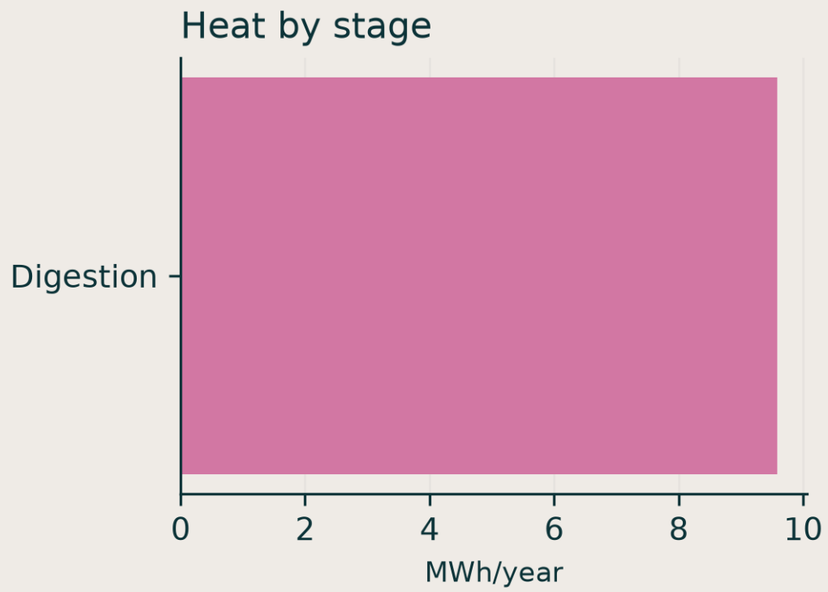
Front-end split, solid products, dry gas volumes and specific yields

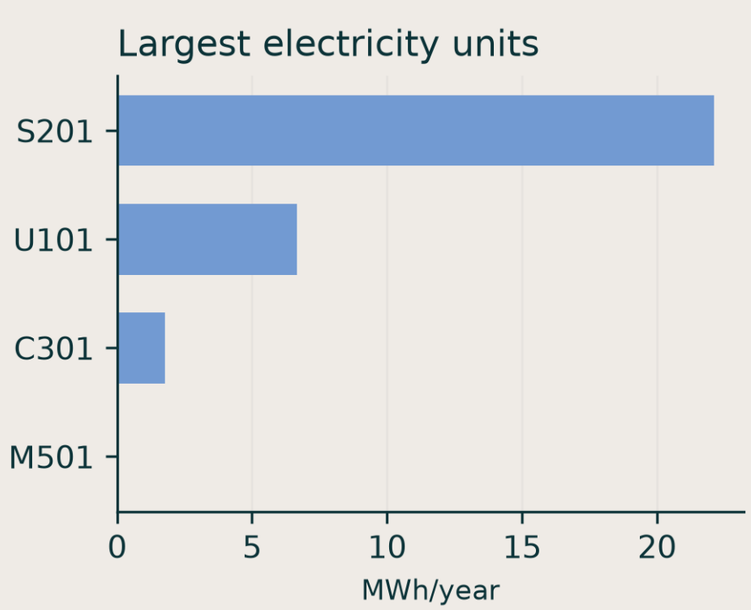
Scenario: matched_protein_bypass rank 8; 280e23dac98c416b887fdcac52b99f40; 506.3 kg/h; 8760 h/year
Units: kg/h; t/year; dry Nm³/year; kg/t feed
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

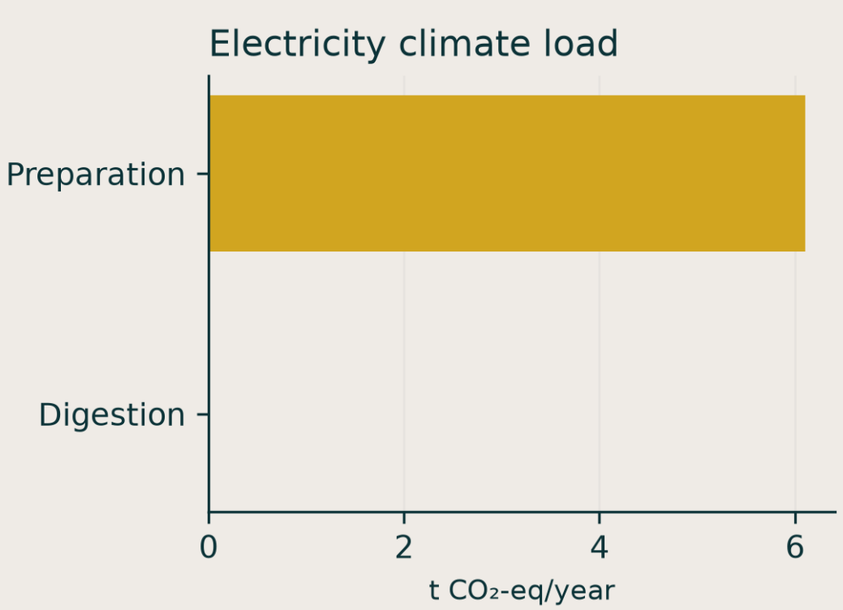
Energy and climate hotspots

Protein X · Biogas X · N X · P X · K X









Stage and unit utility hotspots and included climate load

Scenario: matched_protein_bypass rank 8; 280e23dac98c416b887fdcac52b99f40; 506.3 kg/h; 8760 h/year
Units: MWh/year; t CO_2-eq/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

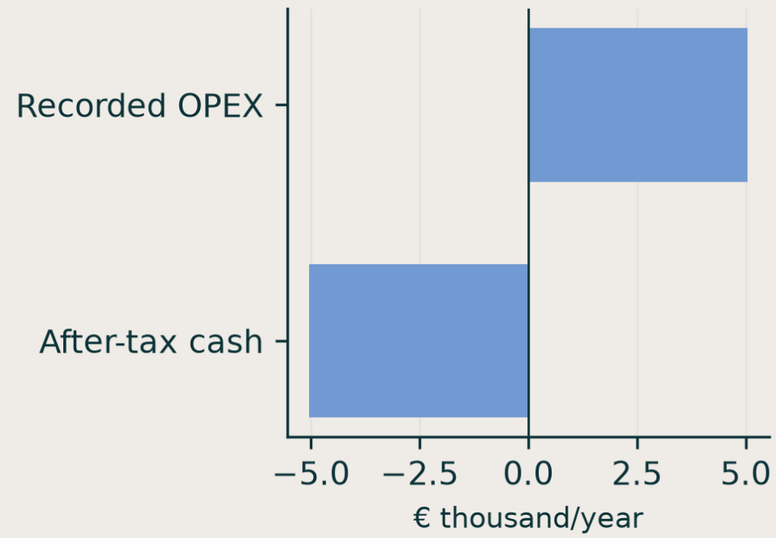
Economic accounts

Protein X · Biogas X · N X · P X · K X

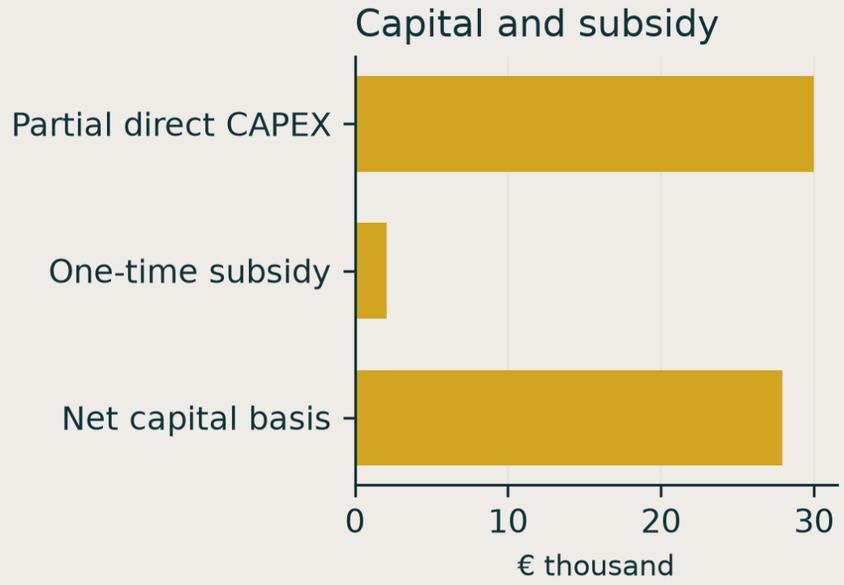
Product revenue

No modeled output

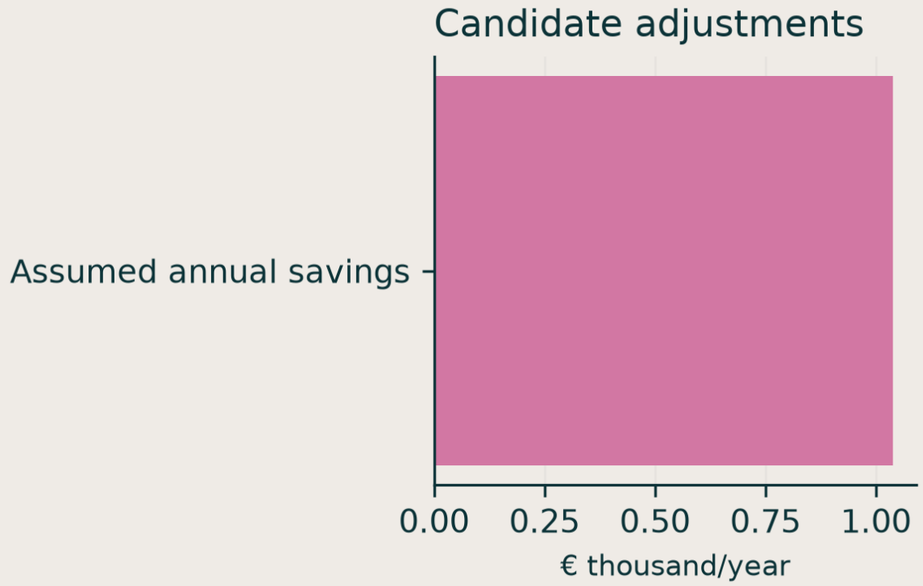
Annual budget



Capital and subsidy



Candidate adjustments

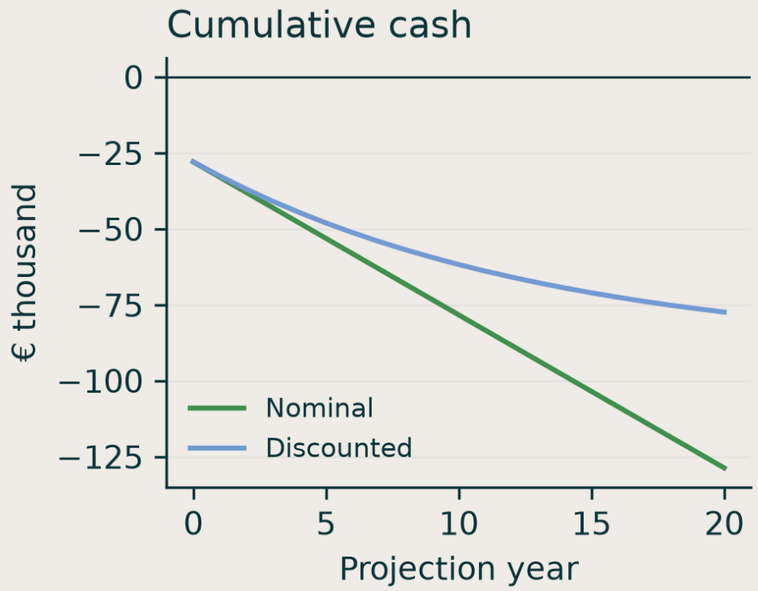


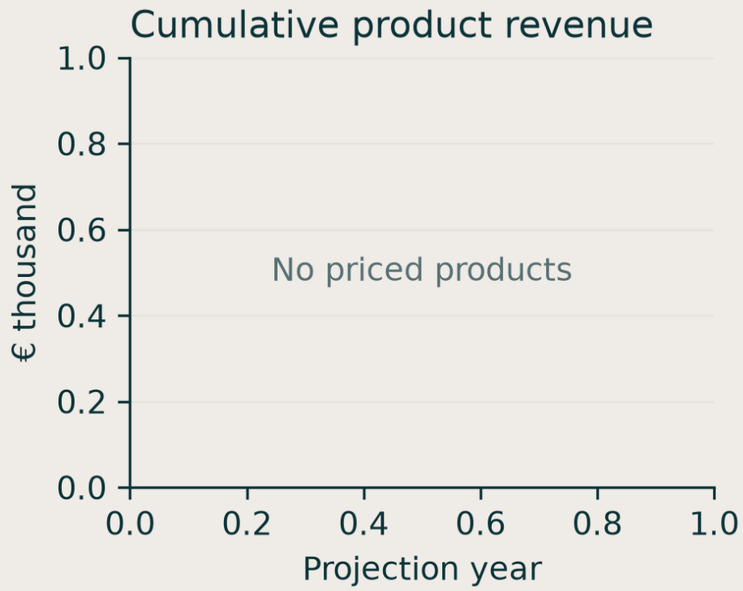
Product revenue, partial annual budget, capital subsidy and candidate overlays

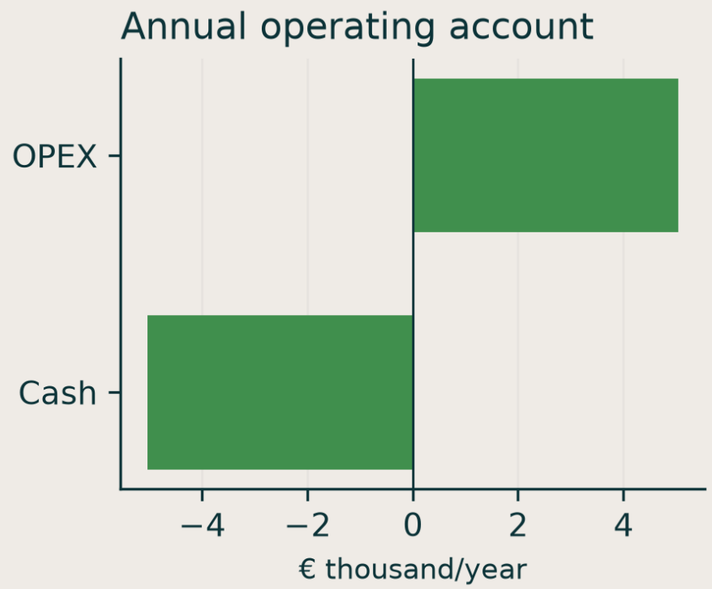
Scenario: matched_protein_bypass rank 8; 280e23dac98c416b887fdcac52b99f40; 506.3 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

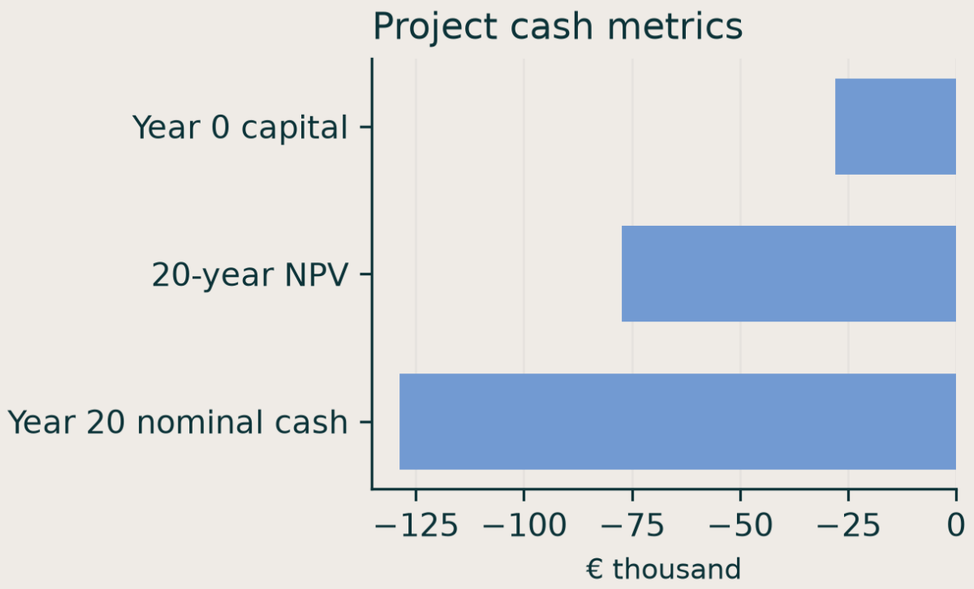
20-year financial projection

Protein X · Biogas X · N X · P X · K X









Constant-account cash flow and cumulative product revenue over 20 years

Scenario: matched_protein_bypass rank 8; 280e23dac98c416b887fdcac52b99f40; 506.3 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Balance and evidence

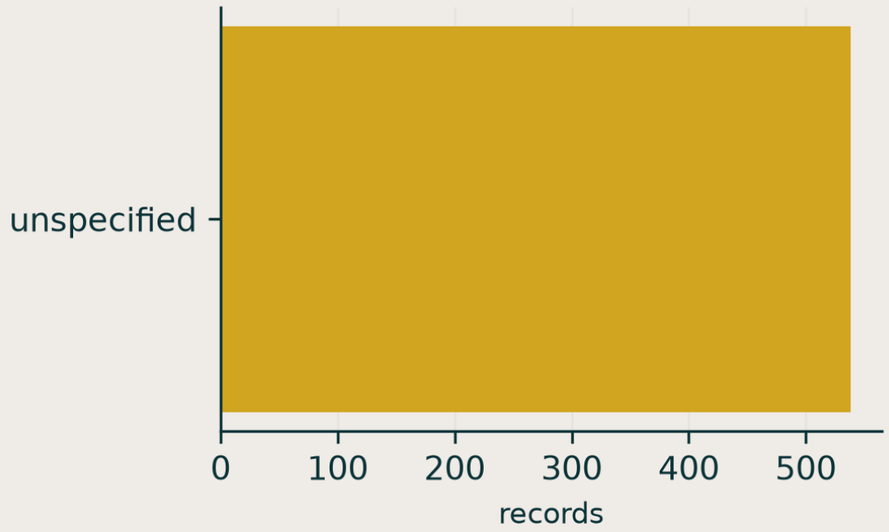
Protein X · Biogas X · N X · P X · K X

Element closure

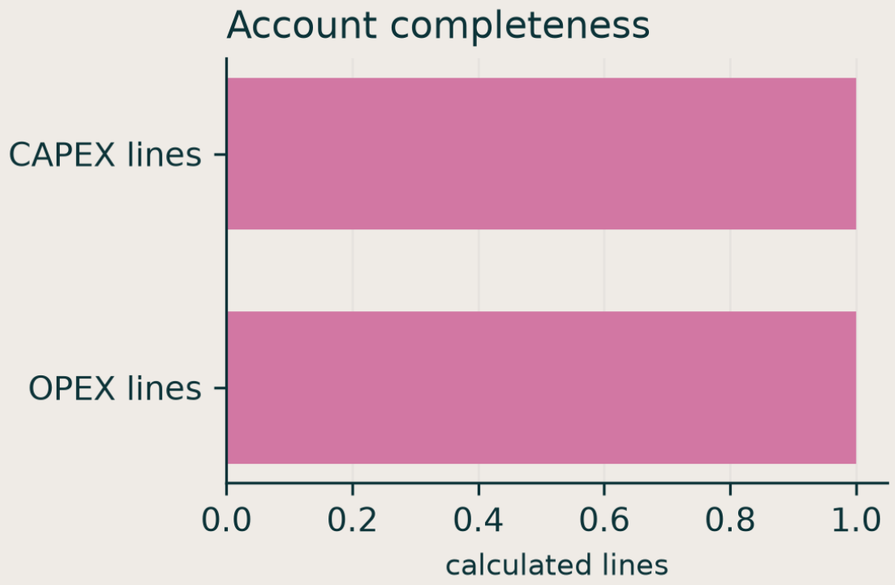
C	0 kg/year
N	0 kg/year
P	0 kg/year
K	0 kg/year

Numerical residuals

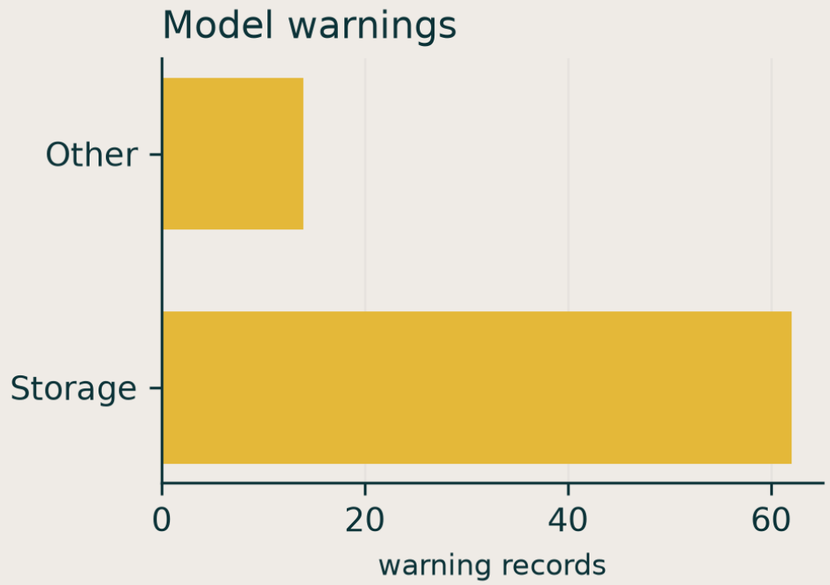
Parameter evidence



Account completeness



Model warnings



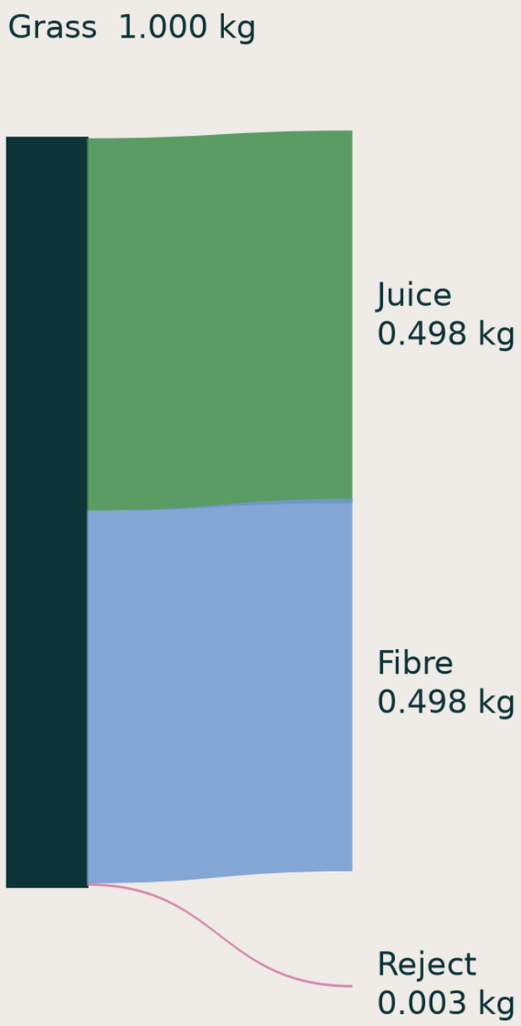
Numerical closure, evidence counts, calculated account lines and warnings

Scenario: matched_protein_bypass rank 8; 280e23dac98c416b887fdcac52b99f40; 506.3 kg/h; 8760 h/year
Units: kg/year; records
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

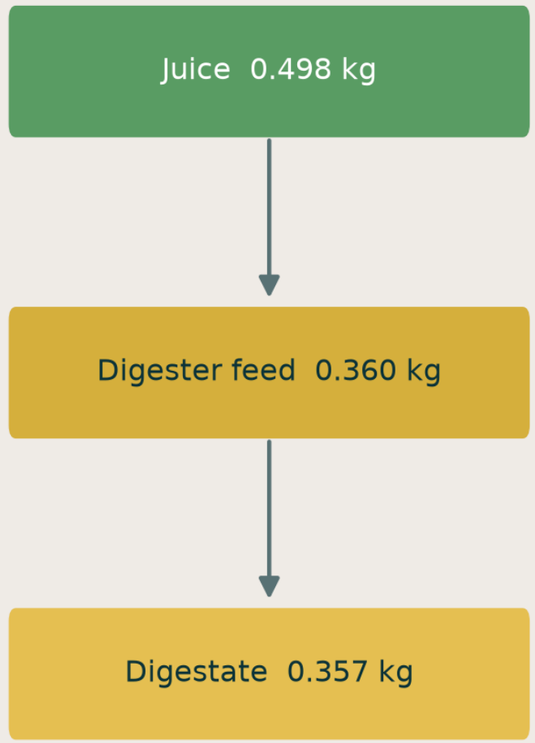
Grass to products

Protein X · Biogas ✓ · N X · P X · K X

Grass split



Juice pathway



Selected intermediate streams

Recovered outputs

Nutrient solids 0.0090 kg

Raw gas 0.0021 Nm³

Raw export 0.0021 Nm³

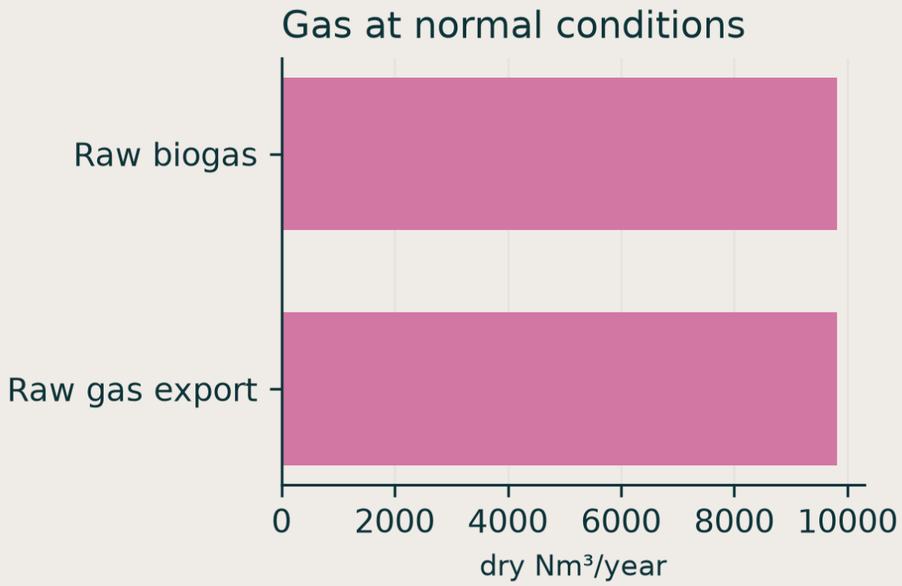
per kg wet grass

Mass-proportional grass split plus modeled downstream stream/product map

Scenario: matched_protein_bypass rank 9; f7f84c6a5ef34ec58cd2cb457319d4f1; 523.4 kg/h; 8760 h/year
Units: kg/kg wet grass; dry Nm³/kg wet grass
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Material and products

Protein X · Biogas ✓ · N X · P X · K X

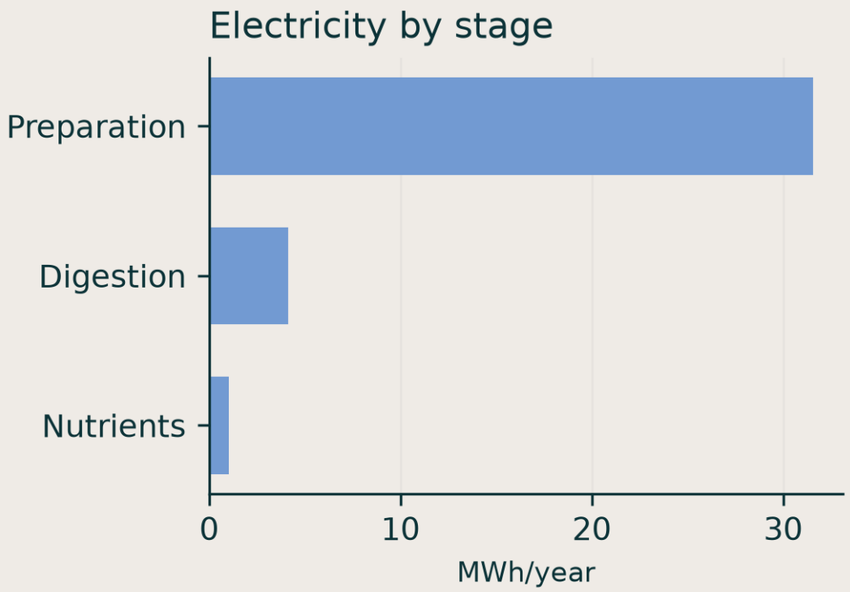


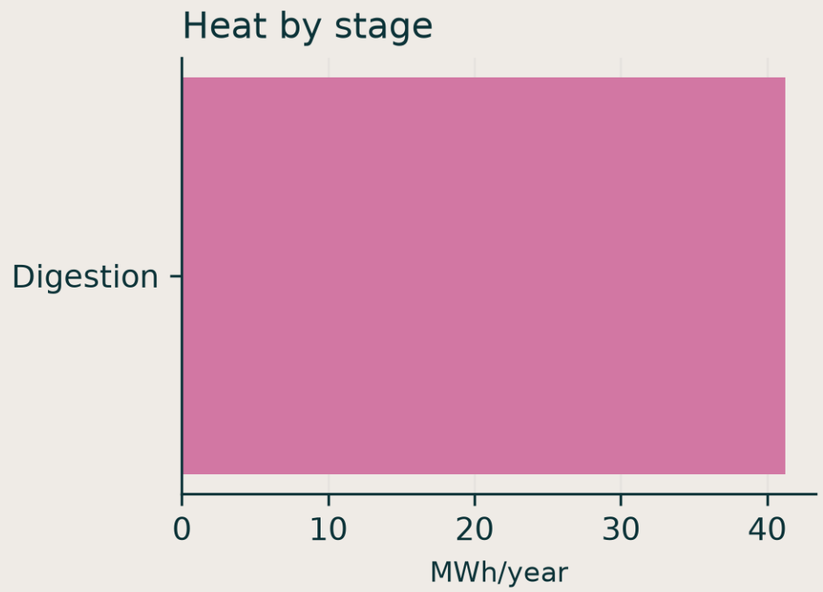
Front-end split, solid products, dry gas volumes and specific yields

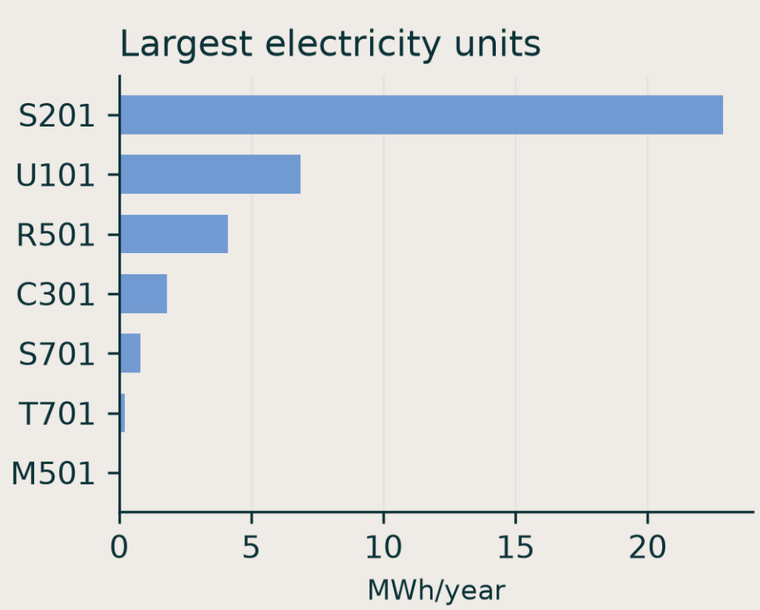
Scenario: matched_protein_bypass rank 9; f7f84c6a5ef34ec58cd2cb457319d4f1; 523.4 kg/h; 8760 h/year
Units: kg/h; t/year; dry Nm³/year; kg/t feed
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

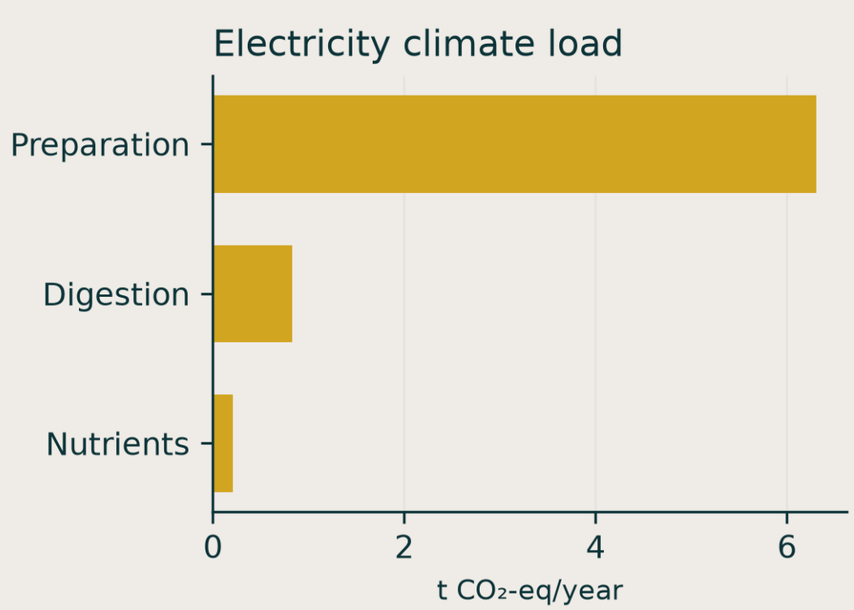
Energy and climate hotspots

Protein X · Biogas ✓ · N X · P X · K X







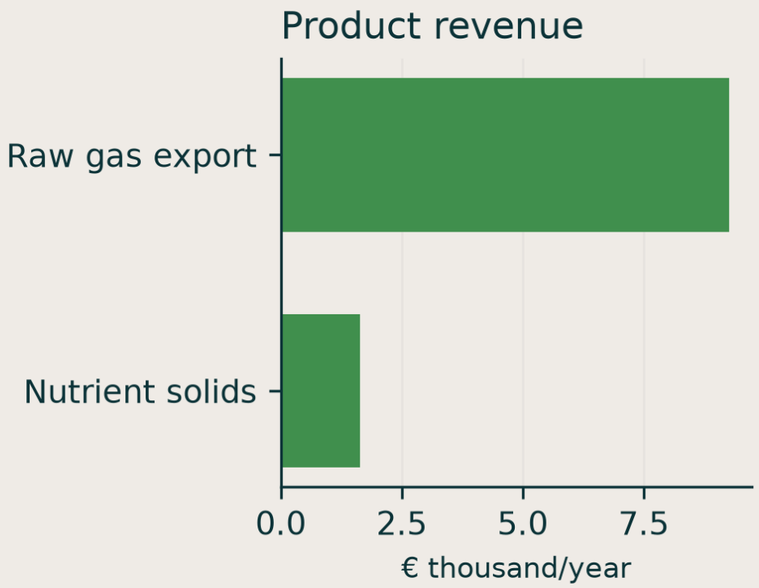


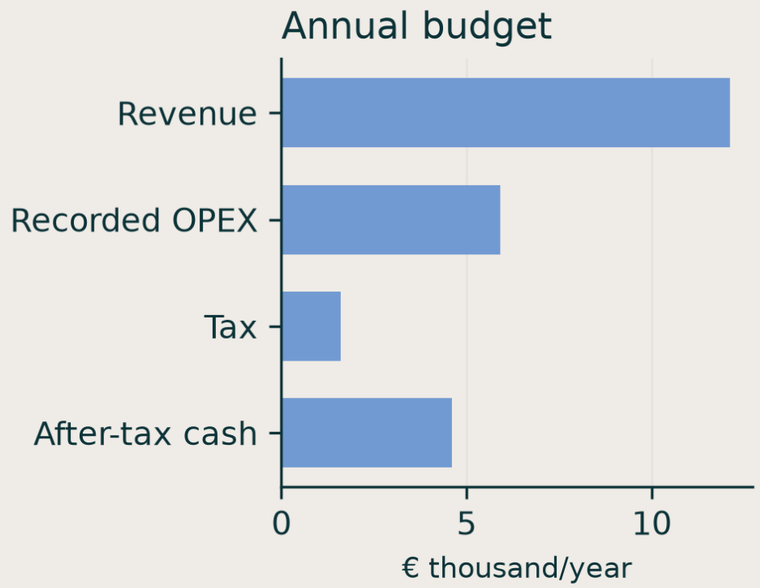
Stage and unit utility hotspots and included climate load

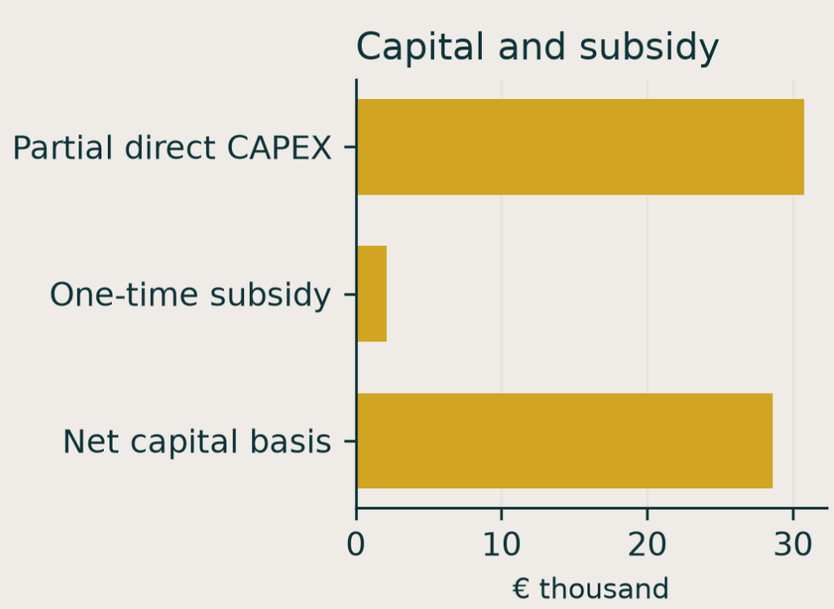
Scenario: matched_protein_bypass rank 9; f7f84c6a5ef34ec58cd2cb457319d4f1; 523.4 kg/h; 8760 h/year
Units: MWh/year; t CO_2-eq/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

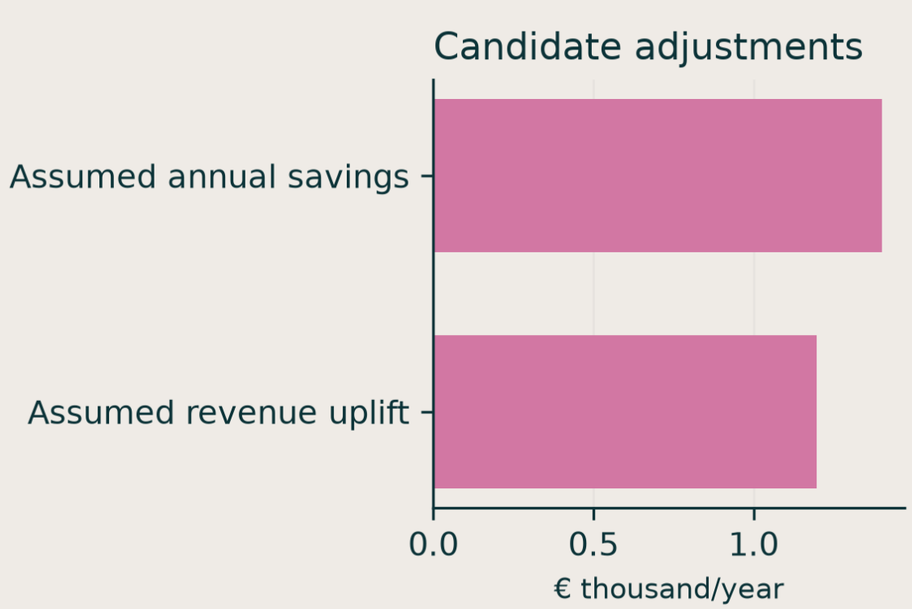
Economic accounts

Protein X · Biogas ✓ · N X · P X · K X







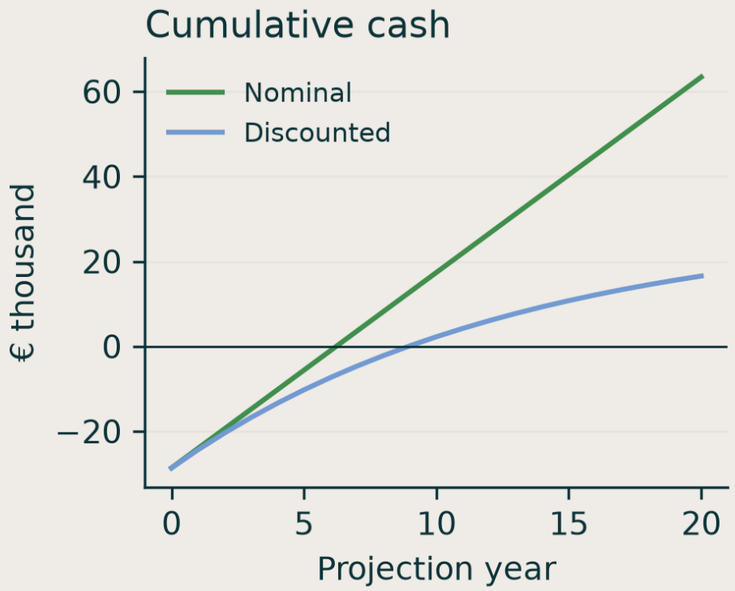


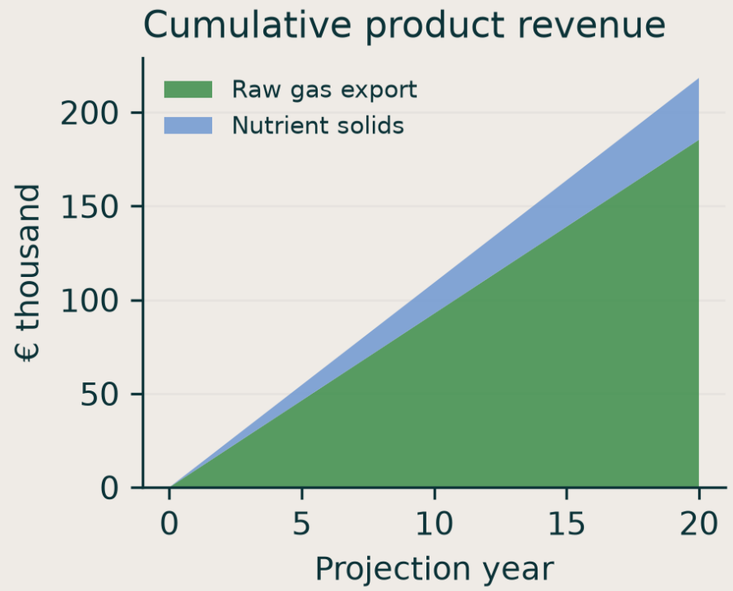
Product revenue, partial annual budget, capital subsidy and candidate overlays

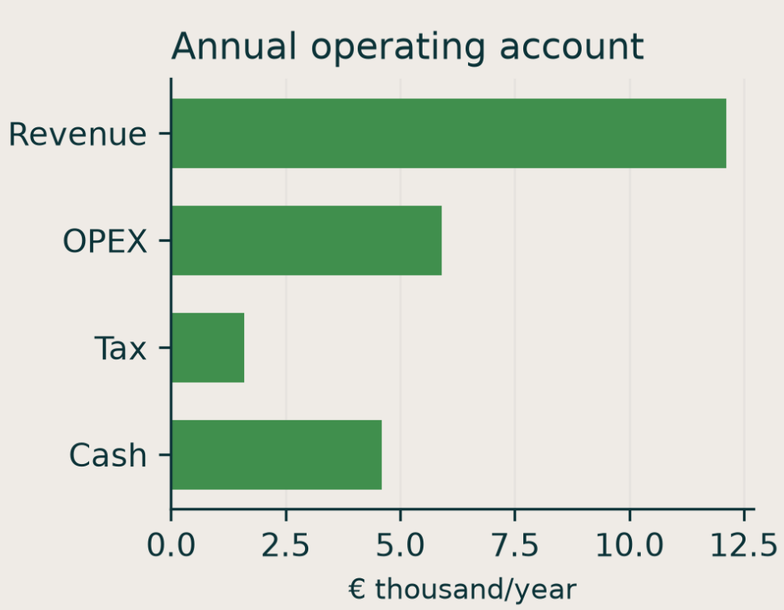
Scenario: matched_protein_bypass rank 9; f7f84c6a5ef34ec58cd2cb457319d4f1; 523.4 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

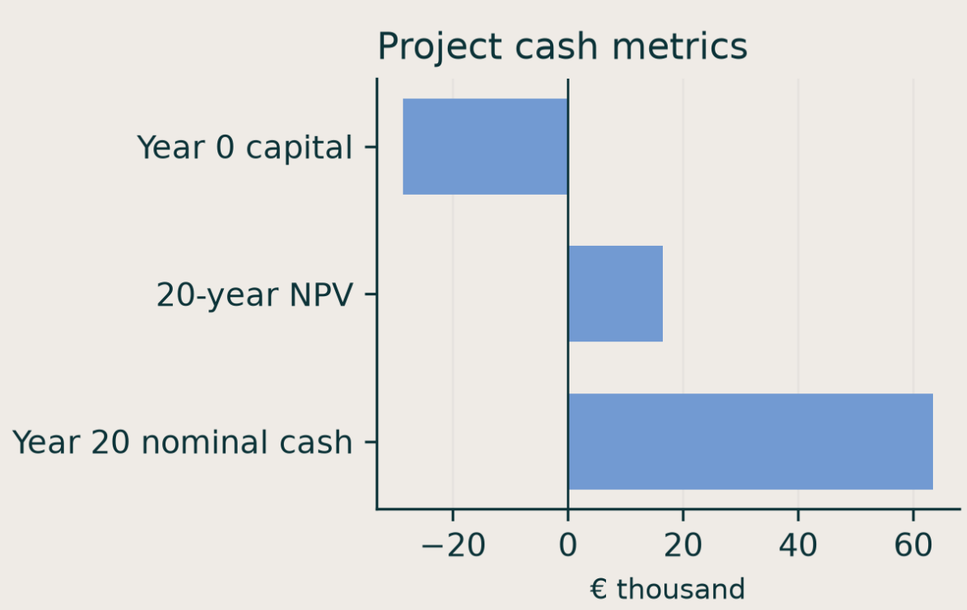
20-year financial projection

Protein X · Biogas ✓ · N X · P X · K X









Constant-account cash flow and cumulative product revenue over 20 years

Scenario: matched_protein_bypass rank 9; f7f84c6a5ef34ec58cd2cb457319d4f1; 523.4 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Balance and evidence

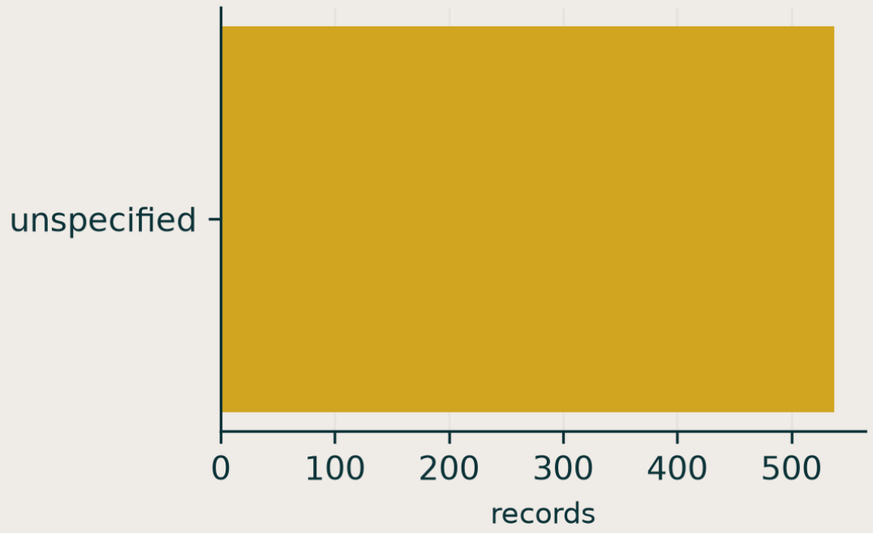
Protein X · Biogas ✓ · N X · P X · K X

Element closure

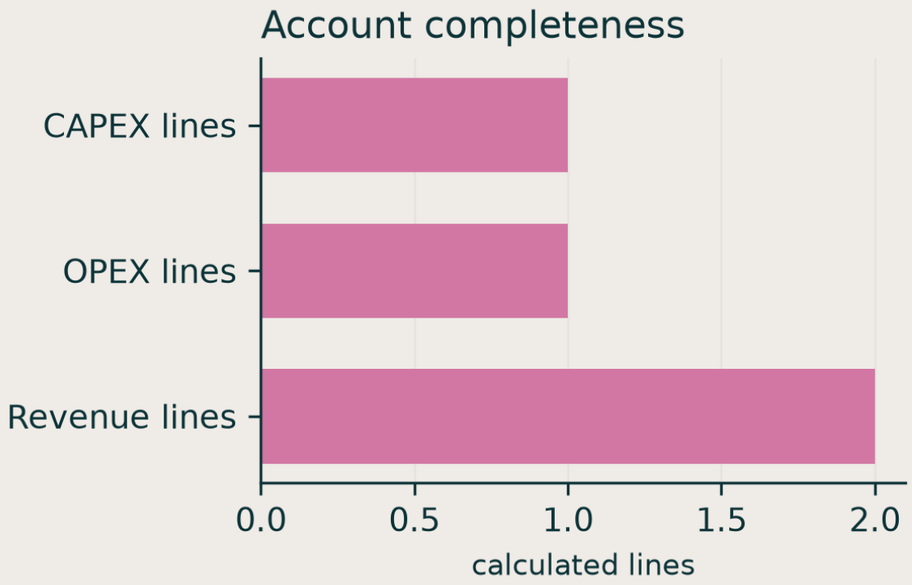
C	1.9e-12 kg/year
N	0 kg/year
P	0 kg/year
K	2.4e-13 kg/year

Numerical residuals

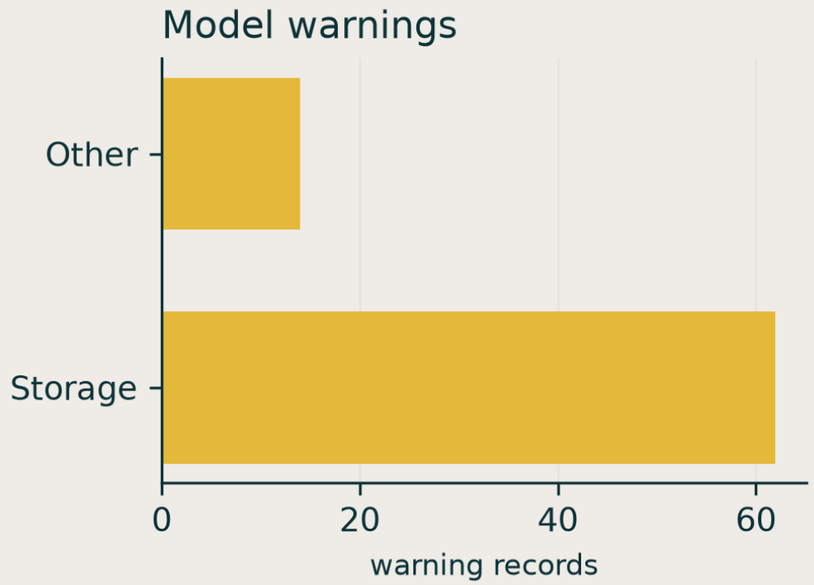
Parameter evidence



Account completeness



Model warnings



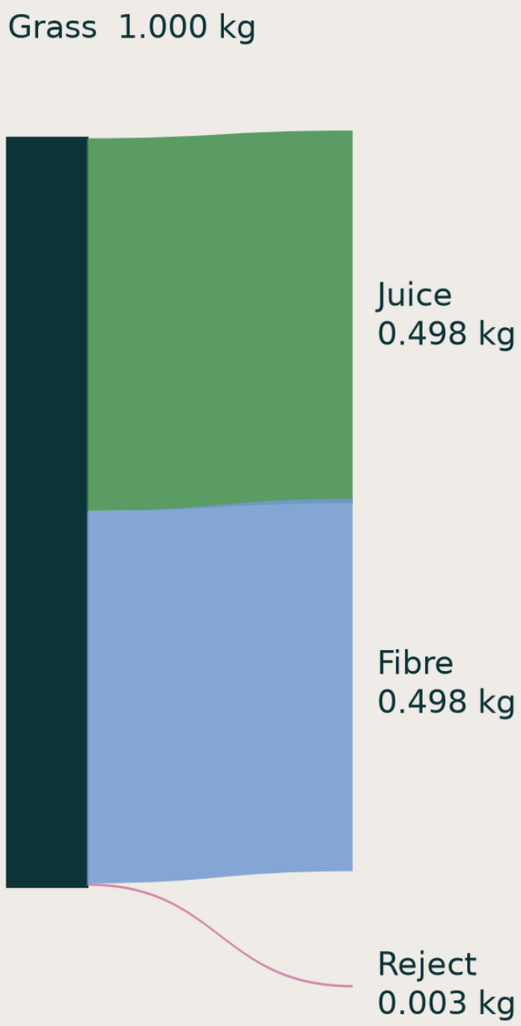
Numerical closure, evidence counts, calculated account lines and warnings

Scenario: matched_protein_bypass rank 9; f7f84c6a5ef34ec58cd2cb457319d4f1; 523.4 kg/h; 8760 h/year
Units: kg/year; records
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

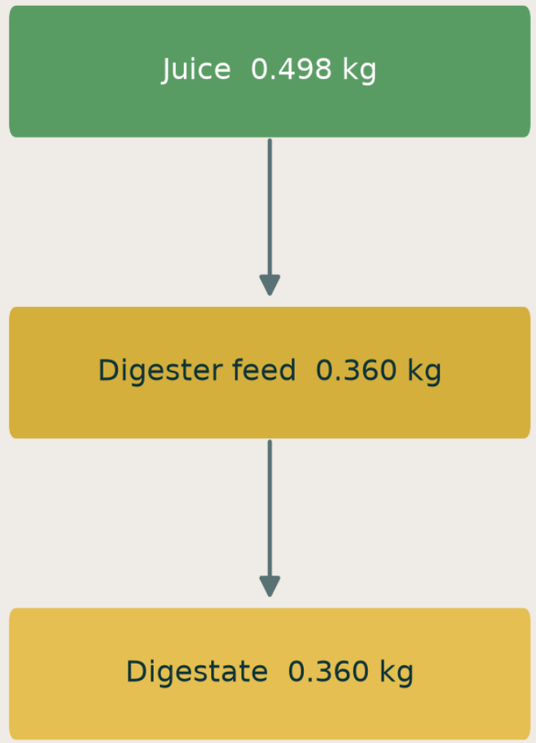
Grass to products

Protein X · Biogas X · N X · P X · K X

Grass split



Juice pathway



Selected intermediate streams

Recovered outputs

per kg wet grass

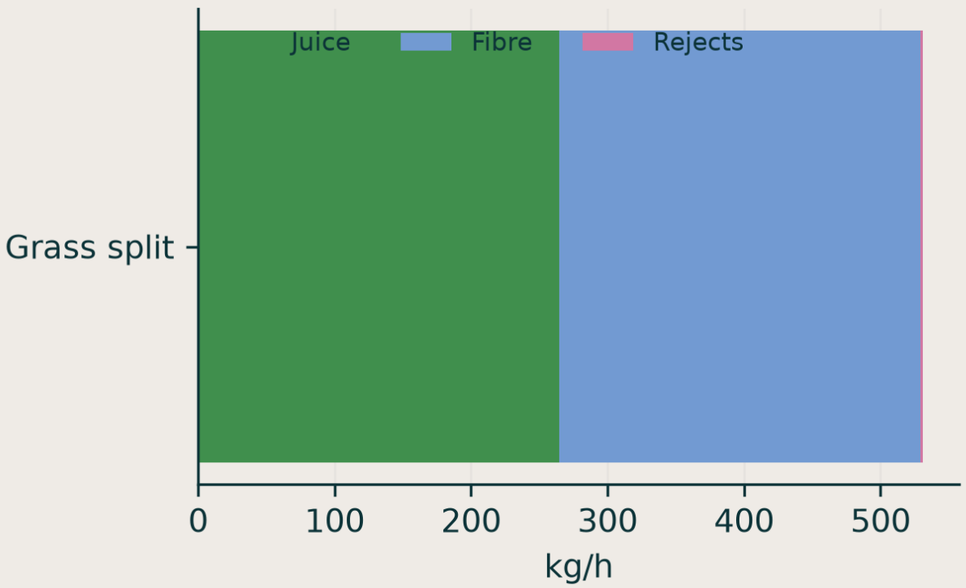
Mass-proportional grass split plus modeled downstream stream/product map

Scenario: matched_protein_bypass rank 10; 3b45eb5419a94dccbfcebe8c7304abee; 531.1 kg/h; 8760 h/year
Units: kg/kg wet grass; dry Nm³/kg wet grass
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Material and products

Protein X · Biogas X · N X · P X · K X

Feed destinations



Solid products

No modeled output

Gas at normal conditions

No modeled output

Specific product yields

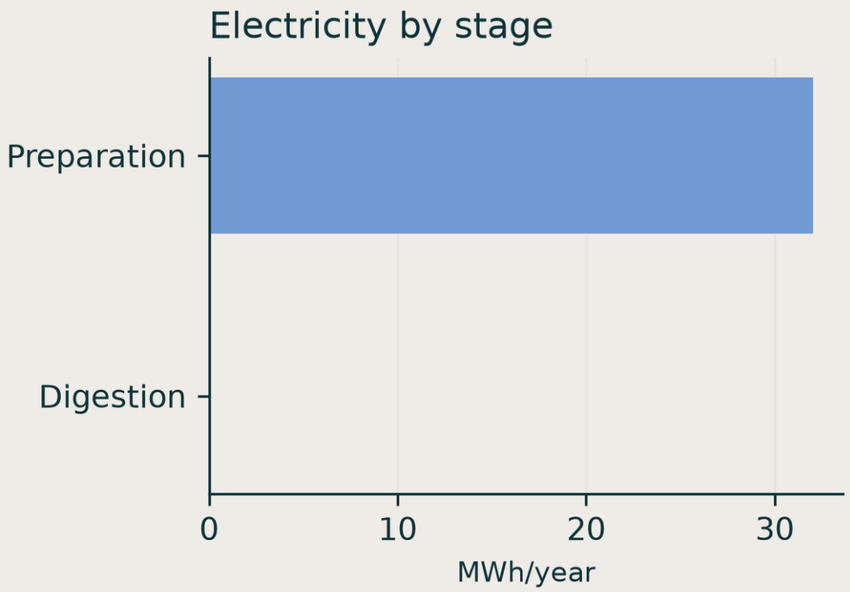
No modeled output

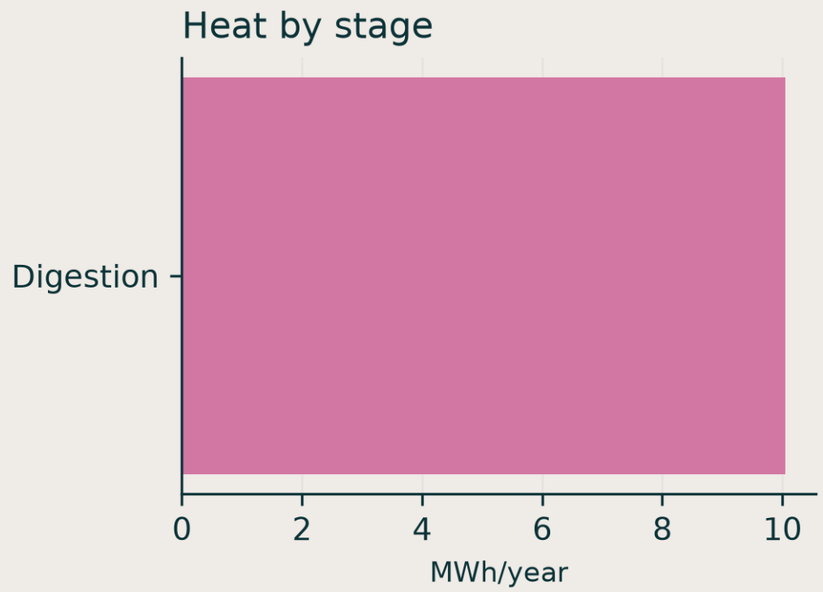
Front-end split, solid products, dry gas volumes and specific yields

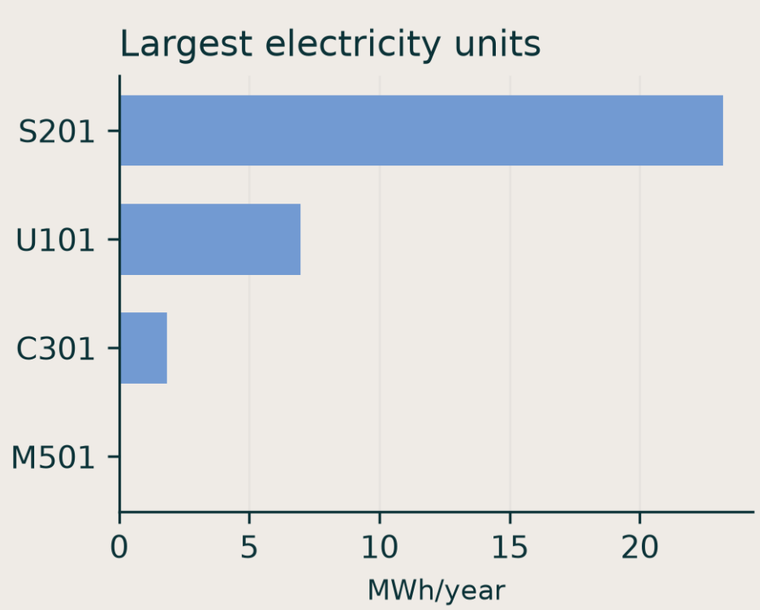
Scenario: matched_protein_bypass rank 10; 3b45eb5419a94dccbfcebe8c7304abee; 531.1 kg/h; 8760 h/year
Units: kg/h; t/year; dry Nm³/year; kg/t feed
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

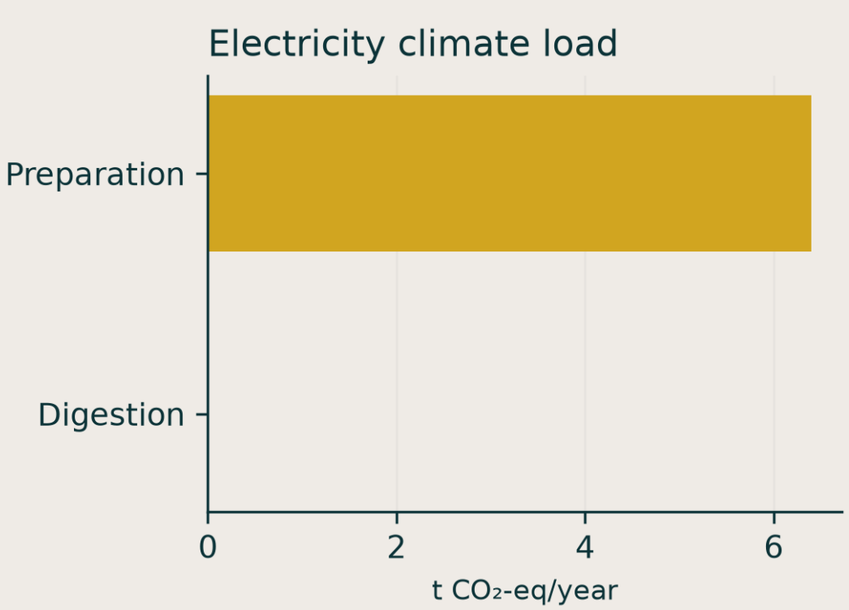
Energy and climate hotspots

Protein X · Biogas X · N X · P X · K X









Stage and unit utility hotspots and included climate load

Scenario: matched_protein_bypass rank 10; 3b45eb5419a94dccbfcebe8c7304abee; 531.1 kg/h; 8760 h/year
Units: MWh/year; t CO_2-eq/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

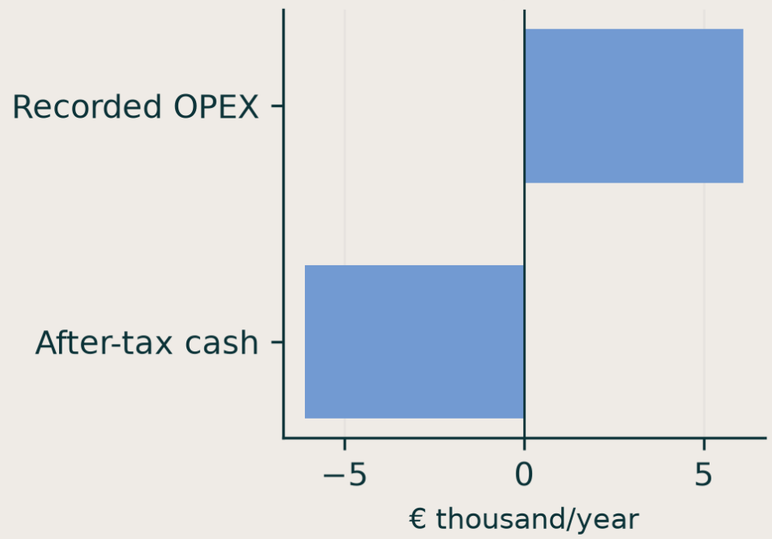
Economic accounts

Protein X · Biogas X · N X · P X · K X

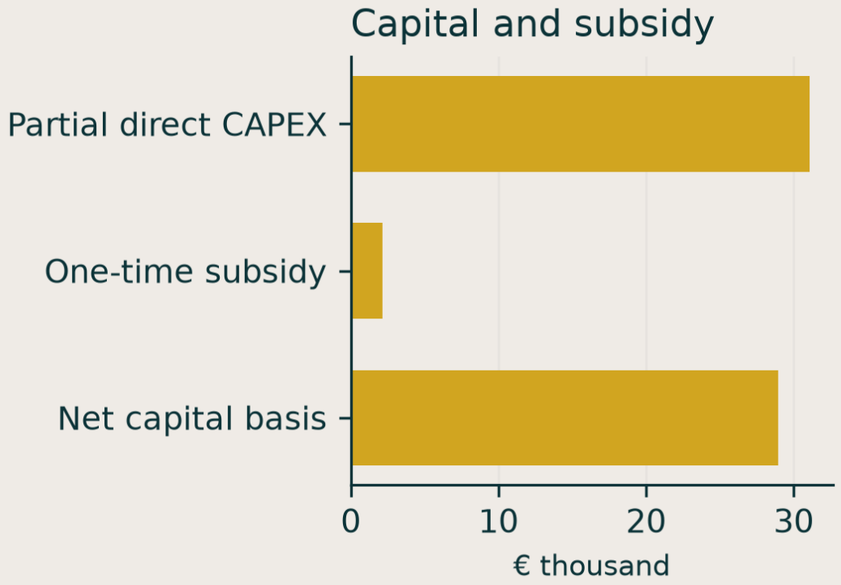
Product revenue

No modeled output

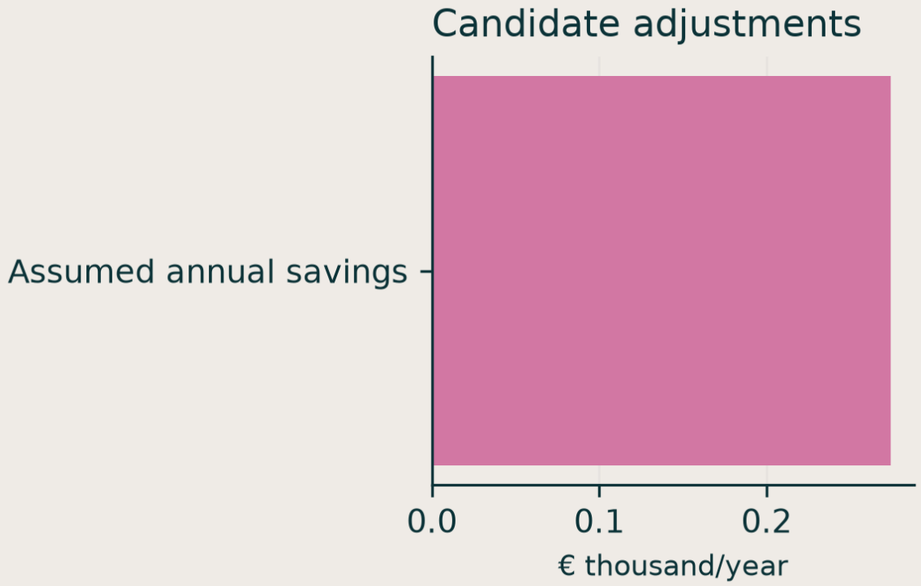
Annual budget



Capital and subsidy



Candidate adjustments

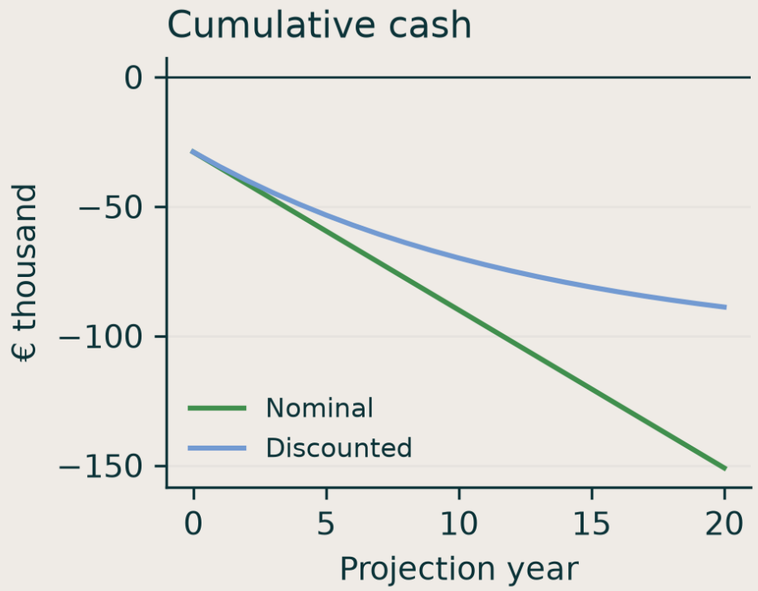


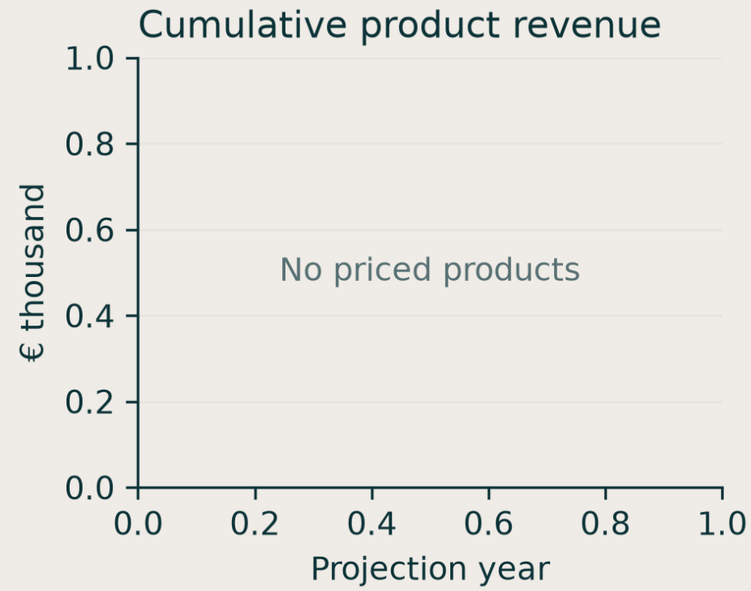
Product revenue, partial annual budget, capital subsidy and candidate overlays

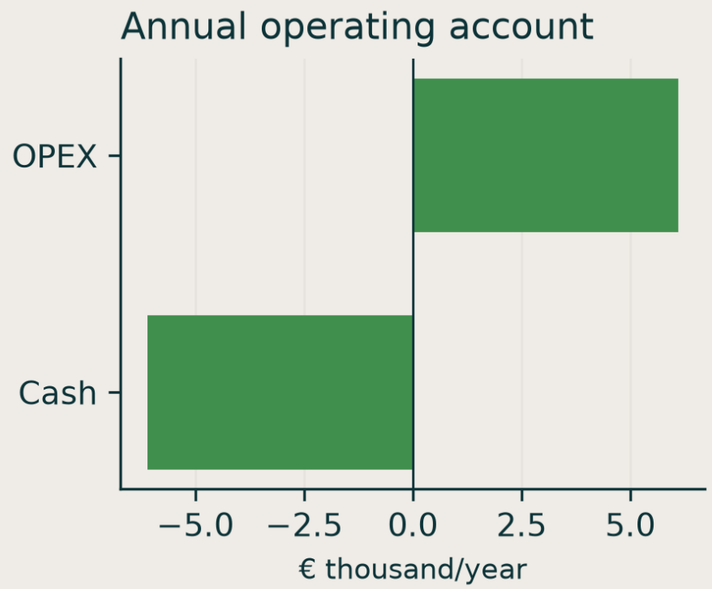
Scenario: matched_protein_bypass rank 10; 3b45eb5419a94dccbfcebe8c7304abee; 531.1 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

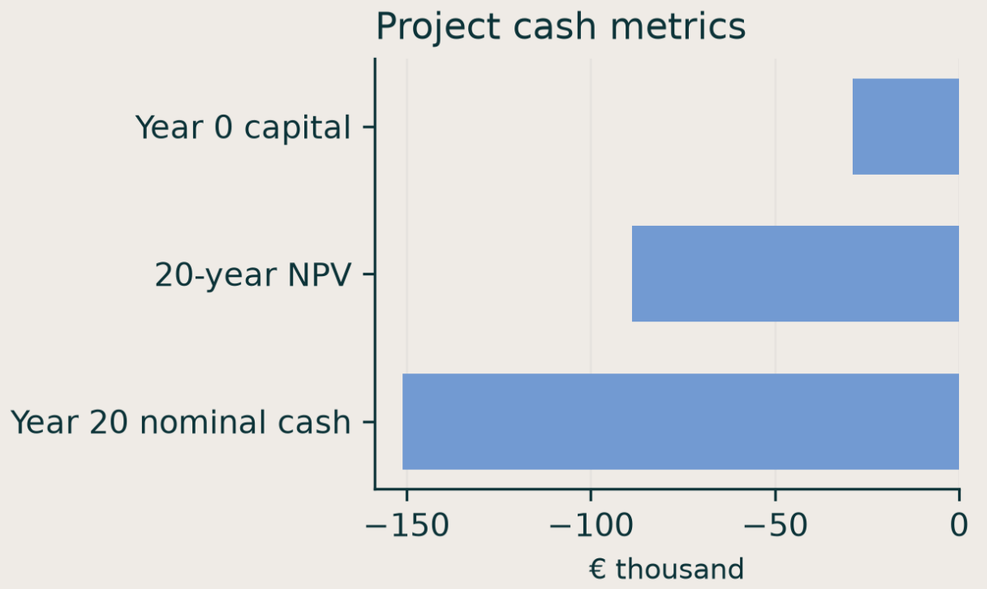
20-year financial projection

Protein X · Biogas X · N X · P X · K X









Constant-account cash flow and cumulative product revenue over 20 years

Scenario: matched_protein_bypass rank 10; 3b45eb5419a94dccbfcebe8c7304abee; 531.1 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Balance and evidence

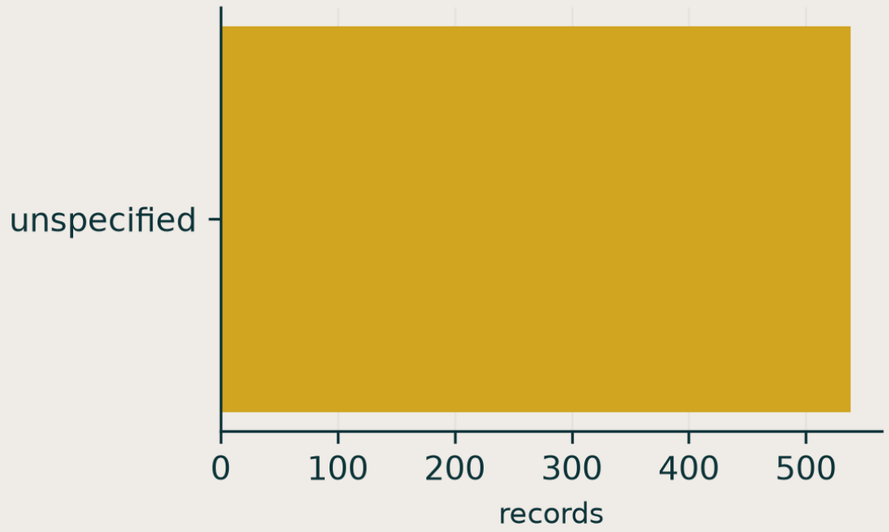
Protein X · Biogas X · N X · P X · K X

Element closure

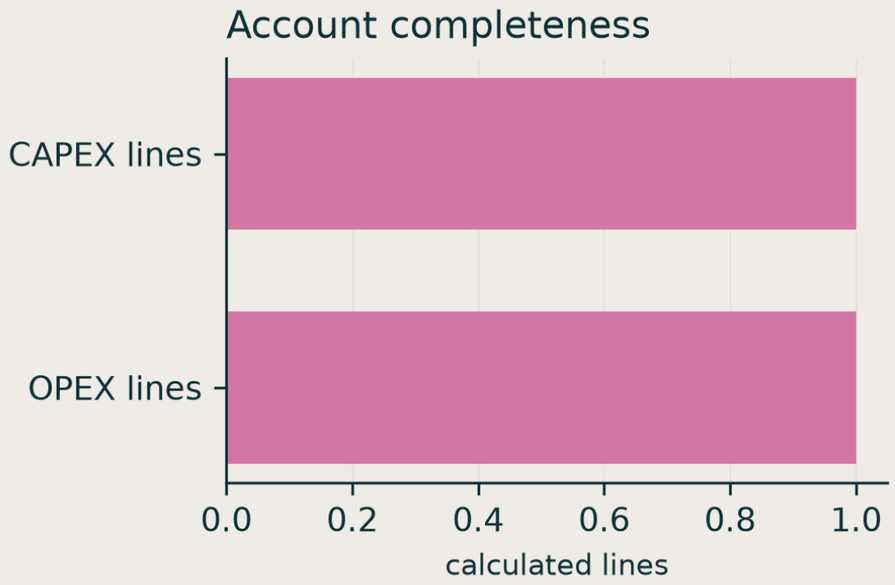
C	0 kg/year
N	0 kg/year
P	0 kg/year
K	0 kg/year

Numerical residuals

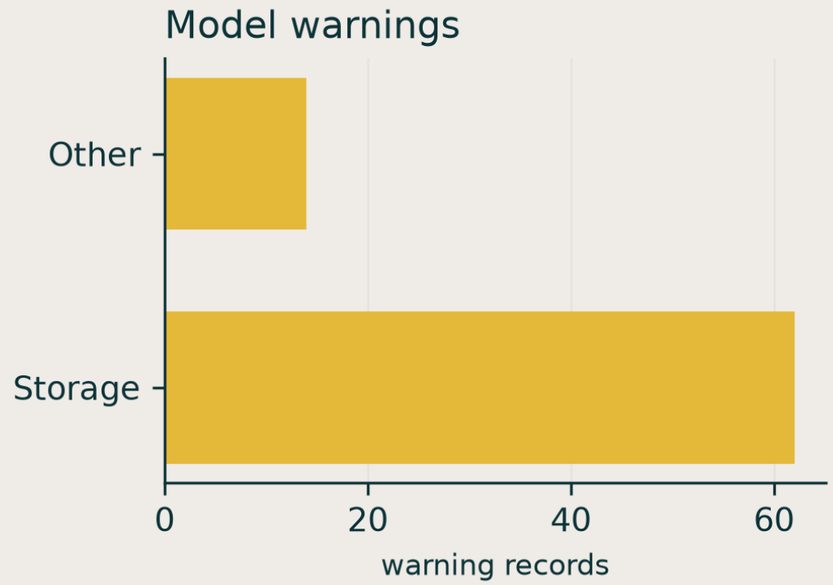
Parameter evidence



Account completeness



Model warnings



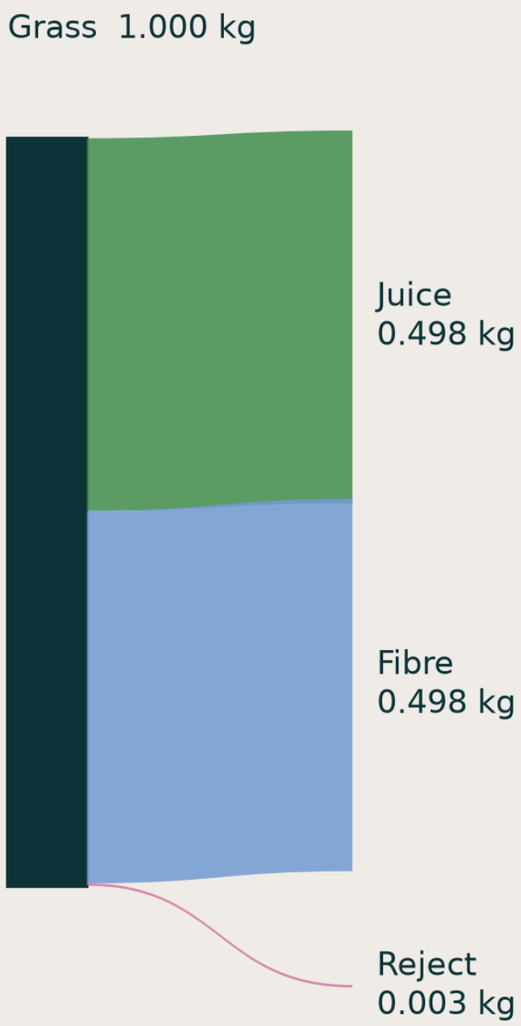
Numerical closure, evidence counts, calculated account lines and warnings

Scenario: matched_protein_bypass rank 10; 3b45eb5419a94dccbfcebe8c7304abee; 531.1 kg/h; 8760 h/year
Units: kg/year; records
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

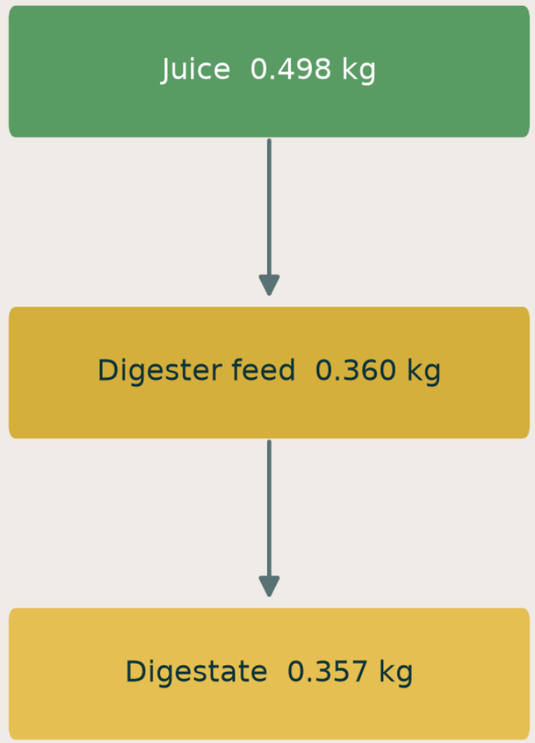
Grass to products

Protein X · Biogas ✓ · N ✓ · P X · K X

Grass split

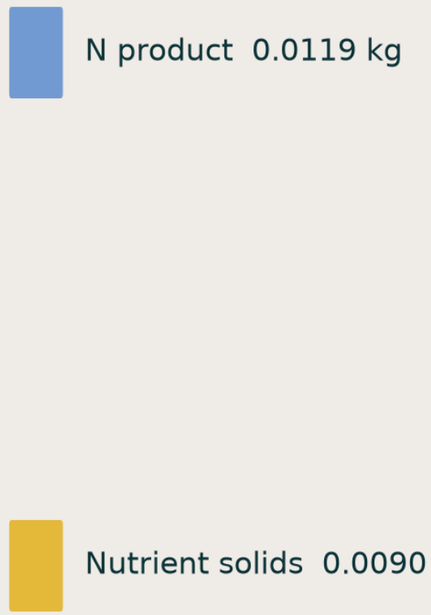


Juice pathway



Selected intermediate streams

Recovered outputs



Raw gas 0.0021 Nm³

Raw export 0.0021 Nm³

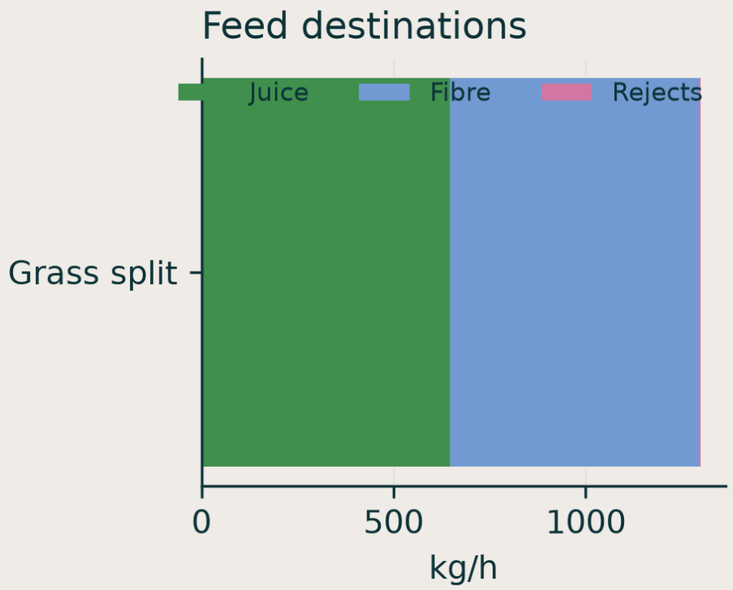
per kg wet grass

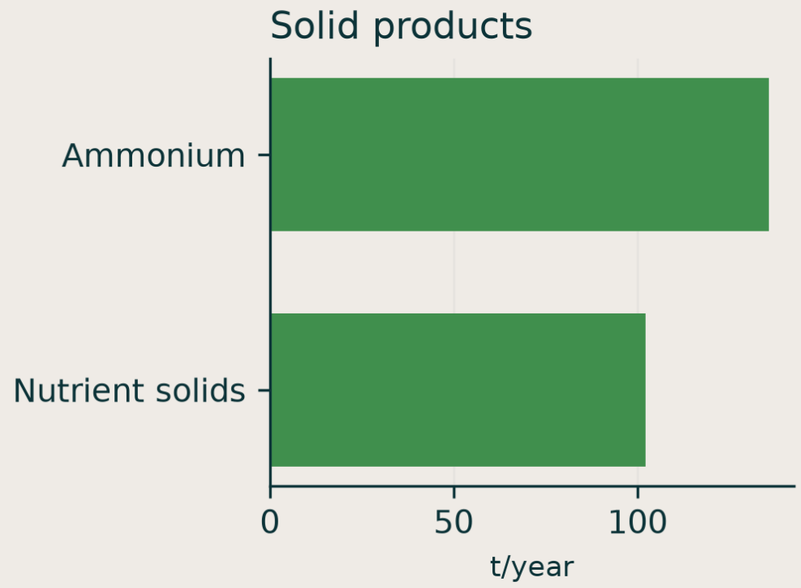
Mass-proportional grass split plus modeled downstream stream/product map

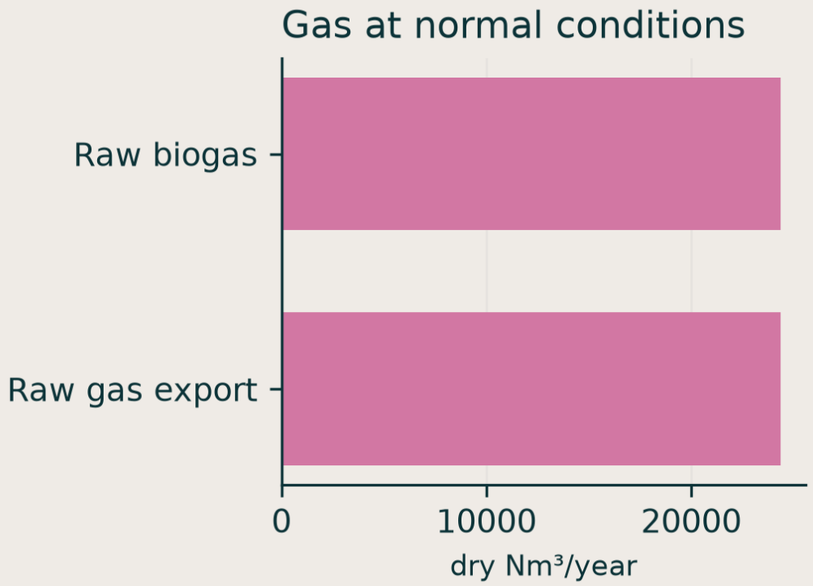
Scenario: protein_bypass rank 0; 20fb5b811d0e4cb9b33a5a0c441a3b2c; 1338.0 kg/h; 8760 h/year
Units: kg/kg wet grass; dry Nm³/kg wet grass
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

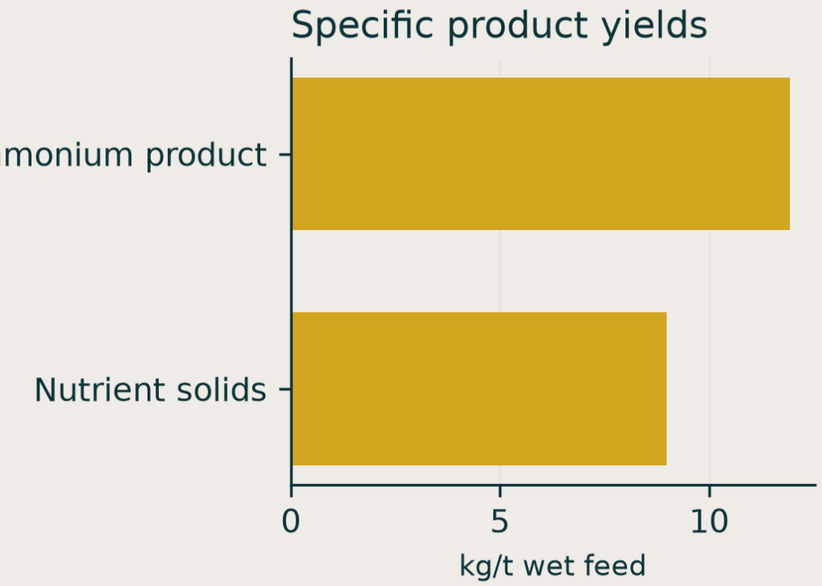
Material and products

Protein X · Biogas ✓ · N ✓ · P X · K X







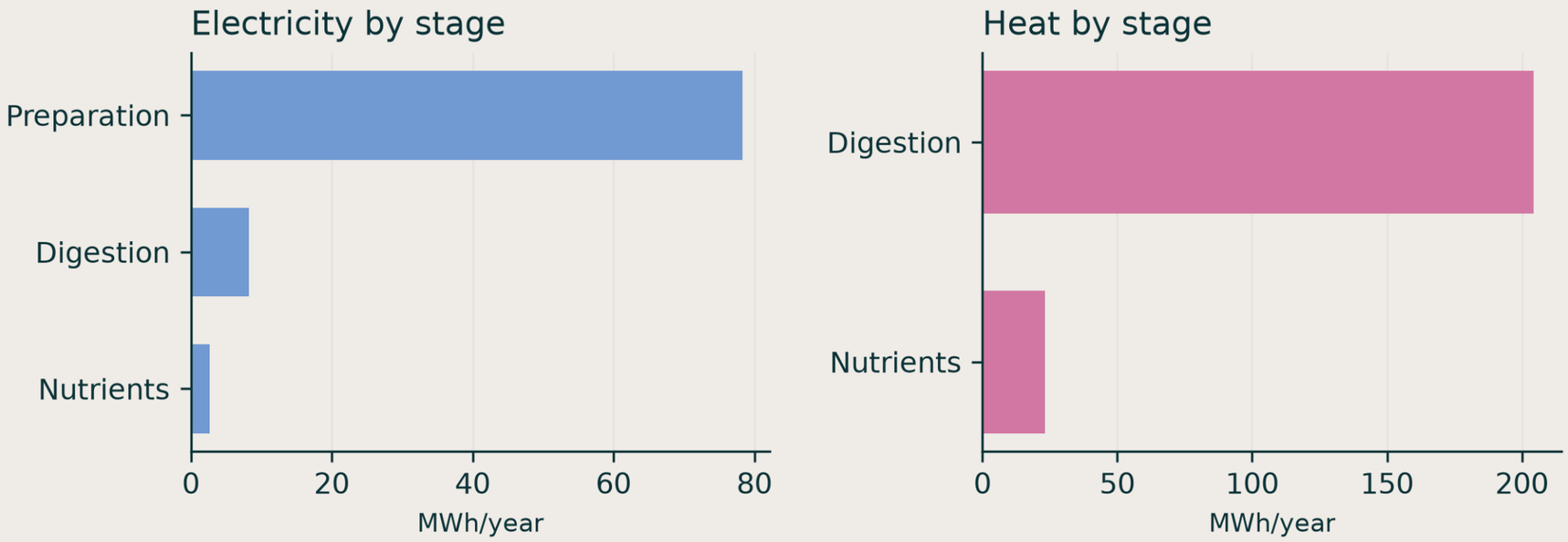


Front-end split, solid products, dry gas volumes and specific yields

Scenario: protein_bypass rank 0; 20fb5b811d0e4cb9b33a5a0c441a3b2c; 1338.0 kg/h; 8760 h/year
Units: kg/h; t/year; dry Nm³/year; kg/t feed
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Energy and climate hotspots

Protein X · Biogas ✓ · N ✓ · P X · K X

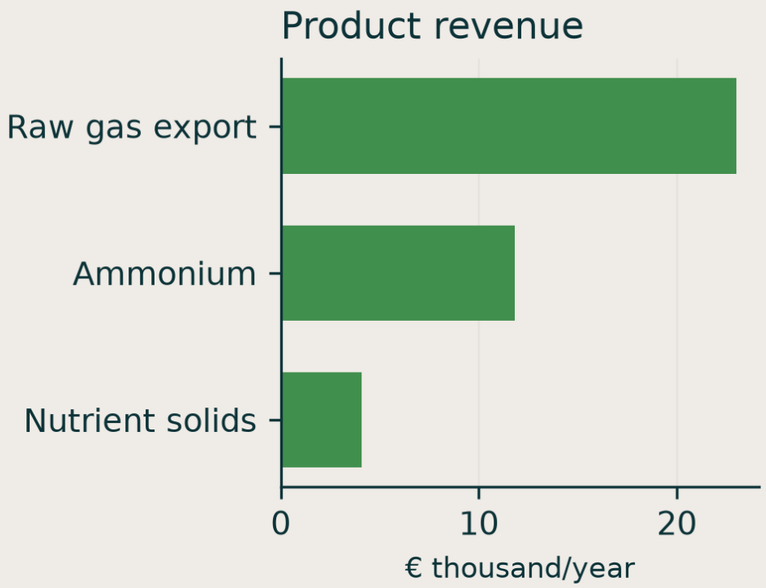


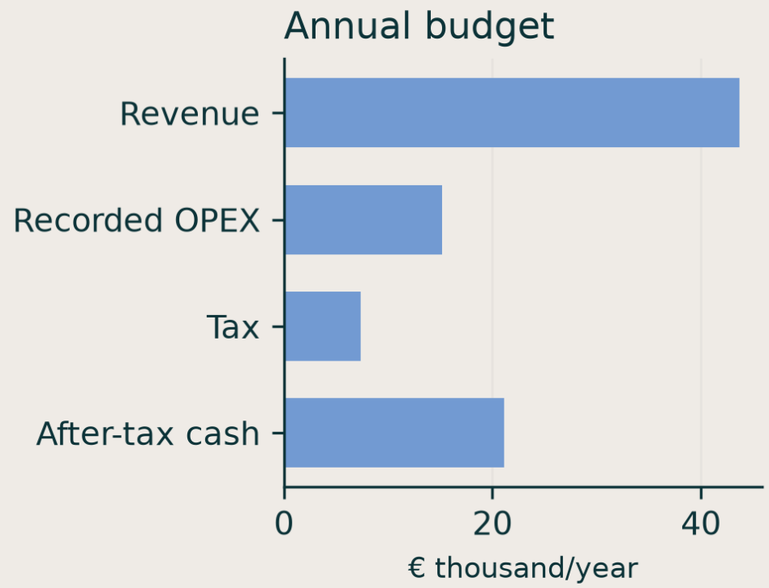
Stage and unit utility hotspots and included climate load

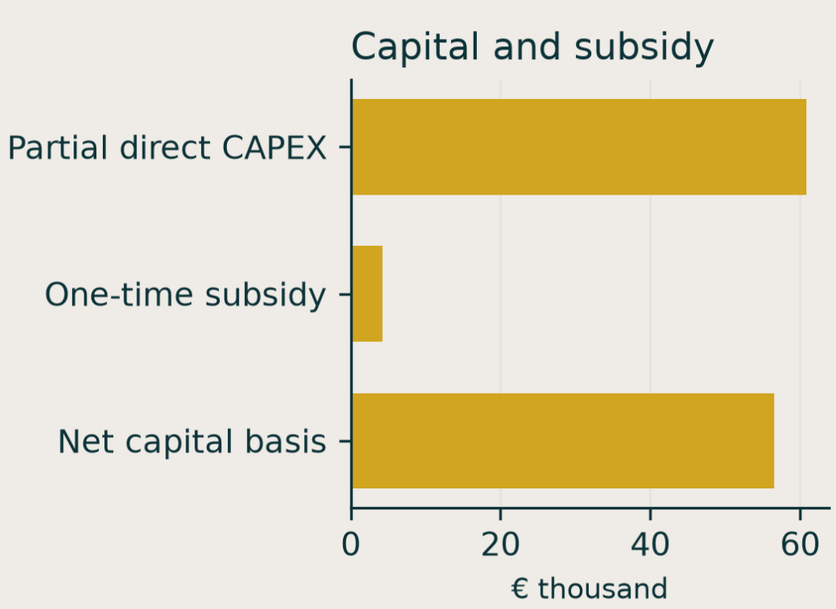
Scenario: protein_bypass rank 0; 20fb5b811d0e4cb9b33a5a0c441a3b2c; 1338.0 kg/h; 8760 h/year
Units: MWh/year; t CO_2-eq/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

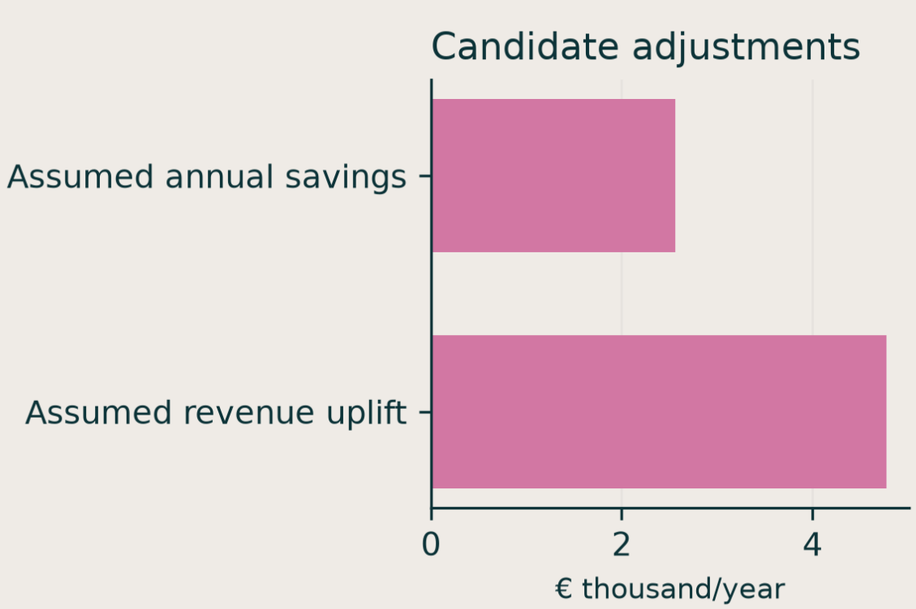
Economic accounts

Protein X · Biogas ✓ · N ✓ · P X · K X







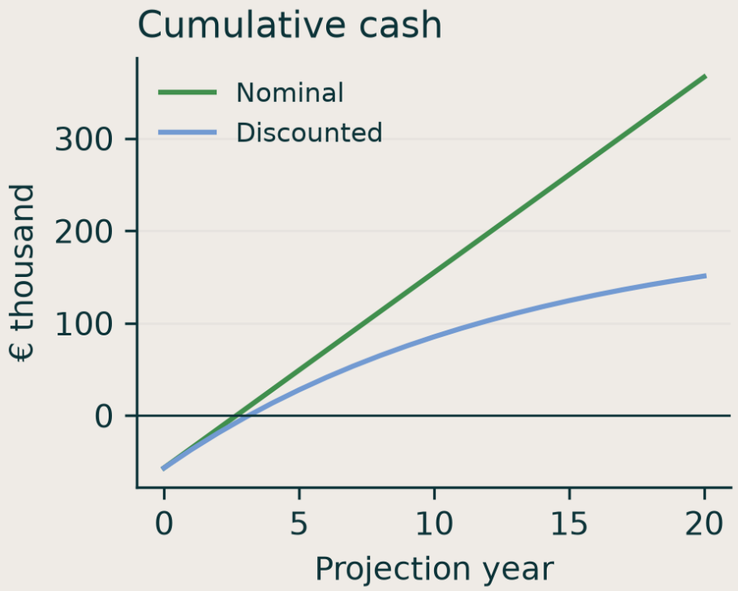


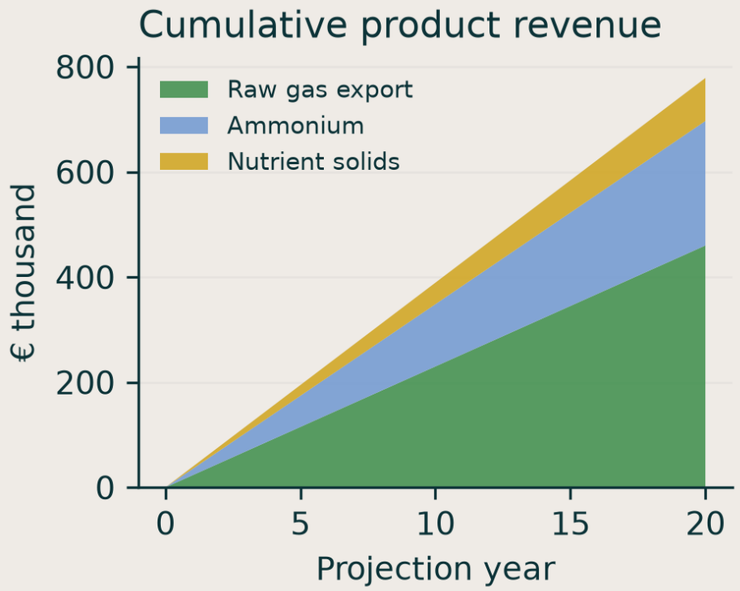
Product revenue, partial annual budget, capital subsidy and candidate overlays

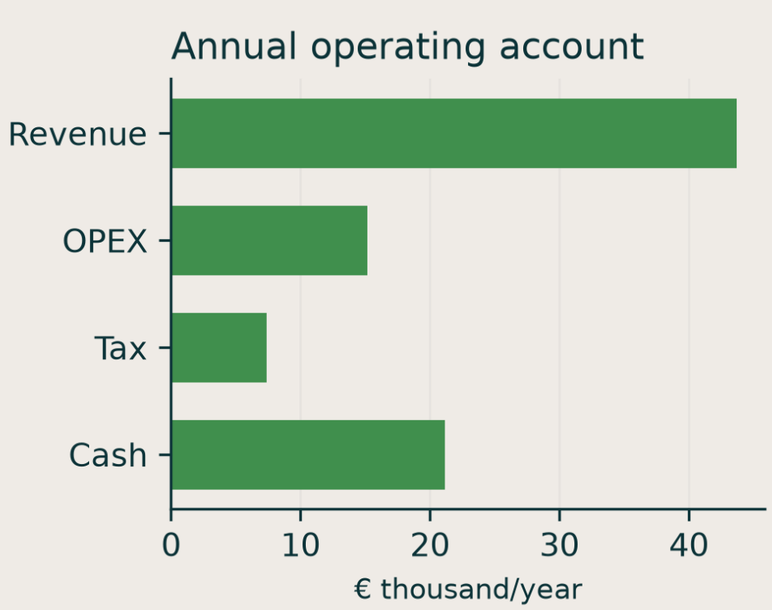
Scenario: protein_bypass rank 0; 20fb5b811d0e4cb9b33a5a0c441a3b2c; 1338.0 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

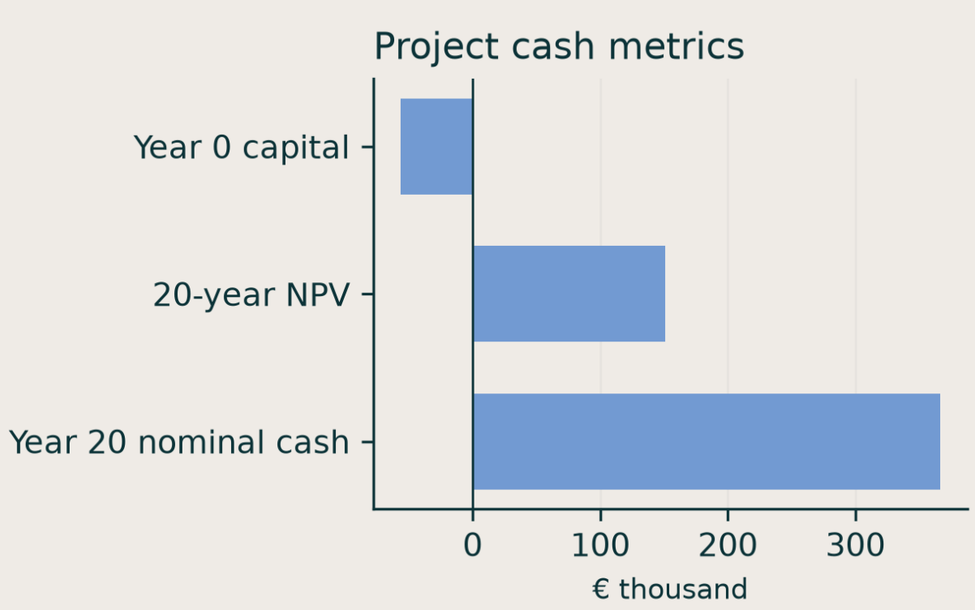
20-year financial projection

Protein X · Biogas ✓ · N ✓ · P X · K X









Constant-account cash flow and cumulative product revenue over 20 years

Scenario: protein_bypass rank 0; 20fb5b811d0e4cb9b33a5a0c441a3b2c; 1338.0 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Balance and evidence

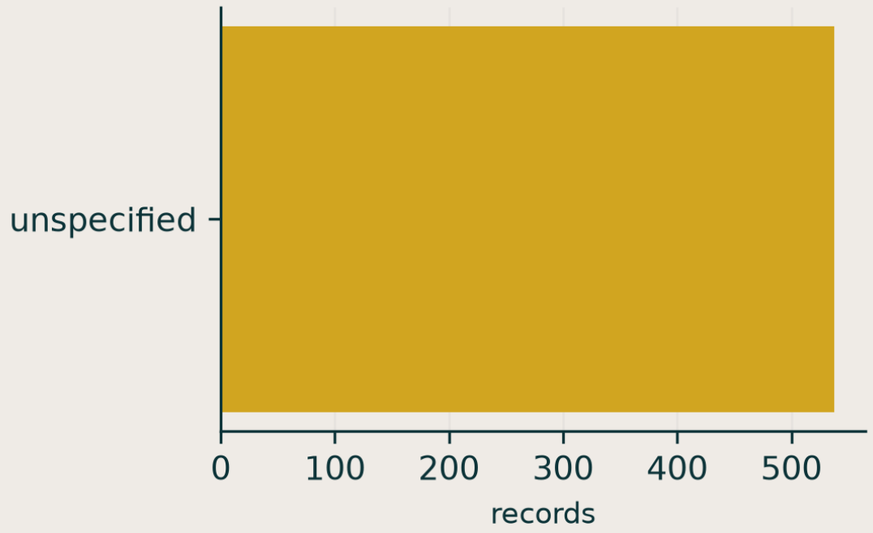
Protein X · Biogas ✓ · N ✓ · P X · K X

Element closure

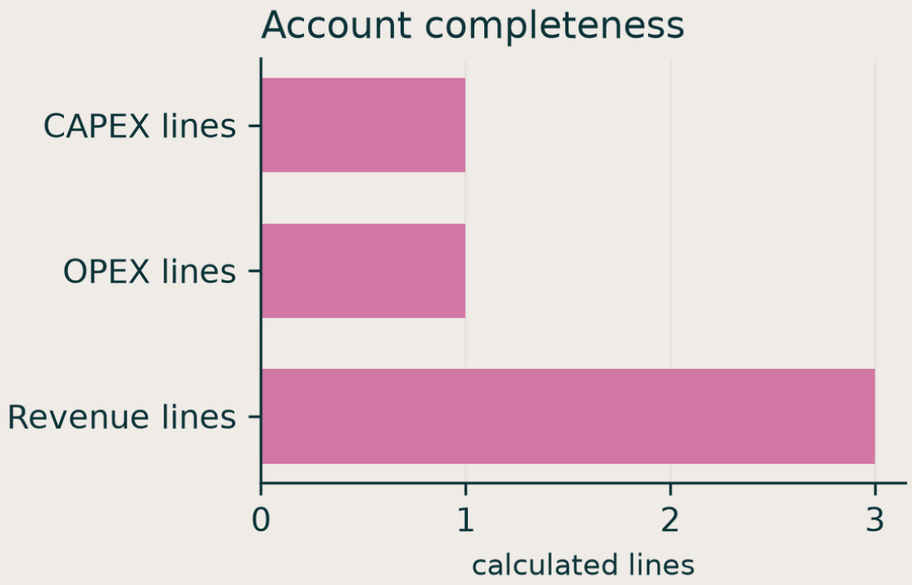
C	0 kg/year
N	0 kg/year
P	0 kg/year
K	0 kg/year

Numerical residuals

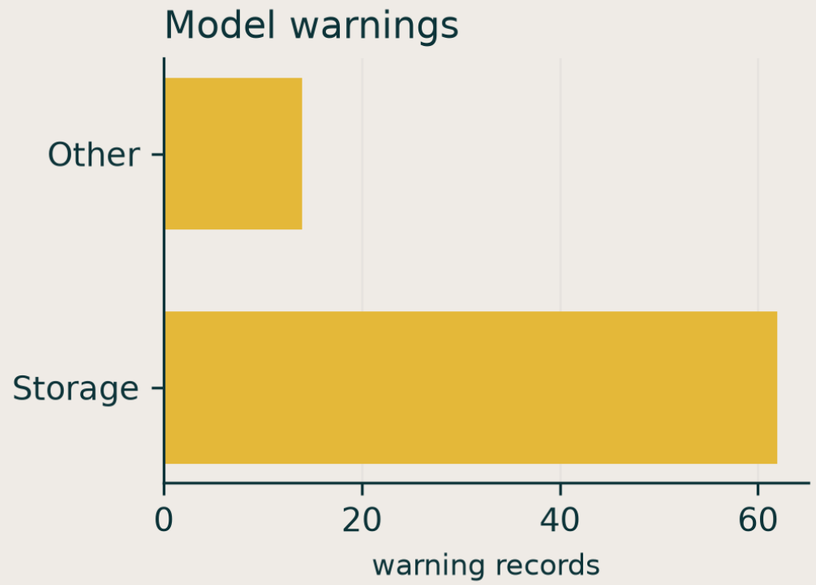
Parameter evidence



Account completeness



Model warnings



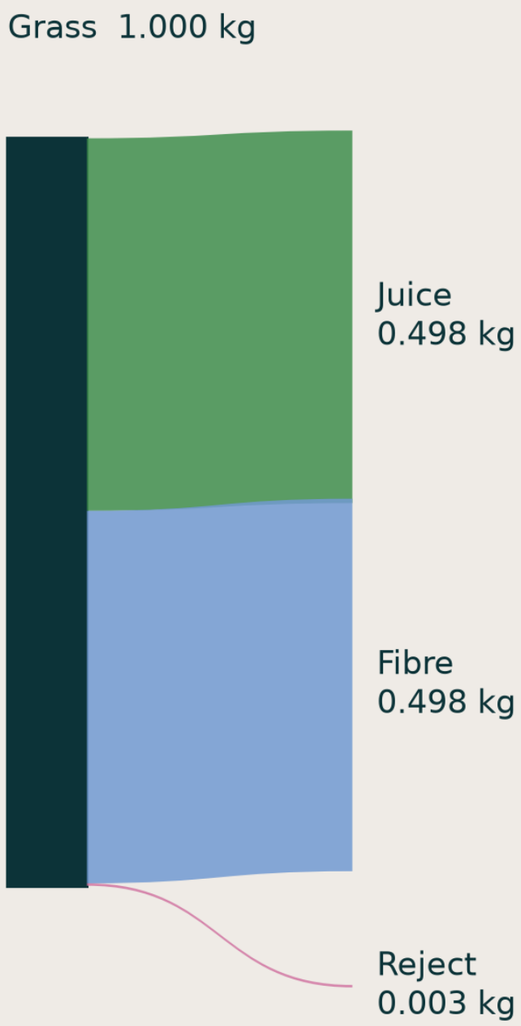
Numerical closure, evidence counts, calculated account lines and warnings

Scenario: protein_bypass rank 0; 20fb5b811d0e4cb9b33a5a0c441a3b2c; 1338.0 kg/h; 8760 h/year
Units: kg/year; records
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

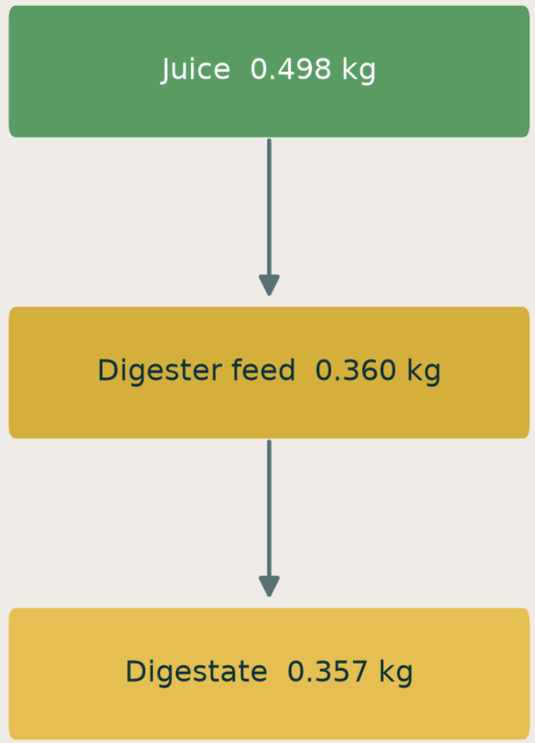
Grass to products

Protein X · Biogas ✓ · N ✓ · P ✓ · K X

Grass split



Juice pathway



Selected intermediate streams

Recovered outputs

Raw gas 0.0021 Nm³

Raw export 0.0021 Nm³

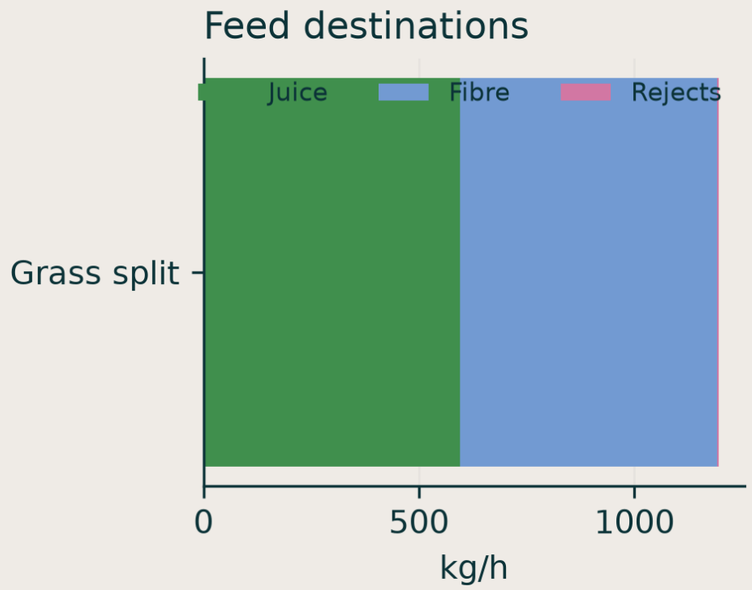
per kg wet grass

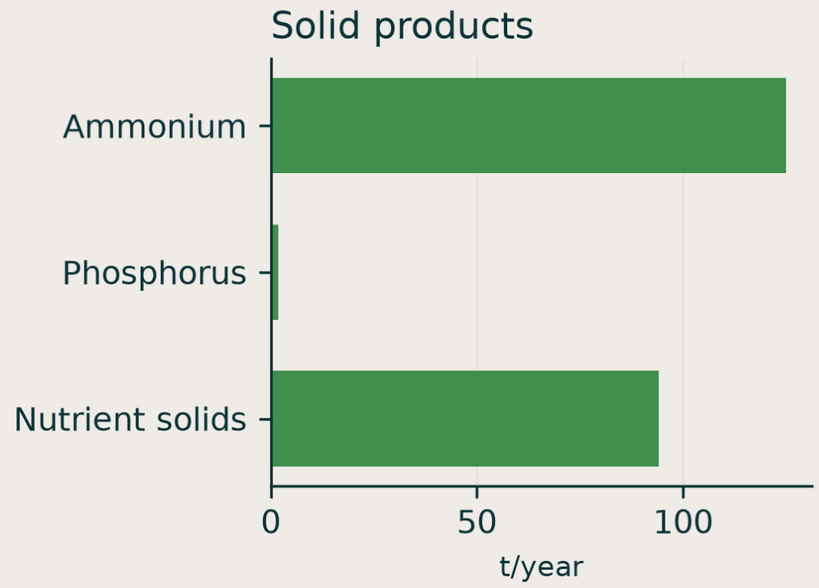
Mass-proportional grass split plus modeled downstream stream/product map

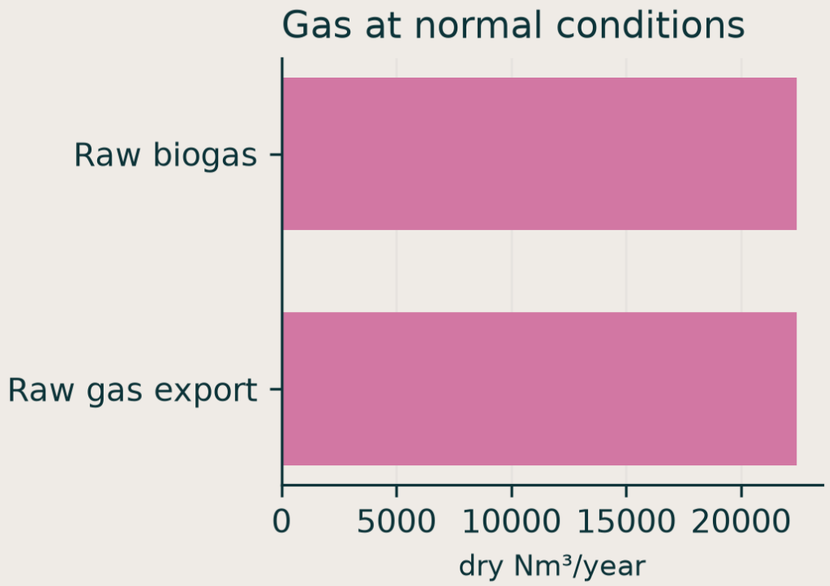
Scenario: protein_bypass rank 1; 070effa38aec44e7bfb28e0665a2b0d0; 1195.4 kg/h; 8760 h/year
Units: kg/kg wet grass; dry Nm³/kg wet grass
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

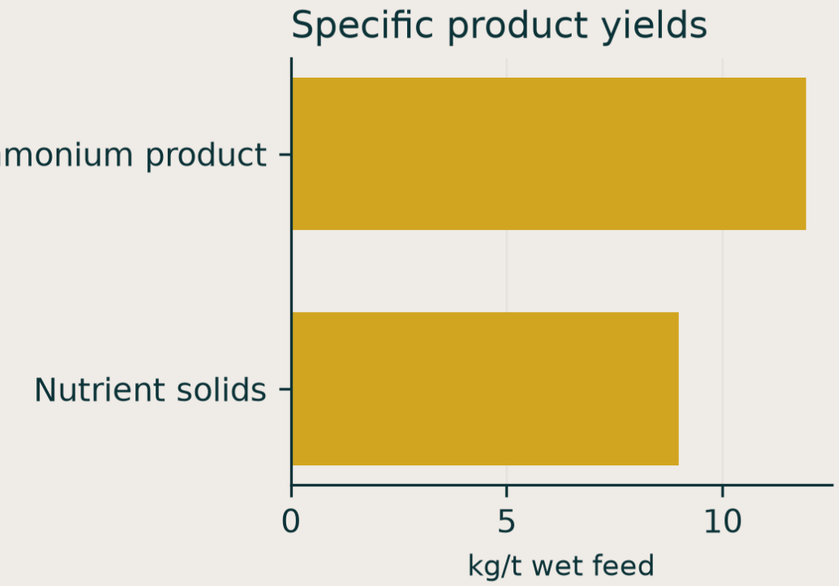
Material and products

Protein X · Biogas ✓ · N ✓ · P ✓ · K X







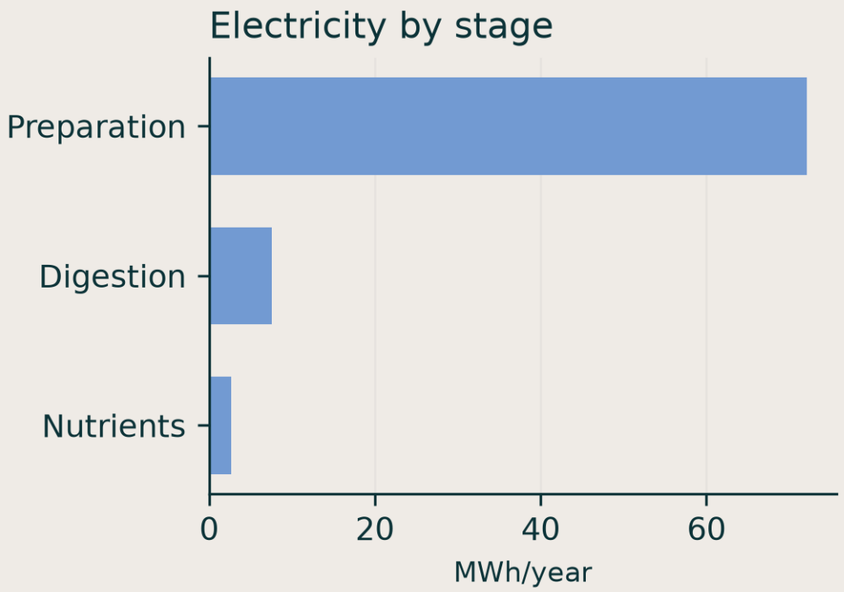


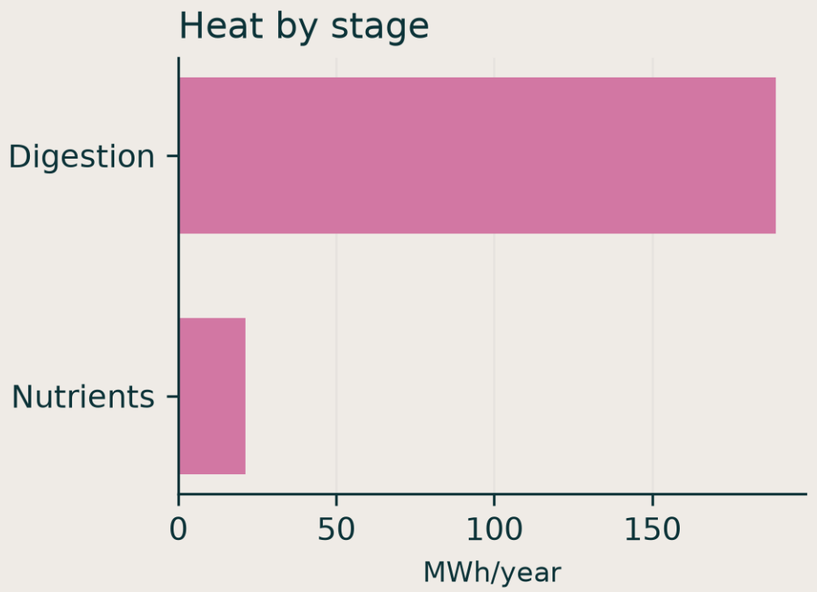
Front-end split, solid products, dry gas volumes and specific yields

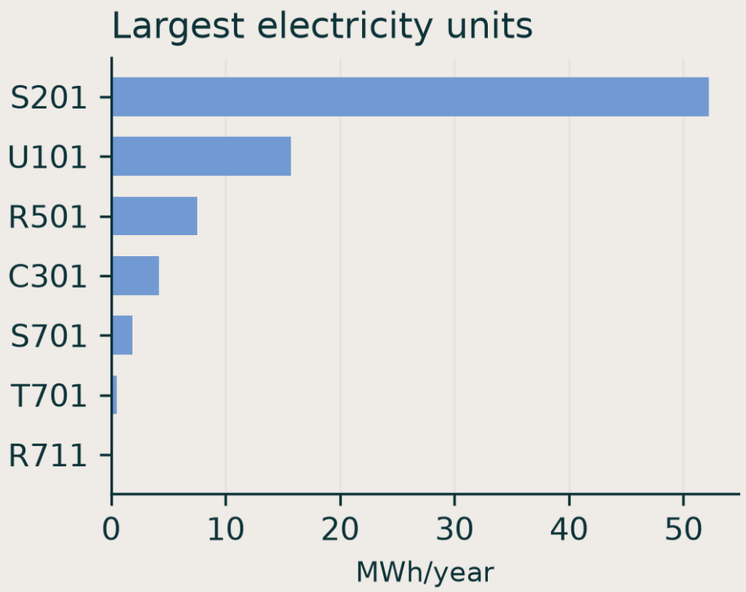
Scenario: protein_bypass rank 1; 070effa38aec44e7bfb28e0665a2b0d0; 1195.4 kg/h; 8760 h/year
Units: kg/h; t/year; dry Nm³/year; kg/t feed
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

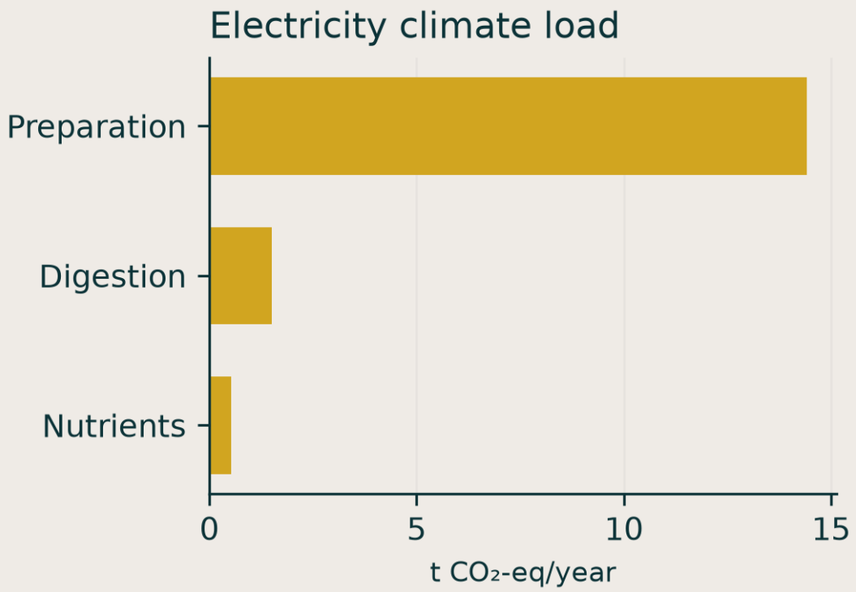
Energy and climate hotspots

Protein X · Biogas ✓ · N ✓ · P ✓ · K X







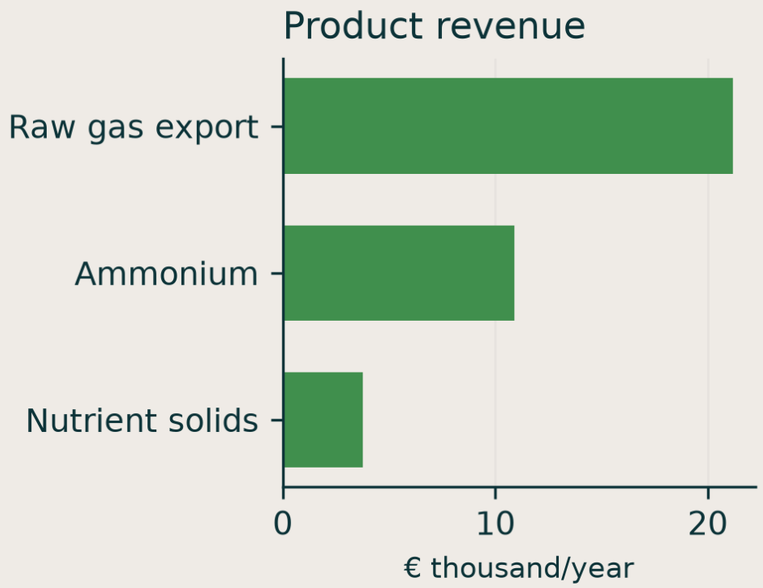


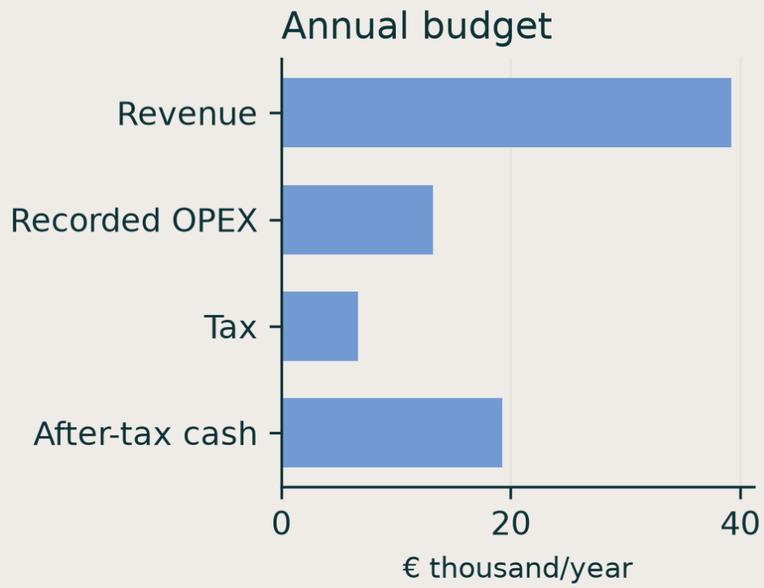
Stage and unit utility hotspots and included climate load

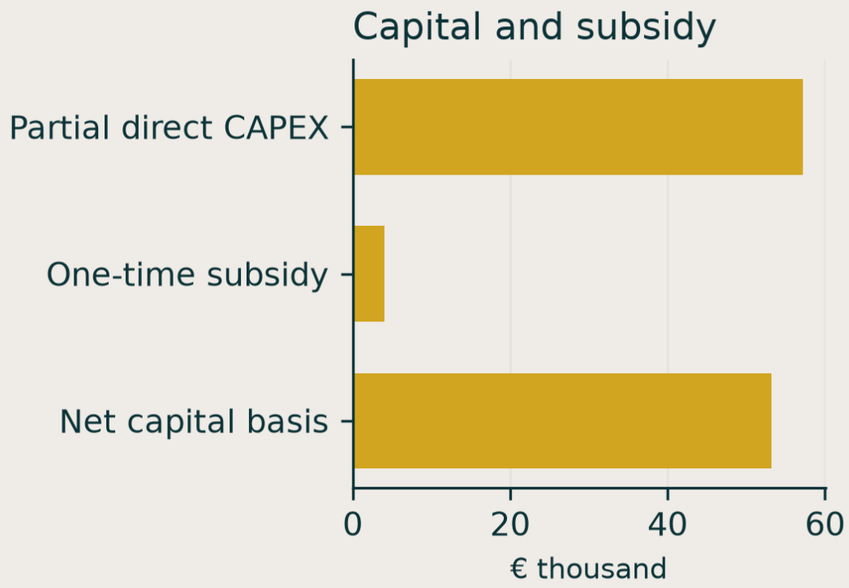
Scenario: protein_bypass rank 1; 070effa38aec44e7bfb28e0665a2b0d0; 1195.4 kg/h; 8760 h/year
Units: MWh/year; t CO_2-eq/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

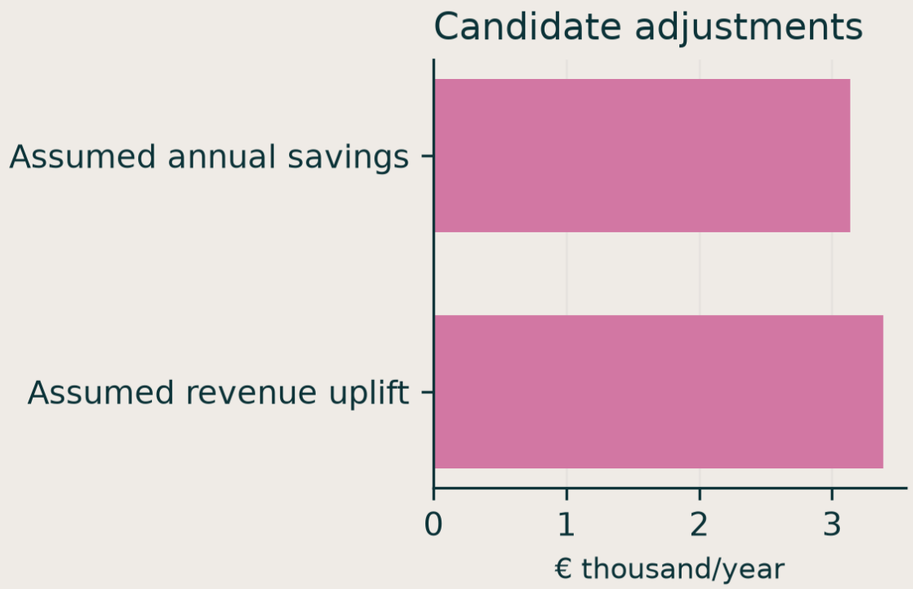
Economic accounts

Protein X · Biogas ✓ · N ✓ · P ✓ · K X







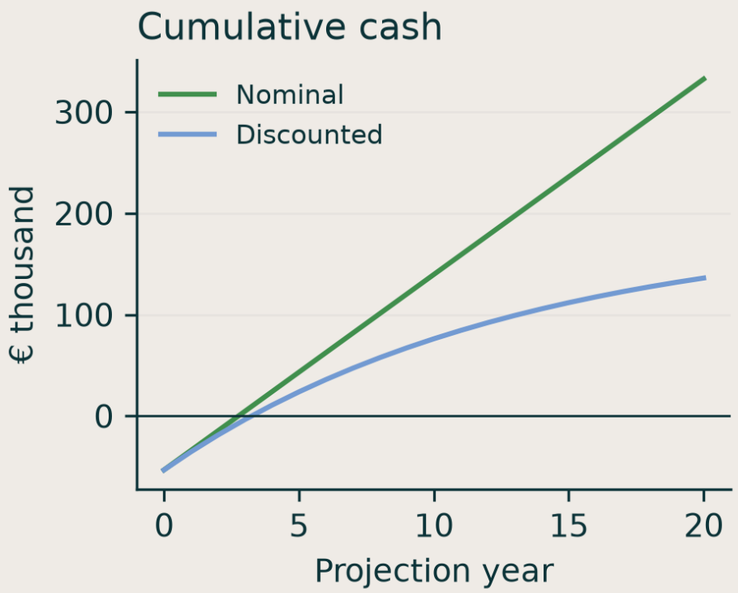


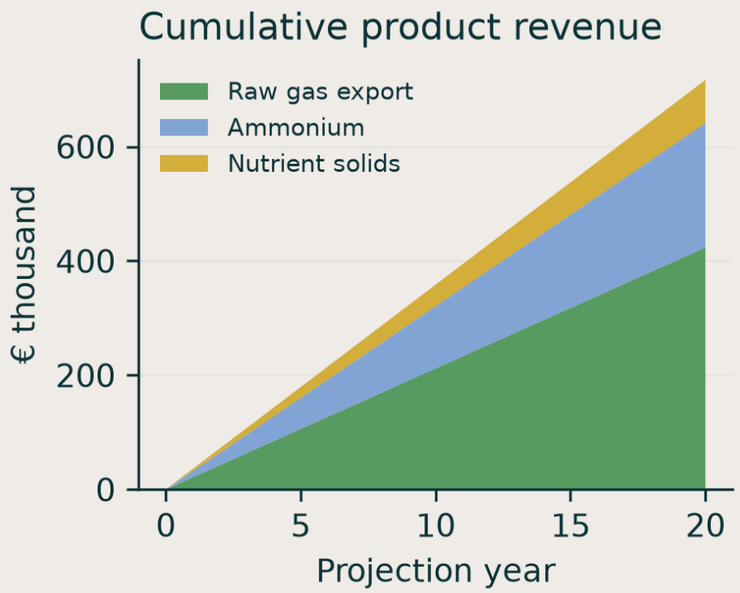
Product revenue, partial annual budget, capital subsidy and candidate overlays

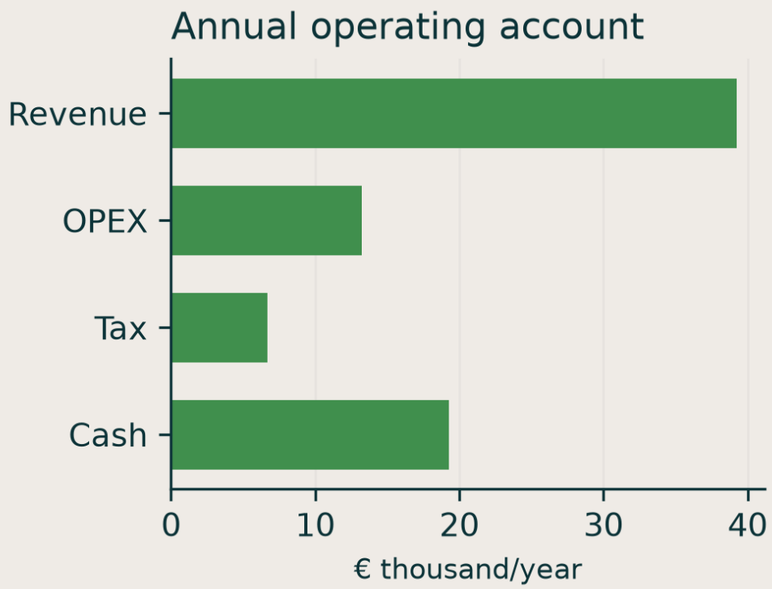
Scenario: protein_bypass rank 1; 070effa38aec44e7bfb28e0665a2b0d0; 1195.4 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

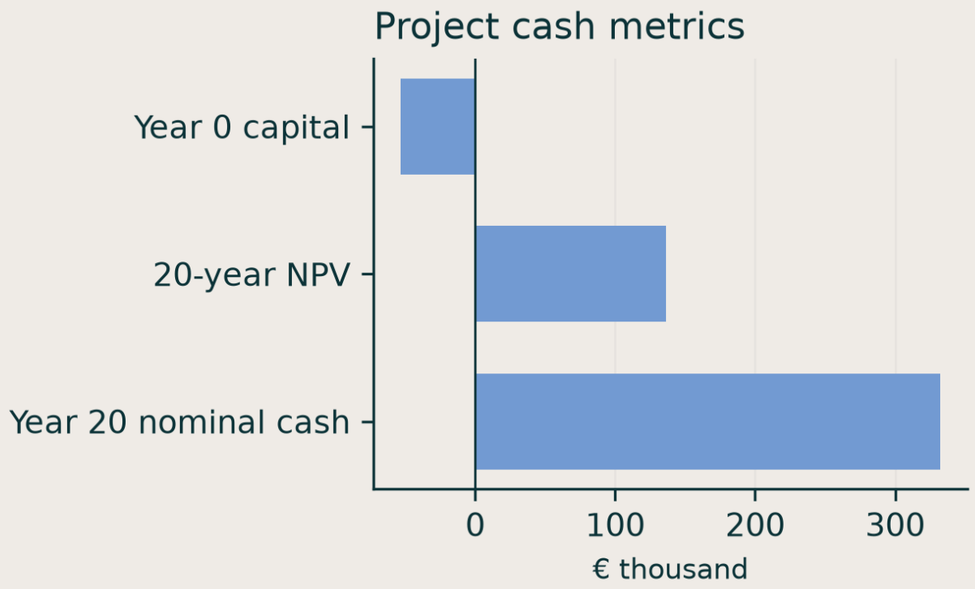
20-year financial projection

Protein X · Biogas ✓ · N ✓ · P ✓ · K X









Constant-account cash flow and cumulative product revenue over 20 years

Scenario: protein_bypass rank 1; 070effa38aec44e7bfb28e0665a2b0d0; 1195.4 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Balance and evidence

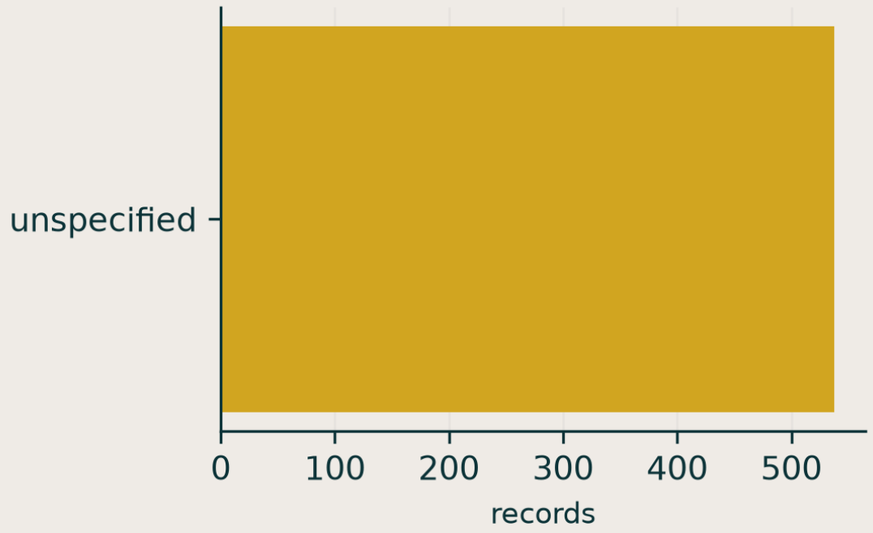
Protein X · Biogas ✓ · N ✓ · P ✓ · K X

Element closure

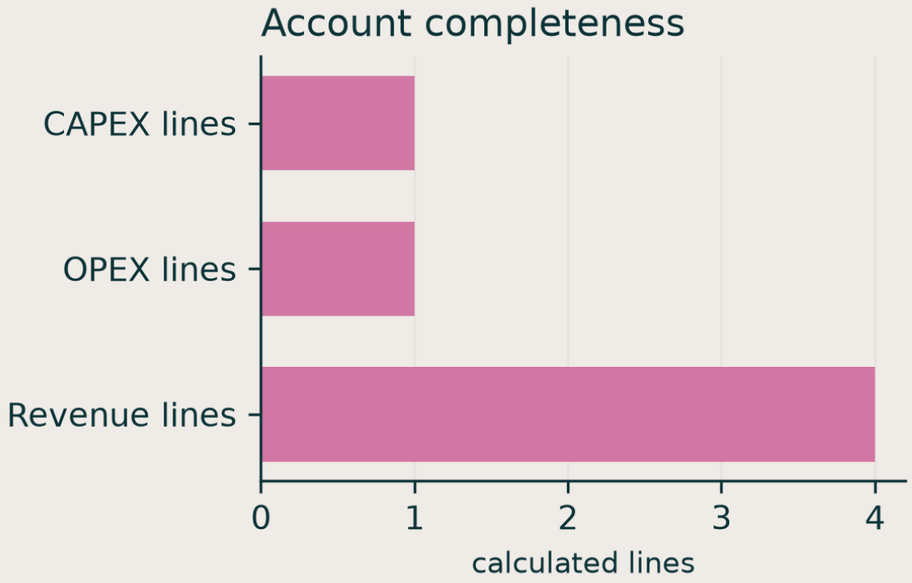
C	3.9e-12 kg/year
N	0 kg/year
P	0 kg/year
K	9.7e-13 kg/year

Numerical residuals

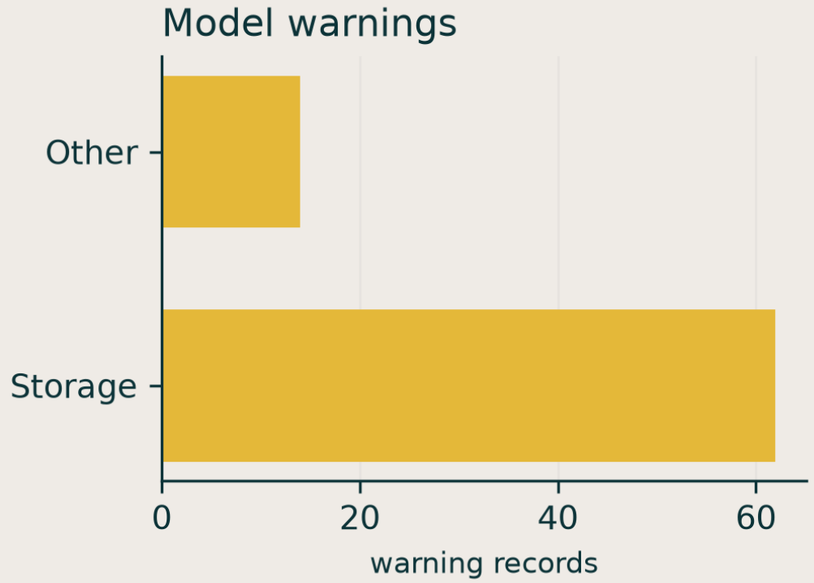
Parameter evidence



Account completeness



Model warnings



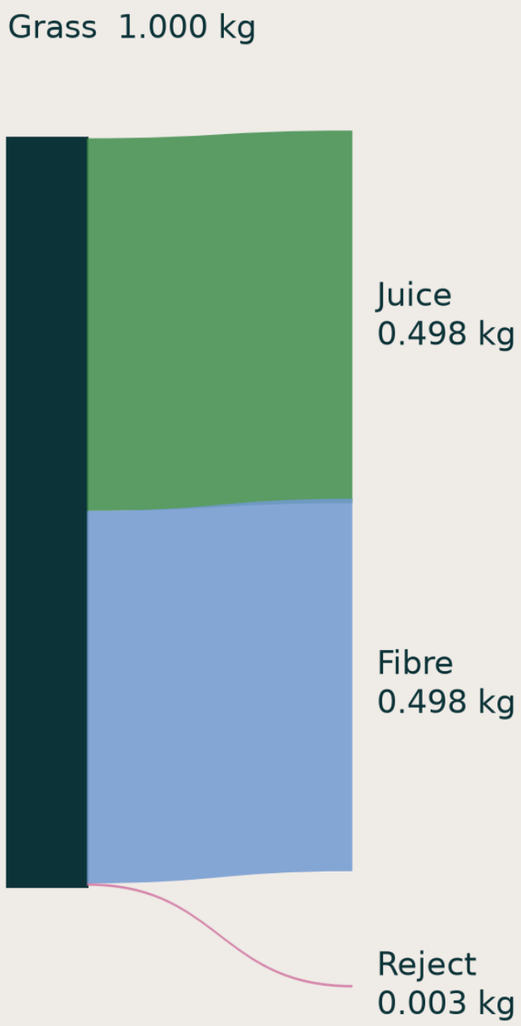
Numerical closure, evidence counts, calculated account lines and warnings

Scenario: protein_bypass rank 1; 070effa38aec44e7bfb28e0665a2b0d0; 1195.4 kg/h; 8760 h/year
Units: kg/year; records
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

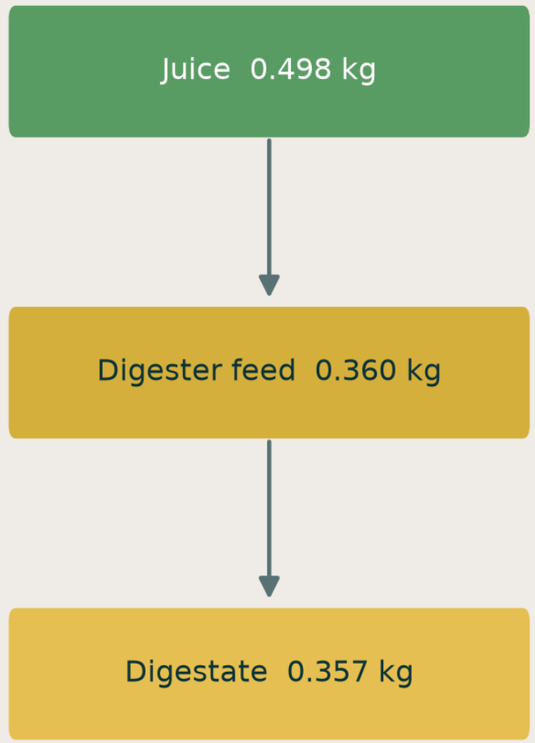
Grass to products

Protein X · Biogas ✓ · N ✓ · P X · K X

Grass split

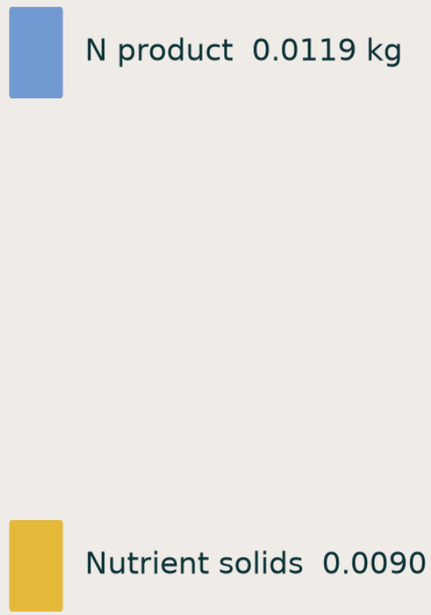


Juice pathway



Selected intermediate streams

Recovered outputs



Raw gas 0.0021 Nm³

Raw export 0.0021 Nm³

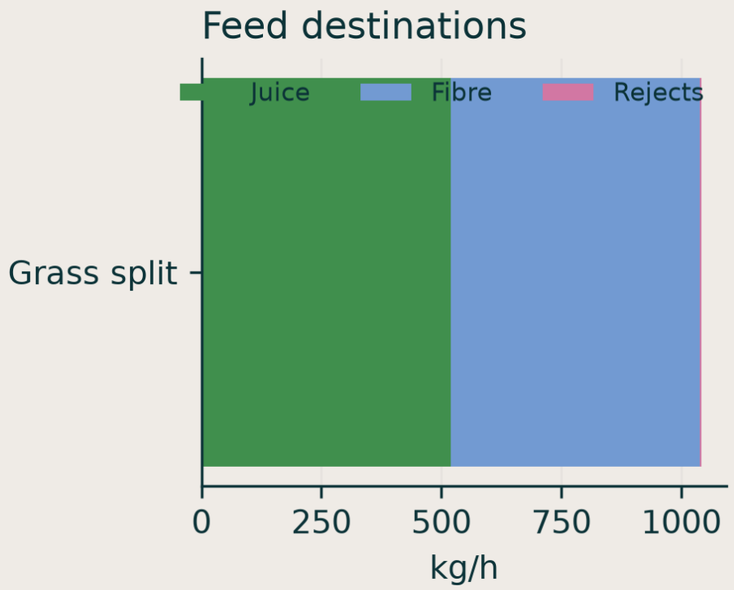
per kg wet grass

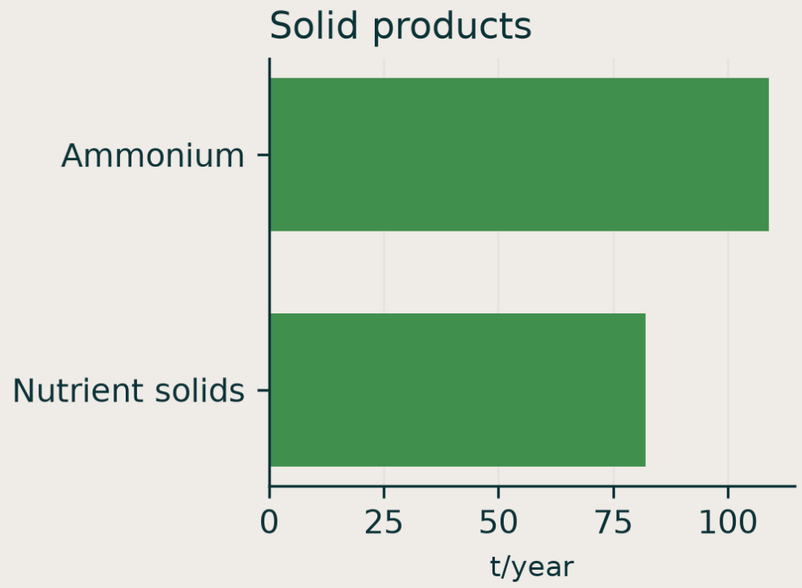
Mass-proportional grass split plus modeled downstream stream/product map

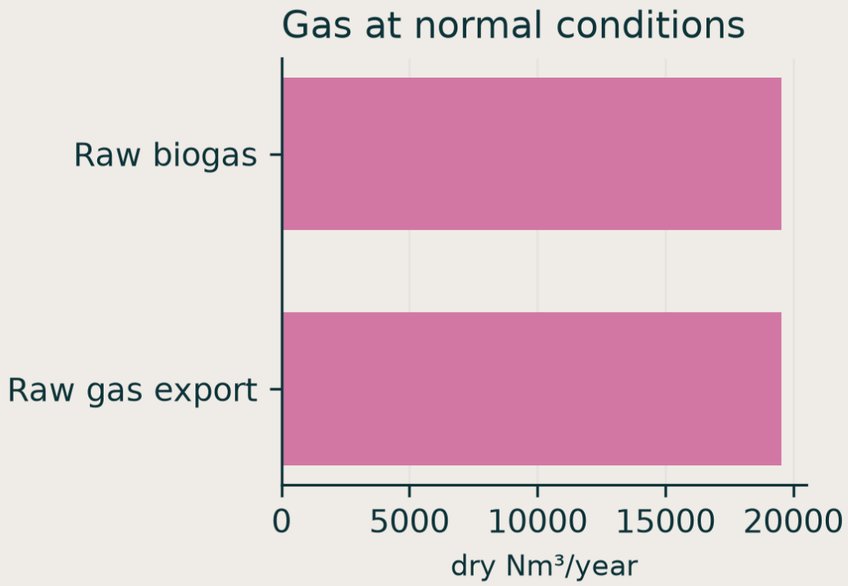
Scenario: protein_bypass rank 2; 8561366cffcf4fdf94f571a2dbe6324b; 1257.3 kg/h; 8760 h/year
Units: kg/kg wet grass; dry Nm³/kg wet grass
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

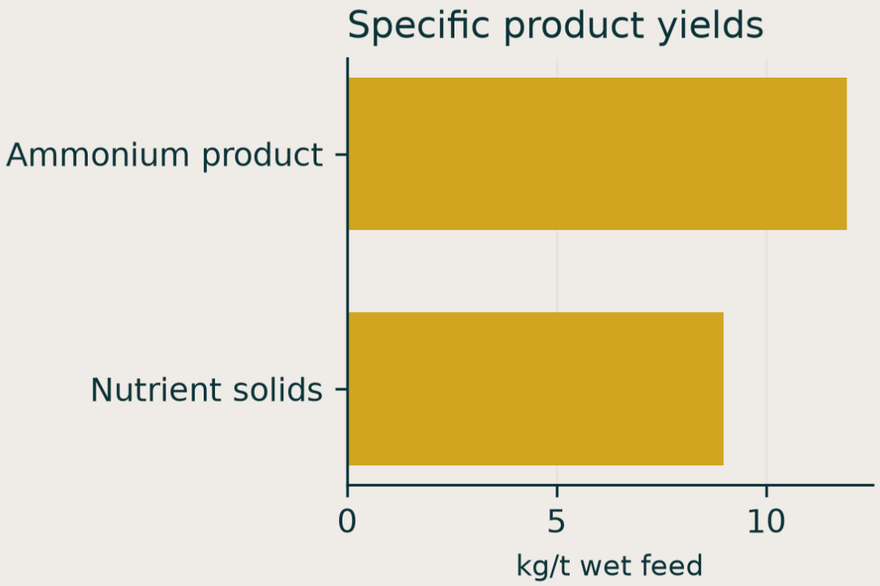
Material and products

Protein X · Biogas ✓ · N ✓ · P X · K X







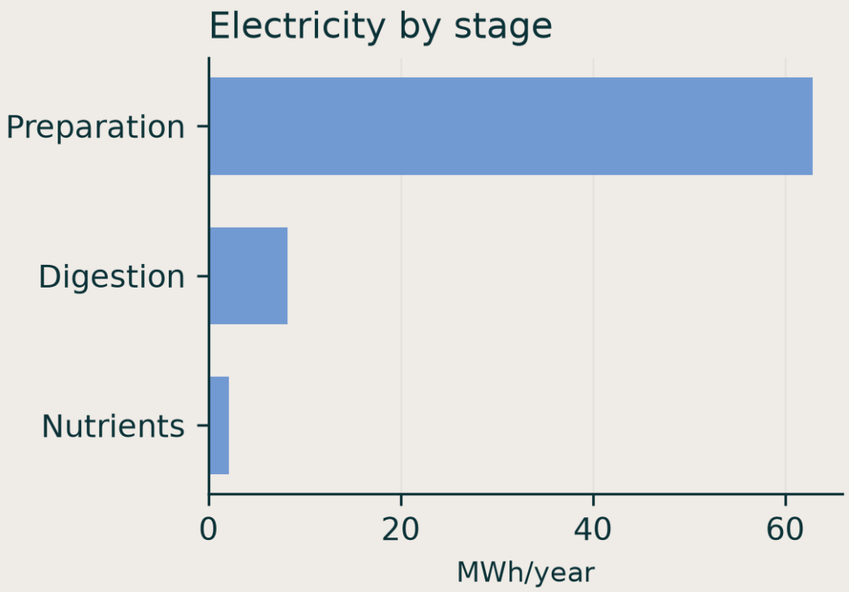


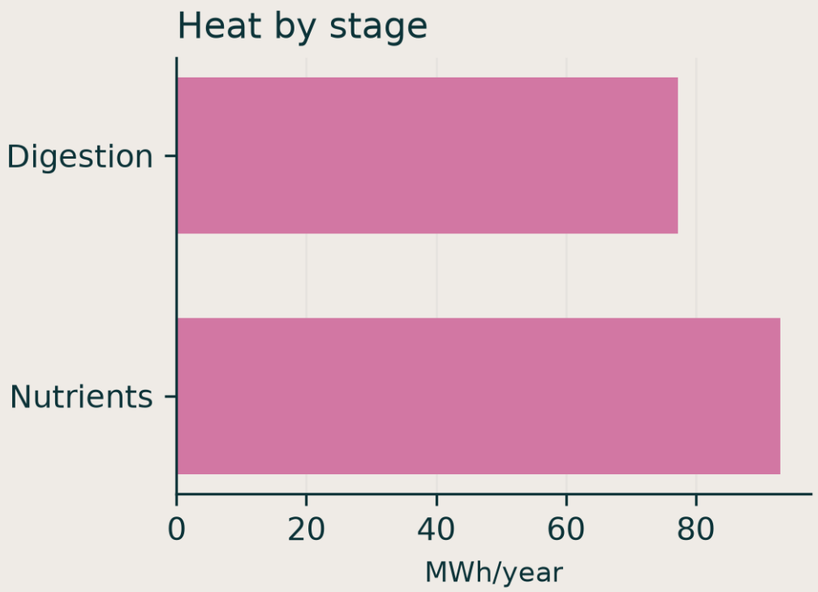
Front-end split, solid products, dry gas volumes and specific yields

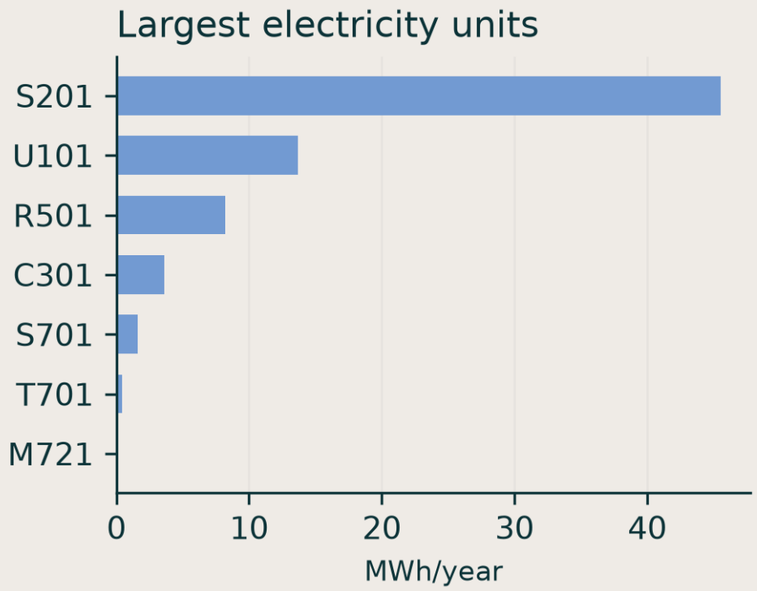
Scenario: protein_bypass rank 2; 8561366cffcf4fdf94f571a2dbe6324b; 1257.3 kg/h; 8760 h/year
Units: kg/h; t/year; dry Nm³/year; kg/t feed
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

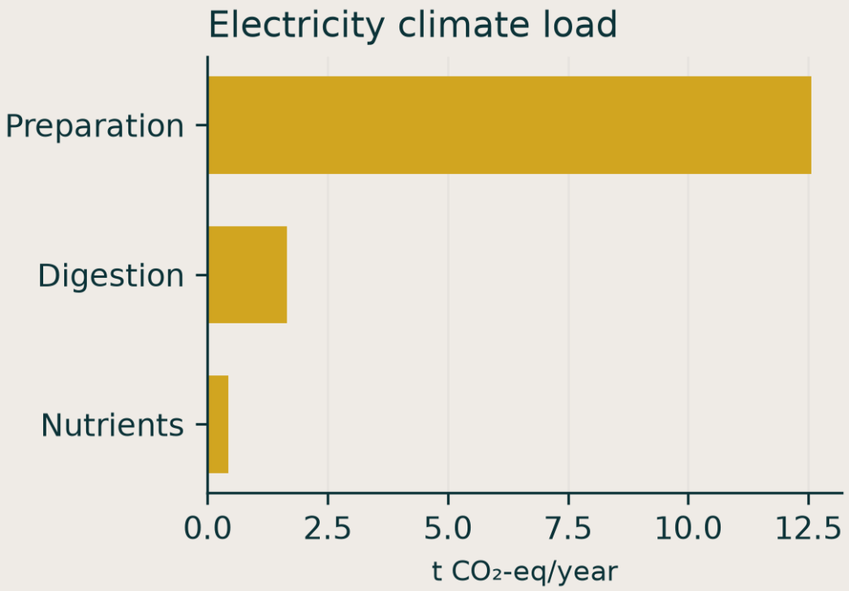
Energy and climate hotspots

Protein X · Biogas ✓ · N ✓ · P X · K X







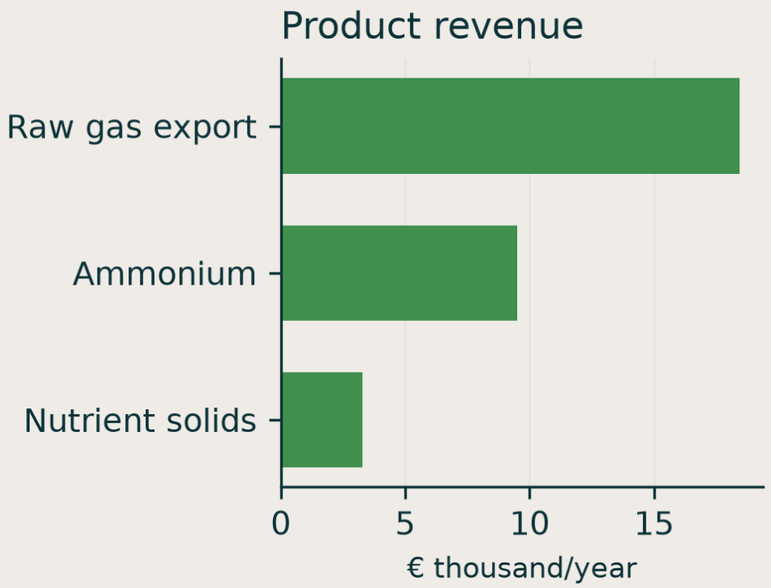


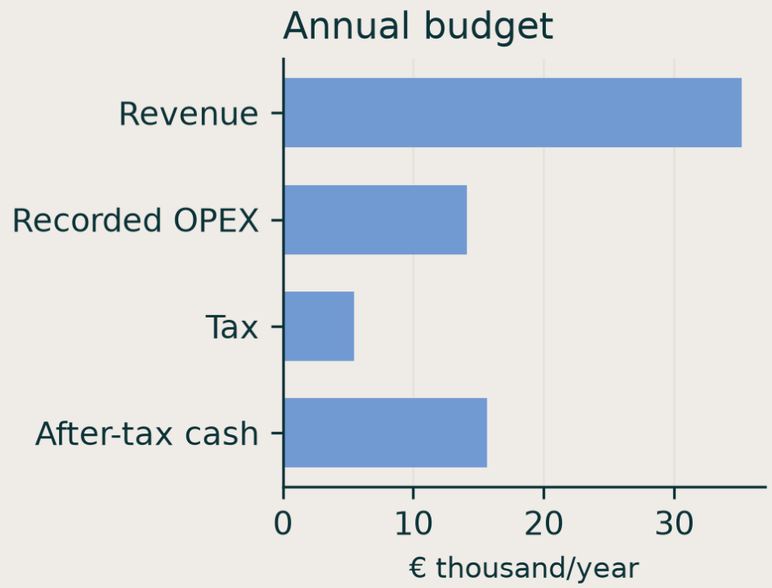
Stage and unit utility hotspots and included climate load

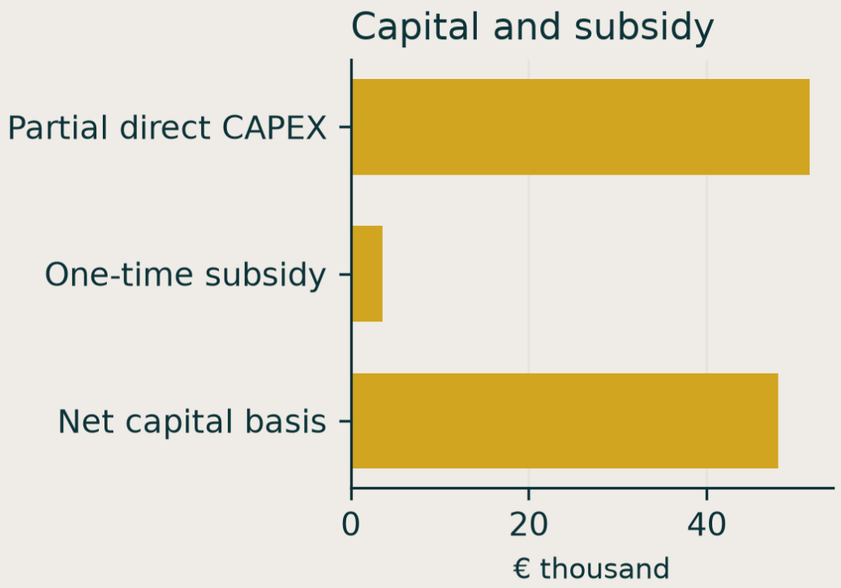
Scenario: protein_bypass rank 2; 8561366cffcf4fdf94f571a2dbe6324b; 1257.3 kg/h; 8760 h/year
Units: MWh/year; t CO_2-eq/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

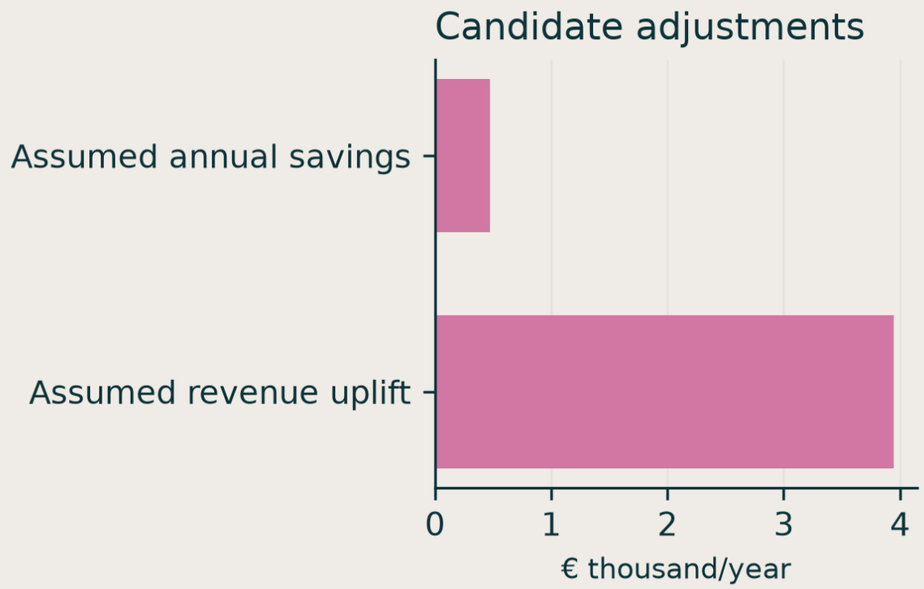
Economic accounts

Protein X · Biogas ✓ · N ✓ · P X · K X







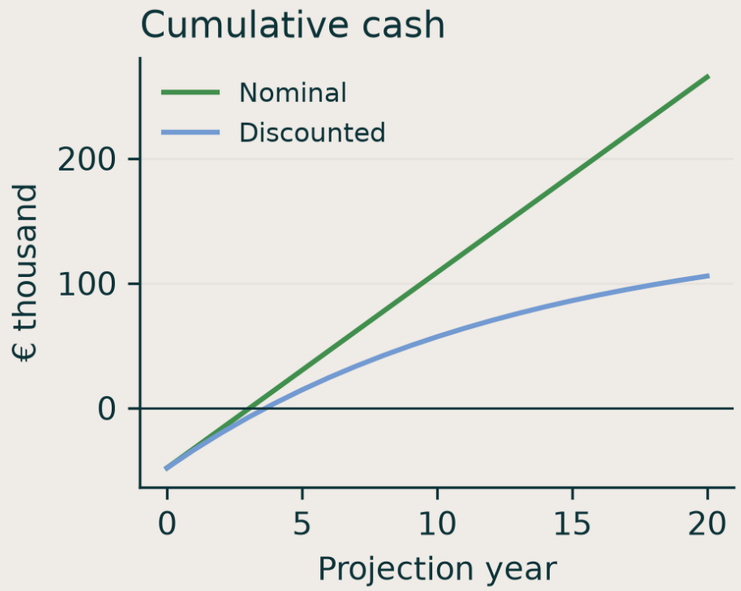


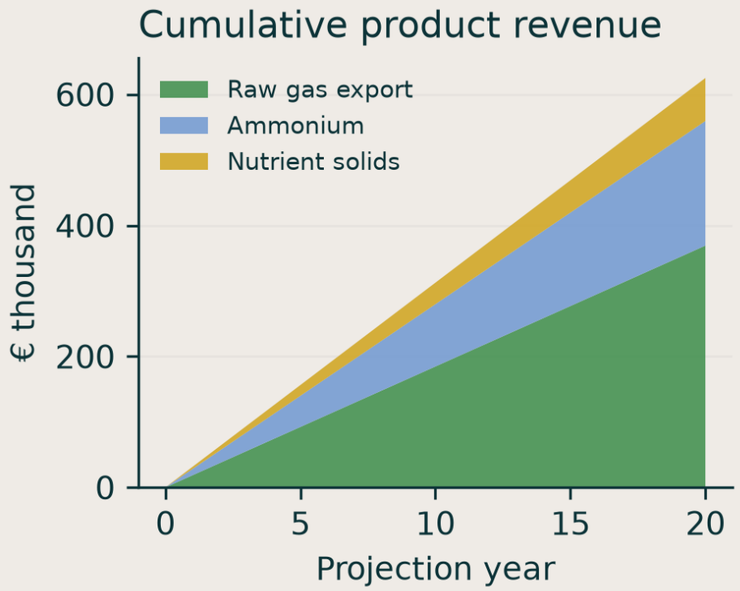
Product revenue, partial annual budget, capital subsidy and candidate overlays

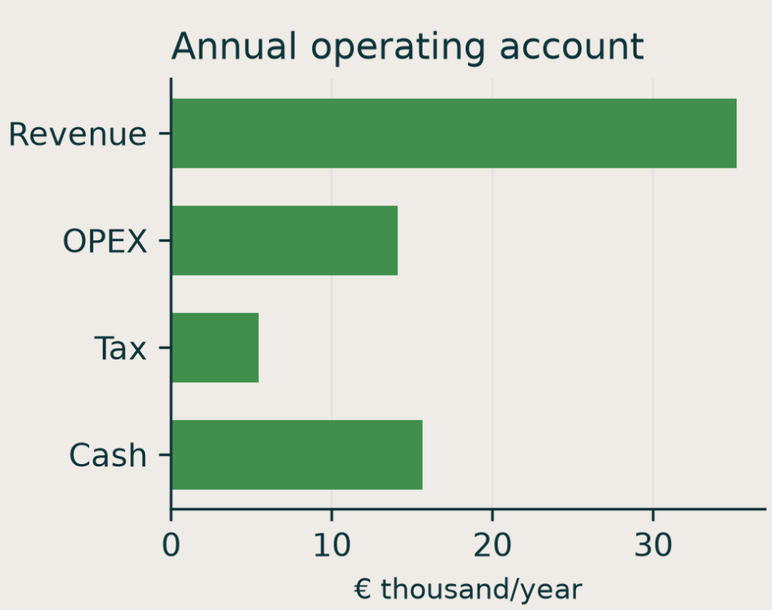
Scenario: protein_bypass rank 2; 8561366cffcf4fdf94f571a2dbe6324b; 1257.3 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

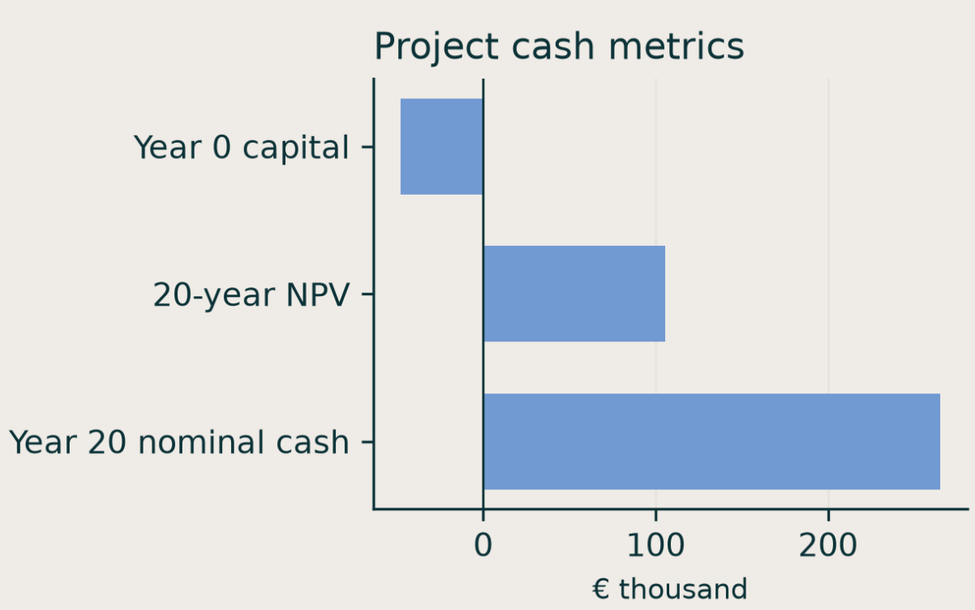
20-year financial projection

Protein X · Biogas ✓ · N ✓ · P X · K X









Constant-account cash flow and cumulative product revenue over 20 years

Scenario: protein_bypass rank 2; 8561366cffcf4fdf94f571a2dbe6324b; 1257.3 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Balance and evidence

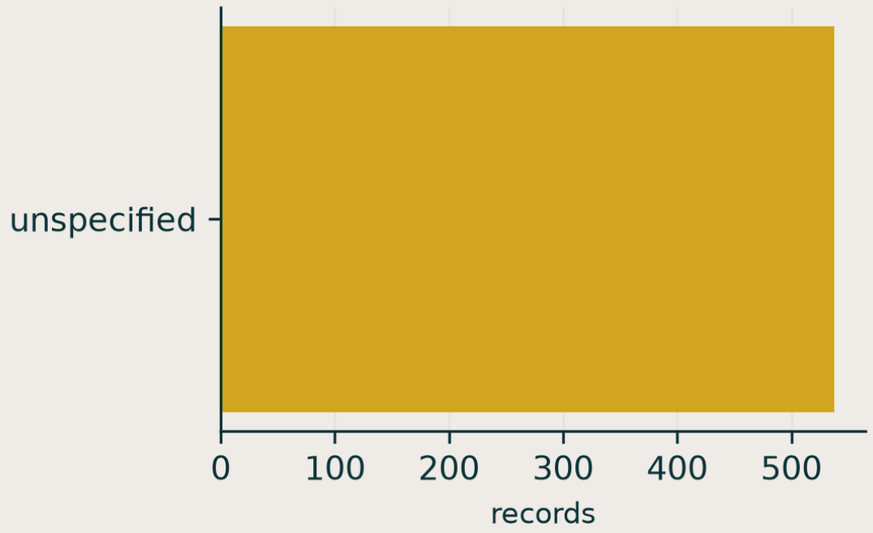
Protein X · Biogas ✓ · N ✓ · P X · K X

Element closure

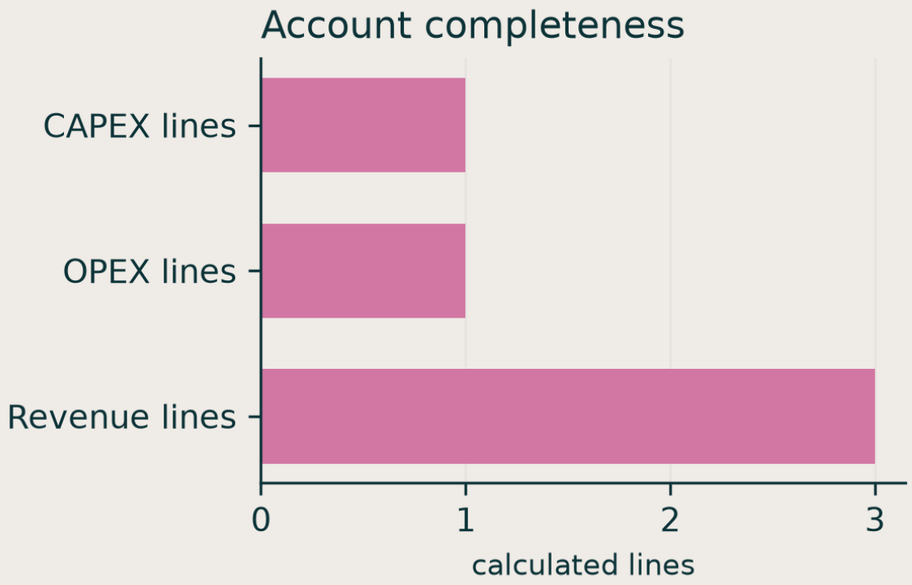
C	0 kg/year
N	0 kg/year
P	0 kg/year
K	0 kg/year

Numerical residuals

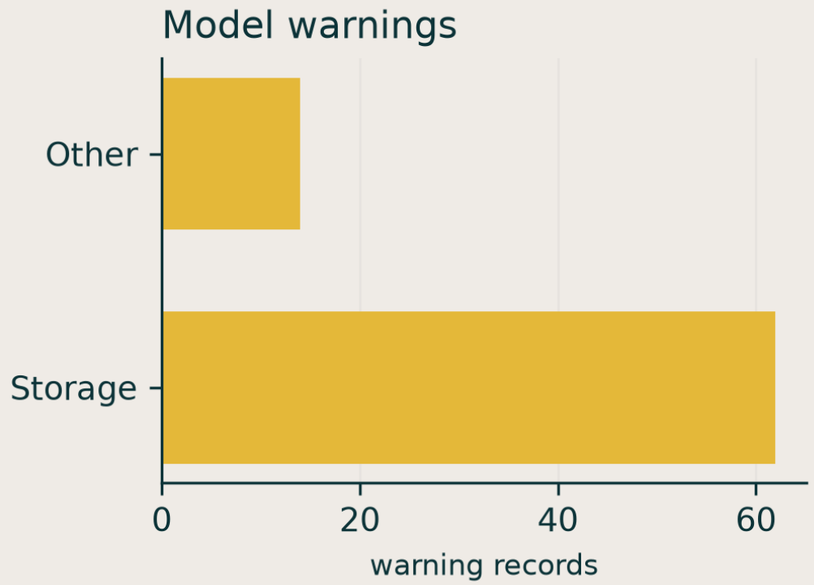
Parameter evidence



Account completeness



Model warnings



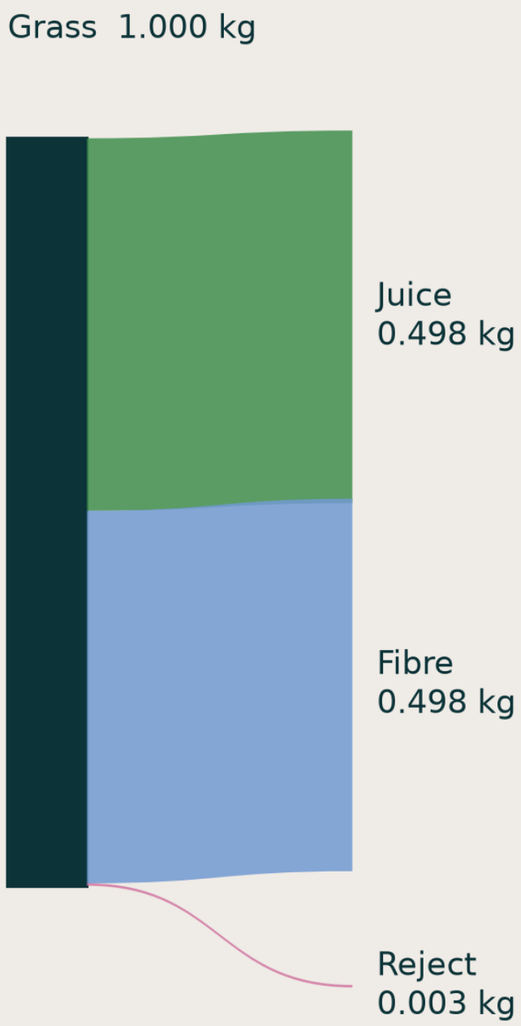
Numerical closure, evidence counts, calculated account lines and warnings

Scenario: protein_bypass rank 2; 8561366cffcf4fdf94f571a2dbe6324b; 1257.3 kg/h; 8760 h/year
Units: kg/year; records
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

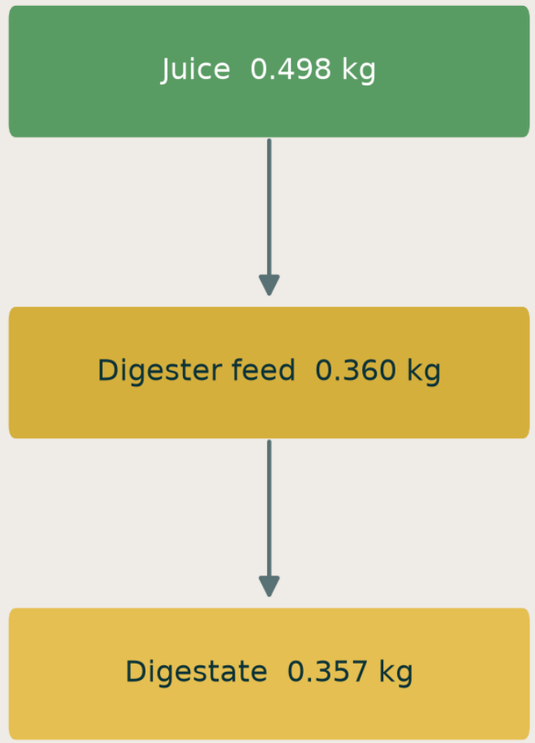
Grass to products

Protein X · Biogas ✓ · N ✓ · P ✓ · K X

Grass split



Juice pathway



Selected intermediate streams

Recovered outputs

Raw gas 0.0021 Nm³

Raw export 0.0021 Nm³

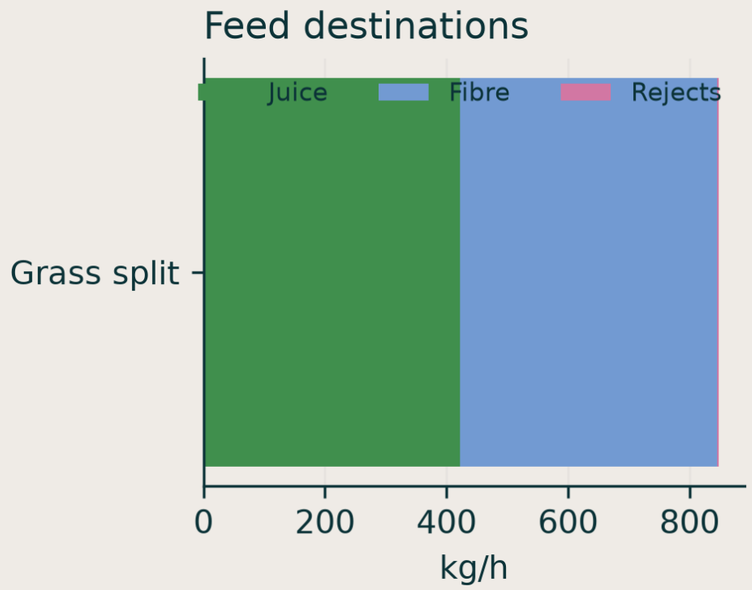
per kg wet grass

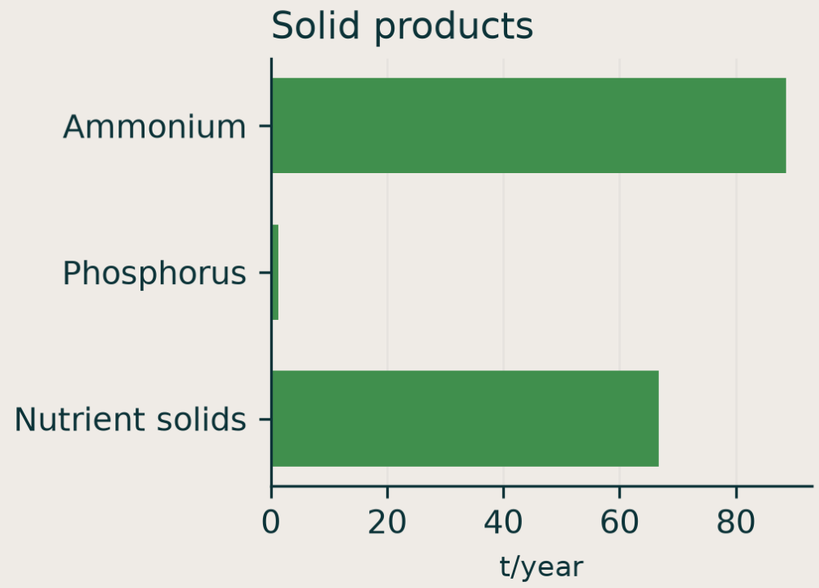
Mass-proportional grass split plus modeled downstream stream/product map

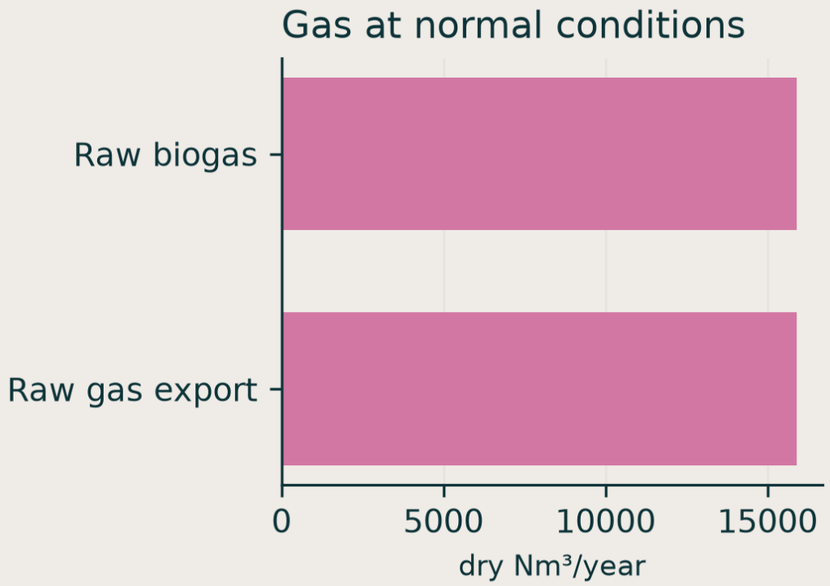
Scenario: protein_bypass rank 3; ecb05c86f525455cb46f2e5bef5cf1a8; 847.8 kg/h; 8760 h/year
Units: kg/kg wet grass; dry Nm³/kg wet grass
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

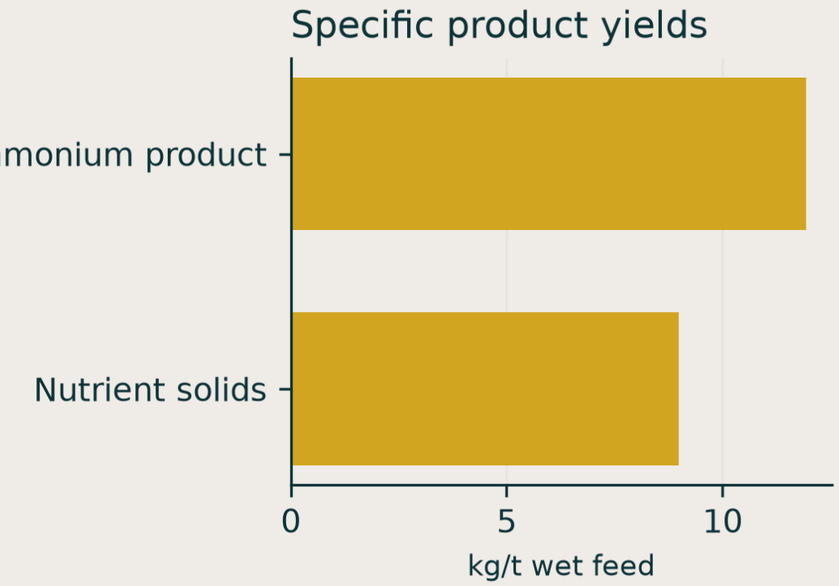
Material and products

Protein X · Biogas ✓ · N ✓ · P ✓ · K X







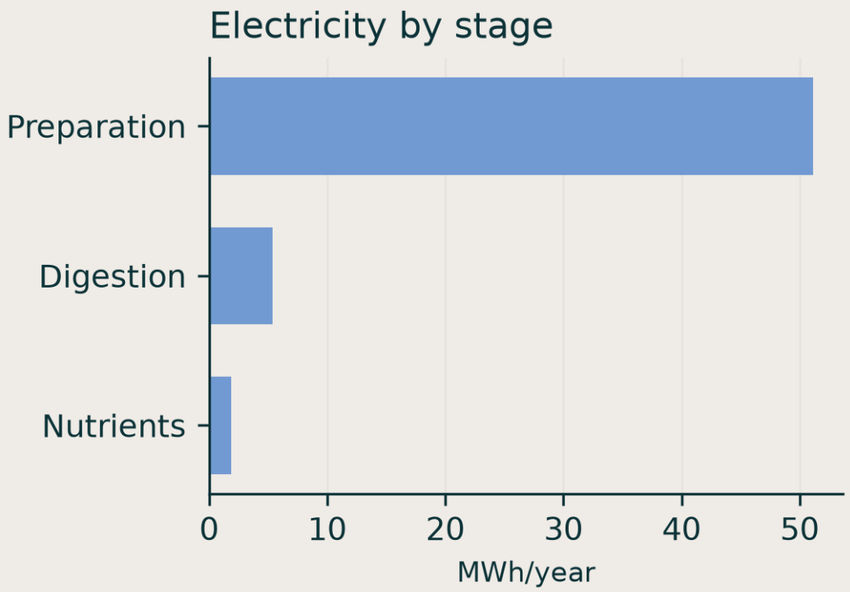


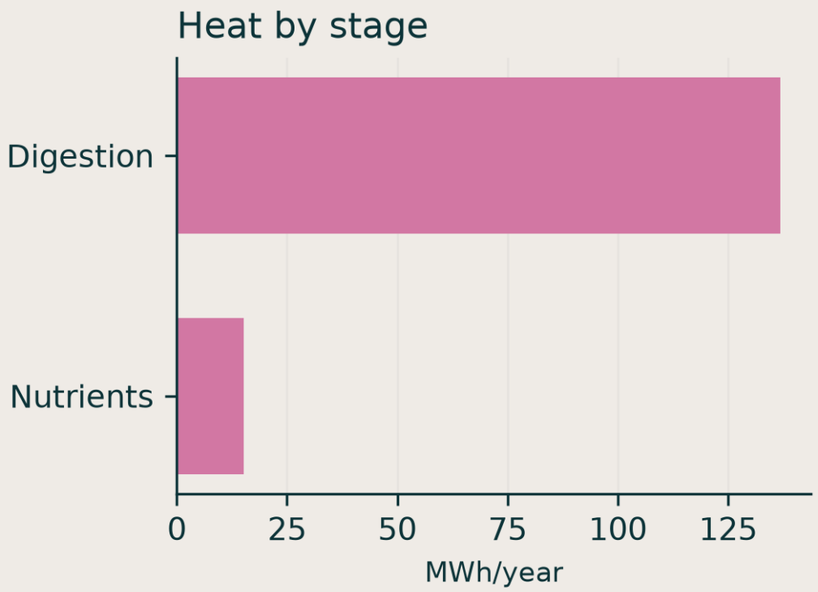
Front-end split, solid products, dry gas volumes and specific yields

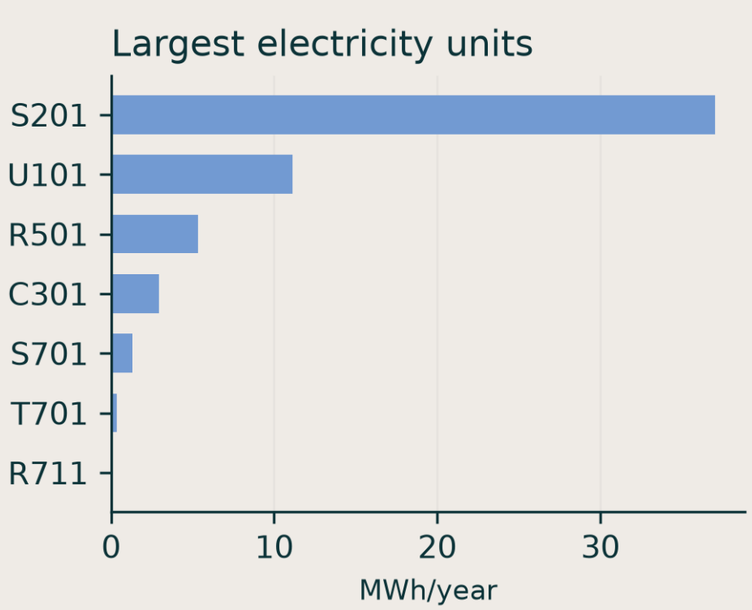
Scenario: protein_bypass rank 3; ecb05c86f525455cb46f2e5bef5cf1a8; 847.8 kg/h; 8760 h/year
Units: kg/h; t/year; dry Nm³/year; kg/t feed
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

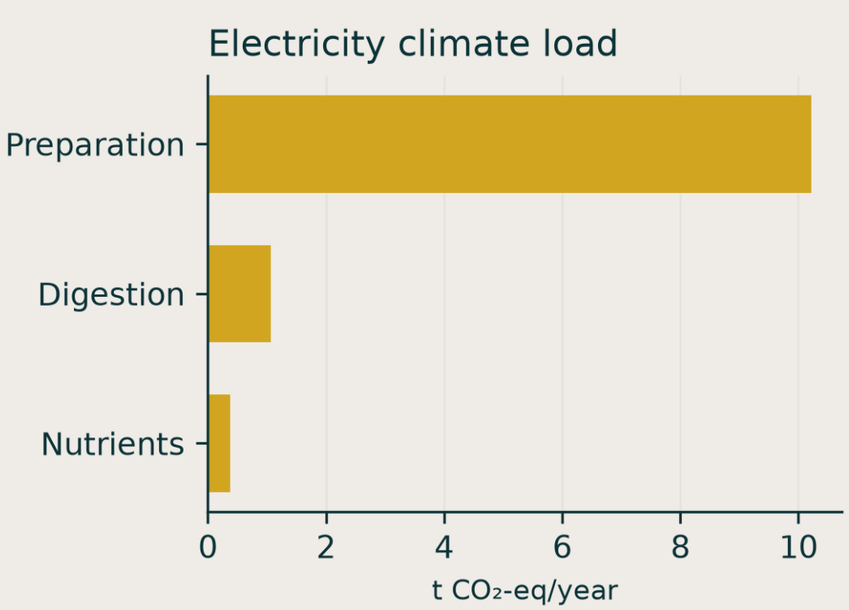
Energy and climate hotspots

Protein X · Biogas ✓ · N ✓ · P ✓ · K X







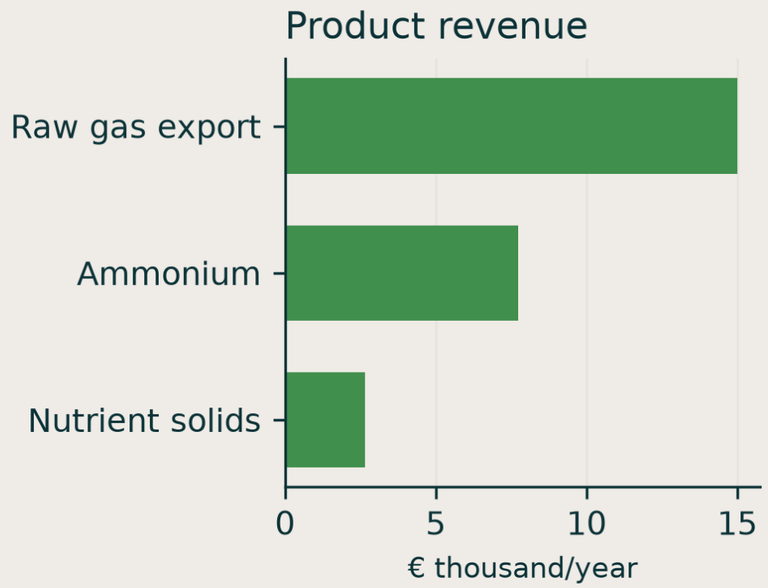


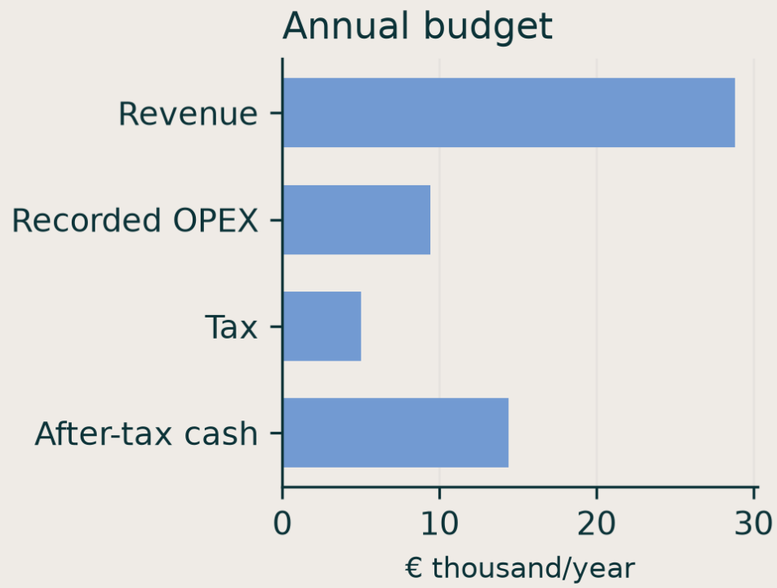
Stage and unit utility hotspots and included climate load

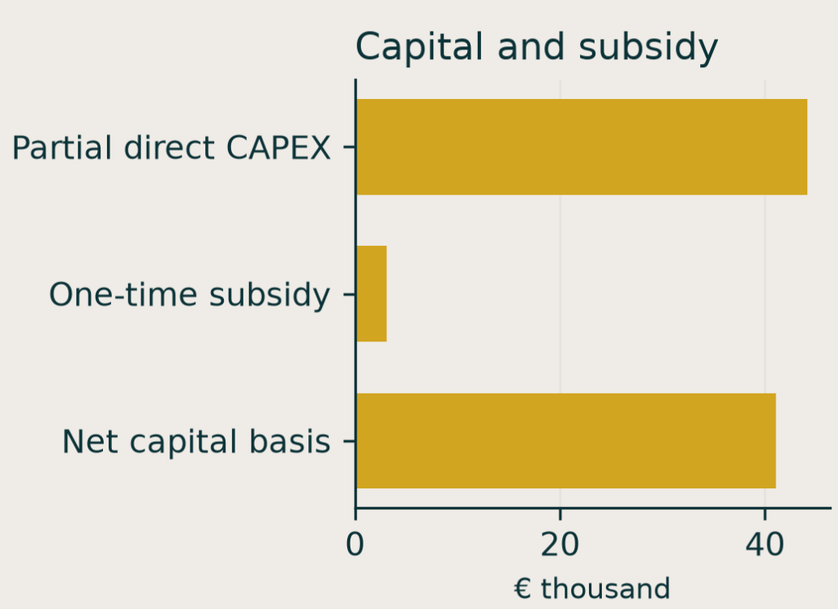
Scenario: protein_bypass rank 3; ecb05c86f525455cb46f2e5bef5cf1a8; 847.8 kg/h; 8760 h/year
Units: MWh/year; t CO_2-eq/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

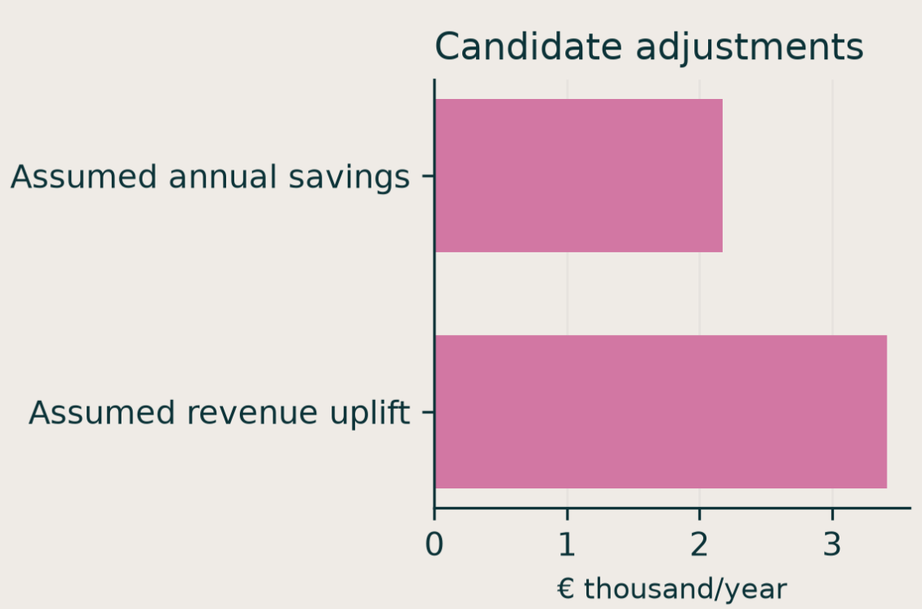
Economic accounts

Protein X · Biogas ✓ · N ✓ · P ✓ · K X







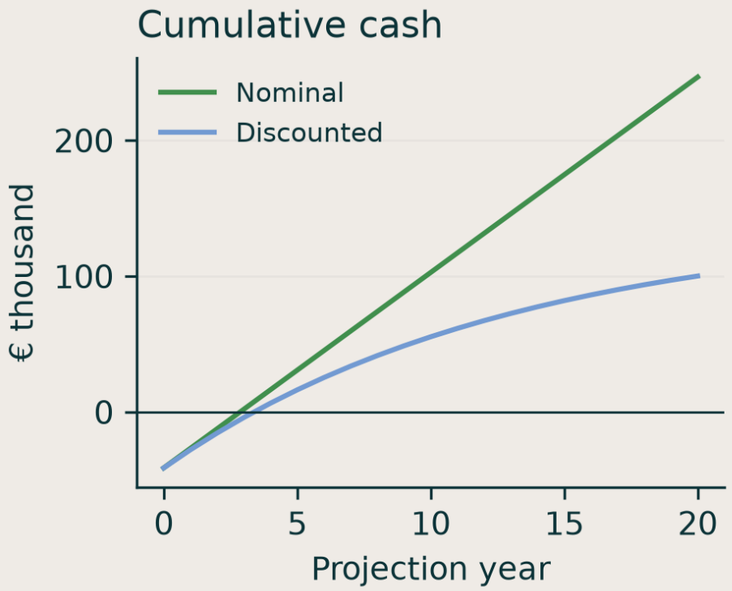


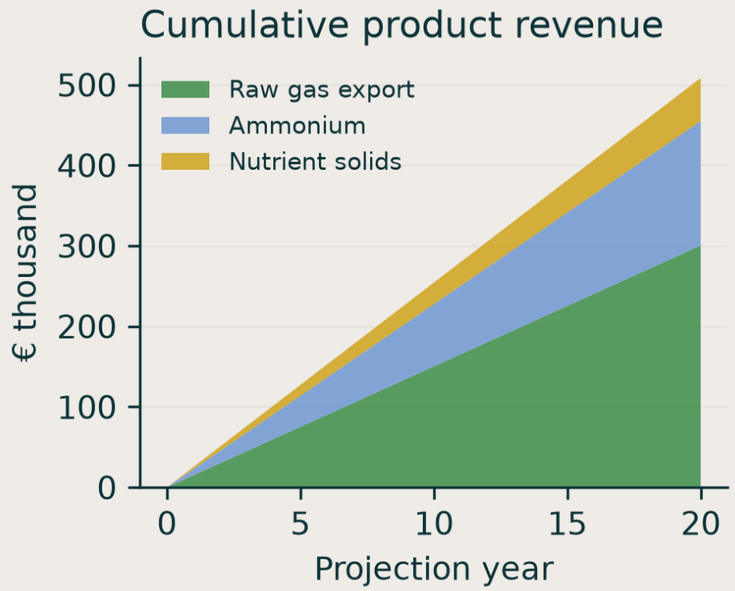
Product revenue, partial annual budget, capital subsidy and candidate overlays

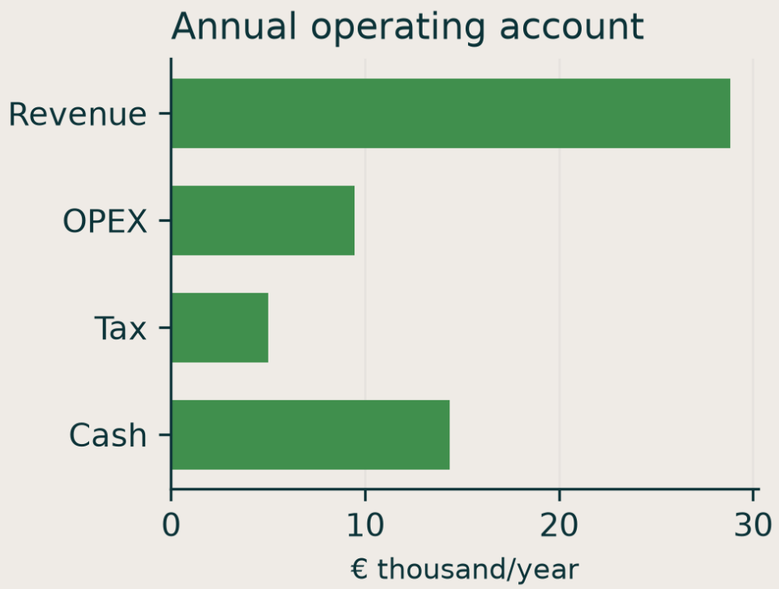
Scenario: protein_bypass rank 3; ecb05c86f525455cb46f2e5bef5cf1a8; 847.8 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

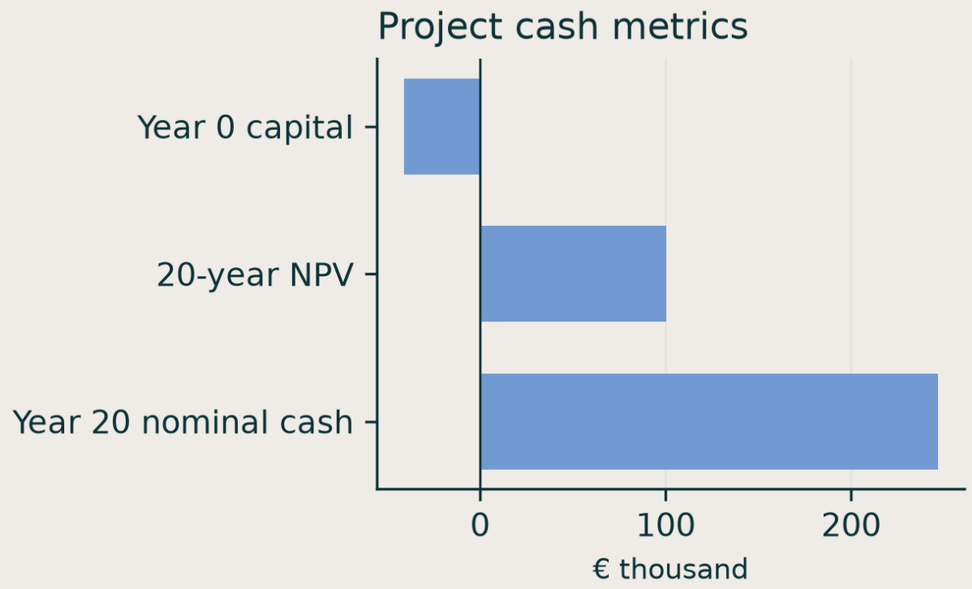
20-year financial projection

Protein X · Biogas ✓ · N ✓ · P ✓ · K X









Constant-account cash flow and cumulative product revenue over 20 years

Scenario: protein_bypass rank 3; ecb05c86f525455cb46f2e5bef5cf1a8; 847.8 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Balance and evidence

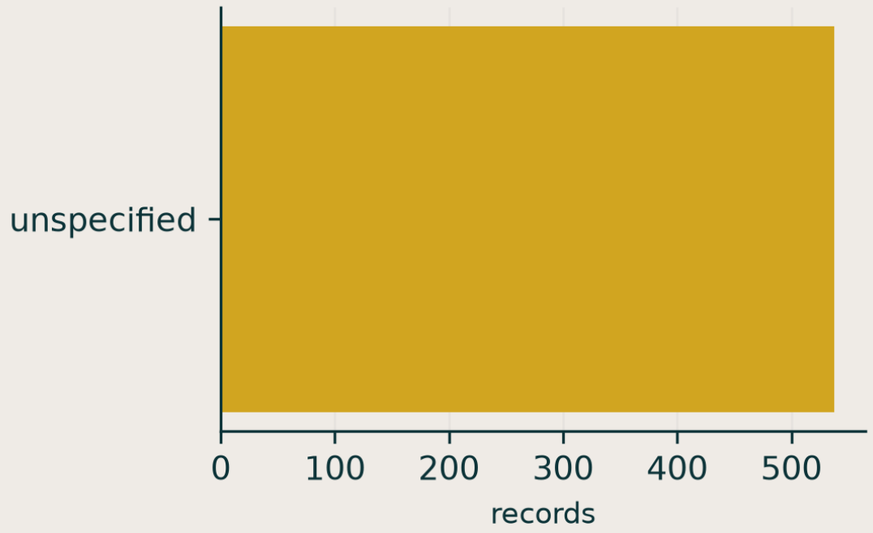
Protein X · Biogas ✓ · N ✓ · P ✓ · K X

Element closure

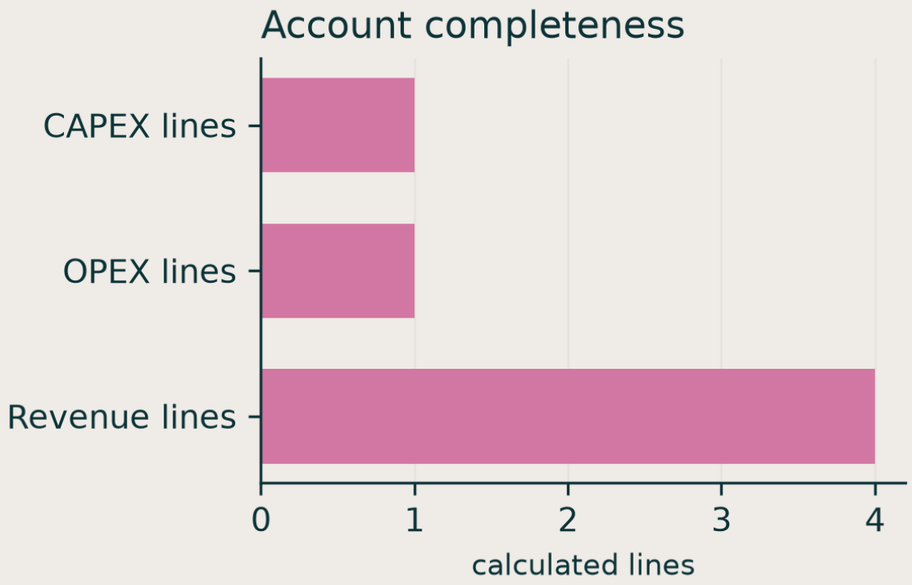
C	0 kg/year
N	0 kg/year
P	1.2e-13 kg/year
K	0 kg/year

Numerical residuals

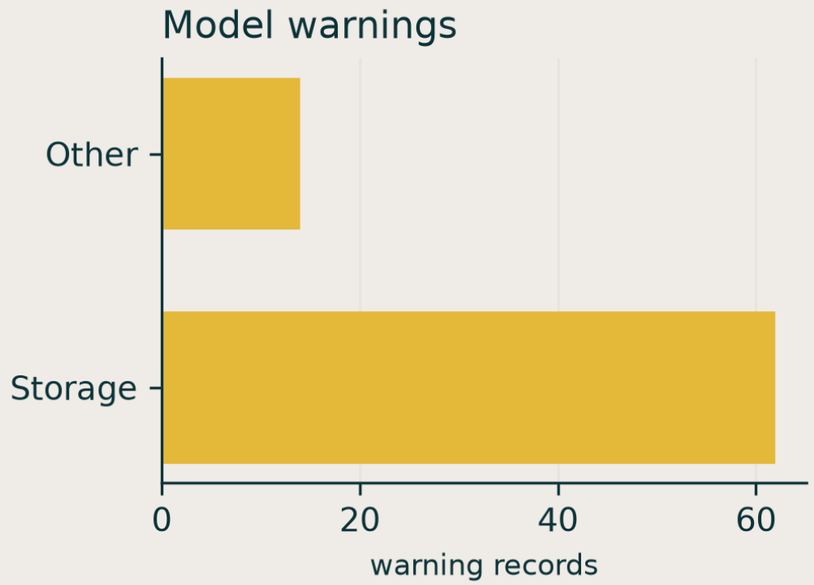
Parameter evidence



Account completeness



Model warnings



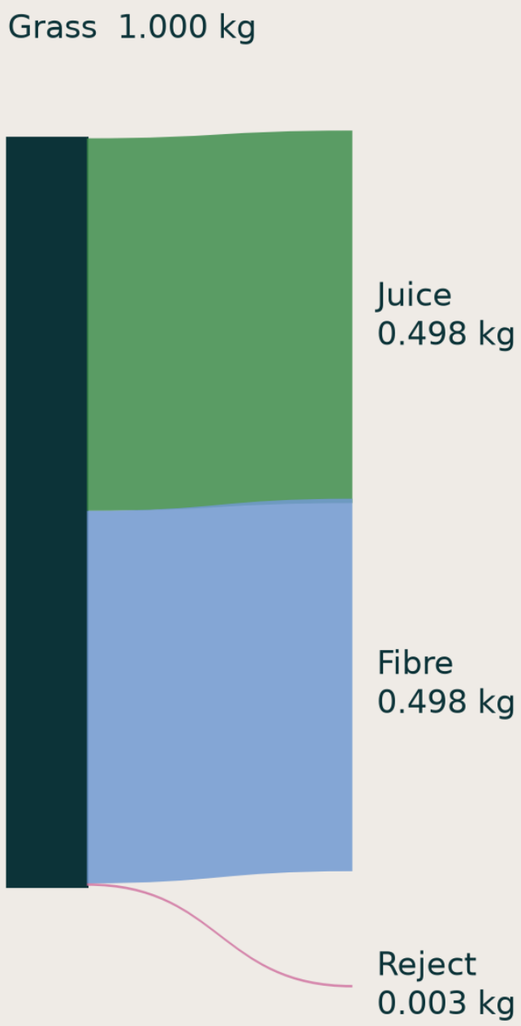
Numerical closure, evidence counts, calculated account lines and warnings

Scenario: protein_bypass rank 3; ecb05c86f525455cb46f2e5bef5cf1a8; 847.8 kg/h; 8760 h/year
Units: kg/year; records
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

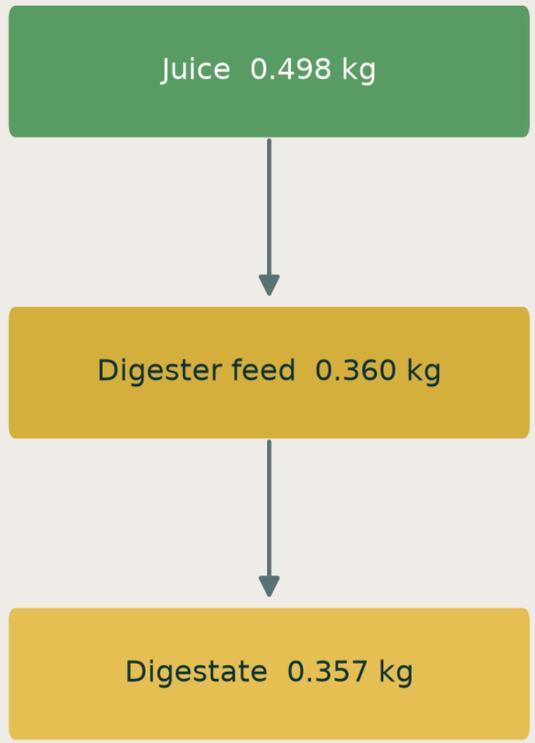
Grass to products

Protein X · Biogas ✓ · N ✓ · P X · K X

Grass split

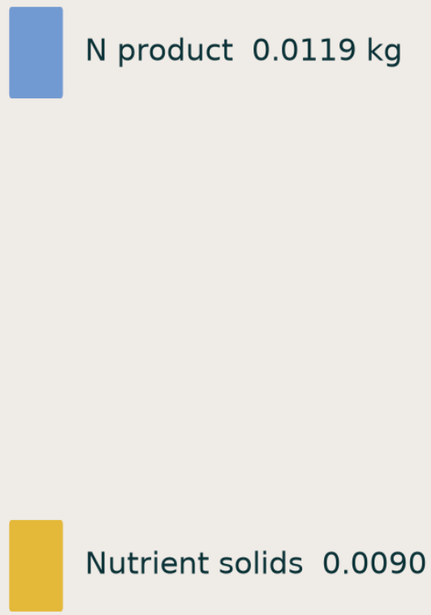


Juice pathway



Selected intermediate streams

Recovered outputs



Raw gas 0.0021 Nm³

Raw export 0.0021 Nm³

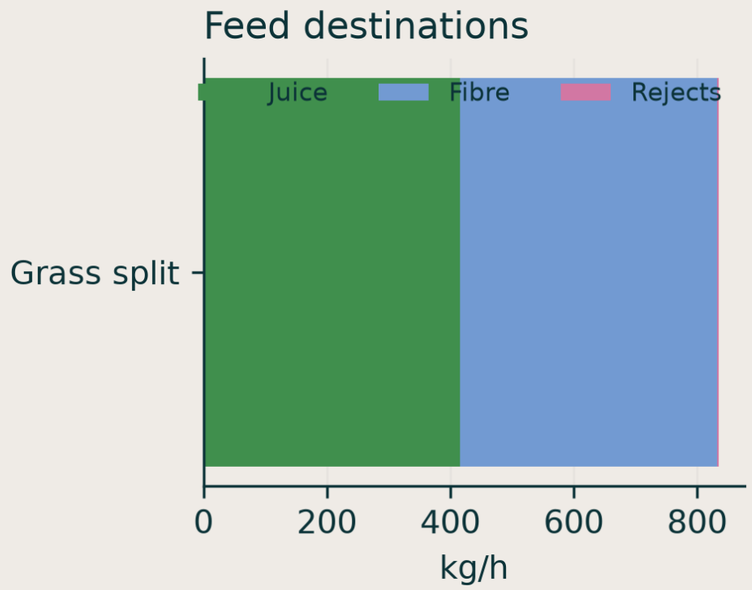
per kg wet grass

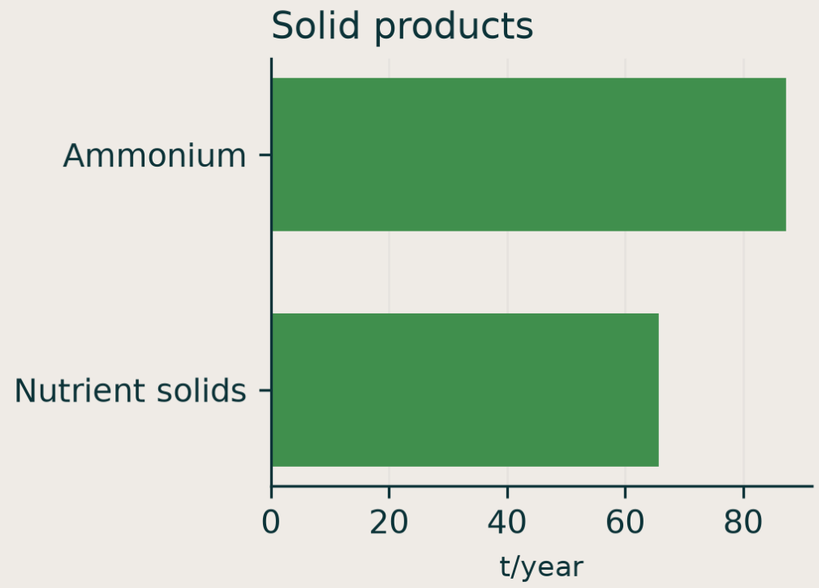
Mass-proportional grass split plus modeled downstream stream/product map

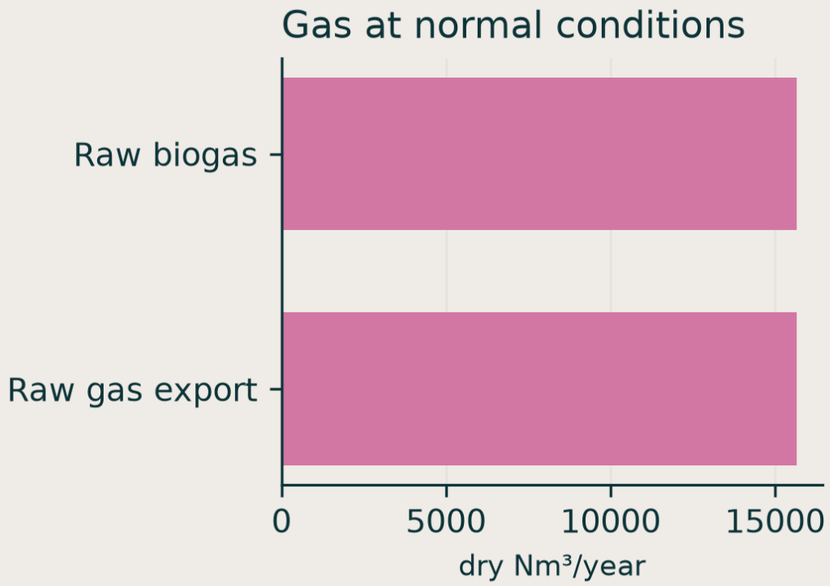
Scenario: protein_bypass rank 4; c42790274f2e484c9ca2cbe93c863b0e; 834.8 kg/h; 8760 h/year
Units: kg/kg wet grass; dry Nm³/kg wet grass
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

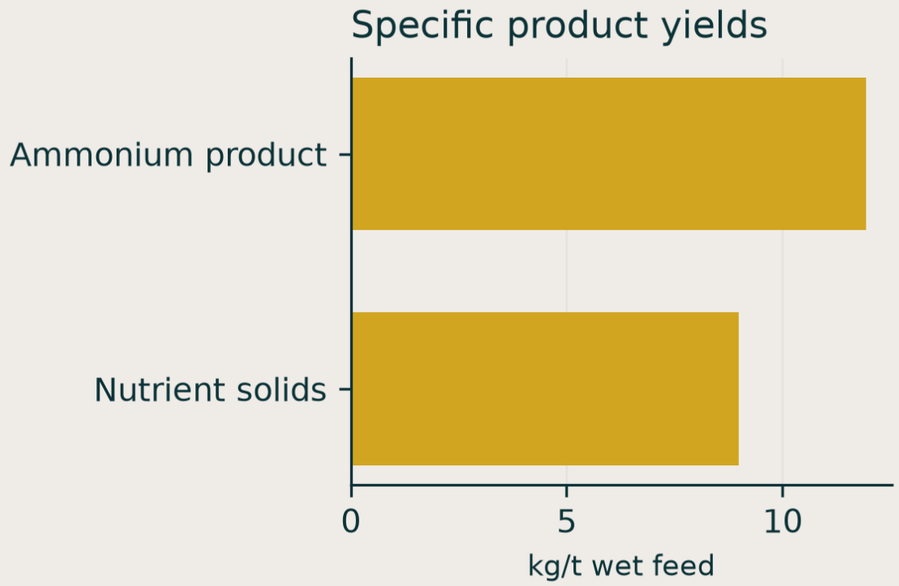
Material and products

Protein X · Biogas ✓ · N ✓ · P X · K X







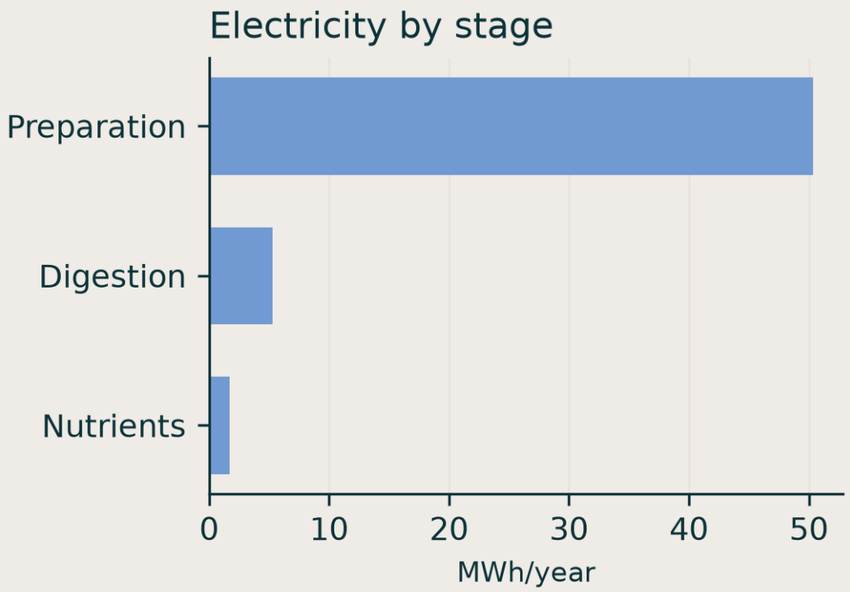


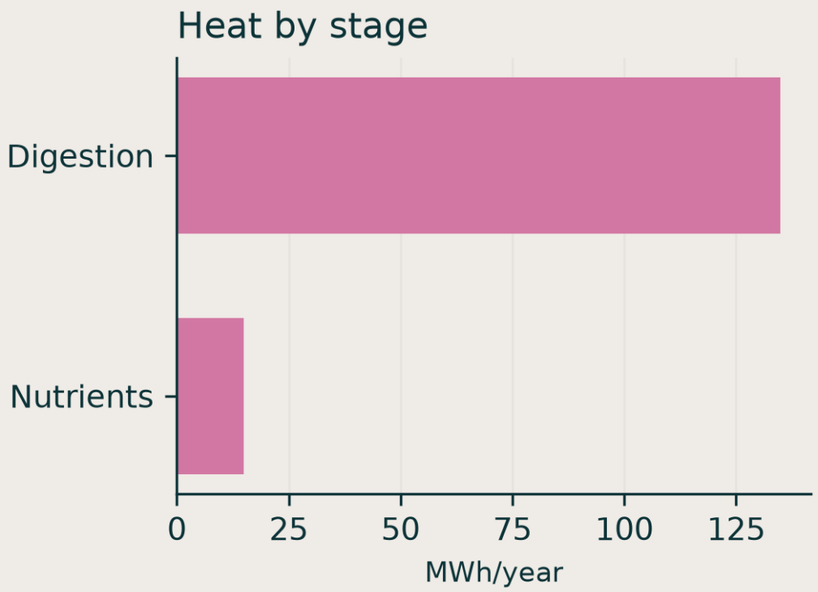
Front-end split, solid products, dry gas volumes and specific yields

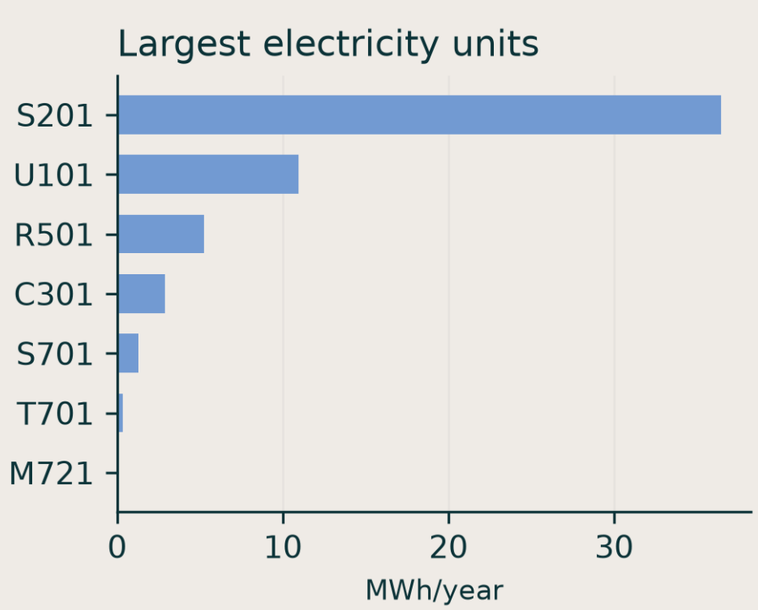
Scenario: protein_bypass rank 4; c42790274f2e484c9ca2cbe93c863b0e; 834.8 kg/h; 8760 h/year
Units: kg/h; t/year; dry Nm³/year; kg/t feed
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

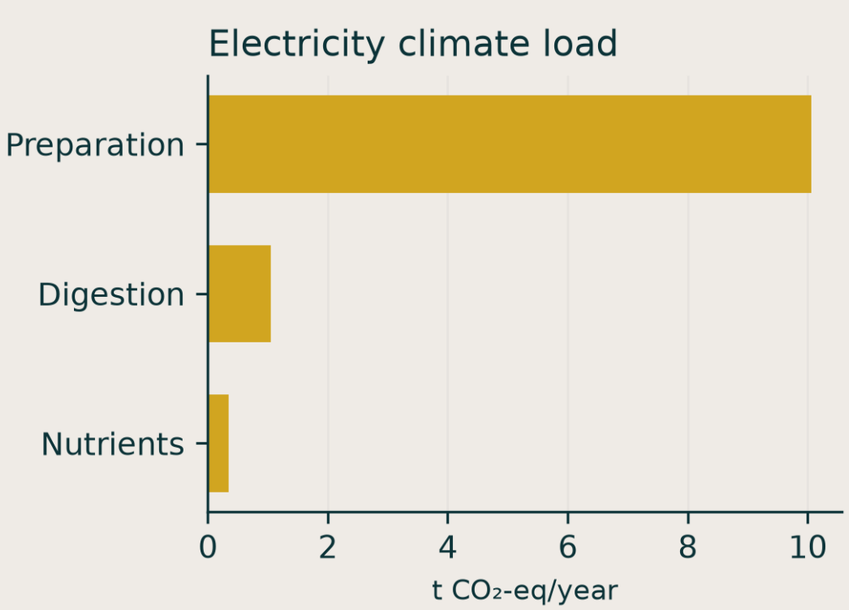
Energy and climate hotspots

Protein X · Biogas ✓ · N ✓ · P X · K X







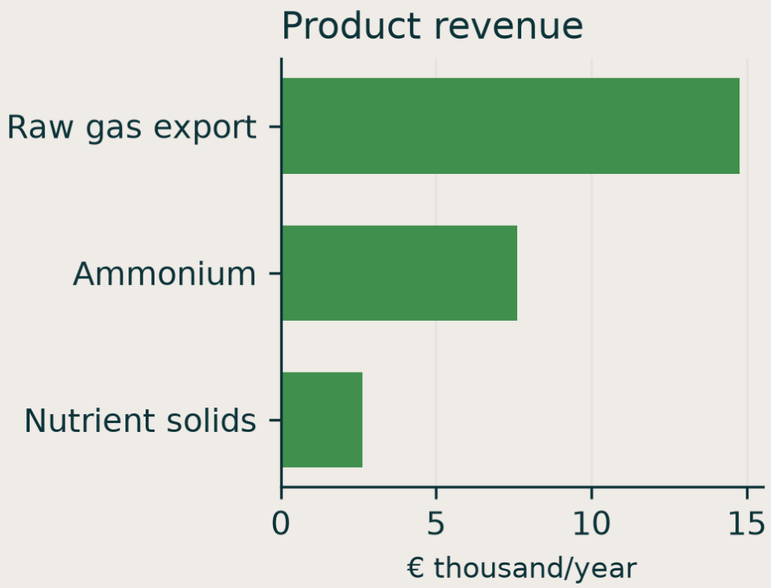


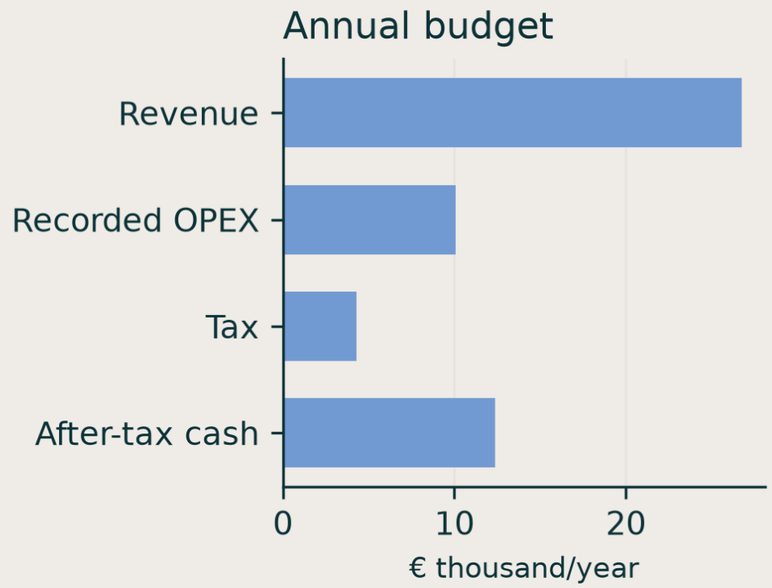
Stage and unit utility hotspots and included climate load

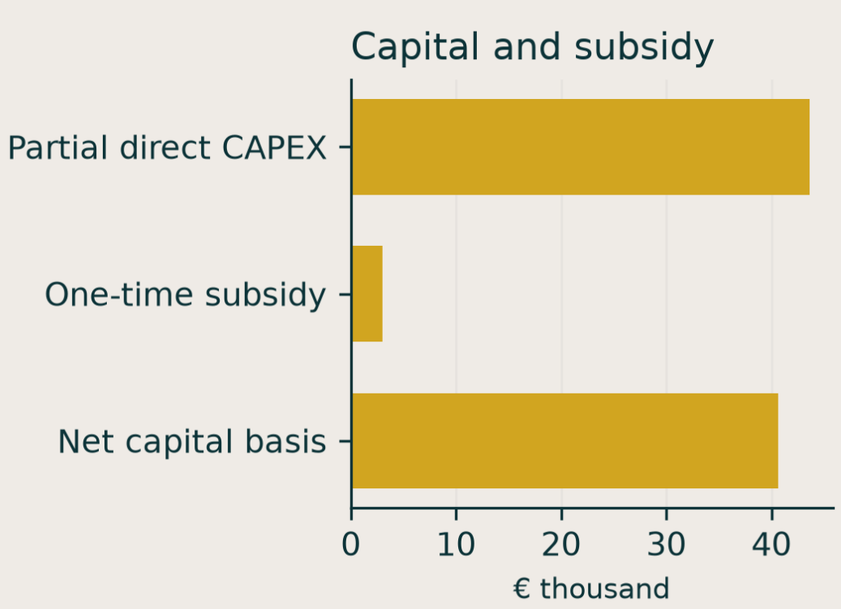
Scenario: protein_bypass rank 4; c42790274f2e484c9ca2cbe93c863b0e; 834.8 kg/h; 8760 h/year
Units: MWh/year; t CO_2-eq/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

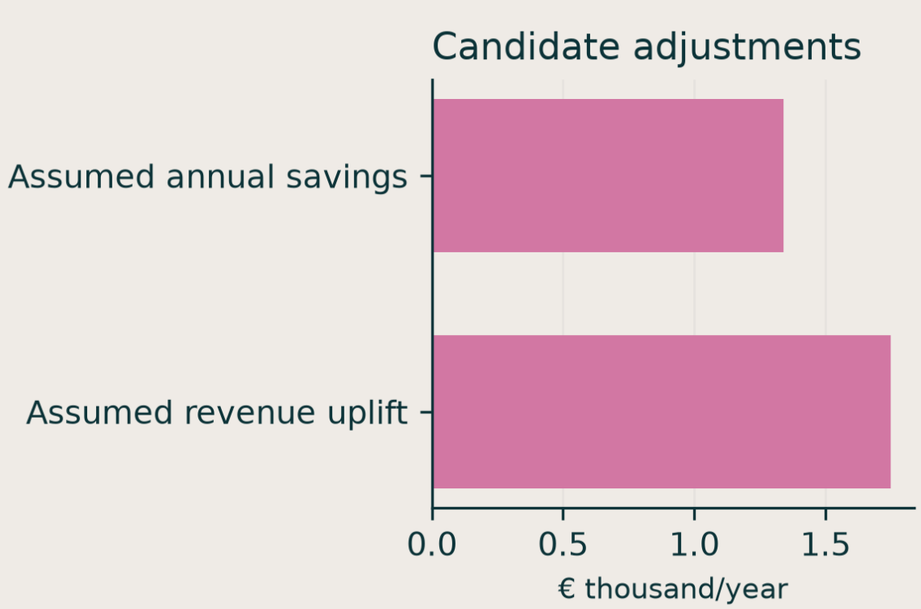
Economic accounts

Protein X · Biogas ✓ · N ✓ · P X · K X







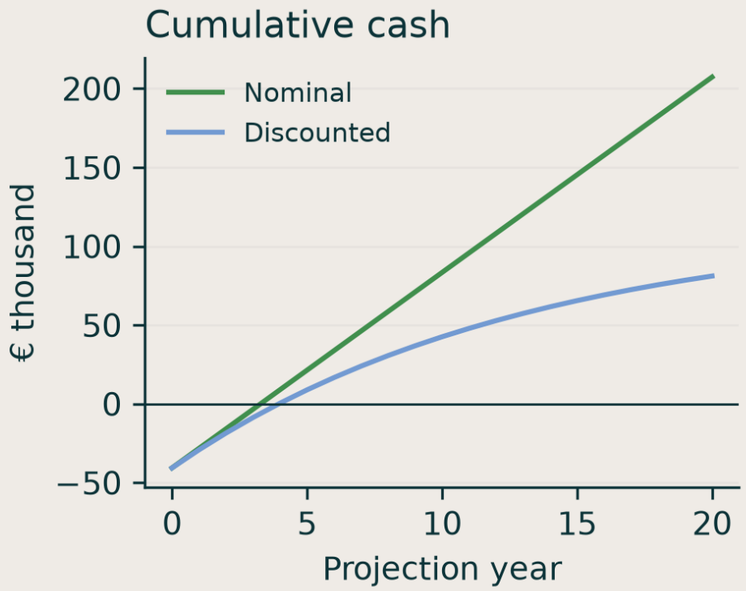


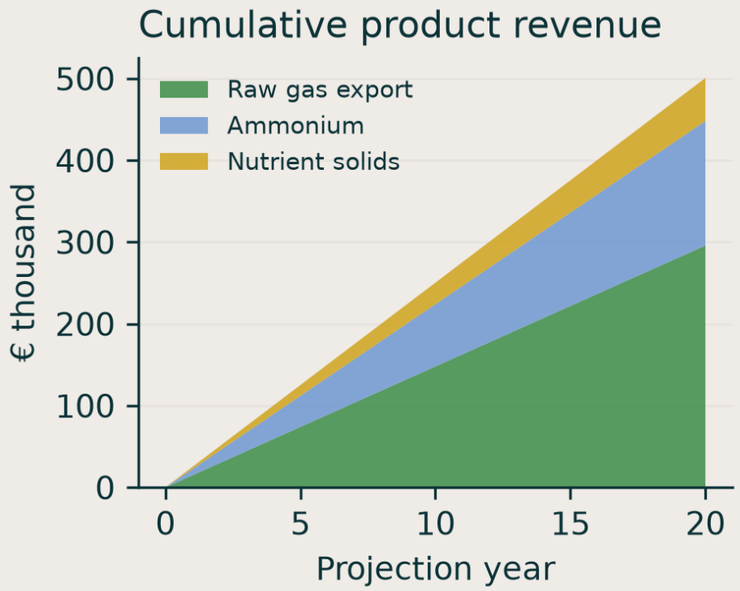
Product revenue, partial annual budget, capital subsidy and candidate overlays

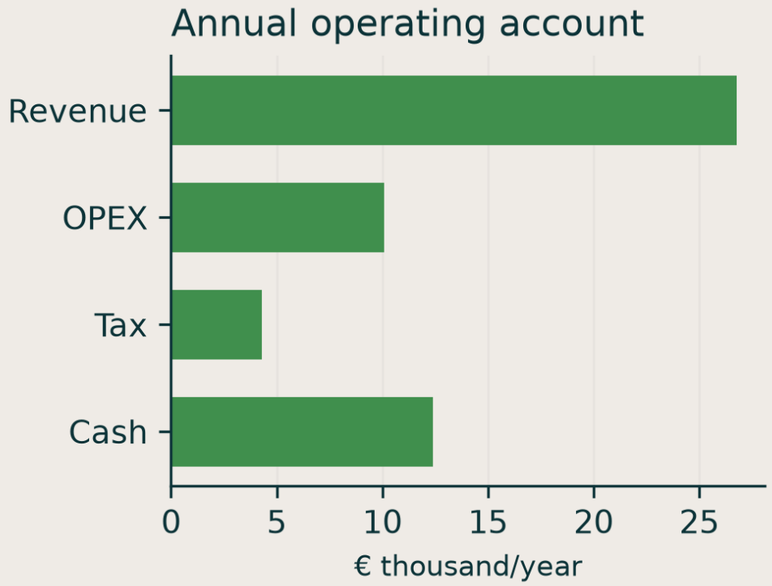
Scenario: protein_bypass rank 4; c42790274f2e484c9ca2cbe93c863b0e; 834.8 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

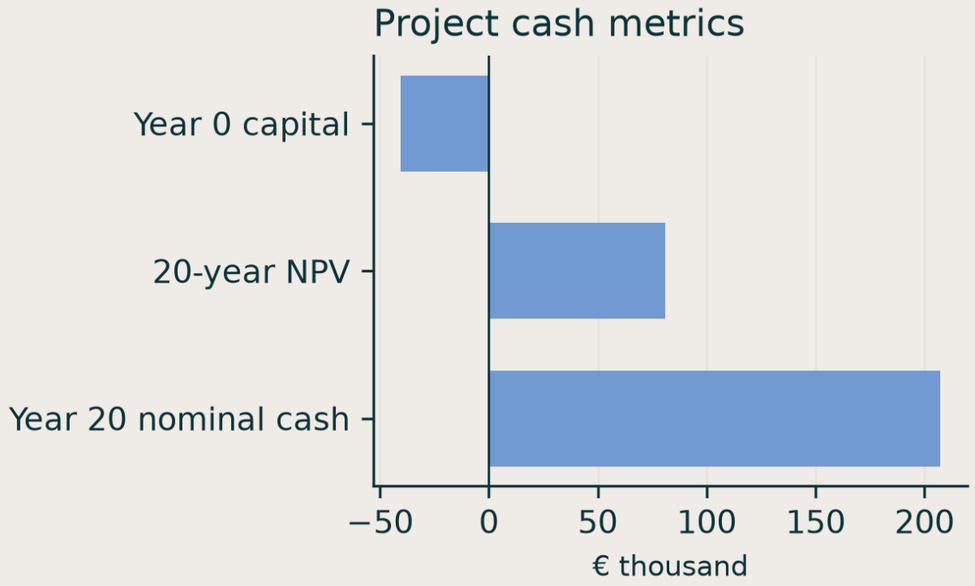
20-year financial projection

Protein X · Biogas ✓ · N ✓ · P X · K X









Constant-account cash flow and cumulative product revenue over 20 years

Scenario: protein_bypass rank 4; c42790274f2e484c9ca2cbe93c863b0e; 834.8 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Balance and evidence

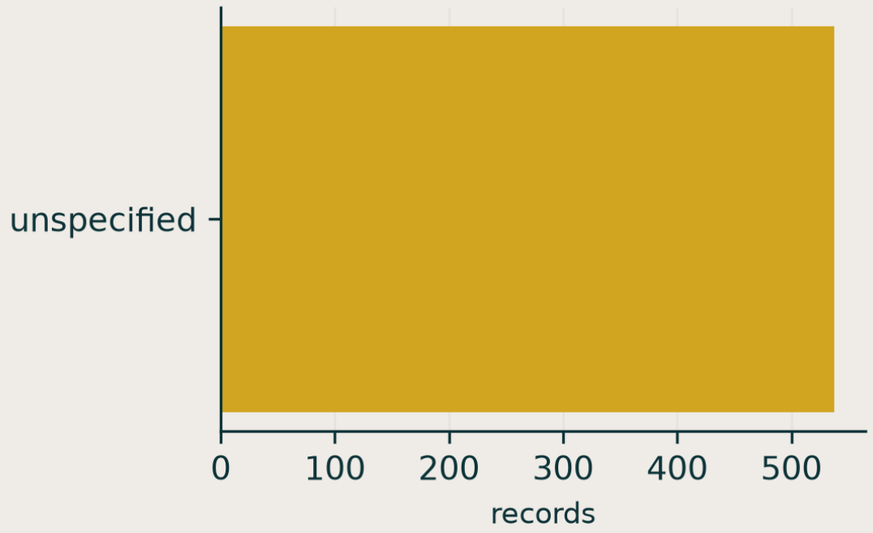
Protein X · Biogas ✓ · N ✓ · P X · K X

Element closure

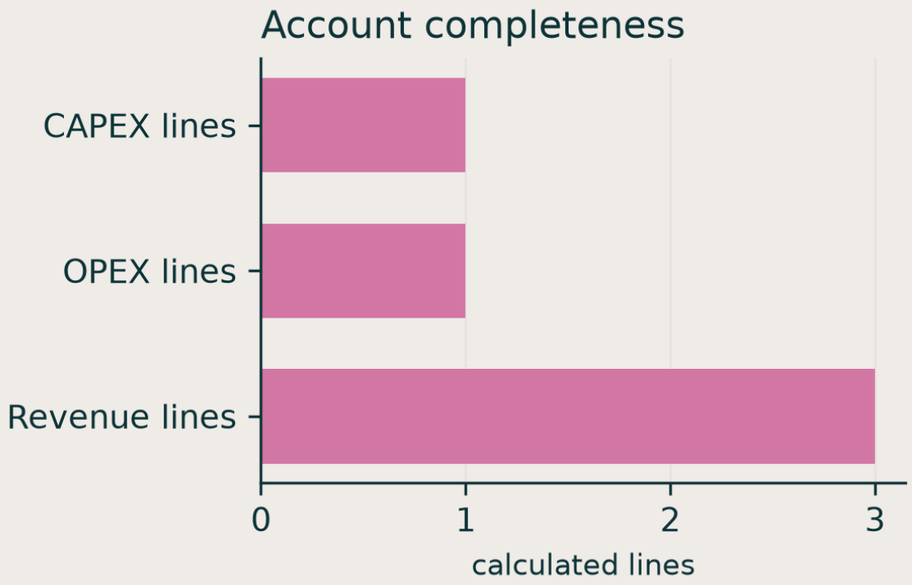
C	0 kg/year
N	0 kg/year
P	0 kg/year
K	4.9e-13 kg/year

Numerical residuals

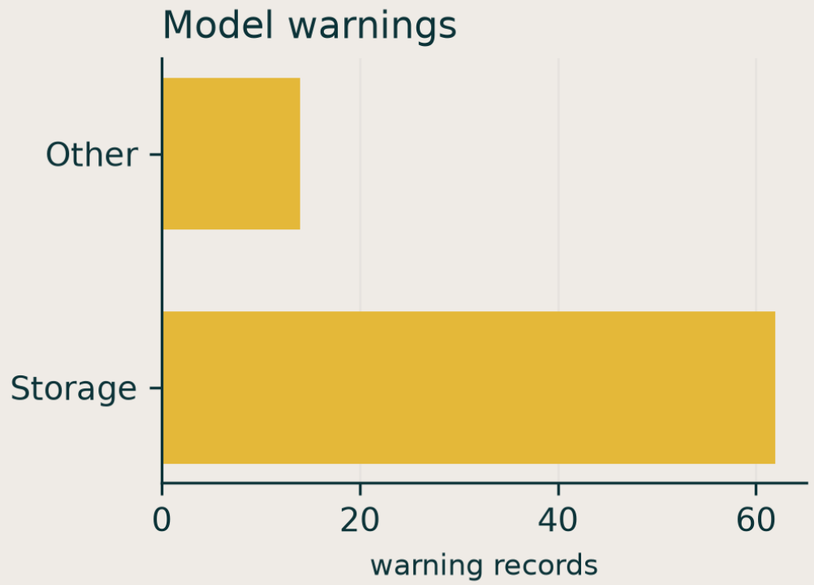
Parameter evidence



Account completeness



Model warnings



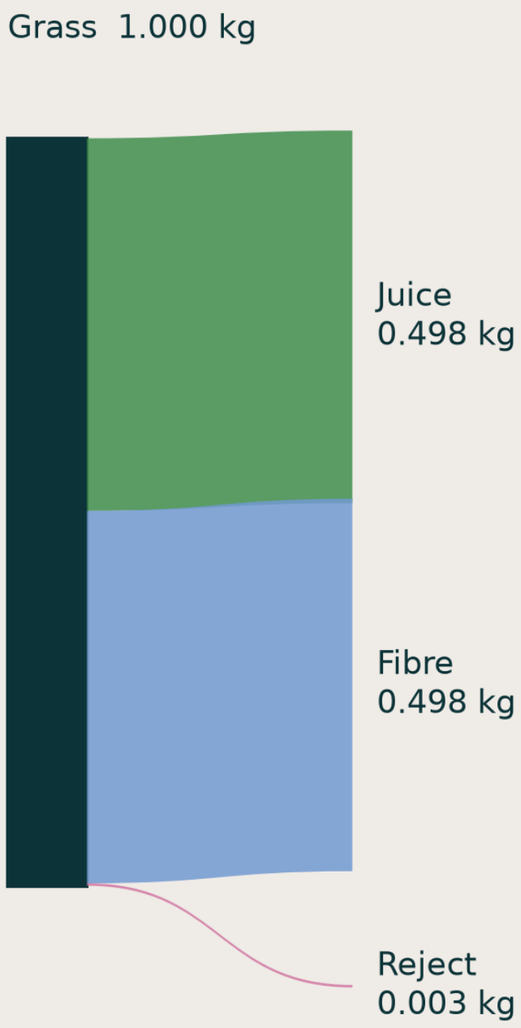
Numerical closure, evidence counts, calculated account lines and warnings

Scenario: protein_bypass rank 4; c42790274f2e484c9ca2cbe93c863b0e; 834.8 kg/h; 8760 h/year
Units: kg/year; records
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

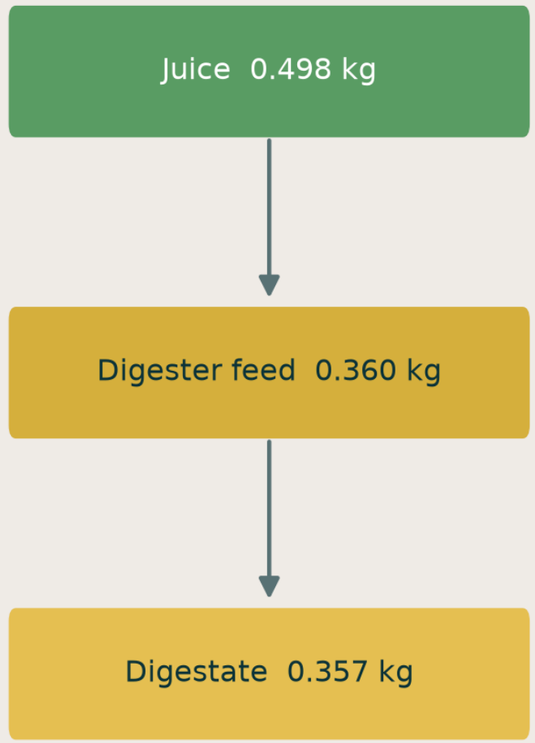
Grass to products

Protein X · Biogas ✓ · N ✓ · P ✓ · K X

Grass split



Juice pathway



Selected intermediate streams

Recovered outputs

Raw gas 0.0021 Nm³

Raw export 0.0021 Nm³

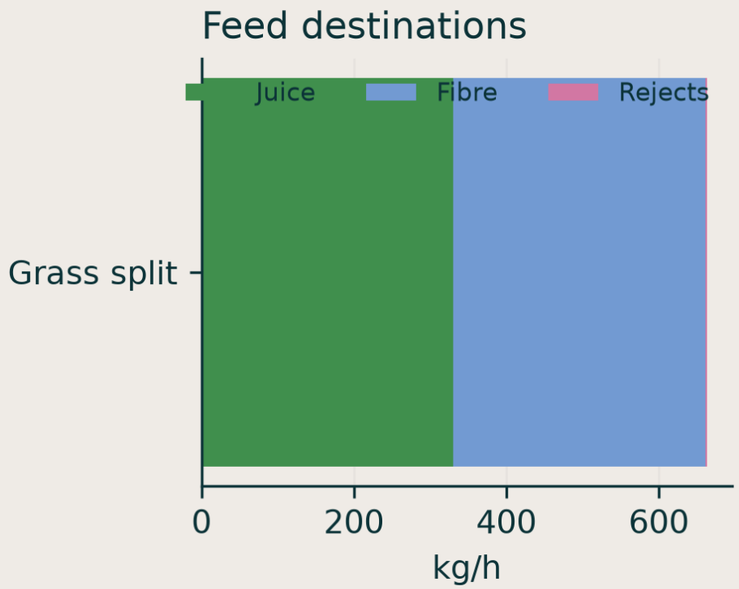
per kg wet grass

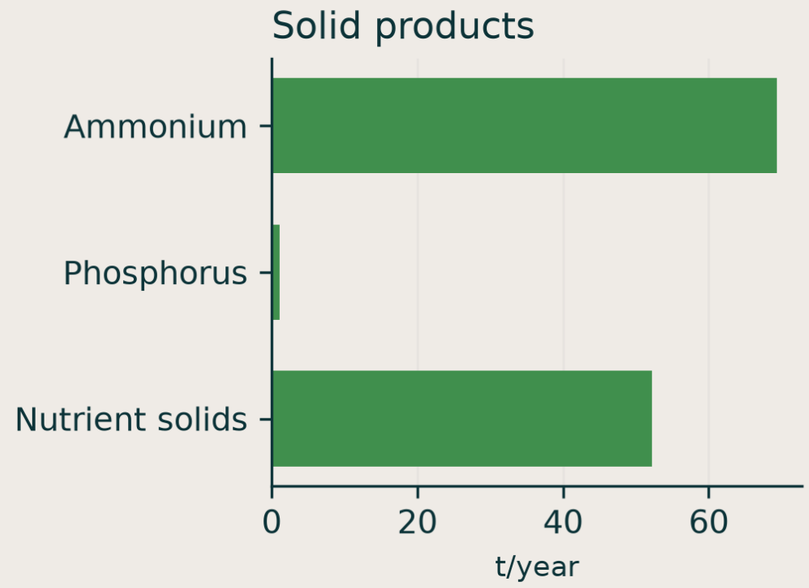
Mass-proportional grass split plus modeled downstream stream/product map

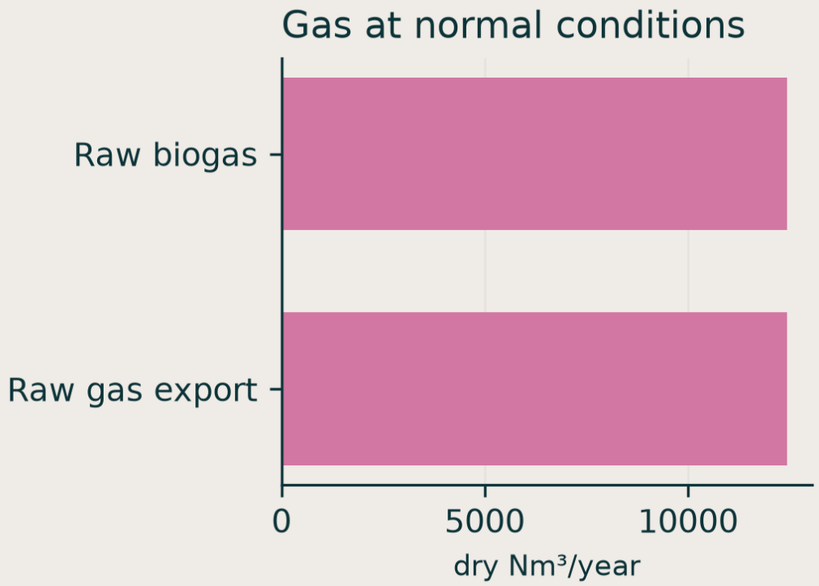
Scenario: protein_bypass rank 5; 41e61fe4fcc44e438b85bff810789f54; 663.0 kg/h; 8760 h/year
Units: kg/kg wet grass; dry Nm³/kg wet grass
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

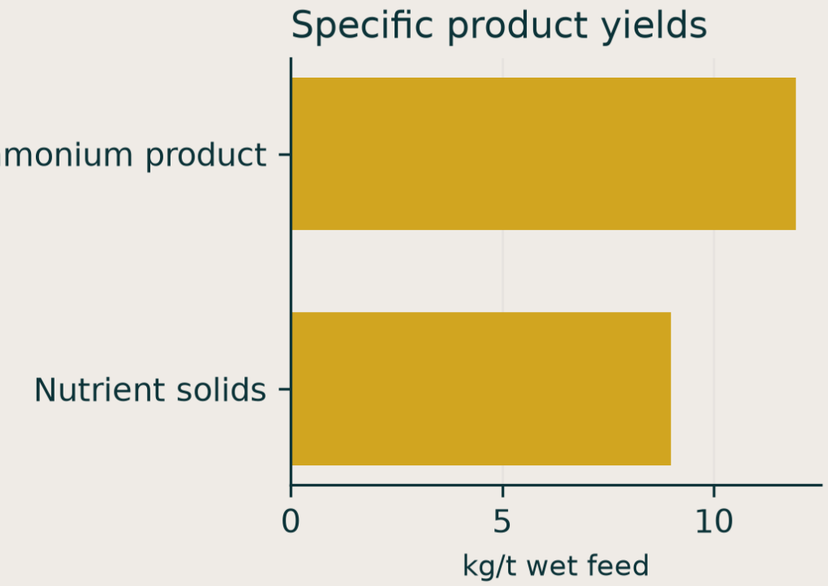
Material and products

Protein X · Biogas ✓ · N ✓ · P ✓ · K X







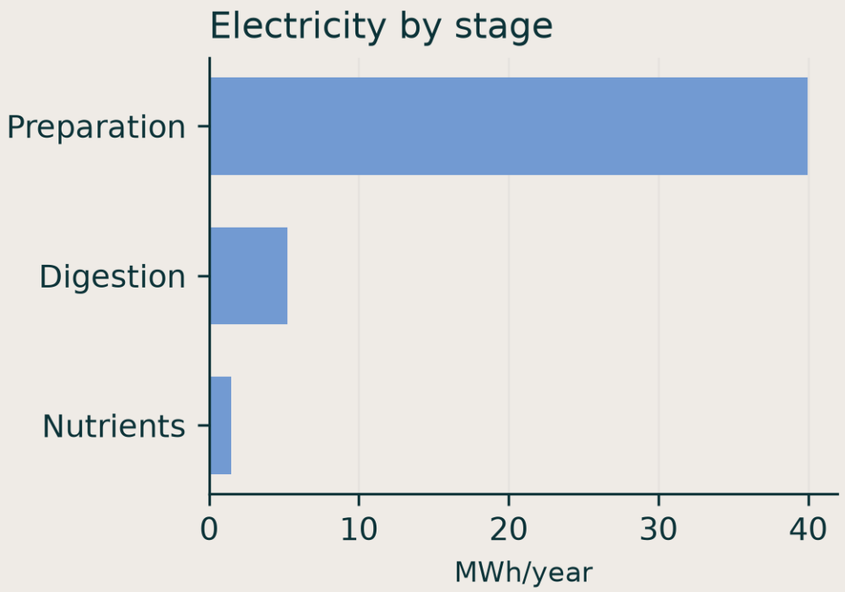


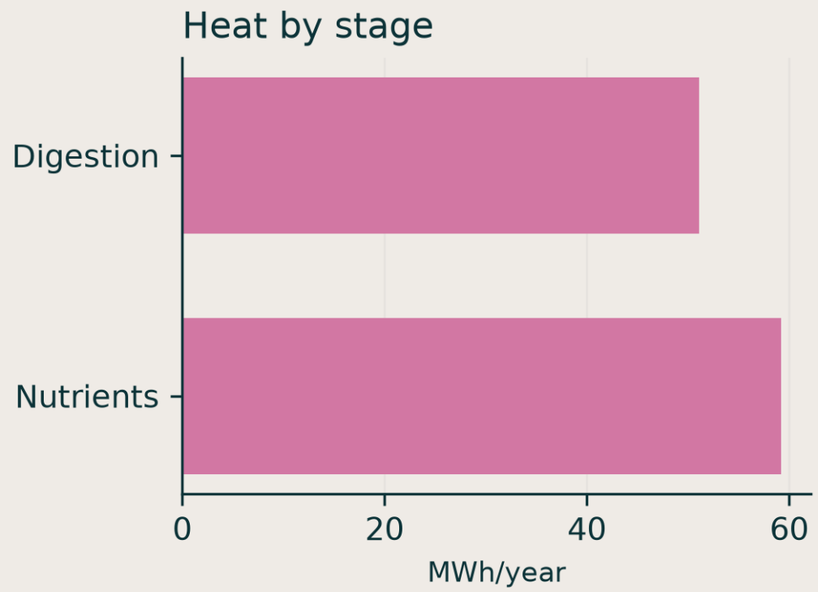
Front-end split, solid products, dry gas volumes and specific yields

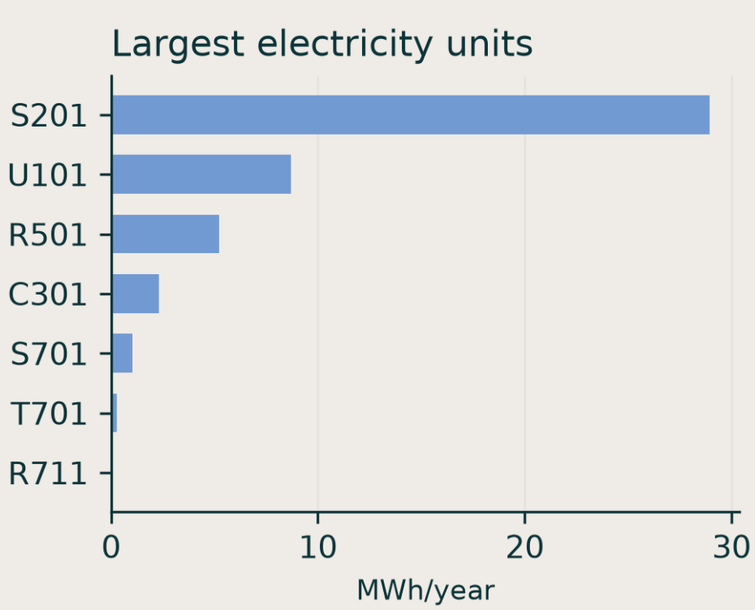
Scenario: protein_bypass rank 5; 41e61fe4fcc44e438b85bff810789f54; 663.0 kg/h; 8760 h/year
Units: kg/h; t/year; dry Nm³/year; kg/t feed
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

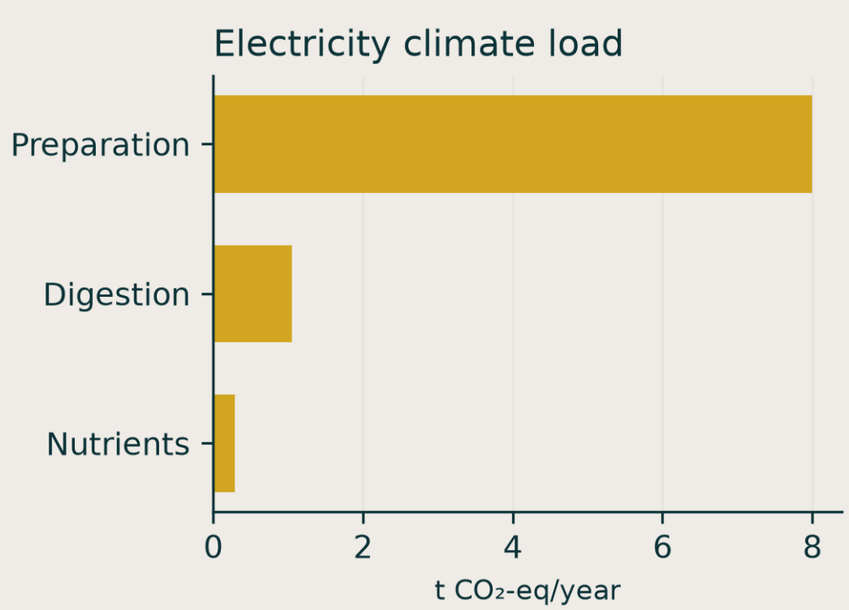
Energy and climate hotspots

Protein X · Biogas ✓ · N ✓ · P ✓ · K X







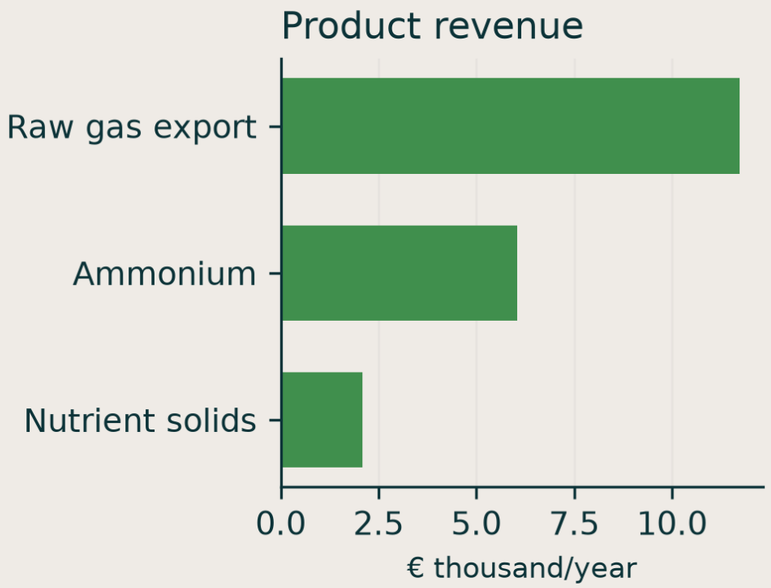


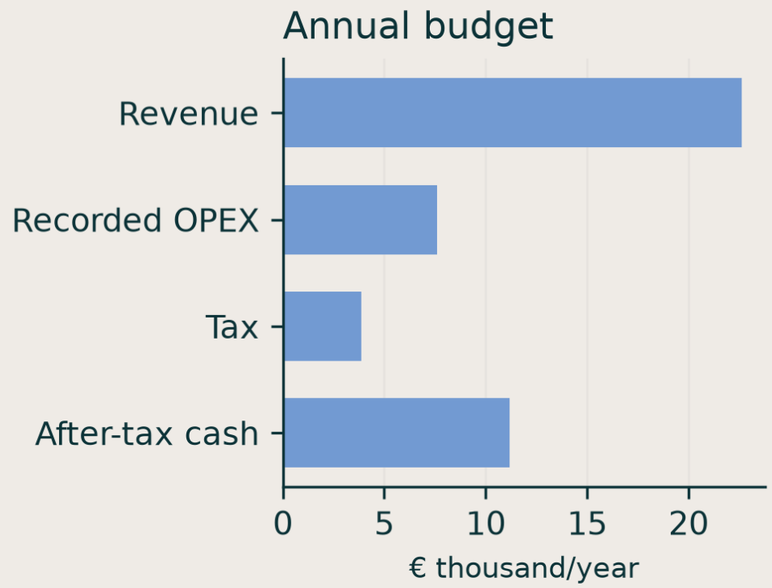
Stage and unit utility hotspots and included climate load

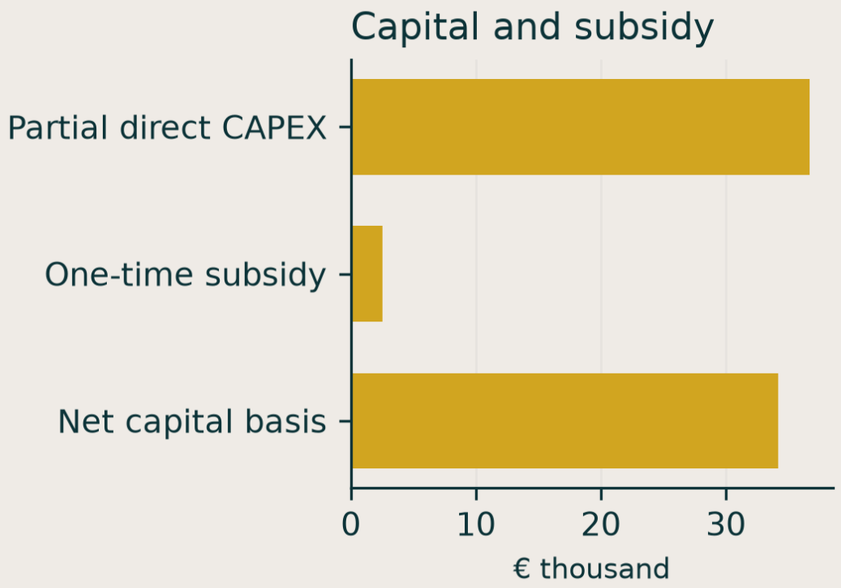
Scenario: protein_bypass rank 5; 41e61fe4fcc44e438b85bff810789f54; 663.0 kg/h; 8760 h/year
Units: MWh/year; t CO_2-eq/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

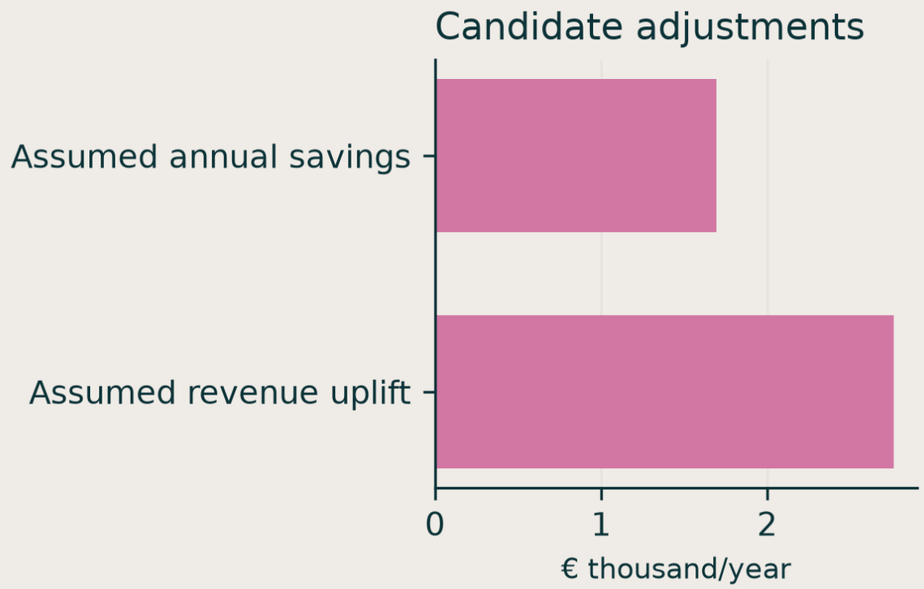
Economic accounts

Protein X · Biogas ✓ · N ✓ · P ✓ · K X







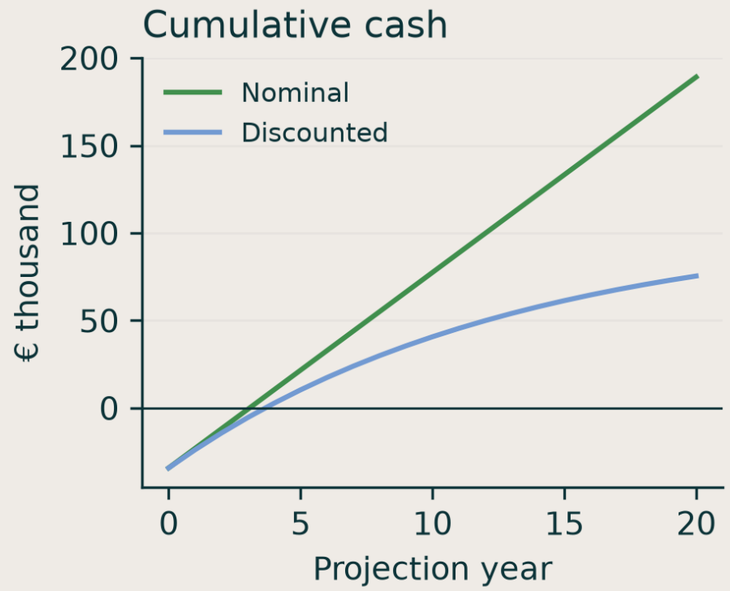


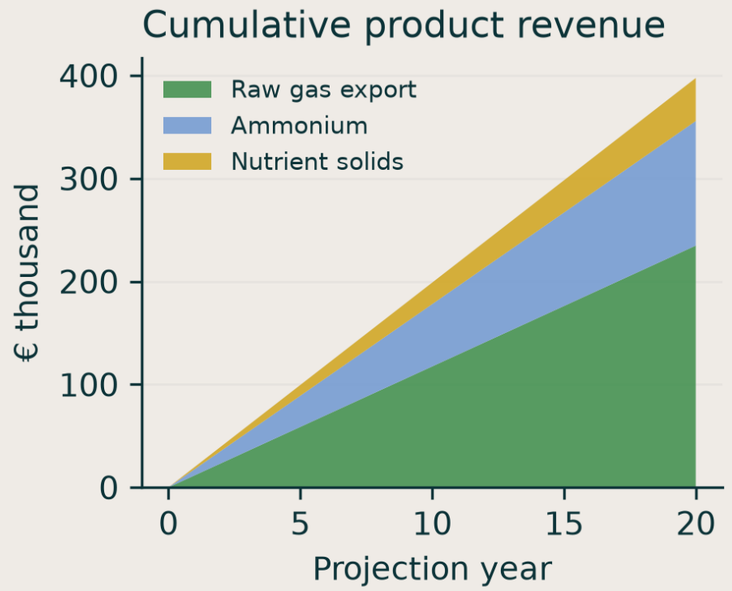
Product revenue, partial annual budget, capital subsidy and candidate overlays

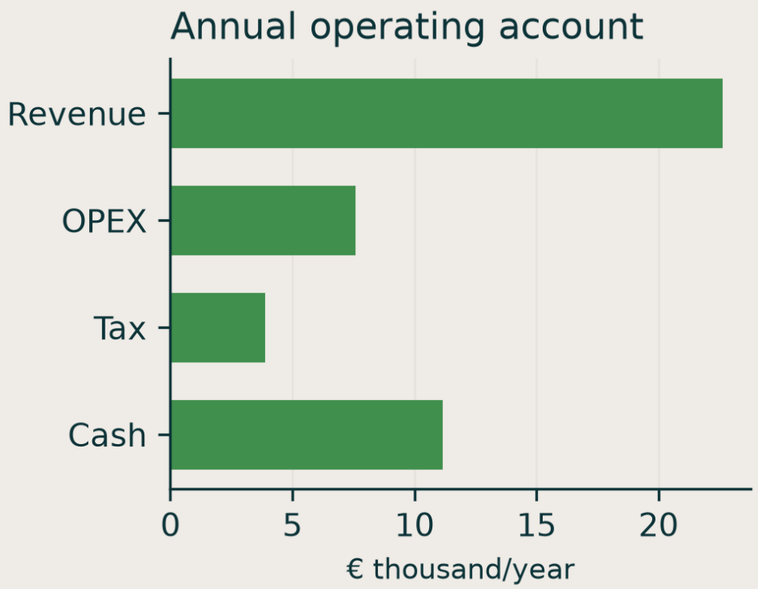
Scenario: protein_bypass rank 5; 41e61fe4fcc44e438b85bff810789f54; 663.0 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

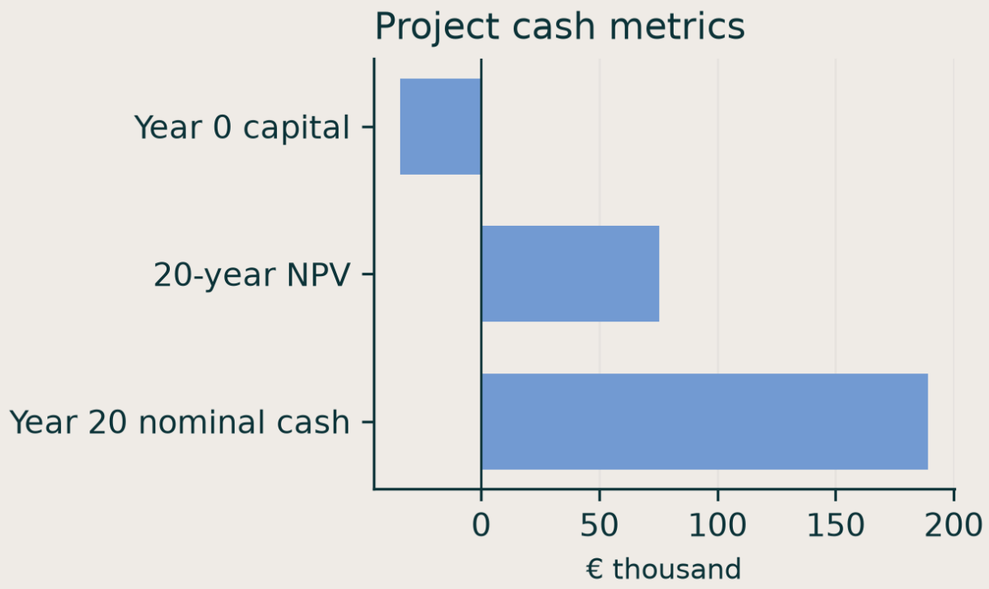
20-year financial projection

Protein X · Biogas ✓ · N ✓ · P ✓ · K X









Constant-account cash flow and cumulative product revenue over 20 years

Scenario: protein_bypass rank 5; 41e61fe4fcc44e438b85bff810789f54; 663.0 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Balance and evidence

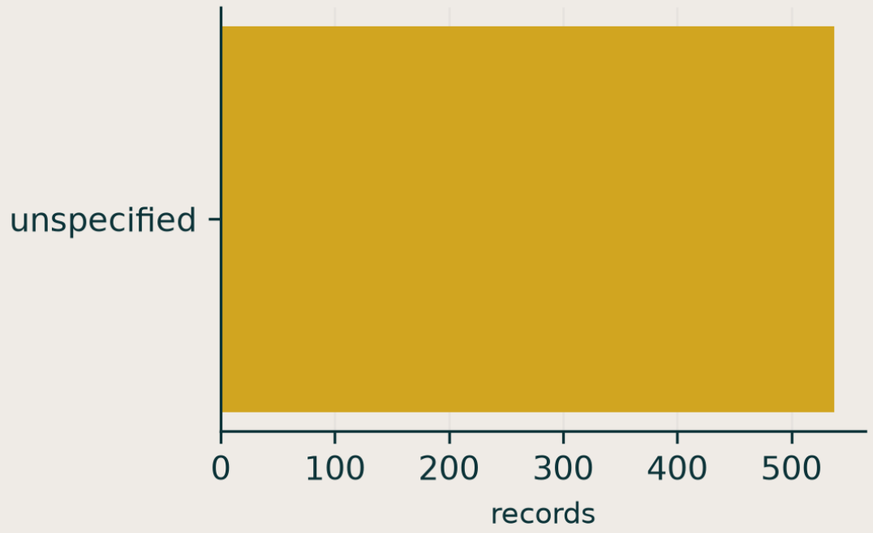
Protein X · Biogas ✓ · N ✓ · P ✓ · K X

Element closure

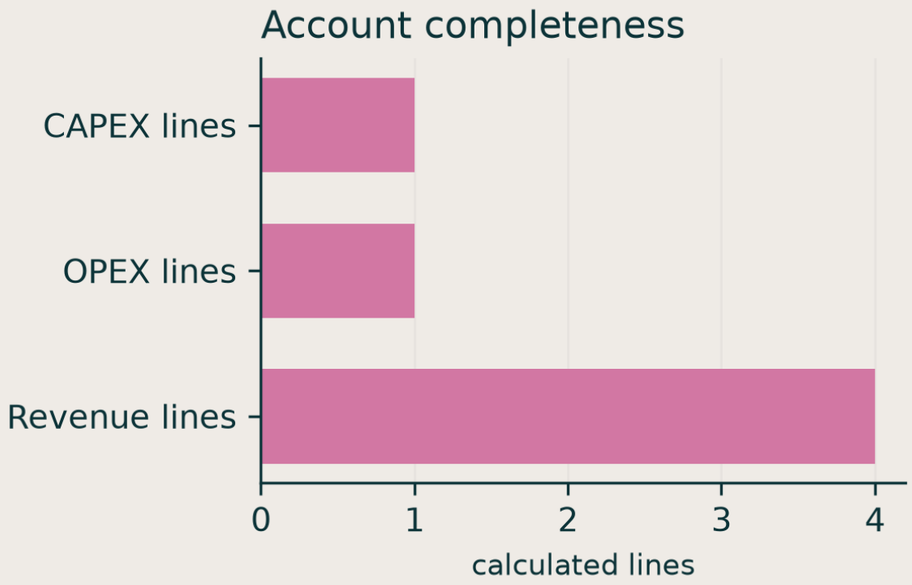
C	0 kg/year
N	0 kg/year
P	0 kg/year
K	0 kg/year

Numerical residuals

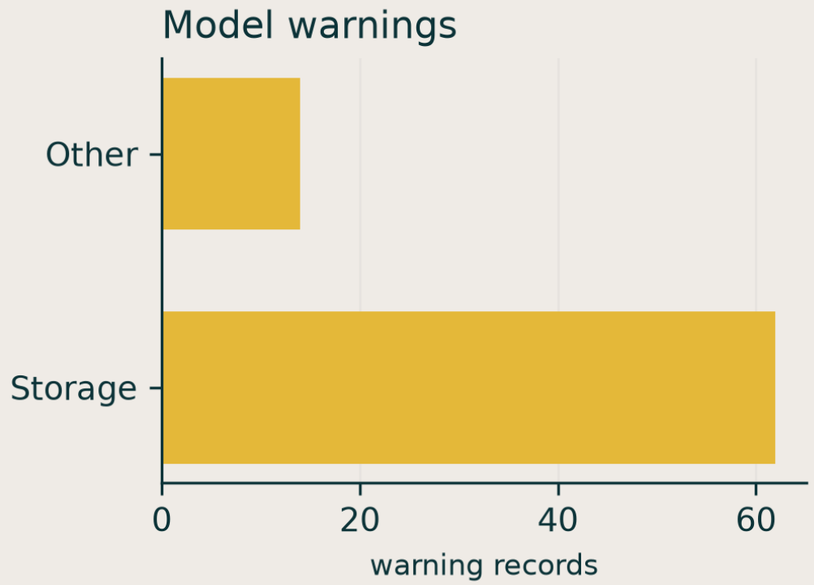
Parameter evidence



Account completeness



Model warnings



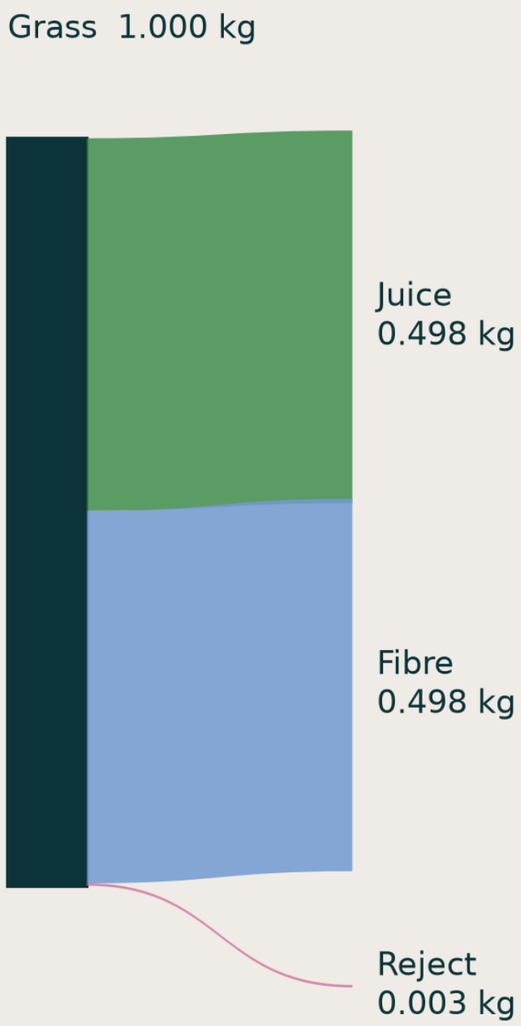
Numerical closure, evidence counts, calculated account lines and warnings

Scenario: protein_bypass rank 5; 41e61fe4fcc44e438b85bff810789f54; 663.0 kg/h; 8760 h/year
Units: kg/year; records
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

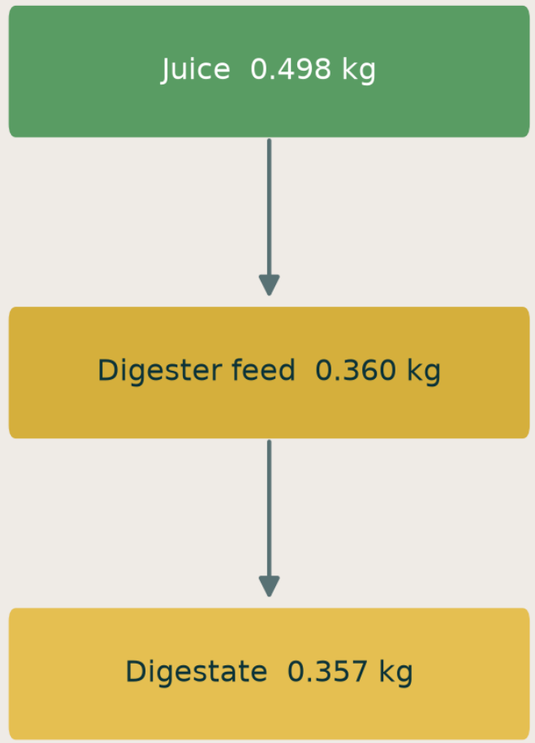
Grass to products

Protein X · Biogas ✓ · N ✓ · P X · K X

Grass split

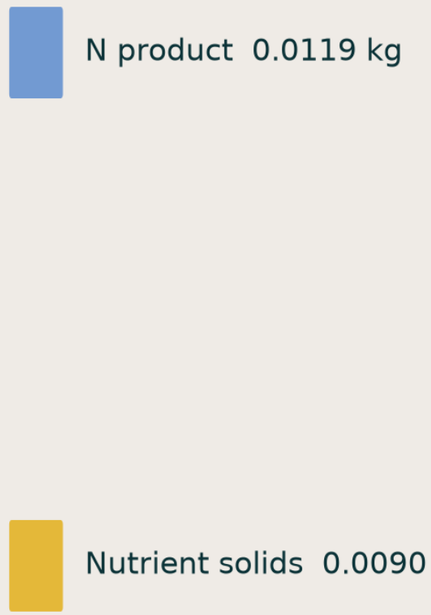


Juice pathway



Selected intermediate streams

Recovered outputs



Raw gas 0.0021 Nm³

Raw export 0.0021 Nm³

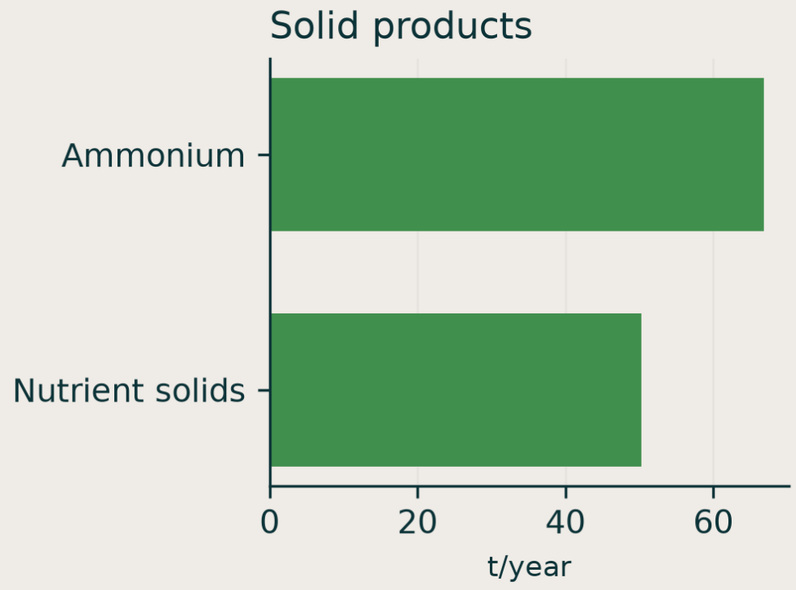
per kg wet grass

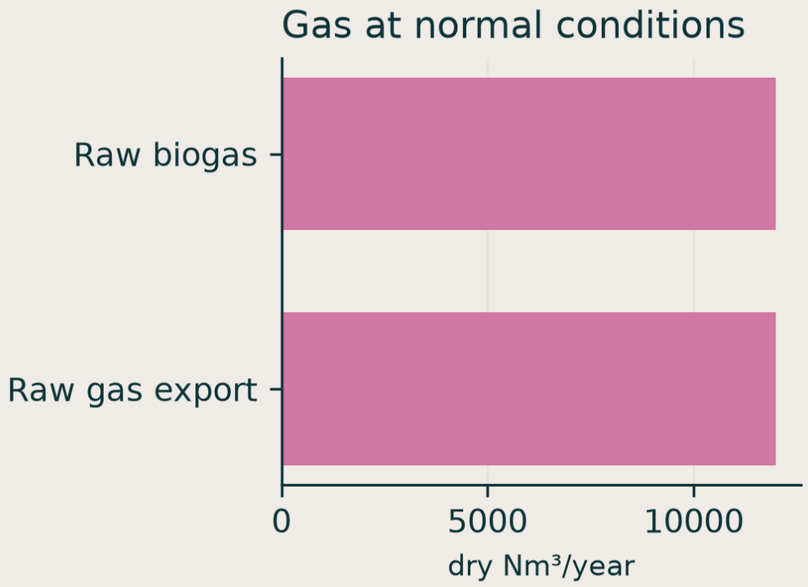
Mass-proportional grass split plus modeled downstream stream/product map

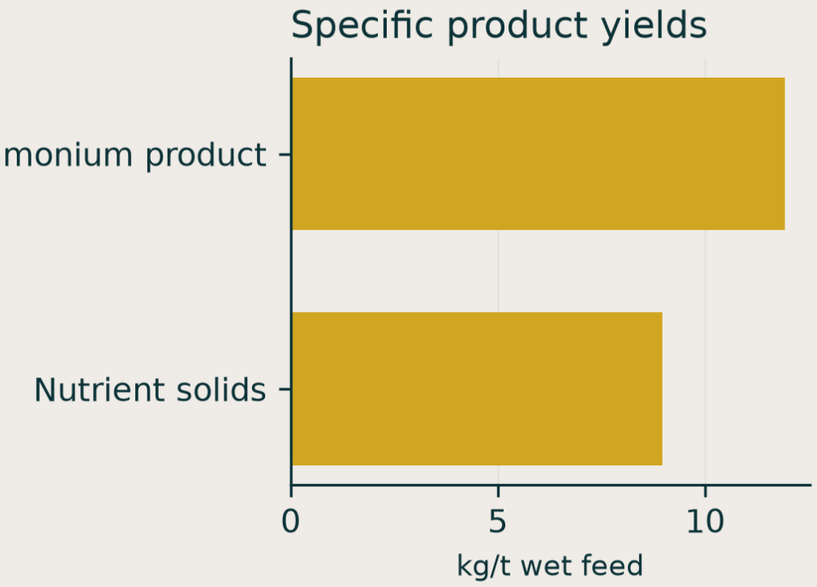
Scenario: protein_bypass rank 6; 999a369dc9d444c0be7a245352a35db1; 639.6 kg/h; 8760 h/year
Units: kg/kg wet grass; dry Nm³/kg wet grass
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Material and products

Protein X · Biogas ✓ · N ✓ · P X · K X





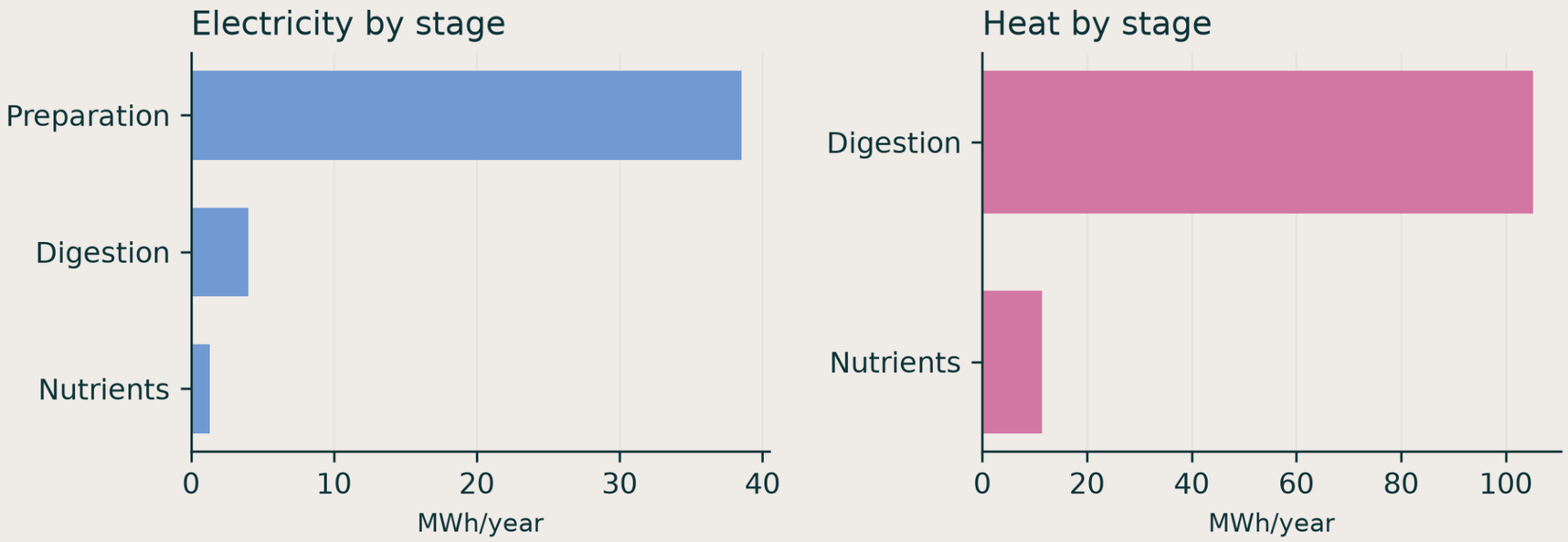


Front-end split, solid products, dry gas volumes and specific yields

Scenario: protein_bypass rank 6; 999a369dc9d444c0be7a245352a35db1; 639.6 kg/h; 8760 h/year
Units: kg/h; t/year; dry Nm³/year; kg/t feed
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

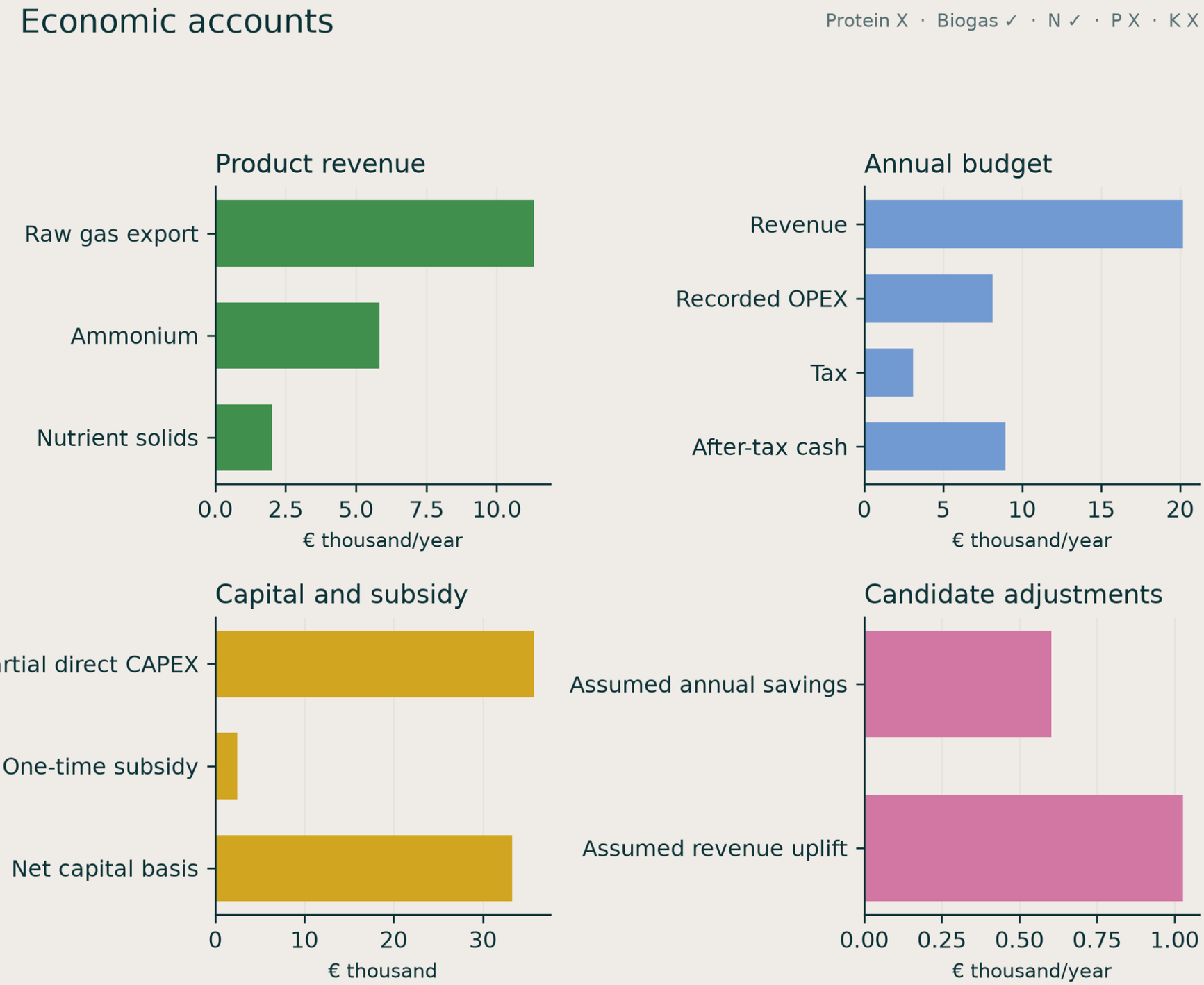
Energy and climate hotspots

Protein X · Biogas ✓ · N ✓ · P X · K X



Stage and unit utility hotspots and included climate load

Scenario: protein_bypass rank 6; 999a369dc9d444c0be7a245352a35db1; 639.6 kg/h; 8760 h/year
Units: MWh/year; t CO_2-eq/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

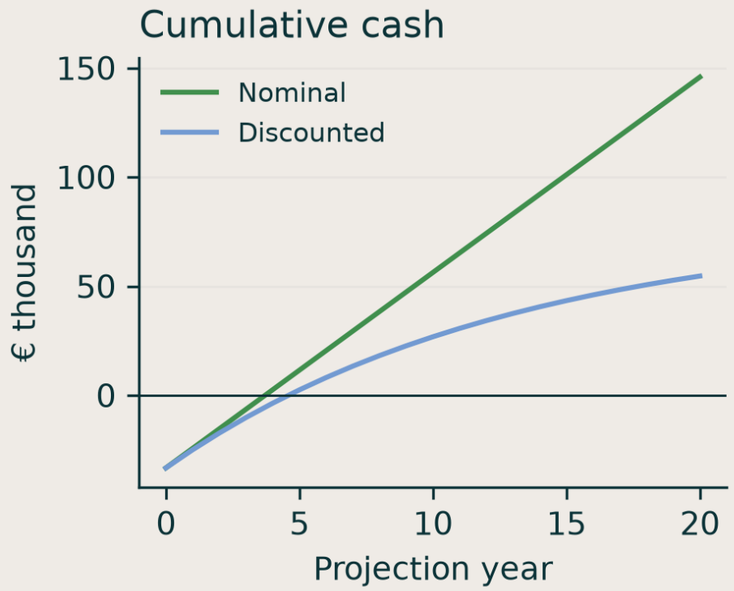


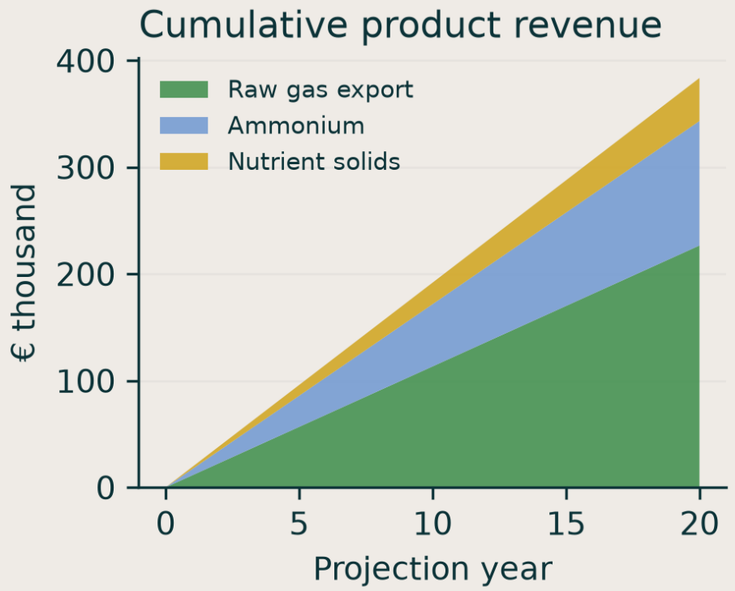
Product revenue, partial annual budget, capital subsidy and candidate overlays

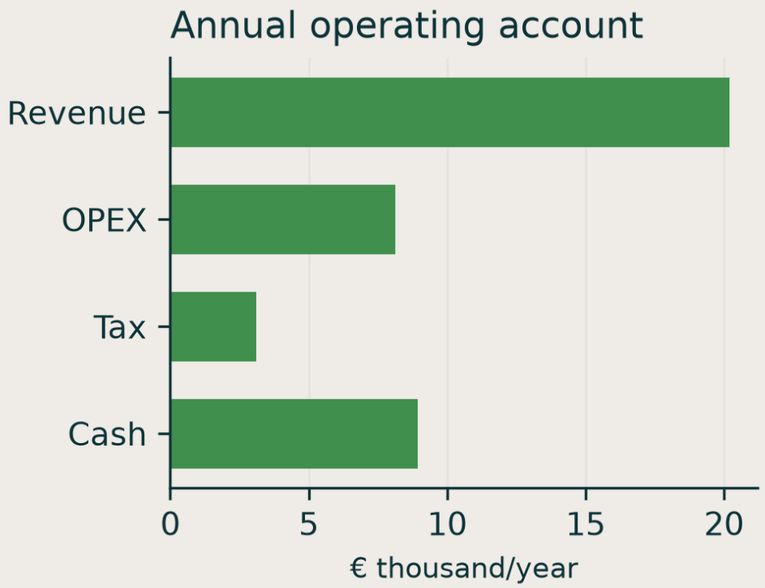
Scenario: protein_bypass rank 6; 999a369dc9d444c0be7a245352a35db1; 639.6 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

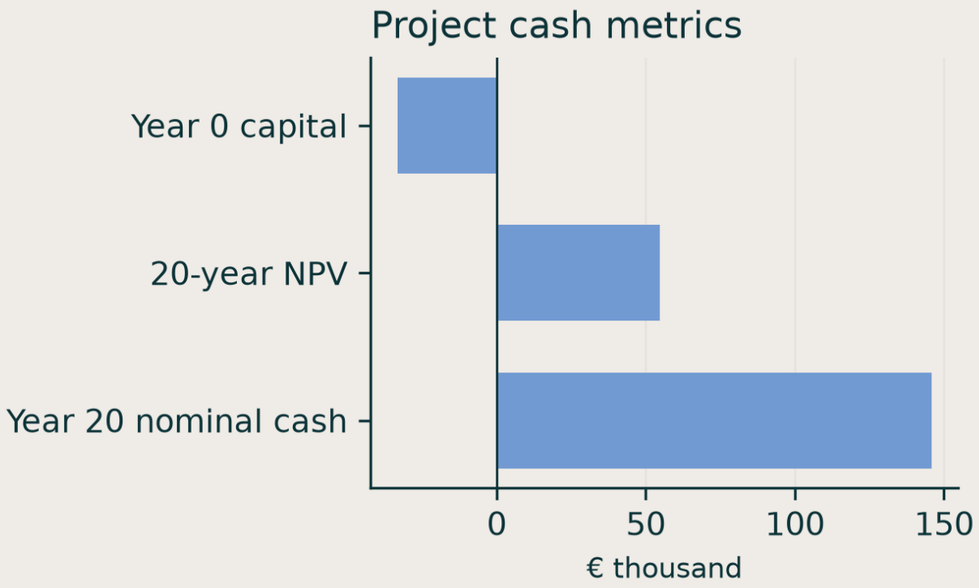
20-year financial projection

Protein X · Biogas ✓ · N ✓ · P X · K X









Constant-account cash flow and cumulative product revenue over 20 years

Scenario: protein_bypass rank 6; 999a369dc9d444c0be7a245352a35db1; 639.6 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Balance and evidence

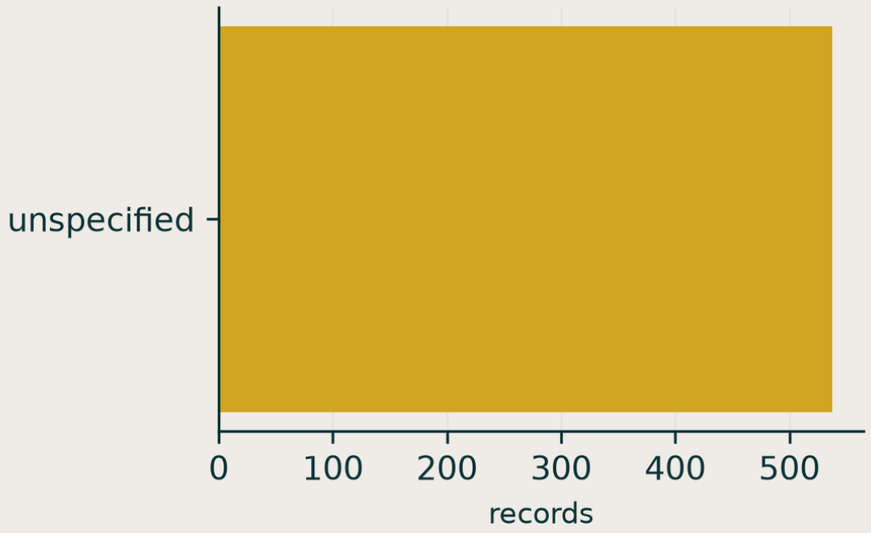
Protein X · Biogas ✓ · N ✓ · P X · K X

Element closure

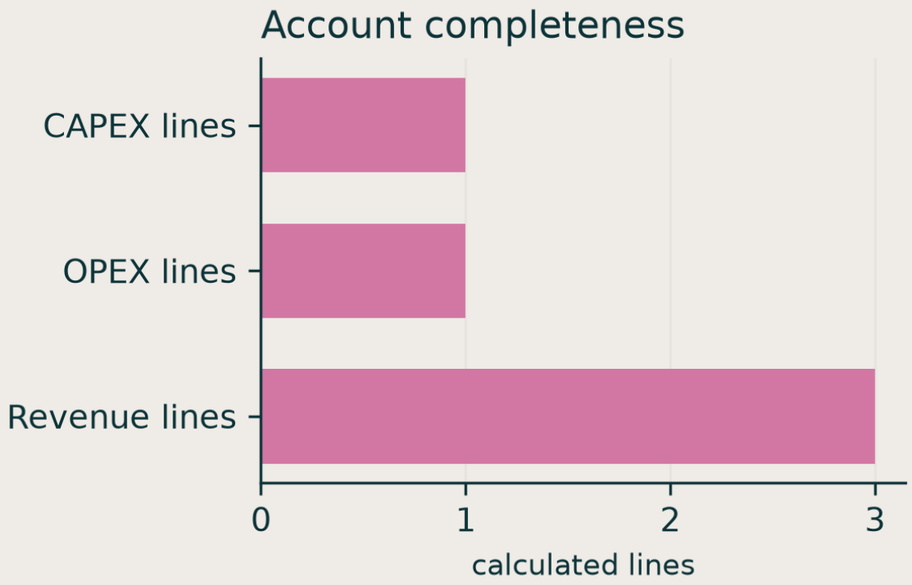
C	0 kg/year
N	0 kg/year
P	0 kg/year
K	0 kg/year

Numerical residuals

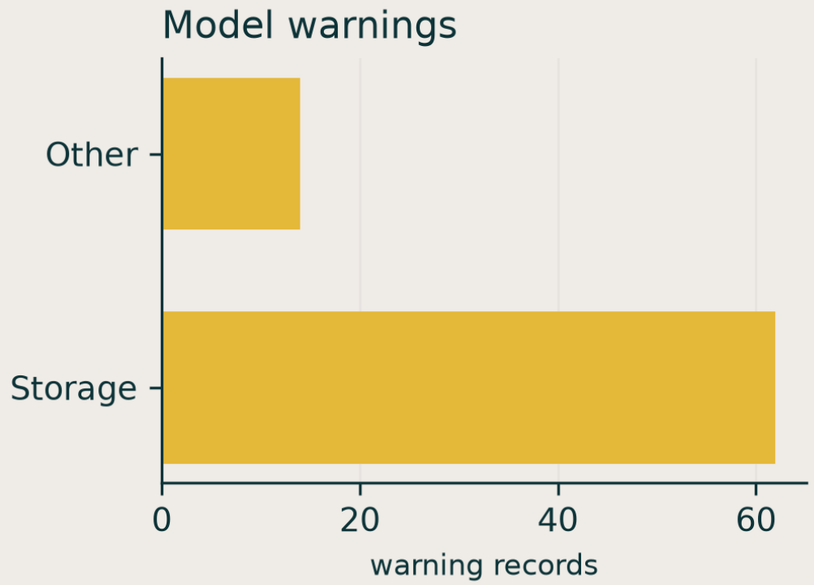
Parameter evidence



Account completeness



Model warnings



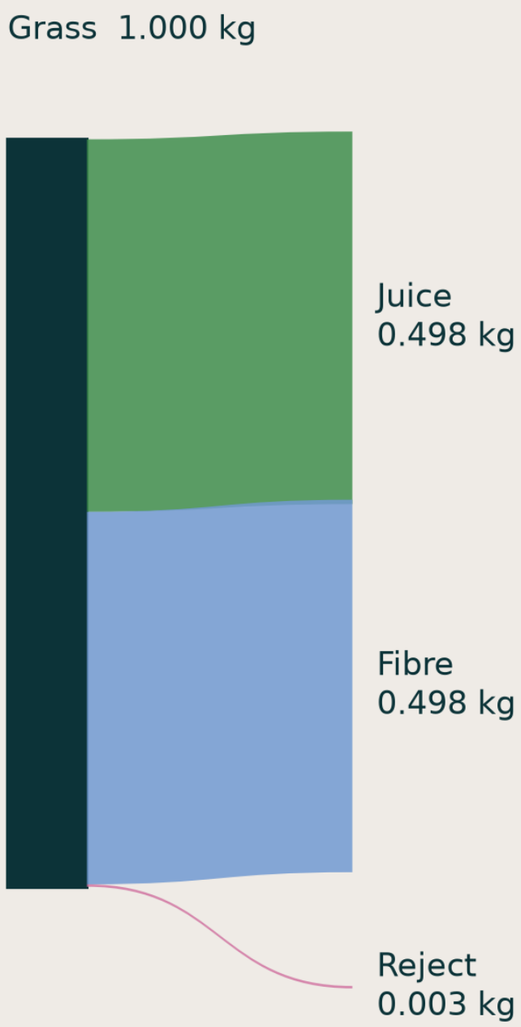
Numerical closure, evidence counts, calculated account lines and warnings

Scenario: protein_bypass rank 6; 999a369dc9d444c0be7a245352a35db1; 639.6 kg/h; 8760 h/year
Units: kg/year; records
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

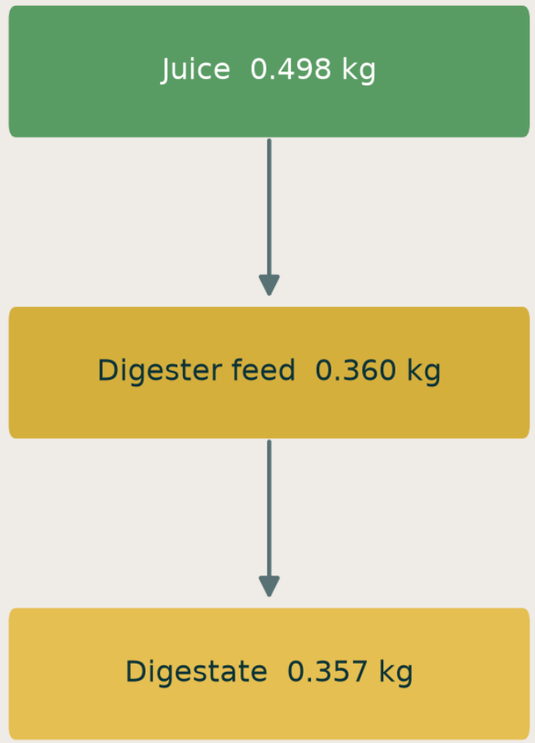
Grass to products

Protein X · Biogas ✓ · N X · P X · K X

Grass split



Juice pathway



Selected intermediate streams

Recovered outputs

Raw gas 0.0021 Nm³

Raw export 0.0021 Nm³

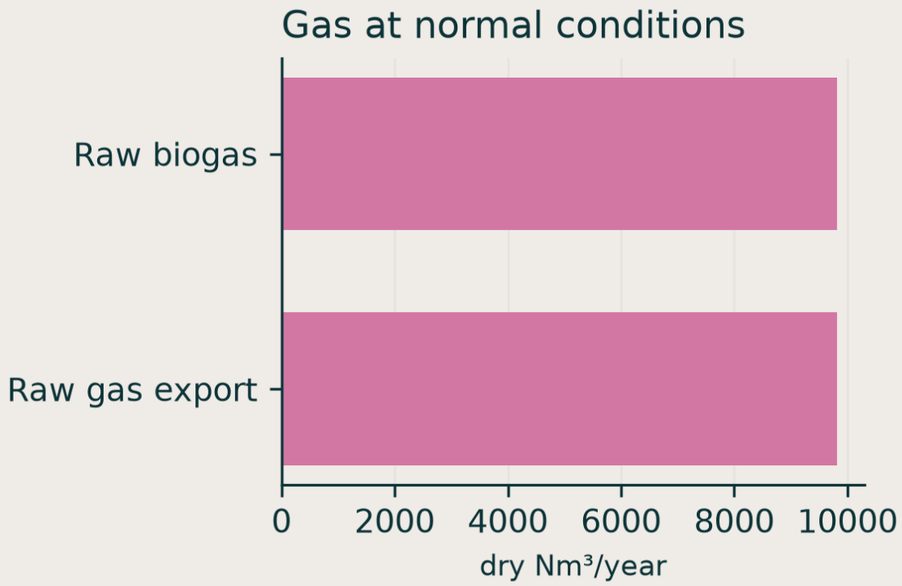
per kg wet grass

Mass-proportional grass split plus modeled downstream stream/product map

Scenario: protein_bypass rank 7; 68657b79ad57489e881d453dfbcf85d7; 523.4 kg/h; 8760 h/year
Units: kg/kg wet grass; dry Nm³/kg wet grass
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Material and products

Protein X · Biogas ✓ · N X · P X · K X

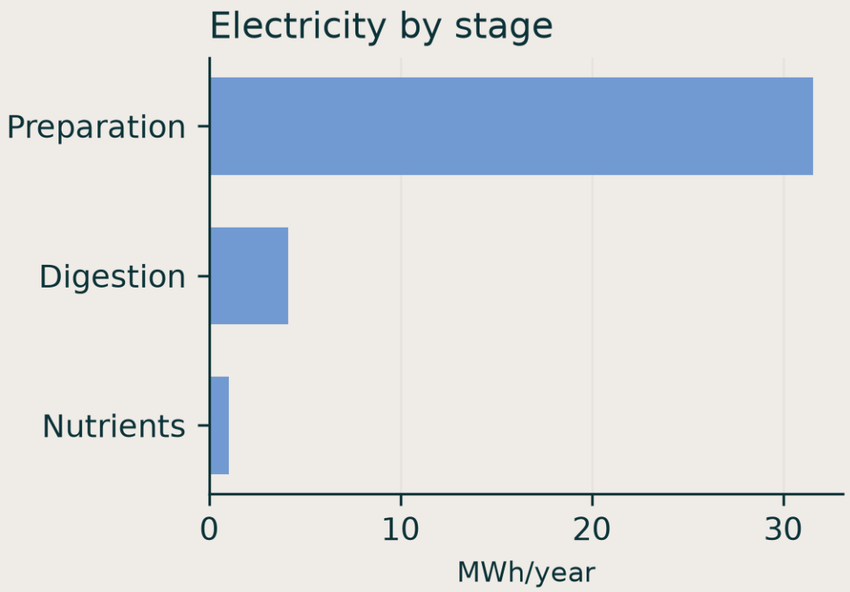


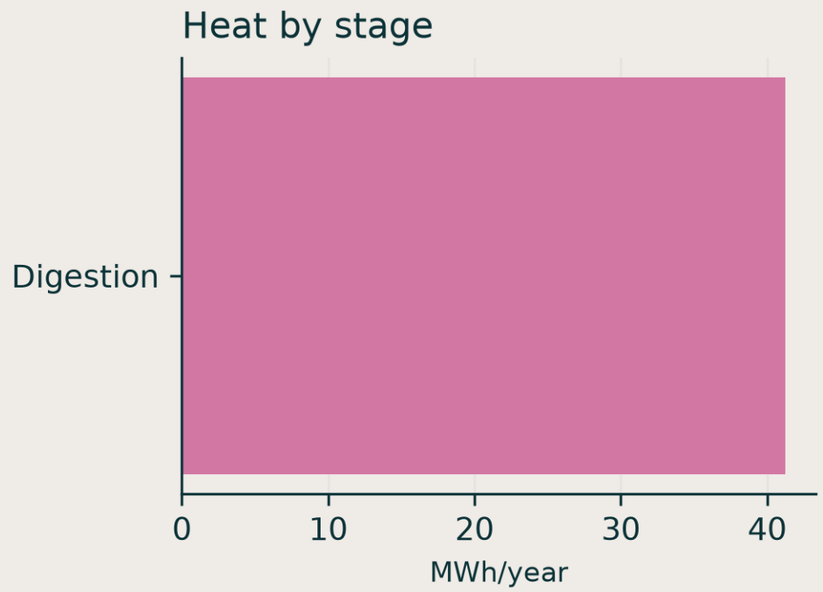
Front-end split, solid products, dry gas volumes and specific yields

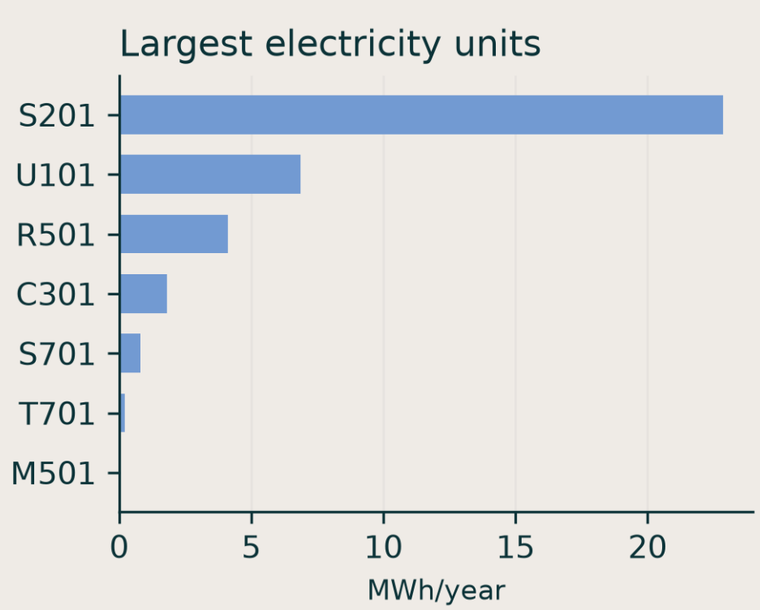
Scenario: protein_bypass rank 7; 68657b79ad57489e881d453dfbcf85d7; 523.4 kg/h; 8760 h/year
Units: kg/h; t/year; dry Nm³/year; kg/t feed
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

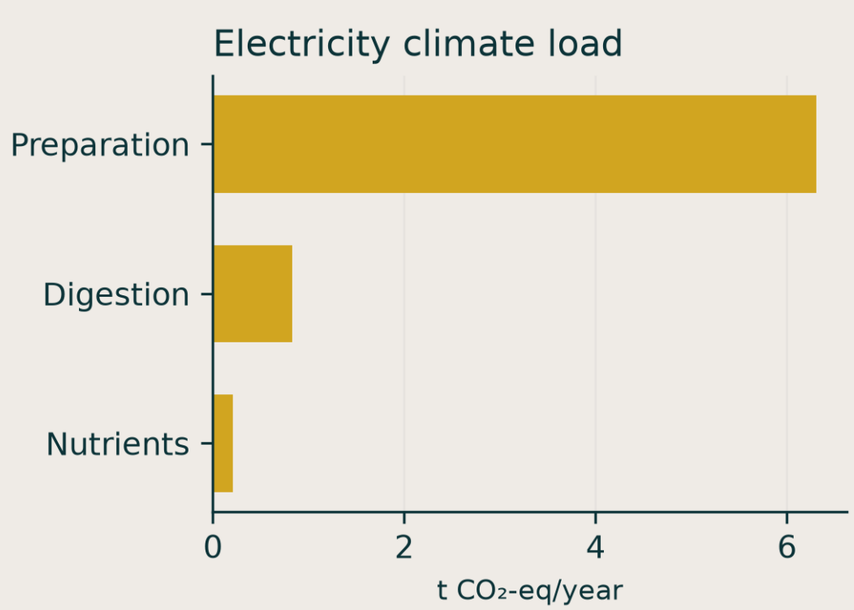
Energy and climate hotspots

Protein X · Biogas ✓ · N X · P X · K X







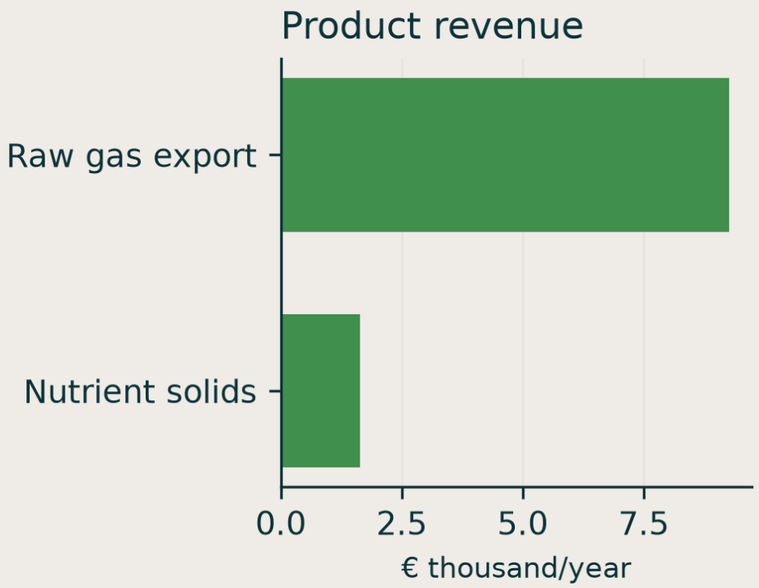


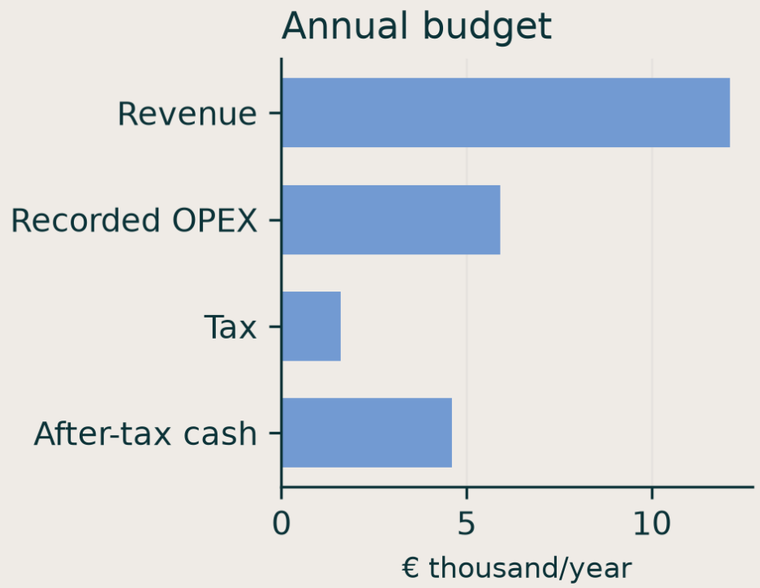
Stage and unit utility hotspots and included climate load

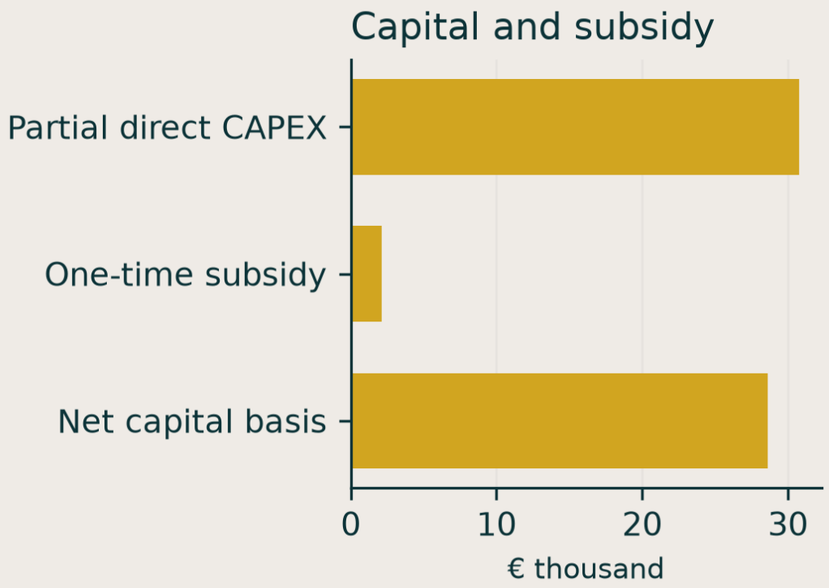
Scenario: protein_bypass rank 7; 68657b79ad57489e881d453dfbcf85d7; 523.4 kg/h; 8760 h/year
Units: MWh/year; t CO_2-eq/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

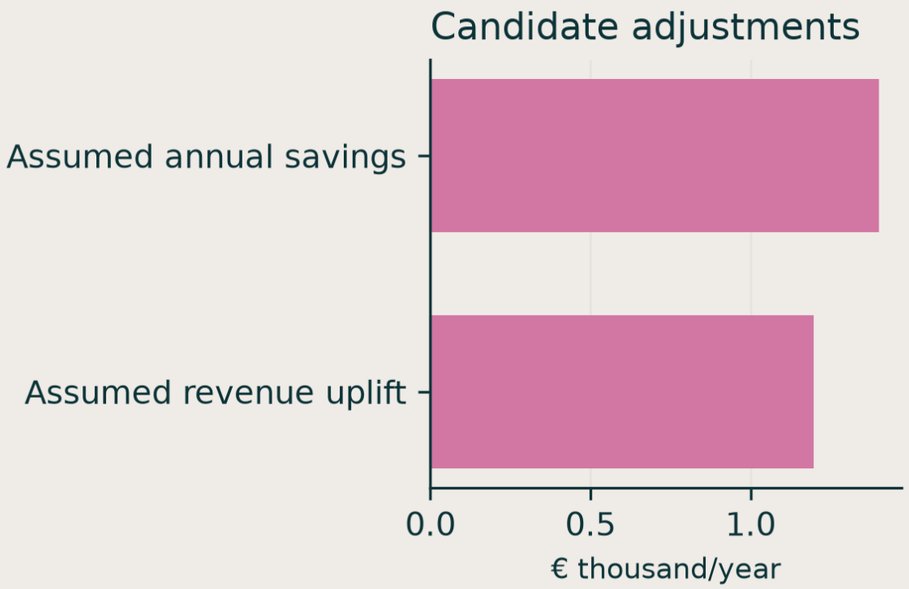
Economic accounts

Protein X · Biogas ✓ · N X · P X · K X







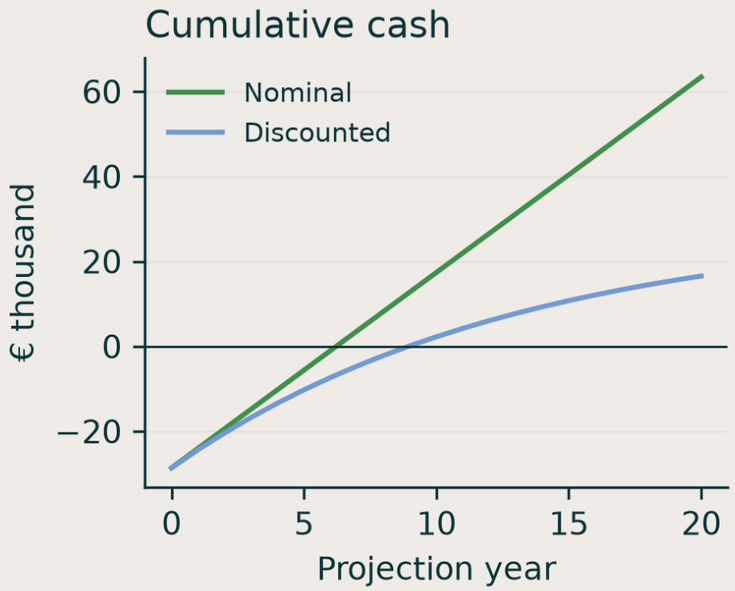


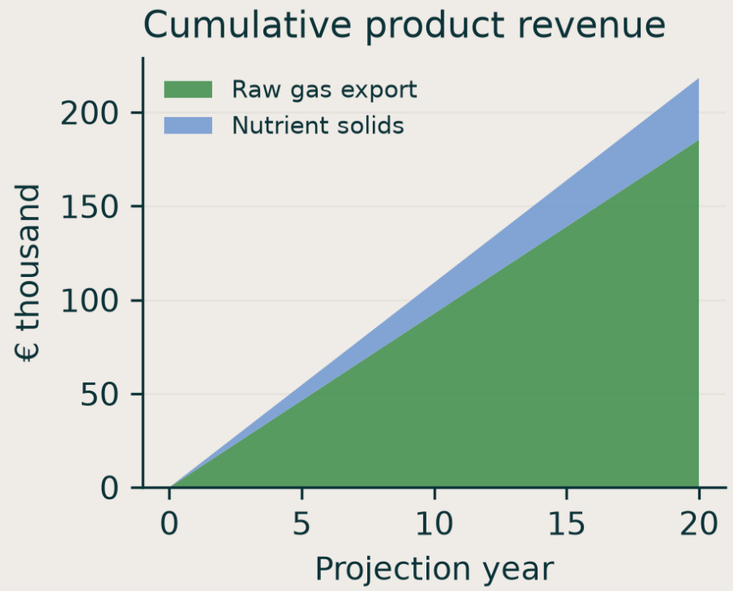
Product revenue, partial annual budget, capital subsidy and candidate overlays

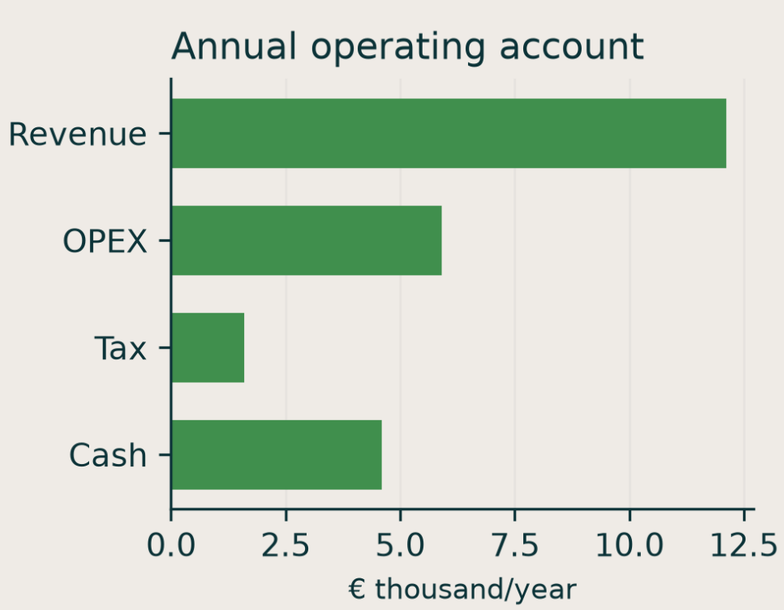
Scenario: protein_bypass rank 7; 68657b79ad57489e881d453dfbcf85d7; 523.4 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

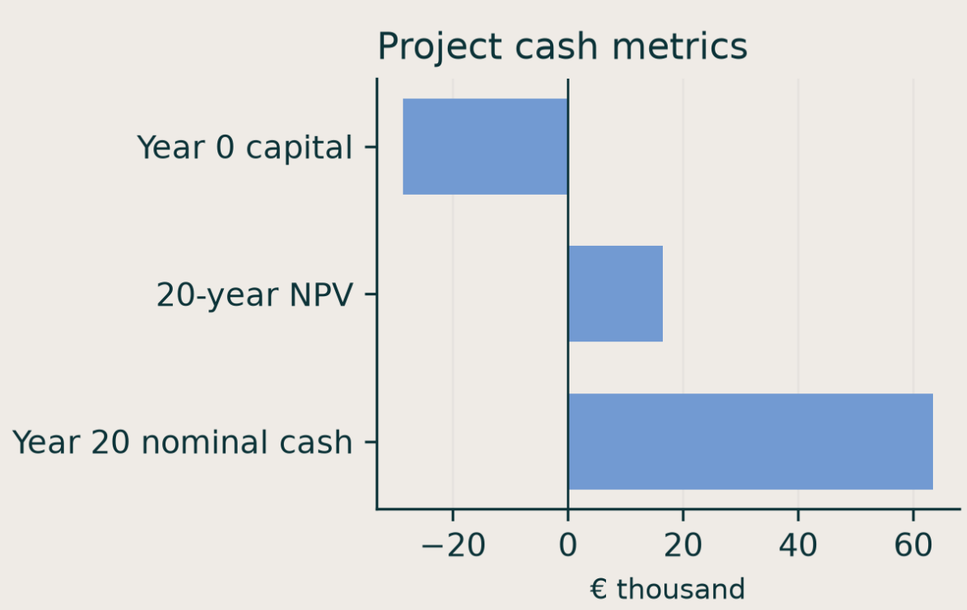
20-year financial projection

Protein X · Biogas ✓ · N X · P X · K X









Constant-account cash flow and cumulative product revenue over 20 years

Scenario: protein_bypass rank 7; 68657b79ad57489e881d453dfbcf85d7; 523.4 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Balance and evidence

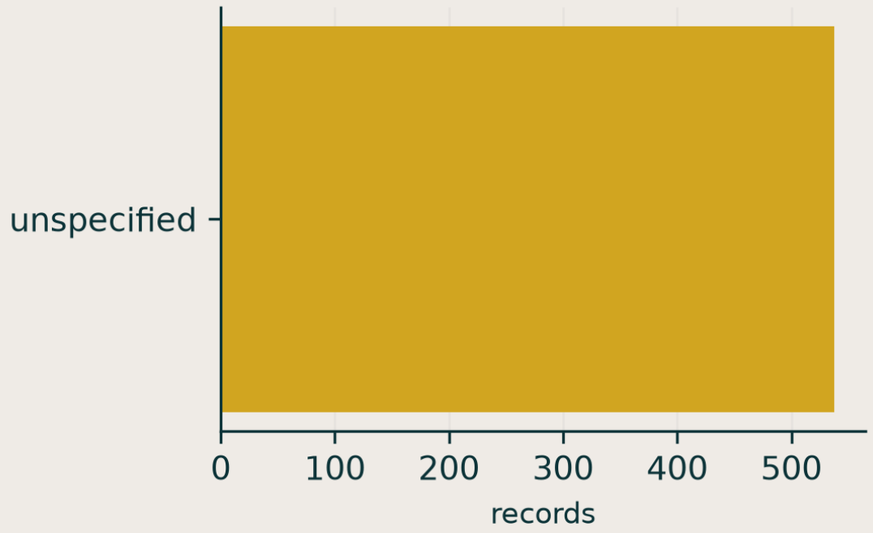
Protein X · Biogas ✓ · N X · P X · K X

Element closure

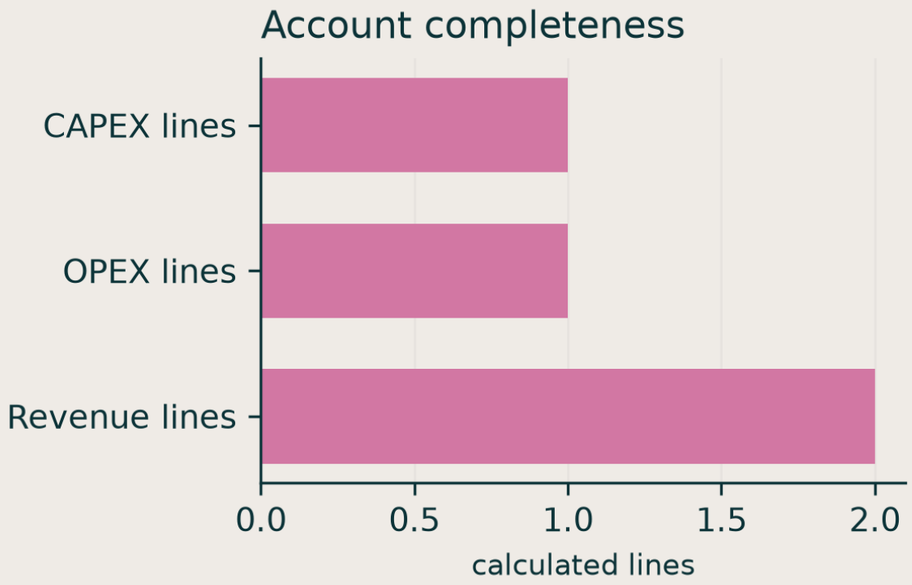
C	1.9e-12 kg/year
N	0 kg/year
P	0 kg/year
K	2.4e-13 kg/year

Numerical residuals

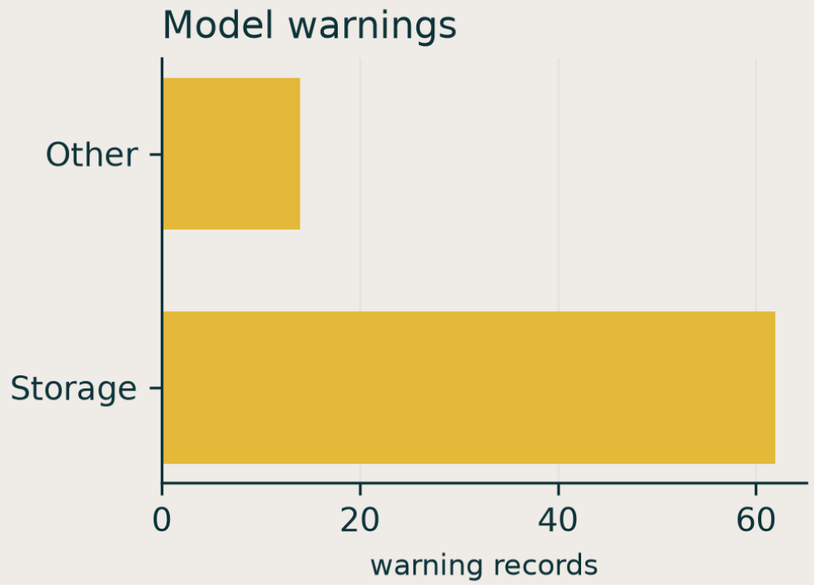
Parameter evidence



Account completeness



Model warnings



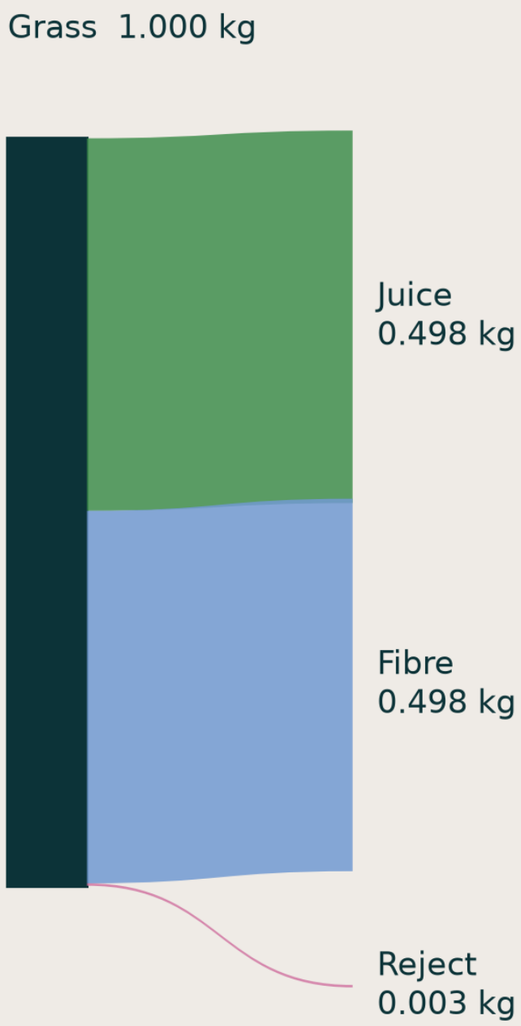
Numerical closure, evidence counts, calculated account lines and warnings

Scenario: protein_bypass rank 7; 68657b79ad57489e881d453dfbcf85d7; 523.4 kg/h; 8760 h/year
Units: kg/year; records
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

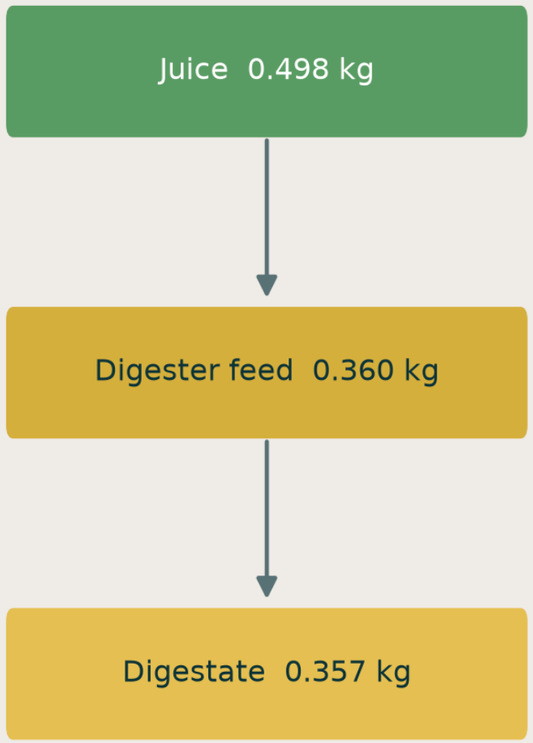
Grass to products

Protein X · Biogas ✓ · N X · P X · K X

Grass split



Juice pathway



Selected intermediate streams

Recovered outputs

Raw gas 0.0021 Nm³

Upgraded 0.0013 Nm³

per kg wet grass

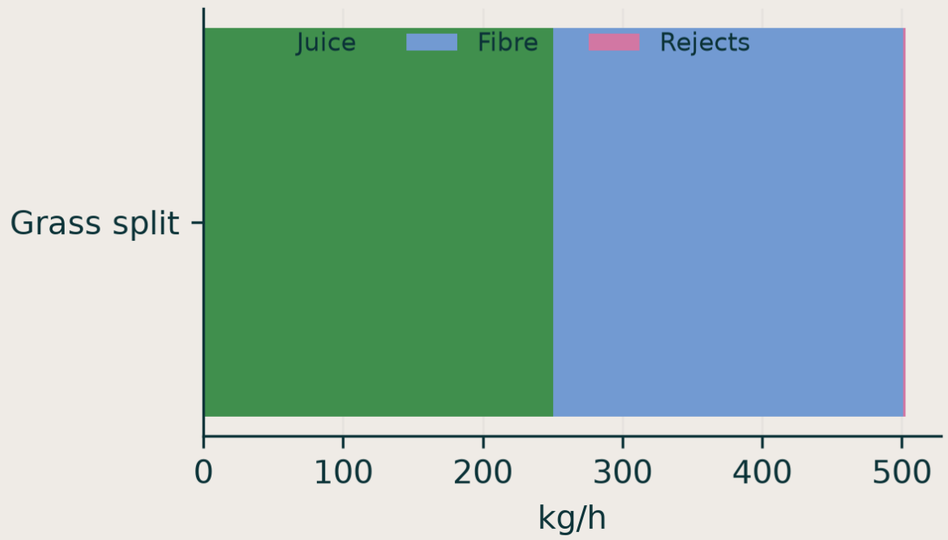
Mass-proportional grass split plus modeled downstream stream/product map

Scenario: protein_bypass rank 8; 102cd2e2817a4768a41d565264b4a682; 502.8 kg/h; 8760 h/year
Units: kg/kg wet grass; dry Nm³/kg wet grass
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Material and products

Protein X · Biogas ✓ · N X · P X · K X

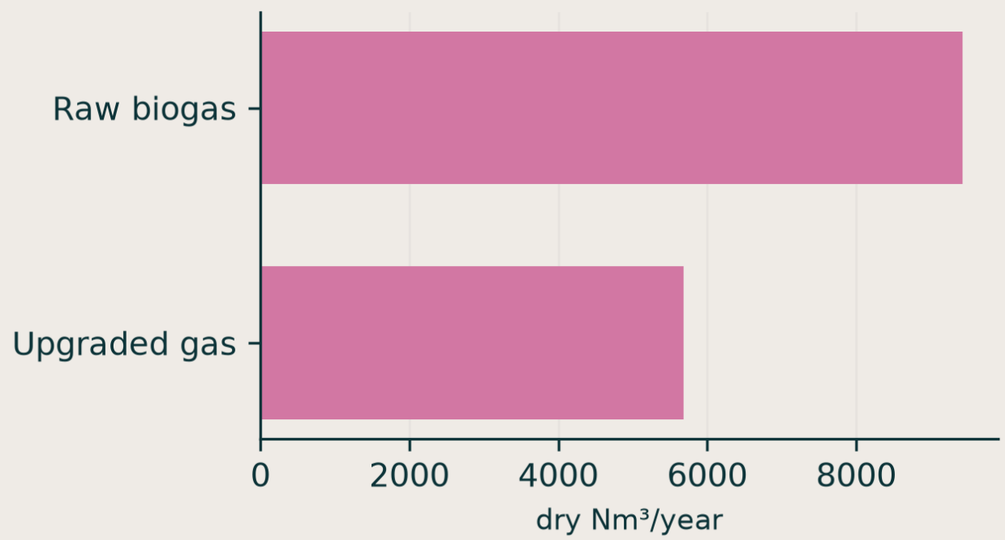
Feed destinations



Solid products

No modeled output

Gas at normal conditions



Specific product yields

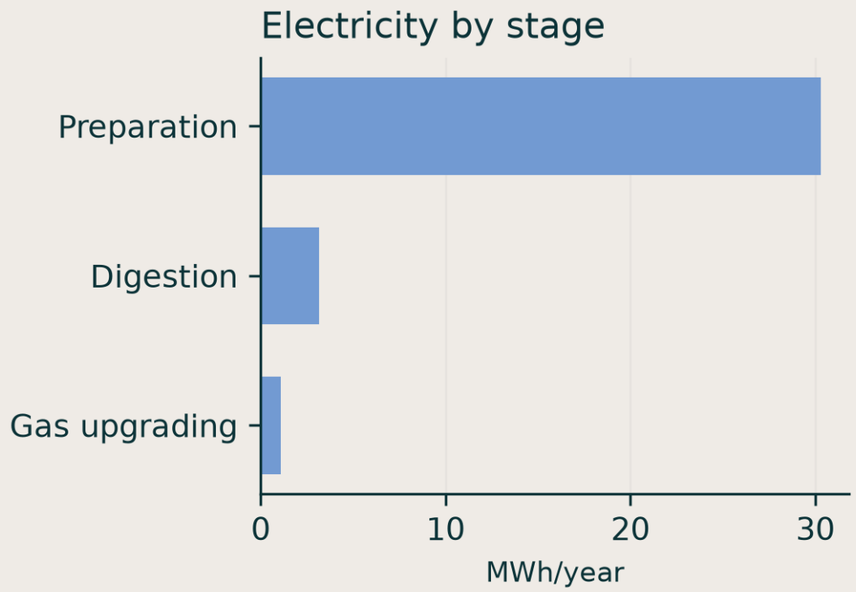
No modeled output

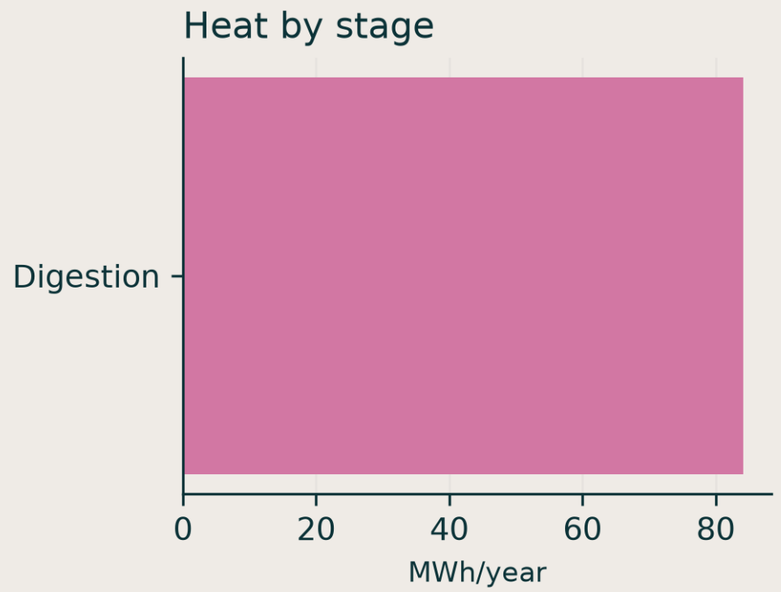
Front-end split, solid products, dry gas volumes and specific yields

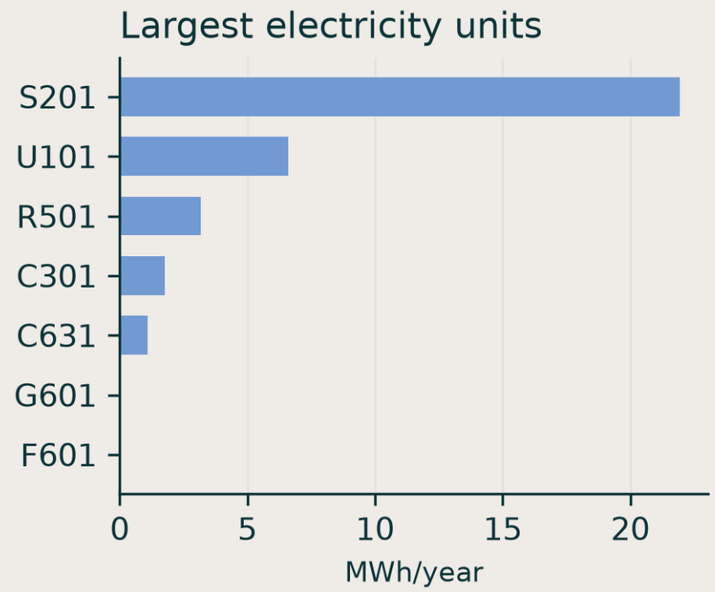
Scenario: protein_bypass rank 8; 102cd2e2817a4768a41d565264b4a682; 502.8 kg/h; 8760 h/year
Units: kg/h; t/year; dry Nm³/year; kg/t feed
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

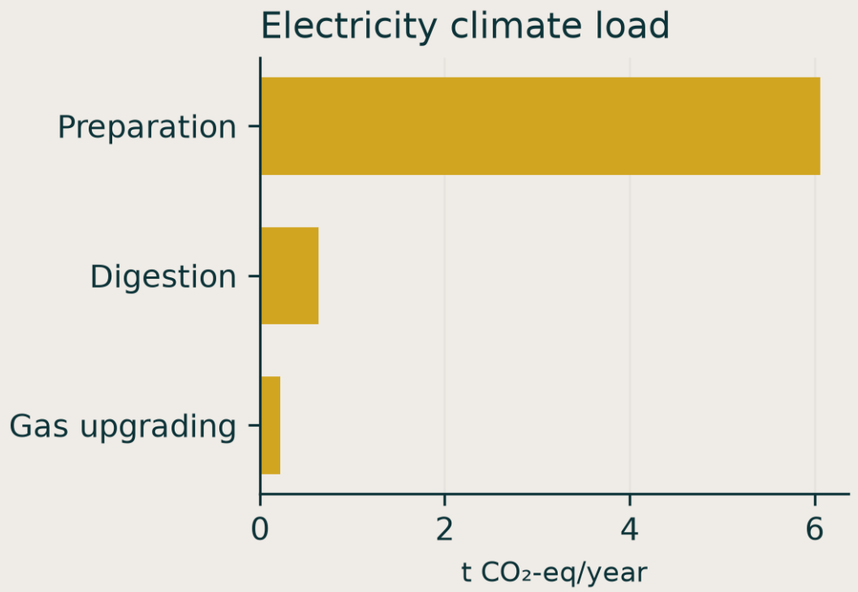
Energy and climate hotspots

Protein X · Biogas ✓ · N X · P X · K X







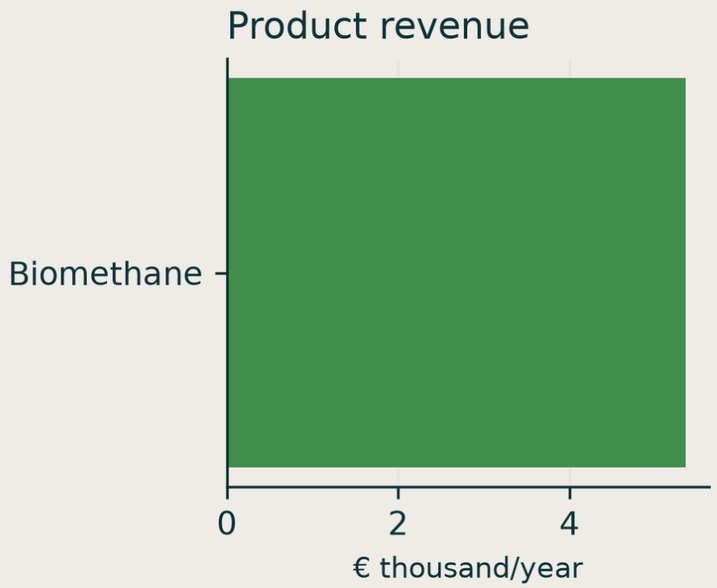


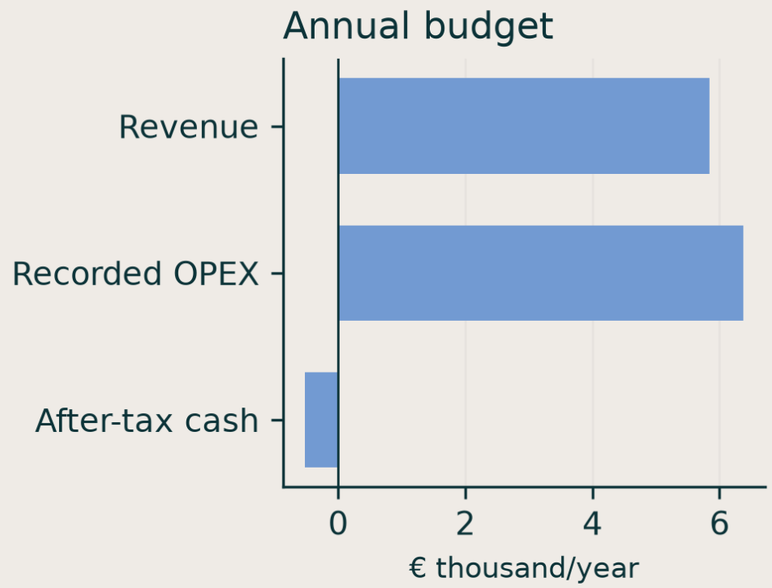
Stage and unit utility hotspots and included climate load

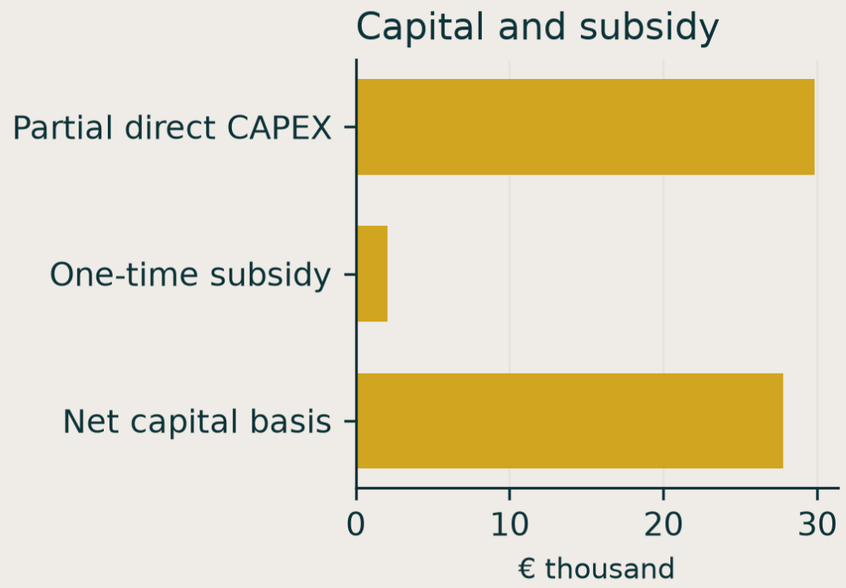
Scenario: protein_bypass rank 8; 102cd2e2817a4768a41d565264b4a682; 502.8 kg/h; 8760 h/year
Units: MWh/year; t CO_2-eq/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

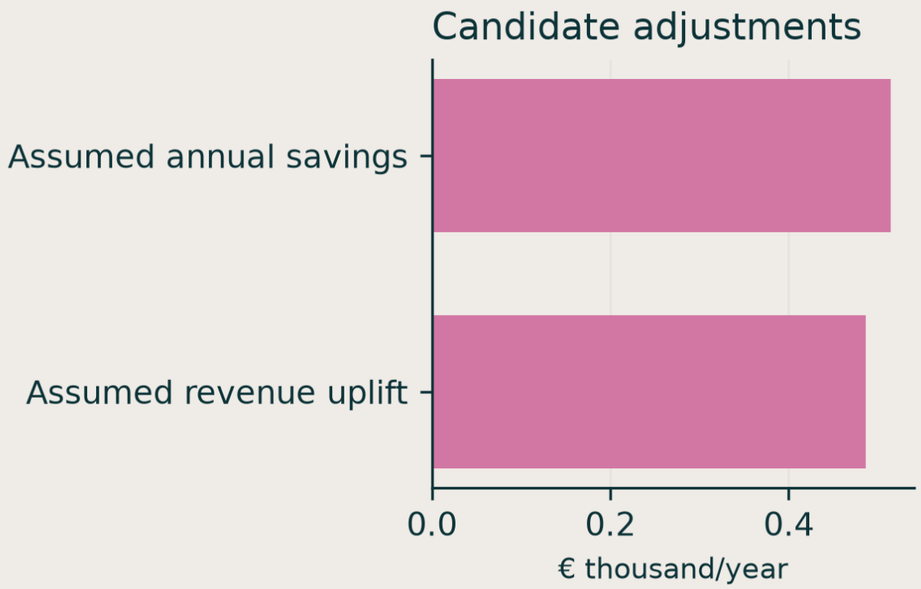
Economic accounts

Protein X · Biogas ✓ · N X · P X · K X







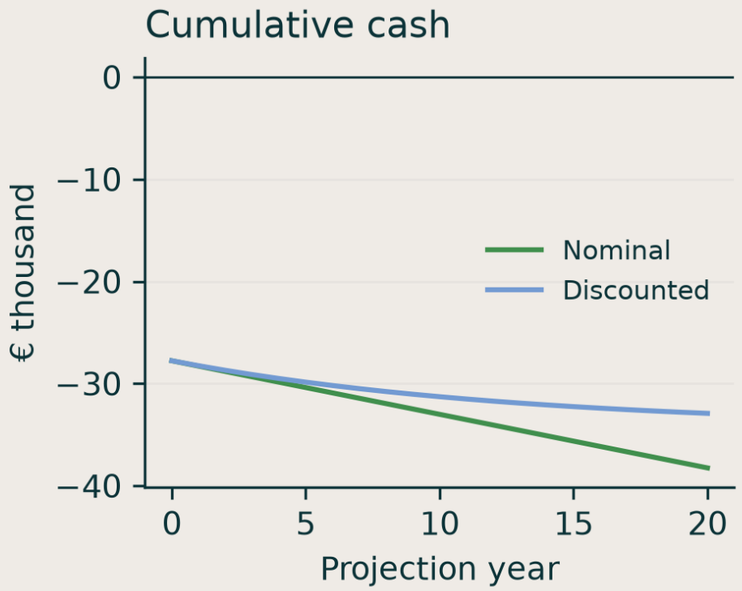


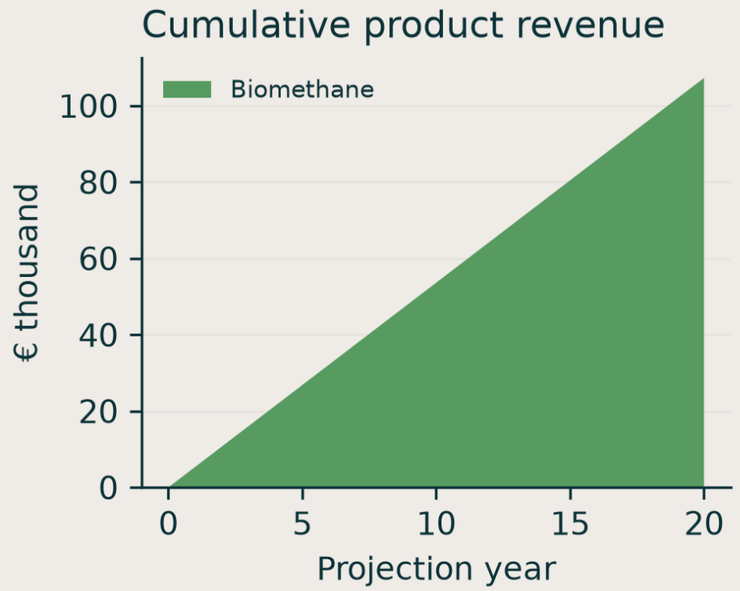
Product revenue, partial annual budget, capital subsidy and candidate overlays

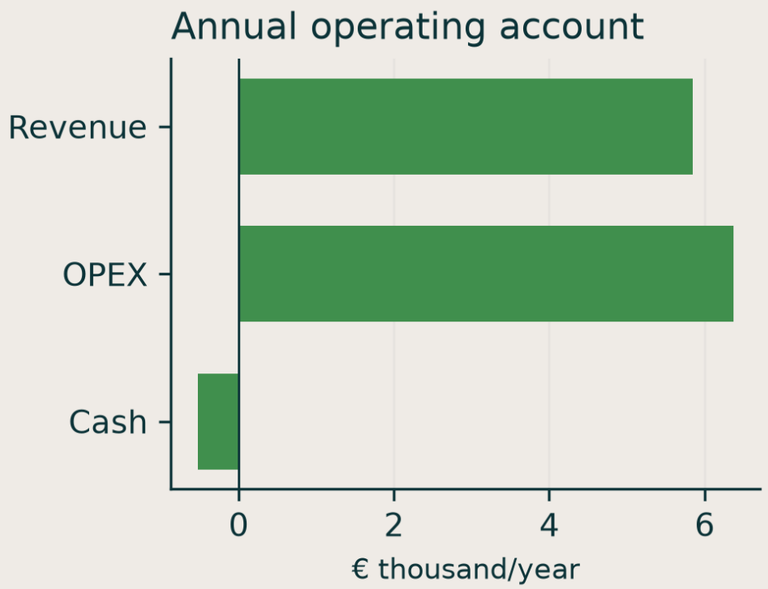
Scenario: protein_bypass rank 8; 102cd2e2817a4768a41d565264b4a682; 502.8 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

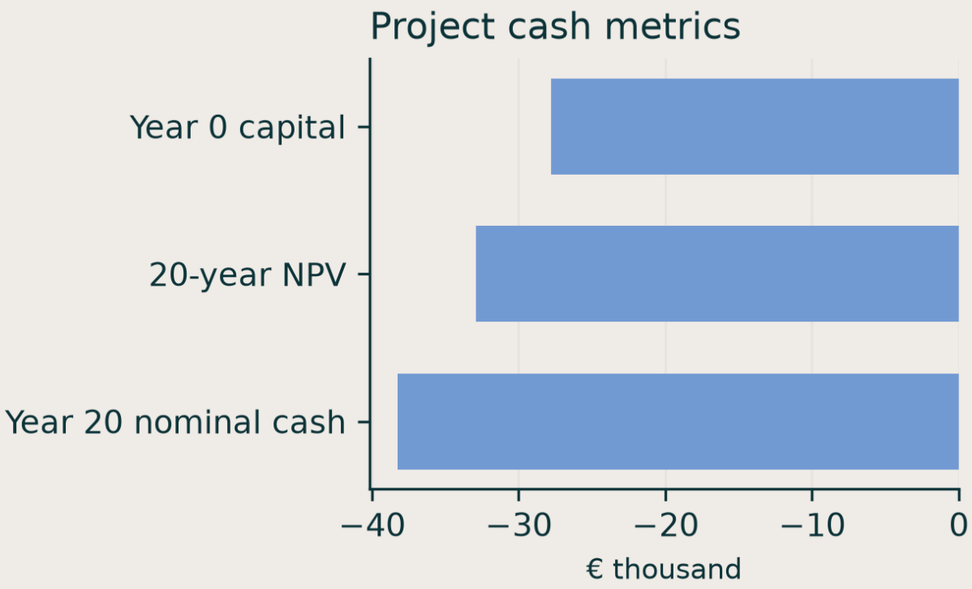
20-year financial projection

Protein X · Biogas ✓ · N X · P X · K X









Constant-account cash flow and cumulative product revenue over 20 years

Scenario: protein_bypass rank 8; 102cd2e2817a4768a41d565264b4a682; 502.8 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Balance and evidence

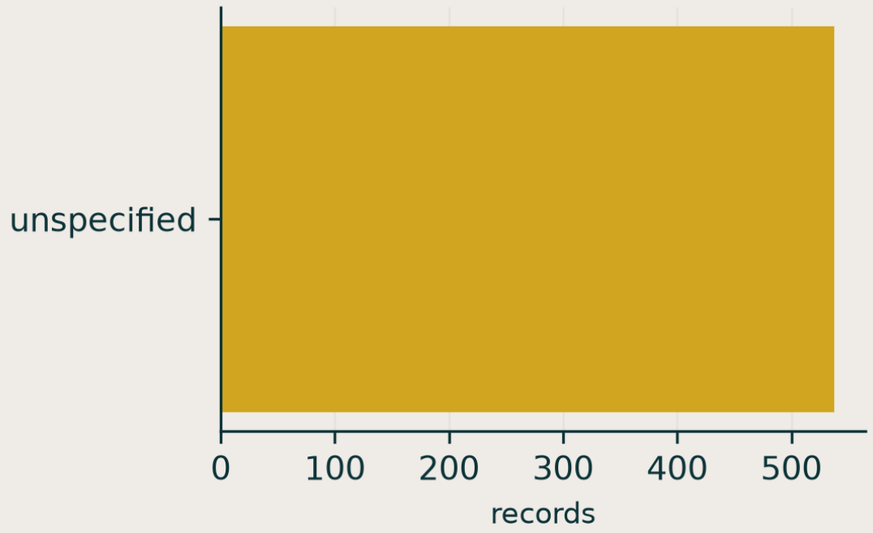
Protein X · Biogas ✓ · N X · P X · K X

Element closure

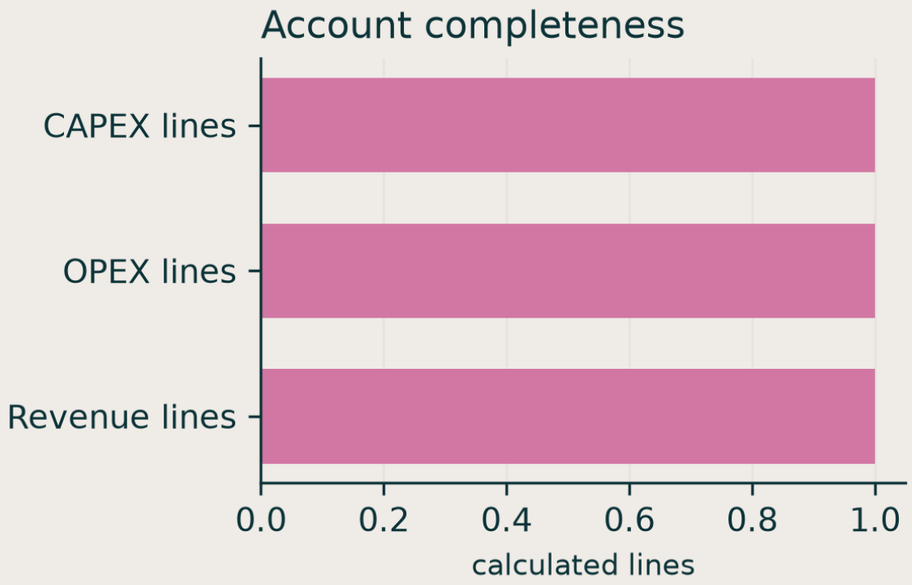
C	0 kg/year
N	0 kg/year
P	0 kg/year
K	0 kg/year

Numerical residuals

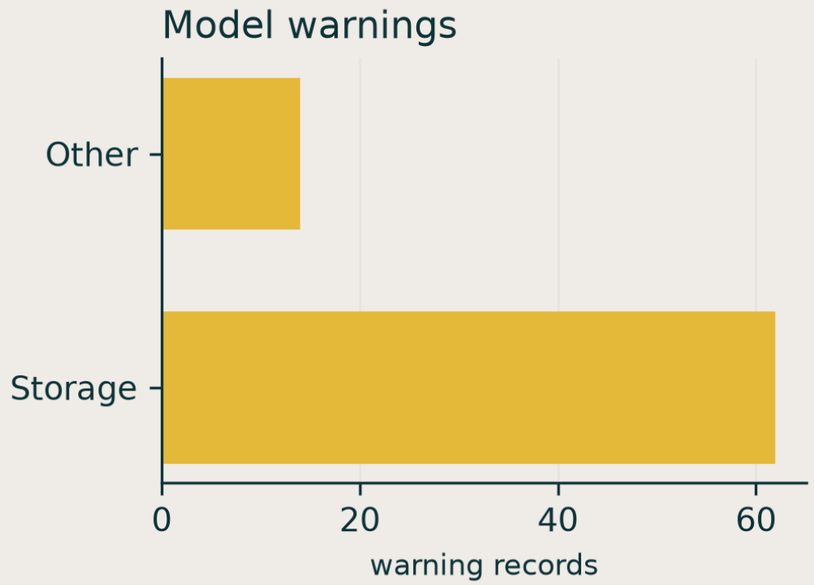
Parameter evidence



Account completeness



Model warnings



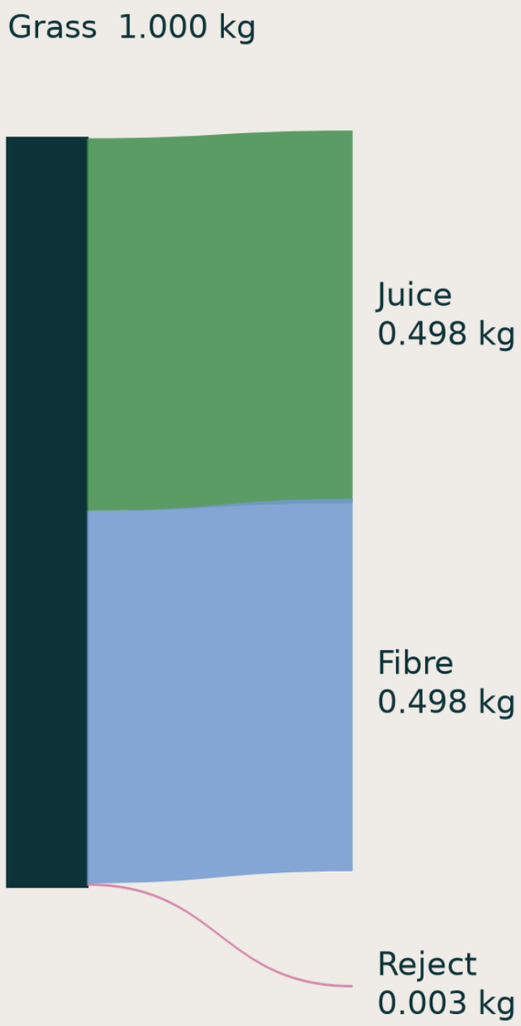
Numerical closure, evidence counts, calculated account lines and warnings

Scenario: protein_bypass rank 8; 102cd2e2817a4768a41d565264b4a682; 502.8 kg/h; 8760 h/year
Units: kg/year; records
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

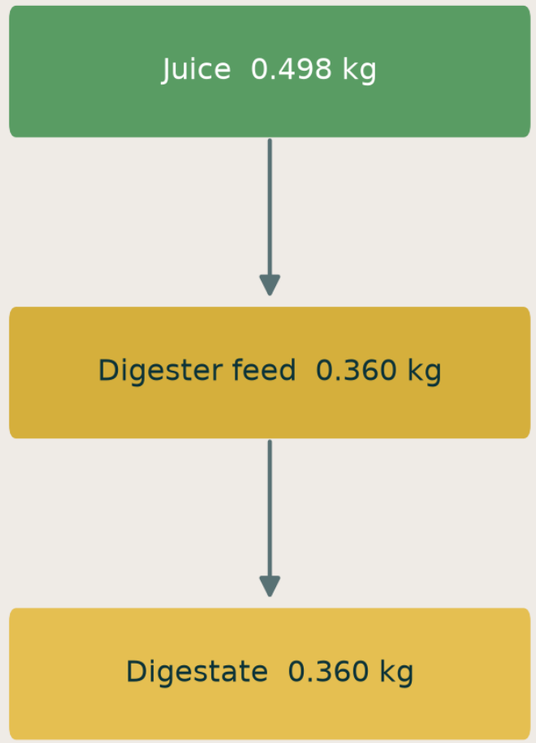
Grass to products

Protein X · Biogas X · N X · P X · K X

Grass split



Juice pathway



Selected intermediate streams

Recovered outputs

per kg wet grass

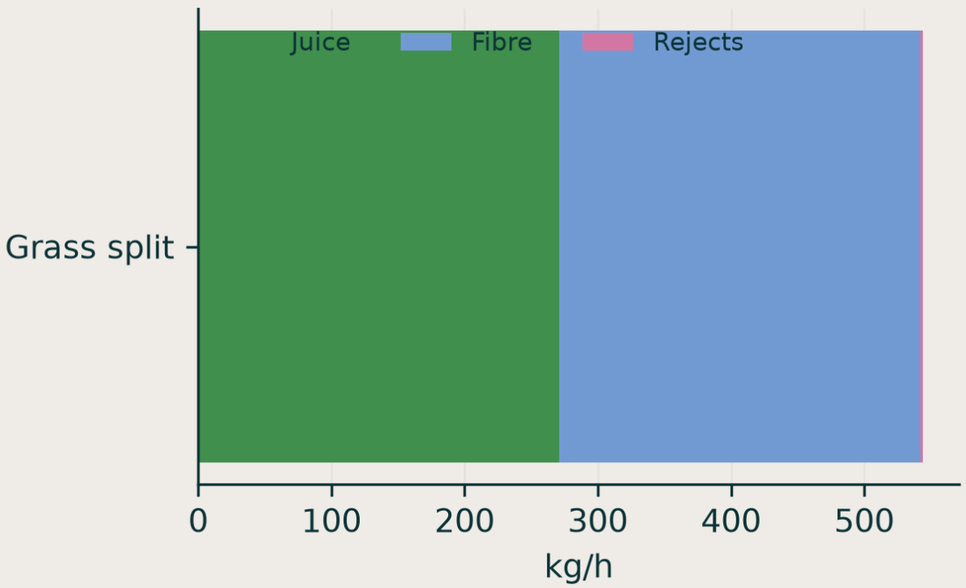
Mass-proportional grass split plus modeled downstream stream/product map

Scenario: protein_bypass rank 9; 6aa672cdb3cb45bfb5fcc01fd37205e1; 544.2 kg/h; 8760 h/year
Units: kg/kg wet grass; dry Nm³/kg wet grass
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Material and products

Protein X · Biogas X · N X · P X · K X

Feed destinations



Solid products

No modeled output

Gas at normal conditions

No modeled output

Specific product yields

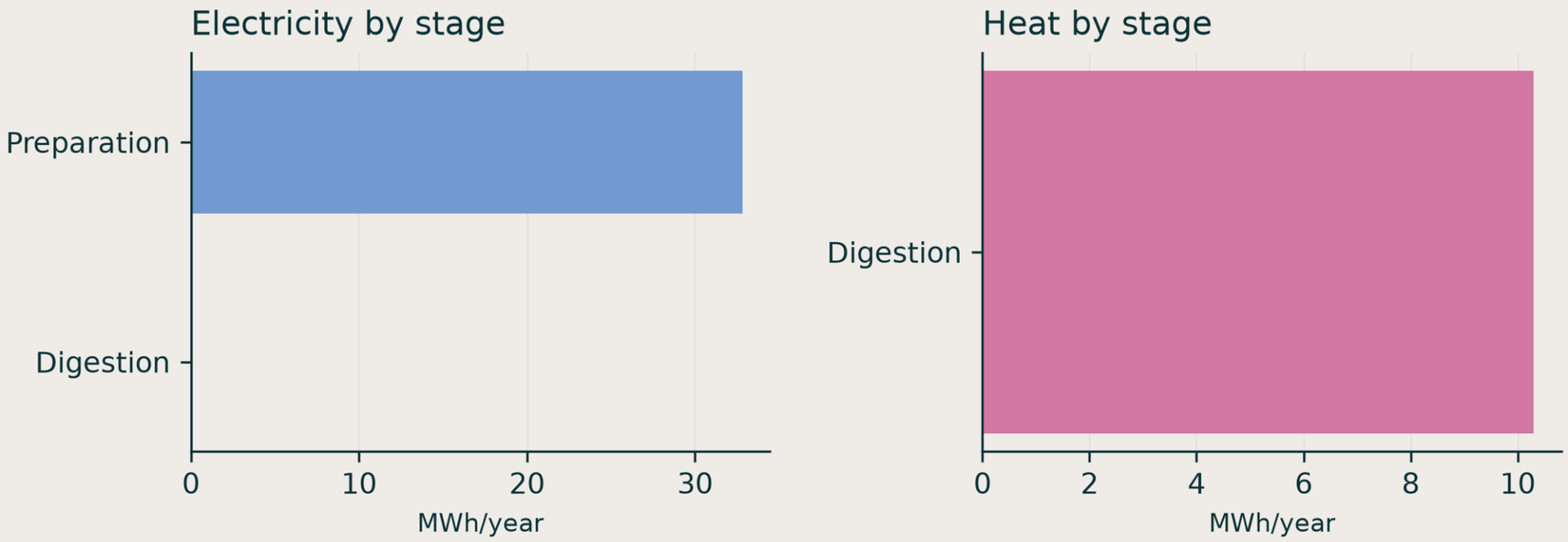
No modeled output

Front-end split, solid products, dry gas volumes and specific yields

Scenario: protein_bypass rank 9; 6aa672cdb3cb45bfb5fcc01fd37205e1; 544.2 kg/h; 8760 h/year
Units: kg/h; t/year; dry Nm³/year; kg/t feed
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Energy and climate hotspots

Protein X · Biogas X · N X · P X · K X



Stage and unit utility hotspots and included climate load

Scenario: protein_bypass rank 9; 6aa672cdb3cb45bfb5fcc01fd37205e1; 544.2 kg/h; 8760 h/year
Units: MWh/year; t CO_2-eq/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

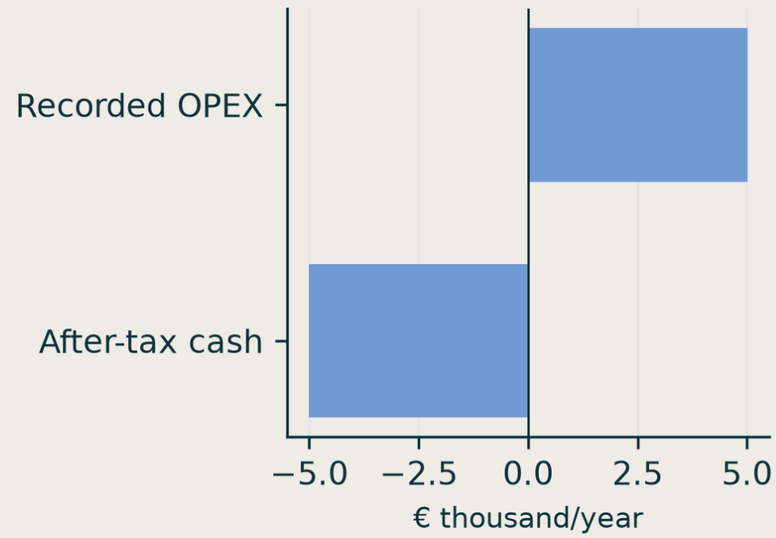
Economic accounts

Protein X · Biogas X · N X · P X · K X

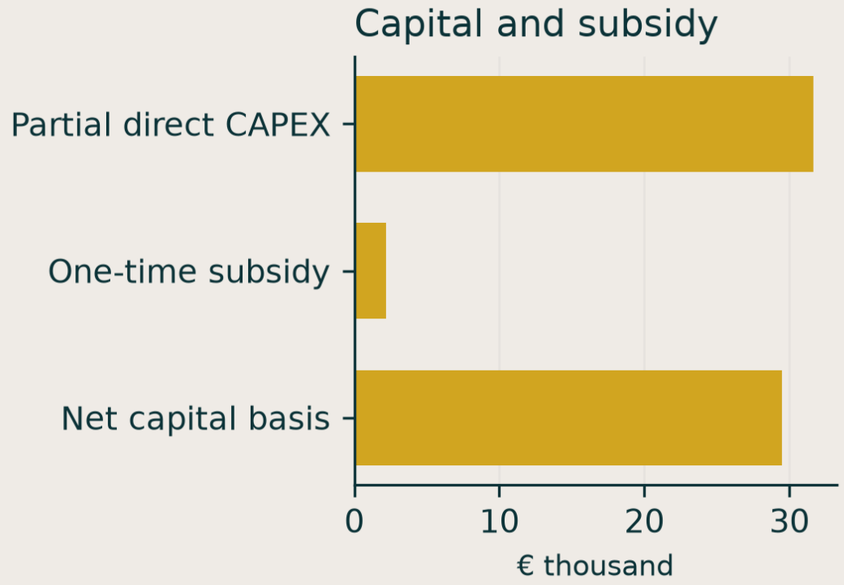
Product revenue

No modeled output

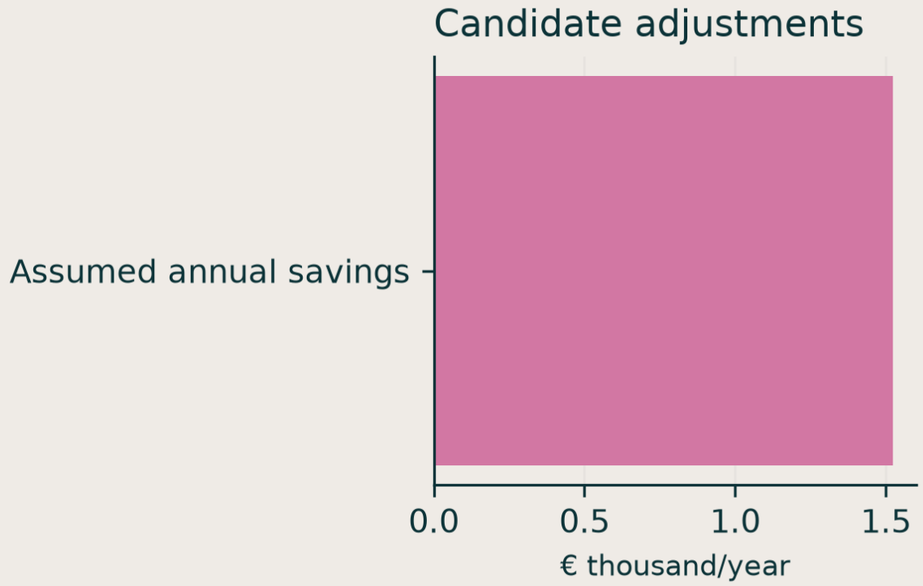
Annual budget



Capital and subsidy



Candidate adjustments

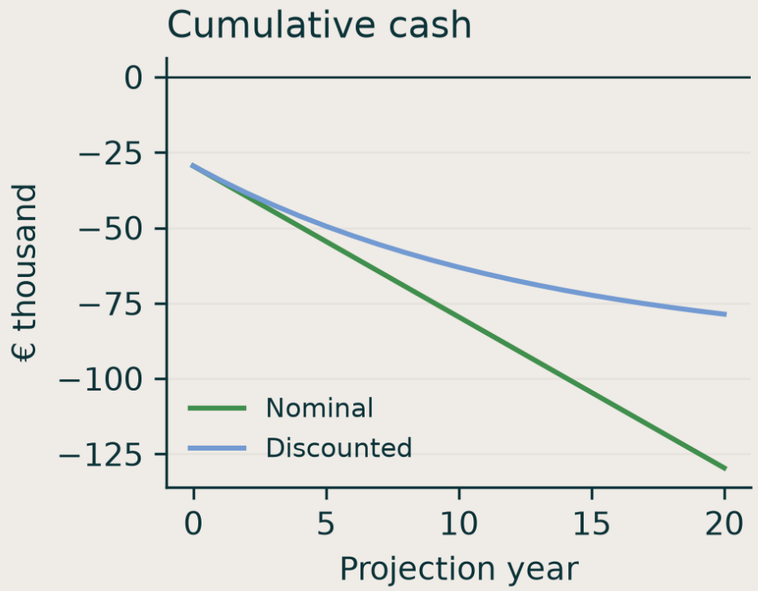


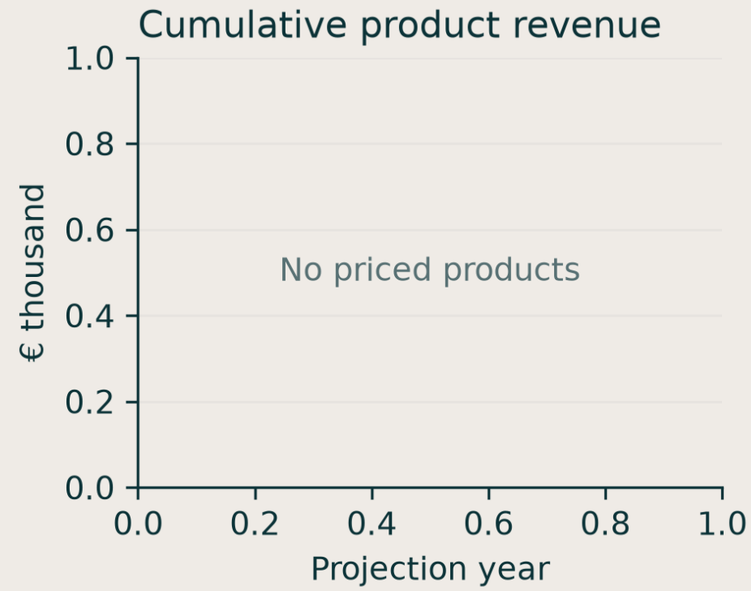
Product revenue, partial annual budget, capital subsidy and candidate overlays

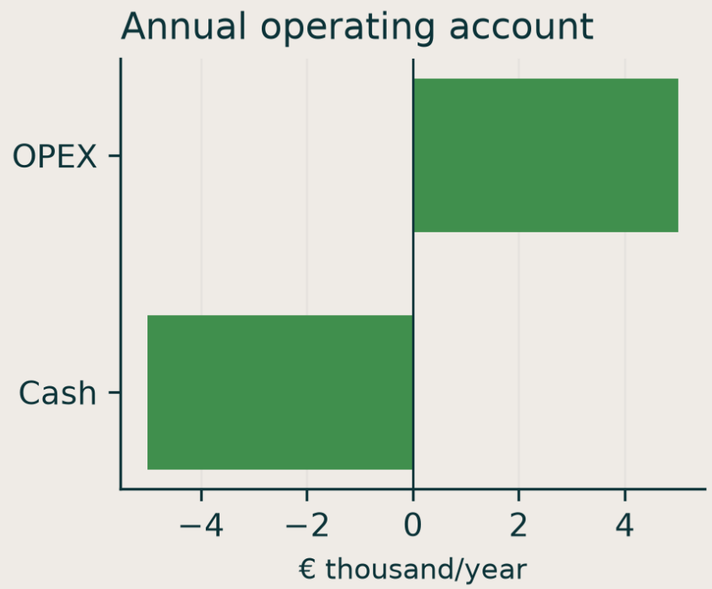
Scenario: protein_bypass rank 9; 6aa672cdb3cb45bfb5fcc01fd37205e1; 544.2 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

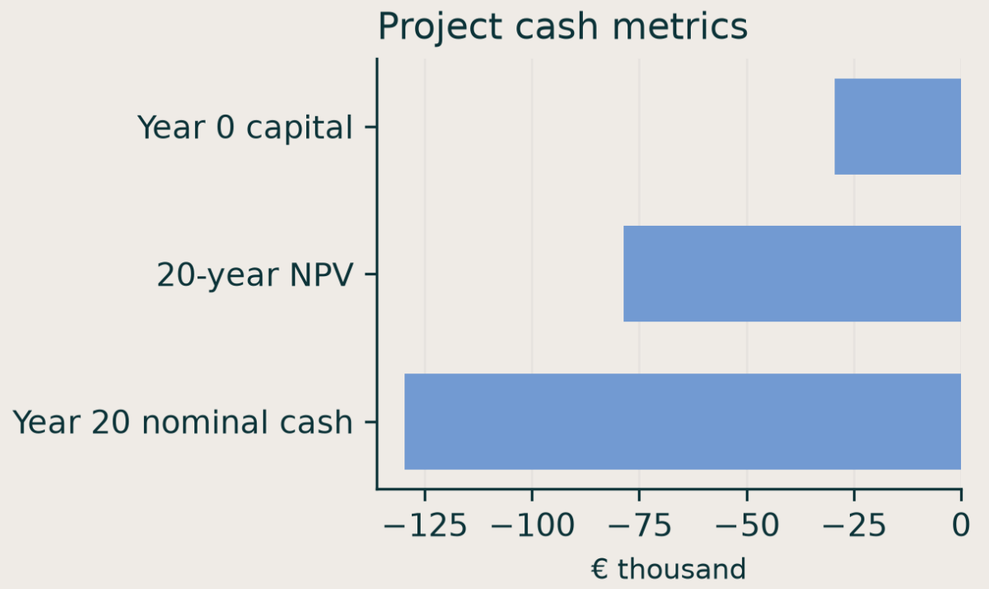
20-year financial projection

Protein X · Biogas X · N X · P X · K X









Constant-account cash flow and cumulative product revenue over 20 years

Scenario: protein_bypass rank 9; 6aa672cdb3cb45bfb5fcc01fd37205e1; 544.2 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Balance and evidence

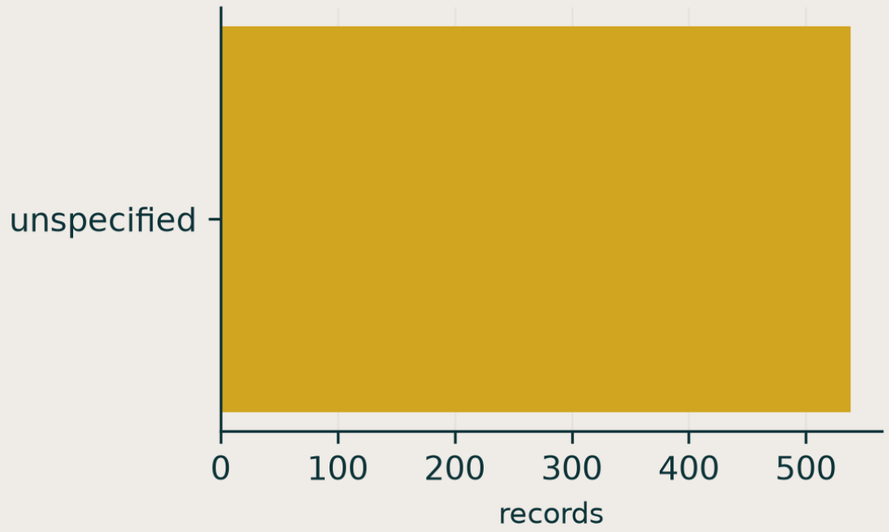
Protein X · Biogas X · N X · P X · K X

Element closure

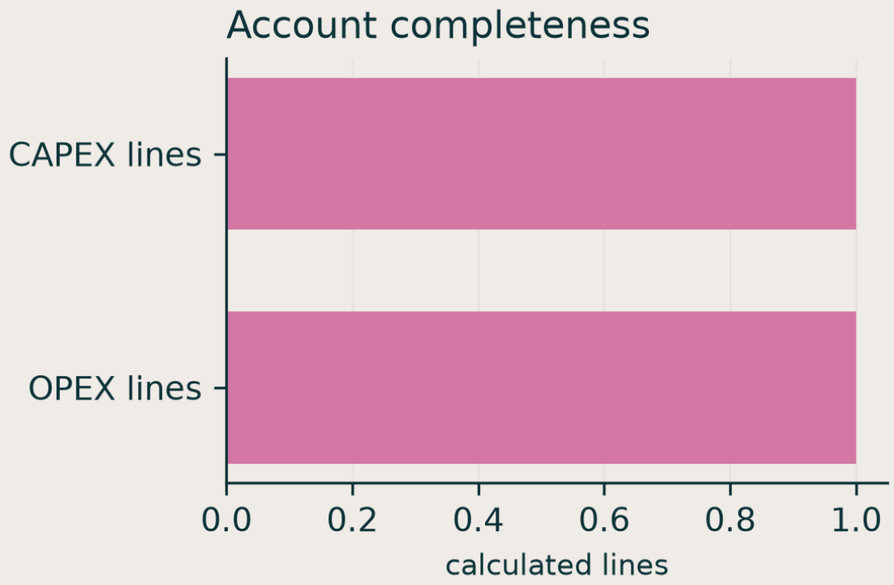
C	0 kg/year
N	0 kg/year
P	0 kg/year
K	0 kg/year

Numerical residuals

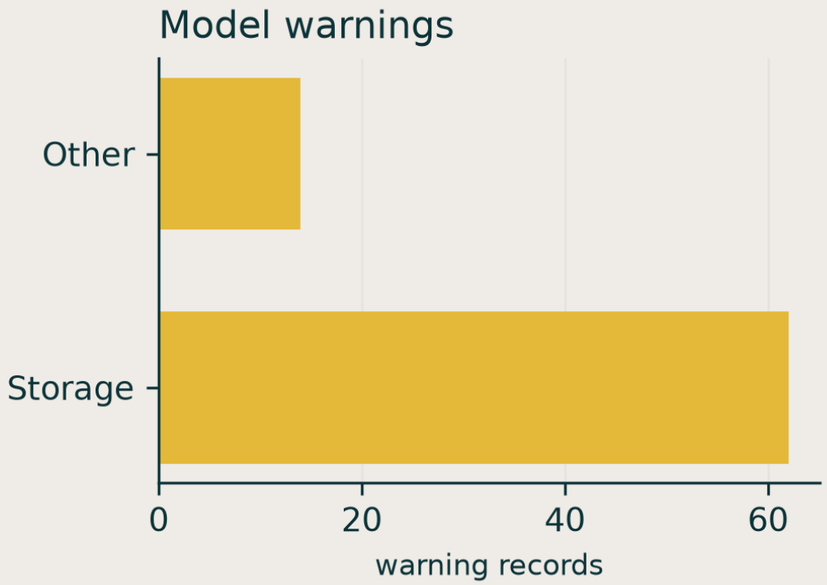
Parameter evidence



Account completeness



Model warnings



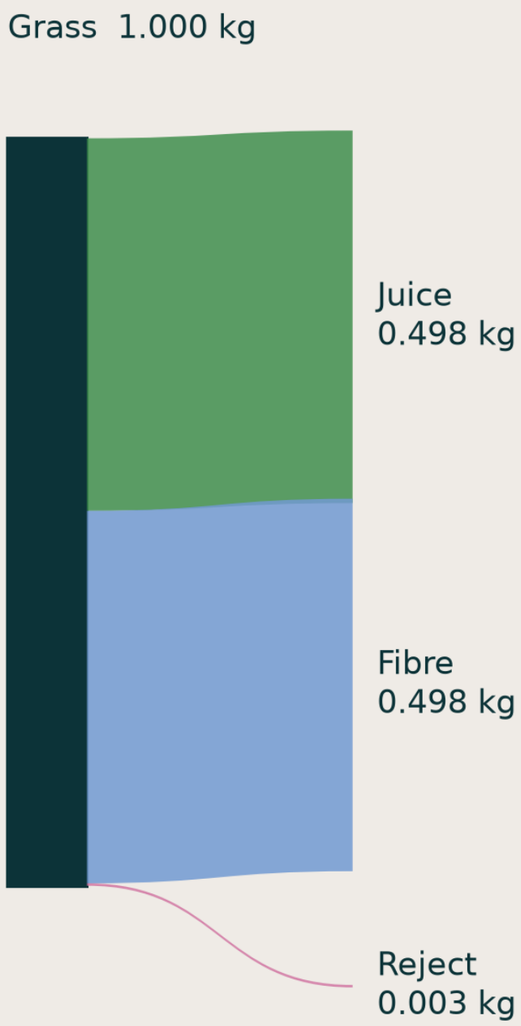
Numerical closure, evidence counts, calculated account lines and warnings

Scenario: protein_bypass rank 9; 6aa672cdb3cb45bfb5fcc01fd37205e1; 544.2 kg/h; 8760 h/year
Units: kg/year; records
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

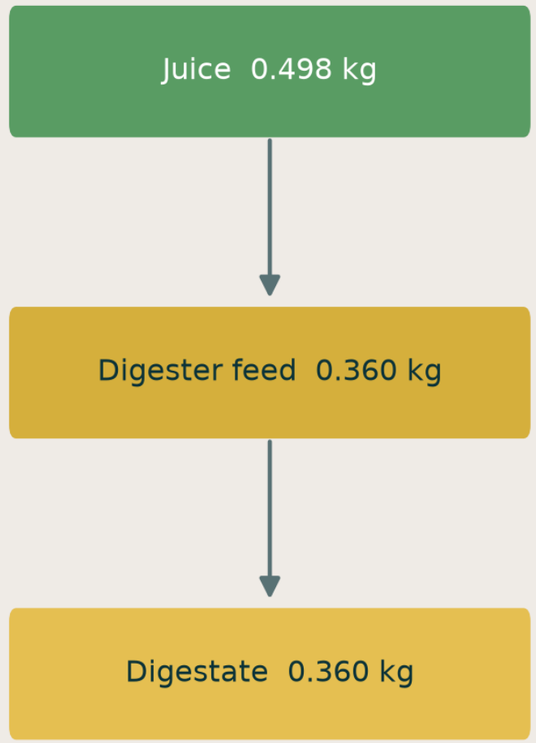
Grass to products

Protein X · Biogas X · N X · P X · K X

Grass split



Juice pathway



Selected intermediate streams

Recovered outputs

per kg wet grass

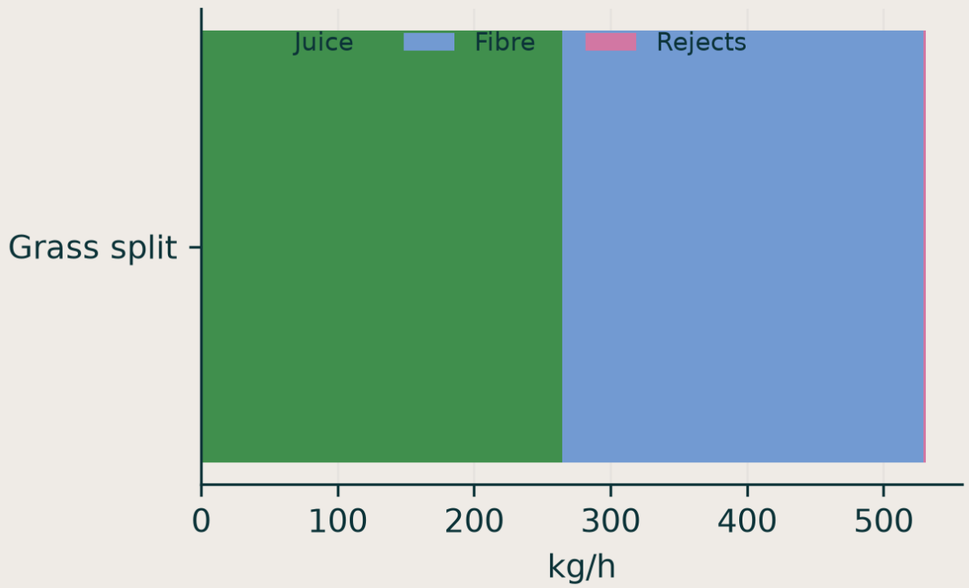
Mass-proportional grass split plus modeled downstream stream/product map

Scenario: protein_bypass rank 10; af77a6a1831d4f95895a5a6082b208cc; 531.1 kg/h; 8760 h/year
Units: kg/kg wet grass; dry Nm³/kg wet grass
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Material and products

Protein X · Biogas X · N X · P X · K X

Feed destinations



Solid products

No modeled output

Gas at normal conditions

No modeled output

Specific product yields

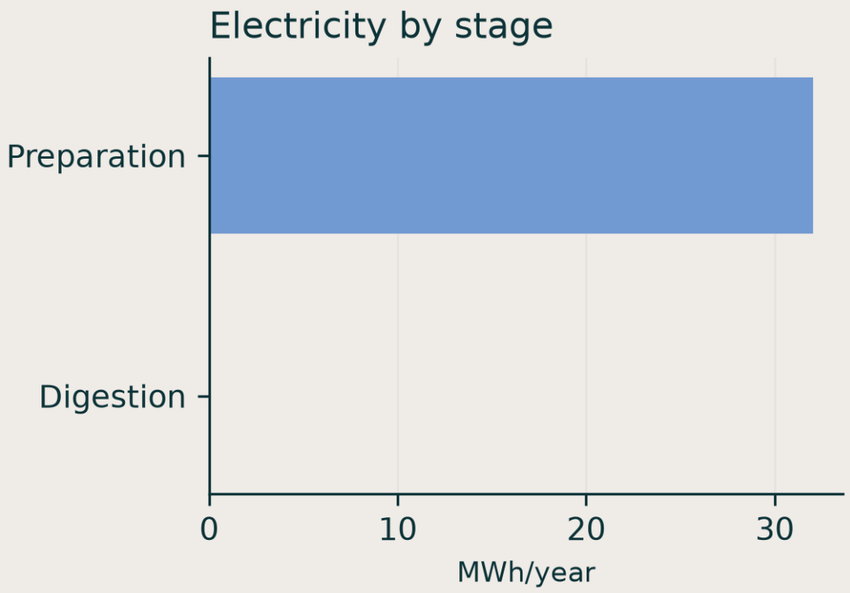
No modeled output

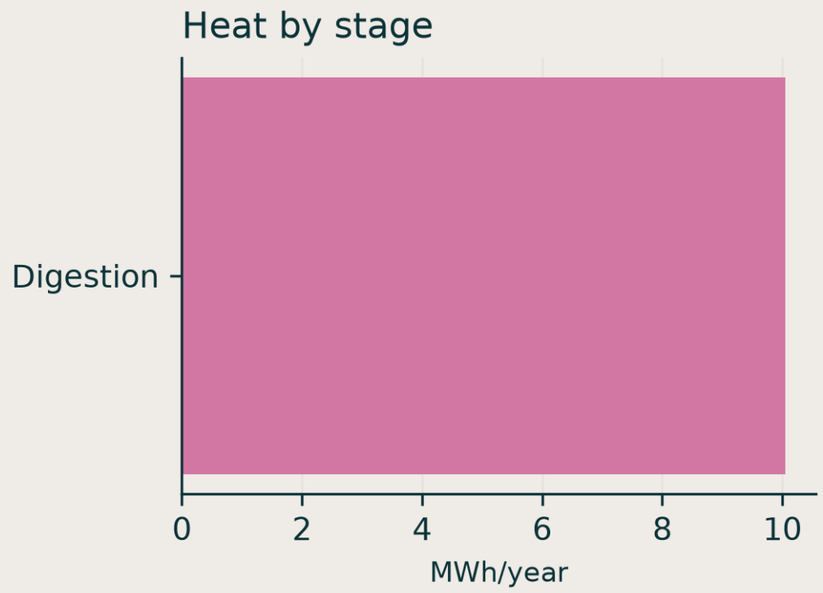
Front-end split, solid products, dry gas volumes and specific yields

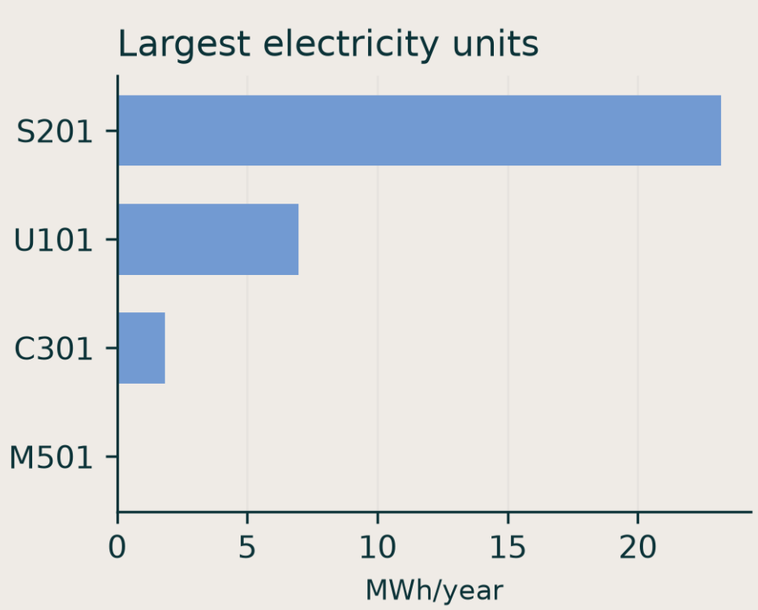
Scenario: protein_bypass rank 10; af77a6a1831d4f95895a5a6082b208cc; 531.1 kg/h; 8760 h/year
Units: kg/h; t/year; dry Nm³/year; kg/t feed
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

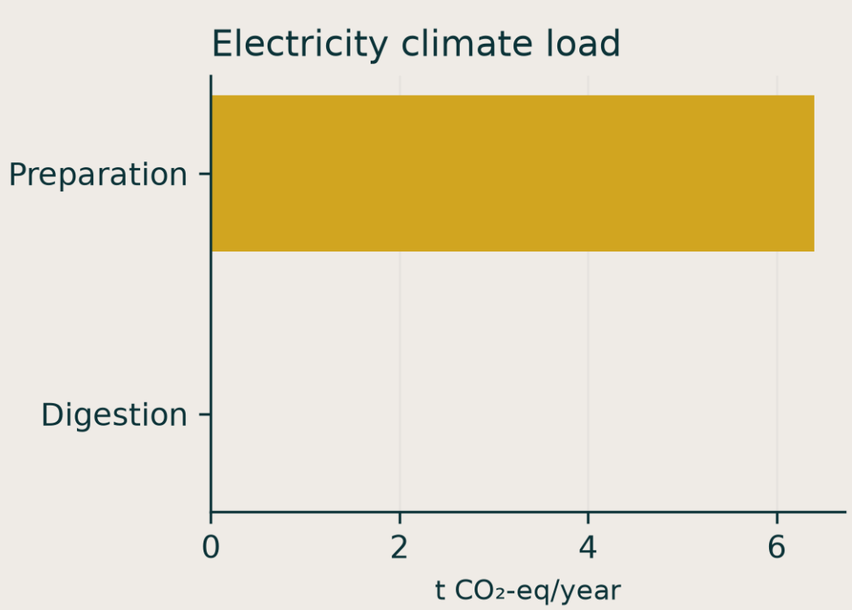
Energy and climate hotspots

Protein X · Biogas X · N X · P X · K X









Stage and unit utility hotspots and included climate load

Scenario: protein_bypass rank 10; af77a6a1831d4f95895a5a6082b208cc; 531.1 kg/h; 8760 h/year
Units: MWh/year; t CO_2-eq/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

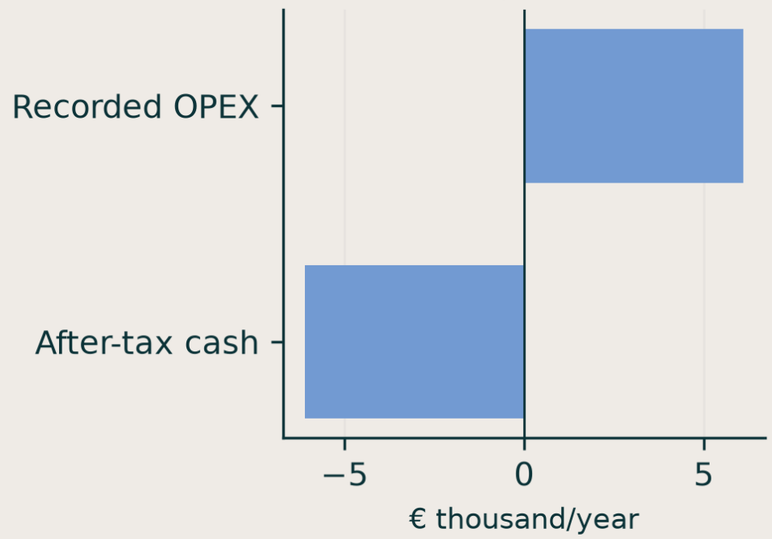
Economic accounts

Protein X · Biogas X · N X · P X · K X

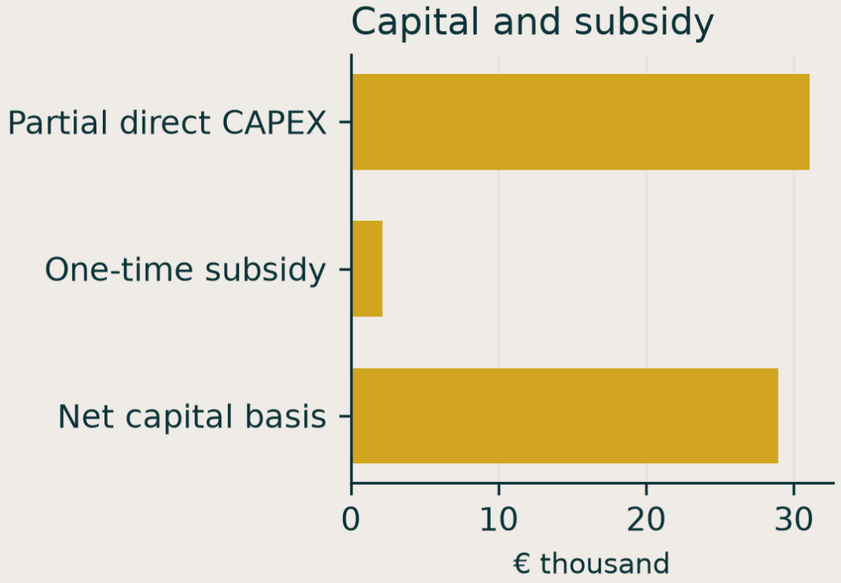
Product revenue

No modeled output

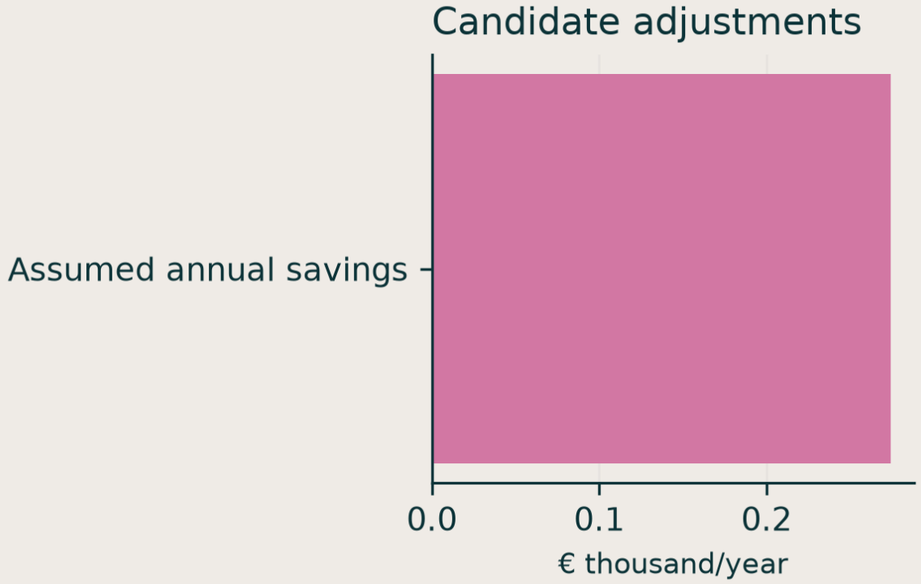
Annual budget



Capital and subsidy



Candidate adjustments

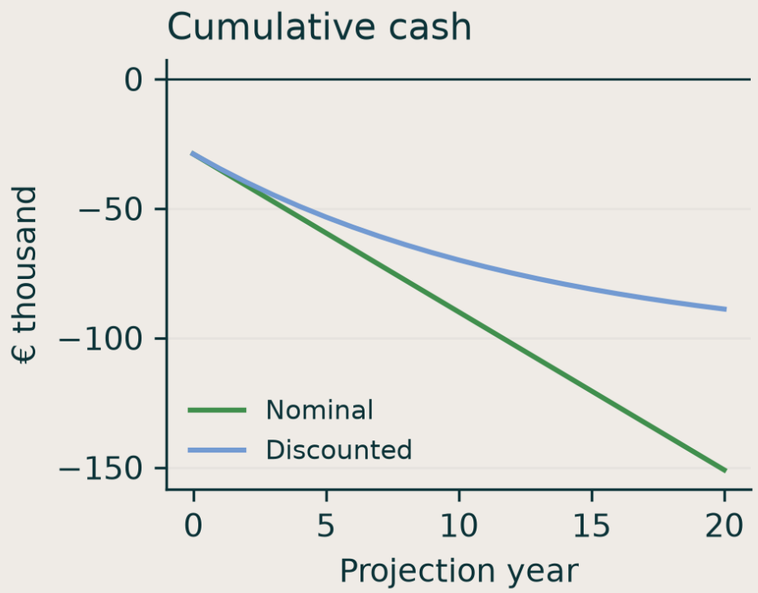


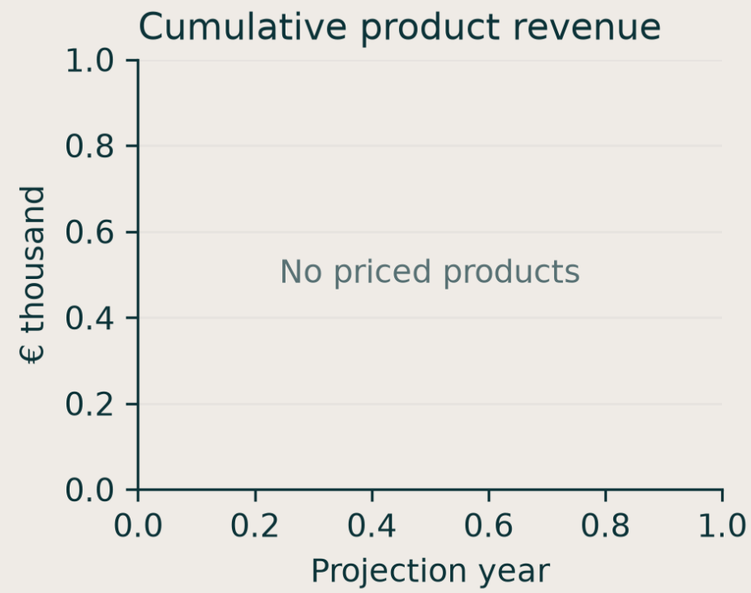
Product revenue, partial annual budget, capital subsidy and candidate overlays

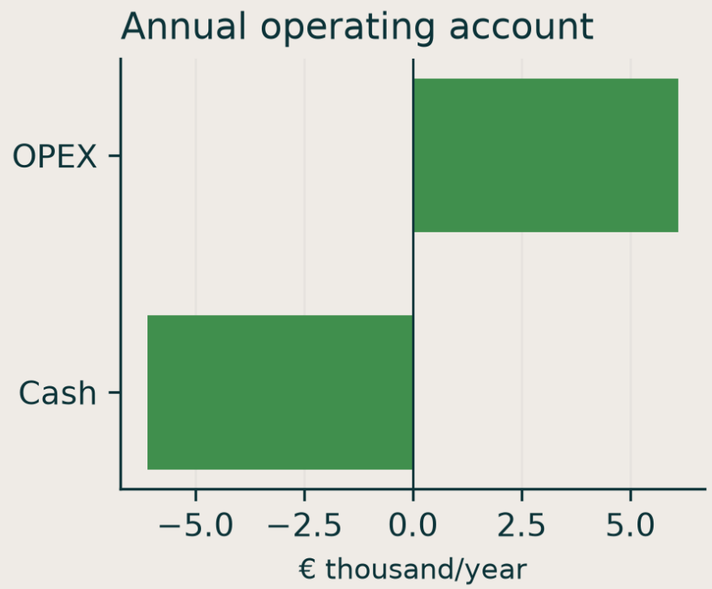
Scenario: protein_bypass rank 10; af77a6a1831d4f95895a5a6082b208cc; 531.1 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

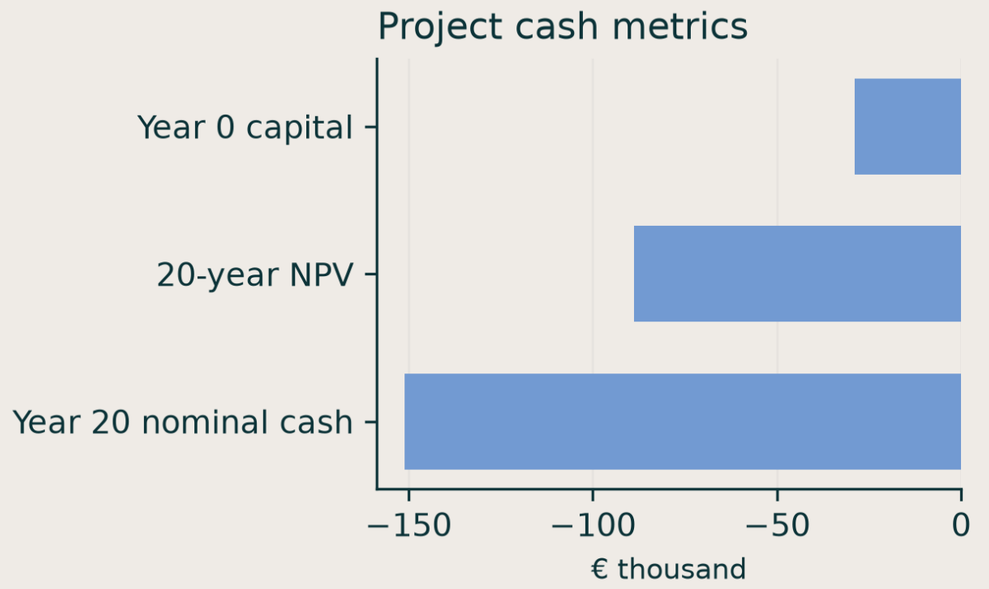
20-year financial projection

Protein X · Biogas X · N X · P X · K X









Constant-account cash flow and cumulative product revenue over 20 years

Scenario: protein_bypass rank 10; af77a6a1831d4f95895a5a6082b208cc; 531.1 kg/h; 8760 h/year
Units: € thousand; € thousand/year
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Balance and evidence

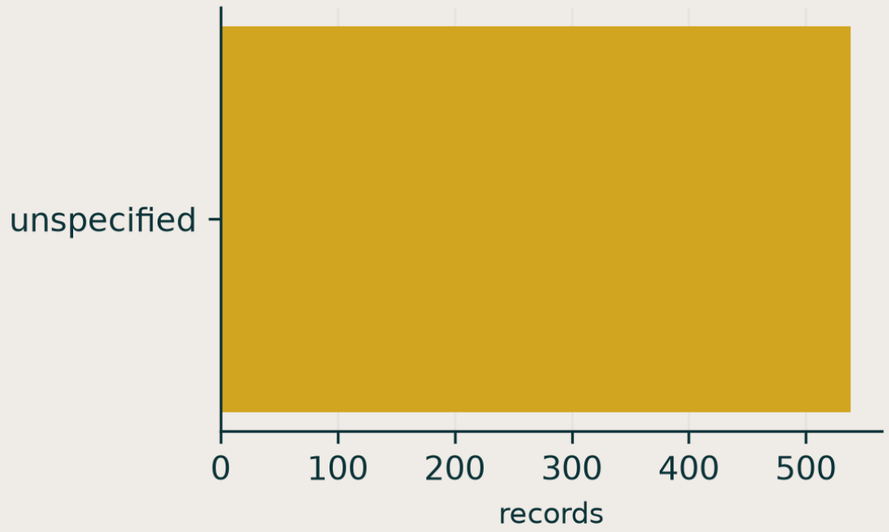
Protein X · Biogas X · N X · P X · K X

Element closure

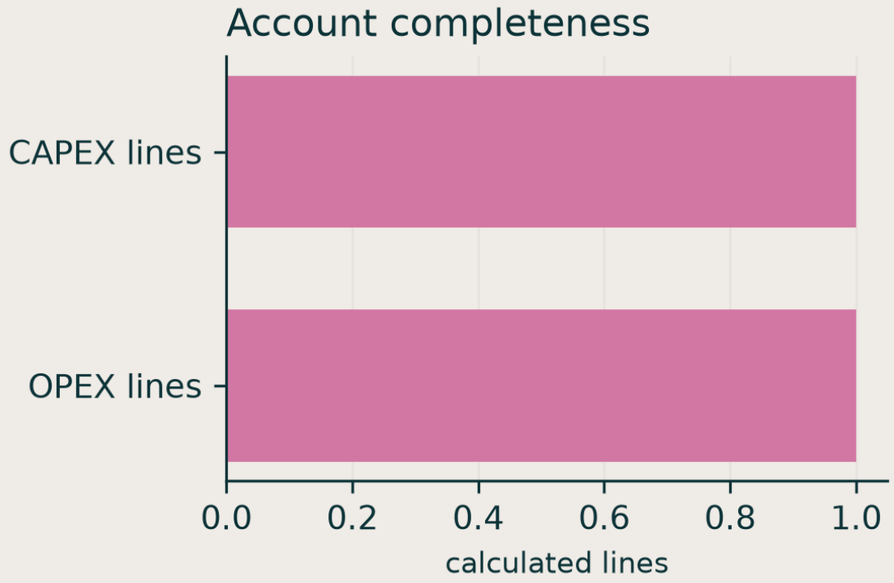
C	0 kg/year
N	0 kg/year
P	0 kg/year
K	0 kg/year

Numerical residuals

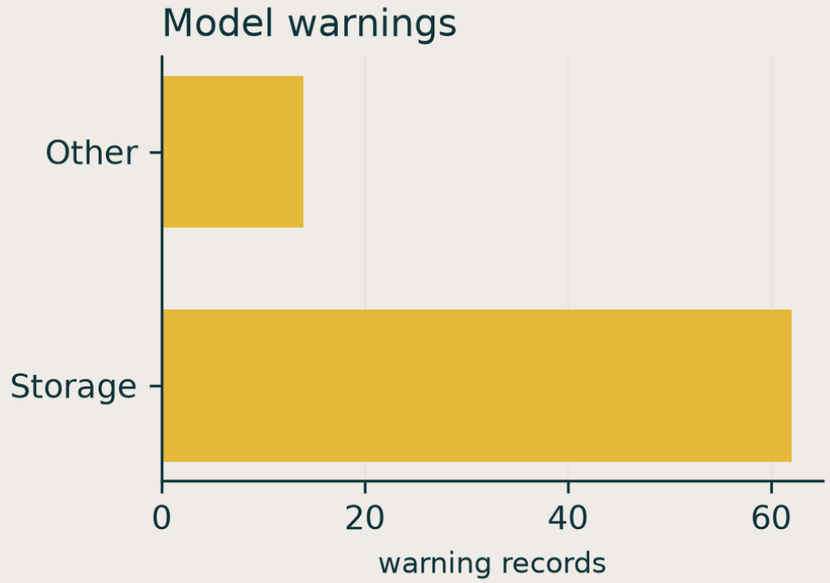
Parameter evidence



Account completeness



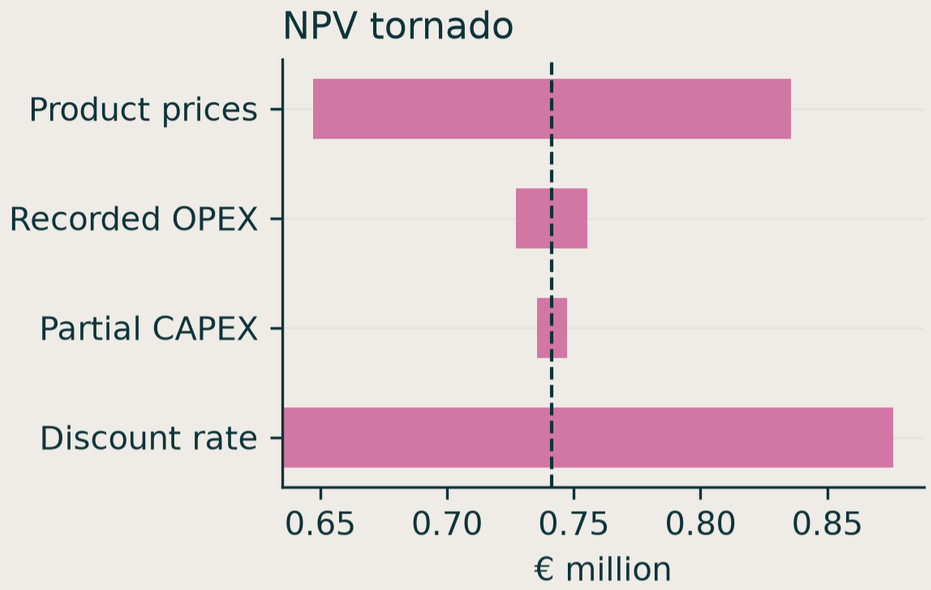
Model warnings

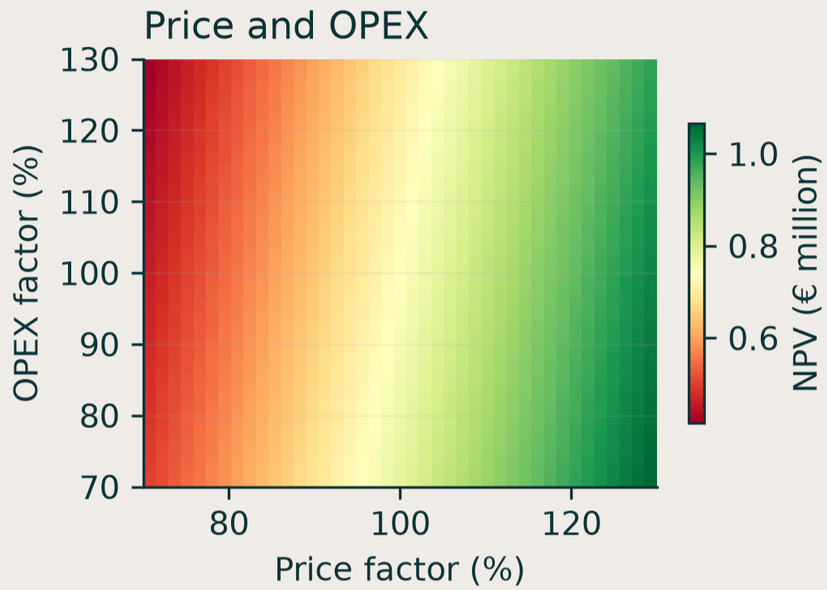


Numerical closure, evidence counts, calculated account lines and warnings

Scenario: protein_bypass rank 10; af77a6a1831d4f95895a5a6082b208cc; 531.1 kg/h; 8760 h/year
Units: kg/year; records
Source: Saved BioSTEAM decision reports and Greenhub/results/runs/{run_id}/streams.json; candidate economic overlay

Economic sensitivity

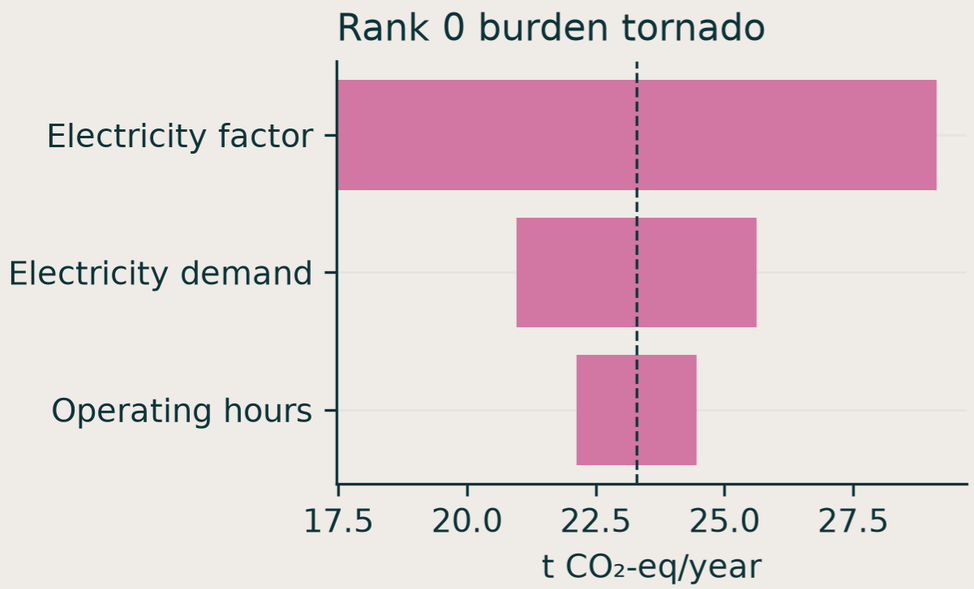


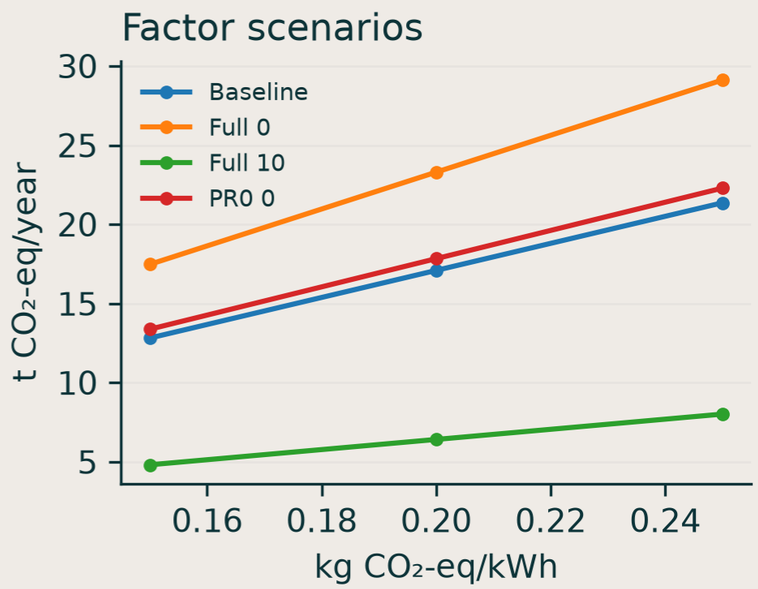


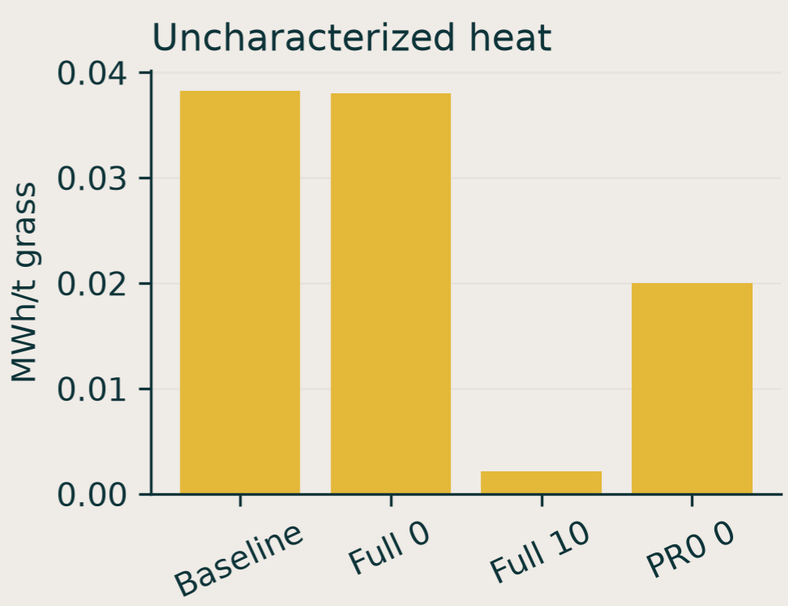
Deterministic one-factor NPV sensitivity and two-factor price/OPEX surface for full-front rank 0

Scenario: F1 non-PEF saved BioSTEAM batch and selected exact reruns
Units: € million; %
Source: Full-front rank 0 recorded revenue, OPEX and partial capital

Environmental sensitivity



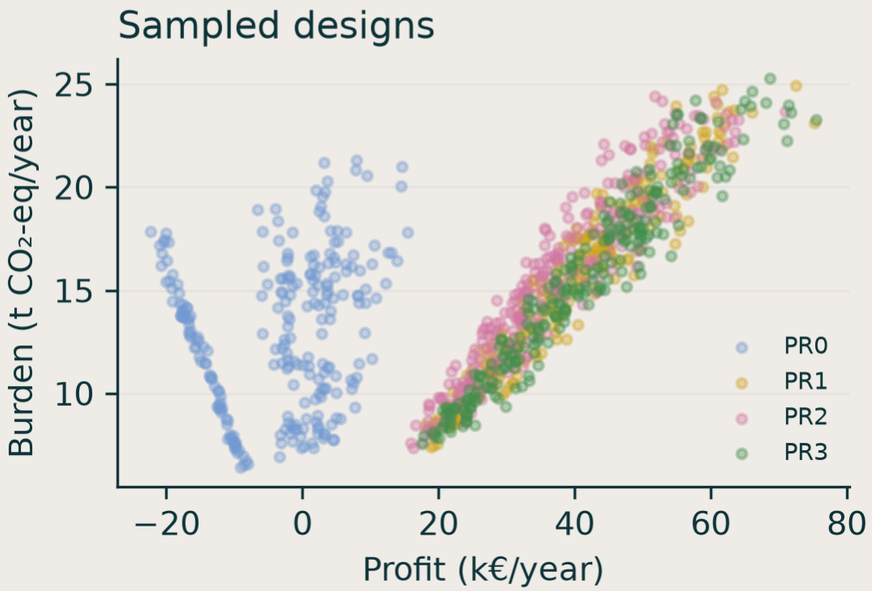


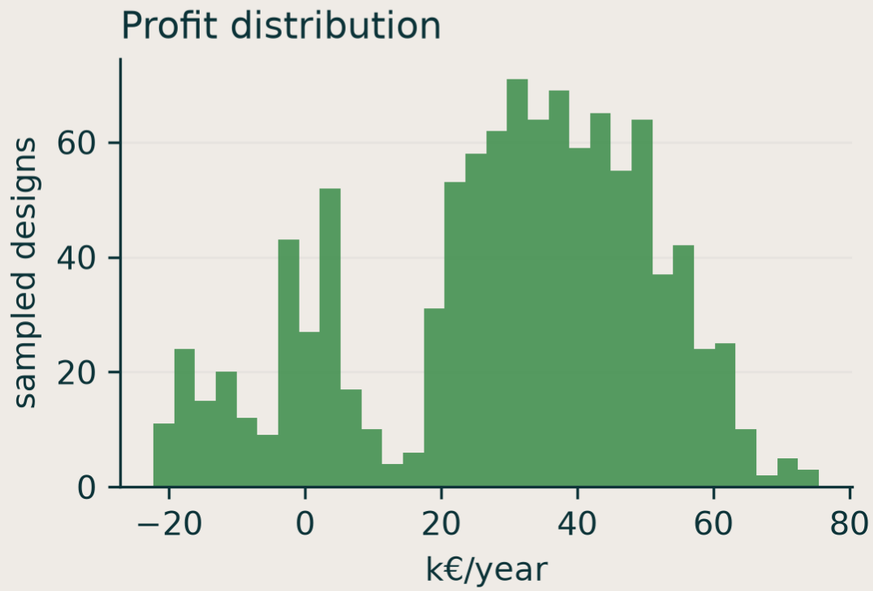


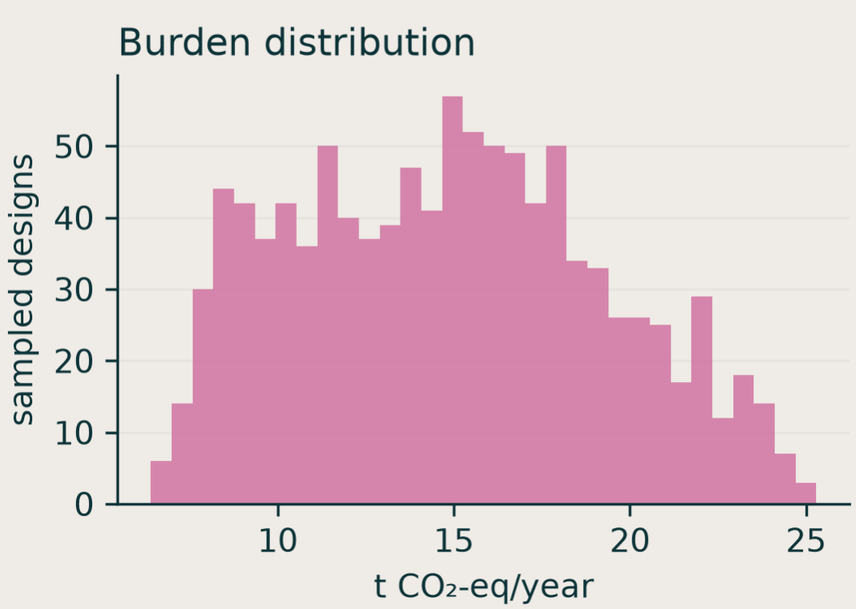
Deterministic electricity-factor, demand and operating-hours sensitivity of partial attributional burden

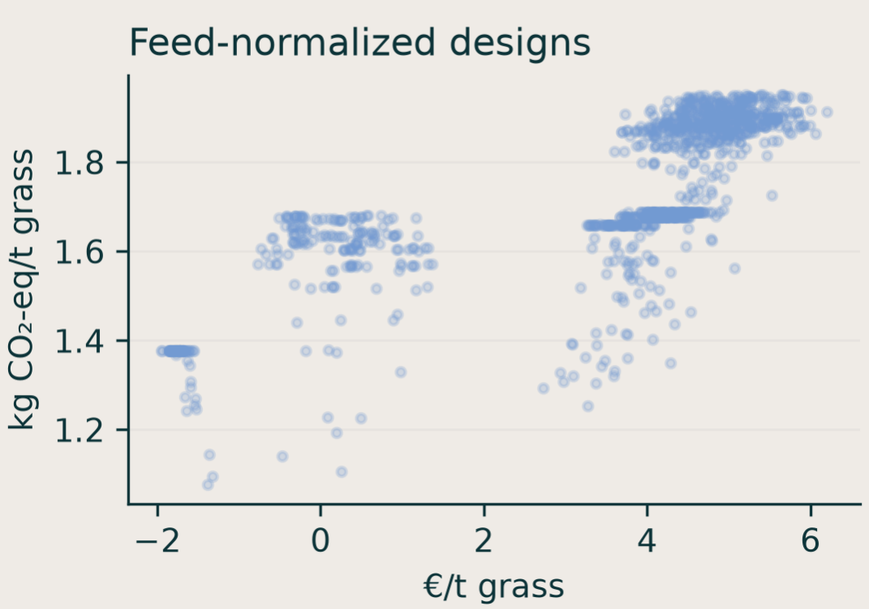
Scenario: F1 non-PEF saved BioSTEAM batch and selected exact reruns
Units: t CO_2-eq/year; kg CO_2-eq/t; MWh/t
Source: Exact F1 electricity and heat inventories

Non-PEF design sweep





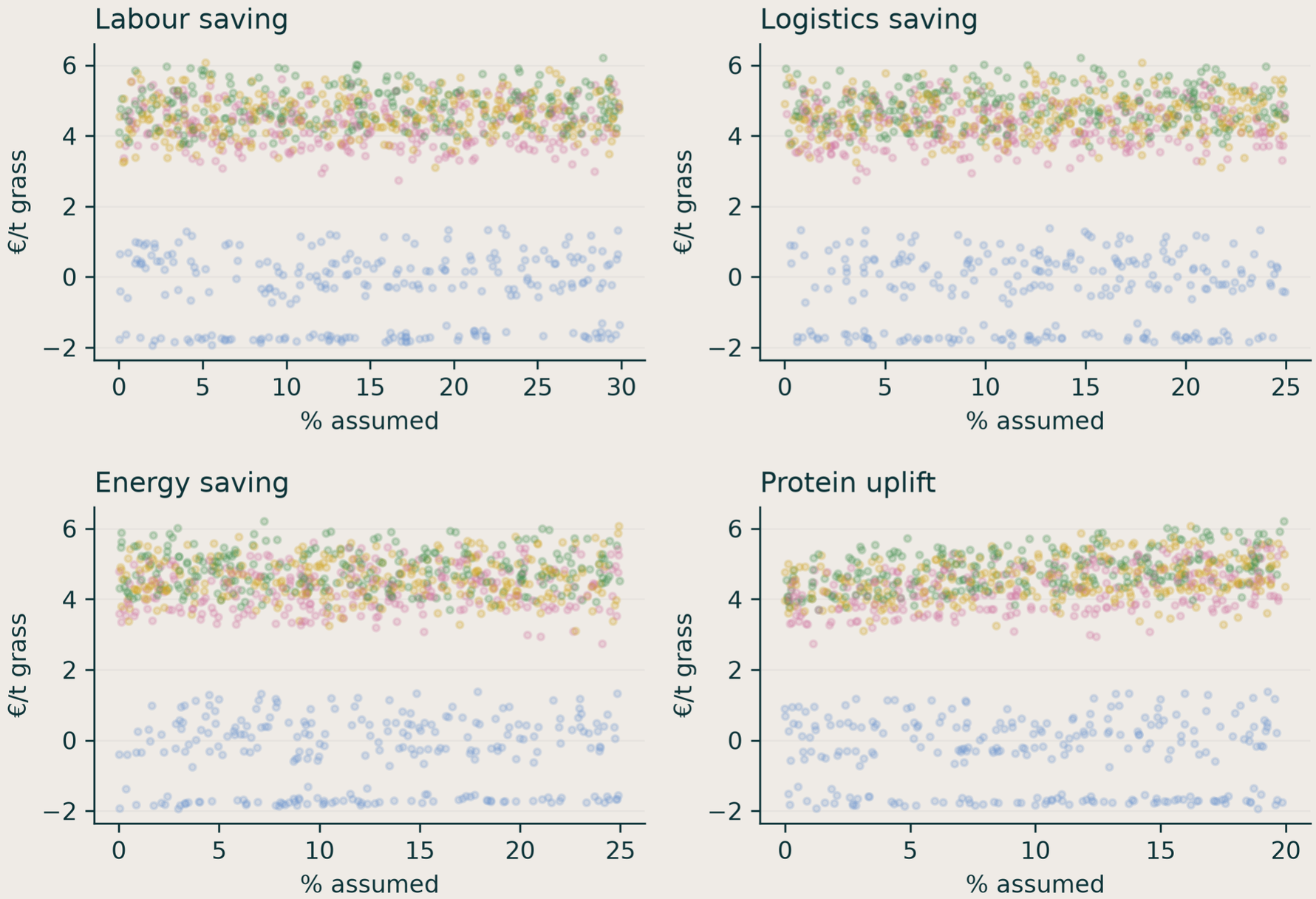




Distribution of 1049 eligible saved non-PEF random design evaluations

Scenario: F1 non-PEF saved BioSTEAM batch and selected exact reruns
Units: k€/year; t CO₂-eq/year; €/t; kg CO₂-eq/t
Source: C:\my\Spaak\Greenhub\Model\Greenhub\results\actual_pareto_2000\decision_batches\eed439a280ad488aa235a19b0efc32af.json

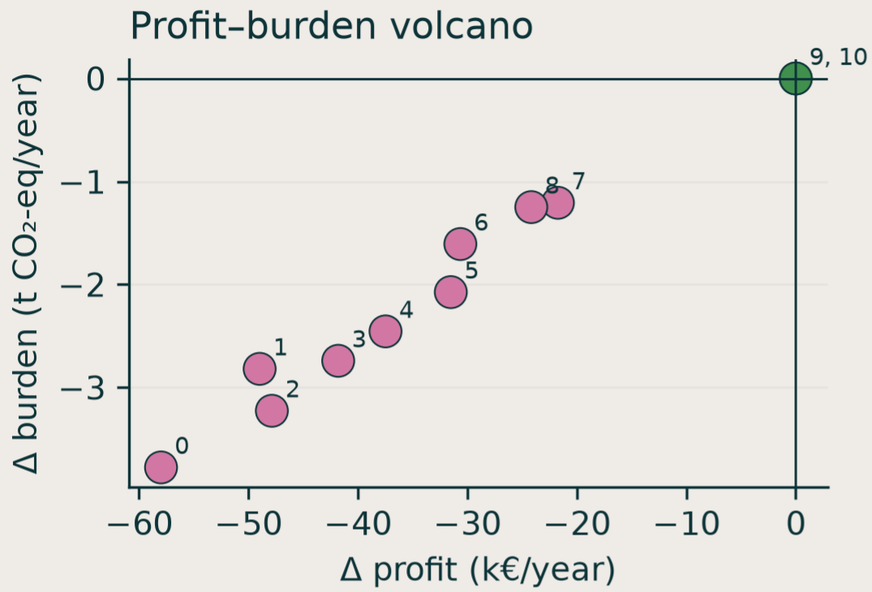
Sampled parameter response

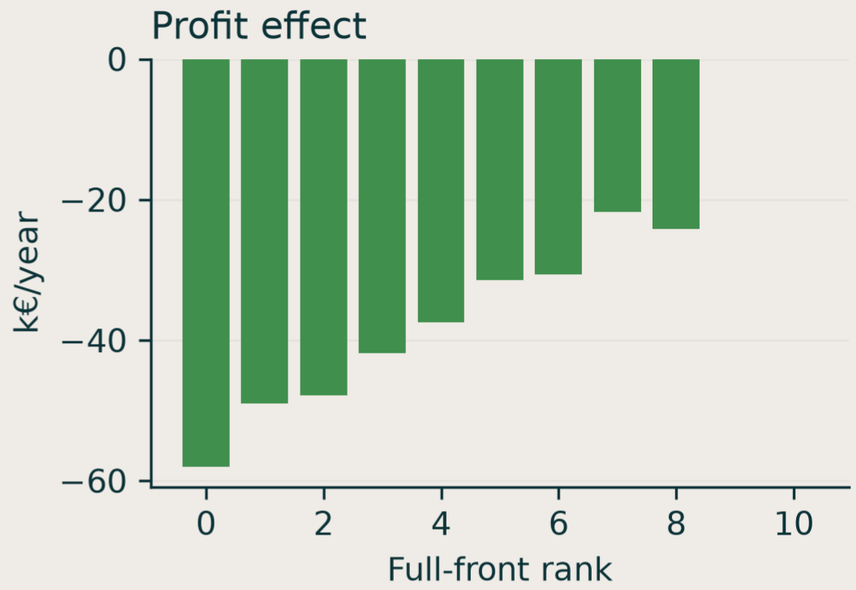


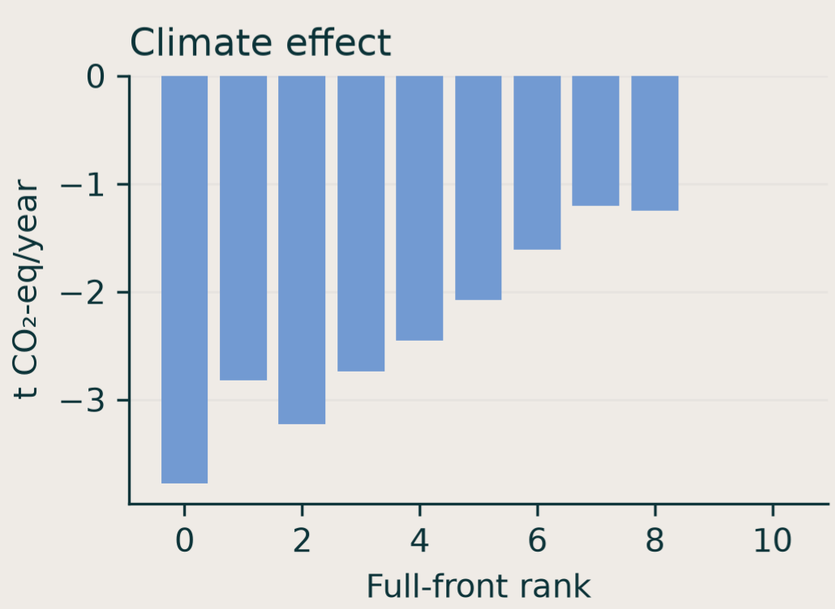
Observed screening profit intensity against sampled candidate economic parameters

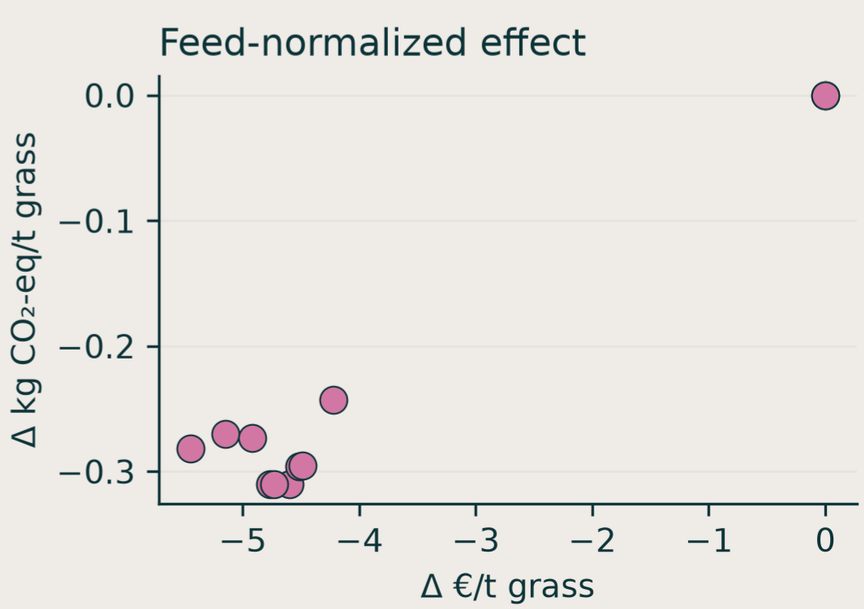
Scenario: F1 non-PEF saved BioSTEAM batch and selected exact reruns
Units: %; €/t wet grass
Source: C:\my\Spaak\Greenhub\Model\Greenhub\results\actual_pareto_2000\decision_batches\eed439a280ad488aa235a19b0efc32af.json

Protein bypass trade-offs





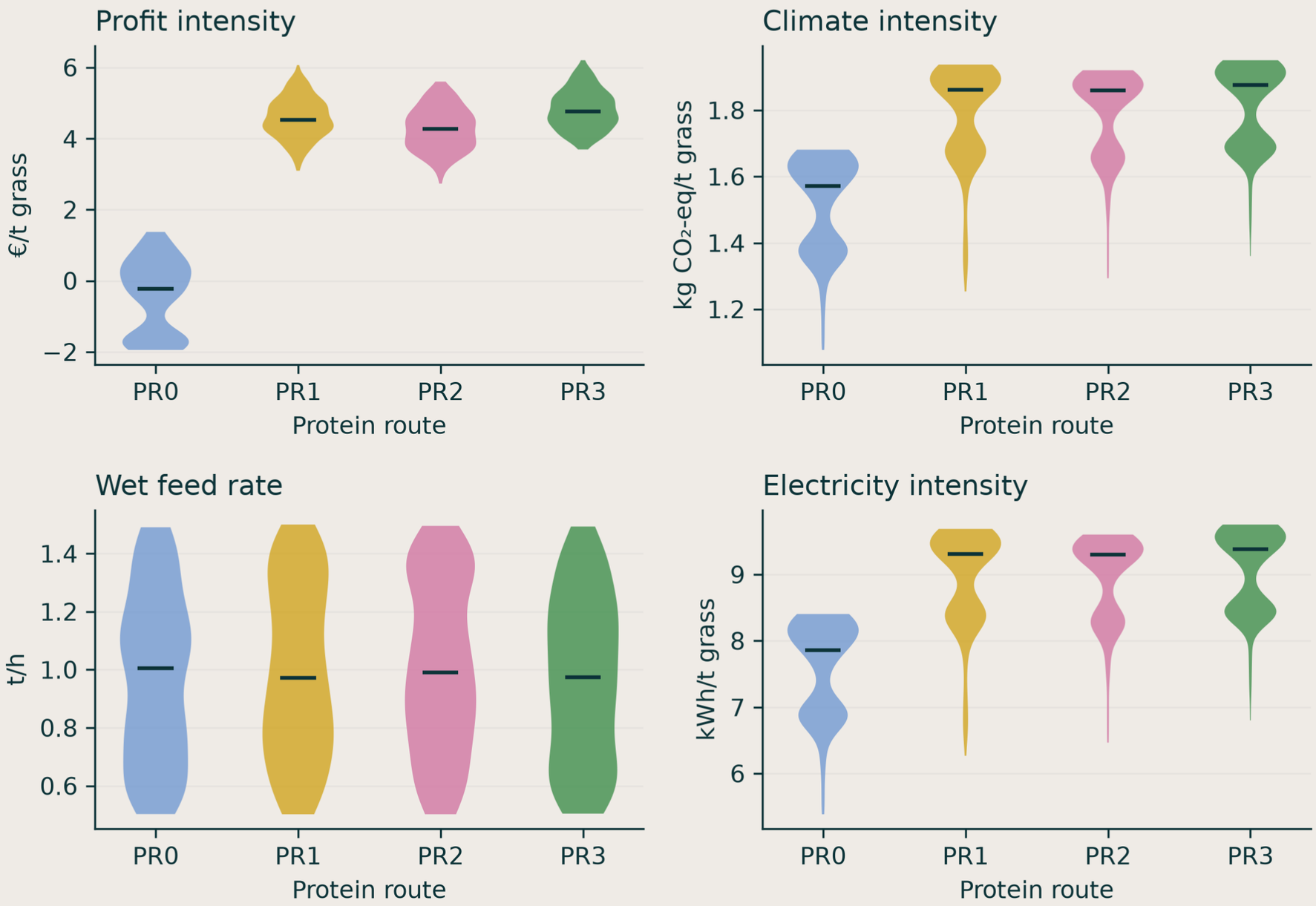




Paired protein-bypass economic and climate changes at fixed feed and economic assumptions

Scenario: F1 non-PEF saved BioSTEAM batch and selected exact reruns
Units: k€/year; t CO_2-eq/year; €/t; kg CO_2-eq/t
Source: Full-front and matched protein-bypass exact BioSTEAM reruns

Route design distributions



Distribution of actual sampled F1 design outcomes by protein technology route

Scenario: F1 non-PEF saved BioSTEAM batch and selected exact reruns
Units: €/t; kg CO_2-eq/t; t/h; kWh/t
Source: C:\my\Spaak\Greenhub\Model\Greenhub\results\actual_pareto_2000\decision_batches\eed439a280ad488aa235a19b0efc32af.json